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SOLID STATE

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NEET SYLLABUS

Solid State : Classification of solids based on different binding forces; molecular, ionic covalent and metallic solids, amorphous and crystalline solids (elementary idea), unit cell in two dimensional and three dimensional lattices, calculation of density of unit cell, packing in solids, packing efficiency, voids, number of atoms per unit cell in a cubic unit cell, point defects, (electrical and magnetic properties, Band theory of metals, conductors, semiconductors and insulators.) (Explained in Physics)

OBJECTIVES

After studying this unit, we will be able to :

- *describe general characteristics of solids.*
- *distinction between amorphous and crystalline solids.*
- *study crystal lattice and unit cell.*
- *correlate the density of a substance with its unit cell properties.*
- *explain close packing of particles.*
- *study type of voids and ionic crystals.*
- *describe the imperfections in solids and their effect on properties.*

*"Chemistry without knowledge of solids ; would be a sword without a handle ;
a light without brilliance ; a bell without sound"*

Alwin Mittasch

SOLID STATE

1.0 INTRODUCTION

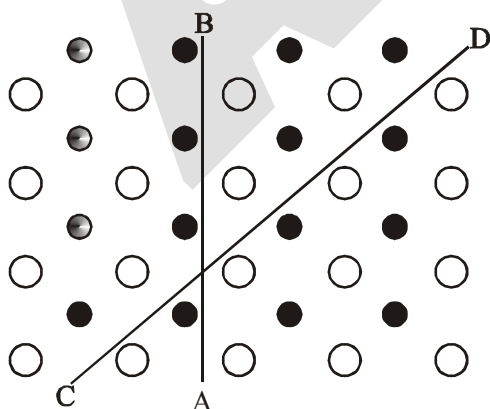
The solid are characterised by incompressibility, rigidity and mechanical strength. The molecules, atoms or ions in solids are closely packed means they are held together by strong forces and can not move randomly. Thus solids have definite volume, shape, slow diffusion, low vapour pressure and possesses the unique property of being rigid. Such solids are known as **true solids** like NaCl, KCl, Sugar, Ag, Cu etc. On the other hand the solid which loses shapes on long standing, flows under its own weight and easily distorted by even mild distortion forces are called **pseudo solids** such as glass, pitch etc.

Some solids such as NaCl, Sugar, Sulphur etc. have properties not only of rigidity and incompressibility but also of having typical geometrical forms. These solids are called as **crystalline solids**. In such solids there is definite arrangements of particles (atoms, ions or molecules) throughout the entire three dimensional network of a crystal in long-range order. This three dimensional arrangement is called **crystal lattice or space lattice**. Other solids such as glass, rubber, plastics etc. have rigidity and incompressibility to a certain extent but they do not have definite geometrical forms or do not have long range order are known as **amorphous solids**.

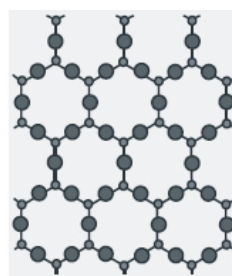
Distinction between Crystalline and Amorphous Solids

Property	Crystalline solids	Amorphous solids
Shape	Definite characteristic geometrical shape	Irregular shape
Melting point	Melt at a sharp and characteristic temperature	Gradually soften over a range of temperature
Cleavage property	When cut with a sharp edged tool, they split into two pieces and the newly generated surfaces are plain and smooth	When cut with a sharp edged tool, they cut into two pieces with irregular surfaces
Heat of fusion	They have a definite and characteristic heat of fusion	They do not have definite heat of fusion
Anisotropic nature	Anisotropic in nature (direction dependent) True solids	Isotropic in nature (direction independent) Pseudo solids or super cooled liquids
Arrangement of particles	Long range order	Only short range order.
Examples	NaCl, Metals, Diamond	Rubber, Plastics

GOLDEN KEY POINTS



Anisotropy in crystals is due to different arrangement of particles along different directions.



(a)



(b)

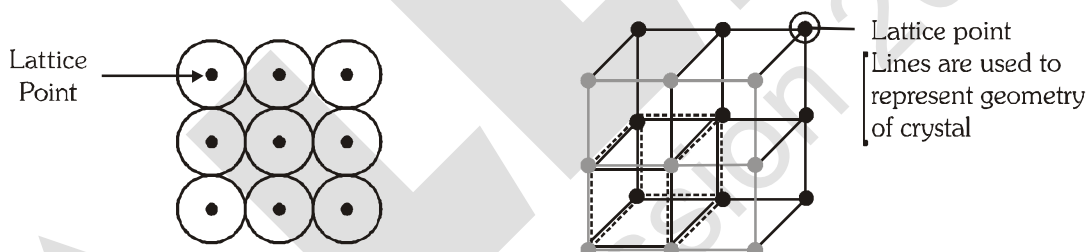
Two dimensional structure of
(a) quartz and (b) quartz glass

Table (a) : Classification of crystalline solids

S. No.	Type of Solid	Constituent Particles	Bonding/ Attractive Forces	Examples	Physical Nature	Electrical Conductivity	Melting Point
1.	Molecular solids (i) Non polar	Molecules	Dispersion or London forces	Ar, CCl ₄ , H ₂ , I ₂ , CO ₂	Soft	Insulator	Very low
	(ii) Polar		Dipole-dipole interactions	HCl, SO ₂	Soft	Insulator	Low
	(iii) Hydrogen bonded		Hydrogen bonding	H ₂ O (ice)	Hard	Insulator	Low
2.	Ionic solids	Ions	Coulombic or electrostatic	NaCl, MgO, ZnS, CaF ₂	Hard but brittle	Insulators in solid state but conductors in molten and in aqueous state	High
3.	Metallic solids	Positive metal ions in a sea of delocalised e ⁻	Metallic bonding	Fe, Cu, Ag, Mg	Hard but malleable and ductile	Conductors in solid and in molten state	Fairly high
4.	Covalent or network solids	Atoms	Covalent bonds	SiO ₂ , SiC, AlN, C(diamond), C(graphite)	Hard Soft	Insulators Conductor	Very high

1.1 Space Lattice/Crystalline Lattice/3-D Lattice

- Space lattice is a regular arrangement of lattice points (atoms or ions or molecules) showing how the particles are arranged at different sites in 3D-view.



- "The three dimensional distribution of component particles in a crystal can be found by X-ray diffraction of different faces of the crystal.
- On the basis of the classification of symmetry, the crystals have been divided into seven systems. These seven systems with the characteristics of their axes (Interfacial angles and intercepts) where some examples of each are given in the following table (b). These crystal systems differ in length of unit cell edges (a, b and c) and the angles between the unit cell edges. In cubic and trigonal (rhombohedral) systems, the three unit edges are of equal lengths but for the rest five systems it is not so. The interfacial angles are all 90° in the cubic, tetragonal and orthorhombic systems but it is not so for the rest four systems.

1.2 Unit Cell (U.C.)

Unit cell of the crystalline substance is defined as the smallest repeating portion which shows the complete geometry of the crystalline substance like brick in wall. A unit cell is the smallest portion of the whole crystal. A unit cell is characterized by the edge lengths a, b and c along the three axes of the unit cell and the angles α , β and γ between the pair of edges bc, ca and ab respectively.

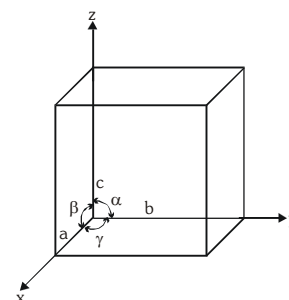
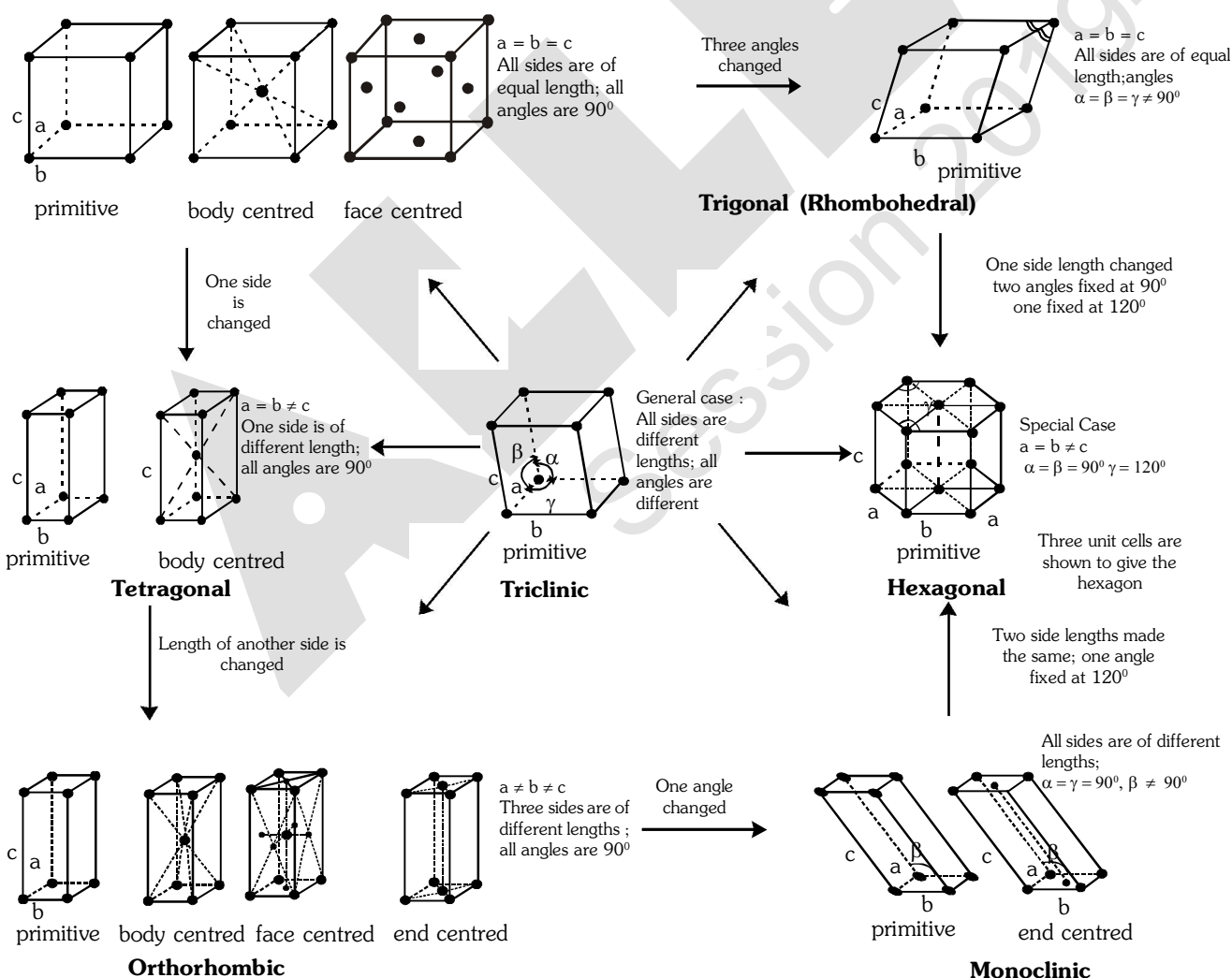


Table (b) : The Seven Crystal Systems (Bravais Lattice)

S. No.	System	Name of Lengths	Edge Angles	Bravais Lattices	Examples
1.	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	Primitive, Face-centred, Body centred = 3	NaCl, Zinc blende, Cu
2.	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Primitive, Body centred = 2	White tin, SnO_2 , TiO_2 , CaSO_4
3.	Orthorhombic or Rhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Primitive, Face-centred, Body centred, End centred = 4	Rhombic sulphur, KNO_3 , BaSO_4 , Match Box, Duster
4.	Rhombohedral or Trigonal	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	Primitive = 1	Calcite (CaCO_3), HgS (cinnabar)
5.	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	Primitive = 1	Graphite, ZnO, CdS
6.	Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$	Primitive, End centred = 2	Monoclinic sulphur, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$
7.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	Primitive = 1	$\text{K}_2\text{Cr}_2\text{O}_7$, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, H_3BO_3
				Total = 14	



1.3 Co-ordination Number

The number of nearest neighbouring particles around a specific particle in a given crystalline substance is called as co-ordination number of that crystalline substance.

1.4 Packing Efficiency or Packing fraction (P.E.)

"Packing efficiency is defined as the ratio of volume occupied by the atoms to the total volume of the crystalline substance"

$$\text{P.E.} = \frac{\text{Volume occupied by atoms present in a crystal}}{\text{Volume of crystal}}$$

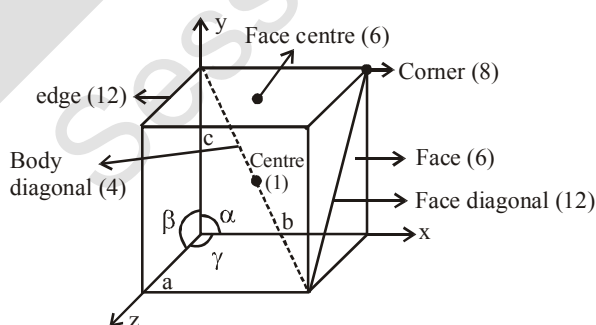
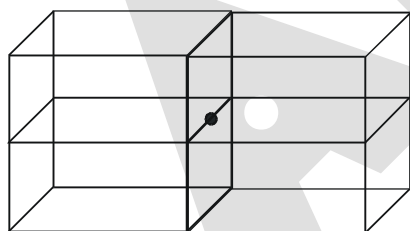
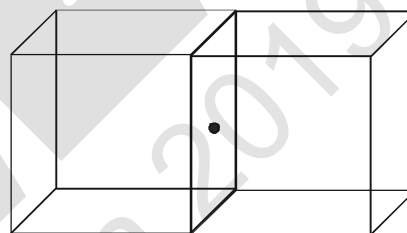
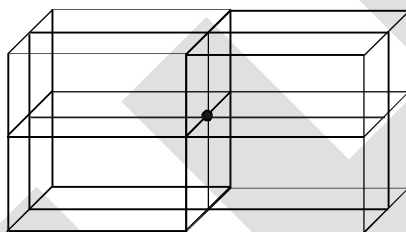
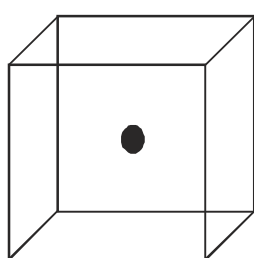
$$\text{P.E.} = \frac{\text{Volume occupied by atoms present in unit cell}}{\text{Volume of unit cell}} \quad \text{or} \quad \text{P.E.} = \frac{z \times (4/3)\pi r^3}{V}$$

Where z = number of atoms present in unit cell / Number of formula units for ionic crystals.

GOLDEN KEY POINTS

● In a cube

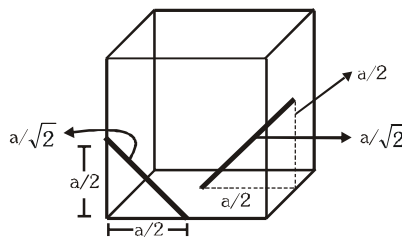
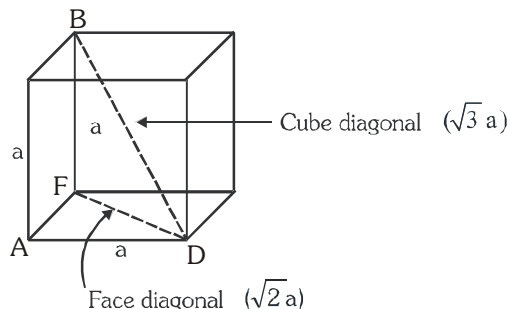
- | | |
|---------------------------------|----------------------------------|
| 1. Number of corners = 8 | 2. Number of faces = 6 |
| 3. Number of edges = 12 | 4. Number of body centre = 1 |
| 5. Number of body diagonals = 4 | 6. Number of face diagonals = 12 |



Contribution of an atom at different lattice points of cube :

- A corner of a cube is common in 8 cubes. So $\frac{1}{8}$ th part of an atom is present at this corner of cube.
- A face of a cube is common in 2 cubes. So $\frac{1}{2}$ th part of an atom is present at the face of a cube.
- An edge of a cube is common in four cubes, so $\frac{1}{4}$ th part of the atom is present at the edge of a cube.
- A cube centre is not common in any another cube, so one complete atom is present at the cube centre.

1.5 Length of Face Diagonal and Cube Diagonal



$$\text{Distance between 2 adjacent face centres} = \frac{a}{\sqrt{2}} = \frac{a\sqrt{2}}{2}$$

$$\text{Distance between 2 adjacent edge centres} = \frac{a}{\sqrt{2}} = \frac{a\sqrt{2}}{2}$$

Consider the triangle AFD (with the help of pythagororous theorem)

$$FD = \sqrt{AF^2 + AD^2} = \sqrt{a^2 + a^2} = \sqrt{2}a \quad (\text{length of face diagonal.})$$

Consider the triangle $\angle BFD$ (with the help of pythagororous theorem)

$$BD = \sqrt{BF^2 + FD^2} = \sqrt{a^2 + (\sqrt{2}a)^2} = \sqrt{3}a \quad (\text{length of cube diagonal})$$

Illustrations

Illustration 1. Which of the following is a non-crystalline solid ?

- (1) CsCl (2) NaCl (3) CaF_2 (4) Glass

Solution. **Ans. (4)**

Illustration 2. Which of the following statements is incorrect about amorphous solids?

- (1) They are anisotropic
(2) They are rigid and incompressible
(3) They melt over a wide range of temperature
(4) There is no orderly arrangement of particles

Solution **Ans. (1)**

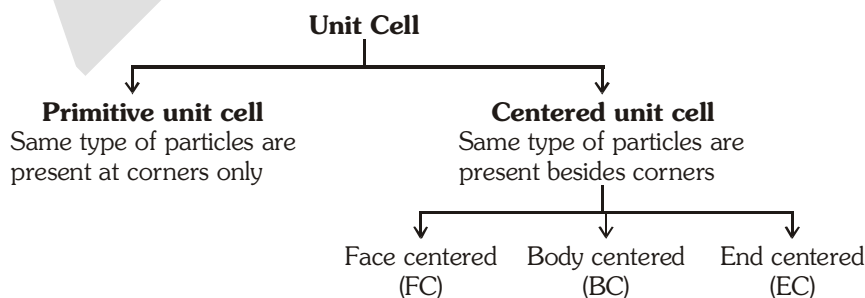
Illustration 3. **Assertion :** Crystalline solids are anisotropic.

Reason : The constituent particles are very closely packed.

- (1) A (2) B (3) C (4) D

Solution. **Ans. (2)**

1.6 (I) CLASSIFICATION OF UNIT CELL (As per Bravais)



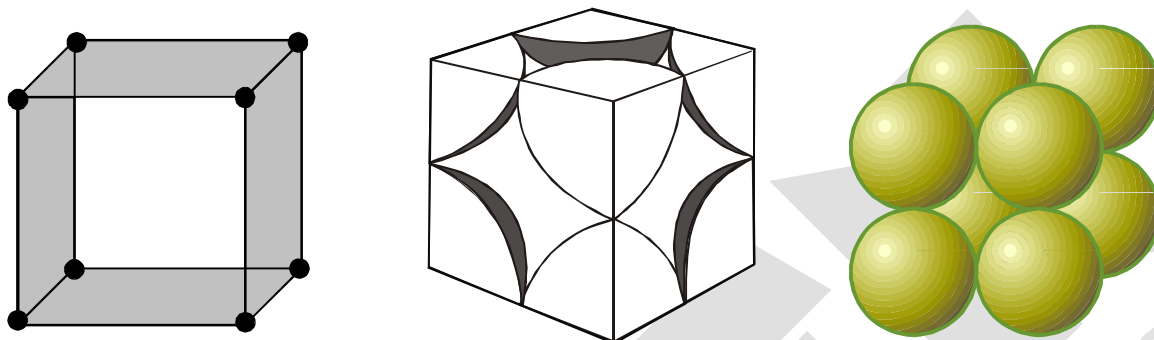
- In end centered same type of particles are present at corners and any two opposite face centres.
- End centered type of Bravais lattice is present only in orthorhombic and monoclinic type unit cell.

(II) TYPES OF CUBIC UNIT CELL

The distance between successive lattice planes of the same type is called the spacing of planes or interplanar distance between the planes. On the basis of this aspect, the lattices may be divided in following classes :

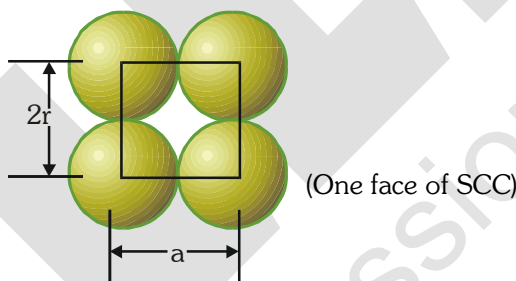
(A) SIMPLE/PRIMITIVE/BASIC CUBIC UNIT CELL

A unit cell having lattice point only at corners is called primitive or simple unit cell. i.e. in this case there is one atom at each of the eight corners of the unit cell considering an atom at one corner as the centre, it will be found that this atom is surrounded by six equidistant neighbours (atoms) and thus the co-ordination number will be six. If 'a' is the side of the unit cell, then the distance between the nearest neighbours shall be equal to 'a'.



(a) **Relationship between edge length 'a' and atomic radius 'r' :** $a = 2r$ i.e.

$$r = \frac{a}{2}$$



(b) **Number of atoms present in unit cell (z) :** In this case one atom lies at each corner.

Hence simple cubic unit cell contains a total of $\frac{1}{8} \times 8 = 1$ atom / unit cell.

(c) **Packing efficiency (P. E.) :**

$$\text{P.E.} = \frac{\text{Volume occupied by atoms present in unit cell}}{\text{Volume of unit cell}} = \frac{z \times \frac{4}{3} \pi r^3}{V} \left[\because \text{Volume of atom} = \frac{4}{3} \pi r^3 \right]$$

$$\text{For SC : P.E.} = \frac{1 \times \frac{4}{3} \times \pi \times \left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.524 \text{ or } 52.4\% \quad \left[\because r = \frac{a}{2} \text{ and } V = a^3, z = 1 \right]$$

In SC, 52.4% of total volume is occupied by atoms

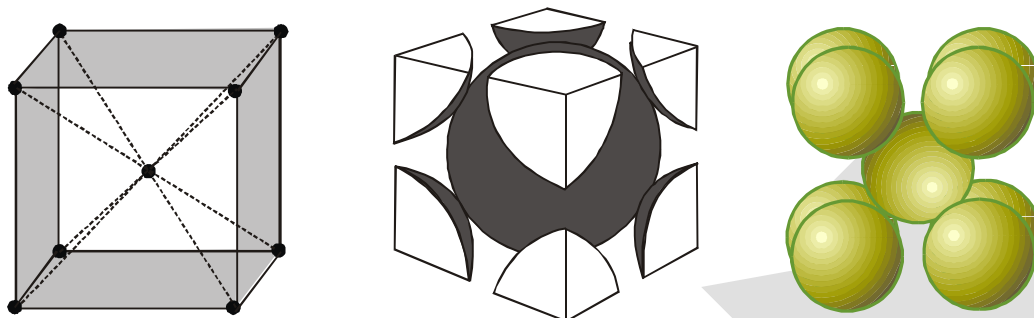
% void space = 47.6

(B) BODY CENTRED CUBIC UNIT CELL (BCC)

A unit cell having lattice point at the body centre in addition to the lattice point at every corner is called as body centered unit cell : where body diagonal particles are touching particle.

Here the central atom is surrounded by eight equidistant atoms and hence the co-ordination number is eight.

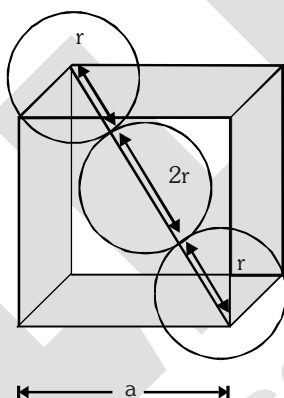
The nearest distance between two atoms will be $\frac{a\sqrt{3}}{2}$



(a) Relationship between edge length 'a' and atomic radius 'r'

In BCC, along cube diagonal all atoms touches each other and the length of cube diagonal is $\sqrt{3}a$.

So, $\sqrt{3}a = 4r$ i.e. $r = \frac{\sqrt{3}a}{4}$



(b) Number of atom present in unit cell (z)

$$z = \left(\frac{1}{8} \times 8\right) + (1 \times 1) = 1 + 1 = 2 \text{ atoms/unit cell}$$

(Corner) (Body centre)

In this case one atom lies at the each corner of the cube.

Thus contribution of the 8 corners is $\left(\frac{1}{8} \times 8\right) = 1$, while that of the body centred is 1 in the unit cell.

Hence total number of atoms per unit cell is $1 + 1 = 2$ atoms

(c) Packing efficiency :

$$\text{P.E.} = \frac{z \times \frac{4}{3} \pi r^3}{V} = \frac{2 \times \frac{4}{3} \times \pi \left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{\sqrt{3}\pi}{8} = 0.68 \quad \left[\because z = 2, r = \frac{\sqrt{3}a}{4}, V = a^3\right]$$

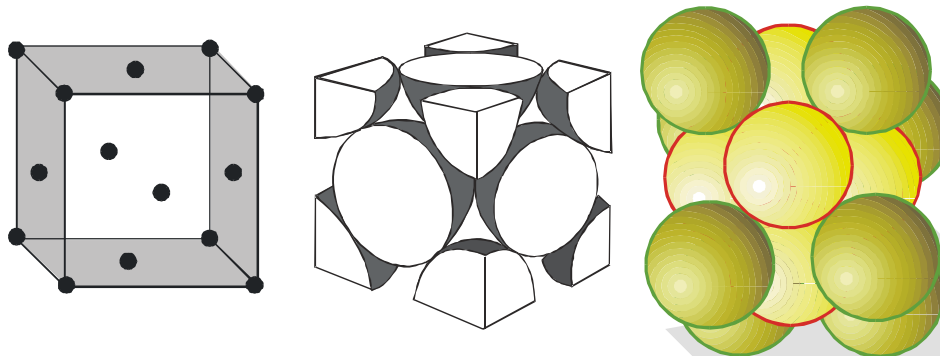
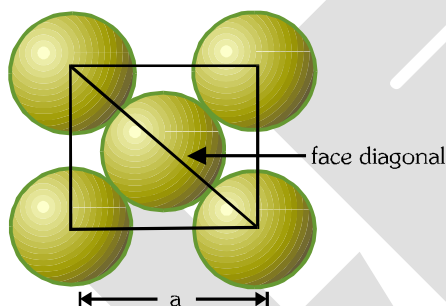
In B.C.C. , 68% of total volume is occupied by atoms.

% void space = 32

(C) FACE CENTRED CUBIC UNIT CELL (FCC)

A unit cell having lattice point at every face centre in addition to the lattice point at every corner called as face centred unit cell. i.e. in this case there are eight atoms at the eight corners of the unit cell and six atoms at the centre of six faces. The co-ordination number will be 12 and the distance between the two

nearest atoms will be $\frac{a\sqrt{2}}{2}$

**(a) Relationship between edge length 'a' and atomic radius 'r' :**

In FCC, along the face diagonal all atoms touches each other and the length of face diagonal is $\sqrt{2}a$.

$$\text{So } 4r = \sqrt{2}a \quad \text{i.e. } r = \frac{\sqrt{2}a}{4} = \frac{a}{2\sqrt{2}} \quad \boxed{r = \frac{a}{2\sqrt{2}}}$$

(b) Number of atoms per unit cell (z) :

$$z = \left(\frac{1}{8} \times 8\right) + \left(6 \times \frac{1}{2}\right) = 1 + 3 = 4 \text{ atoms/unit cell}$$

Corner faces

In this case one atom lies at the each corner of the cube and one atom lies at the centre of each face of the cube. It may noted that only $\frac{1}{2}$ of each face sphere lie within the unit cell and there are six such faces.

The total contribution of 8 corners is $\left(\frac{1}{8} \times 8\right) = 1$, while that of 6 face centred atoms is $\left(\frac{1}{2} \times 6\right) = 3$ in the unit cell. Hence total number of atoms per unit cell is $1 + 3 = 4$ atoms.

(c) Packing efficiency :

$$\text{P.E.} = \frac{z \times \frac{4}{3} \pi r^3}{V} = \frac{4 \times \frac{4}{3} \pi \times \left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74 \text{ or } 74\% \quad [\because z=4, r=\frac{a}{2\sqrt{2}}, V=a^3]$$

i.e. In FCC, 74% of total volume is occupied by atoms.

% void space = 26

SUMMARY TABLE

Unit cell	No. of atoms per unit cell	Relation between r & a	Co-ordination Number	Volume occupied by particles (%)
Simple cube	$8 \times \frac{1}{8} = 1$	$r = \frac{a}{2}$	6	$\frac{\pi}{6} \times 100 = 52.4$
Body centred cube (BCC)	$8 \times \frac{1}{8} + 1 = 2$	$r = \frac{a\sqrt{3}}{4}$	8	$\frac{\pi\sqrt{3}}{8} \times 100 = 68$
Face centred cube (FCC)	$8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$	$r = \frac{a\sqrt{2}}{4}$	12	$\frac{\pi}{3\sqrt{2}} \times 100 = 74$

Illustrations

Illustration 4. If the radius of an atom of an element is 75 pm and the lattice is body-centred cubic, the edge of the unit cell will be

- (1) 32.475 pm (2) 173.2 pm (3) 37.5 pm (4) 212.1 pm

Solution. **Ans. (2)**

Illustration 5. **Assertion :** A particle present at the corner of the face centred unit cell has 1/8th of its contribution of the unit cell.

Reason : In any space lattice, the corner of the unit cell is always shared by the eight unit cells.

- (1) A (2) B (3) C (4) D

Solution. **Ans (3)**

Illustration 6. In a face centred cubic arrangement of A and B atoms where A are present at the corner and B at the face centres, A atoms are missing from 4 corners in each unit cell ? What is the simplest formula of the compound?

Solution. No. of A atoms = $4 \times \frac{1}{8} = \frac{1}{2}$, No. of B atoms = $6 \times \frac{1}{2} = 3$ Formula = $A_{1/2}B_3 = AB_6$

1.7 DENSITY OF THE CRYSTAL (d or ρ)

If the length of edge of the unit cell is known we can calculate the density of the crystal as follow :

Let length of edge of the unit cell be 'a' cm.

∴ Volume of the cubic unit cell = $V \text{ cm}^3 = a^3 \text{ cm}^3$

$$\text{Density of the unit cell} = \frac{\text{Mass of unit cell}}{\text{Volume of unit cell}}$$

Let mass of N particles present in a lattice = m g

$$\text{mass of 1 particles present in a lattice} = \frac{m}{N} \text{ g}$$

$$\text{mass of z particles present in lattice} = \frac{z \times m}{N} \text{ g}$$

$$d = \frac{z \times m}{N \times a^3}$$

where z = number of particles or number of formula units (for ionic crystals)

m = mass of lattice in g

d = density (g/cm^3)

a = edge length in cm

if m = M (molar mass)

then $N = N_A$

$$d = \frac{z \times M}{N_A \times a^3}$$

GOLDEN KEY POINTS

- If number of particles in a lattice = N
number of SC unit cell = N
number of bcc unit cell = $\frac{N}{2}$
number of fcc unit cell = $\frac{N}{4}$
- For calculation use $N_A = 6 \times 10^{23}$
- $1 \text{ pm} = 10^{-12} \text{ m} = 10^{-10} \text{ cm}$
- $1 \text{ \AA} = 10^{-10} \text{ m} = 10^{-8} \text{ cm}$

BEGINNER'S BOX-1

- Aluminium (Molecular weight = 27) crystallises in a cubic unit cell with edge length $a = 100 \text{ pm}$, with density, $d = 180 \text{ g/cm}^3$, then type of unit cell is
(1) SC (2) BCC (3) FCC (4) HCP
- An element has BCC unit cell, with edge length $10 \text{ \AA}$, if density is 0.2 g/cm^3 , then molar mass of the compound is :-
(1) 240 (2) 60 (3) 35 (4) 280
- In a tetragonal crystal :-
(1) $a = b = c$, $\alpha = \beta = \gamma = 90^\circ \neq \gamma$ (2) $\alpha = \beta = \gamma = 90^\circ$, $a = b \neq c$
(3) $\alpha = \beta = \gamma = 90^\circ$, $a \neq b \neq c$ (4) $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$, $a = b \neq c$
- Edge length of a cube is 400 pm , its body diagonal would be :-
(1) 500 pm (2) 600 pm (3) 566 pm (4) 693 pm
- A metal crystallises into two cubic phases fcc and bcc whose unit lengths are 3.5 and $3.0 \text{ \AA}$ respectively, the ratio of densities of fcc and bcc is :-
(1) 1.26 (2) 1.75 (3) 2.10 (4) 1.90

1.8 CLOSE PACKING OF IDENTICAL SOLID SPHERES

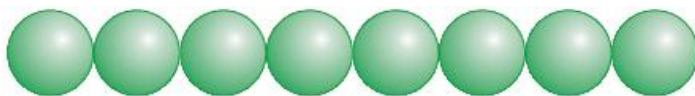
The solids which have non-directional bonding, their structures are determined on the basis of geometrical consideration. For such solids, it is found that the lowest energy structure is that in which each particle is surrounded by the greatest possible number of neighbours. In order to understand the structure of such solids, let us consider the particles as hard sphere of equal size in three directions. Although there are many ways to arrange the hard spheres but the one in which maximum available space is occupied will be economical which is known as closed packing now we describe the different arrangements of spherical particles of equal size.

When the spheres are packed in a plane it gives two types of packing.

Types of Packing :-

(A) Single line arrangement

(1) Close packing of atoms in one dimension

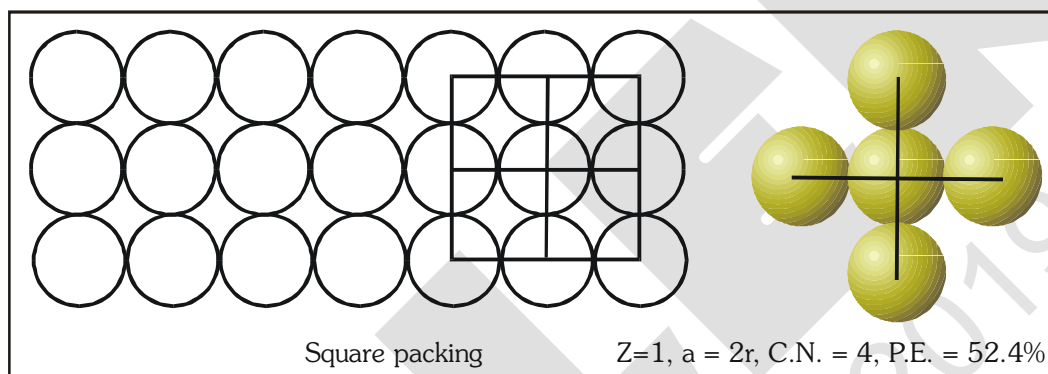


coordination number = 2

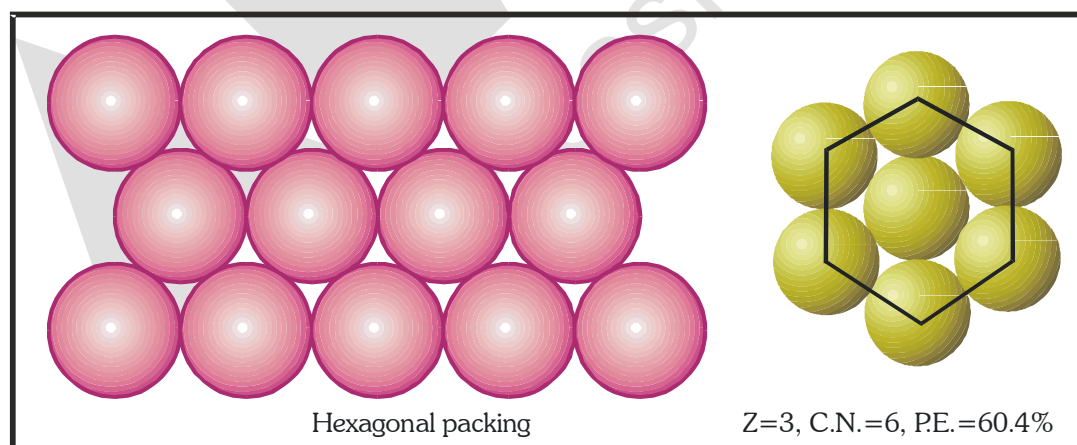
(B) Single layer arrangement

(2) Close packing of atoms in two dimensions :- When the spheres are placed in a plane, it gives two type of packing.

(a) Square close packing :- The centres of the spheres lie one below another. This type of arrangement is called square packing. In such packing one sphere touches four other spheres. In this case 52.4% of the volume is occupied. The remaining 47.6% of the volume is empty and is called void volume.



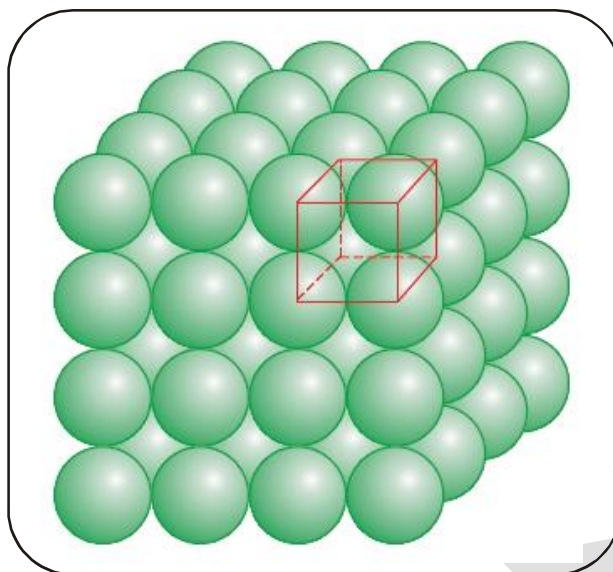
(b) Hexagonal close packing :- Another type of arrangement of atoms is shown below. This type of packing is called hexagonal packing. In such packing one sphere touches six other spheres. In this case 60.4% of the volume is occupied. The remaining 39.6% of the volume is empty and is called void volume. Therefore this type of packing is more compact than the square packing.



(C) Multi layer arrangement :-

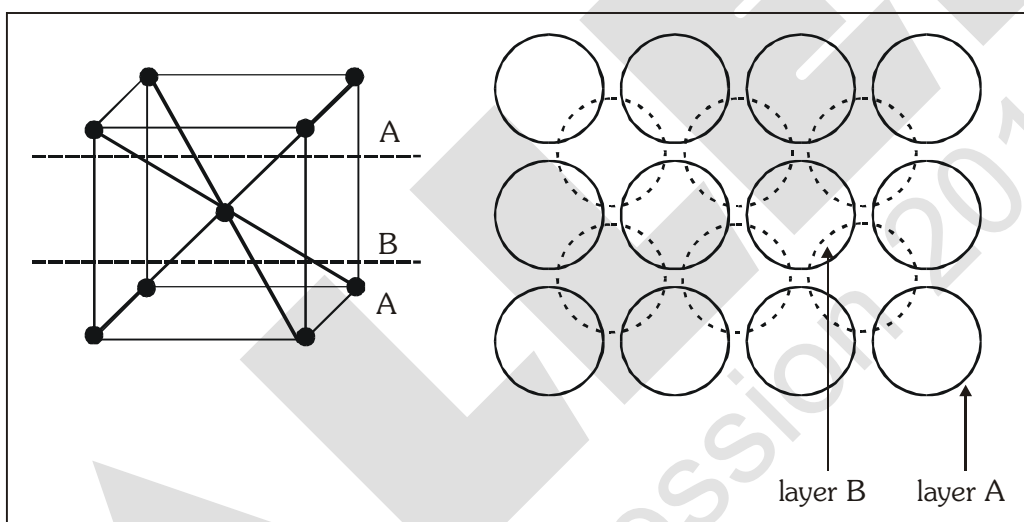
(3) Close packing of atoms in three dimensions

(a) Simple cubic close packing (Three dimensional close packing from two dimensional closed packed layers) When first square arrangement of layer A is exactly placed above occupies 2nd square arrangement of layer A and so on. So this type of arrangement is known as **AAA..... arrangement** and its unit cell is simple cubic unit cell.



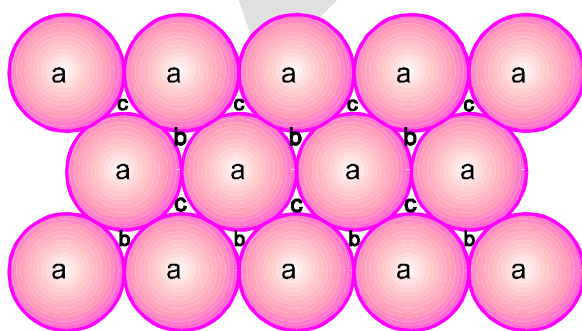
Simple cubic lattice formed by A A A arrangement (type of packing)
 $Z=1$, C.N.=6, P.E. = 52.4%. It is not compact close packing.

(b) Body centred cubic close packing :-

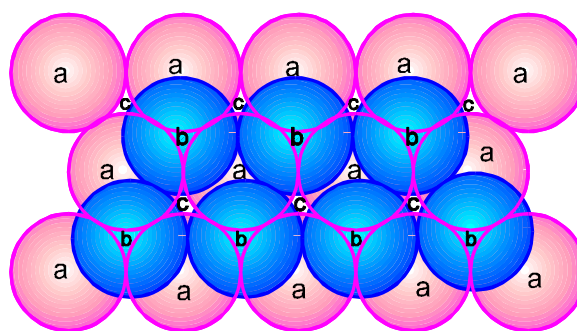


Body centred cubic lattice formed by ABAB arrangement (type of packing)
 $Z=2$, C.N.=8, P.E.=68%. It is not compact close packing.

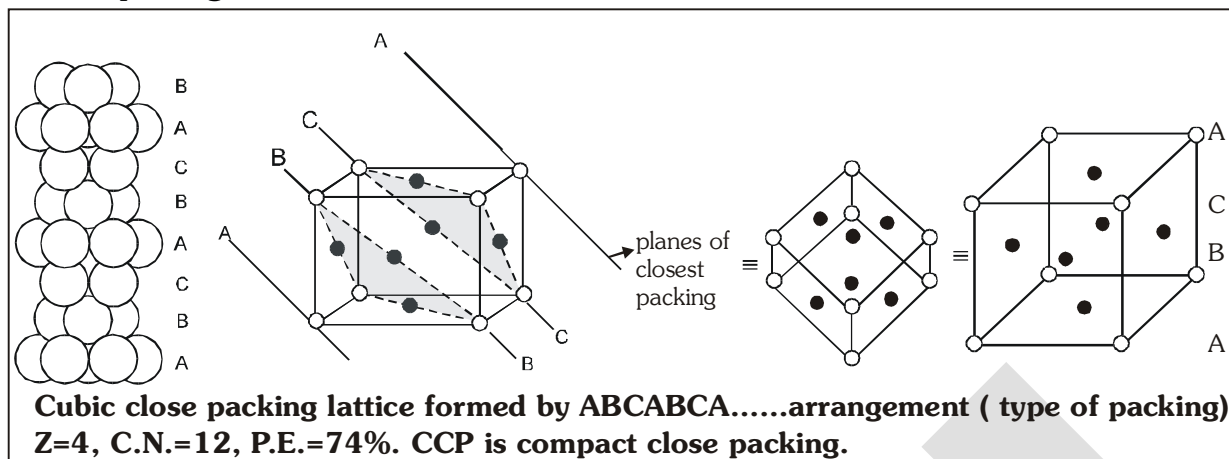
(c) Cubic close packing and hexagonal close packing :- (Three dimensional close packing from two dimensional hexagonal close packed layers)



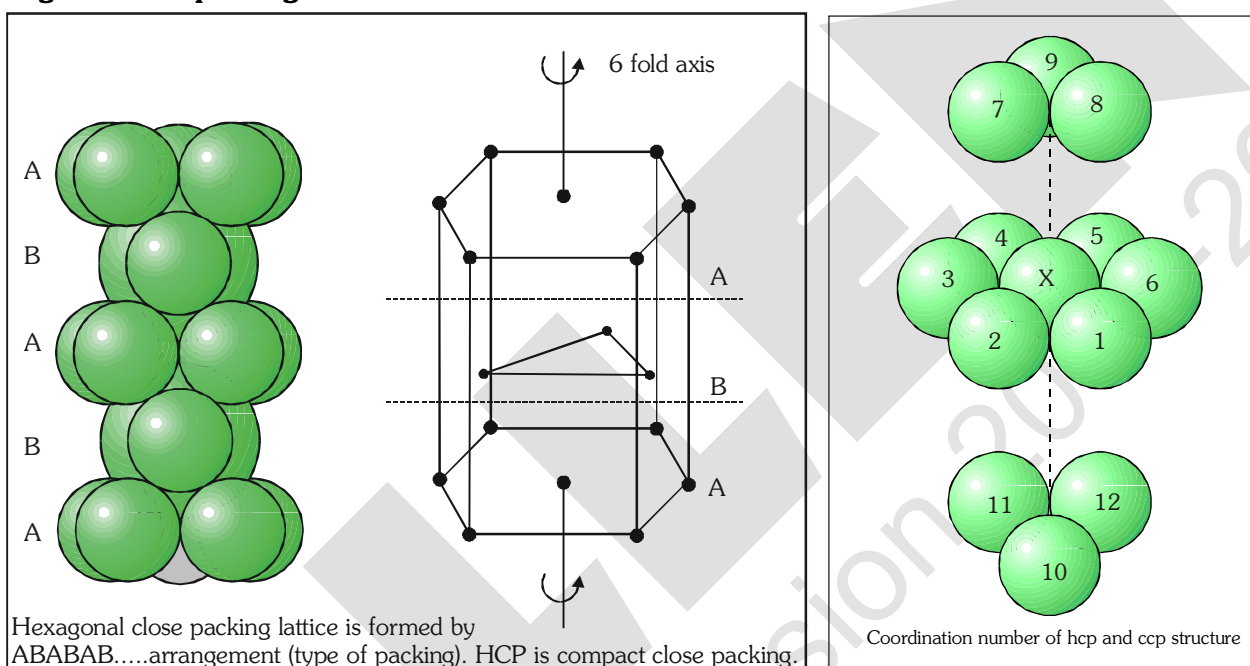
Layer A $\rightarrow$ [b, c] type voids



Layer B $\rightarrow$ [c, a] type voids



Hexagonal close packing :-



Number of particles in hexagonal unit cell (z) = $12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 \times 1 = 6$

C.N. = 12, P.E. = 74%, type of packing ABAB.....

GOLDEN KEY POINTS

- Some examples of metals with their lattice types and coordination number are given in the following table.

Li	Be		Body Centre Cubic (bcc)												
Na	Mg	Al	Face Cenral Cubic	Hexagonal Closed Packed (hcp)											
K	Ca	Cs	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn				
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd				
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au					

● MULTILAYER CLOSE PACKING

Contents	SCC	BCC	CCP/FCC	HCP
Type of packing	AAAA... packing but not close packing	ABAB..... packing but not close packing	ABCABC close packing	ABAB close packing
No. of atoms	1	2	4	6
Co-ordination no.	6	8	12	12
Packing efficiency	52.4%	68%	74%	74%
Examples	Po	IA, Ba V & Cr group Fe, Mn	Ca, Sr, Al Co group, Ni group, Copper group, all inert gases except He	Remaining d-block elements Be & Mg

- In close packing, number of particles = N, number of OHV = N, Number of THV = 2N

Illustrations

Illustration 7. The arrangement of the first two layers, one above the other, in HCP and CCP arrangements is
 (1) Exactly same in both cases (2) Partly same and partly different
 (3) Different from each other (4) Nothing definite

Solution. **Ans. (1)**

Illustration 8. **Assertion :** ABAB.... pattern of close packing gives ccp arrangement.
Reason : In FCC arrangement each sphere associated with two tetrahedral voids.
 (1) A (2) B (3) C (4) D

Solution. **Ans. (4)**

Illustration 9. "There is no difference in the arrangement of atoms in CCP and HCP structure" Do you agree with this statement ? Explain why.

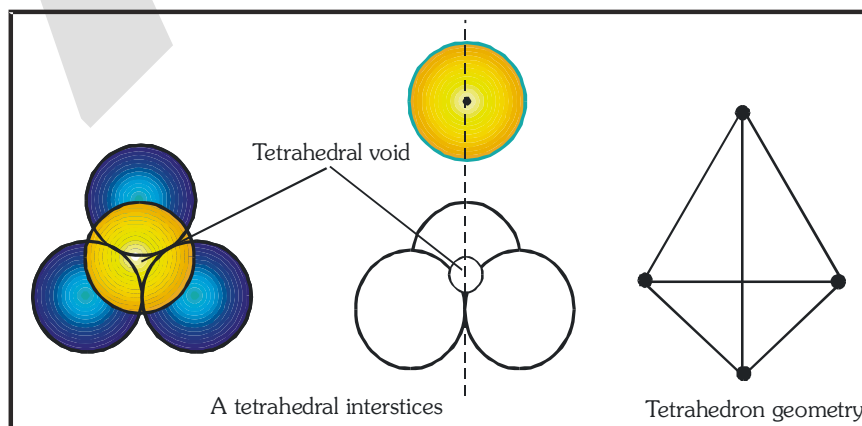
Solution. Statement is incorrect
 CCP ABC ABC ABC
 HCP AB AB AB

1.9 INTERSTICES OR VOIDS OR HOLES IN CRYSTALS

It has been shown that the particles are closely packed in the crystals even though there is some empty space left in between the spheres. This is known as interstices (or interstitial site or hole or empty space or voids). In three dimensional close packing (CCP & HCP) the interstices are of two types : (i) tetrahedral interstices and (ii) octahedral interstices.

(A) Tetrahedral Interstices

We have seen that in hexagonal close packing (HCP) and cubic close packing (CCP) each sphere of second layer touches with three spheres of first layer. Thus they leave a small space in between which is known as **tetrahedral site or interstices** or the vacant space between 4 touching spheres is called as tetrahedral void. Since a sphere touches three spheres in the below layer and three spheres in the above layer hence there are two tetrahedral sites associated with one sphere.



It may be noted that a tetrahedral site does not mean that the site is tetrahedral in geometry but it means that this site is surrounded by four spheres and by joining the centres of these four spheres forms a regular tetrahedron.

In FCC, one corner and its three face centres form a tetrahedral void

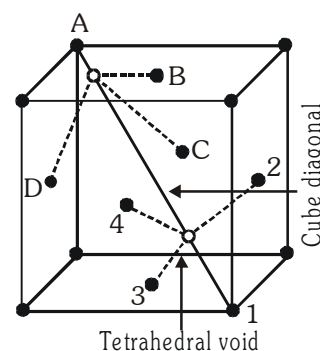
In FCC, two tetrahedral voids are obtained along one cube diagonal.

So in FCC, 8 tetrahedral voids are present.

In FCC, total number of atoms = 4

In FCC, total number of tetrahedral voids = 8

So, we can say that, in 3D close packing 2 tetrahedral voids are attached with one atom.

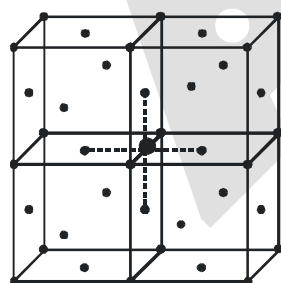
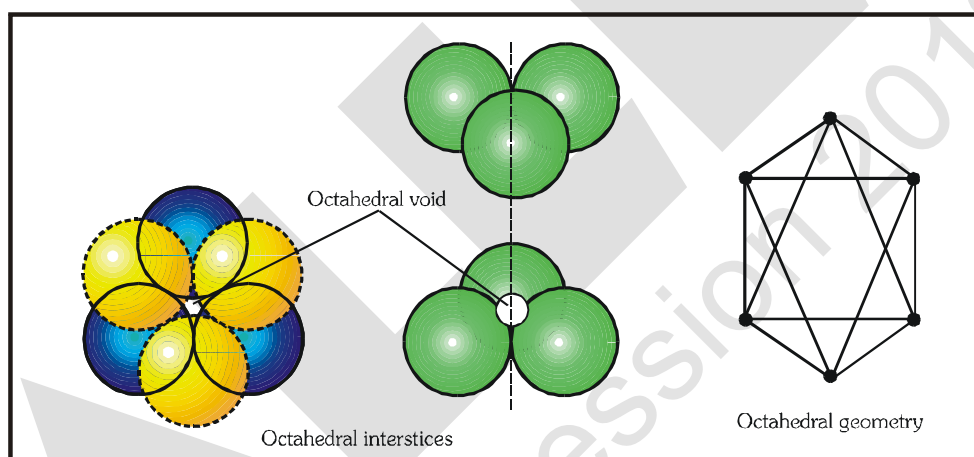


(B) Octahedral Interstices

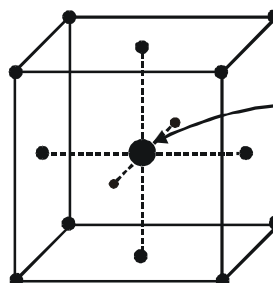
Hexagonal close packing (hcp) and cubic close packing (ccp) also form another type of interstices which is called octahedral site. The vacant space between 6 touching spheres is called as octahedral void.

In the figure, two layers of close packed spheres are shown. The spheres of first layer are shown by full circles while that of second layer by dotted circles. Two triangles are drawn by joining the centres of three touching spheres of both the layers :

In fcc, 6 face centres form a octahedral void. On super imposing these triangles on one another one octahedral site is created. It may be noted that an octahedral site does not mean that the hole is octahedral in shape but it means that this site is surrounded by six nearest neighbour lattice points arranged octahedrally.



Octahedral void (at the edge)
(1/4 th part of octahedral void is obtained at each edge)



Octahedral void (at the body centre)

$$\text{In FCC, total number of octahedral voids} = (1 \times 1) + (12 \times \frac{1}{4}) = 1 + 3 = 4$$

(Cube centre) (edge centre)

In FCC, number of atoms = 4 and number of octahedral voids = 4 : number of tetrahedral voids = 8

So we can say that, in 3D close packing one octahedral void is attached with one atom.

Illustrations

Illustration 10. The number of octahedral sites in a cubical close packed array of N spheres is

- (1) $N/2$ (2) $2N$ (3) N (4) $4N$

Solution. **Ans. (3)**

Illustration 11. Atom A is every element of FCC, atom B is present at every Octahedral void, atom C is present at 25% of Tetrahedral void. Find out the possible molecular formula of the compound ?

Solution.

Atom A is every element of FCC = 4 atoms of A

Atom B is present at every octahedral void = 4 atoms of B

Atom C is present at 25% of tetrahedral void = $8 \times \frac{25}{100} = 2$ atoms of C

So, the possible molecular formula is $A_4 B_4 C_2 = A_2 B_2 C$.

1.10 STUDY OF IONIC CRYSTALS

Limiting Radius Ratio

An ionic crystal contains a large number of cations and anions. Generally cations are smaller in size than that of anions. The cations are surrounded by anions and they touch each other. These ions are arranged in space in such a way to produce maximum stability.

So the limiting radius ratio tells about :

- (i) Stability of ionic crystal
- (ii) Coordination number
- (iii) Geometry of voids

(a) Triangular : All anions touches each other and co-ordination number is 3

$$\cos \theta = \frac{R}{R+r}$$

$$\cos 30^\circ = \frac{R}{r+R}$$

$$\sqrt{3}R + \sqrt{3}r = 2R$$

$$\sqrt{3}r = (2 - \sqrt{3})R$$

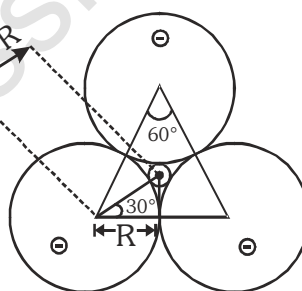
$$\frac{r}{R} = \frac{2 - \sqrt{3}}{\sqrt{3}} = \frac{2 - 1.73}{1.73} = \frac{0.27}{1.73} = 0.155$$

$$\text{L.R.R.} = 0.155 = \frac{r}{R} < 1$$

(b) Tetrahedral void : All anions touches each other and co-ordination number of cation is 4.

$$\text{Face diagonal AC} = \sqrt{2}a = 2R$$

$$R = \frac{a}{\sqrt{2}} \quad \text{or} \quad a = \sqrt{2}R$$



Triangle ACD -

$$AD^2 = AC^2 + CD^2$$

$$AD^2 = (\sqrt{2}a)^2 + (a)^2 = 2a^2 + a^2 = 3a^2$$

$$AD = \sqrt{3} \text{ a}$$

According to cube diagonal AD

$$\therefore \frac{\sqrt{3}a}{2} = r + R$$

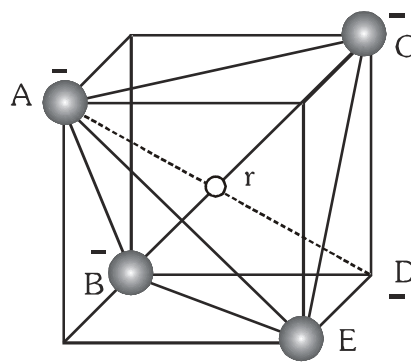
$$\therefore \sqrt{3}a = 2r + 2R = AD$$

Put the value of $a = \sqrt{2}R$

$$\sqrt{3} \times \sqrt{2} R = 2r + 2R$$

$$\frac{\sqrt{3} \times \sqrt{2}R}{2R} = \frac{2r + 2R}{2R} \quad \text{or} \quad \frac{\sqrt{3}}{\sqrt{2}} = \frac{r}{R} + 1$$

$$\frac{r}{R} = \frac{\sqrt{3}}{\sqrt{2}} - 1 = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{2}} = \frac{1.73 - 1.414}{1.414} = 0.225$$



- (c) **Octahedral void** : All the anions are touch each other and co-ordination number is 6.

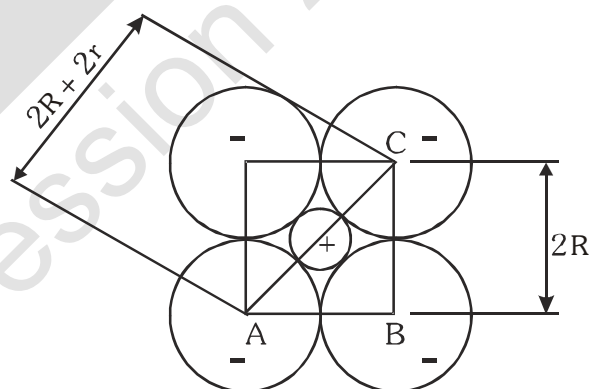
In $\triangle ABC$ $AC^2 = AB^2 + BC^2$
 $(2R + 2r)^2 = (2R)^2 + (2R)^2$

$$(2R + 2r) = \sqrt{8R^2}$$

$$2R + 2r = 2\sqrt{2}R$$

$$2R(\sqrt{2} - 1) = 2r$$

$$\frac{r}{R} = \sqrt{2} - 1 = 0.414$$



- (d) Cubic void :** All the anions are touch each other and co-ordination number is 8.

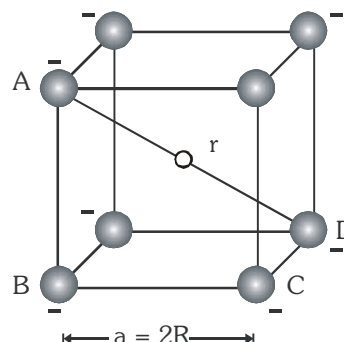
According to cube diagonal

$$AD = \sqrt{3}a = 2r + 2R \quad (a = 2R = BC)$$

$$\sqrt{3} \times 2R = 2r + 2R$$

Dividing by $2r$ on both sides.

$$\sqrt{3} = \frac{r}{R} + 1 \Rightarrow \frac{r}{R} = \sqrt{3} - 1 = 1.732 - 1 = 0.732$$



GOLDEN KEY POINTS

- The preferred direction of the structure with increase in the radius ratio is as follows :
Plane triangular $\xrightarrow{0.225}$ Tetrahedral $\xrightarrow{0.414}$ octahedral $\xrightarrow{0.732}$ Cubic

- LIMITING RADIUS RATIO FOR VARIOUS TYPES OF VOIDS**

Limiting radius ratio = r/R	Coordination Number	Structural Arrangement (Type of voids)	Example
$0.155 \leq r/R < 0.225$	3	Plane Trigonal	Boron Oxide(B_2O_3)
$0.225 \leq r/R < 0.414$	4	Tetrahedral	ZnS, SiO_2
$0.414 \leq r/R < 0.732$	6	Octahedral	NaCl, MgO
$0.732 \leq r/R < 1.000$	8	Cubical	CsCl

Illustrations

Illustration 12. Assertion : In crystal lattice, the size of the cation is larger in a tetrahedral hole than in an octahedral hole.

Reason : The cations occupy more space than atoms in crystal packing.

- (1) A (2) B (3) C (4) D

Solution. Ans. (4)

Illustration 13. Each unit cell of NaCl consists of 14 Cl^- ions and

- (1) 13 Na^+ (2) 14 Na^+ (3) 6 Na^+ (4) All are wrong

Solution. Ans. (1)

Illustration 14. A solid A^+B^- has NaCl type close packed structure. If the anion has a radius of 250 pm, what should be the ideal radius for the cation? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal A^+B^- ? Give reason for your answer.

Solution In Na^+Cl^- crystal each Na^+ ion is surrounded by 6 Cl^- ions and vice versa. Thus Na^+ ion is placed in octahedral hole.

The limiting radius ratio for octahedral site = 0.414 or $\frac{A^+}{B^-} = \frac{r}{R} = 0.414$

Given that radius of anion (B^-) $R = 250$ pm

i.e. radius of cation (A^+) $r = 0.414 R = 0.414 \times 250$ pm = 103.5 pm

Thus ideal radius for cation (A^+) is $r = 103.5$ pm.

We know that (r/R) for tetrahedral hole is 0.225.

$\therefore \frac{r}{R} = 0.225$: So $r = 56.25$ pm

Thus ideal radius for cation is 56.25 pm for tetrahedral hole. But the radius of C^+ is 180 pm. It is much larger than ideal radius i.e. 56.25 pm. Therefore we can not slipped cation C^+ into the tetrahedral site.

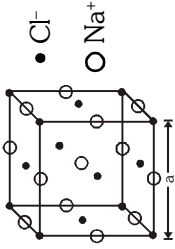
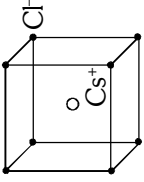
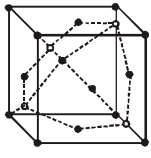
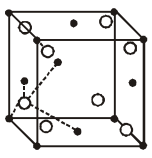
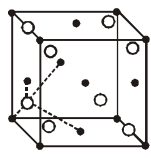
Illustration 15. Assertion :- In rock salt structure, the sodium ions occupy octahedral voids.

Reason :- The radius ratio r^+/r^- in case of NaCl lies between 0.225 to 0.414.

- (1) A (2) B (3) C (4) D

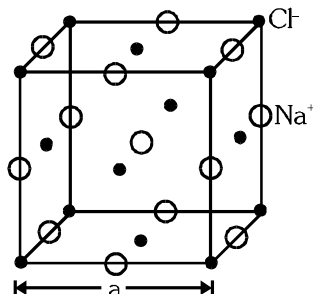
Solution. Ans. (3)

1.11 SOME IONIC CRYSTALS

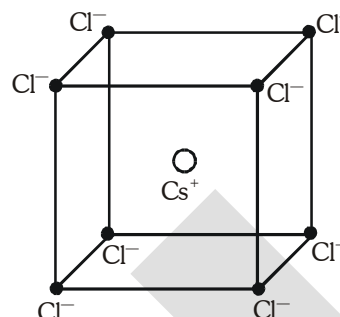
S. No.	Type of Ionic Crystal	Geometry	Co-ordination Number	No. of formula's per U.C.	Examples
1.	NaCl (1 : 1) (Rock Salt Type)	$\begin{array}{c} \text{Na}^+ \rightarrow \text{At every OHV} \\ \text{C.C.P.} \swarrow \searrow \text{Cl}^- \rightarrow \text{Every element of C.C.P.} \end{array}$	6 : 6	$4\text{Na}^+ + 4\text{Cl}^-$ 4NaCl (4)	Halides of (Li, Na, K, Rb) Oxides and sulphides of alkaline, earth metals (Except BeS) AgCl, AgBr, NH_4X 
2.	CsCl Type (1 : 1)	$\begin{array}{c} \text{Cs}^+ \rightarrow \text{At body centre of} \\ \text{B.C.C.} \swarrow \searrow \text{Cl}^- \rightarrow \text{At every corner} \end{array}$	8 : 8	$1\text{Cs}^+ + 1\text{Cl}^-$ 1CsCl (1)	Halides of 'Cs' TlCl, TlBr, CaS 
3.	ZnS Type (1 : 1) (Zinc Blende Type) (Sphalerite)	$\begin{array}{c} \text{Zn}^{+2} \rightarrow \text{At 50\% of T.H.V. or} \\ \text{C.C.P.} \swarrow \searrow \text{S}^{+2} \rightarrow \text{Every element of C.C.P.} \end{array}$	4 : 4	$4\text{Zn}^{+2} + 4\text{S}^{-2}$ 4ZnS (4)	BeS, BeO, CaO, AgI, CuCl, CuBr, CuI 
4.	CaF ₂ Type (1 : 2) (Fluorite Type)	$\begin{array}{c} \text{Ca}^{+2} \rightarrow \text{Every element of C.C.P.} \\ \text{C.C.P.} \swarrow \searrow \text{F}^- \rightarrow \text{At every T.H.V.} \end{array}$	$4\text{Ca}^{+2}, 8\text{F}^-$ 8 : 4	$4\text{Ca}^{+2} + 8\text{F}^-$ 4CaF_2 (4)	BaCl ₂ , BaF ₂ SrCl ₂ , SrF ₂ CaCl ₂ , CaF ₂ 
5.	Na ₂ O Type (2 : 1) (Antifluorite)	$\begin{array}{c} \text{Na}^+ \rightarrow \text{At every T.H.V.} \\ \text{C.C.P.} \swarrow \searrow \text{O}^{2-} \rightarrow \text{Every element of C.C.P.} \end{array}$	$8\text{Na}^+, 4\text{O}^{2-}$ 4 : 8	$8\text{Na}^+ + 4\text{O}^{2-}$ $4\text{Na}_2\text{O}$ (4)	Li ₂ O, Li ₂ S Na ₂ O, Na ₂ S K ₂ O, K ₂ S 
6.	ZnS Type (1 : 1) (Wurtzite) another geometry of ZnS	$\begin{array}{c} \text{Zn}^{+2} \rightarrow 50\% \text{ of T.H.V. or} \\ \text{H.C.P.} \swarrow \searrow \text{S}^{2-} \rightarrow \text{Every element of H.C.P.} \end{array}$	4 : 4	$6\text{Zn}^{+2} + 6\text{S}^{-2}$ 6ZnS (6)	Same as sphalerite

● **NaCl TYPE****For NaCl :**Distance between two nearest ions ($r^+ + r^-$) :-

$$2r^+ + 2r^- = a \quad \text{i.e.} \quad r^+ + r^- = \frac{a}{2}$$

**CsCl TYPE****For CsCl :**Distance between two nearest ions ($r^+ + r^-$) :-

$$2r^+ + 2r^- = \sqrt{3}a \quad \text{i.e.} \quad r^+ + r^- = \frac{\sqrt{3}}{2}a$$

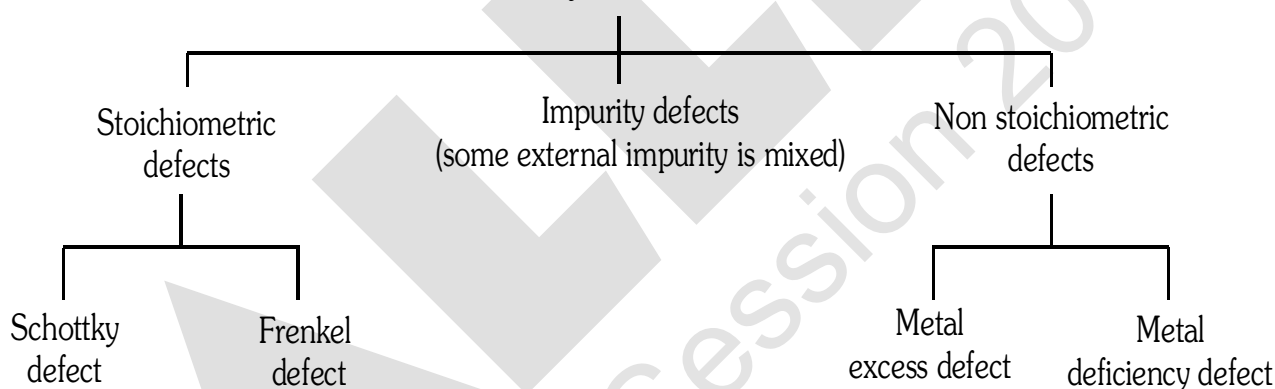


In both structures cation and anion can exchange their position.

1.12 DEFECTS OR IMPERFECTIONS IN SOLIDS

Ideal crystal : The crystal in which all the lattice points are occupied by the component particles or groups of particles is called an ideal crystal. Ideal crystals are grown up at absolute kelvin temperature (zero K).

Solid state is characterised by vibratory motion about the mean position of constituent particles. At absolute zero, all the types of motions cease, and therefore crystals tend to have a perfectly ordered arrangement. As the temperature increases, molecular motions (vibratory amplitudes) increase and therefore the ions may lose their usual sites and either get missed or occupy interstitial positions in the crystal, i.e., deviations from ordered arrangement take place. Any deviation from the perfectly ordered arrangement gives rise to a defect or imperfection in the crystal. Defect in crystals are produced either due to thermal effects or by adding certain impurities in the pure crystals (doping). Defects in crystals may be discussed as

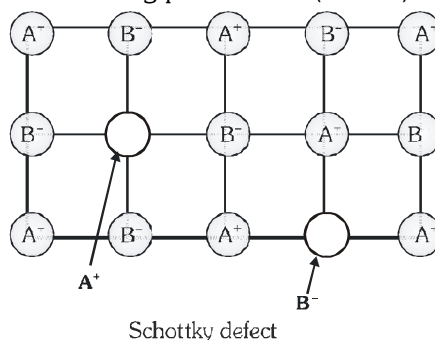
Crystal defects**(A) Stoichiometric Defects**

Stoichiometric compounds are those in which the number of positive and negative ions are exactly in the ratio as shown by their chemical formulae. Two types of defects are observed in these compounds.

(a) Schottky defect,

(b) Frenkel defect

- (a) Schottky defect :** This type of defect is produced when equal number of cations and anions are missing from their respective positions leaving behind a pair of holes. The crystal as a whole remains neutral because the number of missing positive ions (cations) and negative ions (anions) is the same

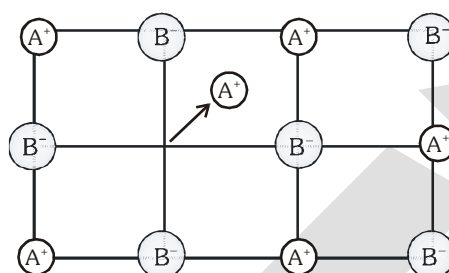


- It is kind of vacancy defect.
- Schottky defect appears generally in ionic compounds in which radius ratio $[r^+/r^-]$ is not far below unity. For this defect, the cations and anion should not differ much in size.
- For schottky defect, co-ordination numbers of the ions should be high, Examples of ionic solids showing this defect are NaCl, CsCl, KCl, AgBr etc.

Consequences of schottky defect :

- Density of the crystal decreases
- The crystal begins to conduct electricity to small extent by ionic mechanism.
- The presence of too many 'voids' lowers lattice energy or the stability of the crystal.

(b) Frenkel defect : This type of defect is created when an ion leaves its appropriate site in the lattice and occupies an interstitial site. A hole or vacancy is thus produced in the lattice.



Frenkel defect

The electroneutrality of the crystal is maintained since the number of positive and negative ions is the same. Since positive ions are small, in size, they usually leave their positions in the lattice and occupy interstitial positions. It is also called dislocation defect.

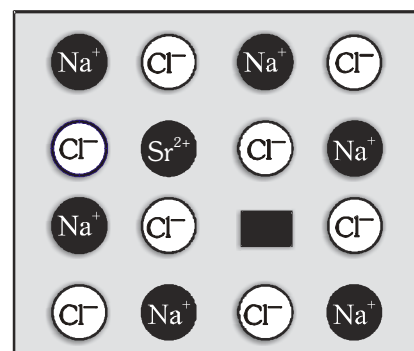
Frenkel defect is exhibited in ionic compounds in which the radius ratio $[r^+/r^-]$ is low. The cations and anions differ much in their sizes and the ions have low co-ordination numbers. Examples are ZnS, AgBr, AgI, AgCl.

Consequences of Frenkel defect :

- It is a kind of dislocation defect.
- The closeness of like charges tends to increase the dielectric constant of the crystal.
- The crystal showing a frenkel defect conducts electricity to a small extent by ionic mechanism.
- The density of the crystal remains the same.

(B) Impurity Defect

If molten NaCl containing a little amount of SrCl_2 is crystallised, some of the sites of Na^+ ions are occupied by Sr^{2+} . Each Sr^{2+} replaces two Na^+ ions. It occupies the site of one ion and the other site remains vacant. The cationic vacancies thus produced are equal in number to that of Sr^{2+} ions. Another similar example is the solid solution of CdCl_2 and AgCl .



Induction of cation vacancy in NaCl by substitution of Na^+ by Sr^{2+}

(C) Non-stoichiometric Defects

These types of defects are observed in the compounds of transitional elements. These defects arise either due to the presence of excess metal ions or excess non-metal ions.

(a) **Metal excess defect due to anion vacancies :** A compound may have excess metal ion if an anion (negative ion) is absent from its appropriate lattice site creating a 'void' which is occupied by an electron. Ionic crystal which are likely to possess Schottky defect, may also develop this type of metal excess defect. When alkali metal halides are heated in an atmosphere of vapours of the alkali metal, anion vacancies are created. The anions (halide ions) diffuse to the surface of the crystal from their appropriate lattice sites to combine with the newly generated metal cations. The e^- lost by the metal atom diffuse through the crystal are known as F-centres. The main consequence of metal excess defect is the development of colour in the crystal. For example, when NaCl crystal is heated in an atmosphere of Na vapours, it becomes yellow.

Similarly, KCl crystal when heated in an atmosphere of potassium vapours, it appears violet.

(b) **Metal excess defect due to interstitial cations :** This type of defect is shown by crystals which are likely to exhibit Frenkel defect. An excess positive ion is located in the interstitial site. To maintain electrical neutrality, an electron is present in the interstitial site. An example showing this type of defect is ZnO. When ZnO is heated, it loses oxygen reversibly. The excess Zn^{++} ions are accommodated in the interstitial sites, the electrons are enclosed in the neighbouring interstitials. The yellow colour of ZnO when hot is due to these trapped electrons. These electrons also explain electrical conductivity of the ZnO crystal.

(c) **Metal Deficiency Defect**

There are many solids which are difficult to prepare in the stoichiometric composition and contain less amount of the metal as compared to the stoichiometric proportion. A typical example of this type is FeO which is mostly found with a composition of $Fe_{0.95}O$. It may actually range from $Fe_{0.93}O$ to $Fe_{0.96}O$. In crystals of FeO some Fe^{2+} cations are missing and the loss of positive charge is made up by the presence of required number of Fe^{3+} ions.

GOLDEN KEY POINTS

- On increasing temperature coordination number decreases.
- On increasing pressure; coordination number increases.
- If in a crystal complete lattice of ion along a line are disappeared, such defect is called line defect.
- Schottky defects and Frenkel defects are point defects.
- Structure of diamond :** Solids which follow the structure of diamonds known as diamond cubic. Diamond has CCP structure where carbon atoms are present at corners and face centre with four more atoms present at alternate tetrahedral voids or 50% THV so number of effective atoms

$$z = \left(8 \times \frac{1}{8} + 6 \times \frac{1}{2} \right) + (4 \times 2) \times \frac{1}{2} = 8.$$

The carbon atoms present in THV touches its four surrounding atoms so coordination number is equal to four.

$$2r = \frac{\sqrt{3}}{4}a ; a = \frac{8r}{\sqrt{3}} ; PE = \frac{8 \times \frac{4\pi}{3} r^3}{\left(\frac{8r}{\sqrt{3}} \right)^3} = 0.34 ; \text{So \% PE} = 34$$

Illustrations

Illustration 16. In the Schottky defect

- (1) cations are missing from the lattice sites and occupy the interstitial sites
- (2) equal number of cations and anions are missing
- (3) anions are missing and electrons are present in their place
- (4) equal number of extra cations and electrons are present in the interstitial sites

Ans. (2)

Solution
22

@neetfocusquizz

Illustration 17. Assertion :- Schottky type defects is shown by crystals with high co-ordination number.

Reason :- In Schottky defect equal number of cations and anions are missing from their lattice sites.

- (1) A (2) B (3) C (4) D

Solution **Ans. (2)**

Illustration 18. Schottky defect lowers the density of ionic crystals while Frenkel defect does not. Discuss.

Solution. In Schottky defect, certain cations and anions are missing from the crystal lattice. Therefore, the density of the crystal decreases. However, in Frenkel defect the ions do not leave the lattice but they simply change their positions from lattice points to the interstitial spaces. As a result, the density of the crystal remains unchanged.

BEGINNER'S BOX-2

- The edge length of face centred cubic unit cell having rock salt structure is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is—
(1) 144 pm (2) 288 pm (3) 618 pm (4) 398 pm
- In a solid 'AB' having the NaCl structure, 'A' atoms occupy the every CCP and B atoms are in OHV. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is—
(1) AB_2 (2) A_2B (3) A_4B_3 (4) A_3B_4
- How many unit cells are present in a cube shaped ideal crystal of NaCl of mass 1.00 g ?
[Atomic mass : Na = 23, Cl = 35.5]
(1) 5.14×10^{21} (2) 1.28×10^{21} (3) 1.71×10^{21} (4) 2.57×10^{21}
- A solid AB has the NaCl structure. If radius of the cation A^+ is 120 pm, the maximum value of the radius of the anion B^- is
(1) 290 pm (2) 350 pm (3) 250 pm (4) 400 pm
- If the distance between Na^+ and Cl^- ions in sodium chloride crystal is x pm, then length of the edge of the unit cell is
(1) 2x pm (2) 4x pm (3) x/2 pm (4) x/4 pm
- HCP crystal is made of A ions and B ions are arranged in $2/3^{rd}$ of octahedral voids. The simplest formula of unit cell is
(1) A_4B_6 (2) A_6B_4 (3) A_2B_3 (4) A_3B_2
- A FCC crystal of O^{2-} (oxide) ions has M^{+n} ions in 50% of octahedral voids, then the value of n is
(1) 2 (2) 3 (3) 4 (4) 1
- If the edge length of a KCl unit cell is 488 pm, what is the length of KCl bond if it crystallises in the FCC structure ?
(1) 122 pm (2) 244 pm (3) 488 pm (4) 974 pm
- In a solid lattice, the cation has left a lattice site and is located at interstitial position, the lattice defect is
(1) interstitial defect (2) vacancy defect (3) Frenkel defect (4) Schottky defect

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5					
	Ans.	3	2	2	4	1					
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	
	Ans.	1	4	4	1	1	4	3	2	3	

EXERCISE-I (Conceptual Questions)**Build Up Your Understanding****INTRODUCTION**

- $a \neq b \neq c$, $\alpha = \gamma = 90^\circ$, $\beta \neq 90^\circ$ represents
 - tetragonal system
 - orthorhombic system
 - monoclinic system
 - triclinic system
- The most unsymmetrical crystal system is :
 - Cubic
 - Hexagonal
 - Triclinic
 - Orthorhombic
- Bravais lattices are of
 - 10 types
 - 8 types
 - 7 types
 - 14 types
- The crystal system of a compound with unit cell dimensions $a = 0.387$, $b = 0.387$ and $c = 0.504$ nm and $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$ is :
 - Cubic
 - Hexagonal
 - Orthorhombic
 - Rhombohedral

ANALYSIS OF CUBIC CRYSTAL

- In a simple cubic cell, each point on a corner is shared by
 - 2 unit cells
 - 1 unit cell
 - 8 unit cells
 - 4 unit cells
- In a face centred cubic cell, an atom at the face contributes to the unit cell
 - 1 part
 - 1/2 part
 - 1/4 part
 - 1/8 part
- In a body centred cubic cell, an atom at the body centre is shared by
 - 1 unit cell
 - 2 unit cell
 - 3 unit cells
 - 4 unit cells
- Which of the following type of cubic lattice has maximum number of atoms per unit cell ?
 - Simple cubic
 - Body centred cubic
 - Face centred cubic
 - All have same
- The number of atoms present in a unit cell of a monoatomic element of a simple cubic lattice, body-centred cubic and face centred cubic respectively are
 - 8, 9 and 14
 - 1, 2 and 4
 - 4, 5 and 6
 - 2, 3 and 5

- Which one of the following is a primitive unit cell ?
 - Simple cubic
 - BCC
 - FCC
 - bcc and fcc both
- In a body centred cubic unit cell, a metal atom at the centre of the cell is surrounded by how many other metal atoms :
 - 8
 - 6
 - 12
 - 4
- A compound is formed by elements A and B. This crystallises in the cubic structure when atoms A are at the corners of the cube and atoms B are at the centre of the body. The simplest formula of the compound is
 - AB
 - AB_2
 - A_2B
 - AB_4
- A compound formed by elements A and B has cubic structure in which A atoms are at the corners of the cube and B atoms are at the face centres. The formula of the compound will be
 - A_4B_3
 - A_2B
 - AB_3
 - A_2B_3
- Sodium metal crystallises in BCC lattice with the cell edge length (a) = 42.29 Å. What is the radius of sodium atom ?
 - 1.86 Å
 - 1.90 Å
 - 18.3 Å
 - 1.12 Å
- An element has BCC structure having unit cells 12.08×10^{23} . The number of atoms in these cells is
 - 12.08×10^{23}
 - 24.16×10^{23}
 - 48.38×10^{23}
 - 12.08×10^{22}
- A metal has BCC structure and the edge length of its unit cell is 3.04 Å. The volume of the unit cell in cm^3 will be
 - $1.6 \times 10^{-21} \text{ cm}^3$
 - $2.81 \times 10^{-23} \text{ cm}^3$
 - $6.02 \times 10^{-23} \text{ cm}^3$
 - $6.6 \times 10^{-24} \text{ cm}^3$

CRYSTAL DENSITY

- An element having BCC geometry has atomic mass 50. Calculate the density of the unit cell, if its edge length is 290 pm.
 - 6.81 g cm^{-3}
 - 3.40 g cm^{-3}
 - 13.62 g cm^{-3}
 - 1.23 g cm^{-3}
- A metal (atomic mass = 50) has a body centred cubic crystal structure. The density of metal is 5.96 g cm^{-3} . Find the volume of the unit cell.
 - $13.9 \times 10^{-24} \text{ cm}^3$
 - $27.8 \times 10^{-24} \text{ cm}^3$
 - $6.95 \times 10^{-24} \text{ cm}^3$
 - $55.6 \times 10^{-24} \text{ cm}^3$

19. An element crystallises in BCC structure. The edge length of its unit cell is 288 pm. If the density of the crystal is 7.2 g cm^{-3} , what is the atomic mass of the element?
 (1) 51.8 (2) 103.6
 (3) 25.9 (4) 207.2
20. An element, density 6.8 g cm^{-3} occurs in BCC structure with cell edge 290 pm. Calculate the number of atoms present in 200 g of the element.
 (1) 2.4×10^{42} (2) 1.2×10^{42}
 (3) 1.2×10^{24} (4) 2.4×10^{24}
21. An element A crystallises in fcc structure. 200 g of this element has 4.12×10^{24} atoms. The density of A is 7.2 g cm^{-3} . Calculate the edge length of the unit cell.
 (1) $26.97 \times 10^{-24} \text{ cm}$ (2) 299.9 pm
 (3) $5 \times 10^{-12} \text{ cm}$ (4) 2.99 cm
22. Density of Li atom is 0.53 g cm^{-3} . The edge length of Li is 3.5 Å. Find out the number of Li atoms in an unit cell. ($N_A = 6.023 \times 10^{23}$), ($M = 6.94 \text{ g mol}^{-1}$)
 (1) 2 (2) 8 (3) 4 (4) 6

CLOSE PACKING OF IDENTICAL SOLID SPHERES

23. The coordination number of hexagonal closest packed (HCP) structure is
 (1) 12 (2) 10 (3) 8 (4) 6
24. The ABAB close packing and ABC ABC close packing are respectively called as
 (1) hexagonal close packing(hcp) and cubic close packing (ccp)
 (2) ccp and hcp
 (3) body centred cubic (bcc) packing and hexagonal close packing (hcp)
 (4) hcp and bcc
25. The space occupied in BCC arrangement is
 (1) 74% (2) 70% (3) 68% (4) 60.4%
26. The vacant space in BCC unit cell is
 (1) 32% (2) 10% (3) 23% (4) 46%
27. The space occupied by spheres of equal size in three dimensions in both HCP and CCP arrangement is
 (1) 74% (2) 70% (3) 60.4% (4) 52.4%
28. The empty space in the HCP and CCP is about
 (1) 26% (2) 30% (3) 35% (4) 40%
29. Which one of the following is not a close packing ?
 (1) hcp (2) ccp (3) bcc (4) fcc

30. Close packing is maximum in the crystal lattice of
 (1) Simple cubic
 (2) Face centred
 (3) Body centred
 (4) Simple cubic and body centred
31. Which of the following has HCP structure ?
 (1) Al (2) Mg (3) Cu (4) Ni
32. All noble gases crystallise in the CCP structure except
 (1) Helium (2) Neon (3) Argon (4) Krypton
33. If the coordination number of an element in its crystal lattice is 8, then packing is :
 (1) FCC (2) HCP
 (3) BCC (4) None of the above

HOLES / VOIDS IN CRYSTALS

34. A tetrahedral void in a crystal implies that
 (1) shape of the void is tetrahedral
 (2) molecules forming the void are tetrahedral in shape
 (3) the void is surrounded tetrahedrally by four spheres
 (4) the void is surrounded by six spheres
35. In a closest packed lattice, the number of octahedral sites as compared to tetrahedral ones will be
 (1) Equal (2) Half
 (3) Double (4) One fourth
36. The coordination number of a cation occupying an octahedral hole is
 (1) 4 (2) 6 (3) 8 (4) 12
37. The size of an octahedral void formed in a closest packed lattice as compared to tetrahedral void is
 (1) Equal (2) Smaller
 (3) Larger (4) Not definite
38. The coordination number of a cation occupying a tetrahedral hole is
 (1) 4 (2) 6 (3) 8 (4) 12
39. Number of tetrahedral voids per atom in a crystal lattice is :
 (1) 1 (2) 2 (3) 4 (4) 8
40. A compound contains P and Q elements. Atoms Q are in CCP arrangement while P occupy all tetrahedral sites. Formula of compound is :
 (1) PQ (2) PQ_2 (3) P_2Q (4) P_3Q
41. If 'Z' is the number of atoms in the unit cell that represents the closest packing sequence ABCABC---, the number of tetrahedral voids in the unit cell is equal to
 (1) Z (2) 2Z (3) Z/2 (4) Z/4

IONIC CRYSTAL

42. The limiting radius ratio for tetrahedral shape is
(1) 0 to 0.155 (2) 0.155 to 0.225
(3) 0.225 to 0.414 (4) 0.414 to 0.732
43. For an octahedral arrangement the lowest radius ratio limit is
(1) 0.155 (2) 0.732 (3) 0.414 (4) 0.225
44. If the radius ratio is in the range of 0.414 - 0.732 then the co-ordination number will be :
(1) 2 (2) 4 (3) 6 (4) 8
45. In NaCl crystal r^+/r^- ratio is :
(1) 0.4 (2) 0.98 (3) 1.0 (4) 0.52
46. Which one of the following statements is incorrect about rock salt type ?
(1) It has fcc arrangement of Na^+
(2) Na^+ and Cl^- ions have a co-ordination number of 6 : 6
(3) A unit cell of NaCl consists of four NaCl units
(4) All halides of alkali metals have rock-salt type structure
47. In sodium chloride, Cl^- ions form ccp arrangement. Which site does Na^+ ions will occupy in this structure ?
(1) Cubic (2) Tetragonal
(3) Octahedral (4) Trigonal bipyramidal
48. The positions of Cl^- ions in NaCl structure are
(1) Corners of the cube
(2) Centres of faces of the cube
(3) Corners as well as centres of the faces of the cube
(4) Edge centres of the cube
49. The number of NaCl units present in a unit cell of NaCl are
(1) 1 (2) 2 (3) 4 (4) 8
50. The tetrahedral voids formed by ccp arrangement of Cl^- ions in rock salt structure are
(1) Occupied by Na^+ ions
(2) Occupied by Cl^- ions
(3) Occupied by either Na^+ or Cl^- ions
(4) Vacant
51. The structure of MgO is similar to NaCl. The co-ordination number of Mg is
(1) 2 (2) 6 (3) 4 (4) 8
52. The co-ordination number of Cs^+ and Cl^- ions in CsCl structure is
(1) 4 : 4 (2) 6 : 6 (3) 8 : 8 (4) 4 : 8
53. A unit cell of CsCl consists of
(1) one CsCl unit (2) two CsCl units
(3) four CsCl units (4) eight CsCl units
54. The NaCl structure can be converted into CsCl structure
(1) by application of high pressure
(2) by heating to 760 K
(3) both by heat and pressure
(4) the conversion is not possible
55. TlCl has structure similar to CsCl. The co-ordination number of Tl^+ is
(1) 4 (2) 6 (3) 10 (4) 8
56. The co-ordination number of Zn^{2+} and S^{2-} ions in the zinc blende (ZnS) type structure is
(1) 4 : 4 (2) 6 : 6 (3) 8 : 8 (4) 4 : 8
57. The co-ordination number of calcium fluoride (CaF_2) type structure is
(1) 1 : 2 (2) 4 : 4 (3) 4 : 8 (4) 8 : 4
58. The number of formula units in a unit cell of fluorite is
(1) 2 (2) 4 (3) 6 (4) 8
59. 4 : 4 Co-ordination is found in
(1) ZnS (2) CuCl (3) AgI (4) All
60. Antifluorite structure is derived from fluorite structure by
(1) Heating fluorite crystal lattice.
(2) Subjecting fluorite structure to high pressure.
(3) Interchanging the positions of positive and negative ions in the lattice.
(4) All of the above
61. A binary solid (A^+B^-) has a zinc blend structure with B^- ions constituting the lattice and A^+ ions occupying 25% tetrahedral holes. The formula of solid is.
(1) AB (2) A_2B (3) AB_2 (4) AB_4
62. The radius of Na^+ is 95 pm and that of Cl^- ion is 181 pm. Hence the co-ordination number of Na^+ will be
(1) 4 (2) 6 (3) 8 (4) 12
63. The ionic radii of Rb^+ and I^- are 1.46 and 2.16 Å respectively. The most probable type of structure exhibited by it is
(1) CsCl type (2) NaCl type
(3) ZnS type (4) CaF_2 type
64. A binary solid (A^+B^-) has a rock salt structure if the edge length is 400 pm and radius of cation (A^+) is 75 pm, the radius of anion (B^-) is
(1) 100 pm (2) 125 pm
(3) 250 pm (4) 325 pm

- 65.** A solid XY has a bcc structure. If the distance of closest approach between the two ions is 173 pm, the edge length of the cell is
- (1) 200 pm (2) $\frac{\sqrt{3}}{\sqrt{2}}$ pm
(3) 142.2 pm (4) $\sqrt{2}$ pm
- 66.** If the distance between Na^+ and Cl^- ions in NaCl crystal is 'a' pm, what is the length of the cell edge?
- (1) 2a pm (2) a/2 pm
(3) 4a pm (4) a/4 pm
- 67.** Potassium fluoride has NaCl-type structure. What is the distance between K^+ and F^- ions if its cell edge length is 'a' cm ?
- (1) 2a cm (2) a/2 cm
(3) 4a cm (4) a/4 cm
- 68.** KF has NaCl structure. What is the distance between K^+ and F^- in KF if density is 2.48 g cm^{-3} ?
- (1) 268.8 pm (2) 537.5 pm
(3) $155.3 \times 10^{-24} \text{ cm}$ (4) 5.375 cm
- 69.** In a NaCl type crystal distance between Na^+ and Cl^- ion is $2.814 \text{ \AA}$ and the density of solid is 2.167 g cm^{-3} then find out the value of Avogadro constant.
- (1) 6.05×10^{23} (2) 3.02×10^{23}
(3) 12.10×10^{23} (4) 9.6×10^{24}
- 70.** The density of crystalline sodium chloride is 5.85 g cm^{-3} . What is the edge length of the unit cell.
- (1) $4.06 \times 10^{-8} \text{ cm}$. (2) $1.32 \times 10^{-14} \text{ cm}$
(3) 7.8×10^{-23} (4) 9.6×10^{-24}
- 71.** A unit cell of sodium chloride has four formula units. The edge length of unit cell is 0.6 nm. What is the density of sodium chloride ?
- (1) 7.60 g cm^{-3} (2) 1.80 g cm^{-3}
(3) 9.60 g cm^{-3} (4) 6.38 g cm^{-3}

IMPERFECTIONS IN SOLIDS

- 72.** At zero kelvin, most of the ionic crystals possess
(1) Frenkel defect (2) Schottky defect
(3) Metal excess defect (4) No defect
- 73.** As a result of Schottky defect
(1) there is no effect on the density
(2) density of the crystal increases
(3) density of the crystal decreases
(4) any of the above three can happen
- 74.** Schottky as well as Frenkel defects are observed in the crystal of
(1) NaCl (2) AgBr (3) AgCl (4) MgCl_2
- 75.** Which one of the following is correct?
(1) Schottky defect lowers the density
(2) Frenkel defect increases the dielectric constant of the crystals
(3) Stoichiometric defects make the crystals good electrical conductors
(4) All the three
- 76.** Frenkel defect is generally observed in
(1) AgBr (2) AgI (3) ZnS (4) All of these
- 77.** Frenkel defect is found in crystals in which the radius ratio is :
(1) low (2) 1.3
(3) 1.5 (4) slightly less than unity
- 78.** F-centres in an ionic crystal are
(1) lattice sites containing electrons
(2) interstitial sites containing electrons
(3) lattice sites that are vacant
(4) interstitial sites containing cations
- 79.** The correct statement regarding F-centre is
(1) Electrons are held in the lattice site of crystals
(2) F-centre imparts colour to the crystal
(3) Conductivity of the crystal increases due to F-centre
(4) All the three above

EXERCISE-I (Conceptual Questions)

ANSWER KEY

[illegible]

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2006**

1. CsBr crystallises in a body centred cubic lattice. The unit cell length is 436.6 pm. Given that the atomic mass of Cs=133 and that of Br=80 amu and Avogadro number being $6.023 \times 10^{23} \text{ mol}^{-1}$ the density of CsBr is :
- (1) 42.5 g cm^{-3} (2) 0.425 g cm^{-3}
 (3) 8.25 g cm^{-3} (4) 4.25 g cm^{-3}

AIIMS 2006

2. The Ca^{2+} and F^- are located in CaF_2 crystal, respectively at face centred cubic lattice points and in –
- (1) Tetrahedral voids
 (2) Half of tetrahedral voids
 (3) Octahedral voids
 (4) half of octahedral voids

AIPMT 2007

3. NaCl is doped with $10^{-4} \text{ mol\%}$ SrCl_2 , the concentration of cation vacancies is –

$$\left[\text{Hint : Concentration of vacancies} = \frac{10^{-4}}{100} N_A \right]$$

- (1) $6.02 \times 10^{15} \text{ mol}^{-1}$ (2) $6.02 \times 10^{16} \text{ mol}^{-1}$
 (3) $6.02 \times 10^{17} \text{ mol}^{-1}$ (4) $6.02 \times 10^{14} \text{ mol}^{-1}$
4. The fraction of total volume occupied by the atoms present in a simple cube is –
- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3\sqrt{2}}$ (3) $\frac{\pi}{4\sqrt{2}}$ (4) $\frac{\pi}{4}$

AIPMT 2008

5. Percentage of free space in a body centred cubic unit is :
- (1) 32% (2) 34%
 (3) 28% (4) 30 %
6. If 'a' stands for the edge length of the cubic systems: simple cubic, body centred cubic and face centred cubic, then the ratio of radii of the spheres in these system will be respectively.

(1) $\frac{1}{2}a : \sqrt{3}a : \frac{1}{\sqrt{2}}a$ (2) $\frac{1}{2}a : \frac{\sqrt{3}}{2}a : \frac{\sqrt{2}}{2}a$
 (3) $1a : \sqrt{3}a : \sqrt{2}a$ (4) $\frac{1}{2}a : \frac{\sqrt{3}}{4}a : \frac{1}{2\sqrt{2}}a$

7. Which of the following statement is not correct ?
- (1) Molecular solids are generally volatile
 (2) The numbers of carbon atoms in a unit cell of diamond is 8
 (3) The number of Bravais lattices in which a crystal can be categorized is 14
 (4) The fraction of the total volume occupied by the atoms in a primitive cell is 0.48

AIPMT 2009

8. Lithium metal crystallises in a body centred cubic crystal. If the length of the side of the unit cell of lithium is 351 pm, the atomic radius of the lithium will be :-
- (1) 300.5 pm (2) 240.8 pm
 (3) 151.8 pm (4) 75.5 pm
9. Copper crystallises in a face-centred cubic lattice with a unit cell length of 361 pm. What is the radius of copper atom in pm ?
- (1) 108 (2) 128
 (3) 157 (4) 181

AIPMT 2010

10. AB crystallizes in a body centred cubic lattice with edge length 'a' equal to 387 pm. The distance between two oppositely charged ions in the lattice is :-
- (1) 300 pm (2) 335 pm
 (3) 250 pm (4) 200 pm

AIIMS 2010

11. In a trigonal crystal, which statement is incorrect
- (1) All the axial length and axial angle are equal
 (2) All the three axial lengths are equal
 (3) All the three axial angles are equal
 (4) Two axial angles are same and one is different

AIPMT Mains. 2011

12. A solid compound XY has NaCl structure. If the radius of the cation is 100 pm the radius of the anion (Y^-) will be :-
- (1) 165.7 pm
 (2) 275.1 pm
 (3) 322.5 pm
 (4) 241.5 pm

AIIMS 2011

- 13.** Schottky defect is–
- (1) vacancy of ions
 - (2) Delocalization of ions
 - (3) Interstitial vacancy of ions
 - (4) vacancy of only cation

AIPMT Pre. 2012

- 14.** The number of octahedral void(s) per atom present in a cubic-close-packed structure is:
(1) 2 (2) 4 (3) 1 (4) 3
- 15.** A metal crystallizes with a face-centered cubic lattice. The edge of the unit cell is 408 pm. The diameter of the metal atom is:
(1) 144 pm (2) 204 pm
(3) 288 pm (4) 408 pm

AIPMT Mains 2012

- 16.** Structure of a mixed oxide is cubic close-packed (CCP). The cubic unit cell of mixed oxide is composed of oxide ions. One fourth of the tetrahedral voids are occupied by divalent metal A and the octahedral voids are occupied by a monovalent metal B. The formula of the oxide is:
- (1) $A_2B_3O_4$ (2) AB_2O_2
(3) ABO_2 (4) A_2BO_2

AIIMS 2012

- 17.** In BCC structure, contribution of corner and central atom is :-
- (1) $\frac{1}{8}, 1$ (2) $\frac{1}{4}, \frac{1}{8}$
- (3) $\frac{1}{8}, \frac{1}{2}$ (4) $1, \frac{1}{2}$

NEET-UG 2013

- 18.** The number of carbon atoms per unit cell of diamond unit cell is :-
(1) 1 (2) 4 (3) 8 (4) 6
- 19.** A metal has a FCC lattice. The edge length of the unit cell is 404 pm. The density of the metal is 2.72 g cm^{-3} . The molar mass of the metal is :-
(1) 20 g mol^{-1} (2) 40 g mol^{-1}
(3) 30 g mol^{-1} (4) 27 g mol^{-1}

AIIMS 2013

- 20.** In a solid, atom M occupy CCP lattice and $1/3^{\text{rd}}$ of tetrahedral voids are occupied by atom N. Find formula of solid formed by M and N :-
- (1) M_3N_2 (2) M_2N_3
(3) M_4N_3 (4) M_3N_4

AIPMT 2014

- 21.** If a is the length of the side of a cube, the distance between the body centered atom and one corner atom in the cube will be :
- (1) $\frac{2}{\sqrt{3}}a$ (2) $\frac{4}{\sqrt{3}}a$ (3) $\frac{\sqrt{3}}{4}a$ (4) $\frac{\sqrt{3}}{2}a$

AIIMS 2014

- 22.** In any compound, atom B occupies in HCP packing and A is $\frac{1}{3}$ rd of tetrahedral voids. Then empirical formula of compound is :-
 (1) B_2A_3 (2) B_3A_2 (3) AB (4) AB_2

AIPMT 2015

- 23.** A given metal crystallizes out with a cubic structure having edge length of 361 pm. If there are four metal atoms in one unit cell, what is the radius of one atom?
- (1) 127 pm (2) 80 pm
(3) 108 pm (4) 40 pm

Re-AIPMT 2015

- 24.** The vacant space in BCC lattice unit cell is :
(1) 23% (2) 32% (3) 26% (4) 48%
- 25.** The correct statement regarding defects in crystalline solids is :-
(1) Frenkel defect is a dislocation defect
(2) Frenkel defect is found in halides of alkaline metals
(3) Schottky defects have no effect on the density of crystalline solids
(4) Frenkel defects decrease the density of crystalline solids

AIIMS 2015

- 26.** Defects in NaCl & AgCl crystals are respectively
 (1) Schottky, Frenkel (2) Frenkel, Schottky
 (3) Both Schottky (4) Both Frenkel

27. What will be value of $r_{Cs^+} + r_{Cl^-}$ in CsCl crystal having edge length 'a'

(1) $\frac{a\sqrt{3}}{2}$ (2) $a\sqrt{3}$ (3) $\frac{a\sqrt{3}}{4}$ (4) $\frac{a}{2\sqrt{2}}$

NEET-I 2016

28. Lithium has a BCC structure. Its density is 530 kg m^{-3} and its atomic mass is 6.94 g mol^{-1} . Calculate the edge length of a unit cell of Lithium metal. ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)
 (1) 154 pm (2) 352 pm
 (3) 527 pm (4) 264 pm
29. The ionic radii of A^+ and B^- ions are $0.98 \times 10^{-10} \text{ m}$ and $1.81 \times 10^{-10} \text{ m}$. The coordination number of each ion in AB is :-
 (1) 6 (2) 4 (3) 8 (4) 2

NEET-II 2016

30. In calcium fluoride, having the fluorite structure, the coordination numbers for calcium ion (Ca^{2+}) and fluoride ion (F^-) are
 (1) 8 and 4 (2) 4 and 8
 (3) 4 and 2 (4) 6 and 6

AIIMS 2016

31. In a crystalline solid, X atoms occupy hcp and $\frac{1}{4}$ th of tetrahedral voids are occupied by Y atoms. Then empirical formula of crystalline solid is :-
 (1) X_2Y (2) XY_2 (3) X_4Y (4) XY
32. A metal crystallizes in bcc lattice having a radius $4\text{\AA}$. Then what will be edge length of its unit cell?
 (1) $\frac{16}{\sqrt{3}}\text{\AA}$ (2) $16\sqrt{3}\text{\AA}$ (3) $\frac{16}{\sqrt{2}}\text{\AA}$ (4) $16\sqrt{2}\text{\AA}$

NEET(UG) 2017

33. Which is the **incorrect** statement ?
 (1) Density decreases in case of crystals with Schottky's defect
 (2) NaCl(s) is insulator, silicon is semiconductor, silver is conductor, quartz is piezo electric crystal

- (3) Frenkel defect is favoured in those ionic compounds in which sizes of cation and anions are almost equal
 (4) $\text{FeO}_{0.98}$ has non stoichiometric metal deficiency defect

AIIMS 2017

34. Order of packing efficiency is :-
 (1) HCP > fcc > BCC > simple cubic
 (2) HCP $\approx$ FCC > simple cubic > BCC
 (3) BCC > HCP > simple cubic > fcc
 (4) HCP $\approx$ FCC > BCC > simple cubic
35. In a cube X atoms are present at corners and Y atoms are at face centres of a cube. The number of X and Y are :-
 (1) 8, 8 (2) 6, 8 (3) 8, 6 (4) 6, 6

NEET-II 2018

36. Iron exhibits bcc structure at room temperature. Above 900°C , it transforms to fcc structure. The ratio of density of iron at room temperature to that at 900°C (assuming molar mass and atomic radii of iron remains constant with temperature) is
 (1) $\frac{\sqrt{3}}{\sqrt{2}}$ (2) $\frac{4\sqrt{3}}{3\sqrt{2}}$ (3) $\frac{3\sqrt{3}}{4\sqrt{2}}$ (4) $\frac{1}{2}$

AIIMS 2018

37. F-centre is produced due to :-
 (1) Unpaired electron occupied anionic vacancies
 (2) Monovalent ion is replaced by bivalent ion
 (3) Ions present in cationic vacancies
 (4) Disslocation of ions
38. Which of the following is correct for ZnO :-
 (1) Impart color due to charge transfer
 (2) On heating it turns yellow
 (3) Zn can reduce hydrogen at room temprature
 (4) It sublimes on heating
39. Among the following the molecular solid is :-
 (1) $\text{SO}_{2(\text{solid})}$ (2) SiC (3) Graphite (4) ZnS
40. In any compound, atoms of A occupies in HCP packing and atoms of B is at $\frac{1}{3}$ rd of tetrahedral voids. Then emperical formula of compound is :-
 (1) A_2B_3 (2) A_3B_2 (3) B_3A (4) BA_2

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	3	1	1	4	4	3	2	2	4	4	1	3	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	3	4	1	4	2	1	2	1	1	1	2	1	1
Que.	31	32	33	34	35	36	37	38	39	40					
Ans.	1	1	3,4	4	3	3	1	2	1	2					

EXERCISE-III (Analytical Questions)

Check Your Understanding

- What is the % of volume occupied by atom in BCC
(1) 58% (2) 68%
(3) 78% (4) 88%
- Coordination number in HCP is :-
(1) 6 (2) 8
(3) 12 (4) 18
- The lattice structure of group I elements is
(1) FCC (2) BCC
(3) HCP (4) CCP
- Elements Na and Mg are Crystalizes in bcc and fcc respectively, then number of atoms of Na and Mg per unit cell are respectively :-
(1) 2, 4 (2) 9, 14
(3) 14, 9 (4) 4, 2
- In a face centred cubic lattice, atom A occupies the corner positions only and atom B occupies the face centre positions. If one atom of B is missing from one of the face centred points, the formula of the compound is :-
(1) A_2B_3 (2) A_2B_5
(3) A_2B (4) AB_2
- An alloy of copper, silver and gold is found to have copper constituting the fcc lattice. If silver atoms occupy the edge centres and gold is present at body centre, the alloy has a formula
(1) Cu_4Ag_2Au (2) Cu_4Ag_4Au
(3) Cu_4Ag_3Au (4) $CuAgAu$
- Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :-
(1) 7.01% (2) 4.08%
(3) 6.05% (4) 5.08%

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7
Ans.	2	3	2	1	2	3	2

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion :-** Metals are generally good conductors of electricity.

Reason :- Electrical conductivity of metals is due to schottky defect.

- (1) A (2) B (3) C (4) D

2. **Assertion:-** Graphite is an example of tetragonal crystal system.

Reason:- For a tetragonal system $a=b \neq c$, $\alpha=\beta=90^\circ$, $\gamma=120^\circ$.

- (1) A (2) B (3) C (4) D

3. **Assertion :-** BCC & HCP has same packing efficiency

Reason :- Both have same number of atoms per unit Cell and same arrangement.

- (1) A (2) B (3) C (4) D

4. **Assertion :-** The number of tetrahedral void is double the number of octahedral voids.

Reason :- The size of the tetrahedral void is half of that of the octahedral void.

- (1) A (2) B (3) C (4) D

5. **Assertion :-** In CsCl crystal, the Co-ordination number of Cs^+ ions is 6.

Reason :- Cl^- ion in CsCl, adopt body centred cubic arrangement.

- (1) A (2) B (3) C (4) D

6. **Assertion :-** Na_2O adopts structure similar to that of CaF_2 but positions of positive and negative ions are reversed.

Reason :- The structure of Na_2O is also called antifluorite structure.

- (1) A (2) B (3) C (4) D

7. **Assertion :-** FeO has metal deficiency defect.

Reason :- Some Fe^{+3} replace Fe^{+2} ion in crystal structure.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7
Ans.	3	4	4	3	4	2	1

CHEMICAL KINETICS

E

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NEET SYLLABUS

CHEMICAL KINETICS : Rate of a reaction (average and instantaneous), order and molecularity of a reaction; rate law and specific rate constant, integrated rate equations and half life (only for zero and first order reactions); concept of collision theory (elementary idea, no mathematical treatment). Activation energy, Arrhenius equation; factors affecting rate of reaction; concentration, temperature, catalyst.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- define the average and instantaneous rate of a reaction.
- write the rate of reaction in form of change in concentration of reactants or products.
- compare elementary and complex reactions.
- compare molecularity and order of a reaction.
- describe rate constant
- explain the dependence of rate of reaction on concentration, temperature and catalyst.
- derive integrated rate equations for zero, first and n^{th} order reactions.
- determine rate constants for various reactions.
- describe collision theory of reaction rates.

"Kinetic research conducts to the knowledge of philosophical truth, and forms the mind to philosophical enlargement and accuracy of thought, more happily than almost any other investigation in which the human intellect can be employed"

Alexander Tilloch

CHEMICAL KINETICS

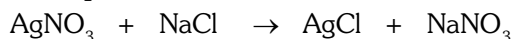
2.0 INTRODUCTION

Chemical Kinetics is the branch of physical chemistry which deals with the study of rate of reactions, the mechanism by which the reactions proceed and factors affecting rate of reaction.

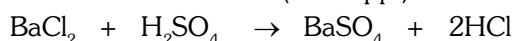
On the basis of rate, chemical reaction are broadly divided into three categories :-

(a) Very fast or instantaneous reactions : Generally these reactions involve ionic species and known as ionic reactions. These reactions take about 10^{-14} or 10^{-16} seconds for completion. So, it is almost impossible to determine the rate of these reactions.

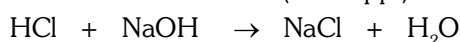
Examples :



(white ppt)



(white ppt)

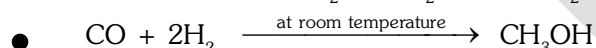


(b) Very slow reactions : These reactions proceed very slowly, may take days or months to show any measurable change at room temperature.

Examples :

- Rusting of iron.

- Reaction between H_2 and O_2 to form H_2O at ordinary temperature in absence of catalyst.



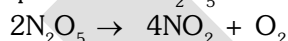
(c) Moderate or slow reactions : This type of reactions proceed with a measurable rates at normal temperature and we can measure the rate of these reactions easily. Mostly these reactions are molecular in nature.

Examples :

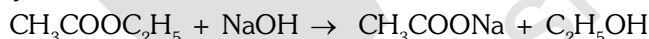
- Decomposition of H_2O_2



- Decomposition of N_2O_5



- Hydrolysis of ester



- Inversion of cane sugar in aqueous solution

- Reaction of NO with chlorine



2.1 RATE OF REACTION

Rate of reaction is defined as the change in concentration or pressure of reactant or product per unit time. It is always a positive quantity.

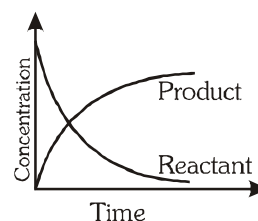
$$\text{Rate of reaction} = \frac{\text{Change in concentration of reactant or product}}{\text{Time taken in change}}$$

$$r = \pm \frac{\Delta C}{\Delta t}$$

Where ΔC = change in concentration in a small interval Δt
 [+] sign is used when we refer for product concentration.
 [-] sign is used when we refer for reactant concentration.

$$\text{For gaseous reactions} \quad r = \pm \frac{\Delta P}{\Delta t} \quad (\text{unit of rate} = \text{pressure time}^{-1})$$

$$\text{and} \quad r = \frac{1}{RT} \times \left[\pm \frac{\Delta P}{\Delta t} \right] \quad (\text{unit of rate} = \text{M time}^{-1})$$



Types of Rate of Reaction :-**(A) Average Rate of Reaction**

The rate of reaction over a certain measurable period of time during the course of reaction is called average rate of reaction. It is denoted by $\bar{r}$.

For a reaction $A \longrightarrow B$

$$r_{\text{average}} = \bar{r} = \left(\frac{[A]_2 - [A]_1}{t_2 - t_1} \right) = - \frac{\Delta[A]}{\Delta t}$$

Where $[A]_1$ = Concentration of reactant A at time t_1 ,

$[A]_2$ = Concentration of reactant A at time t_2 .

(B) Instantaneous Rate of Reaction

The rate of reaction at any particular instant during the course of reaction is called instantaneous rate of reaction.

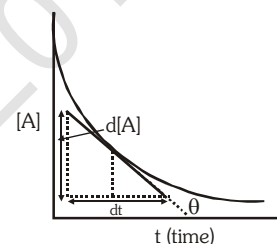
For a reaction $A \longrightarrow B$

Mathematically ; Instantaneous rate = $\lim_{\Delta t \rightarrow 0} (\text{Average rate})$

$$r_{\text{inst}} = \lim_{\Delta t \rightarrow 0} \left(- \frac{\Delta[A]}{\Delta t} \right) = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta[B]}{\Delta t} \right) \quad \text{or} \quad \boxed{r_{\text{inst}} = (-) \frac{d[A]}{dt} = (+) \frac{d[B]}{dt}}$$

Hence, Slope of the tangent at time t in plot of concentration with time gives instantaneous rate of reaction.

$$\text{Instantaneous rate of reaction} = \lim_{\Delta t \rightarrow 0} \left(\pm \frac{\Delta C}{\Delta t} \right) = \pm \frac{dC}{dt}$$

**2.2 RATE OF REACTION IN THE FORM OF STOICHIOMETRY OF A CHEMICAL REACTION**

Let us consider a reaction : $m_1A + m_2B \rightarrow n_1P + n_2Q$

Where, Rate of disappearance of A = $-\frac{d[A]}{dt}$

Rate of disappearance of B = $-\frac{d[B]}{dt}$

Rate of appearance of P = $\frac{d[P]}{dt}$

Rate of appearance of Q = $\frac{d[Q]}{dt}$

$$\boxed{\text{Rate of reaction} = \frac{1}{m_1} \left(- \frac{d[A]}{dt} \right) = \frac{1}{m_2} \left(- \frac{d[B]}{dt} \right) = \frac{1}{n_1} \frac{d[P]}{dt} = \frac{1}{n_2} \frac{d[Q]}{dt}}$$

- Rate of reaction is always positive; negative sign represents decrease in concentration of reactant.

Units of Rate of Reaction

Unit of rate of reaction = $\text{mol L}^{-1} \text{time}^{-1}$ i.e. ($\text{mol L}^{-1} \text{s}^{-1}$ or $\text{mol L}^{-1} \text{min}^{-1}$ or $\text{mol L}^{-1} \text{h}^{-1}$)

Illustrations

Illustration 1. For the reaction : $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$. If the concentration of NO_2 increases by $1.6 \times 10^{-2} \text{ M}$ in 4 s. Calculate the following -

- (i) rate of formation of NO_2 (ii) rate of formation of O_2
(iii) rate of disappearance of N_2O_5 (iv) rate of reaction

Solution.

(i) Rate of formation of NO_2 : $\frac{d[\text{NO}_2]}{dt} = \frac{1.6 \times 10^{-2}}{4} = 4.0 \times 10^{-3} \text{ M/s}$

(ii) Rate of formation of O_2 : $\frac{d[\text{O}_2]}{dt} = \frac{1}{4} \frac{d[\text{NO}_2]}{dt} = \frac{1}{4} \times 4 \times 10^{-3} = 1.0 \times 10^{-3} \text{ M/s}$

(iii) Rate of disappearance of N_2O_5 : $-\frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{1}{2} \frac{d[\text{NO}_2]}{dt} = \frac{1}{2} \times 4 \times 10^{-3} = 2.0 \times 10^{-3} \text{ M/s}$

(iv) Rate of reaction = $\frac{1}{2} \left(-\frac{d[\text{N}_2\text{O}_5]}{dt} \right) = \frac{1}{2} \times 2 \times 10^{-3} = 1.0 \times 10^{-3} \text{ M/s}$

or $\frac{1}{4} \frac{d[\text{NO}_2]}{dt} = \frac{1}{4} \times 4 \times 10^{-3} = 1.0 \times 10^{-3} \text{ M/s}$

or $\frac{d[\text{O}_2]}{dt} = 1.0 \times 10^{-3} \text{ M/s}$

Illustration 2. Express the rate of reaction for the following changes :

- (a) $2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$
(b) $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$

Solution.

(a) Rate of reaction = $\frac{1}{2} \left(-\frac{d[\text{HI}]}{dt} \right) = \frac{d[\text{H}_2]}{dt} = \frac{d[\text{I}_2]}{dt}$

(b) Rate of reaction = $\frac{1}{2} \left(-\frac{d[\text{SO}_2]}{dt} \right) = \frac{-d[\text{O}_2]}{dt} = \frac{1}{2} \frac{d[\text{SO}_3]}{dt}$

BEGINNER'S BOX-1

1. The rate of a reaction is expressed as :

$$+\frac{1}{2} \frac{\Delta[\text{C}]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{D}]}{\Delta t} = \frac{1}{4} \left(-\frac{\Delta[\text{A}]}{\Delta t} \right) = \left(-\frac{\Delta[\text{B}]}{\Delta t} \right)$$

Then reaction is

- (1) $4\text{A} + \text{B} \rightarrow 2\text{C} + 3\text{D}$
(3) $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$

- (2) $\text{B} + 3\text{D} \rightarrow 4\text{A} + 2\text{C}$
(4) $\text{B} + \text{D} \rightarrow \text{A} + \text{C}$

2. In the reaction, $\text{A} + 2\text{B} \rightarrow 6\text{C} + 2\text{D}$ if the initial rate $(-)\frac{\Delta[\text{A}]}{\Delta t}$ at $t = 0$ is $2.6 \times 10^{-2} \text{ M s}^{-1}$, what will

be the value of $(-)\frac{\Delta[\text{B}]}{\Delta t}$ at $t = 0$?

- (1) $8.5 \times 10^{-2} \text{ M s}^{-1}$ (2) $2.6 \times 10^{-2} \text{ M s}^{-1}$ (3) $5.2 \times 10^{-2} \text{ M s}^{-1}$ (4) $7.5 \times 10^{-2} \text{ M s}^{-1}$

3. In the following reaction, how is the rate of appearance of the underlined product related to the rate of disappearance of the underlined reactant $\text{BrO}_3^-(\text{aq}) + 5\text{Br}^-(\text{aq}) + 6\text{H}^+(\text{aq}) \rightarrow \underline{3\text{Br}_2}(\ell) + 3\text{H}_2\text{O}(\text{aq})$

- (1) $(-)\frac{\Delta[\text{BrO}_3^-]}{\Delta t} = \frac{\Delta[\text{Br}_2]}{\Delta t}$ (2) $(-)\frac{1}{3} \frac{\Delta[\text{BrO}_3^-]}{\Delta t} = \frac{\Delta[\text{Br}_2]}{\Delta t}$ (3) $(-)\frac{\Delta[\text{BrO}_3^-]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{Br}_2]}{\Delta t}$ (4) None of these

2.3 RATE LAW

The experimental expression of rate of reaction in terms of concentration of reactants is known as rate law. In this expression the rate of a reaction is proportional to the product of molar concentration of reactants with each term raised to the power or exponent that has to be found experimentally.

In a chemical reaction :- $aA + bB \longrightarrow \text{Product}$

The rate law is :- $\text{Rate} \propto [A]^x[B]^y$

The values of exponents x and y are found experimentally which may or may not be same as stoichiometric coefficients.

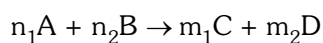
Above relationship can be written as :-

$$\text{Rate} = k[A]^x[B]^y$$

Where k is a proportionality constant known as rate constant.

Rate constant :

In a chemical reaction –



according to law of mass action

$$\text{Rate} = k[A]^{n_1}[B]^{n_2}$$

but according to rate law (experimental concept)

$$\text{Rate} = k[A]^x[B]^y$$

if $[A] = [B] = 1 \text{ mol/L}$

then, $\text{Rate} = k$

Rate of reaction at unit concentration of reactants is called as rate constant or specific reaction rate.

Rate constant does not depend on concentration of reactant but it depends on temperature and catalyst.

2.4 ORDER OF REACTION –

The sum of powers of concentration of reactants in rate law expression is known as order of reaction.

For the reaction $aA + bB \rightarrow \text{Product}$

Rate law is $\text{Rate} = k[A]^x[B]^y$

Here

$x = \text{order of reaction with respect to A}$

$y = \text{order of reaction with respect to B}$

$x + y = n \text{ (overall order of reaction)}$

- Order of reaction may be zero, positive, negative or fractional.
- Order of reaction is an experimental quantity.

Units of rate constant :

$$\text{Rate} = k[A]^n$$

$$k = \frac{r}{[A]^n} = \frac{\text{unit of rate}}{[\text{unit of concentration}]^n} = \frac{\frac{\text{mol}}{\text{L}} \times \text{time}^{-1}}{\left[\frac{\text{mol}}{\text{L}}\right]^n}$$

- Unit of $k = \left[\frac{\text{mol}}{\text{L}}\right]^{1-n} \times \text{time}^{-1}$
- For gaseous reaction unit of k may be $= (\text{atm})^{1-n} \times \text{time}^{-1}$

2.5 MECHANISM OF REACTION :

(a) Elementary reactions :

Those reactions which completes in single step and which have exponents in rate law equal to stoichiometric coefficients of the reactants.

If $A + B \longrightarrow \text{Products}$; is an elementary reaction
then rate law will be -

$$\text{Rate} = k[A][B]$$

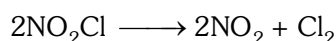
- Zero order reactions can never be elementary reactions.
- For elementary reactions fractional order is not possible.

(b) Complex reactions :

Those reactions which complete in multisteps. For these reactions a mechanism is proposed.

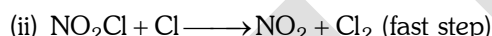
- For complex reactions the overall rate of reaction is controlled by the slowest step which is called as rate determining step (R.D.S.).
- In rate law expression rate of reaction depends on concentration of reactants of slowest step which must be free from intermediate.
- If R.D.S. contains intermediate, its value is solved using K_{eq} of fast step (assumed as reversible)

Example-1



Experimentally, the rate law is $\text{Rate} = k[\text{NO}_2\text{Cl}]$

The mechanism of the reaction is given as –



So the rate law from slowest step $\text{Rate} = k[\text{NO}_2\text{Cl}]$

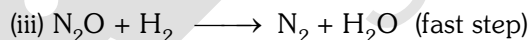
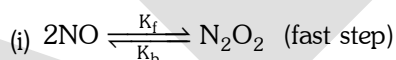
In this way the predicted rate law derived from two step mechanism agrees with experimental rate law.

Example-2



experimentally the rate law is, $\text{Rate} = k [\text{NO}]^2[\text{H}_2]$

The mechanism of the reaction is given as –



The Rate law from slowest step is :

$$\text{Rate} = k[\text{N}_2\text{O}_2] [\text{H}_2]$$

The rate law expression should be free from intermediate species N_2O_2 .

From fast reversible step –

$$k_f[\text{NO}]^2 = k_b[\text{N}_2\text{O}_2]$$

$$\therefore [\text{N}_2\text{O}_2] = \frac{k_f}{k_b} [\text{NO}]^2$$

and rate law becomes

$$\text{Rate} = k \left(\frac{k_f}{k_b} \right) [\text{NO}]^2 [\text{H}_2]$$

therefore, $\text{Rate} = k'[\text{NO}]^2[\text{H}_2]$

This derived rate law agrees with experimental rate law.

2.6 MOLECULARITY

Total number of molecules, atoms or ions (reacting species) participating in an elementary reaction is called as molecularity of reaction.

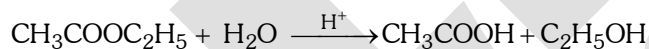
- Molecularity is a theoretical quantity.
- Molecularity can be an integer (1, 2 or 3) but it cannot be zero or negative or fractional.
- In elementary reaction molecularity is equal to its order.
- In complex reaction molecularity of each step of mechanism is defined separately.
- Total molecularity of complex reaction is meaningless.
- In complex reactions generally molecularity of slowest step is same as order of reaction which can be considered as molecularity of reaction. (Except when slowest step contain intermediate)
- Maximum value of molecularity or order is 3 because chances of effective collision of more than three molecules is very rare.

2.7 PSEUDO FIRST ORDER REACTION

A chemical reaction in which value of order of reaction is one but molecularity is more than one are known as pseudo unimolecular/pseudo first order reaction.

Example-1

Hydrolysis of ester in acidic medium.



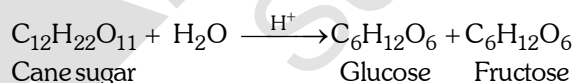
$$\text{Rate} = k [\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]$$

Water is in excess then its concentration remain constant during the reaction and $[\text{H}_2\text{O}]$ is taken as constant therefore,

$$\text{Rate} = k' [\text{CH}_3\text{COOC}_2\text{H}_5] \quad \text{where } k' = k [\text{H}_2\text{O}]$$

Example-2

Inversion of cane sugar.



$$\text{Rate} = k [\text{C}_{12}\text{H}_{22}\text{O}_{11}][\text{H}_2\text{O}]$$

Water is in excess then its concentration remain constant during the reaction and $[\text{H}_2\text{O}]$ is taken as constant therefore,

$$\text{Rate} = k' [\text{C}_{12}\text{H}_{22}\text{O}_{11}] \quad \text{where } k' = k [\text{H}_2\text{O}]$$

GOLDEN KEY POINTS

- Reactant taken in excess can't affect order of reaction.
- In certain complex reaction product is also considered in order calculation.
- Order of reaction is determined experimentally.

Illustrations

Illustration 3. Give the following data for the reaction : $A + B \rightarrow \text{Product}$.

Experiment No.	[A]	[B]	ROR ($\text{mol L}^{-1}\text{s}^{-1}$)
1	1	2	4
2	2	2	4
3	2	4	16

What is the rate law equation ?

Solution. We know that rate law is given by : $r = k[A]^x[B]^y$ (1)

put the values of experiment No. 1 and 2 in equation 1, we get

$$4 = k[1]^x[2]^y \quad \text{.....(2)}$$

$$\text{and } 4 = k[2]^x[2]^y \quad \text{.....(3)}$$

Now, Equation (2) is divided by (3), we get

$$1 = \left(\frac{1}{2}\right)^x \quad \text{i.e., } x = 0$$

Now, put the values of experiment no. 2 and 3 in equation (1), we get

$$4 = (2)^x (2)^y \quad \text{.....(4)}$$

$$16 = (2)^x (4)^y \quad \text{.....(5)}$$

Equation (4) is divided by (5), we get

$$\frac{1}{4} = \left(\frac{1}{2}\right)^y \quad \text{i.e., } y = 2$$

So, order of reaction with respect to 'A' is 0 and with respect to 'B' is 2 and the rate of reaction is $r = k[A]^0[B]^2 = k[B]^2$ i.e., order of reaction = 2

Illustration 4. The decomposition of dimethyl ether leads to the formation of CH_4 , H_2 and CO and the reaction rate is given by :

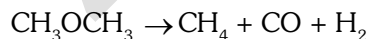
$$\text{Rate} = k[\text{CH}_3\text{OCH}_3]^{3/2}$$

The rate of reaction is followed by increase in pressure in a closed vessel and the rate can also be expressed in terms of the partial pressure of dimethyl ether i.e.,

$$\text{Rate} = k(P_{\text{CH}_3\text{OCH}_3})^{3/2}$$

If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constant?

Solution.



$$\text{Rate} = k[\text{CH}_3\text{OCH}_3]^{3/2} = k(P_{\text{CH}_3\text{OCH}_3})^{3/2}$$

$$\text{Unit of rate} = \text{bar min}^{-1}$$

$$\text{Unit of } k = \frac{\text{Rate}}{(P_{\text{CH}_3\text{OCH}_3})^{3/2}} = \frac{\text{bar min}^{-1}}{\text{bar}^{3/2}} = \text{bar}^{-1/2} \text{ min}^{-1}$$

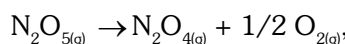
Illustration 5. For a reaction ; $3A \rightarrow \text{Products}$, it is found that the rate of reaction doubles if concentration of A is increased four times, calculate order of reaction.

Solution. Rate = $k[\text{Reactant}]^n$ if $[\text{Reactant}] = a$; rate = r_1
 $r_1 = k[a]^n$ if $[\text{Reactant}] = 4a$; rate = $2r_1$

$$2r_1 = k[4a]^n ; \frac{1}{2} = \left[\frac{1}{4}\right]^n \Rightarrow n = \frac{1}{2}$$

Illustration 6. For the decomposition, $N_2O_{5(g)} \rightarrow N_2O_{4(g)} + 1/2 O_{2(g)}$, the initial pressure of N_2O_5 is 114 mm and after 20 s the pressure of reaction mixture becomes 133 mm of Hg. Calculate the rate of reaction in terms of (a) change in atm s^{-1} and (b) change in molarity s^{-1} . Given that reaction is carried out at 127°C .

Solution.



Initial pressure P 0 0

Pressure at $t = 20$ s $(P - P')$ P' $P'/2$

Given $P = 114$ mm

After 20 s, total pressure = $(P - P') + P' + (P'/2) = 133$ mm

$$\Rightarrow P'/2 = 19 \text{ mm} \quad P' = 38 \text{ mm}$$

Thus rate of reaction in terms of change in pressure = $38/20 = 1.9 \text{ mm s}^{-1}$

$$= \frac{1.9}{760} \text{ atm s}^{-1} = 2.5 \times 10^{-3} \text{ atm s}^{-1}$$

Also we have $PV = nRT$ or $\frac{n}{V} = C = \frac{P}{RT}$

$\therefore$ change in concentration in 20 second = $\Delta C = \frac{P'}{RT}$ (change in pressure)

$$= \frac{38}{760 \times 0.0821 \times 400} = 1.52 \times 10^{-3} \text{ M}$$

$\therefore$ rate of reaction in terms of change in concentration = $\frac{1.52 \times 10^{-3}}{20} = 7.61 \times 10^{-5} \text{ M s}^{-1}$

BEGINNER'S BOX-2

1. $A_2 + B_2 \rightarrow 2AB$; R.O.R. = $k[A_2]^a[B_2]^b$

$[A_2]$	$[B_2]$	Rate of reaction $[\text{Ms}^{-1}]$
0.2	0.2	0.04
0.1	0.4	0.04
0.2	0.4	0.08

order of reaction with respect to A_2 and B_2 are respectively :

(1) $a=1, b=1$ (2) $a=2, b=0$ (3) $a=2, b=1$ (4) None

2. For a reaction the initial rate is given as : $R_0 = k[A]_0^2[B]_0$ by what factor, the initial rate of reaction will increase if initial concentration of A is 1.5 times and B is tripled ?

(1) 4.5 (2) 2.25 (3) 6.75 (4) None of these

3. For $A_{(g)} + B_{(g)} \rightarrow C_{(g)}$; rate = $k[A]^{1/2}[B]^2$, if initial concentration of A and B are increased by factor of 4 and 2 respectively, then the initial rate is changed by the factor :-

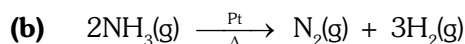
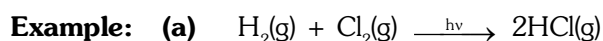
(1) 4 (2) 6 (3) 8 (4) None of these

2.8 STUDY OF DIFFERENT ORDER REACTIONS :

(A) Zero order reactions

Reactions in which rate of reaction remains independent of concentration of the reactant are said to be zero order reactions.

- Zero order reactions are relatively uncommon but they occur under special conditions. Some enzyme catalysed reactions and reactions which occur on metal surfaces are a few examples of zero order reactions.

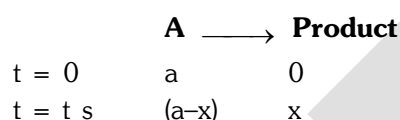


(c) Reaction between Acetone and Bromine.

(d) Dissociation of HI on gold surface.

(e) Adsorption of gases on metal surface :- At low P, rate of adsorption is proportional to surface area covered which is proportional to P or concentration of gas hence order is 1 whereas at high P, complete surface gets covered by gas & rate becomes independent of P & concentration hence order is 0.

Differential Rate Equation



$$-\frac{d[A]}{dt} = k[A]^0$$

$$\frac{dx}{dt} = k[A]^0$$

Calculation of Rate Constant

Let us take the reaction



$$-\frac{d[A]}{dt} = k[A]^0 = k$$

$$- \int d[A] = \int k dt$$

$$- [A]_t = kt + C$$

$$\text{at } t = 0 \quad [A]_t = [A]_0$$

$$- [A]_0 = k \times 0 + C$$

$$C = - [A]_0$$

On substituting the value of C

$$- [A]_t = kt - [A]_0$$

$$[A]_t = [A]_0 - kt \quad [\text{Integrated rate equation}]$$

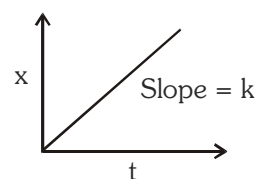
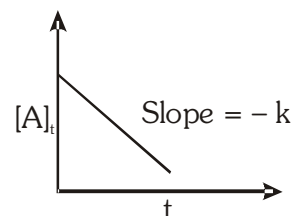
$$y = c - mx$$

$$[A]_0 - [A]_t = kt$$

$$k = \frac{[A]_0 - [A]_t}{t} = \frac{x}{t}$$

$$x = kt$$

x = Amount of reactant that will change in product.



- For zero order reaction, rate of reaction is equal to rate constant

Unit of rate constant

$k = \text{mol L}^{-1} \text{s}^{-1} = \text{unit of rate of reaction.}$

Half-life period – The time in which half of the initial amount of reactant is consumed.

$$\text{At } t = t_{1/2}, \quad [A]_t = \frac{[A]_0}{2}$$

$$\therefore kt_{1/2} = [A]_0 - \frac{[A]_0}{2} \quad \text{or} \quad t_{1/2} = \frac{[A]_0}{2k}$$

The half life period for a zero order reaction is directly proportional to the initial concentration of the reactants.

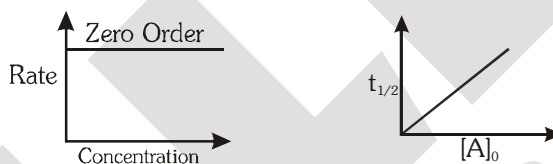
Time for completion of reaction

$$[A]_t = [A]_0 - kt$$

For completion $[A]_t = 0$

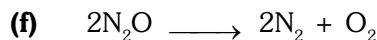
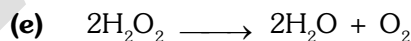
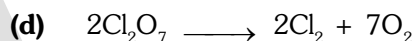
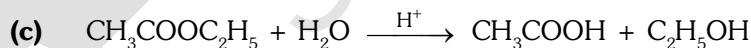
$$k = \frac{[A]_0}{t}$$

$$t_{100\%} = \frac{[A]_0}{k}$$

Graphical representation**(B) First order reactions**

Reactions in which the rate of reaction is directly proportional to concentration of reactant.

Example :

**Differential rate equation**

	A	Product
$t = 0$	a	0
$t = t \text{ s}$	(a-x)	x

$$-\frac{d[A]}{dt} = k[A]$$

$$\frac{dx}{dt} = k(a-x)$$

Calculation of rate constant

$$-\int \frac{d[A]}{[A]} = k \int dt$$

$$-\ln[A]_t = kt + c \dots\dots\dots(i)$$

At $t = 0$; $[A]_t = [A]_0 \therefore C = -\ln[A]_0$

Putting the value of C in equation (i)

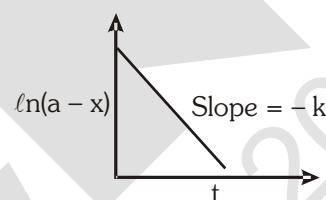
$$-\ln[A]_t = kt - \ln[A]_0$$

$$\ln[A]_t = \ln[A]_0 - kt$$

$$y = c - mx$$

$$\ln \frac{[A]_0}{[A]_t} = kt \dots\dots\dots(ii)$$

$$kt = \ln \left(\frac{a}{a-x} \right)$$



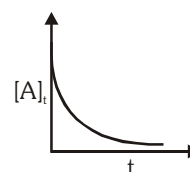
$$2.303 \log_{10} \frac{[A]_0}{[A]_t} = kt$$

$$K = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

from equation(ii)

$$\frac{[A]_0}{[A]_t} = e^{kt}$$

$$\frac{[A]_t}{[A]_0} = e^{-kt} \Rightarrow [A]_t = [A]_0 e^{-kt} \text{ Wilhelmy equation}$$



Unit of rate constant $[k = \text{time}^{-1}]$

Half-life Period : The time in which half of the initial amount of reactant is consumed.

At $t = t_{1/2}$; $x = a/2$; $a-x = a/2$

$$t_{1/2} = \frac{1}{k} \ln \left(\frac{a}{a/2} \right) \quad \text{or} \quad t_{1/2} = \frac{\ln 2}{k} = \frac{2.303}{k} (\log 2) \quad \text{or} \quad t_{1/2} = \frac{0.693}{k}$$

Half-life period for first order reaction is independent of the initial concentration of reactant.

Time for 3/4th of the Reaction ($t_{3/4}$) : The time in which 3/4th of the initial amount of reactant is consumed.

At $t = t_{3/4}$; $x = 3a/4$; $a-x = a/4$

$$t_{3/4} = \frac{2.303}{k}(\log 4) = \frac{2.303}{k} \times 2 \log 2 = 2 \times t_{1/2}$$

$t_{3/4}$ for first order reaction is independent of the initial concentration of reactant.

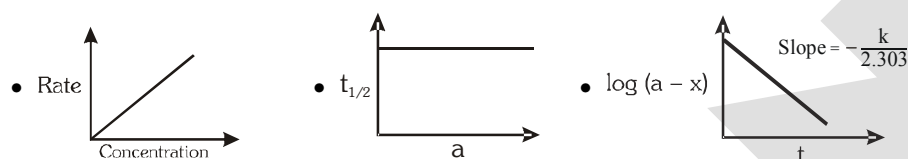
Interval Formula

$$k = \frac{2.303}{t_2 - t_1} \log \frac{a - x_1}{a - x_2}$$

where x_1 and x_2 are the amount consumed at time t_1 and t_2 respectively.

Time required for the completion of definite fraction of the first order reaction is independent of the initial concentration of the reactant.

Graphical Representation



(C) GENERAL INTEGRATED RATE EQUATION (n^{th} order kinetics)

$$kt = \frac{1}{(n-1)} \left[\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}} \right] \quad [n \neq 1]$$

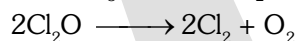
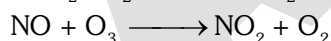
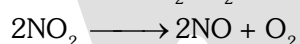
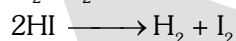
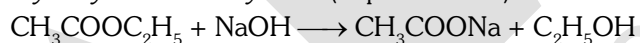
If $t = t_{1/2}$; $x = \frac{a}{2}$

Therefore $kt_{1/2} = \frac{1}{(n-1)} \left[\left(\frac{2}{a} \right)^{n-1} - \left(\frac{1}{a} \right)^{n-1} \right]$

$$kt_{1/2} = \frac{1}{(n-1)} \left[\frac{2^{n-1} - 1}{a^{n-1}} \right] \quad (n \neq 1) ; \quad t_{1/2} \propto \frac{1}{a^{n-1}}$$

(i) Second order reactions

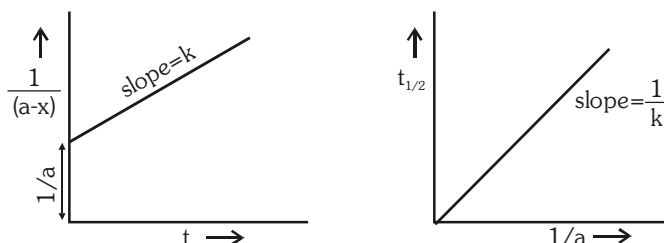
Hydrolysis of ester by alkali (Saponification)



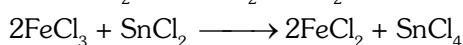
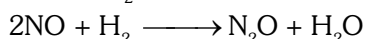
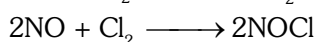
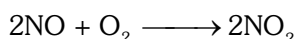
For second order : $n = 2$

$$kt = \frac{1}{(2-1)} \left[\frac{1}{(a-x)} - \frac{1}{a} \right] = \frac{1}{(a-x)} - \frac{1}{a}$$

Half life $t_{1/2} = \frac{1}{ak}$

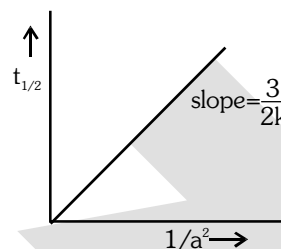
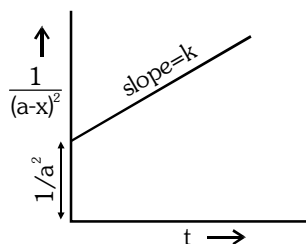


(ii) **Third order reactions**



for third order $n = 3$
$$kt = \frac{1}{(3-1)} \left[\frac{1}{(a-x)^2} - \frac{1}{a^2} \right] = \frac{1}{2} \left[\frac{1}{(a-x)^2} - \frac{1}{a^2} \right]$$

Half life
$$t_{1/2} = \frac{3}{2a^2k}$$



Examples of fractional order reaction

Reaction	Order
$\text{H}_2 + \text{Br}_2 \longrightarrow 2\text{HBr}$	1.5
$\text{CO} + \text{Cl}_2 \longrightarrow \text{COCl}_2$	2.5
$\text{COCl}_2 \longrightarrow \text{CO} + \text{Cl}_2$	1.5
$\text{CH}_3\text{CHO} \longrightarrow \text{CH}_4 + \text{CO}$	1.5
$\text{CHCl}_3 + \text{Cl}_2 \longrightarrow \text{CCl}_4 + \text{HCl}$	1.5

Illustrations

Illustration 7. 90% of a first order reaction was completed in 10 hours. When will 99.9% of the reaction complete ?

Solution.
$$k_1 = \frac{2.303}{t} \log \frac{a}{a-x}, \quad a = 100, \quad x = 90, \quad t = 10$$

So
$$k_1 = \frac{2.303}{10} \log \frac{100}{10} = 2.303 \times 10^{-1} \text{ hour}^{-1}$$

Now for 99.9% completion : $a = 100$ and $x = 99.9$

$$t = \frac{2.303}{k_1} \log \frac{100}{0.1} = \frac{2.303}{2.303 \times 10^{-1}} \times 3 = 30 \text{ hours}$$

Illustration 8. 20% of a first order reaction was completed in 5 min. When will 60% of the reaction complete ?

Solution. $t = 5 \text{ min}, \quad a = 100, \quad x_1 = 20$

$$k = \frac{2.303}{t} \log \left(\frac{a}{a-x_1} \right) = \frac{2.303}{5} \log \frac{100}{80} = \frac{2.303}{5} \log \frac{5}{4} = \frac{2.303}{5} [\log 5 - \log 4]$$

$$k = \frac{2.303}{5} [0.6989 - 0.6020] = \frac{2.303}{5} [0.0969] = 0.0446$$

Now $x_2 = 60$

$$k = \frac{2.303}{t} \log \left(\frac{a}{a-x_2} \right)$$

$$t = \frac{2.303}{0.0446} \log \frac{100}{40} = \frac{2.303}{0.0446} [\log 5 - \log 2] = \frac{2.303}{0.0446} \times 0.3979 = 20.55 \text{ min.}$$

Illustration 9. For the first order reaction, half life is 6 min. Calculate the rate constant of the reaction ?

Solution. $t_{1/2} = \frac{0.693}{k_1} \Rightarrow k_1 = \frac{0.693}{t_{1/2}} = \frac{0.693}{6} = 0.1155 \text{ min}^{-1}$

Illustration 10. A first order reaction is 90% complete in 40 min. Calculate the Half life of the reaction.

Solution. $a = 100, x = 90$

$$k_1 = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{40} \log \frac{100}{10} = \frac{2.303}{40} = 5.757 \times 10^{-2} \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{k_1} = \frac{0.693}{5.757 \times 10^{-2}} = 12.03 \text{ min.}$$

Illustration 11. In a second order reaction both the reactants have equal initial concentration, the time taken for 60% completion of reaction is 3000 s. What will be the time taken for 20% of the reaction?

Solution. $k_2 = \frac{1}{t} \frac{x}{a(a-x)}, \quad \text{Let } a = 1,$

$$k_2 = \frac{1}{t} \frac{x}{a(a-x)} = \frac{1}{3000} \left(\frac{0.6}{1-0.6} \right) = \frac{1}{3000} \times \frac{0.6}{0.4} = \frac{1}{2000}$$

So time for the 20% completion is :

$$t = \frac{1}{k_2} \frac{x}{a(a-x)} = 2000 \times \frac{0.20}{0.80} = 500 \text{ s.}$$

Illustration 12. If the initial concentration of reactants are doubled then half life becomes half. Calculate order of the reaction ?

Solution. $\left(\frac{t_1}{2} \right)_1 = \left(\frac{a_2}{a_1} \right)^{n-1} \Rightarrow \left(\frac{t_1}{2} \right)_1 = \left(\frac{2a}{a} \right)^{n-1}$

$$2 = (2)^{n-1} \Rightarrow n - 1 = 1 \Rightarrow n = 2$$

so, the order of the reaction will be 2.

BEGINNER'S BOX-3

1. Which of the following expressions is correct for zero order and first order reactions respectively (where a is initial concentration) ?

(1) $t_{1/2} \propto a; t_{1/2} \propto \frac{1}{a}$ (2) $t_{1/2} \propto a; t_{1/2} \propto a^0$ (3) $t_{1/2} \propto a^0; t_{1/2} \propto a$ (4) $t_{1/2} \propto a; t_{1/2} \propto \frac{1}{a^2}$

2. For the zero order reaction, $A \rightarrow B + C$; initial concentration of A is 0.1 M. If $[A] = 0.08 \text{ M}$ after 10 minutes, then it's half-life and completion time are respectively :

(1) 10 min; 20 min (2) $2 \times 10^{-3} \text{ min}; 4 \times 10^{-3} \text{ min}$
(3) 25 min, 50 min (4) 250 min, 500 min

3. For an elementary reaction, $X(g) \rightarrow Y(g) + Z(g)$ the half life period is 10 min. In what period of time the concentration of X will be reduced to 10% of original concentration?

(1) 20 min (2) 33 min (3) 15 min (4) 25 min

4. A first order reaction is 75% completed in 100 min. How long will it take for it's 87.5% completion?

(1) 125 min (2) 150 min (3) 175 min (4) 200 min

5. The rate constant for a first order reaction which has half life 480 s is :-

(1) $1.44 \times 10^{-3} \text{ s}^{-1}$ (2) $1.44 \times \text{s}^{-1}$ (3) $0.72 \times 10^{-3} \text{ s}^{-1}$ (4) $2.88 \times 10^{-3} \text{ s}^{-1}$

2.9 COLLISION THEORY OF CHEMICAL REACTIONS

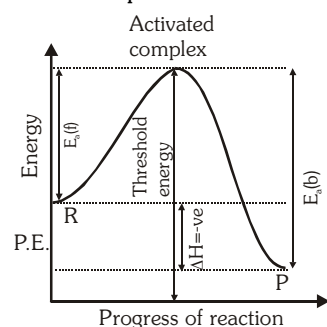
This theory was given by Max Trautz and William Lewis. According to it, for a reaction to occur there must be collisions in between reacting molecules. Total number of collisions per second in unit volume is called collision frequency (z). Generally its value is very high for gaseous reactions (10^{25} to 10^{28} collisions/sec-cm³). But only a small fraction of these collisions are capable to convert reactant into product. These collisions are called as effective collisions.

For effective collision following two conditions must be satisfied at a time :

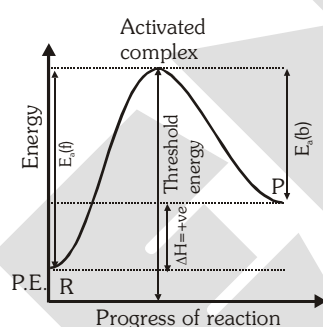
(a) Reacting molecules must possess a minimum amount of energy.

(b) Proper orientation of collision

- **Threshold energy** : The minimum energy which must be possessed by reacting molecules for a chemical reaction to occur.
- **Activation energy** : The minimum extra amount of energy required by reactant molecules for converting into products.



Exothermic reaction



Endothermic reaction

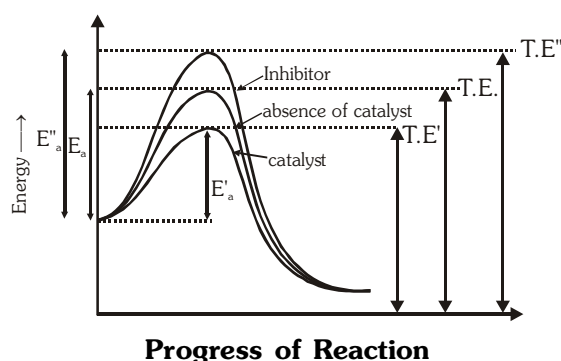
1. $E_a(f)$ = Activation energy for forward reaction
2. $E_a(b)$ = Activation energy for backward reaction

If not specified in questions then consider E_a for forward reaction.

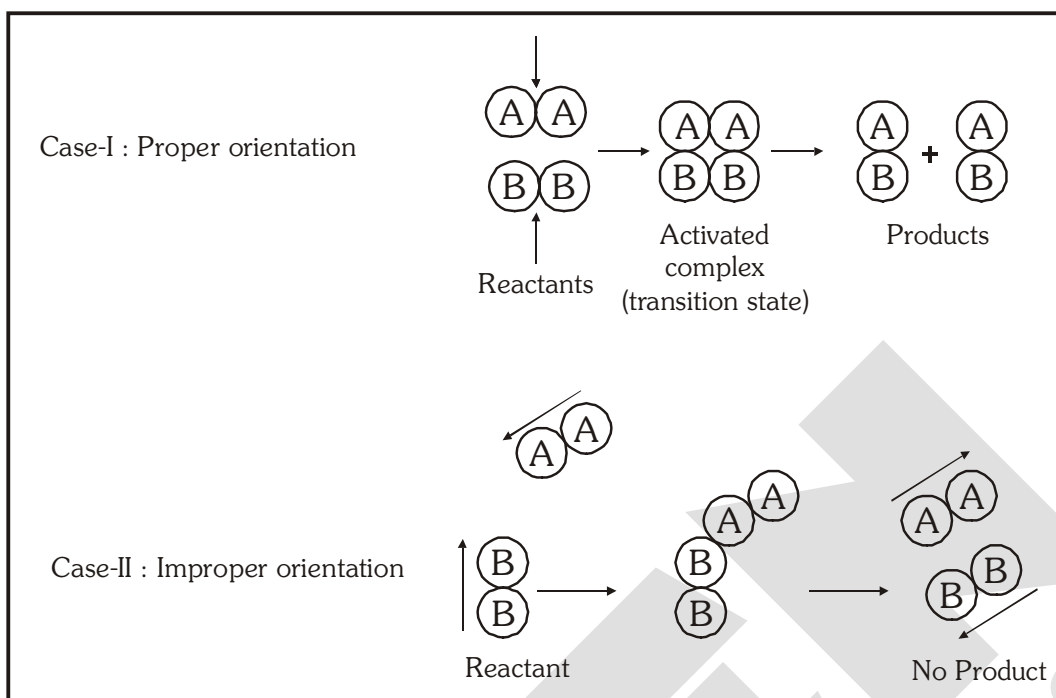
$$\Delta H = E_{a(f)} - E_{a(b)} \quad ; \quad \Delta H = H_p - H_R$$

ACTIVATION ENERGY MAINLY DEPENDS UPON :

- Nature of reactant** : For different reactants, number of bonds and bond energies are different, therefore activation energy will also be different.
Reactions which have less E_a , take place at faster rate.
- Presence of catalyst** : Catalyst provide an alternative path of reaction mechanism for the reaction.
 - In presence of catalyst threshold energy decreases, activation energy decreases and rate of reaction increases.
 - In presence of negative catalyst (inhibitor) threshold energy increases, activation energy increases, rate of reaction decreases.



(b) Orientation :



• Limitations :

- This theory is mainly applicable for gaseous reactions and also for solutions in which reacting species are molecules.
- This theory is mainly applicable for simple bimolecular reactions but fails for complex reactions.
- It considers molecules to be hard sphere and ignore structural aspect of molecules.

GOLDEN KEY POINTS

- Activated complex is most unstable complex formed in the transition state with effective collision.
- On increasing temperature E_a for reaction does not decrease but number of active molecules which are crossing the energy barrier increases therefore rate of reaction increases.
- According to Arrhenius ; rate of reaction $\propto e^{-\frac{E_a}{RT}}$ E_a = Activation energy
 R = Gas constant
 T = Temperature (in K)

Illustrations

Illustration 13 The E_a for an exothermic reaction $A \rightarrow B$ is 80 kJ mol^{-1} . Heat of reaction is 20 kJ mol^{-1} . E_a for the reaction $B \rightarrow A$ will be ?

Solution $\Delta H = E_{a(f)} - E_{a(b)} \Rightarrow -20 = 80 - E_{a(b)}$
 $E_{a(b)} = 100 \text{ kJ mol}^{-1}$

Illustration 14 For the reaction $A + B \rightleftharpoons C + D$ the activation energy is 32 kJ mol^{-1} . For the reverse reaction the E_a is 58 kJ mol^{-1} . Determine (i) Nature of reaction (ii) ΔH

Solution $\Delta H = E_{a(f)} - E_{a(b)}$
 $\Delta H = 32 - 58$
 $\Delta H = -26 \text{ kJ mol}^{-1}$ (exothermic)

Illustration 15 For an endothermic reaction

- $E_a > \Delta H$
- $E_a < \Delta H$
- $E_a = \Delta H$
- any one of the above

Solution **Ans. (1)**

Illustration 16 For an exothermic reaction

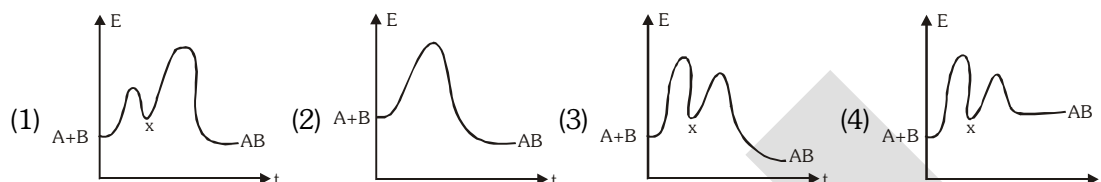
- (1) $E_a > \Delta H$ (2) $E_a < \Delta H$ (3) $E_a = \Delta H$ (4) any one of the above

Solution **Ans. (4)**

Illustration 17 An exothermic chemical reaction is occurring in two steps as follows

- (i) $A + B \rightarrow X$ (slow) (ii) $X \rightarrow AB$ (fast)

the process of reaction can be best described by



Solution **Ans. (3)**

2.10 FACTORS AFFECTING RATE OF REACTION :

1. Nature of reactant :

(a) Physical state of reactant :

increasing order of rate of reaction – Solid < liquid < gas

(Intermolecular attractive force decreases which provides more freedom for collisions)

(b) Physical size of particles (if reactant is solid) :

$$\text{Rate of reaction} \propto \frac{1}{\text{physical size}} \propto \text{surface area}$$

(c) **Chemical nature of reactant** : For different reacting species number of bonds broken and their bond energies are different. Therefore requirement of activation energy is also different. Now reactions having less value of activation energy will take place at faster rate.

2. Concentration of reactant : Rate of reaction $\propto$ concentration of reactant

3. **Pressure** : Effect of pressure on Rate of reaction is negligible when reactants are solid or liquid. But if reactants are in gaseous state then rate of reaction increases on increasing pressure because number of effective collisions increases.

4. **Temperature** : On increasing temperature rate of reaction increases whether the reaction is exothermic or endothermic. When temperature increases KE of molecules increases, number of activated molecules increases thus rate of reaction increases.

Relation between rate constant and Temperature :-

(a) Generally it is found that for every 10 °C rise in temperature Rate of reaction becomes 2 to 3 times.

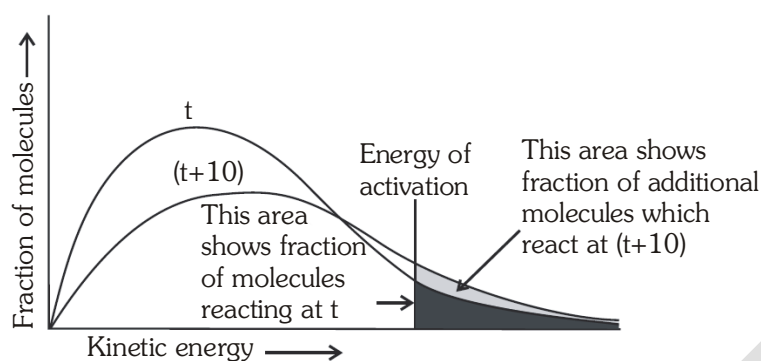
Temperature coefficient (μ) : It is defined as ratio of rate constant of a reaction at two different temperatures which will differ by 10 °C.

$$\mu = \frac{k_{T+10}}{k_T} = 2 \text{ to } 3 ; \quad \frac{r_2}{r_1} = \frac{k_2}{k_1} = \mu^{\Delta T/10}$$

If temperature of reaction is not specified then consider 25 °C.

(If μ is not given consider it as minimum 2)

MAXWELL AND BOLTZMANN ENERGY DISTRIBUTION CURVE



(b) Arrhenius equation

$$k = Ae^{-E_a/RT} \quad \dots(1)$$

A = Arrhenius constant / pre-exponential factor / Frequency factor

E_a = Activation energy

R = gas constant

T = Temperature (Kelvin)

- k increases with increase in temperature
If $T \rightarrow \infty$; $k = A$

- $\frac{k}{A} = e^{-E_a/RT}$ = fraction of molecules having energy $\geq E_a$

$e^{-E_a/RT}$ = Boltzman factor

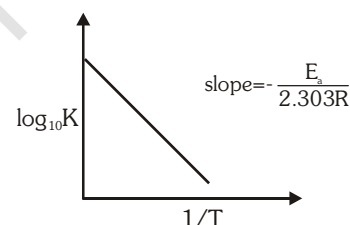
On taking logarithm for equation (1) on both sides.

$$\ln k = \ln A + \ln e^{-E_a/RT}$$

$$2.303 \log_{10} k = 2.303 \log_{10} A - \frac{E_a}{RT}$$

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303RT} \quad \dots(2)$$

$y = c - mx$



- E_a of reaction can be determined by measuring rate constant at two different temperatures
At temperature T_1 :

$$\log_{10} k_1 = \log_{10} A - \frac{E_a}{2.303RT_1} \quad \dots(3)$$

At temperature T_2 :

$$\log_{10} k_2 = \log_{10} A - \frac{E_a}{2.303RT_2} \quad \dots(4)$$

Equation (4) – Equation (3) gives -

$$\log k_2 - \log k_1 = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log_{10} \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

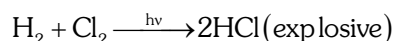
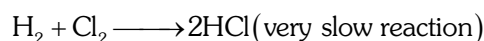
- From Arrhenius equation - $k = Ae^{-E_a/RT}$

$$\ln k = \ln A - \frac{E_a}{RT}$$

$$\frac{d}{dT}(\ln k) = \frac{d}{dT}(\ln A) + \frac{d}{dT}\left(-\frac{E_a}{RT}\right) = 0 - \frac{E_a}{R} \frac{d}{dT}(T^{-1}) = \frac{E_a}{R}(T^{-2})$$

$$\boxed{\frac{d}{dT} \ln k = \frac{E_a}{RT^2}} \text{ differential form of Arrhenius equation}$$

- 5. Presence of catalyst :** In presence of catalyst E_a of reaction decreases and rate of reaction increases.
- 6. Exposure to radiation :** Rate of some reactions also increases when reaction are carried out in the presence of radiation. (only for photochemical reaction)
e.g. formation of HCl



GOLDEN KEY POINTS

● PHOTO CHEMICAL REACTION

Reactions which occur in presence of light are photochemical reaction i.e., reaction which do not occur in dark.

Example : (i) $H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$ **(ii)** Photo synthesis **(iii)** Vision

- In photochemical reaction, each molecule of the reacting substance absorbs one quantum (single photon) of radiation and is activated to form the product.
- For every reaction, a certain amount of energy is required.
- Higher the intensity of light greater will be the rate of reaction.
- For these reactions, the value of ΔG may be positive or negative.
- Temperature have no significant effect.

Mechanism : $Cl_2 + h\nu \longrightarrow Cl_2^*$ (activated molecule)

Primary step of reaction : $Cl_2^* \longrightarrow Cl + Cl$

Secondary step of reaction : $H_2 + Cl \longrightarrow HCl + H$

$Cl_2 + H \longrightarrow HCl + Cl$

Final step of reaction : $Cl + Cl \longrightarrow Cl_2$

● Quantum yield (or quantum efficiency) :

The quantum efficiency yield (ϕ) of a photochemical reaction may be expressed as

$$\phi = \frac{\text{Number of molecules reacted or product formed}}{\text{Number of photons absorbed}}$$

● Application of Photochemical Reaction

- | | | |
|---------------------------|-------------------------------------|---|
| (i) In photography | (ii) In photo printing | (iii) In photosynthesis |
| (iv) In vision | (v) In electronic industries | (vi) In many polymerization reactions. |

Illustrations

Illustration 18 A reaction whose temperature is increased from 10 °C to 50 °C then increase in rate of reaction will be—

Solution. We know that on increasing temperature by 10°C the rate of reaction becomes 2 to 3 times.

Note : If the value of temperature coefficient is not given then we will take 2.

$$\begin{aligned} 10\text{ }^{\circ}\text{C} &\text{—} & 20\text{ }^{\circ}\text{C} &= 2 \text{ times} \\ 20\text{ }^{\circ}\text{C} &\text{—} & 30\text{ }^{\circ}\text{C} &= 2^2 \text{ times} \\ 30\text{ }^{\circ}\text{C} &\text{—} & 40\text{ }^{\circ}\text{C} &= 2^3 \text{ times} \\ 40\text{ }^{\circ}\text{C} &\text{—} & 50\text{ }^{\circ}\text{C} &= 2^4 \text{ times} \\ && &= 16 \text{ times} \end{aligned}$$

Illustration 19 For a reaction, temperature coefficient = 2, then calculate the activation energy (in kJ) of the reaction.

Solution. Let $T_1 = 25\text{ }^{\circ}\text{C}$, $T_2 = 35\text{ }^{\circ}\text{C}$

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

Given : Temperature coefficient = $\frac{k_2}{k_1} = 2$

$$T_1 = 25 + 273 = 298\text{ K}, T_2 = 35 + 273 = 308\text{ K}, R = 8.314$$

$$\log 2 = \frac{E_a}{2.303 \times 8.314} \times \left(\frac{10}{298 \times 308} \right)$$

$$E_a = 52.897\text{ kJ}$$

Illustration 20 An exothermic reaction $A \longrightarrow B$ has an activation energy of 17 kJ mol⁻¹ of A. The heat of the reaction is 40 kJ. Calculate the activation energy for the reverse reaction $B \longrightarrow A$.

Solution. For the reaction $A \longrightarrow B$.

Activation energy $E_a = 17\text{ kJ}$

$$\Delta H = -40\text{ kJ}$$

$$\Delta H = E_a(f) - E_a(b)$$

$$E_a(b) = 17 - (-40) = 57\text{ kJ}$$

Illustration 21 For first order gaseous reaction $\log k$ when plotted against $\frac{1}{T}$, gives a straight line with a slope of -8000. Calculate the activation energy of the reaction.

Solution. Arrhenius equation $k = Ae^{-E_a/RT}$

$$\log k = \log A - \frac{E_a}{2.303R} \times \frac{1}{T}$$

when curve is plotted between $\log k$ and $\frac{1}{T}$, a straight line is obtained. Slope of this line = $-\frac{E_a}{2.303R}$

$$\text{Then, } \frac{E_a}{2.303R} = 8000 \text{ or } E_a = 8000 \times 2.303 \times 1.987 = 36608\text{ Cal}$$

Illustration 22 If temperature of a reaction is increased from 10°C to 100°C then how many times rate of reaction will become?

Solution 2^9 times

Illustration 23 If temperature of a reaction is increased from t_1 to t_2 then rate of reaction becomes ?

Solution $(\mu)^{t_2-t_1/10} \Rightarrow (\mu)^{\Delta T/10}$ times ; $r_{\text{new}} = r_{\text{old}} \times (\mu)^{\Delta T/10}$

Illustration 24 A reaction is carried out at 10°C. If temperature is increased by 50°C then how many times rate of reaction will become?

Solution 32 times

BEGINNER'S BOX-4

- For an endothermic reaction where ΔH represents the enthalpy of reaction in kJ mol^{-1} , the minimum value for the energy of activation will be
 (1) less than ΔH (2) more than ΔH
 (3) equal to ΔH (4) zero
- The activation energy of the reaction, $A + B \rightarrow C + D + 38 \text{ kcal}$ is 20 kCal, what would be the activation energy of the reaction, $C + D \rightarrow A + B$
 (1) 20 kCal (2) - 20 kCal
 (3) 18 kCal (4) 58 kCal
- $\frac{k_{35^\circ}}{k_{34^\circ}} > 1$, this means that
 (1) Rate increases with the rise in temperature (2) Rate decreases with rise in temperature
 (3) rate does not change with rise in temperature (4) None of the above
- The plot of $\ln k$ versus $1/T$ is linear with slope of
 (1) $-\frac{E_a}{R}$ (2) $\frac{E_a}{R}$
 (3) $\frac{E_a}{2.303R}$ (4) $-\frac{E_a}{2.303R}$

ANSWER KEY

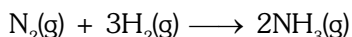
BEGINNER'S BOX-1	Que.	1	2	3							
	Ans.	1	3	3							
BEGINNER'S BOX-2	Que.	1	2	3							
	Ans.	1	3	3							
BEGINNER'S BOX-3	Que.	1	2	3	4	5					
	Ans.	2	3	2	2	1					
BEGINNER'S BOX-4	Que.	1	2	3	4						
	Ans.	2	4	1	1						

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

RATE OF REACTION

1. Consider the chemical reaction :



The rate of this reaction can be expressed in terms of concentration of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ or $\text{NH}_3(\text{g})$. Identify the correct relationship amongst the rate expressions.

$$(1) \text{Rate} = -\frac{d[\text{N}_2]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$$

$$(2) \text{Rate} = -\frac{d[\text{N}_2]}{dt} = -\frac{3d[\text{H}_2]}{dt} = \frac{2d[\text{NH}_3]}{dt}$$

$$(3) \text{Rate} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt} = \frac{1}{2} \frac{d[\text{NH}_3]}{dt}$$

$$(4) \text{Rate} = -\frac{d[\text{N}_2]}{dt} = -\frac{d[\text{H}_2]}{dt} = \frac{d[\text{NH}_3]}{dt}$$

2. In the formation of sulphur trioxide by the contact process
- $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$
- ; the rate of reaction is expressed as

$$-\frac{d[\text{O}_2]}{dt} = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

The rate of disappearance of (SO_2) will be

- (1) $5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (2) $-2.25 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (3) $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (4) $50.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

3. In a catalytic reaction involving the formation of ammonia by Haber's process
- $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
- the rate of appearance of
- NH_3
- was measured as
- $2.5 \times 10^{-4} \text{ mole L}^{-1} \text{ s}^{-1}$
- . The rate of disappearance of
- H_2
- will be -

- (1) $2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (2) $1.25 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (3) $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (4) $5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

4. Which of the following statement is correct for a reaction
- $\text{X} + 2\text{Y} \rightarrow \text{Product}$

- (1) The rate of disappearance of X = twice the rate of disappearance of Y.
 (2) The rate of disappearance of X = $\frac{1}{2}$ rate of appearance of products
 (3) The rate of appearance of products = $\frac{1}{2}$ the rate of disappearance of Y
 (4) The rate of appearance of products = $\frac{1}{2}$ the rate of disappearance of X

5. For the reaction,
- $\text{N}_2\text{O}_5 \longrightarrow 2\text{NO}_2 + \frac{1}{2} \text{O}_2$

$$\text{Given } -\frac{d[\text{N}_2\text{O}_5]}{dt} = K_1[\text{N}_2\text{O}_5], \frac{d[\text{NO}_2]}{dt} = K_2[\text{N}_2\text{O}_5]$$

$$\frac{d[\text{O}_2]}{dt} = K_3[\text{N}_2\text{O}_5]$$

The relation between K_1 , K_2 and K_3 is -

- (1) $2K_1 = K_2 = 4K_3$ (2) $K_1 = K_2 = K_3$
 (3) $2K_1 = 4K_2 = K_3$ (4) None

6. Rate of formation of
- SO_3
- according to the reaction
- $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$
- is
- $1.6 \times 10^{-3} \text{ kg min}^{-1}$
- . Hence rate at which
- SO_2
- reacts is :-

- (1) $1.6 \times 10^{-3} \text{ kg min}^{-1}$
 (2) $8.0 \times 10^{-4} \text{ kg min}^{-1}$
 (3) $3.2 \times 10^{-3} \text{ kg min}^{-1}$
 (4) $1.28 \times 10^{-3} \text{ kg min}^{-1}$

7. For a general chemical change
- $2\text{A} + 3\text{B} \rightarrow \text{products}$
- , the rate of disappearance of A is
- r_1
- and of B is
- r_2
- . The rates
- r_1
- and
- r_2
- are related as :-

- (1) $3r_1 = 2r_2$ (2) $r_1 = r_2$
 (3) $2r_1 = 3r_2$ (4) $r_1^2 = 2r_2^2$

8. In a reaction
- $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$
- the rate of appearance of
- NH_3
- is
- $2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- . The Rate of reaction & rate of disappearance of
- H_2
- will be (
- $\text{In mol L}^{-1} \text{ sec}^{-1}$
-)

- (1) $3.75 \times 10^{-4}, 1.25 \times 10^{-4}$
 (2) $1.25 \times 10^{-4}, 2.5 \times 10^{-4}$
 (3) $1.25 \times 10^{-4}, 3.75 \times 10^{-4}$
 (4) $5.0 \times 10^{-4}, 3.75 \times 10^{-4}$

9. Which of the following statement is not correct for the reaction:
- $4\text{A} + \text{B} \rightarrow 2\text{C} + 2\text{D}$

- (1) The rate of disappearance of B is twice the rate of appearance of C
 (2) The rate of disappearance of B is one fourth the rate of disappearance of A
 (3) The rate of formation of D is one half the rate of consumption of A
 (4) The rate of formation of C and D are equal

10. For gaseous reaction, rate =
- $k[\text{A}][\text{B}]$
- . If volume of container is reduced to
- $\frac{1}{4}$
- of initial, then the rate of the reaction will be..... times of initial:-

- (1) $\frac{1}{8}$ (2) 8 (3) $\frac{1}{16}$ (4) 16

11. $2A_{(g)} + B_{(g)} \rightleftharpoons \text{Product}$ is an elementary reaction, If pressure is increased three times of the initial pressure, the velocity of forward reaction will be ----- of the previous velocity:-

- (1) 9 times (2) 27 times
(3) $\frac{1}{9}$ times (4) $\frac{1}{27}$ times

RATE LAW / ORDER / MOLECULARITY

12. The rate of certain hypothetical reaction $A + B + C \rightarrow \text{products}$ is given by

$$r = \frac{-d[A]}{dt} = k[A]^{\frac{1}{2}}[B]^{\frac{1}{3}}[C]^{\frac{1}{4}}$$

The order of the reaction -

- (1) 1 (2) $\frac{1}{2}$ (3) 2 (4) $\frac{13}{12}$

13. Which of the following rate law has an overall order of 0.5 for reaction involving substances x,y and z ?

- (1) Rate = $k(C_x)(C_y)(C_z)$
(2) Rate = $k(C_x)^{0.5}(C_y)^{0.5}(C_z)^{0.5}$
(3) Rate = $k(C_x)^{1.5}(C_y)^{-1}(C_z)^0$
(4) Rate = $k(C_x)(C_z)^0 / (C_y)^2$

14. A chemical reaction involves two reacting species. The rate of reaction is directly proportional to the concentration of one of them and inversely proportional to the concentration of the other. The order of reaction is -

- (1) 1 (2) 2
(3) Zero (4) Unpredictable

15. For the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, the experimental data suggests, Rate = $k[H_2][Br_2]^{\frac{1}{2}}$. The order for this reaction is -

- (1) 2 (2) $1\frac{1}{2}$ (3) 1 (4) $2\frac{1}{2}$

16. Select the rate law that corresponds to the data shown for the following reaction $A + B \rightarrow C$

Exp.	[A]	[B]	Initial rate
1.	0.012	0.035	0.10
2.	0.024	0.070	1.6
3.	0.024	0.035	0.20
4.	0.012	0.070	0.80

- (1) Rate = $k[B]^3$ (2) Rate = $k[B]^4$
(3) Rate = $k[A][B]^3$ (4) Rate = $k[A]^2[B]^2$

17. In a certain gaseous reaction between X and Y, $X + 3Y \rightarrow XY_3$. The initial rates are reported as follows -

[X]	[Y]	Rate
0.1 M	0.1 M	0.002 Ms^{-1}
0.2 M	0.1 M	0.002 Ms^{-1}
0.3 M	0.2 M	0.008 Ms^{-1}
0.4 M	0.3 M	0.018 Ms^{-1}

The rate law is -

- (1) $r = k[X][Y]^3$ (2) $r = k[X]^q[Y]^2$
(3) $r = k[X][Y]$ (4) $r = [X]^q[Y]^3$

18. Select the law that corresponds to data shown for the following reaction $2A + B \rightarrow C + D$:-

Exp.	[A]	[B]	Initial rate ($\text{mol L}^{-1} \text{ min}^{-1}$)
1.	0.1	0.1	7.5×10^{-3}
2.	0.3	0.2	9.0×10^{-2}
3.	0.3	0.4	3.6×10^{-1}
4.	0.4	0.1	3.0×10^{-2}

- (1) Rate = $k[A]^2[B]$ (2) Rate = $k[A][B]^2$
(3) Rate = $k[A][B]^3$ (4) Rate = $k[A][B]$

19. For a hypothetical reaction ; $A + B \rightarrow C$ the following data were obtained in three different experiments :-

[A] (mol L^{-1})	[B] (mol L^{-1})	Rate of reaction ($\text{mol L}^{-1} \text{ min}^{-1}$)
0.01	0.01	1.0×10^{-4}
0.01	0.03	9.0×10^{-4}
0.03	0.03	2.70×10^{-3}

The rate law will be :-

- (1) $r = k[A]^2[B]$ (2) $r = k[A][B]^2$
(3) $r = k[A][B]$ (4) None of these

20. Calculate the order of the reaction w.r.t. A and B:

[A] (mol L^{-1})	[B] (mol L^{-1})	Rate
0.05	0.05	1.2×10^{-3}
0.10	0.05	2.4×10^{-3}
0.05	0.10	1.2×10^{-3}

- (1) 1 and 0 (2) 1 and 1
(3) 0 and 1 (4) None

21. For a chemical reaction $A + B \rightarrow \text{product}$, the order is one with respect to each A and B. Value of x and y from the given data is :-

Rate ($\text{mol L}^{-1}\text{s}^{-1}$)	[A]	[B]
0.10	0.20 M	0.05 M
0.40	x	0.05 M
0.80	0.40 M	y

- (1) 0.20, 0.80 (2) 0.80, 0.40
(3) 0.80, 0.20 (4) 0.40, 0.20
22. Time required to complete a half fraction of a reaction varies inversely to the concentration of reactant then the order of reaction is -
(1) Zero (2) 1
(3) 2 (4) 3
23. The rate law for the single step reaction $2A + B \rightarrow 2C$, is given by -
(1) Rate = $K[A][B]$ (2) Rate = $K[A]^2[B]$
(3) Rate = $K[2A][B]$ (4) Rate = $K[A]^2[B]^0$
24. For a reaction of the type $A + B \rightarrow \text{products}$, it is observed that doubling the concentration of A causes the reaction rate to be four times as great, but doubling the amount of B does not effect the rate. The rate equation is -
(1) Rate = $K[A][B]$
(2) Rate = $K[A]^2$
(3) Rate = $K[A]^2[B]$
(4) Rate = $K[A]^2[B]^2$
25. Point out incorrect statement.
(1) Rate law is an experimental value
(2) Law of mass action is a theoretical proposal
(3) Rate law is more informative than law of mass action for developing mechanism
(4) Rate law is always different from the expression of law of mass action.
26. For an elementary process $2X + Y \rightarrow Z + W$, the molecularity is -
(1) 2 (2) 1
(3) 3 (4) Unpredictable
27. For a reaction $A + B \rightarrow \text{products}$, the rate of the reaction was doubled when the concentration of A was doubled, the rate was again doubled when the concentration of A & B were doubled the order of the reaction with respect to A & B are:-
(1) 1, 1 (2) 2, 0
(3) 1, 0 (4) 0, 1

28. For the reaction $A + B \rightarrow \text{products}$, it is found that the order of A is 1 and the order of B is $\frac{1}{2}$.

When the concentration of both A and B are increased four times, the rate will increase by a factor of :-

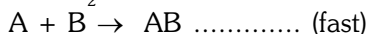
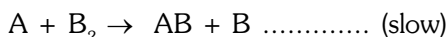
- (1) 16 (2) 8 (3) 6 (4) 4

29. The rate law for a reaction $A + B \rightarrow \text{product}$ is rate = $K[A]^1[B]^2$. Then which one of the following statement is false :-
(1) If [B] is held constant while [A] is doubled, the reaction will proceed twice as fast
(2) If [A] is held constant while [B] is reduced to one quarter, the rate will be halved
(3) If [A] and [B] are both doubled, the reaction will proceed 8 times as fast
(4) This is a third order reaction
30. For a chemical reaction $A \rightarrow B$, the rate of reaction doubles when the concentration of A is increased 8 times. The order of reaction w.r.t. A is :-
(1) 3 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) Zero
31. The specific rate constant of a first order reaction depends on the :-
(1) Concentration of the reactant
(2) Concentration of the product
(3) Time
(4) Temperature

PSEUDO FIRST ORDER REACTION / MECHANISM OF REACTION

32. Following mechanism has been proposed for a reaction $2A + B \rightarrow D + E$
 $A + B \rightarrow C + D$ (slow)
 $A + C \rightarrow E$ (fast)
 The rate law expression for the reaction is -
 (1) $r = K[A]^2[B]$ (2) $r = K[A][B]$
 (3) $r = K[A]^2$ (4) $r = K[A][C]$
33. The chemical reaction $2O_3 \rightarrow 3O_2$ proceeds as follows
 $O_3 \rightleftharpoons O_2 + O$ (fast)
 $O + O_3 \rightarrow 2O_2$ (slow)
 The rate law expression should be -
 (1) $r = K[O_3]^2$ (2) $r = K[O_3]^2[O_2]^{-1}$
 (3) $r = K[O_3][O_2]$ (4) Unpredictable

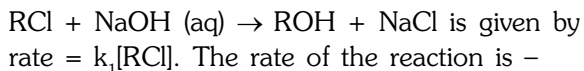
34. The hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below –



The order of the over all reaction is –

- (1) 2 (2) 1 (3) $1\frac{1}{2}$ (4) Zero

35. The rate for the reaction



- (1) Doubled on doubling the concentration of NaOH
 (2) Halved on reducing the concentration of RCl to half
 (3) Decreased on increasing the temperature of reaction
 (4) Unaffected by increasing the temperature of the reaction

36. For reaction $NO_2 + CO \rightarrow CO_2 + NO$, the rate expression is, Rate = $k[NO_2]^2$

The number of molecules of CO involved in the slowest step will be –

- (1) 0 (2) 1
 (3) 2 (4) 3

37. The rate law of the reaction $A + 2B \rightarrow \text{product}$

is given by $\frac{d[P]}{dt} = K[A]^2[B]$. If A is taken in large

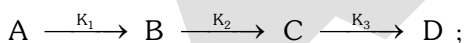
excess, the order of the reaction will be –

- (1) Zero (2) 1
 (3) 2 (4) 3

38. The acid hydrolysis of ester is –

- (1) Psuedo 1st order reaction
 (2) Bimolecular reaction
 (3) Pseudo unimolecular reaction
 (4) All

39. In the sequence of reaction



$K_3 > K_2 > K_1$, then the rate determining step of the reaction is :-

- (1) $A \rightarrow B$ (2) $C \rightarrow D$
 (3) $B \rightarrow C$ (4) $A \rightarrow D$

40. The reaction mechanism for the reaction $P \rightarrow R$ is as follows :-

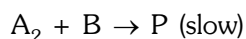
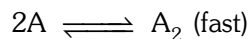


the rate law for the main reaction ($P \rightarrow R$) is

[where K_1 is an equilibrium constant]

- (1) $k_1[P][Q]$ (2) $k_1k_2[P]$
 (3) $k_1k_2[P]^2$ (4) $k_1k_2[a]$

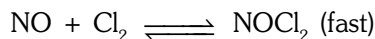
41. The reaction $2A + B \rightarrow P$, follows the mechanism



The order of the reaction is :-

- (1) 1.5 (2) 3 (3) 1 (4) 2

42. For the reaction $2NO + Cl_2 \rightarrow 2NOCl$ the following mechanism has been proposed



$NOCl_2 + NO \rightarrow 2NOCl$ (slow) the rate law for the reaction is :-

- (1) Rate = $K[NO]^2[Cl_2]$ (2) Rate = $K[NO][Cl_2]^2$
 (3) Rate = $K[NOCl_2]$ (4) Rate = $K[NOCl]^2$

ZERO / FIRST / nth ORDER REACTIONS

43. The rate constant is numerically the same for three reactions of first, second and third order respectively. Which one is true at a moment for rate of all three reactions if concentration of reactants is same and greater than 1 M.

- (1) $r_1 = r_2 = r_3$ (2) $r_1 > r_2 > r_3$
 (3) $r_1 < r_2 < r_3$ (4) All

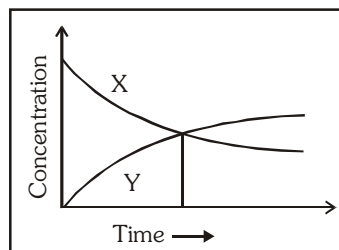
44. K for a zero order reaction is $2 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$. If the concentration of the reactant after 25 s is 0.5 M, the initial concentration must have been.

- (1) 0.5 M (2) 1.25 M
 (3) 12.5 M (4) 1.0 M

45. The decomposition of N_2O_5 occurs as, $2N_2O_5 \rightarrow 4NO_2 + O_2$, and follows first order kinetics; hence

- (1) The reaction is bimolecular
 (2) The reaction is unimolecular
 (3) $t_{1/2} \propto a^0$
 (4) $t_{1/2} \propto a^2$

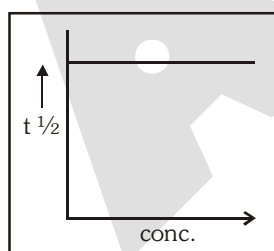
46. The accompanying figure depicts the change in concentration of species X and Y for the reaction $X \rightarrow Y$ as a function of time the point of intersection of the two curves represents.



- (1) $t_{1/2}$
 (2) $t_{3/4}$
 (3) $t_{2/3}$
 (4) Data are insufficient to predict

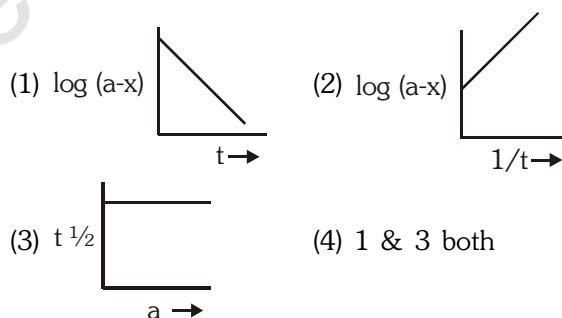
47. If the first order reaction involves gaseous reactants and gaseous products the unit of its rate is –
 (1) atm. (2) atm - s
 (3) atm - s⁻¹ (4) atm² s²
48. Plot of log(a - x) vs time t is straight line. This indicates that the reaction is of –
 (1) Second order (2) First order
 (3) Zero order (4) third order
49. The rate constant of a first order reaction is $4 \times 10^{-3} \text{ s}^{-1}$. At a reactant concentration of 0.02 M, the rate of reaction would be –
 (1) $8 \times 10^{-5} \text{ M s}^{-1}$ (2) $4 \times 10^{-3} \text{ M s}^{-1}$
 (3) $2 \times 10^{-1} \text{ M s}^{-1}$ (4) $4 \times 10^{-1} \text{ M s}^{-1}$
50. In a first order reaction the concentration of the reactant is decreased from 1.0 M to 0.25 M in 20 min. The rate constant of the reaction would be –
 (1) 10 min^{-1} (2) 6.931 min^{-1}
 (3) 0.6931 min^{-1} (4) 0.06931 min^{-1}
51. In a first order reaction the $a/(a - x)$ was found to be 8 after 10 minute. The rate constant is
 (1) $\frac{(2.303 \times 3 \log 2)}{10}$ (2) $\frac{(2.303 \times 2 \log 3)}{10}$
 (3) $10 \times 2.303 \times 2 \log 3$ (4) $10 \times 2.303 \times 3 \log 2$
52. 75 % of a first order reaction was found to complete in 32 min. When will 50 % of the same reaction complete –
 (1) 24 min (2) 16 min
 (3) 8 min (4) 4 min
53. A first order reaction has a half life period of 69.3 s. At 0.10 mol L⁻¹ reactant concentration, the rate will be –
 (1) 10^{-4} M s^{-1} (2) 10^{-3} M s^{-1}
 (3) 10^{-1} M s^{-1} (4) $6.93 \times 10^{-1} \text{ M s}^{-1}$
54. For a first order reaction $A \rightarrow$ products, the rate of reaction at $[A] = 0.2 \text{ M}$ is $1 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$. The half life period for the reaction is –
 (1) 832 min. (2) 440 s
 (3) 416 min. (4) 14 min.
55. 99 % of a first order reaction was completed in 32 min. when will 99.9 % of the reaction complete ?
 (1) 50 min. (2) 46 min.
 (3) 49 min. (4) 48 min.
56. The half life for the first order reaction $\text{N}_2\text{O}_5 \rightarrow 2\text{NO}_2 + \frac{1}{2}\text{O}_2$ is 24 hrs. at 30°C. Starting with 10 g of N_2O_5 how many grams of N_2O_5 will remain after a period of 96 hours ?
 (1) 1.25 g (2) 0.63 g
 (3) 1.77 g (4) 0.5 g
57. What is the half life of a radioactive substance if 87.5% of any given amount of the substance disintegrate in 40 minutes ?
 (1) 160 min (2) 10 min
 (3) 20 min (4) 13 min 20 sec.
58. For a given reaction of first order it takes 20 minute for the concentration to drop from 1 M to 0.6 M. The time required for the concentration to drop from 0.6 M to 0.36 M will be :
 (1) More than 20 min
 (2) Less than 20 min
 (3) Equal to 20 min
 (4) Infinity
59. A first order reaction is carried out with an initial concentration of 10 mol per litre and 80% of the reactant changes into the product. Now if the same reaction is carried out with an initial concentration of 5 mol per litre for the same period the percentage of the reactant changing to the product is.
 (1) 40 (2) 80
 (3) 160 (4) Cannot be calculated
60. In the following first order reactions,
 $A + \text{Reagent} \rightarrow \text{Product}$, $B + \text{Reagent} \rightarrow \text{Product}$
 Calculate the ratio of K_1 / K_2 , 50% of B has been reacted, when 94% of A has been reacted –
 (1) 4.06 (2) 0.246
 (3) 2.06 (4) 0.06
61. The reaction $L \rightarrow M$ is started with 10 g/L. After 30 minute and 90 minute, 5 g/L and 1.25 g/L are left respectively. The order of reaction is
 (1) 0 (2) 2
 (3) 1 (4) 3
62. The doubling the initial concentration of a reactant doubles $t_{1/2}$ of the reaction, then order of the reaction is –
 (1) 3 (2) 2
 (3) 1 (4) 0
63. The half life period for catalytic decomposition of AB_3 at 50 mm is found to be 4 hrs and at 100 mm it is 2 hrs. The order of reaction is –
 (1) 3 (2) 1
 (3) 2 (4) 0

64. The rate constant for a reaction is $10.8 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$. The reaction obeys –
 (1) First order (2) Zero order
 (3) Second order (4) All are wrong
65. A substance 'A' decomposes in solution following the first order kinetics. Flask I contains 1 L of 1M solution of A and flask II contains 100 ml of 0.6 M solution. After 8 hr. the concentration of A in flask I becomes 0.25 M, what will be the time for concentration of A in flask II to become 0.3 M.
 (1) 0.4 hr.
 (2) 2.4 hr.
 (3) 4.0 hr.
 (4) Unpredictable as rate constant is not given
66. The rate constant (K) for the reaction $2A + B \rightarrow$ product, was found to be $2.5 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ after 15 s, $2.60 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ after 30 s and $2.55 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ after 50 s. The order of reaction is
 (1) 2 (2) 3 (3) Zero (4) 1
67. The rate constant for a second order reaction is $8 \times 10^{-5} \text{ M}^{-1} \text{ min}^{-1}$. How long will it take a 1M solution to be reduced to 0.5 M.
 (1) $8.665 \times 10^3 \text{ min}$ (2) $8 \times 10^{-3} \text{ min}$
 (3) $1.25 \times 10^4 \text{ min}$ (4) $4 \times 10^{-5} \text{ min}$
68. A graph between $t_{1/2}$ and concentration for n^{th} order reaction is a straight line. Reaction of this nature is completed 50% in 10 minutes when concentration is 2 mol L^{-1} . This is decomposed 50% in t minutes at 4 mol L^{-1} , n and t are respectively



- (1) 0, 20 min. (2) 1, 10 min.
 (3) 1, 20 min. (4) 0, 5 min.
69. In the first order reaction, 75% of the reactant disappeared in 1.388 h. Calculate the rate constant of the reaction :-
 (1) 1 s^{-1} (2) $2.8 \times 10^{-4} \text{ s}^{-1}$
 (3) $17.2 \times 10^{-3} \text{ s}^{-1}$ (4) $1.8 \times 10^{-3} \text{ s}^{-1}$

70. In the case of first order reaction, the ratio of time required for 99.9 % completion to 50 % completion is :-
 (1) 2 (2) 5 (3) 10 (4) None
71. From different sets of data of $t_{1/2}$ at different initial concentrations say 'a' for a given reaction, the $[t_{1/2} \times a]$ is found to be constant. The order of reaction is :-
 (1) 0 (2) 1 (3) 2 (4) 3
72. The reaction
 $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
 is first order with respect to N_2O_5 . Which of the following graph would yield a straight line :-
 (1) $\log(P_{\text{N}_2\text{O}_5})$ v/s time with negative slope
 (2) $P_{\text{N}_2\text{O}_5}^{-1}$ v/s time
 (3) $P_{\text{N}_2\text{O}_5}$ v/s time
 (4) $\log(P_{\text{N}_2\text{O}_5})$ v/s time with positive slope
73. Which of the following statement is not correct for the reaction whose rate is $r = k$ (rate constant)
 (1) rate of reaction is independent of concentration of reactant
 (2) $t_{1/2}$ of reaction is not depends upon concentration of reactant
 (3) rate constant is independent of concentration of reactant
 (4) this is zero order reaction
74. Which of the following curves represents a 1st order reaction :-



75. The following data were obtained at a certain temperature for the decomposition of ammonia
- | p (mm) | 50 | 100 | 200 |
|-----------|------|------|------|
| $t_{1/2}$ | 3.64 | 1.82 | 0.91 |
- The order of the reaction is :-
 (1) 0 (2) 1
 (3) 2 (4) 3

76. A reaction is found to have the rate constant $\times s^{-1}$ by what factor the rate is increased if initial concentration of A is tripled
 (1) 3
 (2) 9
 (3) $\times$
 (4) Remains same
77. Which is incorrect :-
 (1) Half life of a first order reaction is independent of initial concentration
 (2) Rate of reaction is constant for first order reaction
 (3) Unit of K for second order reaction is $\text{mol}^{-1} \text{L s}^{-1}$
 (4) Half life of zero order is proportional to initial concentration
78. Hydrolysis of ester in alkaline medium is :-
 (1) First order reaction with molecularity one
 (2) Second order reaction with molecularity two
 (3) First order reaction with molecularity two
 (4) Second order reaction with molecularity one
79. The expression which gives $\frac{1}{4}$ th life of 1st order reaction is :-
 (1) $\frac{K}{2.303} \log \frac{4}{3}$
 (2) $\frac{2.303}{K} \log 3$
 (3) $\frac{2.303}{K} \log \frac{3}{4}$
 (4) $\frac{2.303}{K} \log \frac{4}{3}$
80. The rate constant of a zero order reaction is $0.2 \text{ mol dm}^{-3} \text{h}^{-1}$. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3} . Then its initial concentration would be :-
 (1) 6.05 mol dm^{-3}
 (2) 0.15 mol dm^{-3}
 (3) 0.25 mol dm^{-3}
 (4) 4.00 mol dm^{-3}
81. A reaction is of first order. After 100 minutes 75 gm of the reactant A are decomposed when 100 gm are taken initially, calculate the time required when 150 gm of the reactant A are decomposed, the initial weight taken is 200 gm :-
 (1) 100 minutes
 (2) 200 minutes
 (3) 150 minutes
 (4) 175 minutes
82. For which of the following, the unit of rate and rate constant of the reaction are identical :-
 (1) First order reaction
 (2) Zero order reaction
 (3) Second order reaction
 (4) Fractional order of reaction
83. $2A \rightarrow \text{Product}$, follows the first order kinetics. If the half life period of the reaction at $[A]_{\text{initial}} = 0.2 \text{ mol L}^{-1}$ is 20 min. then the value of rate constant would be:-
 (1) 4 s^{-1}
 (2) 20 s^{-1}
 (3) $4 \text{ lit. mol}^{-1} \text{ min.}^{-1}$
 (4) 0.034 min^{-1}
84. Correct statement about first order reaction is:-
 (1) $t_{\text{completion}} = \text{finite}$
 (2) $t_{1/2} \propto \frac{1}{a}$
 (3) Unit of K is $\text{mole lit}^{-1} \text{ s}^{-1}$
 (4) $t_{1/2} \times K = \text{at constant temperature}$

COLLISION THEORY AND FACTORS AFFECTING RATE OF REACTION

85. According to collision theory of reaction rates -
 (1) Every collision between reactants leads to chemical reaction
 (2) Rate of reaction is proportional to velocity of molecules
 (3) All reactions which occur in gaseous phase are zero order reaction
 (4) Rate of reaction is directly proportional to collision frequency.
86. Activation energy of a reaction is -
 (1) The energy released during the reaction
 (2) The energy evolved when activated complex is formed
 (3) Minimum extra amount of energy needed to overcome the potential barrier of reaction
 (4) The energy needed to form one mole of the product
87. The minimum energy for molecules to enter into chemical reaction is called.
 (1) Kinetic energy
 (2) Potential energy
 (3) Threshold energy
 (4) Activation energy
88. The rate constant k_1 of a reaction is found to be double that of rate constant k_2 of another reaction. The relationship between corresponding activation energies of the two reactions at same temperature (E_1 & E_2) can be represented as
 (1) $E_1 > E_2$
 (2) $E_1 < E_2$
 (3) $E_1 = E_2$
 (4) $E_1 = 4E_2$

89. At room temperature the reaction between NO and O_2 to give NO_2 is fast while that between CO and O_2 is slow it is due to –
 (1) CO is smaller in size than that of NO
 (2) CO is poisonous
 (3) The activation energy for the reaction $2NO + O_2 \rightarrow 2NO_2$ is less than $2CO + O_2 \rightarrow 2CO_2$
 (4) NO_2 is poisonous
90. Chemical reaction occurs as a result of collision between reacting molecules. Therefore the reaction rate is given by
 (1) Total number of collisions occurring in a unit volume per second
 (2) Fraction of molecules which possess energy less than the threshold energy.
 (3) Total number of effective collisions
 (4) Temperature
91. An endothermic reaction $A \rightarrow B$ have an activation energy 15 kCal/mol and the heat of the reaction is 5 k cal/mol. The activation energy of the reaction $B \rightarrow A$ is –
 (1) 20 kCal/mol (2) 15 kCal/mol
 (3) 10 kCal/mol (4) Zero
92. A large increase in the rate of a reaction for a rise in temperature is due to –
 (1) Increase in the number of collisions
 (2) Increase in the number of activated molecules
 (3) Lowering of activation energy
 (4) Shortening of the mean free path
93. Rate of which reactions increases with temperature
 (1) Of any reaction
 (2) Of exothermic reaction
 (3) Of endothermic reaction
 (4) Can't be predicted
94. The rate of a chemical reaction doubles for every $10^\circ C$ rise in temperature. If the temp is increased by $60^\circ C$ the rate of reaction increases by :
 (1) 20 times (2) 32 times
 (3) 64 times (4) 128 times
95. According to the arrhenius equation a straight line is to be obtained by plotting the logarithm of the rate constant of chemical reaction ($\log k$) against.
 (1) (T) (2) $\log T$
 (3) $1/T$ (4) $\log 1/T$

96. Which plot will give the value of activation energy.
 (1) k v/s T (2) $1/k$ v/s T
 (3) $\ln k$ v/s T (4) $\ln k$ v/s $1/T$
97. Given that k is the rate constant for some order of any reaction at temp T then the value of $\lim_{T \rightarrow \infty} \log k$ _____.
 (1) $A/2.303$ (2) A
 (3) $2.303 A$ (4) $\log A$
98. From the following data; the activation energy for the reaction (Cal/mol) $H_2 + I_2 \rightarrow 2HI$
- | T (in K) | $1/T$ (in K^{-1}) | $\log_{10} k$ |
|----------|----------------------|---------------|
| 769 | 1.3×10^{-3} | 2.9 |
| 667 | 1.5×10^{-3} | 1.1 |
- (1) 4×10^4 (2) 2×10^4
 (3) 8×10^4 (4) 3×10^4
99. The rate constant, the activation energy and the arrhenius parameter of a chemical reaction at $25^\circ C$ are $3 \times 10^{-4} s^{-1}$; $104.4 kJ mol^{-1}$ and $6.0 \times 10^{14} s^{-1}$ respectively, the value of the rate constant as $T \rightarrow \infty$ is.
 (1) $2 \times 10^8 s^{-1}$ (2) $6 \times 10^{14} s^{-1}$
 (3) Infinity (4) $3.6 \times 10^{30} s^{-1}$
100. For an endothermic reaction where ΔH represents the enthalpy of the reaction in kJ/mol ; the minimum value for the energy of activation will be
 (1) Less than ΔH (2) Zero
 (3) More than ΔH (4) Equal to ΔH
101. The rate of reaction increases to 2.3 times when the temperature is raised from 300 K to 310 K. If K is the rate constant at 300 K then the rate constant at 310 K will be equal to –
 (1) $2 k$ (2) k
 (3) $2.3 k$ (4) $3 k^2$
102. If concentration of reactants is increased by 'x' then the k becomes –
 (1) $\ln \frac{k}{x}$ (2) $\frac{k}{x}$
 (3) $k + x$ (4) k

103. If the concentration units are reduced by n times then the value of rate constant of first order will

- (1) Increases by n times
- (2) Decreases by factor of n
- (3) Remain constant
- (4) Decrease $1/n$ times

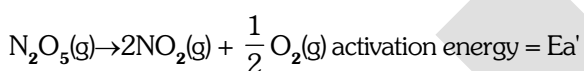
104. Which is used in the determination of reaction rates.

- (1) Reaction Temperature
- (2) Reaction Concentration
- (3) Specific rate constant
- (4) All of these

105. The rate constant of a first order reaction depends on the :-

- (1) Concentration of the reactant
- (2) Concentration of the product
- (3) Time
- (4) Temperature

106. For the decomposition of $N_2O_5(g)$ it is given that—
 $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$ activation energy = E_a



then

- (1) $E_a = 2E_a'$
- (2) $E_a > E_a'$
- (3) $E_a < E_a'$
- (4) $E_a = E_a'$

107. For a reaction in which case the activation energies of forward and reverse reactions are equal :-

- (1) $\Delta H = 0$
- (2) $\Delta S = 0$
- (3) The order is zero
- (4) There is no catalyst

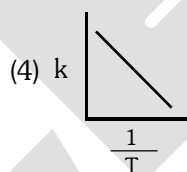
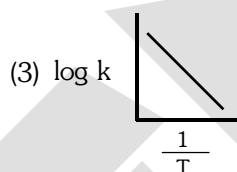
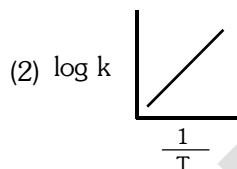
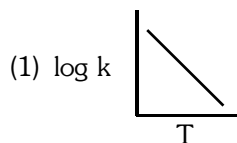
108. The energy of activation of a forward reaction is 50 kCal. The energy of activation of its backward reaction is:-

- (1) Equal to 50 kCal.
- (2) Greater than 50 kCal.
- (3) Less than 50 kCal.
- (4) Either greater or less than 50 kCal.

109. An exothermic reaction $X \rightarrow Y$ has an activation energy 30 kJ mol^{-1} . If energy change (ΔE) during the reaction is -20 kJ , then the activation energy for the reverse reaction is :-

- (1) 10 kJ
- (2) 20 kJ
- (3) 50 kJ
- (4) -30 kJ

110. Which of the following plot is in accordance with the arrhenius equation :-



111. The rate of reaction increases by the increase of temperature because :-

- (1) Collision is increased
- (2) Energy of products decreases
- (3) Fraction of molecules possessing energy $\geq E_T$ (Threshold energy) increases
- (4) Mechanism of a reaction is changed

112. For a certain gaseous reaction rise of temperature from 25°C to 35°C doubles the rate of reaction. What is the value of activation energy :-

(1) $\frac{10}{2.303R \times 298 \times 308}$

(2) $\frac{2.303 \times 10}{298 \times 308R}$

(3) $\frac{0.693R \times 10}{290 \times 308}$

(4) $\frac{0.693R \times 298 \times 308}{10}$

- 113.** The activation energy for the forward reaction $X \rightarrow Y$ is 60 kJ mol^{-1} and ΔH is -20 kJ mol^{-1} . The activation energy for the backward reaction $Y \rightarrow X$ is:-
- (1) 80 kJ mol^{-1} (2) 40 kJ mol^{-1}
 (3) 60 kJ mol^{-1} (4) 20 kJ mol^{-1}

- 114.** For producing the effective collisions, the colliding molecules must possess:-
- (1) A certain minimum amount of energy
 (2) Energy equal to or greater than threshold energy
 (3) Proper orientation
 (4) Threshold energy as well as proper orientation of collision

- 115.** The half life for a reaction is -----of temperature:-
- (1) Independent
 (2) Increased with increase
 (3) Decreased with increase
 (4) Increased or decreased with increase

- 116.** The activation energy for a chemical reaction depends upon :-
- (1) Temperature
 (2) Nature of reacting species
 (3) Concentration of the reacting species
 (4) Collision frequency

EXERCISE-I (Conceptual Questions)
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	3	3	1	4	1	3	1	4	2	4	3	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	2	2	1	3	3	2	2	4	3	3	2	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	2	2	3	2	1	2	4	1	3	2	1	3	4	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	3	2	1	4	1	2	2	4	4	2	4	3	2	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	4	3	2	3	1	3	2	2	3	3	1	2	4	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	1	2	2	4	2	1	2	4	4	4	3	3	2	3	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	3	2	1	3	3	4	4	1	2	3	3	4	3	4	4
Que.	106	107	108	109	110	111	112	113	114	115	116				
Ans.	4	1	4	3	3	3	4	1	4	3	2				

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

1. For the reaction
 $2A + B \rightarrow 3C + D$
 Which of the following does not express the reaction rate ?

(1) $-\frac{1}{3} \frac{d[C]}{dt}$ (2) $-\frac{d[B]}{dt}$ (3) $\frac{d[D]}{dt}$ (4) $-\frac{1}{2} \frac{d[A]}{dt}$

2. Consider the reaction
 $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

The correct equality relationship between $\frac{d[NH_3]}{dt}$

and $-\frac{d[H_2]}{dt}$ is :

(1) $\frac{d[NH_3]}{dt} = -\frac{1}{3} \frac{d[H_2]}{dt}$

(2) $+\frac{d[NH_3]}{dt} = -\frac{2}{3} \frac{d[H_2]}{dt}$

(3) $+\frac{d[NH_3]}{dt} = -\frac{3}{2} \frac{d[H_2]}{dt}$

(4) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

AIPMT 2008

3. The rate constants k_1 and k_2 for two different reactions are $10^{16} e^{-2000/T}$ and $10^{15} e^{-1000/T}$, respectively.

The temperature at which $k_1 = k_2$ is

(1) $\frac{2000}{2.303} K$ (2) 2000 K
 (3) $\frac{1000}{2.303} K$ (4) 1000 K

4. The bromination of acetone that occurs in acid solution is represented by this equation



The kinetic data were obtained for given reaction at different concentrations.

$[CH_3COCH_3]$	$[Br_2]$	$[H^+]$	ROD of Br_2 , Ms^{-1}
0.30	0.05	0.05	5.7×10^{-5}
0.30	0.10	0.05	5.7×10^{-5}
0.30	0.10	0.10	1.2×10^{-4}
0.40	0.05	0.20	3.1×10^{-4}

Based on these data, the rate equation is :

- (1) Rate = $k[CH_3COCH_3][Br_2]$
 (2) Rate = $k[CH_3COCH_3][Br_2][H^+]^2$
 (3) Rate = $k[CH_3COCH_3][Br_2][H^+]$
 (4) Rate = $k[CH_3COCH_3][H^+]$

AIPMT 2009

5. For the reaction, $N_2 + 3H_2 \rightarrow 2NH_3$, if
 $\frac{d[NH_3]}{dt} = 2 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$, the value of

$-\frac{d[H_2]}{dt}$ would be :-

- (1) $1 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$ (2) $3 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (3) $4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$ (4) $6 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

6. For the reaction $A + B \longrightarrow \text{products}$, it is observed that :-

- (a) on doubling the initial concentration of A only, the rate of reaction is also doubled and
 (b) on doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction.

The rate of this reaction is given by :-

- (1) rate = $k[A][B]$ (2) rate = $k[A]^2[B]$
 (3) rate = $k[A][B]^2$ (4) rate = $k[A]^2[B]^2$

7. In the reaction



The rate of appearance of bromine (Br_2) is related to rate of disappearance of bromide ions as following :-

(1) $\frac{d[Br_2]}{dt} = \frac{3}{5} \frac{d[Br^-]}{dt}$

(2) $\frac{d[Br_2]}{dt} = -\frac{3}{5} \frac{d[Br^-]}{dt}$

(3) $\frac{d[Br_2]}{dt} = -\frac{5}{3} \frac{d[Br^-]}{dt}$

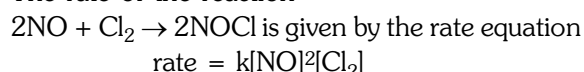
(4) $\frac{d[Br_2]}{dt} = \frac{5}{3} \frac{d[Br^-]}{dt}$

8. Half life period of a first-order reaction is 1386 s. The specific rate constant of the reaction is :-

- (1) $5.0 \times 10^{-2} \text{ s}^{-1}$ (2) $5.0 \times 10^{-3} \text{ s}^{-1}$
 (3) $0.5 \times 10^{-2} \text{ s}^{-1}$ (4) $0.5 \times 10^{-3} \text{ s}^{-1}$

AIPMT 2010

9. The rate of the reaction



the value of rate constant can be increased by :

- (1) Increasing the concentration of NO
 (2) Increasing the concentration of the Cl_2
 (3) Increasing the temperature
 (4) Doing all of these

10. For the reaction $\text{N}_2\text{O}_5(\text{g}) \longrightarrow 2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$

the value of rate of disappearance of N_2O_5 is given as $6.25 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$. The rate of formation of NO_2 and O_2 is given respectively as :-

- (1) $1.25 \times 10^{-2} \text{ mol L}^{-1}\text{s}^{-1}$ and $6.25 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$
- (2) $6.25 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$ and $6.25 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$
- (3) $1.25 \times 10^{-2} \text{ mol L}^{-1}\text{s}^{-1}$ and $3.125 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$
- (4) $6.25 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$ and $3.125 \times 10^{-3} \text{ mol L}^{-1}\text{s}^{-1}$

11. During the kinetic study of the reaction, $2\text{A} + \text{B} \rightarrow \text{C} + \text{D}$, following results were obtained:-
- Ex. [A] (mol L⁻¹) [B] (mol L⁻¹) Initial rate of formation of D (mol L⁻¹ min⁻¹)

I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

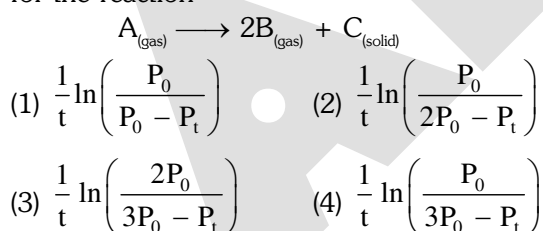
Based on the above data which one of the following is correct ?

- (1) rate = $k[\text{A}][\text{B}]^2$
- (2) rate = $k[\text{A}]^2[\text{B}]$
- (3) rate = $k[\text{A}][\text{B}]$
- (4) rate = $k[\text{A}]^2[\text{B}]^2$

12. For an endothermic reaction, energy of activation is E_a and enthalpy of reaction is ΔH (both of these in kJ/mol). Minimum value of E_a will be :-
- (1) Equal to zero
 - (2) Less than ΔH
 - (3) Equal to ΔH
 - (4) More than ΔH

AIIMS 2010

13. For a first order reaction, the value of rate constant for the reaction



AIPMT Pre. 2011

14. Which one of the following statements for the order of a reaction is incorrect ?
- (1) Order can be determined only experimentally
 - (2) Order is not influenced by stoichiometric coefficient of the reactants
 - (3) Order of reaction is sum of power to the concentration terms of reactants to express the rate of reaction
 - (4) Order of reaction is always whole number

AIPMT Mains 2011

15. The unit of rate constant for a zero order reaction is :-
- (1) s⁻¹
 - (2) mol L⁻¹ s⁻¹
 - (3) L mol⁻¹ s⁻¹
 - (4) L² mol⁻² s⁻¹

16. The half life of a substance in a certain enzyme-catalysed first order reaction is 138 s. The time required for the concentration of the substance to fall from 1.28 mg L⁻¹ to 0.04 mg L⁻¹ :-
- (1) 276 s
 - (2) 414 s
 - (3) 552 s
 - (4) 690 s

17. The rate of the reaction $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$ can be written in three ways

$$-\frac{d[\text{N}_2\text{O}_5]}{dt} = k[\text{N}_2\text{O}_5]$$

$$\frac{d[\text{NO}_2]}{dt} = k'[\text{N}_2\text{O}_5]$$

$$\frac{d[\text{O}_2]}{dt} = k''[\text{N}_2\text{O}_5]$$

The relationship between k and k' and between k and k'' are :-

- (1) k' = k ; k'' = k
- (2) k' = 2k ; k'' = k
- (3) k' = 2k ; k'' = k/2
- (4) k' = 2k ; k'' = 2k

AIIMS 2011

18. For a first order gas phase reaction-



Let P_0 be initial pressure of A and P_t the total pressure at time 't'. Integrated rate equation is-

$$(1) \frac{2.303}{t} \log \left(\frac{P_0}{P_0 - P_t} \right) \quad (2) \frac{2.303}{t} \log \left(\frac{2P_0}{3P_0 - P_t} \right)$$

$$(3) \frac{2.303}{t} \log \left(\frac{P_0}{2P_0 - P_t} \right) \quad (4) \frac{2.303}{t} \log \left(\frac{2P_0}{2P_0 - P_t} \right)$$

AIPMT Pre. 2012

19. In a zero-order reaction for every 10° rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become:
- (1) 64 times
 - (2) 128 times
 - (3) 256 times
 - (4) 512 times
20. In a reaction $\text{A} + \text{B} \rightarrow \text{Product}$, rate is doubled when the concentration of B is doubled and rate increased by a factor of 8 when the concentrations of both the reactants (A and B) are doubled, rate law for the reaction can be written as:
- (1) Rate = $k[\text{A}][\text{B}]$
 - (2) Rate = $k[\text{A}]^2[\text{B}]$
 - (3) Rate = $k[\text{A}][\text{B}]^2$
 - (4) Rate = $k[\text{A}]^2[\text{B}]^2$

AIPMT Mains 2012

21. Activation energy (E_a) and rate constants (k_1 and k_2) of a chemical reaction at two different temperatures (T_1 and T_2) are related by :

$$(1) \ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} + \frac{1}{T_1} \right) \quad (2) \ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$(3) \ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \quad (4) \ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

AIIMS 2012

22. For a reaction $r = k(\text{CH}_3\text{COCH}_3)^{3/2}$ the unit of rate of reaction and rate constant respectively are :-

$$(1) \text{mol L}^{-1}\text{s}^{-1}, \text{mol}^{-\frac{1}{2}} \text{L}^{\frac{1}{2}}\text{s}^{-1}$$

$$(2) \text{mol}^{-1} \text{L}^{-1}\text{s}^{-1}, \text{mol}^{-\frac{1}{2}} \text{L}^{-\frac{1}{2}}\text{s}^{-1}$$

$$(3) \text{mol L}^{-1}\text{s}^{-1}, \text{mol}^{+\frac{1}{2}} \text{L}^{\frac{1}{2}}\text{s}^{-1}$$

$$(4) \text{mol Ls}, \text{mol}^{+\frac{1}{2}} \text{L}^{\frac{1}{2}}\text{s}$$

23. $t_{1/2}$ v/s $\frac{1}{a^2}$ is a straight line graph then determine

the order of reaction is :-

- (1) Zero order (2) First order
(3) Second order (4) Third order

NEET-UG 2013

24. A reaction having equal energies of activation for forward and reverse reactions has :-

- (1) $\Delta H = \Delta G = \Delta S = 0$ (2) $\Delta S = 0$
(3) $\Delta G = 0$ (4) $\Delta H = 0$

25. What is the activation energy of a reaction if its rate doubles when the temperature is raised from 20°C to 35°C ? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) 15.1 kJ mol^{-1} (2) 342 kJ mol^{-1}
(3) 269 kJ mol^{-1} (4) 34.7 kJ mol^{-1}

AIIMS 2013

26. Energy of activation of forward reaction for an endothermic process is 50 kJ and enthalpy change is 20 kJ then activation energy change for backward reaction will be :-

- (1) 30 kJ (2) 20 kJ
(3) 70 kJ (4) 50 kJ

AIIMS 2014

27. In 1st order reaction $A_{(g)} \longrightarrow 2B_{(g)} + C(s)$; if P_i is the initial pressure and P_f is the final pressure of mixture after time t then rate constant is :-

$$(1) k = \frac{2.303}{t} \log \left(\frac{2P_i}{3P_i - P_f} \right)$$

$$(2) k = \frac{2.303}{t} \log \left(\frac{P_i}{2P_i - P_f} \right)$$

$$(3) k = \frac{2.303}{t} \log \left(\frac{P_i}{P_i - 3P_f} \right)$$

$$(4) k = \frac{2.303}{t} \log \left(\frac{P_i}{P_i + P_f} \right)$$

28. If in a 1st order reaction 75% of reaction completes in 4 h. Then how much time is required to complete 87.5% of reaction ?

- (1) 32 h (2) 6 h (3) 8 h (4) 16 h

AIPMT 2015

29. The activation energy of a reaction can be determined from the slope of which of the following graphs ?

$$(1) \frac{\ln k}{T} \text{ v/s. } T \quad (2) \ln k \text{ v/s. } \frac{1}{T}$$

$$(3) \frac{T}{\ln k} \text{ v/s. } \frac{1}{T} \quad (4) \ln k \text{ v/s. } T$$

30. When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :-

- (1) First
(2) Second
(3) More than zero but less than first
(4) Zero

Re-AIPMT 2015

31. The rate constant of the reaction $A \rightarrow B$ is $0.6 \times 10^{-3} \text{ M sec}^{-1}$. If the concentration of A is 5 M, then concentration of B after 20 min is :-

- (1) 0.36 M (2) 0.72 M
(3) 1.08 M (4) 3.60 M

AIIMS 2015

32. For a first order reaction rate constant is given as

$$\log k = 14 - \frac{1.2 \times 10^4}{T}, \text{ then what will be}$$

the value of temperature if its half life period is 6.93×10^{-3} min

- (1) 100 K (2) 1000 K
(3) 720 K (4) 327 K

33. For a reaction $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$;
Initial concentration of N_2O_5 is 7.2 mol L^{-1} and after 20 minute $[\text{NO}_2]$ is 2.4 mol L^{-1} , then what will be the rate of decomposition of N_2O_5 in $\text{mol L}^{-1} \text{ min}^{-1}$
(1) 0.06 (2) 0.03 (3) 1.3 (4) 2.4

34. For a reaction $\text{A} \rightarrow \text{B}$ half life period is 50 min. If reactant concentration is halved then half life period is 25 min. Calculate order of reaction ?
(1) 0 (2) 1 (3) 2 (4) 3

NEET-I 2016

35. The rate of a first-order reaction is $0.04 \text{ mol L}^{-1} \text{ s}^{-1}$ at 10 s and $0.03 \text{ mol L}^{-1} \text{ s}^{-1}$ at 20 s after initiation of the reaction. The half-life period of the reaction is :
(1) 24.1 s (2) 34.1 s
(3) 44.1 s (4) 54.1 s
36. The addition of a catalyst during a chemical reaction alters which of the following quantities ?
(1) Entropy (2) Internal energy
(3) Enthalpy (4) Activation energy

NEET-II 2016

37. The decomposition of phosphine (PH_3) on tungsten at low pressure is a first-order reaction. It is because the
(1) rate is independent of the surface coverage
(2) rate of decomposition is very slow
(3) rate is proportional to the surface coverage
(4) rate is inversely proportional to the surface coverage

AIIMS 2016

38. For a reaction :-
 $6\text{H}^+ + 5\text{Br}^- + \text{BrO}_3^- \longrightarrow 3\text{Br}_2 + 6\text{H}_2\text{O}$
If rate of consumption of BrO_3^- is $x \text{ mol L}^{-1} \text{ s}^{-1}$. Then calculate rate of formation of Br_2 :-
(1) $\frac{x}{3}$ (2) $\frac{2x}{3}$ (3) $\frac{x}{4}$ (4) $3x$

39. On increasing temperature from 200K to 220K, rate of reaction A increases by 3 times and rate of reaction B increases by 9 times then correct relationship between activation energy of A and B is :-
(1) $E_A = 3E_B$ (2) $3E_A = E_B$
(3) $E_B = 2E_A$ (4) $E_A = 2E_B$

40. The slope of $\frac{-E_a}{2.303R}$ is obtained by the plot of:-

- (1) $\log k$ v/s $\log A$
(2) $\log k$ v/s T
(3) $\log k$ v/s $\frac{1}{T}$
(4) k v/s $\frac{1}{T}$

41. For a first order reaction rate constant is $1 \times 10^{-5} \text{ s}^{-1}$ having $E_a = 1800 \text{ kJ/mol}$. Then value of $\ln A$ at $T = 600\text{K}$ is :-

- (1) 151.7 (2) 349.3
(3) 24.7 (4) 11.34

42. Half life of a radioactive decay of C^{14} is 5730 years. How much time it will take so that 25% of C^{14} was found in sample?

- (1) 5730 years (2) 11460 years
(3) 22920 years (4) 930 years

NEET(UG) 2017

43. Mechanism of a hypothetical reaction



- (i) $\text{X}_2 \rightleftharpoons \text{X} + \text{X}(\text{fast})$
(ii) $\text{X} + \text{Y}_2 \rightarrow \text{XY} + \text{Y}(\text{slow})$
(iii) $\text{X} + \text{Y} \rightarrow \text{XY}(\text{fast})$

The overall order of the reaction will be :

- (1) 2 (2) 0
(3) 1.5 (4) 1

44. A first order reaction has a specific reaction rate of 10^{-2} s^{-1} . How much time will it take for 20 g of the reactant to reduce to 5 g ?

- (1) 138.6 s (2) 346.5 s
(3) 693.0 s (4) 238.6 s

AIIMS 2017

45. For a first order gaseous reaction $A \rightarrow 2B + C$, if initial pressure is P_0 and total final pressure of mixture becomes P_t after time t . The rate constant is :-

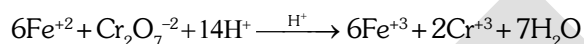
(1) $k = \frac{2.303}{t} \log \frac{2P_0}{P_0 - P_t}$

(2) $k = \frac{2.303}{t} \log \frac{2P_0}{3P_0 - P_t}$

(3) $k = \frac{2.303}{t} \log \frac{2P_0}{P_0 + P_t}$

(4) $k = \frac{2.303}{t} \log \frac{2P_0}{2P_0 + P_t}$

46. When Fe^{+2} reacts with $Cr_2O_7^{2-}$ in acidic medium then Fe^{+3} and Cr^{+3} are formed. If rate of disappearance of $Cr_2O_7^{2-}$ is x then the rate of appearance of Fe^{+3} will be :-



- (1) $x/6$ (2) $3x$ (3) $6x$ (4) $x/3$

47. For a first order reaction : $2A(g) \rightarrow 2B(g) + C(g)$, the initial pressure is P_0 and after time t , total pressure of reaction mixture becomes P_t . The rate constant is :-

(1) $k = \frac{2.303}{t} \log \frac{P_0}{P_0 - P_t}$

(2) $k = \frac{1}{t} \ln \frac{P_0}{3P_0 - 2P_t}$

(3) $k = \frac{1}{t} \ln \frac{P_0}{2P_0 - P_t}$

(4) $k = \frac{1}{t} \ln \frac{2P_0}{3P_0 - P_t}$

48. In the reaction $2A \rightarrow B$, the concentration of A decreases from 0.8 M to 0.4 M in 10 min. The rate of appearance of B is :-

- (1) 0.02 M min^{-1} (2) 0.04 M min^{-1}
(3) 0.08 M min^{-1} (4) 0.2 M min^{-1}

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49. The correct difference between first- and second-order reaction is that

- (1) the rate of a first-order reaction does not depend on reactant concentration; the rate of a second-order reaction does depend on reactant concentrations.
(2) the half-life of a first-order reaction does not depend on $[A]_0$; the half-life of a second-order reaction does depend on $[A]_0$
(3) a first-order reaction can be catalyzed; a second-order reaction cannot be catalyzed.
(4) the rate of a first-order reaction does depend on reactant concentrations; the rate of a second-order reaction does not depend on reactant concentrations

50. When initial concentration of the reactant is doubled, the half-life period of a zero order reaction

- (1) is halved (2) is doubled
(3) is tripled (4) remains unchanged

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51. For a unimolecular reaction $A \rightarrow B$ has rate constant K_1 and for another unimolecular reaction $C \rightarrow D$ has rate constant K_2 . If half life of first reaction is half of half life of second reaction then find $E_{a1} - E_{a2}$ will be :-

- (1) $-RT \ln 2$ (2) $RT \ln 2$
(3) $\frac{RT}{\ln 2}$ (4) $\frac{1}{RT \ln 2}$

52. For zero order reaction relation between $t_{1/2}$ and $t_{1/4}$ is :-

- (1) $t_{1/2} = 2 t_{1/4}$ (2) $t_{1/2} = 1.5 t_{1/4}$
(3) $t_{1/2} = \frac{1}{2} t_{1/4}$ (4) $t_{1/2} = \frac{1}{3} t_{1/4}$

53. For two zero order reaction having same rate at certain temperature; If $E_{a1} - E_{a2} = 2RT$

Then find $\ln \left(\frac{A_1}{A_2} \right)$:-

- (1) +2 (2) -2 (3) $\frac{2}{\ln 2}$ (4) $2 \ln 2$

54. For a first order reaction $t_{80\%}$ is 60 min. Calculate $t_{1/2}$:-

- (1) 25.7 min (2) 25 min
(3) 20 min (4) 257 min

55. The rate of first order reaction are 0.4 M min^{-1} and 0.04 M min^{-1} at 10 min and 30 min respectively. Then half life of reaction is :-

- (1) 6 min (2) 4 min
(3) 8 min (4) 13 min

56. Two different reactions at 300 K having same pre-exponential factor. The activation energy of first reaction is more than that of second reaction by

5.75 kJ. Then find $\frac{K_2}{K_1}$:-

- (1) 1000 (2) 10,000
(3) 10 (4) 1

57. For a reaction, activation energy is 83 kJ calculate

ratio of $\ln \frac{K_{37^\circ\text{C}}}{K_{27^\circ\text{C}}}$ for above gaseous reaction :-

- (1) 1 (2) 2
(3) 3 (4) 4

58. For a 1st order reaction what will be the time required to reach concentration from 1.28 M to 0.04 M if $t_{1/2}$ is 138 s :-

- (1) 690 s (2) 138 s
(3) 69 s (4) 27.6 s

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	3	4	2	3	2	4	3	3	1	4	2	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	3	2	4	2	2,4	1	4	4	4	1	2	2	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	2	1	1	1	4	3	4	3	3	2	2	3	1	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58		
Ans.	3	2	1	2	2	1	1	1	1	1	3	1	1		

EXERCISE-III (Analytical Questions)

Check Your Understanding

1. For the reaction
 $A + 2B \rightarrow \text{Product}$,
 the differential rate equation is :-
 (1) $-\frac{1}{2} \frac{d[A]}{dt} = -\frac{d[B]}{dt} = k[A][B]^2$
 (2) $\frac{1}{2} \frac{d[A]}{dt} = \frac{d[B]}{dt} = k[A][B]^2$
 (3) $-\frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt} = k[A][B]^2$
 (4) $\frac{dA}{dt} = \frac{1}{2} \frac{d[B]}{dt} = k[A][B]^2$
2. The rate of reaction between two reactants A and B decreases by a factor of 4, if the concentration of reactant B is doubled. The order of the reaction with respect to reactant B is :
 (1) -1 (2) -2 (3) 1 (4) 2
3. For the non-stoichiometric reaction
 $2A + B \rightarrow C + D$, the following kinetic data were obtained in three different experiments, measured at 298 K.
- | Initial Conc. (A) | Initial Conc. (B) | Initial rate of formation of C (mol L ⁻¹ s ⁻¹) |
|-------------------|-------------------|---|
| 0.1M | 0.1M | 1.2×10^{-3} |
| 0.1M | 0.2M | 1.2×10^{-3} |
| 0.2M | 0.1M | 2.4×10^{-3} |
- (1) $\frac{dc}{dt} = k[A][B]^2$ (2) $\frac{dc}{dt} = k[A]$
 (3) $\frac{dc}{dt} = k[A][B]$ (4) $\frac{dc}{dt} = k[A]^2[B]$
4. The following mechanism has been proposed for the reaction of NO with Br₂ to form NOBr :
 $\text{NO(g)} + \text{Br}_2\text{(g)} \rightleftharpoons \text{NOBr}_2\text{(g)}$
 $\text{NOBr}_2\text{(g)} + \text{NO(g)} \rightarrow 2\text{NOBr(g)}$
 If the second step is the rate determining step, the order of the reaction with respect to NO(g) is :
 (1) 2 (2) 1 (3) 0 (4) 3
5. Higher order (>3) reactions are rare due to :-
 (1) shifting of equilibrium towards reactants due to elastic collision
 (2) loss of active species on collision
 (3) low probability of simultaneous collision of all the reacting species
 (4) increase in entropy and activation energy as more molecules are involved.

6. For a first order reaction :-
 (1) The degree of dissociation is equal to $(1 - e^{-kt})$
 (2) The pre-exponential factor in the arrhenius equation has the dimension of time⁻¹
 (3) A plot of reciprocal of concentration of the reaction v/s time gives a straight line
 (4) 1 & 2 both
7. Azo isopropane decomposes according to the equation :-
 $(\text{CH}_3)_2\text{CHN} = \text{NCH}(\text{CH}_3)_2 \text{ (g)} \xrightarrow{250-290^\circ\text{C}} \text{N}_2\text{(g)} + \text{C}_6\text{H}_{14}\text{(g)}$
 It is found to be a first order reaction. If initial pressure is P_o and pressure of the mixture at time t is P_t then rate constant k would be :-
 (1) $k = \frac{2.303}{t} \log \frac{P_o}{2P_o - P_t}$
 (2) $k = \frac{2.303}{t} \log \frac{P_o - P_t}{P_o}$
 (3) $k = \frac{2.303}{t} \log \frac{P_o}{P_o - P_t}$
 (4) $k = \frac{2.303}{t} \log \frac{2P_o}{2P_o - P_t}$
8. A reagent undergoes 90% decomposition in 366 min. According to first order reaction, it's half life is :-
 (1) $366 \times 100 \left(\frac{\ln 2}{90} \right)$ (2) $366 \left(\frac{\ln 2}{\ln 10} \right)$
 (3) $\frac{1}{366}$ (4) 183
9. Which of the following statement is false :-
 (1) A fast reaction has a large rate constant and short half life
 (2) For a first order reaction, successive half lives are equal
 (3) For a first order reaction, the half life is independent of concentration
 (4) The half life of a reaction is half the time required for the reaction to go for completion
10. The reaction
 $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{C}_2\text{H}_5\text{OH}$ is :-
 (1) Bimolecular reaction (2) II order reaction
 (3) Both (1) & (2) (4) None
11. Correct expression for the first order reaction is:-
 (1) $C_t = C_0 e^{k_1 t}$ (2) $C_t e^{k_1 t} = C_0$
 (3) $\ln \frac{C_0}{C_t} = -k_1 t$ (4) $\ln \frac{C_t}{C_0} = k_1 t$

12. The rate of first order reaction is $1.5 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$ at 0.5 M concentration of the reactant. The half life of the reaction is
 (1) 7.53 min (2) 0.383 min
 (3) 23.1 min (4) 8.73 min
13. In a first order reaction, the concentration of the reactant, decreases from 0.8 M to 0.4 M in 15 min. The time taken for the concentration to change from 0.1 M to 0.025 M is
 (1) 7.5 min (2) 15 min
 (3) 30 min (4) 60 min
14. The rate equation for the reaction $2A + B \rightarrow C$ is found to be : $\text{rate} = k[A][B]$. The correct statement for this reaction is that the
 (1) rate of formation of C is twice the rate of disappearance of A.
 (2) Half life is a constant
 (3) unit of k must be s^{-1}
 (4) value of k is independent of the initial concentrations of A and B
15. The half-life of a radio isotope is four hours. If the initial mass of the isotope was 200 g, the mass remaining after 24 h undecayed is :
 (1) 3.125 g (2) 2.084 g
 (3) 1.042 g (4) 4.167 g
16. For a first order reaction $A \rightarrow B$ the reaction rate at reactant concentration of 0.01 M is found to be $2.0 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$. The half life period of the reaction is :
 (1) 220 s (2) 30 s (3) 300 s (4) 347 s
17. For reaction $aA \rightarrow xP$. When [A] is 2.2 mM, the rate was found to be 2.4 mM s^{-1} . On reducing concentration of A to half, the rate changes to 0.6 mM s^{-1} . The order of reaction with respect to A is :
 (1) 1.5 (2) 2.0 (3) 2.5 (4) 3.0
18. The $t_{1/2}$ of a reaction is halved as the initial concentration of the reactant is doubled. What is the order of reaction ?
 (1) 1 (2) 0 (3) 2 (4) 3
19. The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C , the rate of the reaction increases by about :-
 (1) 32 times (2) 64 times
 (3) 10 times (4) 24 times

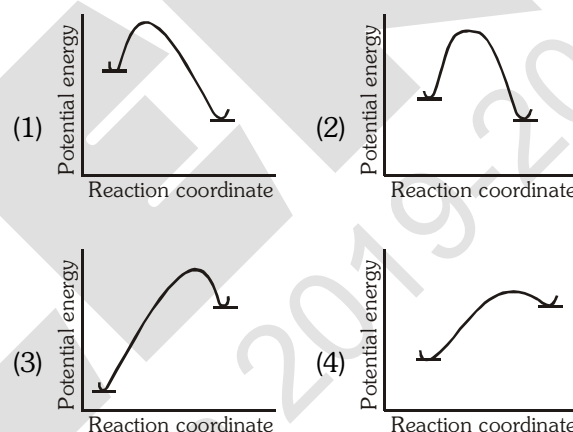
20. For a first order reaction, $(A) \rightarrow \text{products}$, the concentration of A changes from 0.1 M to 0.025 M in 40 min. The rate of reaction when the concentration of A is 0.01 M is :-
 (1) $1.73 \times 10^{-4} \text{ M/min}$ (2) $1.73 \times 10^{-5} \text{ M/min}$
 (3) $3.47 \times 10^{-4} \text{ M/min}$ (4) $3.47 \times 10^{-5} \text{ M/min}$

21. Arrhenius equation may be written as :-

$$(1) \frac{d}{dT} (\ln k) = - \frac{E_a}{RT} \quad (2) \frac{d}{dT} (\ln k) = - \frac{E_a}{RT^2}$$

$$(3) \frac{d}{dT} (\ln k) = + \frac{E_a}{RT^2} \quad (4) \frac{d}{dT} (\ln k) = \frac{E_a}{RT}$$

22. An endothermic reaction with high activation energy for the forward reaction is given by the diagram :



23. A reactant (A) forms two products :



If $E_{a2} = 2 E_{a1}$, then k_1 and k_2 are related as :-

$$(1) k_1 = 2k_2 e^{E_{a2}/RT} \quad (2) k_2 = k_1 e^{E_{a1}/RT}$$

$$(3) k_2 = k_1 e^{E_{a2}/RT} \quad (4) k_1 = k_2 e^{E_{a1}/RT}$$

24. The rate of a reaction doubles when its temperature changes from 300 K to 310 K. Activation energy of such a reaction will be

($R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$ and $\log 2 = 0.301$)

$$(1) 53.6 \text{ kJ mol}^{-1} \quad (2) 48.6 \text{ kJ mol}^{-1}$$

$$(3) 58.5 \text{ kJ mol}^{-1} \quad (4) 60.5 \text{ kJ mol}^{-1}$$

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	1	3	4	1	2	4	3	2	3	3	4	1
Que.	16	17	18	19	20	21	22	23	24						
Ans.	4	2	3	1	3	3	3	4	1						

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion** : The rate of the reaction is the rate of change of concentration of a reactant or a product.
Reason : Rate of reaction remains constant during the complete reaction.
 (1) A (2) B (3) C (4) D
- Assertion** : The molecularity of a reaction is a whole number other than zero, but generally less than 3.
Reason : The order of a reaction is always whole number
 (1) A (2) B (3) C (4) D
- Assertion** : Formation of HI is a bimolecular reaction.
Reason : Two molecules of reactants are involved in this reaction.
 (1) A (2) B (3) C (4) D
- Assertion** : If in a zero order reaction, the concentration of the reactant is doubled, the half-life period is also doubled.
Reason : For a zero order reaction, the rate of reaction is independent of initial concentration.
 (1) A (2) B (3) C (4) D
- Assertion** :- In a first order reaction the rate constant double on doubling the initial concentration of the reaction.
Reason :- The rate constant varies directly with the concentration of the reactants in a first order reaction.
 (1) A (2) B (3) C (4) D
- Assertion** :- Rate of reaction double when concentration of reactant is double if it is a first order reaction.
Reason :- Rate constant also double.
 (1) A (2) B (3) C (4) D
- Assertion** : For a first order reaction $t_{1/2}$ is independent of the initial concentration of reactants.
Reason : For a first order reaction $t_{1/2}$ is twice the $t_{3/4}$.
 (1) A (2) B (3) C (4) D
- Assertion** : Acid catalysed of ethyl acetate is a first order reaction.
Reason : Water does not take part in the reaction.
 (1) A (2) B (3) C (4) D
- Assertion** : Half life period is always independent of initial concentration.
Reason : Half life period is directly proportional to rate concentration.
 (1) A (2) B (3) C (4) D
- Assertion** : In a reversible endothermic reaction activation energy of the forward reaction is higher than that of the backward reaction.
Reason : The threshold energy of the forward reaction is more than that of the backward reaction.
 (1) A (2) B (3) C (4) D
- Assertion** : The reaction having low value of activation energy are generally fast.
Reason : Temperature coefficient for reaction having low activation energy is large.
 (1) A (2) B (3) C (4) D
- Assertion** :- Two different reaction can never have same rate of reaction
Reason :- Rate of reaction always depends only on frequency of collision and Arrhenious factor
 (1) A (2) B (3) C (4) D
- Assertion** :- Rate constant of reaction depends on concentration of reactant.
Reason :- On changing concentration, rate of reaction always increases.
 (1) A (2) B (3) C (4) D

14. Assertion :- On increasing the temperature by 10°C , rate constant almost doubles.

Reason : On increasing temperature by 10°C , collision frequency doubles.

(1) A (2) B (3) C (4) D

15. Assertion :- On increasing temperature rate of reaction always increases.

Reason :- For any reaction equilibrium always shifts in forward direction on increasing temperature.

(1) A (2) B (3) C (4) D

16. Assertion : Catalyst increases rate of reversible reaction.

Reason : It increase the E_a for forward reaction but decreases the activation energy for backward reaction.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	1	2	4	3	3	3	4	3	3	4	4	3	3
Que.	16														
Ans.	3														

IMPORTANT NOTES

SOLUTIONS

E

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NEET SYLLABUS

Solutions : Types of solutions, expression of concentration of solutions of solids in liquids, solubility of gases in liquids, solid solutions, colligative properties- relative lowering of vapour pressure, Raoult's law, elevation of boiling point, depression of freezing point, osmotic pressure, determination of molecular mass using colligative properties abnormal molecular mass, Van Hoff factor.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- describe the formation of different types of solutions.
- express concentration of solution in different units.
- state and explain Henry's law and Raoult's law.
- distinguish between ideal and non-ideal solutions.
- explain deviations of real solutions from Raoult's law.
- describe colligative properties of solutions and correlate with molar masses of solutes.
- explain abnormal colligative properties exhibited by some solutes in solutions.

*"Imagination equals nostalgia for the past, the absent;
it is the liquid solution in which science develops as snapshot of reality"*

Cyril Connoly

SOLUTIONS

3.0 INTRODUCTION

3.1 Definition of Solution :

When two or more chemically non-reacting substances are mixed together forming homogeneous mixture the mixture is called solution.

When the solution is composed of only two chemical substances, it is termed as binary solution, similarly, it is called ternary and quaternary if it is composed by three and four components respectively.

For binary solution : Solution = solute + solvent

- Generally the component present in lesser amount than, other component in solution, is called solute.
- Generally, the component present in greater amount than all other components, is called the solvent.
- Physical state of solvent and solution is same.

Solution	Solute + Solvent	
	(B)	(A)
moles	n	N
mass	$w_{(g)}$	$W_{(g)}$
molar mass	m	M
mole fraction	x_B	x_A

Ex.1 In a syrup (liquid solution) containing 60 g sugar (a solid) and 40 g water (a liquid) same aggregation as solution water is termed as the solvent.

Ex.2 : In a solution of alcohol and water ; having 10 mL alcohol and 20 mL water, water is solvent and alcohol will be solute.

- On the basis of amount of solute, solutions can be classified in two ways.

(a) Dilute Solution

A solution in which relatively a small amount of solute is dissolved in large amount of solvent is called a dilute solution.

(b) Concentrated Solution

A solution in which relatively a large amount of the solute is present is called a concentrated solution.

3.2 CONCENTRATION TERMS

(a) Normality (N)

The *number of equivalents* or *gram equivalents* of solute present in *one litre* of the solution is known as normality (N) of the solution.

$$= \frac{\text{Number of gram equivalents of solute (g)}}{\text{Volume of solution (L)}}$$

$$= \frac{\text{Mass of solute (g)}}{\text{Equivalent mass} \times \text{Volume of solution (L)}}$$

(b) Molarity (M)

The *number of moles* of solute present in *one litre* solution is called its molarity(M).

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{volume of solution(L)}} = \frac{n}{V_{(L)}}$$

(c) Formality (F)

It is the number of *gram formula mass units* present per litre of solution. Formality is generally used for solutions of ionic compounds.

$$\text{Formality} = \frac{\text{Mass of solute (g)}}{\text{Formula mass of solute (g)} \times \text{Volume of solution (L)}}$$

(d) Molality (m)

The *number of moles* of solute present in *1000 gram* of the solvent is called molality of the solution.

$$\text{Molality of a solution} = \frac{\text{Number of moles of solute}}{\text{Amount of solvent (kg)}} = \frac{\text{Number of moles of solute} \times 1000}{\text{Amount of solvent (g)}}$$

(e) Strength of a Solution (S)

The mass of solute in g dissolved in 1L solution is known as its strength in g L^{-1}

$$S = \frac{\text{Mass of solute (g)}}{\text{Volume of solution (L)}}$$

$$S(\text{g L}^{-1}) = \text{Molarity of solution} \times \text{Molar mass of solute}$$

$$S(\text{g L}^{-1}) = \text{Normality of solution} \times \text{Equivalent mass of solute}$$

(f) Concentration in terms of percentage**(i) Percent By Mass (w/W)**

Mass of solute (in g) present in 100 g of solution (g) is called mass percent of the solute.

Where 'w' gram of solute is dissolved in W gram of solvent.

$$\text{Mass percent} = \frac{\text{Mass of solute (g)} \times 100}{\text{Mass of solution (g)}} = \frac{w \times 100}{w + W}$$

Mass percent is independent of temperature.

(ii) Percent By Volume (v/V)

This method is used for solutions of liquid in a liquid. The volume of liquid (solute) in mL present in 100 mL of solution is called volume percent.

$$\text{Volume percent} = \frac{\text{Volume of solute} \times 100}{\text{Volume of solution}}$$

(iii) Percent by strength / percentage mass by volume $\left(\frac{w}{V}\right)$: Mass of solute (in g) present in 100 mL solution is called mass by volume.

$$\% \left(\frac{w}{V} \right) = \frac{\text{mass of solute (g)}}{\text{volume of solution (mL)}} \times 100$$

(g) Parts Per Million (ppm)

This method is used for expressing the concentration of very dilute solutions such as hardness of water, air pollution etc.

$$\text{ppm of substance (by mass)} = \frac{\text{Mass of solute (g)} \times 10^6}{\text{Mass of solution (g)}}$$

$$\text{ppm (by volume)} = \frac{\text{Volume of solute (mL)} \times 10^6}{\text{Volume of solution (mL)}}$$

$$\text{ppm} \left(\text{by } \frac{w}{V} \right) = \frac{\text{mass of solute (g)}}{\text{volume of solution (mL)}} \times 10^6$$

(h) **Mole Fraction**

The ratio of the number of moles of one component to the total number of moles of all the components present in the solution, is called the mole fraction of that component.

$$\text{Mole fraction of solute } X_B = \frac{\text{moles of solute (n)}}{\text{moles of solute (n) + moles of solvent (N)}}$$

$$\text{Mole fraction of solvent } X_A = \frac{\text{moles of solvent (N)}}{\text{moles of solute (n) + moles of solvent (N)}}$$

$$\boxed{X_A + X_B = 1}$$

(i) (i) **Relation Between Molarity and Normality**

$S = \text{Molarity} \times \text{Molar mass of solute}$ and $S = \text{Normality} \times \text{Equivalent mass of solute}$.

So we can write

$\text{Molarity} \times \text{Molar mass of solute} = \text{Normality} \times \text{Equivalent mass of solute}$.

$$\begin{aligned} \text{Normality} &= \frac{\text{Molarity} \times \text{Molar mass of solute}}{\text{Equivalent mass of solute}} \\ &= \frac{\text{Molarity} \times \text{Molar mass of solute}}{(\text{Molar mass of solute} / \text{valency factor})} \end{aligned}$$

$$\text{Normality} = \text{Molarity} \times \text{Valency factor}$$

$$N = M \times n \text{ factor}$$

(ii) **Relation Between Molality (m) and Mole fraction**

$$\frac{x_B}{x_A} = \frac{n_B}{N_A} = \frac{n_B \times M_A \times 1000}{w_A \times 1000}$$

$$\frac{x_B}{x_A} = \frac{\text{molality} \times M_A}{1000}$$

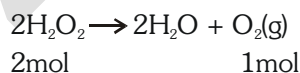
$$\frac{x_B}{1 - x_B} = \frac{\text{molality} \times M_A}{1000} : \text{if } x_B \ll 1 \text{ for very dilute } x_B = \frac{\text{molality} \times M_A}{1000}$$

(j) **VOLUME STRENGTH OF H₂O₂ SOLUTION :**

Volume of O₂ (L) produced by decomposition of unit volume solution of H₂O₂ at NTP/STP; is called its volume strength.

Relationship between volume strength of H₂O₂ and Molarity / Normality / (g/L) / % (w/V)

Let volume strength of H₂O₂ solution be 'X V', it means 1L solution of H₂O₂, can produce X L of O₂ at NTP.



$\therefore$ 1 mol O₂ can be produced by = 2 mol H₂O₂ solution

$$\therefore \frac{X}{22.4} \text{ mol of O}_2 \text{ can be produced by} = \frac{2X}{22.4} \text{ mol H}_2\text{O}_2 \text{ solution}$$

$$= \frac{X}{11.2} \text{ mol of H}_2\text{O}_2 \text{ solution}$$

$$(a) \quad \text{Molarity} = \frac{\text{number of moles of solute}}{\text{volume of solution(L)}} = \frac{2X}{22.4} = \frac{X}{11.2}$$

$$\text{Hence } M = \frac{X}{11.2}$$

$$(b) \quad \text{Normality} = \text{molarity} \times \text{valence factor} = \frac{X}{11.2} \times 2 = \frac{X}{5.6}$$

$$\text{Hence } N = \frac{X}{5.6}$$

$$(c) \quad \text{Strength (g/L)} : S = N \times E$$

$$\text{Hence } S = \frac{X}{5.6} \times 17 \text{ gL}^{-1}$$

$$(d) \quad \% \left(\frac{w}{V} \right) = \frac{X}{5.6} \times \frac{17}{10}$$

GOLDEN KEY POINTS

- If the density of solution is approximately 1g/cc then Molality > Molarity
- ppm unit is used to represent concentration of very dilute solutions like pollutants in environment and salts present in sea water.
- Those concentration terms which involves volume of solution are temperature dependent.
- Molarity, normality, formality, % by volume, % w/V, are temperature dependent.
- Molality, % w/W, mole fraction are temperature independent.

Illustrations

Illustration 1. If 0.4 g of NaOH is present in 40 mL of solution. What is the molarity and normality of solution [Molecular mass of NaOH = 40]

Solution. We know that

$$\text{Molarity} = \frac{\text{Mass of solute} \times 1000}{\text{Molar Mass of solute} \times \text{Volume of solution (mL)}}$$

$$M = \frac{0.4}{40 \times 40} \times 1000 = 0.25 \text{ M}$$

$$\text{Normality} = \frac{\text{Mass of solute}}{\text{Equivalent mass of solute} \times \text{volume of solution (mL)}} \times 1000$$

Equivalent mass of NaOH = 40

$$N = \frac{0.4}{40 \times 40} \times 1000 = 0.25 \text{ N}$$

Illustration 2. The normality of 1.5M H_3PO_4 is –

Solution. Basicity of H_3PO_4 is 3

We know that $N = M \times n \Rightarrow \text{Normality} = 1.5 \times 3 = 4.5 \text{ N}$

Illustraion 3. Find out the mass of H_2SO_4 in 150 mL, $\frac{N}{7}$ H_2SO_4 .

Solution.
$$N = \frac{\text{Mass in gram}}{\text{Equivalent mass} \times \text{Volume(L)}}$$

$$\text{Mass in gram} = \text{Equivalent mass} \times N \times \text{Volume (L)} = 49 \times \frac{1}{7} \times \frac{150}{1000} = \frac{21}{20} = 1.05\text{g}$$

Illustraion 4. Find out the molarity of 93% (w/W) H_2SO_4 (density = 1.84 g/ml).

Solution.
$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution(L)}} = \frac{\text{Mass in gram} \times \text{Density} \times 1000}{\text{Molar mass} \times \text{mass of solution (g)}}$$

$$= \frac{93 \times 1.84 \times 1000}{98 \times 100} = 17.46 \text{ M}$$

Illustraion 5. A 100 cm^3 solution is prepared by dissolving 2g of NaOH in water. Calculate the normality of the solution.

Solution.
$$2\text{g NaOH} = \frac{2}{40} \text{ g eq}; \quad N = \frac{1}{\frac{20}{100}} \times 1000 = \frac{1}{2}$$

$$\text{Normality of solution} = \frac{N}{2}$$

Illustraion 6. Find the percentage by mass and mass fraction of aspirin in the solution prepared by dissolving 3.65 g of aspirin in 25.08 g of water .

Solution. Mass of solution = 3.65 + 25.08 = 28.73 g

$$\text{Mass fraction} = \frac{3.65}{28.73} = 0.127$$

$$\text{Mass percent} = 0.127 \times 100 = 12.7\%$$

Illustraion 7. A solution was prepared by adding 125 cm^3 of isopropyl alcohol to water until the volume of the solution was 175 cm^3 . Find the volume fraction and volume percent of isopropyl alcohol in the solution.

Solution. Volume of solute = 125 cm^3
 Volume of solution = 175 cm^3

$$\therefore \text{Volume fraction} = \frac{125}{175} = 0.714$$

$$\text{Volume percent} = \frac{125}{175} \times 100 = 71.4\%$$

Illustraion 8. Calculate volume strength of H_2O_2 in 5L solution which yields 100 L of O_2 at NTP.

Solution 5L H_2O_2 gives = 100 L O_2
 1L H_2O_2 gives = 20 L O_2
 i.e. 20 V H_2O_2 solution

BEGINNER'S BOX-1

- What is the normality of 2M H_3PO_2 solution?
 (1) 0.5 N (2) 1.0N (3) 2.0 N (4) 3.0 N
- 23 g ethanol is dissolved in 36 g water. Find mole fraction of ethanol?
 (1) 2 (2) 0.5 (3) 0.2 (4) 0.8
- How many gram of HNO_3 is required to prepare 400 mL solution of 0.2 M HNO_3 ?
 (1) 5.04 g (2) 5040 g (3) 25.2 g (4) 2.52 g

4. Calculate normality of 2.1% (w/V) H_2SO_4 solution?
 (1) 2.14 N (2) 4.28 N (3) 0.428 N (4) 0.214 N
5. What is the molarity of 1N H_2SO_4 solution?
 (1) 1 M (2) 2M (3) 0.5 M (4) 3M
6. 20.6 g NaBr is dissolved in 500 mL solution what is the molarity of resulting solution?
 (1) 0.6 (2) 0.4 (3) 1 (4) None
7. Calculate molality of the solution obtained by dissolving 11.7 g NaCl in 500 g water
 (1) 0.1 m (2) 0.3 m (3) 0.2m (4) 0.4m
8. Density of 2.03 M aqueous solution of acetic acid is 1.017 g mL^{-1} molecular mass of acetic acid is 60. Calculate the molality of solution?
 (1) 2.27 (2) 1.27 (3) 3.27 (4) 4.27
9. A molar solution is one that contains one mole of solute in
 (1) 1000 g of the solvent (2) one litre of the solution
 (3) 1000 g of the solution (4) 22.4 litres of the solution
10. Calculate the mole percentage of CH_3OH and H_2O respectively in 60% (by mass) aqueous solution of CH_3OH
 (1) 45.8, 54.2 (2) 54.2, 45.8 (3) 50, 50 (4) 60, 40

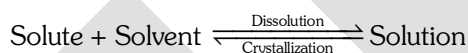
3.3 SOLUBILITY

Maximum amount of solute which can be dissolved in a specified amount of solvent at constant temperature is solubility. Solubility is affected by nature of solute and solvent as well as by temperature and pressure.

(a) Solubility of Solid in Liquid

Polar solutes are soluble in polar solvent and non polar solutes are soluble in non polar solvent due to similar intermolecular forces.

When solid solutes are dissolved in solvent then following equilibrium exists.



Solubility of solids is affected by temperature and pressure according to Le-chatlier's principle. If dissolution is exothermic then solubility decreases with increase in temperature and if endothermic then solubility increases with increase in temperature.

Solubility of solids is not affected by pressure significantly since solids are highly incompressible.

(b) Solubility of Gases in Liquid

Certain gases are highly soluble in water like NH_3 , HCl etc. and certain gases are less soluble in water like O_2 , N_2 , He etc. Solubility of gases is affected by pressure and temperature. Increasing pressure increases solubility and increase in temperature decreases solubility. During dissolution of gas pressure of gas decrease and dissolution of gas is exothermic in nature.

HENRY'S LAW

It can be stated as at constant temperature the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas present above the surface of liquid or solution.

or

Mole fraction of gas in the solution is proportional to the partial pressure of the gas over the solution.

or

The partial pressure of the gas in vapour phase (P) is proportional to the mole fraction of the gas (X) in the solution. (This is most common definition)

$$P = K_H X \quad K_H = \text{Henry's Constant}$$

Henry's Constant is not a universal constant. It depends on nature of gas and temperature. K_H increases with increase in temperature therefore solubility of gas decreases.

Illustrations

Illustration 9. Henry's Law Constant for CO_2 in water is 1.67×10^8 Pa at 298K. Calculate the quantity of CO_2 in 1 L of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

Solution.

$$P = K_H X_{\text{gas}}$$

$$X_{\text{gas}} = \frac{P}{K_H} = \frac{2.5 \times 10^5}{1.67 \times 10^8} = 1.5 \times 10^{-3}$$

$$\frac{n}{n+N} = X$$

$$\frac{n}{N} = X \quad \because n \ll N \therefore n+N \approx N$$

$$\frac{n}{55.55} = 1.5 \times 10^{-3} = 8.3 \times 10^{-2} \text{ mol L}^{-1}$$

$$= 3.65 \text{ g L}^{-1}$$

3.4 VAPOUR PRESSURE

At a constant temperature, the pressure exerted by the vapours of a liquid on its surface when they (liquid and its vapours) are in equilibrium, is known as vapour pressure.

Factor affecting vapour pressure :

- (a) Nature of liquid (b) Temperature

Raoult's Law

(a) **For liquid-liquid system :** For a solution of volatile liquids the partial vapour pressure of any component at constant temperature is equal to vapour pressure of pure component multiplied by mole fraction of that component in the solution.

	Liquid (B)	Liquid (A)
Vapour pressure in pure state	P_B^0	P_A^0
Partial vapour pressure	P_B	P_A
Mole fraction in solution	X_B	X_A
Moles	n moles	N moles
Mass	w g	W g
Molar mass	m	M

$$P_A \propto X_A \text{ so } P_A = P_A^0 X_A \dots (i)$$

$$P_B \propto X_B \text{ so } P_B = P_B^0 X_B \dots (ii)$$

At constant temperature partial vapour pressure of component is directly proportional to mole fraction of component in solution.

According to Dalton's law given below :

$$P_{\text{total}} = P_A + P_B + \dots$$

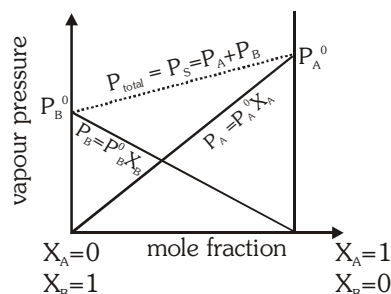
$$P_S = X_A P_A^0 + X_B P_B^0 ; X_A + X_B = 1$$

$$P_S = (1-X_B) P_A^0 + X_B P_B^0$$

$$P_S = P_A^0 - X_B P_A^0 + X_B P_B^0$$

$$P_S = P_A^0 + X_B (P_B^0 - P_A^0)$$

$$\begin{array}{cccc} P_S = P_A^0 + X_B (P_B^0 - P_A^0) \\ \downarrow & \downarrow & \downarrow & \downarrow \\ y & c & x & m \end{array}$$



slope can be positive or negative.

when $P_B^0 > P_A^0$ when $P_A^0 > P_B^0$

• Dalton's Law

Partial pressure of gas = mole fraction $\times$ total pressure of gas

$$P_A = Y_A P_T$$

$$P_B = Y_B P_T$$

$$P_A + P_B = P_T$$

$$\begin{aligned} X_A P_A^0 &= Y_A P_T \\ X_B P_B^0 &= Y_B P_T \end{aligned}$$

 Y_A and Y_B gives mole fraction in vapour phase X_A and X_B gives mole fraction in liquid phase.

Illustrations

Illustration 10. Methanol and C_2H_5OH form an ideal solution. Solution is prepared by mixing 32 g of CH_3OH and 23 g of C_2H_5OH at 300K. At 300K $P_{\text{methanol}}^0 = 90$ mm of Hg and $P_{\text{ethanol}}^0 = 51$ mm of Hg. Calculate

- (i) partial vapour pressure of its constituents and total vapour pressure of solution.
 (ii) Mole fraction of each component in vapour phase.

Solution

$$(i) n_{CH_3OH} = \frac{32}{32} = 1; n_{C_2H_5OH} = \frac{23}{46} = 0.5$$

$$X_{CH_3OH} = \frac{1}{1.5} = \frac{2}{3}; X_{C_2H_5OH} = \frac{0.5}{1.5} = \frac{1}{3}$$

$$p_{CH_3OH} = p_m^0 \cdot X_m = 90 \times \frac{2}{3}$$

$$P_{CH_3OH} = 60 \text{ mm of Hg}$$

$$p_{C_2H_5OH} = p_e^0 \cdot X_e = 51 \times \frac{1}{3}$$

$$P_{C_2H_5OH} = 17 \text{ mm of Hg}$$

$$P_S = p_{CH_3OH} + p_{C_2H_5OH}$$

$$P_S = 77 \text{ mm of Hg}$$

$$(ii) 60 = Y_A \times 77 \quad 17 = Y_B \times 77$$

$$Y_{CH_3OH} = \frac{60}{77} = 0.78; Y_{C_2H_5OH} = 0.22$$

(b) For solid - liquid system : non-volatile solute (B) and volatile solvent (A)

$$P_B^0 = 0$$

$$P_S = X_A P_A^0 + X_B P_B^0$$

$$P_S = X_A P_A^0 = \frac{N}{n+N} P_A^0 \quad \dots(i)$$

For solution of non-volatile solute : At constant temperature vapour pressure of solution containing non-volatile solute is proportional to mole fraction of solvent.

$$\therefore P_S \propto \frac{N}{n+N}; X_A + X_B = 1$$

$$P_S = (1 - X_B) P_A^0; P_S = P_A^0 - X_B P_A^0$$

$$X_B P_A^0 = P_A^0 - P_S \Rightarrow \frac{P_A^0 - P_S}{P_A^0} = X_B = \frac{n}{n+N} \quad \dots(ii)$$

- When a non-volatile solute is added to a volatile liquid its vapour pressure decrease because less number of solvent particles present in solution at surface. (as compare to pure solvent)

∴ less vapour is formed and vapour pressure of solution decreases

$$P_A^0 - P_S = \Delta P \quad (\text{lowering of vapour pressure})$$

$$\frac{P_A^0 - P_S}{P_A^0} = \text{relative lowering of vapour pressure.}$$

$$\frac{P_A^0}{P_A^0 - P_S} = \frac{n + N}{n} \Rightarrow \frac{P_A^0}{P_A^0 - P_S} = 1 + \frac{N}{n}$$

$$\Rightarrow \frac{P_A^0 - P_A^0 + P_S}{P_A^0 - P_S} = \frac{N}{n} \Rightarrow \frac{P_S}{P_A^0 - P_S} = \frac{N}{n}$$

$$\Rightarrow \boxed{\frac{P_A^0 - P_S}{P_S} = \frac{n}{N}} \quad \dots(iii)$$

$$\Rightarrow \Delta P \propto \frac{n}{N}$$

Ex. The vapour pressure of pure liquid A is 40 torr at 310 K. The vapour pressure of this liquid in solution with solid B is 32 torr. Calculate X_A in solution.

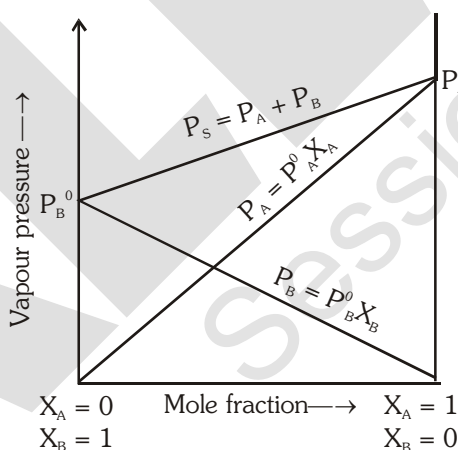
Sol. $32 = X_A \times 40$

$$X_A = 0.8$$

3.5 IDEAL SOLUTIONS (mixture of two liquids A and B)

- A solution which obeys Raoult's law exactly at all concentrations and at all temperatures is called an ideal solution.
- For ideal solutions; A-A interactions = B-B interactions = A-B interactions.

An ideal solution possesses the following characteristics :



- Volume change on mixing should be zero. $\Delta V_{\text{mix}} = 0$, i.e., $(V_{\text{solute}} + V_{\text{solvent}} = V_{\text{solution}})$
- Heat change on mixing should be zero. $\Delta H_{\text{mix}} = 0$ (Heat is neither absorbed nor evolved)
- There should be no chemical reaction between liquid A and liquid B.
- Ideal solution must obey Raoult's law at all concentrations.

$$P_A = P_A^0 \cdot X_A, \quad P_B = P_B^0 \cdot X_B$$

- observed VP = calculated VP
- observed BP = calculated BP
- $\Delta S_{\text{mix}} > 0$
- $\Delta G < 0$

Example

(i) Benzene and toluene

(ii) CCl_4 and SiCl_4

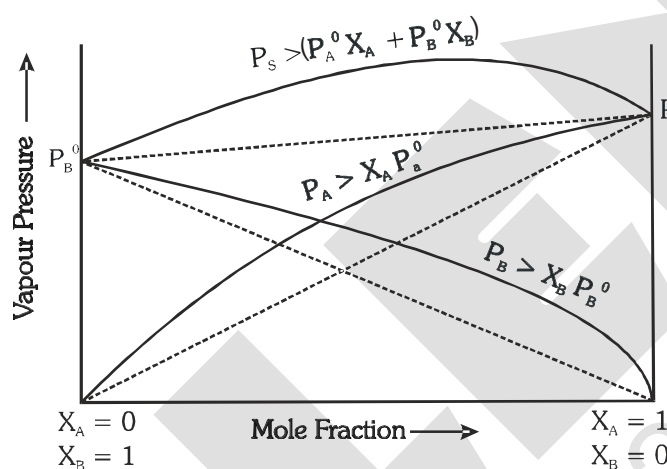
(iii) n-hexane and n-heptane

(iv) $\text{C}_2\text{H}_5\text{Br}$ and $\text{C}_2\text{H}_5\text{Cl}$ (v) PhCl and PhBr

(vi) n-butylchloride and n-butylbromide

3.6 NON-IDEAL SOLUTIONS

- For non ideal solutions ; A-A interactions or B-B interactions $\neq$ A-B interactions.
- Those solutions which do not obey Raoult's law are called non-ideal solutions.
- For such solutions ; $P_A \neq P_A^\circ X_A$; $P_B \neq P_B^\circ X_B$
- Non ideal solutions are formed when the components differ much in their structures and polarities.
- So $\Delta H_{\text{mixing}} \neq 0$ and $\Delta V_{\text{mixing}} \neq 0$
- Non ideal solutions show either positive or negative deviations from Raoult's law.

(a) Non ideal solutions having positive deviation from Raoult's law.

- In these solutions A-B interactions are less than A-A and B-B molecular interactions.
- $P_A > P_A^\circ X_A$
 $P_B > P_B^\circ X_B$
- The total vapour pressure of the solution will be greater than the corresponding vapour pressure expected in case of an ideal solution of same composition. i.e.
- $P_{\text{total}} > (P_A^\circ X_A + P_B^\circ X_B)$
- $\Delta H_{\text{mix}} > 0$; endothermic dissolution ; heat is absorbed.
- $\Delta V_{\text{mix}} > 0$; volume increases after dissolution, i.e., $(V_{\text{solute}} + V_{\text{solvent}} < V_{\text{solution}})$.
- 'A' and 'B' escape easily showing higher vapour pressure than the expected value.
- $(\text{B.P.})_{\text{th}} > (\text{B.P.})_{\text{exp}}$
- $(\Delta S)_{\text{mix}} = +ve$, $\Delta G = -ve$
Entropy change in mixing is positive.

Examples : (i) Ethanol and cyclohexane(v) CCl_4 and Benzene

(ix) Benzene and Acetone

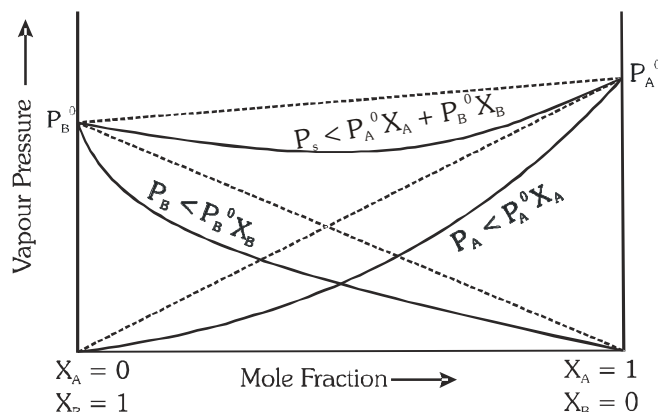
(ii) Ethanol and Water

(vi) CCl_4 and Toluene(x) CS_2 and Acetone

(iii) Ethanol and Acetone

(vii) CCl_4 and CHCl_3 (xi) CS_2 and Acetaldehyde(iv) Methanol and H_2O (viii) CCl_4 and Methanol

(b) Non ideal solutions having negative deviation from Raoult's Law



- In these solutions the A-B interactions are stronger than the A-A and B-B molecular interactions.

$$P_A < P_A^0 X_A \quad ; \quad P_B < P_B^0 X_B$$

- $P_{\text{total}} < (P_A^0 X_A + P_B^0 X_B)$

Total vapour pressure is less than sum of individual vapour pressure.

- $\Delta H_{\text{mix}} < 0$; exothermic dissolution heat is evolved.
- $\Delta V_{\text{mix}} < 0$; volume decreases during dissolution, i.e., $(V_{\text{solvent}} + V_{\text{solute}} > V_{\text{solution}})$.
- Escaping tendency of both components 'A' and 'B' is lowered showing lower vapour pressure than expected ideally.

Examples	(i) CHCl_3 and CH_3COCH_3	(vi) H_2O and HCl
	(ii) CHCl_3 and C_6H_6	(vii) H_2O and HNO_3
	(iii) CHCl_3 and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	(viii) CH_3COOH and CH_3OH
	(iv) CHCl_3 and HNO_3	(ix) CH_3COOH and $\text{C}_5\text{H}_5\text{N}$
	(v) CHCl_3 and CH_3COOH	(x) CH_3COCH_3 and Aniline

AZEOTROPIC MIXTURES :

Some liquids on mixing, form azeotropes which are binary mixtures having the same composition in liquid and vapour phase and boil at a constant temperature, the liquid and vapour have the same composition, and no further separation occurs.

Components form azeotrope can't be separated by fractional distillation but can be separated by azeotropic distillation.

Solutions showing Positive deviation form minimum boiling azeotrope and solutions showing negative deviation form maximum boiling azeotrope.

There are two types of azeotropes :

- (a) Minimum boiling azeotrope (b) Maximum boiling azeotrope

- The solutions which show a large positive deviation from Raoult's law form minimum boiling azeotrope at a specific composition. For example, ethanol-water mixture (obtained by fermentation of sugars) on fractional distillation gives a solution containing approximately 95.57% v/v ethanol.
- The solutions that show large negative deviation from Raoult's law form maximum boiling azeotrope at a specific composition. Nitric acid and water is an example of this class of azeotrope. This azeotrope has the approximate composition, 68% nitric acid and 32% water by mass, with a boiling point of 393.5 K.

GOLDEN KEY POINTS

- Those gases which can react with solvent do not follow Henry's Law. e.g. NH_3 in water.
- Raoult's Law is special case of Henry's Law.
- ΔS is positive and ΔG is negative for ideal as well as non ideal solutions.
- The vapour phase is richer in more volatile component than the less volatile component. This is called as Konowaloff's rule.

Illustrations

Illustration 11. 1 mole heptane (V.P. = 92 mm of Hg) is mixed with 4 mole. Octane (V.P. = 31 mm of Hg), form an ideal solution. Find out the vapour pressure of solution.

Solution.

Total mole = 1 + 4 = 5

Mole fraction of heptane $X_A = 1/5$

Mole fraction of octane $X_B = 4/5$

$$P_s = X_A P_A^0 + X_B P_B^0 = \frac{1}{5} \times 92 + \frac{4}{5} \times 31 = 43.2 \text{ mm of Hg.}$$

Illustration 12. At 88°C benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will be boil at 88°C at 1 atm pressure (benzene – toluene form an ideal solution).

Solution.

$P_s = 760$ torr, because solution boils at 88°C

$$P_s = P_B^0 X_B + P_t^0 X_t$$

$$\therefore 760 = 900 a + 360 - 360 a$$

$$a = 0.74 \text{ where 'a' is mole fraction of } \text{C}_6\text{H}_6 (X_B)$$

BEGINNER'S BOX-2

- Which of the gas will not follow Henry's law?
(1) HCl (2) He (3) O_2 (4) H_2
- If solubility of gas 'X' is 0.5 gL^{-1} at 1 bar then its solubility at 3 bar pressure will be
(1) 0.5 gL^{-1} (2) 1.5 gL^{-1} (3) 3.0 gL^{-1} (4) 2 gL^{-1}
- Among the following that forms an ideal solution?
(1) water and methanol (2) acetone and ethanol (3) benzene and toluene (4) water and HCl
- On mixing 10 mL of acetone with 40 mL of chloroform the total volume of the solution is
(1) $< 50 \text{ mL}$ (2) $> 50 \text{ mL}$ (3) $= 50 \text{ mL}$ (4) cannot be predicted
- The mixture of n-hexane and n-heptane is an example of
(1) ideal solution (2) non-ideal solution (3) dilute solution (4) none

3.7 COLLIGATIVE PROPERTIES OF DILUTE SOLUTIONS (CP)

Those physical properties of a solution which depends upon the relative number of particles of solute and do not depend on nature of solute particles are called colligative properties.

- CP $\propto$ Number of solute particles
 $\propto$ Number of molecules (in the solution of non electrolyte)
 $\propto$ Number of ions (in the solution of electrolytes)
 $\propto$ Number of moles of solute
 $\propto$ Mole fraction of solute

The following four properties are CP :

- Relative lowering in vapour pressure of solvent
- Elevation in boiling point (ΔT_b) of solvent
- Depression in freezing point (ΔT_f) of solvent
- Osmotic pressure (π or P) of solution

(a) Relative Lowering in Vapour Pressure

- When a nonvolatile solute is dissolved in a pure solvent, the vapour pressure of the solvent is lowered i.e. the vapour pressure of a solution is always lower than that of pure solvent, because the escaping tendency of solvent molecules decreases (due to lesser solvent molecules per unit surface area).

If at a certain temperature P^0 is the vapour pressure of pure solvent, and P_s is the vapour pressure of solution then

According to Raoult's law :

$$\text{Relating lowering in vapour pressure} = \frac{P_A^0 - P_s}{P_A^0} = \frac{n_B}{N_A + n_B}$$

$$\text{For a dilute solution } n_B \ll n_A \quad \therefore \frac{P_A^0 - P_s}{P_A^0} \approx \frac{n_B}{N_A} \text{ so } \Delta P \propto \frac{n_B}{N_A}$$

- Relative lowering depends upon relative number of solute particles. Therefore it is called colligative properties.

(b) Elevation in Boiling Point

- The boiling point of a liquid is that temperature at which its vapour pressure becomes equal to the atmospheric pressure.
- When a non-volatile solute is dissolved in a pure solvent, its vapour pressure is decreased and boiling point increases. The difference of boiling points of the solution and pure solvent is called elevation in boiling point (ΔT_b)
- If T_b^0 is the boiling point of pure solvent and T_b is the boiling point of the solution then, $T_b > T_b^0$ and the elevation in boiling point $\Delta T_b = T_b - T_b^0$
- The elevation in boiling point (ΔT_b) is directly proportional to lowering of vapour pressure of the solution i.e.

$$\Delta T_b \propto (P^0 - P_s) \text{ from graph and}$$

$$\Delta T_b \propto \Delta P \propto \frac{n_B}{N_A}$$

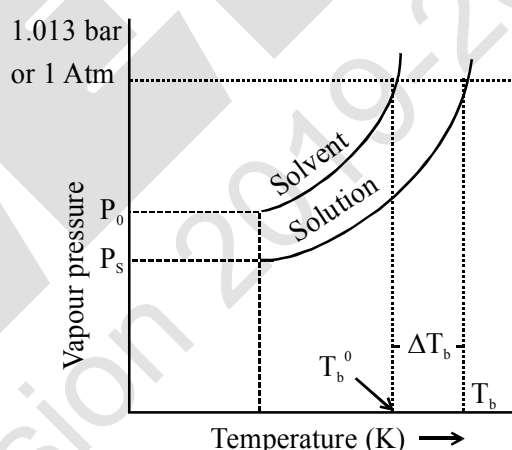
$$\therefore \Delta T_b \propto \frac{n_B}{N_A} = \frac{w_B M_A}{m_B W_A} \text{ for a solvent } P^0 \text{ \& } M_A = \text{constant}$$

$$\therefore \Delta T_b \propto \frac{w_B}{m_B W_A} \text{ or } \Delta T_b = \frac{K w_B}{m_B W_A}$$

where K = elevation constant

$$\text{if } \frac{w_B}{m_B} = 1 \text{ mole and } W_A = 1 \text{ g}$$

then $\Delta T_b = K$ (Elevation constant or molecular elevation constant)



T_b = Boiling point of solution

T_b^0 = Boiling point of solution

The vapour pressure curve for solution lies below the curve for pure water. The diagram shows that ΔT_b denotes the elevation of boiling point of a solvent in solution.

if $\frac{w_B}{m_B} = 1$ and $W_A = 1000$ gram; Then $\Delta T_b = K_b$ (molal elevation constant)

$$\therefore \frac{K}{1000} = K_b \text{ (molal elevation constant or Ebullioscopic constant)}$$

$$\Delta T_b = \frac{K_b \times w_B \times 1000}{m_B \times W_A}; \quad \Delta T_b = \frac{w_B}{m_B} \times \frac{1000}{W_A} \times K_b$$

$$\therefore \boxed{\Delta T_b = \text{molality} \times K_b} \quad \therefore \Delta T_b \propto \text{molality}$$

hence elevation in boiling point (ΔT_b) is a colligative property.

- K_b depends only on nature of solvent which can be explained by thermodynamic relation.

$$\boxed{K_b = \frac{RT_b^{02}}{1000L_v} = \frac{M_w RT_b^{02}}{1000\Delta H_{vap}}}$$

Where T_b^0 = Boiling point of solvent.

M_w = Molar mass of solvent.

ΔH_{vap} = Enthalpy of vapourisation per mole of solvent

L_v = Latent heat of vapourisation per gram of solvent

- The molal elevation constant for some common solvents are given in the following table**

Solvent	B.P. ($^{\circ}\text{C}$)	Molal elevation constant (K_b)
Water	100.0	0.52
Acetone	56.0	1.70
Chloroform	61.2	3.63
Carbon tetra chloride	76.8	5.03
Benzene	80.0	2.53
Ethyl alcohol	78.4	1.20

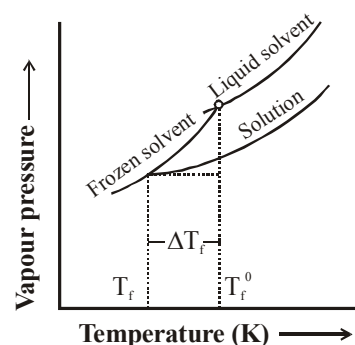
(c) Depression in Freezing Point

- The freezing point of a liquid is that temperature at which the liquid and its solid state exist in equilibrium with each other.
- It may be defined as the temperature at which the liquid and solid states of a substance have the same vapour pressure.
- When a non-volatile is dissolved in a pure solvent the vapour pressure of the solvent is lowered.
- If T_f^0 is the freezing point of pure solvent and T_f is the freezing point of its solution then, $T_f < T_f^0$
- The difference in the freezing point of pure solvent and solution is called depression of freezing point (ΔT_f)

$\Delta T_f = T_f^0 - T_f$; the depression in freezing point is directly proportional to lowering in vapour pressure (ΔP)

$$\Delta T_f \propto \Delta P \propto \frac{n_B}{N_A}; \quad \Delta T_f \propto \frac{n_B}{N_A} \text{ so } \boxed{\Delta T_f = K_f \times \text{molality}}$$

K_f = molal depression constant or Cryoscopic constant.



- K_f depends only on nature of solvent which can be explained by thermodynamic relation

$$K_f = \frac{RT_f^{02}}{1000L_f} = \frac{RT_f^{02}M_w}{1000\Delta H_f}$$

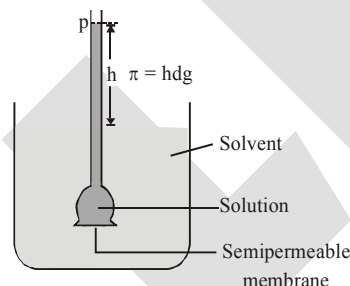
Where T_f^0 = Freezing point of solvent
 M_w = Molar mass of solvent
 ΔH_f = Enthalpy of fusion per mole of solvent
 L_f = latent heat of fusion per gram of solvent

- The molal depression constant for some common solvents are given in the following table

Solvent	F.P.(°C)	Molal depression constant (K_f)
Water	0.0	1.86
Ethyl alcohol	- 114.6	1.99
Chloroform	- 63.5	4.79
Carbon tetra chloride	- 22.8	31.80
Benzene	5.5	5.12
Camphor	179.0	39.70

(d) **Osmosis and Osmotic Pressure**

Osmosis : Osmosis is defined as the spontaneous net flow of solvent molecules through semipermeable membrane from a solvent to a solution or from a dilute solution to a concentrated solution.



Level of solution rises in the thistle funnel due to osmosis of solvent

Osmotic Pressure (p or π)

- The external pressure which must be applied on the solution in order to stop the flow of solvent into the solution through semipermeable membrane is equal to osmotic pressure
- or
- Hydrostatic pressure developed in a vertical column when solution and solvent are separated by SPM.

Osmotic pressure = hydrostatic pressure ; $\pi = hdg$

where h = increase in level in the tube of unit cross section
 d = density of solution
 g = acceleration due to gravity

Van't Hoff law for Dilute Solution

According to it Gas equation $PV = nRT$ is also followed by dilute solution when pressure of gas is replaced by osmotic pressure of solution.

π = osmotic pressure of solution (atm)

V = volume of solution (L)

n = moles of solute

R = (S) gas constant / solution constant = $0.0821 \text{ L atm mol}^{-1}\text{K}^{-1}$; $0.083 \text{ L bar mol}^{-1}\text{K}^{-1}$

$$\pi = \left(\frac{n}{V}\right)RT$$

$$\pi = CRT$$

At constant temperature $\pi \propto C$ $\therefore \pi$ is a colligative property.

$$\begin{array}{ccc} PV = nRT \\ \downarrow \downarrow \downarrow \\ \pi V = nRT \end{array}$$

- On the basis of osmotic pressure solution can be classified by following ways :
 - Isotonic solutions** : Solutions having same osmotic pressure are called isotonic solution.

$$\pi_1 = \pi_2 ; \quad \text{primary condition}$$

$$C_1 RT = C_2 RT \quad (\text{at same temperature})$$

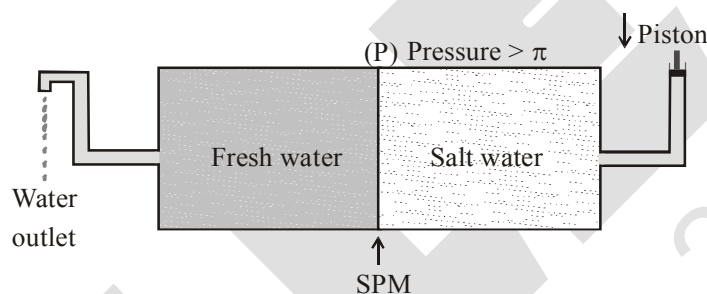
$$C_1 = C_2 \quad (\text{secondary condition}) ; \text{ means}$$

$$\frac{n_1}{V_1} = \frac{n_2}{V_2} ; \text{ such solutions are known as } \mathbf{isotonic}$$

- $\pi_1 > \pi_2$ or $C_1 > C_2$ then solution 1 is called **hypertonic** and solution 2 is called **hypotonic**

Reverse Osmosis

If external pressure greater than osmotic pressure is applied, the flow of solvent molecules can be made to proceed from solution towards pure solvent, i.e., in reverse direction of the ordinary osmosis. Reverse osmosis is used for the desalination of sea water for getting fresh drinking water.



Illustrations

- Illustration 13.** The vapour pressure of benzene at 90°C is 1020 torr. A solution of 5 g of a solute in 58.5 g benzene has vapour pressure 990 torr. The molecular weight of the solute is?

Solution.

$$\frac{P^0 - P_s}{P_s} = \frac{w \times M}{m \times W} \Rightarrow \frac{1020 - 990}{990} = \frac{5 \times 78}{m \times 58.5} \Rightarrow m = 220$$

- Illustration 14.** A 5% $\left(\frac{w}{V}\right)$ solution of cane sugar is isotonic with 0.877% $\left(\frac{w}{V}\right)$ solution of unknown solute. Find out molecular weight of unknown solute.

Solution

$$\frac{5 \times 10}{342} = \frac{0.877 \times 10}{x}$$

$$x = \frac{0.877 \times 342}{5}$$

$$x = 0.877 \times 68.4 = 60$$

Illustraion 15. 0.15 g of a substance dissolved in 15 g of solvent boiled at a temperature higher by 0.216 °C than that of the pure solvent. What is the molar mass of the substance. [K_b for solvent = 2.16 °Cm⁻¹]

Solution. Given $K_b = 2.16^\circ\text{C}$, $w = 0.15$ g, $\Delta T_b = 0.216^\circ\text{C}$, $W = 15$ g

$$\Delta T_b = \text{Molality} \times K_b$$

$$\Delta T_b = \frac{w}{m \times W} \times 1000 \times K_b \Rightarrow 0.216 = \frac{0.15}{m \times 15} \times 1000 \times 2.16$$

$$\Rightarrow m = \frac{0.15 \times 1000 \times 2.16}{0.216 \times 15} = 100$$

Illustraion 16. The rise in boiling point of a solution containing 1.8 g glucose in 100 g of a solvent is 0.1°C. The molal elevation constant of the liquid is –

Solution. $\Delta T_b = \text{Molality} \times K_b$

$$\Delta T_b = \frac{w}{m \times W} \times 1000 \times K_b \Rightarrow K_b = \frac{\Delta T_b \times m \times W}{1000 \times w}$$

given $\Delta T_b = 0.1^\circ\text{C}$, $m = 180$, $W = 100$, $w = 1.8$

$$K_b = \frac{180 \times 0.1 \times 100}{1000 \times 1.8} = 1.0^\circ\text{Cm}^{-1}$$

Illustraion 17. If freezing point of a solution prepared from 1.25 g of a non electrolyte and 20 g of water is 271.9 K, the molar mass of the solute will be

Solution. Given : $(T_f)_s = 271.9$ K, $w = 1.25$ g, $W = 20$ g, $K_f = 1.86$ K kg mol⁻¹

$$\Delta T_f = T_b^0 - (T_f)_s = 273 - 271.9 = 1.1$$

$$\Delta T_f = \text{Molality} \times K_f \quad \text{or} \quad \Delta T_f = \frac{w}{m \times W} \times 1000 \times K_f$$

$$m = \frac{w \times 1000 \times K_f}{\Delta T_f \times W} = \frac{1.25 \times 1000 \times 1.86}{1.1 \times 20} = 105.68$$

Illustraion 18. Molal depression constant for water is 1.86 °Cm⁻¹. The freezing point of a 0.05 molal solution of a non electrolyte in water is –

Solution. $\Delta T_f = \text{Molality} \times K_f = 0.05 \times 1.86 = 0.093^\circ\text{C}$

$$(T_f)_s = T_0 - 0.093 = 0 - 0.093$$

$$(T_f)_s = -0.093^\circ\text{C}$$

Illustraion 19. A cane sugar solution has an osmotic pressure of 2.46 atm at 300 K. What is the strength of the solution.

Solution. $\pi V = nST$ or $\pi = \frac{n}{V}ST = CST$ or $C = \frac{\pi}{ST} = \frac{2.46}{300 \times 0.0821} = 0.1$ M

Illustraion 20. What is the osmotic pressure of 12% solution of cane sugar (molecular weight 342) at 17 °C.

Solution. 12 g sugar is dissolved in 100 mL

$$\text{thus } 342 \text{ g sugar is dissolved in } \frac{100 \times 342}{12 \times 1000} = 2.85 \text{ L}$$

$$\text{Now } \pi V = RT \quad \{ \because n = 1 \}$$

$$\pi = \frac{RT}{V} = \frac{0.0821 \times 290}{2.85} = 8.35 \text{ atm}$$

Illustration 21. A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (weight/volume) solution of an organic, non-volatile solute. Calculate molar mass organic compound.

Solution. Solutions are isotonic

$$\text{so } \pi_1 = \pi_2 \quad \text{or} \quad \frac{n_1}{V_1} RT = \frac{n_2}{V_2} RT \quad \{R \text{ \& } T \text{ are constant}\}$$

$$\text{so, } \frac{n_1}{V_1} = \frac{n_2}{V_2} \quad \text{or} \quad \left(\frac{w_1}{m_1 \times v_1} \right)_{\text{urea}} = \left(\frac{w_2}{m_2 \times v_2} \right)_{\text{organic}}$$

$$\text{or } \frac{8.6}{60 \times 1000} = \frac{0.5}{m_2 \times 100} \Rightarrow m_2 = 34.89$$

GOLDEN KEY POINTS

- (i) $\frac{\Delta P}{P^0} \approx \frac{n}{N} \Rightarrow \Delta P = \frac{wM}{mW} \times P^0$ (ii) $\Delta T_b = \left(\frac{w \times 1000}{m \times W} \right) K_b$ (iii) $\Delta T_f = \left(\frac{w \times 1000}{m \times W} \right) K_f$ (iv) $\pi = CRT$
- Best method to determine molecular weight of biomolecules like protein and polymers is osmotic pressure. Because it is a colligative property also as no temperature change is involved so there is no chance of change in nature of biomolecule. In other methods changes are very less and it is difficult to measure very small changes accurately. But osmotic pressure can be easily measured at room temperature at which biomolecules remain in original state.
- 0.91% $\left(\frac{w}{V} \right)$ NaCl solution is isotonic with blood.
- Intravenous injections need to be isotonic with blood to prevent damage to RBC.
- Magnitude of osmotic pressure is independent with the nature of solvent.
- Berkley Hartley's method and Funnel experiment ($\pi = h d g$) are used to determine osmotic pressure.
- Ostwalds process is used to determine elevation in boiling point.
- Camphor is the best solvent for determination of molar mass of solute by using depression in freezing point since its K_f is very high.

BEGINNER'S BOX-3

1. An example of colligative property
(1) freezing point (2) boiling point (3) vapour pressure (4) osmotic pressure
2. The freezing point order of the solution of glucose is
(1) 10% > 3% > 2% > 1% (2) 1% > 2% > 3% > 10%
(3) 1% > 3% > 10% > 2% (4) 10% > 1% > 3% > 2%
3. In cold countries, ethylene glycol is added to water in the radiators of cars during winters. It result in reducing
(1) viscosity (2) specific heat (3) freezing point (4) boiling point
4. Calculate the molal depression constant of a solvent, which has freezing point 16.6 °C and latent heat of fusion 180.75 Jg⁻¹
(a) 2.68 (2) 3.86 (3) 4.68 (4) 2.86
5. The osmotic pressure of a solution at 0 °C is 4 atm. What will be its osmotic pressure at 546 K under similar condition?
(1) 4 atm (2) 2 atm (3) 8 atm (4) 1 atm

3.8 ABNORMAL COLLIGATIVE PROPERTIES

- It has been observed that difference in the observed and calculated molecular masses of solute is due to association or dissociation of solute molecules in solution. It results in a change in the number of particles in solution.
- Van't hof factor (i)** : Tells about relationship between normal colligative properties and abnormal colligative properties.
- $$i = \frac{\text{number of particles after dissociation or association}}{\text{number of particles before dissociation or association}} = \frac{\text{observed colligative properties}}{\text{calculated colligative properties}}$$

$$= \frac{\text{calculated molecular mass}}{\text{observed molecular mass}}$$
- If $i = 1$ Neither dissociation nor association ; Ex. glucose, urea, sugar
- if $i > 1$ Dissociation will occur ; Ex. NaCl, KCl, BaCl₂
- if $i < 1$ Association will occur ; Ex. CH₃COOH, PhCOOH in benzene

CASE I :

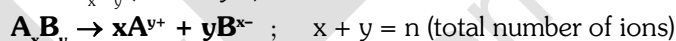
DISSOCIATION OF SOLUTE : Molecules of electrolytes undergo ionization or dissociation in polar solvents to give two or more particles in solution. This dissociation results in an increase in the total number of particles, and therefore the value of colligative properties of such solutions will be higher. As the colligative properties are inversely related to molecular weight, so the molecular weight of ionizable solute will be less than the theoretical value.

Number of solute particles in solution increases.

- observed / experimental C.P. > calculated C.P. (normal)
- observed / experimental molecular weight of solute < calculated (normal) molecular weight of solute

$$\therefore \text{C.P.} \propto \frac{1}{\text{molecular weight of solute}}$$

- Calculation of 'i' : Let solute be A_xB_y (electrolyte)



Initially 1 mol 0 0

After dissociation (1-α) xα yα

$$\begin{aligned} \text{Total number of solute particles} &= 1 - \alpha + x\alpha + y\alpha \\ &= 1 - \alpha + (x+y)\alpha \\ &= (1 - \alpha + n\alpha) \text{ mol} \end{aligned}$$

Observed colligative property is proportional to observed number of solute particles (1-α + nα)

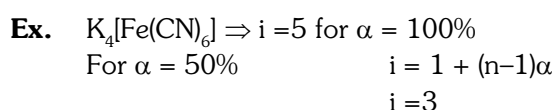
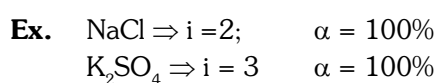
$$i = \frac{\text{number of particles after dissociation}}{\text{number of particles before dissociation}} = \frac{1 - \alpha + n\alpha}{1}$$

$$\boxed{i = 1 + (n-1)\alpha}$$

where α is the degree of dissociation

- For strong electrolytes

$$\text{if } \alpha = 1 \text{ or } 100\% \Rightarrow \text{then } \boxed{i = n}$$



CASE II :

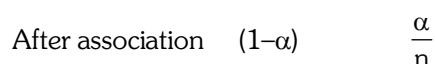
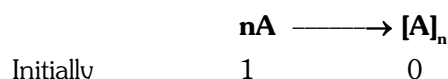
ASSOCIATION OF SOLUTE : The formation of a bigger molecule by the union of two, three or more solute molecules is called association. As a result, the total number of particles in solution becomes less than the number of molecules initially dissolved in the solution and hence the colligative properties will have lower value. As the molar mass of solute is inversely proportional to the colligative properties, so the molar mass of solute will be greater than theoretical value.

Number of solute particles in solution decreases.

- Observed / experimental C.P. < calculated C.P.
- Observed / experimental molecular weight of solute > normal molecular weight of solute

$$\therefore \text{C.P.} \propto \frac{1}{\text{Molecular weight of solute}} ; \quad i < 1 \text{ for association.}$$

- Calculation of i



$$\text{Total number of solute particles} = \left(1 - \alpha + \frac{\alpha}{n}\right) \text{ mol}$$

$$\text{Observed C.P.} \propto \text{observed number of solute particles} \left(1 - \alpha + \frac{\alpha}{n}\right)$$

$$\text{Van't hof factor } (i) = \frac{\text{number of particles after association}}{\text{number of particle before association}} ; i = \frac{1 - \alpha + \frac{\alpha}{n}}{1}$$

$$\boxed{i = 1 - \alpha + \frac{\alpha}{n}}$$

α = degree of association

n = number of solute particles which are associated.

- If $\alpha = 100\%$ or 1 or α is not specified $\boxed{i = \frac{1}{n}}$

Illustrations

Illustration 22. A 5% solution of anhydrous CaCl_2 at 0°C developed 15 atm osmotic pressure. What is the degree of dissociation of CaCl_2 ?

Solution. 5 g. of CaCl_2 are present in 100 mL, so 111 g (M_w of CaCl_2) will be present in $\frac{100 \times 111}{5 \times 1000} = 2.22 \text{ L}$

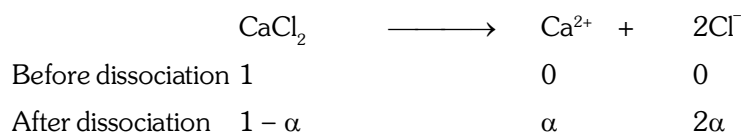
$$\text{Now } \pi V = ST \quad \{ \because n = 1 \} \quad \text{or} \quad \pi = \frac{0.082 \times 273}{2.22} = \frac{22.47}{2.22} = 10.09 \text{ atm}$$

$$\text{We know that Van't Hoff factor } i = \frac{\text{observed colligative property}}{\text{Normal colligative property}} = \frac{15}{10.09}$$

$$\text{and } \alpha = \frac{i-1}{n-1} \quad \text{or} \quad \alpha = \frac{\frac{15}{10.09} - 1}{3-1} = \frac{4.91}{10.09 \times 2} = 0.2433 \quad \text{or } 24.33\%$$

Illustration 23. Calculate the osmotic pressure of 20% (wt/Vol.) anhydrous CaCl_2 solution at 0°C assuming 100% ionisation.

Solution



Given, $w = 20 \text{ g}$, $V = 100 \text{ mL}$, $T = 273 \text{ K}$,

mol. wt. of $\text{CaCl}_2 = 111$

$$\pi_{\text{Normal}} = \frac{w}{mV} \times S \times T = \frac{20 \times 1000 \times 0.0821 \times 273}{111 \times 100} = 40.38 \text{ atm.}$$

$$\text{Now, } i = \frac{1 + 2\alpha}{1} = 1 + 2 = 3 \quad (\because \alpha = 1)$$

$$\pi_{\text{exp}} = i \times \pi_{\text{Normal}}$$

$$\therefore \pi_{\text{exp}} = 40.38 \times 3 = 121.14 \text{ atm}$$

BEGINNER'S BOX-4

- Calculate the percentage degree of dissociation of an electrolyte AB_2 (normal molar mass = 164) in H_2O if observed molar mass is 65.6
(1) 50% (2) 25% (3) 75% (4) None
- Which of the following solution will have highest freezing point
(1) 1M urea (2) 1M Na_2SO_4 (3) 1M NaCl (4) 1M $\text{Al}_2(\text{SO}_4)_3$
- A 1.17% solution of NaCl is isotonic with 7.2% solution of glucose calculate the value of i of NaCl
(1) 1 (2) 2 (3) 3 (4) 4
- Van't hof factor of Hg_2Cl_2 in its aqueous solution will be (Hg_2Cl_2 is 80% ionized in the solution)
(1) 1.6 (2) 2.6 (3) 3.6 (4) 4.6

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	3	1	3	3	2	4	1	2	1

BEGINNER'S BOX-2	Que.	1	2	3	4	5					
	Ans.	1	2	3	1	1					

BEGINNER'S BOX-3	Que.	1	2	3	4	5					
	Ans.	4	2	3	2	3					

BEGINNER'S BOX-4	Que.	1	2	3	4						
	Ans.	3	1	2	2						

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

CONCENTRATION TERMS

- 8 g NaOH is dissolved in one litre of solution, its molarity is :
(1) 0.8 M (2) 0.4 M (3) 0.2 M (4) 0.1 M
- If 18 g of glucose is present in 1000 g of solvent, the solution is said to be :
(1) 1 molar (2) 0.1 molar
(3) 0.5 molar (4) 0.1 molal
- The molarity of a solution of sodium chloride (mol wt. = 58.5) in water containing 5.85 g of sodium chloride in 500 mL of solution is :-
(1) 0.25 (2) 2.0 (3) 1.0 (4) 0.2
- For preparing 0.1 M solution of H_2SO_4 in one litre, we need H_2SO_4 :
(1) 0.98 g (2) 4.9 g
(3) 49.0 g (4) 9.8 g
- Mole fraction of glycerine($\text{C}_3\text{H}_5(\text{OH})_3$) in a solution of 36 g of water and 46 g of glycerine is :
(1) 0.46 (2) 0.36
(3) 0.20 (4) 0.40
- 1000 g aqueous solution of CaCO_3 contains 10 g of calcium carbonate, concentration of the solution is :
(1) 10 ppm (2) 100 ppm
(3) 1000 ppm (4) 10,000 ppm
- What is the normality of 1M H_3PO_4 solution ?
(1) 0.5 N (2) 1.0 N
(3) 2.0 N (4) 3.0 N
- The molarity of 0.2 N Na_2CO_3 solution will be :
(1) 0.05 M (2) 0.2 M
(3) 0.1 M (4) 0.4 M
- Normality of 0.3 M phosphorous acid is:-
(1) 0.15 (2) 0.6
(3) 0.9 (4) 0.1
- The molarity of pure water is :
(1) 100 M (2) 55.5 M
(3) 50 M (4) 18M
- Molarity of 720 g of pure water -
(1) 40M
(2) 4M
(3) 55.5M
(4) Can't be determined

12. Equal weight of NaCl and KCl are dissolved separately in equal volumes of solutions then molarity of the two solutions will be –
(1) Equal
(2) That of NaCl will be less than that of KCl
(3) That of NaCl will be more than that of KCl
Solution
(4) That of NaCl will be half of that of KCl solution
13. The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is :
(1) $\frac{8}{15}$ (2) 0.5 (3) 0.25 (4) 1.0
14. In a solution of 7.8 g benzene (C_6H_6) and 46.0g toluene ($C_6H_5CH_3$) the mole fraction of benzene is:-
(1) $\frac{1}{6}$ (2) $\frac{1}{5}$ (3) $\frac{1}{2}$ (4) $\frac{1}{3}$
15. An X molal solution of a compound in benzene has mole fraction of solute equal to 0.2. The value of X is:-
(1) 14 (2) 3.2 (3) 1.4 (4) 2
16. Mole fraction of ethanol in ethanol water mixture is 0.25. Hence percentage concentration of ethanol by weight of mixture is :-
(1) 25% (2) 75% (3) 46% (4) 54%
17. A 500 g tooth paste sample has 0.02 g fluoride concentration. What is the concentration of fluorine in terms of ppm level:-
(1) 250 (2) 40 (3) 400 (4) 1000
18. The normality of 10 mL of a '20 V' H_2O_2 solution is
(1) 1.79 (2) 3.58 (3) 60.86 (4) 6.086
19. H_2O_2 solution used for hair bleaching is sold as a solution of approximately 5.0 g H_2O_2 per 100 mL of the solution. The molecular mass of H_2O_2 is 34. The molarity of this solution is approximately:-
(1) 0.15 M (2) 1.5 M
(3) 3.0 M (4) 3.4 M
20. Normality of 10% (w\%V) H_2SO_4 solution is nearly
(1) 0.1 (2) 0.2 (3) 0.5 (4) 2
21. What volume of 0.1 N HNO_3 solution can be prepared from 6.3 g of HNO_3 ?
(1) 1 litre (2) 2 litre (3) 0.5 litre (4) 5 litre

- 22.** A molal solution is one that contains one mole of a solute in
- 1000 g of the solvent
 - one litre of the solution
 - one litre of the solvent
 - 22.4 litres of the solution
- 23.** Which of the following statement is true :-
- Molarity is the no. of moles of solute dissolved per litre of solvent.
 - The molarity and normality of a solution of sodium carbonate are same.
 - Molality (m) of a solution is defined as the number of moles of solute dissolved in 1000 gm of solution
 - The ratio of mole fraction of solute and solvent is in the ratio of their respective moles.
- a & c
 - a & d
 - b & c
 - Only d
- 24.** Two bottles of A and B contains 1M and 1m aqueous solution ($d \approx 1 \text{ g mL}^{-1}$) of sulphuric acid respectively-
- A is more concentrated than B
 - B is more concentrated than A
 - Concentration of A = concentration of B
 - It is not possible to compare the concentration
- 25.** Molar concentration of a solution in water is :
- Always equal to normality
 - More than molality of the solution
 - Equal to molality of the solution
 - Less than the molality of the solution
- 26.** The molarity of 98% H_2SO_4 ($d = 1.8 \text{ g mL}^{-1}$) by wt. is
- 6 M
 - 18 M
 - 10 M
 - 4 M

SOLUBILITY (HENRY'S LAW)

- 27.** Henry's law constant for dissolution of CH_4 in benzene at 298 K is $2 \times 10^5 \text{ mm of Hg}$. Then solubility of CH_4 in benzene at 298 K under 760 mm of Hg is :
- 1.2×10^{-5}
 - 3.8×10^{-3}
 - 4×10^{-7}
 - 1×10^{-2}
- 28.** Which of the following gas does not obey Henry's law ?
- NH_3
 - H_2
 - O_2
 - He

VAPOUR PRESSURE (LIQUID-LIQUID MIXTURE)

- 29.** 1 mol of heptane (V. P. = 92 mm of Hg) was mixed with 4 mol of octane (V. P. = 31 mm of Hg). The vapour pressure of resulting ideal solution is :
- 46.2 mm of Hg
 - 40.0 mm of Hg
 - 43.2 mm of Hg
 - 38.4 mm of Hg
- 30.** At 88 °C benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will boil at 88 °C at 1 atm. pressure, benzene - toluene form an ideal solution :
- 0.416
 - 0.588
 - 0.688
 - 0.740
- 31.** If P_A^0 and P_B^0 are 108 and 36 torr respectively. What will be the mole fraction of A in vapour phase if B has mole fraction in solution 0.5.
- 0.25
 - 0.75
 - 0.60
 - 0.35
- 32.** What is correct relation between mole fraction in vapour phase (Y_A) of A in terms of X_A . If mole fraction in solution of A is (X_A) (If P_A^0 is vapour pressure of A in pure state)
- $(1 - X_A)P_A^0$
 - $\frac{X_A}{1 - X_A}P_A^0$
 - $\frac{1 - X_A}{X_A}P_A^0$
 - $\frac{P_A^0 X_A}{P_S}$

IDEAL AND NON-IDEAL SOLUTIONS

- 33.** Which condition is not satisfied by an ideal solution
- $\Delta H_{\text{mixing}} = 0$
 - $\Delta V_{\text{mixing}} = 0$
 - $\Delta S_{\text{mixing}} = 0$
 - Obedience of Raoult's law
- 34.** Among the following, that does not form an ideal solution is :
- C_6H_6 and $\text{C}_6\text{H}_5\text{CH}_3$
 - $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_6\text{H}_5\text{OH}$
 - $\text{C}_6\text{H}_5\text{Cl}$ and $\text{C}_6\text{H}_5\text{Br}$
 - $\text{C}_2\text{H}_5\text{Br}$ and $\text{C}_2\text{H}_5\text{I}$

35. An azeotropic mixture of two liquids has b.p. lower than either of them when it :-
 (1) shows a (+ve) deviation from Raoult's law
 (2) shows no deviation from Raoult's law
 (3) shows (+ve) deviation from Henry's law
 (4) shows (-ve) deviation from Henry's law

COLLIGATIVE PROPERTIES

36. Which is not a colligative property ?
 (1) Osmotic pressure
 (2) Relative lowering in vapour pressure
 (3) Depression in freezing point
 (4) Refractive index
37. The lowering of vapour pressure of a solvent by addition of a non-volatile solute to it is directly proportional to :
 (1) The strength of the solution
 (2) The nature of the solute in the solution
 (3) The atmospheric pressure
 (4) All
38. The relative lowering of vapour pressure is equal to the mole fraction of the nonvolatile solute, This statement was given by :
 (1) Raoult (2) Henry (3) Joule (4) Dalton
39. Which one of the following is the incorrect form of Raoult's law
 (1) $\frac{P_s}{P^0} = \frac{N}{n+N}$ (2) $\frac{P^0}{P^0 - P_s} = 1 + \frac{N}{n}$
 (3) $\frac{P^0 - P_s}{P_s} = \frac{n}{n+N}$ (4) $\frac{P_s}{P^0 - P_s} = \frac{N}{n}$
40. The vapour pressure of a solution having solid as solute and liquid as solvent is :
 (1) Directly proportional to mole fraction of the solvent
 (2) Inversely proportional to mole fraction of the solvent
 (3) Directly proportional to mole fraction of the solute
 (4) Inversely proportional to mole fraction of the solute
41. If P_0 and P_s are the vapour pressure of solvent and its solution respectively. N_1 and N_2 are the mole fraction of solvent and solute respectively then :

$$\begin{array}{ll} (1) P_s = \frac{P_0}{N_2} & (2) P_0 - P_s = P_0 N_2 \\ (3) P_s = P_0 N_2 & (4) \frac{(P_0 - P_s)}{P_s} = \frac{N_1}{(N_1 + N_2)} \end{array}$$

42. One mol of non volatile solute is dissolved in two mol of water. The vapour pressure of the solution relative to that of water is

$$(1) \frac{2}{3} \quad (2) \frac{1}{3} \quad (3) \frac{1}{2} \quad (4) \frac{3}{2}$$

43. The vapour pressure of a dilute aqueous solution of Glucose is 750 mm of mercury at 373 K. The mole fraction of solute is :

$$(1) \frac{1}{10} \quad (2) \frac{1}{7.6} \quad (3) \frac{1}{35} \quad (4) \frac{1}{76}$$

44. The vapour pressure of water at room temperature is 23.8 mm of Hg. The vapour pressure of an aqueous solution of sucrose with mole fraction 0.1 is equal to :

$$(1) 23.9 \text{ mm Hg} \quad (2) 24.2 \text{ mm Hg} \\ (3) 21.42 \text{ mm Hg} \quad (4) 31.44 \text{ mm Hg}$$

45. The vapour pressure of pure A is 10 torr and at the same temperature when 1 g of B is dissolved in 20 gm of A, its vapour pressure is reduced to 9.0 torr. If the molecular mass of A is 200 amu, then the molecular mass of B is :

$$(1) 100 \text{ amu} \quad (2) 90 \text{ amu} \\ (3) 75 \text{ amu} \quad (4) 120 \text{ amu}$$

46. The boiling point of C_6H_6 , CH_3OH , $C_6H_5NH_2$ and $C_6H_5NO_2$ are $80^\circ C$, $65^\circ C$, $184^\circ C$ and $212^\circ C$ respectively. Which will show highest vapour pressure at room temperature :

$$(1) C_6H_6 \quad (2) CH_3OH \\ (3) C_6H_5NH_2 \quad (4) C_6H_5NO_2$$

47. If Raoult's law is obeyed, the vapour pressure of the solvent in a solution is directly proportional to :

$$(1) \text{ Mole fraction of the solvent} \\ (2) \text{ Mole fraction of the solute} \\ (3) \text{ Mole fraction of the solvent and solute} \\ (4) \text{ The volume of the solution}$$

48. The vapour pressure of a pure liquid 'A' is 70 torr at $27^\circ C$. It forms an ideal solution with another liquid B. The mole fraction of B is 0.2 and total vapour pressure of the solution is 84 torr at $27^\circ C$. The vapour pressure of pure liquid B at $27^\circ C$ is -

$$(1) 14 \quad (2) 56 \quad (3) 140 \quad (4) 70$$

49. The boiling point of an aqueous solution of a non volatile solute is $100.15^\circ C$. What is the freezing point of an aqueous solution obtained by diluting the above solution with an equal volume of water ? The values of K_b and K_f for water are 0.512 and $1.86 \text{ K molality}^{-1}$:

$$(1) -0.544^\circ C \quad (2) -0.512^\circ C \\ (3) -0.272^\circ C \quad (4) -1.86^\circ C$$

50. The molal elevation constant is the ratio of the elevation in B.P. to :
- Molarity
 - Molality
 - Mole fraction of solute
 - Mole fraction of solvent
51. Elevation in boiling point was 0.52°C when 6 g of a compound **x** was dissolved in 100 g of water. Molecular weight of **x** is : ($K = 5.2 \text{ K mol}^{-1} 100 \text{ g H}_2\text{O}$)
- 120
 - 60
 - 100
 - 342
52. An aqueous solution containing 1g of urea boils at 100.25°C . The aqueous solution containing 3g of glucose in the same volume will boil at -
- 100.75°C
 - 100.5°C
 - 100°C
 - 100.25°C
53. Pure benzene freezes at 5.45°C at a certain place but a 0.374 m solution of tetrachloroethane in benzene freezes at 3.55°C . The K_f for benzene is-
- $5.08 \text{ K Kg mol}^{-1}$
 - $508 \text{ K Kg mol}^{-1}$
 - $0.508 \text{ K Kg mol}^{-1}$
 - $50.8^\circ\text{C Kg mol}^{-1}$
54. An aqueous solution freezes at -0.186°C ($K_f = 1.86 \text{ K kg mol}^{-1}$; $K_b = 0.512 \text{ K kg mol}^{-1}$). What is the elevation in boiling point ?
- 0.186
 - 0.512
 - $\frac{0.512}{1.86}$
 - 0.0512
55. Camphor is used as solvent to determine the molecular weight of nonvolatile solute by Rast method because for camphor -
- Molal depression constant is high
 - Melting point is high
 - Being cheap
 - All
56. In the depression of freezing point experiment, it is found that :-
- The vapour pressure of solution is more than of pure H_2O .
 - The vapour pressure of solution is less than that of pure solute
 - Only solute molecules solidify at of freezing point
 - Only solvent molecules solidify at freezing point
57. Molal depression constant of water is $1.86 \text{ K Kg mol}^{-1}$. 0.02 mol of urea dissolved in 100 g of water will produce a depression in freezing point of :
- 0.186°C
 - 0.372°C
 - 1.86°C
 - 3.72°C
58. What would be the freezing point of aqueous solution containing 18 g of $\text{C}_6\text{H}_{12}\text{O}_6$ in 1000 g of water ? ($K_f = 1.86 \text{ K molality}^{-1}$)
- -0.186°C
 - -0.372°C
 - -0.54°C
 - -0.72°C
59. A solution of 1.25 g of a non-electrolyte in 20 g of water freezes at 271.94 K . If $K_f = 1.86 \text{ K molality}^{-1}$ then the molecular wt. of the solute is :
- 207.8 g mol^{-1}
 - $179.79 \text{ g mol}^{-1}$
 - 209.6 g mol^{-1}
 - 96.01 g mol^{-1}
60. If a thin slice of sugar beet is placed in concentrated solution of NaCl then
- Sugar beet will lose water from its cells
 - Sugar beet will absorb water from solution
 - Sugar beet will neither absorb nor lose water
 - Sugar beet will dissolve in solution
61. In osmosis phenomenon net flow of :
- Solvent molecules occurs from higher concentration to lower concentration
 - Solvent molecules occurs from lower concentration to higher concentration
 - Solute molecules occurs from higher concentration to lower concentration
 - Solute molecules occurs from lower concentration to higher concentration
62. At constant temperature the osmotic pressure of a solution is :
- Directly proportional to the concentration
 - Inversely proportional to the concentration
 - Directly proportional to the square of concentration
 - Directly proportional to the square root of concentration
63. Which inorganic precipitate acts as semipermeable membrane ?
- Calcium sulphate
 - Barium oxalate
 - Nickel phosphate
 - Copper ferrocyanide

64. If 0.1 M solution of glucose and 0.1 M urea solution are placed on two sides of a semipermeable membrane to equal heights, then it will be correct to say that :
- (1) There will be not net movement across the membrane
 - (2) Glucose will flow towards urea solution
 - (3) Urea will flow towards glucose solution
 - (4) Water will flow from urea solution towards glucose solution.
65. The best colligative property used for the determination of molecular masses of polymers is :
- (1) Relative lowering in vapour pressure
 - (2) Osmotic pressure
 - (3) Elevation in boiling point
 - (4) Depression in freezing point
66. If mole fraction of the solvent in solution decreases then :
- (1) Vapour pressure of solution increases
 - (2) B. P. decreases
 - (3) Osmotic pressure increases
 - (4) All are correct
67. The osmotic pressure of a solution increases if :
- (1) Temperature is lowered
 - (2) Volume is increased
 - (3) Number of solute molecules is increased
 - (4) None
68. Which of the following solutions at the same temperature will be isotonic :
- (1) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in one litre water.
 - (2) 3.42 g of cane sugar in one litre water and 0.18 g of glucose in 0.1 litre water.
 - (3) 3.42 g of cane sugar in one litre water and 0.585g of NaCl in one litre water.
 - (4) 3.42 g of cane sugar in one litre water and 1.17 g of NaCl in one litre water.
69. If density of 2 molal sucrose solution is 1.4 g mL^{-1} at 25°C , find osmotic pressure.
- (1) 4.06 atm
 - (2) 2 atm
 - (3) 40.6 atm
 - (4) 3.4 atm
70. If total concentration of dissolved particles inside red blood cells is 0.3 M (approximate) and the membrane surrounding the cell is semipermeable. Find osmotic pressure inside the cell
- (1) 7.34 atm
 - (2) 1.78 atm
 - (3) 2.34 atm
 - (4) 0.74 atm
71. Equal volume of 0.1 M urea and 0.1 M glucose are mixed. The mixture will have :-
- (1) Lower osmotic pressure
 - (2) Same osmotic pressure
 - (3) Higher osmotic pressure
 - (4) None of these
72. Osmotic pressure of a solution (density is 1 g mL^{-1}) containing 3 g of glucose (molecular weight = 180) in 60 g of water at 15°C is :
- (1) 0.34 atm
 - (2) 0.65 atm
 - (3) 6.25 atm
 - (4) 5.57 atm
73. Osmotic pressure of a sugar solution at 24°C is 2.5 atmosphere. The concentration of the solution in mole per litre is :
- (1) 10.25
 - (2) 1.025
 - (3) 1025
 - (4) 0.1025
74. A solution containing 4 g of a non volatile organic solute per 100 ml was found to have an osmotic pressure equal to 500 cm of mercury at 27°C . The molecular weight of solute is :
- (1) 14.97
 - (2) 149.7
 - (3) 1697
 - (4) 1.497
75. A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (wt./vol) solution of an organic, non volatile solute. The molecular weight of later is :
- (1) 348.9
 - (2) 34.89
 - (3) 3489
 - (4) 861.2
76. If a 6.84% (w/V) solution of cane-sugar (mol. wt. 342) is isotonic with 1.52% (w/V) solution of thiocarbamide, then the molecular weight of thiocarbamide is :
- (1) 152
 - (2) 76
 - (3) 60
 - (4) 180
77. A solution containing 500 g of a protein per litre is isotonic with a solution containing 3.42 g of sucrose per litre. The molecular mass of protein is
- (1) 5
 - (2) 146
 - (3) 34200
 - (4) 50000
78. The osmotic pressure of blood is 7.65 atm. at 310 K. an aqueous solution of Glucose that will be isotonic with blood isw/V :
- (1) 5.41%
 - (2) 54.1%
 - (3) 3.5%
 - (4) 4.53%

ABNORMAL COLLIGATIVE PROPERTIES

79. Equimolal solutions of A and B show depression in freezing point in the ratio of 2 : 1. A remains in normal state in solution. B will be in state in solution :
 (1) Normal (2) Associated
 (3) Hydrolysed (4) Dissociated
80. van't Hoff factor is :
 (1) Less than one in case of dissociation
 (2) More than one in case of association
 (3) Always less than one
 (4) Less than one in case of association
81. The vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is :
 (1) 10 (2) 4 (3) 5 (4) 0.25
82. The experimental molecular weight of an electrolyte will always be less than its calculated value because the value of vant Hoff factor, 'i' is :
 (1) Less than 1 (2) Greater than 1
 (3) One (4) Zero
83. The vant Hoff factor (i) for a dilute aqueous solution of Glucose is :
 (1) Zero (2) 1.0 (3) 1.5 (4) 2.0
84. 0.585% NaCl solution at 27°C has osmotic pressure.
 (1) 2.49 atm (2) 5 atm
 (3) 1.2 atm (4) 3.8 atm
85. The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3 . The vant Hoff's factor will be -
 (1) 1 (2) 2 (3) 3 (4) $\frac{1}{3}$
86. The ratio of the value of any colligative property for KCl solution to that for sugar solution is nearly time :
 (1) 1 (2) 0.5 (3) 2 (4) 2.5
87. The lowering of vapour pressure of 0.1M aqueous solutions of NaCl, $CuSO_4$ and K_2SO_4 are :
 (1) All equal
 (2) In the ratio of 1 : 1 : 1.5
 (3) In the ratio of 3 : 2 : 1
 (4) In the ratio of 1.5 : 1 : 2.5
88. The freezing point of 1 molal NaCl solution assuming NaCl to be 100% dissociated in water is: ($K_f = 1.86 \text{ K Molality}^{-1}$)
 (1) -1.86°C (2) -3.72°C
 (3) $+1.86^\circ \text{C}$ (4) $+3.72^\circ \text{C}$

89. The molal elevation constant of water is 0.51. The boiling point of 0.1 molal aqueous NaCl solution is nearly :
 (1) 100.05°C (2) 100.1°C
 (3) 100.2°C (4) 101.0°C
90. The molal elevation constant of water is $0.52 \text{ K molality}^{-1}$. The boiling point of 1.0 molal aqueous KCl solution (assuming complete dissociation of KCl), should be :
 (1) 100.52°C (2) 101.04°C
 (3) 99.48°C (4) 98.96°C
91. Solute A is ternary electrolyte and solute B is non-electrolyte. If 0.1 M solution of solute B produces an osmotic pressure of 2P, then 0.05M solution of A at the same temperature will produce an osmotic pressure equal to :
 (1) P (2) 1.5 P (3) 2 P (4) 3 P
92. Phenol associates in benzene as

$$C_6H_5OH \rightleftharpoons \frac{1}{2}(C_6H_5OH)_2$$

 If degree of association of phenol is 40%. van't Hoff factor i is :-
 (1) 1 (2) 0.8 (3) 1.4 (4) 0.6
93. The values of observed and calculated molecular weight of calcium nitrate are respectively 65.6 and 164. The degree of dissociation of calcium nitrate will be :
 (1) 25% (2) 50%
 (3) 75% (4) 60%
94. A 0.004M solution of Na_2SO_4 is isotonic with a 0.010M solution of glucose at the 25°C temperature. The apparent degree of dissociation of Na_2SO_4 is
 (1) 25% (2) 50%
 (3) 75% (4) 85%
95. A 5.8% (wt./vol.) NaCl solution will exert an osmotic pressure closest to which one of the following :
 (1) 5.8% (wt./vol) sucrose solution
 (2) 5.8% (wt./vol) glucose solution
 (3) 2 M sucrose solution
 (4) 1 M glucose solution
96. Which salt shows maximum osmotic pressure in its 1M solution :
 (1) $AgNO_3$ (2) Na_2SO_4
 (3) $(NH_4)_3PO_4$ (4) $MgCl_2$
97. Which solution will exert highest osmotic pressure?
 (1) 1M glucose solution (2) 1M urea solution
 (3) 1M Alum solution (4) 1M NaCl solution

- 98.** Which is the correct relation between osmotic pressure of 0.1M NaCl solution and 0.1M Na₂SO₄ solution ?
 (1) The osmotic pressure of Na₂SO₄ is less than NaCl solution
 (2) The osmotic pressure Na₂SO₄ is more than NaCl solution
 (3) Both have same osmotic pressure
 (4) None of the above
- 99.** Which one of the following solutions will have highest osmotic pressure ? (Assume that all the salts are equally dissociated)
 (1) 0.1M Al₂(SO₄)₃
 (2) 0.1M BaCl₂
 (3) 0.1 M Na₂SO₄
 (4) The solution obtained by mixing equal volumes of (2) and (3)
- 100.** The following solutions have equal concentrations. Which one will show minimum osmotic pressure?
 (1) BaCl₂ (2) AgNO₃
 (3) Na₂SO₄ (4) (NH₄)₃PO₄
- 101.** The osmotic pressure of equimolar solutions of BaCl₂, NaCl, and glucose will be in the order :
 (1) Glucose > NaCl > BaCl₂
 (2) BaCl₂ > NaCl > Glucose
 (3) NaCl > BaCl₂ > Glucose
 (4) NaCl > Glucose > BaCl₂
- 102.** Which one of the following pairs of solutions will be expected to be isotonic under the same temperature ?
 (1) 0.1M urea and 0.1 M NaCl
 (2) 0.1M urea and 0.2 M MgCl₂
 (3) 0.1M NaCl and 0.1M Na₂SO₄
 (4) 0.1M Ca(NO₃)₂ and 0.1M Na₂SO₄
- 103.** Two solutions of KNO₃ and CH₃COOH are prepared separately. Molarity of both is 0.1M and osmotic pressures are P₁ and P₂ respectively. The correct relationship between the osmotic pressures is
 (1) P₂ > P₁ (2) P₁ = P₂
 (3) P₁ > P₂ (4) $\frac{P_1}{P_1 + P_2} = \frac{P_2}{P_1 + P_2}$
- 104.** The correct relationship between the boiling points of very dilute solutions of AlCl₃(T₁) and CaCl₂(T₂), having the same molar concentration is :
 (1) T₁ = T₂ (2) T₁ > T₂ (3) T₂ = T₁ (4) T₂ ≥ T₁
- 105.** Which of the following 0.1 M aqueous solutions will have the lowest freezing point :
 (1) Potassium Sulphate (2) Sodium Chloride
 (3) Urea (4) Glucose
- 106.** Which aqueous solution has minimum freezing point ?
 (1) 0.01 M NaCl (2) 0.005 M C₂H₅OH
 (3) 0.005 M MgI₂ (4) 0.005 M MgSO₄
- 107.** If α is the degree of dissociation of K₄[Fe(CN)₆], then abnormal mass of complex in the solution will be :-
 (1) M_{normal} (1+2α)⁻¹ (2) M_{normal} (1+3α)⁻¹
 (3) M_{normal} (1+α)⁻¹ (4) M_{normal} (1+4α)⁻¹
- 108.** Which solution will have least vapour pressure :
 (1) 0.1 M BaCl₂ (2) 0.1 M urea
 (3) 0.1 M Na₂SO₄ (4) 0.1 M Na₃PO₄
- 109.** Which has maximum freezing point :
 (1) 1 molar of NaCl solution
 (2) 1 molar of KCl solution
 (3) 1 molar of CaCl₂ solution
 (4) 1 molar of urea solution
- 110.** The freezing point of 1% aqueous solution of calcium nitrate will be :
 (1) 0°C (2) Above 0°C
 (3) 1°C (4) Below 0°C
- 111.** The following aqueous solution in the correct order of decreasing freezing point is -
 (1) 0.2M BaCl₂, 0.2M KCl, 0.1M Na₂SO₄
 (2) 0.2M KCl, 0.1M Na₂SO₄, 0.2M BaCl₂
 (3) 0.1M Na₂SO₄, 0.2M KCl, 0.2M BaCl₂
 (4) 0.1M Na₂SO₄, 0.2M BaCl₂, 0.2M KCl
- 112.** Which of the following solutions will have highest boiling point ?
 (1) 1% Glucose in water (2) 1% Sucrose in water
 (3) 1% NaCl in water (4) 1% Urea in water
- 113.** The freezing point of equimolar aqueous solution will be highest for :
 (1) C₆H₅NH₃Cl (2) Ca(NO₃)₂
 (3) La(NO₃)₃ (4) C₆H₁₂O₆(Glucose)
- 114.** When mercuric Iodide is added to the aqueous solution of potassium iodide ?
 (1) The boiling point does not change
 (2) Freezing point is raised
 (3) The freezing point is lowered
 (4) Freezing point does not change
- 115.** The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to :
 (1) Ionisation of benzoic acid
 (2) Dimerization of benzoic acid
 (3) Trimerization of benzoic acid
 (4) Solvation of benzoic acid

- 116.** 0.1 M solution of K_2SO_4 is dissolved to the extent of 90%. What would be its osmotic pressure at $27^\circ C$:-
 (1) 6.89 atm (2) 0.689 atm
 (3) 0.344 atm (4) 3.4 atm

- 117.** Which one has same van't Hoff factor i as that of Hg_2Cl_2 :-
 (1) NaCl (2) Na_2SO_4
 (3) $Al(NO_3)_3$ (4) $Al_2(SO_4)_3$

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	4	4	3	4	4	3	2	2	3	3	2	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	2	2	4	1	1	4	1	4	2	2	1	3	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	4	3	2	1	4	1	1	3	1	2	1	4	3	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	1	3	3	2	2	4	1	4	1	4	2	1	4	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	2	1	4	1	2	3	3	2	3	1	2	3	4	2	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	4	1	2	4	2	2	2	2	4	3	2	2	2	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	4	2	3	3	3	3	3	2	1	2	2	4	3	2	1
Que.	106	107	108	109	110	111	112	113	114	115	116	117			
Ans.	1	4	4	4	4	3	3	4	2	2	1	2			

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2006**

- A solution containing 10g per dm^3 of urea (molecular mass = 60g mol^{-1}) is isotonic with a 5% solution of a nonvolatile solute. The molecular mass of this nonvolatile solute is :
 (1) 250g mol^{-1} (2) 300g mol^{-1}
 (3) 350g mol^{-1} (4) 200g mol^{-1}
- 1.00g of a non-electrolyte solute (molar mass 250g mol^{-1}) was dissolved in 51.2g of benzene. If the freezing point depression constant, K_f of benzene is $5.12\text{ K kg mol}^{-1}$, the freezing point of benzene will be lowered by :
 (1) 0.4 K (2) 0.3 K
 (3) 0.5 K (4) 0.2 K
- A solution of acetone in ethanol
 (1) shows a positive deviation from Raoult's law
 (2) behaves like a near ideal solution
 (3) Obeys Raoult's law
 (4) shows a negative deviation from Raoult's law
- During osmosis, flow of water through a semipermeable membrane is :
 (1) from both sides of semipermeable membrane with equal flow rates
 (2) from both sides of semipermeable membrane with unequal flow rates
 (3) from solution having lower concentration only
 (4) from solution having higher concentration only

AIIMS 2006

- A 5% solution (by mass) of cane sugar in water has freezing point of 271 K and freezing point of pure water is 273.15K. The freezing point of a 5% solution (by mass) of glucose in water is –
 (1) 271 K (2) 273.15 K
 (3) 269.07 K (4) 277.23 K

AIPMT 2007

- 0.5 molal aqueous solution of a weak acid (HX) is 20% ionised. If K_f for water is $1.86\text{ K kg mol}^{-1}$, the lowering in freezing point of the solution is :
 (1) -0.56 K (2) -1.12 K
 (3) 0.56 K (4) 1.12 K

AIPMT 2009

- A 0.0020 m aqueous solution of an ionic compound $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}$ freezes at -0.00732°C . Number of moles of ions which 1mol of ionic compound produces on being dissolved in water will be ($K_f = 1.86^\circ\text{C m}^{-1}$) :-
 (1) 1 (2) 2 (3) 3 (4) 4

AIPMT 2010

- An aqueous solution is 1.00 molal in KI. Which change will cause the vapour pressure of the solution of increase ?
 (1) Addition of water
 (2) Addition of NaCl
 (3) Addition of Na_2SO_4
 (4) Addition of 100 molal KI
- A solution of sucrose (molar mass = 342 g mol^{-1}) has been prepared by dissolving 68.5 g of sucrose in 1000 g of water. The freezing point of the solution obtained will be :-
 (K_f for water = $1.86\text{ K kg mol}^{-1}$)
 (1) -0.570°C (2) -0.372°C
 (3) -0.520°C (4) $+0.372^\circ\text{C}$

AIIMS 2010

- For an ideal solution of A and B which statement is incorrect :-
 (1) The enthalpy change of mixing of A and B is zero
 (2) The volume change of solution A and B is zero
 (3) The intermolecular forces of A and B is same as that of A-A and B-B
 (4) The entropy change of mixture of A and B is zero

AIPMT Pre. 2011

- The freezing point depression constant for water is $-1.86^\circ\text{C m}^{-1}$. If 5 g Na_2SO_4 is dissolved in 45.0 g H_2O , the freezing point is changed by -3.82°C . Calculate the van't Hoff factor for Na_2SO_4
 (1) 2.05 (2) 2.63
 (3) 3.11 (4) 0.381
- The van't Hoff factor i for a compound which undergoes dissociation in one solvent and association in other solvent is respectively :
 (1) Less than one and greater than one
 (2) Less than one and less than one
 (3) Greater than one and less than one
 (4) Greater than one and greater than one

13. Mole fraction of the solute in a 1.00 molal aqueous solution is :

(1) 0.1770 (2) 0.0177
(3) 0.0344 (4) 1.7700

AIPMT Mains 2011

14. 200 mL of an aqueous solution of a protein contain its 1.26 g. The Osmotic pressure of this solution at 300 K is found to be 2.57×10^{-3} bar. The molar mass of protein will be :-

(R = $0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$)
(1) 61038 g mol⁻¹ (2) 51022 g mol⁻¹
(3) 122044 g mol⁻¹ (4) 31011 g mol⁻¹

AIPMT Pre 2012

15. p_A and p_B are the vapour pressure of pure liquid components, A and B, respectively of an ideal binary solution. If x_A represents the mole fraction of component A, the total pressure of the solution will be.

(1) $p_B + x_A (p_B - p_A)$ (2) $p_B + x_A (p_A - p_B)$
(3) $p_A + x_A (p_B - p_A)$ (4) $p_A + x_A (p_A - p_B)$

AIPMT Mains 2012

16. Which of the following compounds can be used as antifreeze in automobile radiators ?

(1) Nitrophenol (2) Ethyl alcohol
(3) Methyl alcohol (4) Glycol

17. Vapour pressure of chloroform (CHCl_3) and dichloromethane (CH_2Cl_2) at 25°C are 200 mmHg and 415 mmHg respectively. Vapour pressure of the solution obtained by mixing 25.5 g of CHCl_3 and 40g of CH_2Cl_2 at the same temperature will be: (Molecular mass of CHCl_3 = 119.5 u and molecular mass of CH_2Cl_2 = 85 u)

(1) 347.9 mmHg (2) 280.5 mmHg
(3) 173.9 mmHg (4) 615 mmHg

NEET-UG 2013

18. 6.02×10^{20} molecules of urea are present in 100mL of its solution. The concentration of solution is :-

(1) 0.1 M (2) 0.02 M
(3) 0.01 M (4) 0.001M

AIIMS 2013

19. Boiling point of benzene is 353.23 K. When 1.8 g of non-volatile solute is dissolved in 90 g of benzene. Then boiling point raised to 354.11 K. Given K_b (benzene) = $2.53 \text{ K kg mol}^{-1}$. Then molecular mass of non volatile substance is :-

(1) 58 g mol⁻¹ (2) 120 g mol⁻¹
(3) 116 g mol⁻¹ (4) 60 g mol⁻¹

20. Strength of H_2O_2 is 15.18 g L⁻¹, then it is equal to:-

(1) 1 volume (2) 10 volume
(3) 5 volume (4) 7 volume

AIPMT 2014

21. Of the following 0.10m aqueous solutions, which one will exhibit the largest freezing point depression?

(1) KCl (2) $\text{C}_6\text{H}_{12}\text{O}_6$
(3) $\text{Al}_2(\text{SO}_4)_3$ (4) K_2SO_4

AIIMS 2014

22. Addition of 1 kg of non volatile solute in volatile solvent increases the boiling point from 350 K to 355 K and decreases the freezing point from 250 K to 220 K. If K_b is $15.5 \text{ K Kg mol}^{-1}$, then K_f will be

(1) 2.1 K kg mol⁻¹ (2) 93.0 K kg mol⁻¹
(3) 76 K kg mol⁻¹ (4) 10 K kg mol⁻¹

AIPMT 2015

23. The boiling point of 0.2 mol kg⁻¹ solution of X in water is greater than equimolar solution of Y in water. Which one of the following statements is **true** in this case ?

(1) Molecular mass of X is greater than the molecular mass of Y.
(2) Molecular mass of X is less than the molecular mass of Y.
(3) Y is undergoing dissociation in water while X undergoes no change.
(4) X is undergoing dissociation in water while Y undergoes no change.

24. Which one is **not** equal to zero for an ideal solution:-

(1) ΔS_{mix}
(2) ΔV_{mix}
(3) $\Delta P = P_{\text{observed}} - P_{\text{Raoult}}$
(4) ΔH_{mix}

25. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of the $\text{Al}_2(\text{SO}_4)_3$ (if all are 100% ionised) ?
- (1) $\text{K}_3[\text{Fe}(\text{CN})_6]$
 - (2) $\text{Al}(\text{NO}_3)_3$
 - (3) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 - (4) K_2SO_4

Re-AIPMT 2015

26. What is the mole fraction of the solute in a 1.00 m aqueous solution ?
- (1) 0.0354
 - (2) 0.0177
 - (3) 0.177
 - (4) 1.770

AIIMS 2015

27. Osmotic pressure of a solution containing 1 g protein in 100 mL solution at 300 K is 1.66 bar. What is the molecular weight of protein ($R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$)
- (1) 150
 - (2) 120
 - (3) 130
 - (4) 300
28. On adding a non volatile solute to a solvent, the vapour pressure of solvent decreases and becomes $z \times$ [vapour pressure of solvent] where z is
- (1) mole fraction of solvent
 - (2) mole fraction of solute
 - (3) molality
 - (4) molarity

NEET-I 2016

29. Which of the following statement about the composition of the vapour over an ideal a 1 : 1 molar mixture of benzene and toluene is **correct**? Assume that the temperature is constant at 25°C . (Given : Vapour Pressure Data at 25°C , benzene = 12.8 kPa, Toluene = 3.85 kPa)
- (1) The vapour will contain a higher percentage of benzene
 - (2) The vapour will contain a higher percentage of toluene
 - (3) The vapour will contain equal amounts of benzene and toluene
 - (4) Not enough information is given to make a predication
30. At 100°C the vapour pressure of a solution of 6.5g of a solute in 100 g water is 732 mm. If $K_b = 0.52^\circ\text{C m}^{-1}$, the boiling point of this solution will be :-
- (1) 101°C
 - (2) 100°C
 - (3) 102°C
 - (4) 103°C

31. Consider the following liquid - vapour equilibrium.
- $$\text{Liquid} \rightleftharpoons \text{Vapour}$$
- Which of the following relations is **correct** ?

- (1) $\frac{d \ln G}{dT} = \frac{\Delta H_v}{RT^2}$
- (2) $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{RT}$
- (3) $\frac{d \ln P}{dT} = \frac{-\Delta H_v}{T^2}$
- (4) $\frac{d \ln P}{dT} = \frac{\Delta H_v}{RT^2}$

NEET-II 2016

32. The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is
- (1) 2
 - (2) 3
 - (3) 0
 - (4) 1
33. Which one of the following is **incorrect** for ideal solution ?
- (1) $\Delta P = P_{\text{obs}} - P_{\text{calculated by Raoult's law}} = 0$
 - (2) $\Delta G_{\text{mix}} = 0$
 - (3) $\Delta H_{\text{mix}} = 0$
 - (4) $\Delta U_{\text{mix}} = 0$

AIIMS 2016

34. 0.5 mol NaCl is dissolved in 500g H_2O . Then determine freezing point and boiling point of solution.
- Given $(K_f)_{\text{H}_2\text{O}} = 1.8 \text{ K kg mol}^{-1}$, $(K_b)_{\text{H}_2\text{O}} = 0.5 \text{ K kg mol}^{-1}$
- (1) -1.8°C , 100.5°C
 - (2) -3.6°C , 101°C
 - (3) -3.6°C , 99°C
 - (4) 0°C , 100°C
35. 1g solute is dissolved in 50 g benzene shows depression in freezing point by 0.4 K. Then calculate molar mass of solute if $K_f = 5.12 \text{ K kg mol}^{-1}$
- (1) 512
 - (2) 256
 - (3) 728
 - (4) 738
36. If the density of methanol is 0.8 kg L^{-1} , what is the volume needed for making 2.5 L of its 0.4 M solution ?
- (1) 0.4 L
 - (2) 4.0 L
 - (3) 0.04 L
 - (4) 40 L

37. When nonvolatile solute is added in a solvent then relative lowering in vapour pressure depends upon:-

- (1) $\frac{\text{Number of moles of solute}}{\text{Number of moles of solvent}}$
- (2) $\frac{\text{Number of moles of solvent}}{\text{Number of moles of solute}}$
- (3) $\frac{\text{Number of moles of solvent}}{\text{Total number of moles of solution}}$
- (4) $\frac{\text{Number of moles of solution}}{\text{Number of moles of solute}}$

NEET(UG) 2017

38. If molality of a dilute solution is doubled, the value of molal depression constant (K_f) will be :-

- (1) halved
- (2) tripled
- (3) unchanged
- (4) doubled

39. Which of the following is dependent on temperature?

- (1) Molarity
- (2) Mole fraction
- (3) Weight percentage
- (4) Molality

AIIMS 2017

40. 5g of an unknown solute is dissolved in 295 g solvent. If molarity and density of solution are 0.05 M and 1.5 g cc⁻¹ respectively. The molecular weight of unknown solute is :-

- (1) 375
- (2) 425
- (3) 400
- (4) 500

41. The depression in freezing point of water containing ethylene glycol is 7.4°C. Calculate the mass of ethylene glycol added in 500 g water.

($K_f = 1.86 \text{ K kg mol}^{-1}$).

- (1) 23.3 g
- (2) 233.3 g
- (3) 123.3 g
- (4) 12.33 g

42. The molality of a solution containing 3g CH₃COOH in 50 g benzene is :-

- (1) 0.1 m
- (2) 0.5 m
- (3) 0.2 m
- (4) 1 m

AIIMS 2018

43. 45 g non volatile solute A is dissolved in 600 g of water if depression in freezing point is 2.2 K. Then what will be molar mass of solute :-

- (1) 126.8
- (2) 31.7
- (3) 63.4
- (4) 158.5

44. 1 g of a protein polymer having molar mass 16×10^4 is present in 800 mL solution at 300K then osmotic pressure (in pascal) is [$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]:-

- (1) 0.4
- (2) 0.02
- (3) 0.04
- (4) 0.2

45. For 1% H₂SO₄ solution find it's normality (approx)

- (1) 0.1
- (2) 0.2
- (3) 0.4
- (4) 0.02

46. 0.04 molal aqueous monoprotic acid has depression in freezing point equal to 0.1 K then find vant Hoff factor for acid :-

- (1) 1
- (2) 2
- (3) 2.5
- (4) 1.34

47. Ethylene glycol $\left(\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array} \right)$ is used as an antifreezing

agent. Mass of ethylene glycol which should be added to 2 litre of water to prevent it from freezing at -24°C will be

(K_f for water = $1.86 \text{ K kg mol}^{-1}$)

- (1) 1600 g
- (2) 1200 g
- (3) 800 g
- (4) 400 g

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	2	3	4	2	1	2	4	2	3	2	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	3	1	3	3	2	4	1	3	2	1	1	1	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	2	2	2	2	3	1	3	1	4	3	4	3	4	2
Que.	46	47													
Ans.	4	1													

EXERCISE-III (Analytical Questions)

Check Your Understanding

- Which of the following solutions has the highest normality :-
 (1) 8 g of KOH per litre
 (2) 1 N phosphoric acid
 (3) 6 g of NaOH per 100 mL
 (4) 0.5 M H_2SO_4
- An aqueous solution of glucose is 10% in strength. The volume in which 2 g mole of it is dissolved will be :-
 (1) 18 litre (2) 3.6 litre
 (3) 0.9 litre (4) 1.8 litre
- One litre of N/2 HCl solution is heated in a beaker. It was observed that when the volume of the solution was reduced to 600 mL, 3.25 g of HCl is lost, calculate the normality of the new solution
 (1) 1.70 (2) 0.685
 (3) 0.342 (4) 0.171
- The mole fraction of the solute in one molal aqueous solution is :-
 (1) 0.027 (2) 0.036
 (3) 0.018 (4) 0.009
- Density of a 2.05 M solution of acetic acid in water is 1.02 g mL^{-1} . The molality of the solution is :
 (1) 1.14 mol kg^{-1} (2) 3.28 mol kg^{-1}
 (3) 2.28 mol kg^{-1} (4) 0.44 mol kg^{-1}
- What will be density (in g mL^{-1}) of 3.60 molar sulphuric acid having 29% by mass. (Molar mass = 98 g mol^{-1})
 (1) 1.88 (2) 1.22
 (3) 1.45 (4) 1.64
- The molality of a urea solution in which 0.0100g of urea, $[(\text{NH}_2)_2\text{CO}]$ is added to 0.3000 dm^3 of water at STP is :-
 (1) 0.555 m (2) $5.55 \times 10^{-4} \text{ m}$
 (3) 33.3 m (4) $3.33 \times 10^{-2} \text{ m}$
- The concentration of a solution of H_2O_2 is 6.8%(w/V) then the volume concentration of the solution is:-
 (1) 22.4 (2) 11.2
 (3) 20 (4) 5
- A gaseous mixture was prepared by taking equal mole of CO and N_2 . If the total pressure of the mixture was found 1 atmosphere, the partial pressure of the nitrogen (N_2) in the mixture is :
 (1) 0.5 atm (2) 0.8 atm
 (3) 0.9 atm (4) 1 atm
- The vapour pressure of two liquids 'P' and 'Q' are 80 and 60 torr, respectively. The total vapour pressure of solution obtained by mixing 3 mole of P and 2 mol of Q would be :-
 (1) 68 torr (2) 140 torr
 (3) 72 torr (4) 20 torr
- Which of the following plots does not represent the behaviour of an ideal binary liquid solution :-
 (1) Plot of P_A versus X_A (mole fraction of A in liquid phase) is linear
 (2) Plot of P_B versus X_B is linear
 (3) Plot of p_{total} versus X_A (or X_B) is linear
 (4) Plot of p_{total} versus X_A is non linear
- The vapour pressure of two pure liquids (A) and (B) are 100 and 80 torr respectively. The total pressure of the solution obtained by mixing 2 mol of (A) and 3 mol of (B) would be
 (1) 20 torr (2) 36 torr
 (3) 88 torr (4) 180 torr
- For a solution of two liquids A and B, it was proved that $P = X_A (P_A^\circ - P_B^\circ) + P_B^\circ$. The solution is:-
 (1) Ideal
 (2) Non ideal
 (3) Both (1) and (2)
 (4) None of the above
- Mole fraction of A vapours above solution in mixture of A and B ($X_A = 0.4$) will be :-
 ($P_A^\circ = 100 \text{ mm}$, $P_B^\circ = 200 \text{ mm}$)
 (1) 0.4 (2) 0.8
 (3) 0.25 (4) None
- The vapour pressure of pure benzene and toluene are 160 and 60 torr respectively. The mole fraction of toluene in vapour phase in contact with equimolar solution of benzene and toluene is:
 (1) 0.50 (2) 0.6
 (3) 0.27 (4) 0.73
- The vapour pressure of ethanol and methanol are 42.0 mm and 88.5 mm Hg respectively. An ideal solution is formed at the same temperature by mixing 46.0 g of ethanol with 16.0 g of methanol. The mole fraction of methanol in the vapour is :
 (1) 0.467 (2) 0.502
 (3) 0.513 (4) 0.556

17. A solution has a 1 : 4 mole ratio of pentane to hexane. The vapour pressures of the pure hydrocarbons at 20°C are 440 mm Hg for pentane and 120 mm Hg for hexane. The mole fraction of pentane in the vapour phase would be :-
 (1) 0.200 (2) 0.478
 (3) 0.549 (4) 0.786
18. A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be
 (1) 300 (2) 700
 (3) 360 (4) 350
19. A mixture of liquid showing positive deviation in Raoult's law is :-
 (1) $(\text{CH}_3)_2\text{CO} + \text{C}_2\text{H}_5\text{OH}$
 (2) $(\text{CH}_3)_2\text{CO} + \text{CHCl}_3$
 (3) $(\text{C}_2\text{H}_5)_2\text{O} + \text{CHCl}_3$
 (4) $(\text{CH}_3)_2\text{CO} + \text{C}_6\text{H}_5\text{NH}_2$
20. An ideal solution was obtained by mixing methanol and ethanol. If the partial vapour pressure of methanol and ethanol are 2.619 K Pa and 4.556 K Pa respectively, the composition of vapour (in terms of mole fraction) will be -
 (1) 0.635 MeOH, 0.365 EtOH
 (2) 0.365 MeOH, 0.635 EtOH
 (3) 0.574 MeOH, 0.326 EtOH
 (4) 0.173 MeOH, 0.827 EtOH
21. Azeotropic mixture are :
 (1) Mixture of two solids
 (2) Those which boil at different temperatures
 (3) Those which can be fractionally distilled
 (4) Constant boiling mixtures
22. An azeotropic mixture of two liquids boil at a lower temperature than either of them when
 (1) It is saturated
 (2) It does not deviate from Raoult's law
 (3) It shows negative deviation from Raoult's law
 (4) It shows positive deviation from Raoult's law
23. The azeotropic mixture of water (B.P 100.15°C) and HCl (B.P. -85°C) boils at 108.5°C. When this mixture is distilled, it is possible to obtain :
 (1) Pure HCl
 (2) Pure water
 (3) Pure water as well as HCl
 (4) Neither HCl nor H_2O in their pure states
24. Colligative properties depend on the :-
 (1) Relative no. of solute molecules in solution and the nature of the solvent
 (2) Relative no. of solute molecules in solvent and the nature of solute
 (3) Relative no. of solute molecules and the nature of solute and solvent
 (4) Relative no. of solute molecules, irrespective of the nature of solvent and solute
25. The vapour pressure of a solvent decreases by 10 mm. of Hg when a non volatile solute was added to the solvent. The mole fraction of the solute in the solution is 0.2. What should be the mole fraction of the solvent if the decrease in vapour pressure is to be 20 mm of Hg :-
 (1) 0.2 (2) 0.4 (3) 0.6 (4) 0.8
26. The vapour pressure of a solution of 5 g. of non electrolyte in 100 g. of water at a particular temperature is 2985 Nm^{-2} . The vapour pressure of pure water at that temperature is 3000 Nm^{-2} . The molecular weight of the solute is :-
 (1) 180 (2) 90
 (3) 270 (4) 200
27. How many gram of a non volatile solute having a molecular weight of 90 are to be dissolved in 97.5 g water in order to decrease the vapour pressure of water by 2.5 percent :-
 (1) 25 (2) 18
 (3) 12.5 (4) 9
28. The vapour pressure of a pure liquid solvent (X) is decreased to 0.60 atm. from 0.80 atm on addition of a non volatile substance (Y). The mole fraction of (Y) in the solution is :-
 (1) 0.20 (2) 0.25
 (3) 0.5 (4) 0.75
29. 18 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is added to 178.2 g of water. The vapour pressure of this aqueous solution at 100 °C is :
 (1) 759.00 torr (2) 7.60 torr
 (3) 76.00 torr (4) 752.40 torr
30. Water is added to the solution such that the mole fraction of water in the solution becomes 0.9. The boiling point of the solution is
 (1) 354.7 K (2) 375.5 K
 (3) 376.2 K (4) 380.4 K

- 31.** Glucose is added to 1 litre water to such an extent that $\frac{\Delta T_f}{K_f}$ becomes equal to $\frac{1}{1000}$, the wt. of glucose added is :-
 (1) 180 g (2) 18 g
 (3) 1.8 g (4) 0.18 g
- 32.** A solution of urea boils at 100.18°C at the atmospheric pressure. If K_f and K_b for water are 1.86 and 0.512 K kg mol⁻¹ respectively, the above solution will freeze at :-
 (1) -6.54°C (2) -0.654°C
 (3) 6.54°C (4) 0.654°C
- 33.** Ethylene glycol is used as an antifreeze in a cold climate. Mass of ethylene glycol which should be added to 4 kg of water to prevent it from freezing at -6°C will be : (K_f for water = 1.86 K kgmol⁻¹, and molar mass of ethylene glycol = 62 g mol⁻¹)
 (1) 400.00 g (2) 304.60 g
 (3) 800.00 g (4) 204.30 g
- 34.** K_f for water is 1.86 K kg mol⁻¹. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol (C₂H₆O₂) must you add to get the freezing point of the solution lowered to -2.8°C ?
 (1) 27 g (2) 72 g
 (3) 93 g (4) 39 g
- 35.** Osmosis of A into solution B will not take place if:-
 (1) A is hypertonic
 (2) A is hypotonic
 (3) A is isotonic
 (4) Either 1 or 3 may correct
- 36.** Insulin (C₂H₁₀O₅)_n is dissolved in a suitable solvent and the osmotic pressure (π) of solutions of various concentrations (g cm⁻³) C is measured at 20 °C. The slope of a plot of π against C is found to be 4.65×10^{-3} . The molecular weight of the insulin is
 (1) 4.8×10^5 (2) 9×10^5
 (3) 3×10^5 (4) 5.16×10^6
- 37.** The average osmotic pressure of human blood is 7.8 bar at 37°C. What is the concentration of an aqueous NaCl solution that could be used in the blood stream
 (1) 0.16 mol L⁻¹ (2) 0.32 mol L⁻¹
 (3) 0.60 mol L⁻¹ (4) 0.45 mol L⁻¹
- 38.** Camphor is often used in molecular mass determination because :-
 (1) It has a very high cryoscopic constant
 (2) It is volatile
 (3) It is solvent for organic substances
 (4) It is readily available
- 39.** A 5.25% solution of a substance is isotonic with a 1.5% solution of urea (molar mass=60g mol⁻¹) in the same solvent. If the densities of both the solutions are assumed to be equal to 1.0 gcm⁻³, molar mass of the substance will be.
 (1) 115.0 g mol⁻¹ (2) 105.0 g mol⁻¹
 (3) 210.0 g mol⁻¹ (4) 90.0 g mol⁻¹
- 40.** A 5% solution of cane sugar (molar mass 342) is isotonic with 1% of a solution of an unknown solute. The molar mass of unknown solute in g mol⁻¹ is :-
 (1) 136.2 (2) 171.2
 (3) 68.4 (4) 34.2
- 41.** The degree of dissociation (α) of a weak electrolyte, A_xB_y, is related to van't Hoff factor (i) by the expression :-
 (1) $\alpha = \frac{x+y-1}{i-1}$ (2) $\alpha = \frac{x+y+1}{i-1}$
 (3) $\alpha = \frac{i-1}{(x+y-1)}$ (4) $\alpha = \frac{i-1}{x+y+1}$
- 42.** The van't Hoff factor for 0.1 M Ba(NO₃)₂ solution is 2.74. The degree of dissociation is :-
 (1) 91.3% (2) 87%
 (3) 100% (4) 74%
- 43.** Arrange the following aqueous solutions in the order of their increasing boiling points :-
 (i) 10⁻⁴ M NaCl (ii) 10⁻⁴ M Urea
 (iii) 10⁻³ M MgCl₂ (iv) 10⁻² M NaCl
 (1) (i) < (ii) < (iv) < (iii) (2) (ii) < (i) = (iii) < (iv)
 (3) (ii) < (i) < (iii) < (iv) (4) (iv) < (iii) < (i) = (ii)
- 44.** The relationship between the values of osmotic pressure of solutions obtained by dissolving 6.00 gL⁻¹ of CH₃COOH (π_1) and 7.45 gL⁻¹ of KCl (π_2) is :-
 (1) $\pi_1 > \pi_2$ (2) $\pi_1 < \pi_2$
 (3) $\pi_1 = \pi_2$ (4) None of these

45. Among 0.1M solutions of urea, Na_3PO_4 and $\text{Al}_2(\text{SO}_4)_3$:-
 (a) The vapour pressure and freezing point are the lowest for urea
 (b) The vapour pressure and freezing point are the highest for urea
 (c) The elevation in boiling point is the highest for $\text{Al}_2(\text{SO}_4)_3$
 (d) The depression in freezing point is the highest for $\text{Al}_2(\text{SO}_4)_3$
 (1) Only a (2) b & c both
 (3) b, c and d (4) a, b, c and d
46. When equimolar aqueous solutions of glucose, sodium chloride and barium nitrate are compared the vapour pressure of the solutions will be in the following order :-
 (1) Glucose > NaCl > $\text{Ba}(\text{NO}_3)_2$
 (2) Glucose = NaCl = $\text{Ba}(\text{NO}_3)_2$
 (3) $\text{Ba}(\text{NO}_3)_2$ > NaCl > Glucose
 (4) NaCl > $\text{Ba}(\text{NO}_3)_2$ > Glucose
47. What is the freezing point of a solution containing 8.1 g of HBr in 100g water assuming the acid to be 90% ionised (K_f for water = $1.86 \text{ K molality}^{-1}$)
 (1) 0.85°C (2) -3.53°C
 (3) 0°C (4) -0.35°C
48. A 0.2 molal aqueous solution of a weak acid (HX) is 20% ionised. The elevation in boiling point of this solution is (given $K_b = 0.52^\circ\text{C kg mol}^{-1}$ for H_2O)
 (1) 0.81 (2) 0.125
 (3) 0.48 (4) 1.3
49. The substance when dissolved in water would decrease the vapour pressure of water to the greatest extent is :-
 (1) 0.1 M KCl (2) 0.1 M urea
 (3) 0.1 M BaCl_2 (4) 0.1 M NaCl
50. The molar mass of NaCl determined by the osmotic pressure method will be :-
 (1) Higher than the theoretical value
 (2) Lower than the theoretical value
 (3) The same as the theoretical value
 (4) None of these
51. Consider separate solution of 0.500 M $\text{C}_2\text{H}_5\text{OH}(\text{aq})$, 0.100 M $\text{Mg}_3(\text{PO}_4)_2(\text{aq})$, 0.250 M $\text{KBr}(\text{aq})$ and 0.125 M $\text{Na}_3\text{PO}_4(\text{aq})$ at 25°C . Which statement is true about these solutions, assuming all salts to be strong electrolytes ?
 (1) 0.125 M $\text{Na}_3\text{PO}_4(\text{aq})$ has the highest osmotic pressure.
 (2) 0.500 M $\text{C}_2\text{H}_5\text{OH}(\text{aq})$ has the highest osmotic pressure.
 (3) They all have the same osmotic pressure.
 (4) 0.100 M $\text{Mg}_3(\text{PO}_4)_2(\text{aq})$ has the highest osmotic pressure.

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	3	3	2	2	1	1	3	4	3	1	3	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	4	1	2	4	4	4	1	3	1	3	2	4	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	2	3	3	4	4	1	1	3	3	3	2	3	2	3
Que.	46	47	48	49	50	51									
Ans.	1	2	2	3	2	3									

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** The molality of the solution doesn't change with change in temperature.
Reason :- The molality is expressed in units of moles per 1000 mL solution.
 (1) A (2) B (3) C (4) D
- Assertion :-** The sum of the mole fraction of all the components is unity.
Reason :- Mole fraction is a temperature dependent mode of concentration.
 (1) A (2) B (3) C (4) D
- Assertion :-** A solution which contains one gram equivalent of solute per litre of the solution is called normal solution.
Reason :- A normal solution mean a solutions in which the solute doesn't associate or dissociate.
 (1) A (2) B (3) C (4) D
- Assertion :-** The molarity and normality of a solution of sodium carbonate are same.
Reason :- Normality is the product of molarity & valency factor.
 (1) A (2) B (3) C (4) D
- Assertion :-** Ideal solutions are one which obeys Raoult's law at all temperature and concentration.
Reason :- Very dilute solution can be treated as ideal solution.
 (1) A (2) B (3) C (4) D
- Assertion :-** Cooking time is reduced in pressure cookers.
Reason :- Boiling point inside the pressure cooker is raised.
 (1) A (2) B (3) C (4) D
- Assertion :-** In positive deviation of non-ideal solution attraction between solute solvent is less than the attraction between solute-solute and solvent-solvent.
Reason :- In negative deviation of non-ideal solution attraction between solute-solvent is more than the attraction between solute-solute and solvent-solvent.
 (1) A (2) B (3) C (4) D
- Assertion :-** Isotonic solutions don't show net phenomenon of osmosis.
Reason :- Isotonic solutions have equal osmotic pressure at constant temp.
 (1) A (2) B (3) C (4) D
- Assertion :-** When Benzoic acid is dissolved in benzene its vant Hoff factor is less than one.
Reason :- In benzene, benzoic acid has tendency to form dimer.
 (1) A (2) B (3) C (4) D
- Assertion :-** Non-ideal solutions form azeotropic mixture.
Reason :- Boiling point of azeotropic mixture is only higher than boiling points of both the components.
 (1) A (2) B (3) C (4) D
- Assertion :-** van't Hoff factor for benzoic acid in benzene is one.
Reason :- Benzoic acid behaves as a weaker electrolyte in benzene.
 (1) A (2) B (3) C (4) D
- Assertion :-** van't Hoff factor is always more or equal to one.
Reason :- van't Hoff factor is the ratio of experimental colligative properties to observed colligative properties.
 (1) A (2) B (3) C (4) D
- Assertion :-** Azeotropic mixture can't be separated by fractional distillation.
Reason :- Azeotropic mixtures are constant boiling mixtures.
 (1) A (2) B (3) C (4) D
- Assertion :-** ΔH_{mix} and ΔV_{mix} for the preparation of ideal solution is zero.
Reason :- A-B interaction in ideal solution are same as between A-A and B-B in the two liquids before mixing.
 (1) A (2) B (3) C (4) D

15. Assertion :- 0.1M solution of NaCl have low freezing point than 0.1M urea solution.

Reason :- Van't Hoff factor (No. of particles) for NaCl is more than urea.

- (1) A (2) B (3) C (4) D

16. Assertion :- A non volatile solute is mixed in a solution then elevation in boiling point and depression in freezing point both are 2K.

Reason :- Elevation in boiling point and depression in freezing point both depend on melting point of non-volatile solute

- (1) A (2) B (3) C (4) D

17. Assertion : Elevation in boiling point of 0.1 M KCl and 0.1 M CaCl_2 is same.

Reason : Because colligative properties depends on molarity only.

- (1) A (2) B (3) C (4) D

18. Assertion :- On mixing aniline and Acetone shows +ve deviation.

Reason :- Interaction between Acetone and aniline is weaker than interaction of Acetone-Acetone and aniline-aniline.

- (1) A (2) B (3) C (4) D

19. Assertion :- When two volatile liquids are mixed then vapour pressure of one increases and other decreases.

Reason :- Vapour pressure of solution decreases.

- (1) A (2) B (3) C (4) D

20. Assertion :- At high altitude cooking food is difficult.

Reason :- Water boils at high temperature.

- (1) A (2) B (3) C (4) D

21. Assertion :- After adding non-volatile solute freezing point of solvent depressed.

Reason :- Vapour pressure of solvent decreases and become equal to solid solvent, at temperature lower than the freezing point of pure solvent.

- (1) A (2) B (3) C (4) D

22. Assertion : On increasing pressure solubility of gases in liquid solvent increases.

Reason : Dissolution of gases in liquid is exothermic.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	3	4	2	1	2	1	1	3	4	4	1	1	1
Que.	16	17	18	19	20	21	22								
Ans.	4	4	4	4	3	1	2								

ELECTROCHEMISTRY

E

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NEET SYLLABUS

Electrochemistry : Conductance in electrolytic solutions, specific and molar conductivity variation of conductivity with concentration, Kohlrausch's Law, Galvanic cells, Redox reactions, EMF of a cell, standard electrode potential, Relation between Gibbs energy change and EMF of a cell, electrolysis and electrolytic cells, Laws of electrolysis (elementary idea), dry cell, lead accumulator, fuel cells, corrosion.

OBJECTIVES

After studying this unit, we will be able to :

- describe an electrochemical cell and differentiate between galvanic and electrolytic cells;
- apply Nernst equation for calculating the emf of galvanic cell and define standard potential of the cell;
- derive relation between standard potential of the cell, Gibbs energy of cell reaction and its equilibrium constant;
- define resistivity (ρ), conductivity (κ) and molar conductivity (Λ_m) of ionic solutions;
- differentiate between ionic (electrolytic) and electronic conductivity;
- describe the methods for measurement of conductivity of electrolytic solutions and calculation of their molar conductivity;
- justify the variation of conductivity and molar conductivity of solutions with change in their concentration and define Λ_m^∞ (molar conductivity at zero concentration or infinite dilution);
- enunciate Kohlrausch law and learn its applications;
- understand quantitative aspects of electrolysis;
- describe the construction of some primary and secondary batteries and fuel cells;
- explain corrosion as an electrochemical process.

In arranging the bodies in order of their electrical nature, there is formed an electro-chemical system which, in my opinion is more fit than any other to give an idea of Chemistry.

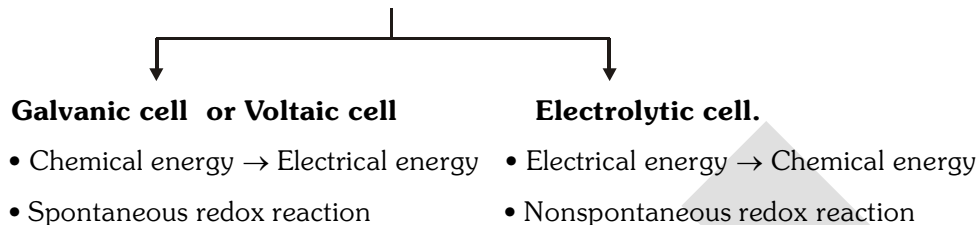
Jons Jacob Berzelius

ELECTROCHEMISTRY

4.0 INTRODUCTION

Electrochemistry is the branch of physical chemistry which deals with the study of interconversion of chemical energy and electrical energy

ELECTROCHEMICAL CELL



4.1 CONDUCTORS

Substances which allow electric current to flow through them are called conductors.

Examples - Metals, Aqueous solution of acids, bases and salts, fused salts and impure water etc.

Conductors are of two types - (i) Metallic conductors or electronic conductors
(ii) Electrolytic conductors or ionic conductors

(i) Metallic conductors : The conductors which conduct electric current by the movement of electrons without undergoing any chemical change are known as metallic conductors.

Metals - Cu, Ag, Fe, Al etc, non metals - graphite

(ii) Electrolytic conductors : Those substances which conduct the electric current through ions in aqueous solution or in molten state are called electrolytic conductors.

- **Strong electrolyte :** Electrolytes which are completely ionized in aqueous solution are called strong electrolytes.

Ex. : Salts, strong acids and strong bases.

- **Weak electrolyte :-** Electrolytes which are not completely ionized in aqueous solution are called weak electrolytes.

Ex. : CH_3COOH , HCN , NH_3 , amine, etc.

Difference between metallic and electrolytic conductor

Metallic conductor	Electrolytic conductor
(i) Charge carriers are free electrons.	Charges carriers are free ions.
(ii) Flow of electricity takes place without any chemical change.	Flow of electricity takes place by chemical changes at electrodes.
(iii) No transfer of matter takes place.	Transfer of matter takes place in the form of ions.
(iv) Resistance is due to vibration of Kernels.	Resistance is due to inter ionic attraction and viscosity of medium.
(v) The resistance increases with the increase in temperature.	The resistance decreases with the increase in temperature.
(vi) Faraday's laws of electrolysis are not followed.	Faraday's laws of electrolysis are followed.

4.2 ELECTROLYTIC CONDUCTANCE

- (a) **Resistance (R)** :- Metallic and electrolytic conductors obey ohm's law according to which the resistance of a conductor is the ratio of the applied potential difference (V) to the current (I) flowing .

$$R = \frac{V}{I}$$

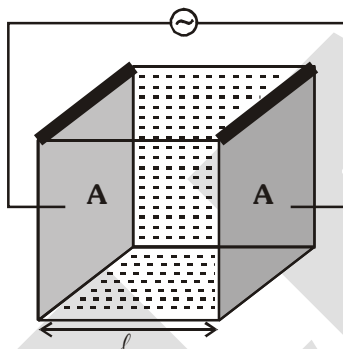
R is expressed in ohms.

- (b) **Conductance (G)** :- It is the property by virtue of which it favours flow of current. The conductance of a conductor is equal to reciprocal of resistance $G = \frac{1}{R}$

unit of G is **mho** or **ohm⁻¹** or **Siemens (S)**.

- (c) **Specific resistance/Resistivity (ρ)** :- The resistance (R) of a conductor is directly proportional to its length (ℓ) and inversely proportional to its area of cross section (A).

$$R \propto \frac{\ell}{A} \quad ; \quad R = \rho \frac{\ell}{A}$$



$$R = \rho \cdot \frac{\ell}{A} \quad ; \quad \rho = R \cdot \frac{A}{\ell}$$

If $\ell = 1 \text{ cm}$, $A = 1 \text{ cm}^2$, therefore $\rho = R$

$$\therefore V = A \times \ell = 1 \text{ cm}^3$$

Therefore resistance offered by 1 cm^3 electrolytic solution is known as resistivity.

Unit of $\rho \rightarrow \text{ohm cm}$

- (d) **Specific conductance/ Conductivity (κ)** :-

It is defined as the reciprocal of specific resistance, $\kappa = \frac{1}{\rho}$

$$R = \rho \frac{\ell}{A}$$

$$\frac{1}{\rho} = \frac{1}{R} \cdot \frac{\ell}{A}$$

$$\kappa = G \times G^*$$

Specific conductance = Conductance $\times$ Cell constant

Hence specific conductivity of a solution is defined as the conductance offered by 1 cm^3 of electrolytic solution. Unit of κ is $\text{ohm}^{-1} \text{ cm}^{-1}$

Cell constant :

$$G^* = \frac{\ell}{A} \quad ; \text{ Its unit is } \text{cm}^{-1}$$

(e) **Molar conductivity or Molar conductance** :- (Λ_m , λ_m or μ) : It is defined as the conductance of all the ions produced by one mole of electrolyte present in the given volume of solution.

$$\Lambda_m = \kappa \times V$$

V = Volume of solution containing 1 mol of electrolyte.

If concentration of solution is M - mol per litre then

$$\Lambda_m = \frac{\kappa \times 1000}{M} \quad \text{Unit} \rightarrow \text{ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

(f) **Equivalent conductivity or Equivalent conductance** (Λ_{eq} or λ_{eq}) :- It is defined as the conductance of all the ions produced by one gram equivalent of the electrolyte in the given volume of solution.

$$\therefore \Lambda_{eq} = \kappa \times V$$

V = Volume of solution containing 1 g-eq of electrolyte.

If concentration of solution is N - gram equivalent per litre then

$$\Lambda_{eq} = \frac{\kappa \times 1000}{N} \quad \text{Unit} \rightarrow \text{ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

● **Relation between Λ_{eq} and Λ_m**

$$\Lambda_m = \frac{\kappa \times 1000}{M} \quad \text{and} \quad \Lambda_{eq} = \frac{\kappa \times 1000}{N}$$

We know that

Normality = Valency Factor $\times$ Molarity . So

$$\Lambda_{eq} = \frac{\Lambda_m}{V.F.}$$

Illustrations

Illustration 1. The resistance of a 1N solution of salt is 50 Ω . Calculate the equivalent conductance of the solution, if the two platinum electrodes in solution are 2.1 cm apart and each having an area of 4.2 cm^2 .

Solution. $\kappa = \frac{1}{\rho} = \frac{1}{R} \left(\frac{\ell}{A} \right) = \frac{1}{50} \times \frac{2.1}{4.2} = \frac{1}{100}$ and $\Lambda_{eq} = \frac{\kappa \times 1000}{N} = \frac{1}{100} \times \frac{1000}{1} = 10 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$

Illustration 2. Which of the following have maximum molar conductivity.

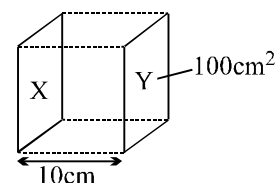
- (i) 0.08 M solution and its specific conductivity is $2 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}$.
(ii) 0.1 M solution and its resistivity is 50 $\Omega \text{ cm}$.

Solution. (i) $\Lambda_m = \frac{\kappa \times 1000}{M} = 2 \times 10^{-2} \times \frac{1000}{0.08} = 250 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
(ii) $\Lambda_m = \frac{\kappa \times 1000}{M}$, $\therefore \kappa = \frac{1}{\rho}$, $\therefore \Lambda_m = \frac{1}{50} \times \frac{1000}{0.1} = 200 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

So, the molar conductivity of 0.08 M solution will be greater than 0.1 M solution

Illustration 3. The conductance of a salt solution (AB) measured by two parallel electrodes of area 100 cm^2 separated by 10 cm was found to be 0.0001 Ω^{-1} . If volume enclosed between two electrodes contain 0.5 mol of salt. What is the molar conductivity ($\text{Scm}^2 \text{ mol}^{-1}$) of salt at same concentration.

- (A) 0.01 (B) 0.02 (C) 2×10^{-5} (D) none of these



Solution

$$G^* = \frac{\ell}{a} = \frac{10}{100} = 0.1; \quad G = 0.0001 \text{ S}; \quad V = 100 \times 10 = 1000 \text{ cm}^3 = 1 \text{ litre}$$

$$\kappa = G G^* = 0.1 \times 0.0001 = 10^{-5}$$

$$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{(0.1 \times 0.0001) \times 1000}{0.5} = 0.02 \text{ S cm}^2 \text{ mol}^{-1}$$

4.3 FACTORS AFFECTING ELECTROLYTIC CONDUCTANCE

- (a) **Inter ionic attraction :-** If inter ionic attraction between ions of solute is more, then the conductance will be less.
- (b) **Polarity of solvent :-** If solvent has high-dielectric constant then the ionization and conductance will be higher.
- (c) **Viscosity of medium :-** On increasing the viscosity of medium, the conductance decreases.
- (d) **Temperature :-** As the temperature of electrolytic solution is increased, the conductance increases because K.E. of the ions increases and all types of attraction forces decrease and the viscosity of medium decreases.
- (e) **Hydrated size :** Due to hydration of ions conductance decreases.
- (f) **Dilution :-**
- On increasing the dilution conductance (G) increases.
For strong electrolyte on dilution interionic force of attraction decreases therefore conductance increases.
For weak electrolyte with dilution degree of dissociation (α) increases therefore conductance increases.
 - On dilution specific conductance decreases because on dilution number of ions in 1 ml solution decreases.
 - On dilution equivalent and molar conductance increases because with dilution normality or molarity decreases

DETERMINATION OF MOLAR CONDUCTANCE OF ELECTROLYTES AT INFINITE DILUTION

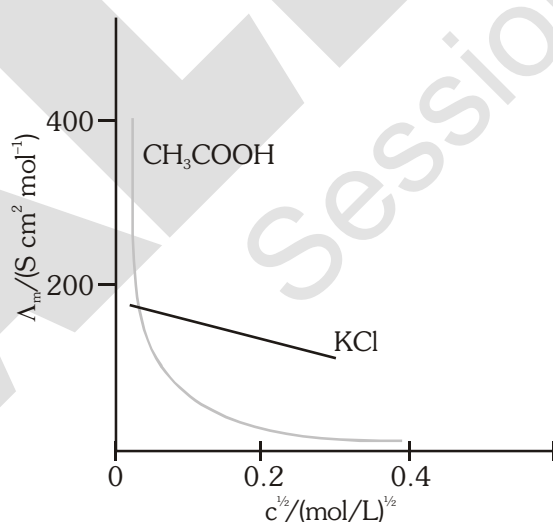
Debye Huckel Onsager equation –

$$\Lambda_m = \Lambda_m^\infty - b\sqrt{C} \quad [\text{only for strong electrolyte}]$$

where Λ_m = molar conductance at concentration C ,

Λ_m^∞ = molar conductance at ∞ dilution,

b = constant and its value is same for a particular type of electrolyte.



If a graph is plotted between Λ_m vs $\sqrt{C}$ a straight line is obtained for strong electrolyte. On extrapolation this line touches Λ_m axis. Therefore Λ_m^∞ for strong electrolyte can be found out from intercept. For weak electrolyte the slope of curve is not constant so it can not be extrapolated to obtain Λ_m^∞ .

4.4 KOHLRAUSCH'S LAW OF INDEPENDENT MIGRATION OF IONS

At infinite dilution when dissociation of electrolyte is complete; each ion makes a definite contribution towards equivalent conductivity of any electrolyte irrespective of the nature of other ion associated with it.

This law states that the equivalent conductivity of any electrolyte at infinite dilution (Λ_{eq}^{∞}) is the sum of equivalent ionic conductivities of the cation and anion given by the electrolytes at infinite dilution.

$$\Lambda_{eq}^{\infty} = \lambda_c^{\infty} + \lambda_a^{\infty}$$

Where λ_c^{∞} = equivalent conductance of cation at infinite dilution.

λ_a^{∞} = equivalent conductance of anion at infinite dilution.

For $A_x B_y \rightarrow xA^{y+} + yB^{x-}$

In terms of molar conductances of cation and anion

$$\Lambda_{eq}^{\infty} = \frac{1}{y} \lambda_{m(c)}^{\infty} + \frac{1}{x} \lambda_{m(a)}^{\infty}$$

Where $\lambda_{m(c)}^{\infty}$ = Limiting molar conductance of cation .

$\lambda_{m(a)}^{\infty}$ = Limiting molar conductance of anion .

y = charge on cation

x = charge on anion

$$\Lambda_m^{\infty} = x\lambda_{m(c)}^{\infty} + y\lambda_{m(a)}^{\infty}$$

Where x = Stoichiometric coefficient of cation

y = Stoichiometric coefficient of anion

Illustrations

Illustration 4. Calculate Λ_m^{∞} of oxalic acid, given that

$$\Lambda_{eq}^{\infty} Na_2C_2O_4 = 400 \Omega^{-1} cm^2 eq^{-1}$$

$$\Lambda_m^{\infty} H_2SO_4 = 700 \Omega^{-1} cm^2 mol^{-1}$$

$$\Lambda_{eq}^{\infty} Na_2SO_4 = 450 \Omega^{-1} cm^2 eq^{-1}$$

Solution

$$\Lambda_m^{\infty} H_2C_2O_4 = 700 + 800 - 900 = 600 \Omega^{-1} cm^2 mol^{-1}$$

4.5 APPLICATIONS OF KOHLRAUSCH'S LAW :

(a) To calculate conductance of weak electrolyte.

(b) To calculate degree of dissociation :

$$\text{Degree of dissociation } \alpha = \frac{\Lambda_{eq}}{\Lambda_{eq}^{\infty}} = \frac{\text{equivalent conductance at a given concentration}}{\text{equivalent conductance at infinite dilution}}$$

$$\text{or } \alpha = \frac{\Lambda_m}{\Lambda_m^{\infty}} = \frac{\text{molar conductance at a given concentration}}{\text{molar conductance at infinite dilution}}$$

(c) To calculate dissociation constant of weak electrolyte

$$K_a = \frac{C\alpha^2}{(1-\alpha)}$$

If $\alpha \leq 5\%$ then $K_a = C\alpha^2$

(d) To calculate Solubility (S) and K_{sp} of any sparingly soluble salt.

Sparingly soluble salt means salt with very less solubility

Saturated solution of sparingly soluble salt can be considered as infinitely dilute solution.

$$\text{In a saturated solution of sparingly soluble salt. } \Lambda_m = \frac{\kappa \times 1000}{M}$$

M = Solubility (S).

$$\Lambda_m (\text{saturated}) = \Lambda_m^\infty = \frac{\kappa \times 1000}{\text{Solubility}}; S = \frac{\kappa \times 1000}{\Lambda_m^\infty}$$

GOLDEN KEY POINTS

- Ionic conductance (λ) and mobility of ions (u)**

In electrolyte solution the ionic conductance of any ion is directly proportional to their mobility or speed of ion.

$$\lambda \propto u$$

$$\text{for a cation } \lambda_c \propto u_c$$

$$\text{or } \lambda_c = F u_c \quad \text{where } F = 96500 \text{ C}$$

$$\text{for an anion } \lambda_a \propto u_a$$

$$\text{or } \lambda_a = F u_a$$

- Transport number/Hittorf's number(t)**

The current flowing through an electrolytic solution is carried by ions (cation and anion).

The fraction of total current carried by an ion is called its transport number.

$$\text{transport number of cation } (t_c) = \frac{\lambda_c}{\lambda_c + \lambda_a} = \frac{u_c}{u_c + u_a}$$

$$\text{transport number of anion } (t_a) = \frac{\lambda_a}{\lambda_c + \lambda_a} = \frac{u_a}{u_c + u_a}$$

$$\text{and } t_c + t_a = 1$$

$$\text{if } u_c = u_a \text{ then } t_c = t_a = 0.5$$

Illustrations

- Illustration 5.** Equal volumes of 0.015 M CH_3COOH & 0.015 M NaOH are mixed together. What would be the molar conductivity of mixture if conductivity of CH_3COONa is $6.3 \times 10^{-4} \text{ S cm}^{-1}$
- (1) $8.4 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $84 \text{ S cm}^2 \text{ mol}^{-1}$
 (3) $4.2 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $42 \text{ S cm}^2 \text{ mol}^{-1}$

Solution $[\text{Salt}] = \frac{0.015}{2} \text{ M}; \Lambda_m = \frac{6.3 \times 10^{-4} \times 1000}{0.015/2} = 84 \text{ S cm}^2 \text{ mol}^{-1}$

- Illustration 6.** The dissociation constant of n-butyric acid is 1.6×10^{-5} and the molar conductivity at infinite dilution is $380 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$. The specific conductance of the 0.01 M acid solution is
- (1) $1.52 \times 10^{-5} \text{ S m}^{-1}$ (2) $1.52 \times 10^{-2} \text{ S m}^{-1}$
 (3) $1.52 \times 10^{-3} \text{ S m}^{-1}$ (4) None

Solution $K_a = \frac{c\alpha^2}{1-\alpha} \Rightarrow 1.6 \times 10^{-5} = \frac{0.01 \times \alpha^2}{1-\alpha}$

$$\alpha = \sqrt{\frac{1.6 \times 10^{-5}}{0.01}} = \sqrt{1.6 \times 10^{-3}} = 0.04; \alpha = \frac{\Lambda_m}{\Lambda_m^\infty}; \Lambda_m = 0.04 \times 380 \times 10^{-4}$$

$$\Lambda_m = \frac{\kappa \times 10^{-3}}{M}$$

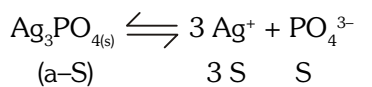
$$\kappa = \frac{0.04 \times 380 \times 10^{-4} \times 0.01}{10^{-3}} = 1.52 \times 10^{-2} \text{ S m}^{-1}$$

Illustration 7. The conductivity of a saturated solution of Ag_3PO_4 is $9 \times 10^{-6} \text{ S m}^{-1}$ and its equivalent conductivity is $1.50 \times 10^{-4} \text{ S m}^2 \text{ eq}^{-1}$. The K_{sp} of Ag_3PO_4 is

- (1) 4.32×10^{-18} (2) 1.8×10^{-9} (3) 8.64×10^{-13} (4) None of these

Solution $\Lambda_{\text{eq}} = \kappa \times \frac{1000}{N}$; $1.50 \times 10^{-4} \times 10^4 = 9 \times 10^{-6} \times 10^{-2} \times \frac{1000}{N}$; $N = 6 \times 10^{-5}$

$$S = M = \frac{N}{n_f} = \frac{6 \times 10^{-5}}{3} = 2 \times 10^{-5} \text{ mol L}^{-1}$$



$$K_{\text{sp}} = (3S)^3 \cdot S = 27S^4$$

$$= 27 \times (2 \times 10^{-5})^4 = 4.32 \times 10^{-18}$$

BEGINNER'S BOX-1

1. The resistance of 1 N solution of CH_3COOH is 250Ω , when measured in a cell of cell constant 1.15 cm^{-1} . The equivalent conductance will be

- (1) $4.6 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$ (2) $9.2 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$
(3) $18.4 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$ (4) $0.023 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$

2. The specific conductivity of N/10 KCl solution at 20°C is $0.012 \Omega^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing this solution at 20°C is 56Ω . The cell constant is

- (1) 4.616 cm^{-1} (2) 0.672 cm^{-1} (3) 2.173 cm^{-1} (4) 3.324 cm^{-1}

3. Molar conductances of BaCl_2 , H_2SO_4 and HCl at infinite dilutions are x_1 , x_2 and x_3 , respectively. Equivalent conductance of BaSO_4 at infinite dilution will be :

- (1) $\frac{[x_1 + x_2 - x_3]}{2}$ (2) $\frac{[x_1 - x_2 - x_3]}{2}$
(3) $2(x_1 + x_2 - 2x_3)$ (4) $\frac{[x_1 + x_2 - 2x_3]}{2}$

4. The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01 M solution is $19.6 \text{ S cm}^2 \text{ eq}^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in $\text{S cm}^2 \text{ eq}^{-1}$) will be

- (1) 39.2 (2) 78.4 (3) 392 (4) 196

5. Which has maximum conductivity :

- (1) $[\text{Cr}(\text{NH}_3)_3 \text{Cl}_3]$ (2) $[\text{Cr}(\text{NH}_3)_4 \text{Cl}_2]\text{Cl}$ (3) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (4) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$

4.6 ELECTRODE POTENTIAL :

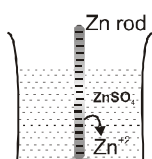
- The potential developed between metal electrode and its ions in solution is known as electrode potential.
- The potential developed between metal electrodes and the solution of its ions at 1 M concentration at 1 bar pressure and 298 K is known as **standard electrode potential**.
- There are two types of electrode potential :-

(a) Oxidation Potential (O.P.)

- The electrode potential for oxidation half reaction



- Tendency to get oxidised.
- **Greater the O.P. greater will be the tendency to get oxidised.**

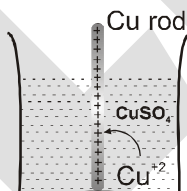


(b) Reduction Potential (R.P.)

- The electrode potential for reduction half reaction



- Tendency to get reduced.
- **Greater the R.P. greater will be the tendency to get reduced.**



- Electrode potential depends upon :
 - Concentration of the solution.
 - Nature of the metal.
 - Pressure temperature conditions.

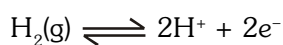
4.7 REFERENCE ELECTRODE :

- The potential of a single electrode cannot be determined but the potential difference between two electrodes can be accurately measured using a reference electrode.
- An electrode is chosen as a reference with respect to which all other electrodes are valued.
- There are two types of reference electrodes

(a) **Primary reference electrode : Standard hydrogen electrode (SHE)**

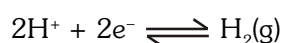
Representation : $\text{Pt}, \text{H}_2(\text{g}) \mid \text{H}^+ (1\text{M})$

When acts as anode

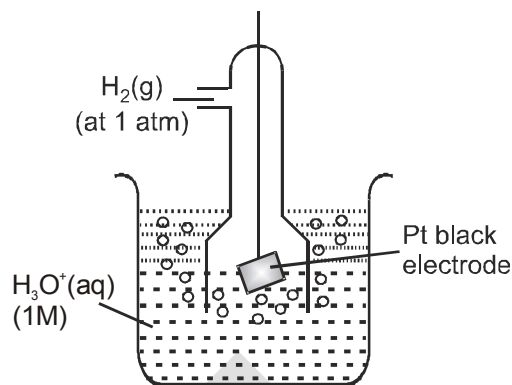


$$E^0_{\text{H}_2(\text{g})/\text{H}^+(\text{aq})} = \text{SOP}$$

When acts as cathode



$$E^0_{\text{H}^+/\text{H}_2(\text{g})} = \text{SRP}$$

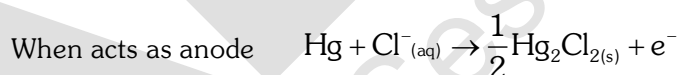
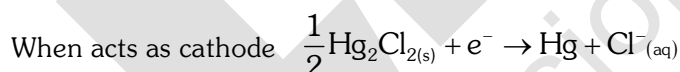


- For SHE electrode potential is arbitrarily assumed to be zero at all temperature.
 $\text{SOP} = -\text{SRP} = 0$ for SHE.
- To calculate standard potential of any other electrode a cell is coupled with standard hydrogen electrode (SHE) and its potential is measured that gives the value of electrode potential of that electrode.

(b) **Secondary reference electrode :**

(i) **Calomel electrode (Hg_2Cl_2) -**

Representation : $\text{Pt} \mid \text{Hg}-\text{Hg}_2\text{Cl}_{2(\text{s})} \mid \text{KCl}_{(\text{aq})}$



(ii) **Silver-silver chloride electrode -**

Representation : $\text{Pt} \mid \text{Ag}-\text{AgCl}_{(\text{s})} \mid \text{KCl}_{(\text{aq})}$



4.8 CELL POTENTIAL (E^0_{cell}) :

$$E^0_{\text{cell}} = \text{SRP of cathode} - \text{SRP of anode}$$

$$E^0_{\text{cell}} = \text{SRP of cathode} + \text{SOP of anode}$$

$$E^0_{\text{cell}} = \text{SOP of anode} - \text{SOP of cathode}$$

4.9 ELECTROCHEMICAL SERIES

Arrangement of different elements on the basis of their SRP values is known as electrochemical series or activity series.

Electrode	Reaction	SRP (at 298 K)
Li	$\text{Li}^+ + e^- \rightarrow \text{Li(s)}$	- 3.05 V
K	$\text{K}^+ + e^- \rightarrow \text{K (s)}$	- 2.93 V
Ba	$\text{Ba}^{+2} + 2e^- \rightarrow \text{Ba}$	- 2.91 V
Ca	$\text{Ca}^{+2} + 2e^- \rightarrow \text{Ca(s)}$	- 2.87 V
Na	$\text{Na}^+ + e^- \rightarrow \text{Na(s)}$	- 2.71 V
Mg	$\text{Mg}^{+2} + 2e^- \rightarrow \text{Mg(s)}$	- 2.36 V
Al	$\text{Al}^{3+} + 3e^- \rightarrow \text{Al}$	- 1.66 V
Mn	$\text{Mn}^{+2} + 2e^- \rightarrow \text{Mn}$	- 1.18 V
H ₂ O	$\text{H}_2\text{O(l)} + e^- \rightarrow \frac{1}{2}\text{H}_2 + \text{OH}^-$	- 0.828 V
Zn	$\text{Zn}^{+2} + 2e^- \rightarrow \text{Zn(s)}$	- 0.76 V
Cr	$\text{Cr}^{+3} + 3e^- \rightarrow \text{Cr(s)}$	- 0.74 V
Fe	$\text{Fe}^{2+} + 2e^- \rightarrow \text{Fe}$	- 0.44 V
Cd	$\text{Cd}^{+2} + 2e^- \rightarrow \text{Cd(s)}$	- 0.40 V
Co	$\text{Co}^{2+} + 2e^- \rightarrow \text{Co}$	- 0.28 V
Ni	$\text{Ni}^{+2} + 2e^- \rightarrow \text{Ni(s)}$	- 0.25 V
Sn	$\text{Sn}^{+2} + 2e^- \rightarrow \text{Sn(s)}$	- 0.14 V
Pb	$\text{Pb}^{+2} + 2e^- \rightarrow \text{Pb(s)}$	- 0.13 V
H ₂	$2\text{H}^+ + 2e^- \rightarrow \text{H}_2\text{(g)}$	0.00 V
Cu	$\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu(s)}$	0.34 V
I ₂	$\text{I}_2 + 2e^- \rightarrow 2\text{I}^-$	0.54 V
Fe	$\text{Fe}^{3+} + e^- \rightarrow \text{Fe}^{2+}$	0.77 V
Hg	$\text{Hg}_2^{2+} + 2e^- \rightarrow \text{Hg(l)}$	0.79 V
Ag	$\text{Ag}^+ + e^- \rightarrow \text{Ag}$	0.80 V
Hg	$\text{Hg}^{2+} + 2e^- \rightarrow \text{Hg(l)}$	0.85 V
Br ₂	$\text{Br}_2 + 2e^- \rightarrow 2\text{Br}^-$	1.09 V
Pt	$\text{Pt}^{+2} + 2e^- \rightarrow \text{Pt}$	1.20 V
O ₂	$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2e^- \rightarrow \text{H}_2\text{O(l)}$	1.23 V
Cl ₂	$\text{Cl}_2 + 2e^- \rightarrow 2\text{Cl}^-$	1.36 V
Au	$\text{Au}^{+3} + 3e^- \rightarrow \text{Au(s)}$	1.40 V
F ₂	$\text{F}_2 + 2e^- \rightarrow 2\text{F}^-$	2.87 V

Increasing strength of reducing agent

Increasing strength of oxidising agent

4.10 APPLICATIONS OF ELECTROCHEMICAL SERIES

(i) Oxidising and reducing power –

$$\text{Oxidising power} \propto \text{SRP} \propto \frac{1}{\text{SOP}}$$

$$\text{Reducing power} \propto \text{SOP} \propto \frac{1}{\text{SRP}}$$

(ii) **Reactivity of metals –**

$$\text{Reactivity of metal} \propto \text{SOP} \propto \frac{1}{\text{SRP}}$$

- In ECS reactivity of metal decreases from top to bottom.
- Li is most reactive metal.

(iii) **Reactivity of non-metals –**

$$\text{Reactivity of non-metal} \propto \text{SRP} \propto \frac{1}{\text{SOP}}$$

- In ECS reactivity of non-metal increases from top to bottom.
- F_2 is most reactive non-metal.

(iv) **Displacement reactions in solution –**

More reactive metal / non-metal displaces less reactive metal / non-metal in their solution.

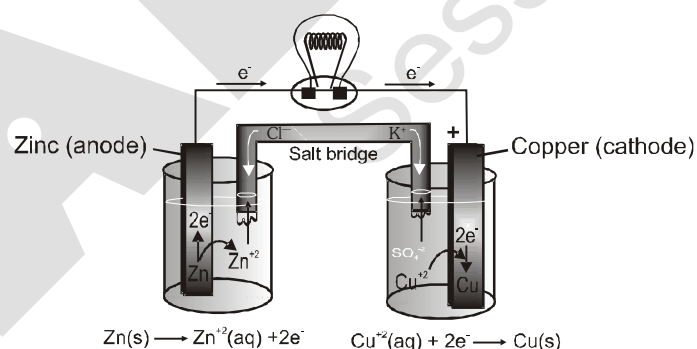
- $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$
 $\text{Ag} + \text{Cu}(\text{NO}_3)_2 \rightarrow \text{No reaction}$
- $2\text{KI} + \text{Cl}_2 \rightarrow 2\text{KCl} + \text{I}_2$
 $\text{I}_2 + 2\text{KCl} \rightarrow \text{No reaction}$

(v) **Metal above hydrogen displaces H_2 from dilute acid solution.**

- $\text{Zn} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{ZnSO}_4 + \text{H}_2(\text{g})$
 $\text{Cu} + \text{H}_2\text{SO}_4(\text{dilute}) \rightarrow \text{No reaction}$

4.11 GALVANIC CELL :

- It has two **half-cells**, each having a beaker containing a metal strip that dips in its aqueous salt solution.
- The metal strips are called **electrodes** and are connected by a conducting wire.
- Two solutions are connected by a **salt bridge**.
- The oxidation and reduction half reactions occur at separate electrodes and electric current flows through the wire.



Salt Bridge and its functions :

Salt bridge is inverted U-tube containing solution of inert electrolyte with agar-agar or gelatin to convert into semi solid form.

Ions of inert electrolyte do not involve in any chemical change.

The electrolyte in salt bridge should be such that speed of it's cation is nearly equal to speed of it's anion.

Ex. KCl , KNO_3 , NH_4NO_3

If Ag^+ , Hg_2^{+2} , Pb^{+2} , Tl^+ ions are present in a cell then KCl is not used because there can be formation of precipitate of AgCl , Hg_2Cl_2 , PbCl_2 or TlCl .

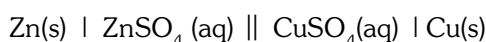
Functions

- It connects the solution of two half cell to complete the circuit.
- It maintains the electrical neutrality of the solution in order to give continuous flow or generation of current.
- If the salt bridge is removed then voltage drops to zero.
- It prevents the liquid-liquid junction potential. The potential difference which arises between two solutions when they brought in contact with each other.

Cell Representation :

We require two half cells to produce an electrochemical cell, which can be represented by following few rules;

- The anode half-cell is always written on the left followed on the right by cathode half cell.
- The separation of two phases (state of matter) is shown by a vertical line.
- The various materials present in the same phase are shown together using commas.
- The salt bridge is represented by a double slash ($\parallel$).
- The significant features of the substance like pressure of a gas, concentration of ions etc. are indicated in brackets immediately after writing the substance.



4.12 NERNST EQUATION :

It gives relation between electrode potential/ EMF of cell and concentration of electrolytic solution.

Walther Nernst derived a relation between free energy change (ΔG) and Reaction quotient (Q).

$$\Delta G = \Delta G^\circ + RT \ln Q \quad \dots(1)$$

where ΔG and ΔG° are free energy change and standard free energy change; 'Q' is reaction quotient.

$$\therefore -\Delta G = nFE \quad \text{and} \quad -\Delta G^\circ = nFE^\circ$$

Thus from Eq. (1), $-nFE = -nFE^\circ + RT \ln Q$

$$E = E^\circ - \frac{2.303 RT}{nF} \log Q$$

Where - E° = standard electrode potential,

R = gas constant,

T = temperature (in K)

F = Faraday (96500 coulomb mol^{-1}),

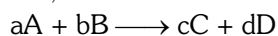
n = number of moles of e^- gained lost or transferred in balanced equation.

At 25°C, above equation may be written as

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

$$E = E^\circ - \frac{0.0591}{n} \log \frac{[P]}{[R]}$$

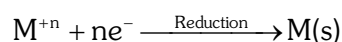
In general, for a redox cell reaction involving the transference of n electrons



$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad \text{or} \quad E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log \frac{[\text{Product}]}{[\text{Reactant}]}$$

4.13 APPLICATIONS OF NERNST EQUATION

(i) Calculation of electrode potential (E_{RP} or E_{OP}) -

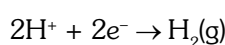


$$E_{RP} = E_{RP}^0 - \frac{0.0591}{n} \log \frac{[M]}{[M^{+n}]}$$

$$E_{RP} = E_{RP}^0 - \frac{0.0591}{n} \log \frac{1}{[M^{+n}]}$$

If $[M^{+n}]$ increases, then E_{RP} increases

(ii) Calculation of electrode potential and pH of hydrogen electrode -



$$E_{RP} = E_{RP}^0 - \frac{0.0591}{2} \log \frac{P_{H_2}}{[H^{+}]^2}$$

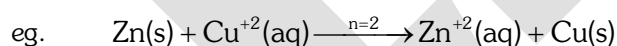
$$\therefore E_{RP}^0 = 0; \quad E_{RP} = E_{RP}^0 - \frac{0.0591}{2} \log \frac{1}{[H^{+}]^2} \quad [P_{H_2} \text{ is taken 1 atm}]$$

$$E_{RP} = 0.0591 \log [H^{+}]$$

$$E_{RP} = -0.0591 \text{ pH}$$

$$E_{OP} = +0.0591 \text{ pH}$$

(iii) Calculation of EMF of cell (E_{cell}) -



$$E_{cell} = E_{cell}^0 - \frac{0.0591}{2} \log \frac{[Zn^{+2}]}{[Cu^{+2}]}$$

If $[Cu^{+2}]$ increases, then E_{cell} increases

If $[Zn^{+2}]$ increases, then E_{cell} decreases

(iv) Prediction and feasibility of a cell reaction -

For a feasible cell reaction

$$\Delta G = -ve \quad (\Delta G = -nFE_{cell})$$

$$E_{cell} = +ve$$

(v) Calculation of equilibrium constant (K_{eq}) and ΔG° -

From Nernst equation -

$$E_{cell} = E_{cell}^0 - \frac{0.0591}{n} \log \frac{[P]}{[R]}$$

At equilibrium, $E_{cell} = 0$ and $\frac{[P]}{[R]} = K_{eq}$

$$E_{\text{cell}}^0 = \frac{0.0591}{n} \log K_{\text{eq}}$$

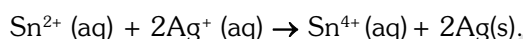
$$E_{\text{cell}}^0 = \frac{2.303RT}{nF} \log K_{\text{eq}}$$

$$n F E_{\text{cell}}^0 = 2.303 RT \log K_{\text{eq}}$$

$$\Delta G^\circ = -2.303RT \log K_{\text{eq}}$$

Illustrations

Illustration 8. Write cell representation for the following redox reaction,



Solution The cell consists of a platinum wire anode dipped in Sn^{2+} solution and a silver cathode dipped in Ag^+ solution therefore $\text{Pt}(\text{s}) | \text{Sn}^{2+}(\text{aq}), \text{Sn}^{4+}(\text{aq}) || \text{Ag}^+(\text{aq}) | \text{Ag}(\text{s})$.

Illustration 9. Calculate the EMF of a Daniel cell when the concentration of ZnSO_4 and CuSO_4 are 0.001 M and 0.1 M respectively. The standard EMF of the cell is 1.1 V.

Solution $\text{Zn}(\text{s}) + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = 1.1 - \frac{0.0591}{2} \log \frac{10^{-3}}{10^{-1}} = 1.159 \text{ V}$$

Illustration 10. Calculate E^0 and E for the cell $\text{Sn} | \text{Sn}^{2+} (1\text{M}) || \text{Pb}^{2+} (10^{-3}\text{M}) | \text{Pb}$, $E^0 (\text{Sn}^{2+} | \text{Sn}) = -0.14\text{V}$, $E^0 (\text{Pb}^{2+} | \text{Pb}) = -0.13 \text{ V}$. Is cell representation correct?

Solution $E_{\text{cell}}^0 = E_{\text{Pb}^{2+}/\text{Pb}}^0 - E_{\text{Sn}^{2+}/\text{Sn}}^0 = -0.13 - (-0.14) = +0.01 \text{ V}$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.0591}{2} \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} = +0.01 - \frac{0.0591}{2} \log \left[\frac{1}{10^{-3}} \right] = -0.078 \text{ V cell with not work}$$

Illustration 11. Calculate the equilibrium constant for the reaction $\text{Fe}^{2+} + \text{Ce}^{4+} \rightleftharpoons \text{Fe}^{3+} + \text{Ce}^{3+}$,

[Given : $E_{\text{Ce}^{4+}/\text{Ce}^{3+}}^0 = 1.44 \text{ V}$; $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = 0.68 \text{ V}$]

$$\text{Take } \frac{2.303 RT}{F} = 0.06 \text{ at } 25^\circ\text{C}, \log 4.68 = 0.67$$

Solution $\text{Fe}^{2+} + \text{Ce}^{4+} \rightleftharpoons \text{Fe}^{3+} + \text{Ce}^{3+}$

$$E^0 = 1.44 - 0.68 = 0.76\text{V} = \frac{0.06}{1} \log K_c; K_c = 4.64 \times 10^{12}$$

Illustration 12. The 0.1 M copper sulphate solution in which copper electrode is dipped at 25°C . Calculate the electrode potential of copper electrode [Given $E_{\text{Cu}^{+2}/\text{Cu}}^0 = 0.34\text{V}$]

Solution. $\text{Cu}^{+2} + 2e^- \rightarrow \text{Cu}(\text{s})$

$$E_{\text{red}} = E_{\text{red}}^0 - \frac{0.059}{n} \log \frac{[\text{Product}]}{[\text{Reactant}]}$$

$$\text{Here } n = 2 \text{ so } E = 0.34 - \frac{0.059}{2} \log 10$$

$$E = 0.34 - 0.03 = 0.31 \text{ V}$$

Illustration 13. Calculate the EMF of the cell $\text{Cr} \mid \text{Cr}^{+3} (0.1\text{M}) \parallel \text{Fe}^{+2} (0.01\text{M}) \mid \text{Fe}$

(Given $E^\circ_{\text{Cr}^{+3}/\text{Cr}} = -0.75\text{V}$, $E^\circ_{\text{Fe}^{+2}/\text{Fe}} = -0.45\text{V}$)

Solution.

Half cell reactions are –

At Anode $[\text{Cr} \rightarrow \text{Cr}^{+3} + 3e^-] \times 2$

At Cathode $[\text{Fe}^{+2} + 2e^- \rightarrow \text{Fe}] \times 3$

Cell reaction $2\text{Cr} + 3\text{Fe}^{+2} \rightarrow 2\text{Cr}^{+3} + 3\text{Fe}$

$E^\circ_{\text{cell}} = \text{Oxidation potential} + \text{Reduction potential} = 0.75 + (-0.45) = 0.30\text{V}$

$$E_{\text{cell}} = E^\circ - \frac{0.059}{n} \log \frac{[\text{Product}]}{[\text{Reactant}]} = 0.30 - \frac{0.059}{6} \log \frac{[\text{Cr}^{+3}]^2}{[\text{Fe}^{+2}]^3} = 0.30 - \frac{0.059}{6} \log \frac{[0.1]^2}{[0.01]^3}$$

$$= 0.30 + \frac{0.24}{6} = 0.34\text{V}$$

Illustration 14. For the cell $\text{Pt(s)} \mid \text{H}_2(0.4\text{ atm}) \mid \text{H}^+(\text{pH}=1) \parallel \text{H}^+(\text{pH}=2) \mid \text{H}_2(0.1\text{ atm}) \mid \text{Pt}$
The measured potential at 25°C is

(1) -0.1V (2) -0.5V (3) -0.041V (4) -0.030V

Solution

Anode reaction : $\text{H}_2(0.4\text{ atm}) \rightarrow 2\text{H}^+(10^{-1}\text{ M}) + 2e^-$

Cathode reaction : $2\text{H}^+(10^{-2}\text{ M}) + 2e^- \rightarrow \text{H}_2(0.1\text{ atm})$

Cell reaction $\text{H}_2(0.4\text{ atm}) + 2\text{H}^+(10^{-2}\text{ M}) \rightarrow 2\text{H}^+(10^{-1}\text{ M}) + \text{H}_2(0.1\text{ atm})$

$$E = 0 - \frac{0.059}{2} \log \frac{(10^{-1})^2(0.1)}{(0.4)(10^{-4})} = -0.041\text{V}$$

Illustration 15. If $E^\circ_{\text{Au}^+/\text{Au}}$ is 1.69V & $E^\circ_{\text{Au}^{3+}/\text{Au}}$ is 1.40V , then $E^\circ_{\text{Au}^{3+}/\text{Au}^+}$ will be

(1) 0.19V (2) 2.945V (3) 1.255V (4) none

Solution

$\text{Au}^+ + e^- \longrightarrow \text{Au(s)}$ $E^\circ = 1.69\text{V}$... (1); ΔG_1°

$\text{Au}^{3+} + 3e^- \longrightarrow \text{Au(s)}$ $E^\circ = 1.40\text{V}$... (2); ΔG_2°

From (2) – (1)

$\text{Au}^{3+} + 2e^- \longrightarrow \text{Au}^+$... (3); ΔG_3°

$$\Delta G_3^\circ = \Delta G_2^\circ - \Delta G_1^\circ$$

$$-2 \times F \times E^\circ = -3 \times F \times 1.40 + 1 \times 1.69 \times F$$

$$E^\circ = 1.255\text{V}$$

GOLDEN KEY POINTS

- The value of electrode potential does not depend on stoichiometry of half reactions as it is an intensive property.
- Gibb's free energy is extensive property.
- In electrolytic cell, cathode is negative terminal whereas in Galvanic cell cathode is positive terminal.
- Cell reaction is spontaneous, if $\Delta G = -ve$ or $E_{\text{cell}} = +ve$
- Cell reaction will be in equilibrium, if e.g. $\Delta G = 0$ or $-nFE_{\text{cell}} = 0$ or $E_{\text{cell}} = 0$

● **Work done by a cell :**

- (i) Let 'n' faraday charge be taken out of a cell of EMF 'E'; then work done by the cell will be calculated as:
 Work = Charge $\times$ Potential = $nF \times E$
- (ii) Work done by cell = Decrease in free energy
 so $-\Delta G = nFE$ or $W_{\max} = + nFE^\circ$, where E° is standard EMF of the cell.

BEGINNER'S BOX-2

1. Calculate E°_{cell} for the following cell

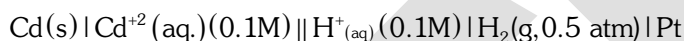


- (1) 0.04 V (2) - 1.56 V (3) 1.56 V (4) 0.84 V

2. Standard EMF of a galvanic cell involving $2e^-$ transfer is found to be 0.591 V at 25°C. Calculate equilibrium constant for the cell reaction

- (1) 20^{10} (2) 10^{20} (3) 10^{200} (4) 200^{10}

3. Calculate cell potential at 298 K for the following galvanic cell



$$E^\circ_{\text{Cd}^{+2}/\text{Cd}} = -0.40\text{V}$$

- (1) 0.38 (2) - 0.38 (3) 0.36 (4) - 0.36

4. Calculate E°_{cell} of the following galvanic cell at 298 K ;



$$E^\circ_{\text{Ca}^{+2}/\text{Ca}} = -2.87\text{V}; E^\circ_{\text{Fe}/\text{Fe}^{+2}} = 0.41\text{V}$$

- (1) 2.46 V (2) - 2.46 V (3) 3.28 V (4) - 3.28 V

5. Given $E^\circ_{\text{Ag}^{+}/\text{Ag}} = 0.80\text{V}$, $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37\text{V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34\text{V}$, $E^\circ_{\text{Hg}^{2+}/\text{Hg}} = 0.79\text{V}$.

Which of the following statement is correct

- (1) AgNO_3 can be stored in copper vessel
 (2) $\text{Mg}(\text{NO}_3)_2$ can not be stored in copper vessel
 (3) CuCl_2 can be stored in silver vessel
 (4) HgCl_2 can be stored in copper vessel

6. $1/2 \text{H}_2(\text{s}) + \text{AgCl(s)} \longrightarrow \text{H}^{+}(\text{aq}) + \text{Cl}^{-}(\text{aq}) + \text{Ag(s)}$ occurs in the galvanic cell :

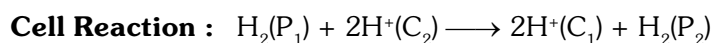
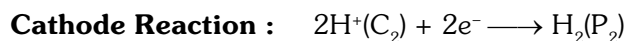
- (1) $\text{Ag(g)} | \text{AgCl(s)} | \text{KCl(sol)} || \text{AgNO}_3(\text{sol}) | \text{Ag(s)}$
 (2) $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{HCl(sol)} || \text{AgNO}_3(\text{sol}) | \text{Ag(s)}$
 (3) $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{HCl(sol)} || \text{Cl}^{-}(\text{aq}) | \text{AgCl(sol)} | \text{Ag(s)}$
 (4) $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{KCl(sol)} || \text{AgCl(sol)} | \text{Ag(s)}$

4.14 SOME SPECIAL CELLS

(A) CONCENTRATION CELLS

A concentration cell is a form of galvanic cell that has two equivalent half cells of the same material having difference only in concentrations. For such cells anode and cathode are same so $E^\circ_{\text{cell}} = 0$.

eg. $\text{Pt}, \text{H}_2(\text{g}) (P_1 \text{ atm}) | \text{H}^{+} (C_1) || \text{H}^{+} (C_2) | \text{H}_2(\text{g}) (P_2 \text{ atm}), \text{Pt}$



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[\text{P}]}{[\text{R}]}$$

$$\text{So, } E_{\text{cell}} = 0 - \frac{0.0591}{2} \log \left[\frac{\text{C}_1}{\text{C}_2} \right]^2 \frac{\text{P}_2}{\text{P}_1}$$

Case 1 : If $\text{P}_1 = \text{P}_2 = 1 \text{ atm}$; $E_{\text{cell}} = -\frac{0.059}{2} \log \left[\frac{\text{C}_1}{\text{C}_2} \right]^2$

$$E_{\text{cell}} = -0.0591 \log \left[\frac{\text{C}_1}{\text{C}_2} \right]$$

If $\text{C}_2 > \text{C}_1$ then $E_{\text{cell}} > 0$ and cell will be spontaneous.

Case 2 : If $\text{C}_1 = \text{C}_2$ $E_{\text{cell}} = -\frac{0.0591}{2} \log \left[\frac{\text{P}_2}{\text{P}_1} \right]$

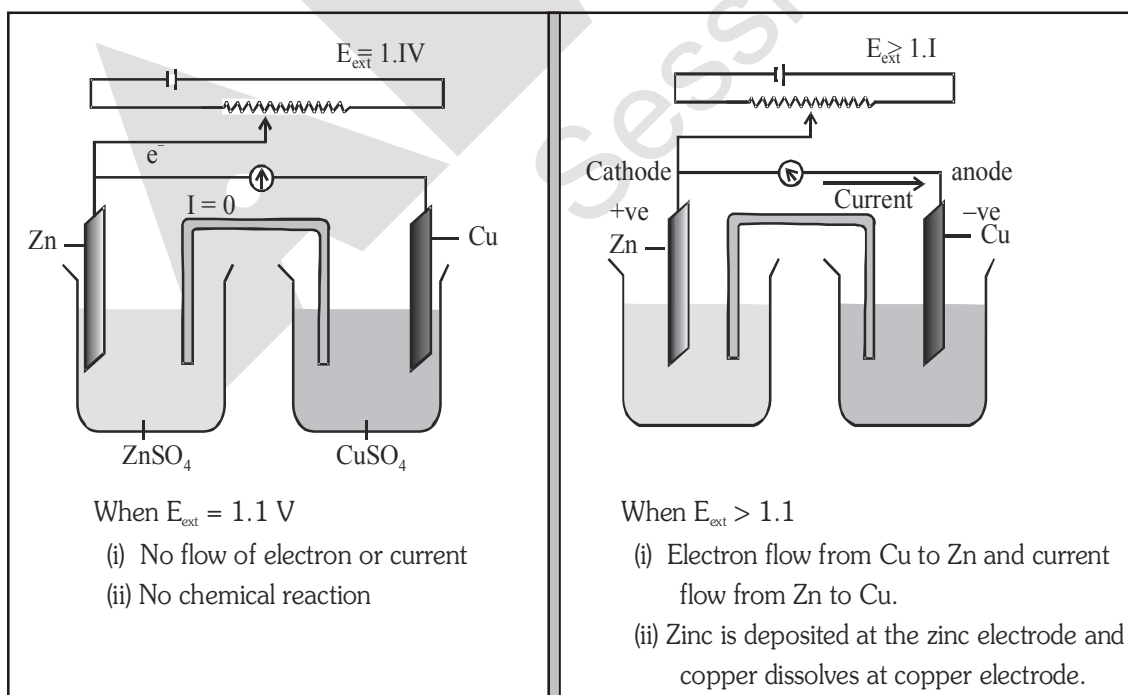
$$\text{P}_1 \neq \text{P}_2$$

If $\text{P}_1 > \text{P}_2$ then $E_{\text{cell}} > 0$ and cell will be spontaneous.

(B) REVERSIBLE CELLS :

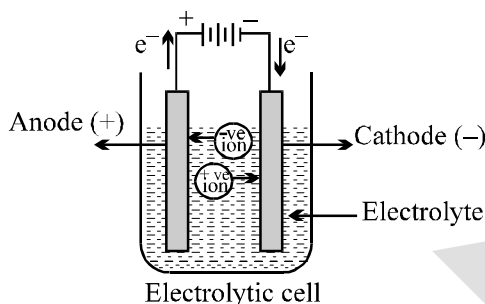
Those cell's in which chemical reactions are reversed by applying external emf in opposite directions known as reversible cells. A daniel cell is said to be reversible if it follows two conditions :

- (i) When $E_{\text{ext}} > 1.1 \text{ V}$ (emf of daniel cell) : cell reactions reversed
 - (ii) When $E_{\text{ext}} = 1.1 \text{ V}$ (emf of daniel cell) ; equilibrium will be established.
- In general secondary cells are reversible in nature.



4.15 ELECTROLYSIS

The process of chemical decomposition of an electrolyte by the passage of electricity is called electrolysis or electrolytic dissociation. It is carried out in **electrolytic cell** where electrical energy is converted into chemical energy. For electrolysis two suitable electrodes are immersed in the solution of an electrolyte. When an electric potential is applied between the electrodes, the positive ions move towards the cathode and negative ions move towards the anode. The gain of electrons (decrease in oxidation number) means reduction takes place at the cathode and loss of electrons (increase in oxidation number) means oxidation takes place at anode.

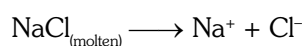


There are two aspects of electrolysis :

(A) Qualitative aspects of electrolysis :-

In qualitative aspect products formed at electrode are identified. At electrode product formation depends up on the following :

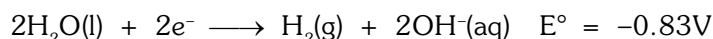
- (i) Nature of electrolyte
- (ii) Nature of electrodes : The metal strip at which current enters is called **anode**; anode is positively charged in electrolytic cell. On the other hand, the electrode at which current leaves is called **cathode**. Cathode is negatively charged. There are two types of electrodes
 - (a) Attackable : The attackable electrodes participate in the electrode reaction. They are made up of reactive metals like Zn, Cu, Ag etc in their respective solutions. In such electrodes, atom of the metal gets oxidised into the corresponding cation, which is passed into the solution. Thus, such anode gets dissolved and their mass decreases.
 - If electrode is active at cathode, metal goes on depositing and at anode metal is dissolved. The process is called electroplating.
 - (b) Non - attackable : Non-attackable electrodes do not participate in the electrode reaction as they are made up of unreactive elements like Pt, graphite etc. Such electrodes do not dissolve and their mass remain same.
 - Among the two cations, that cation is discharged at the cathode which has higher reduction potential. Among the two anions, that anion will be discharged at the anode which has lower reduction potential or higher oxidation potential.
 - The discharge of NO_3^- , PO_4^{3-} and SO_4^{2-} ions at anode does not commonly take place from aqueous solution.
- (iii) Concentration of electrolytic solution.

Examples of electrolysis :**(a) Electrolysis of Molten Sodium Chloride**

(b) **Electrolysis of Sodium Chloride Solution**

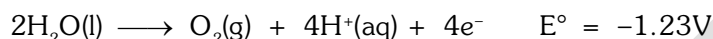
On electrolysis of aqueous solution of NaCl, the possible species involved in half-reactions are Na^+ , Cl^- and H_2O .

The possible cathode half-reactions are



Under the standard conditions, H_2O is to be reduced in preference to Na^+ and Hydrogen is evolved at the cathode.

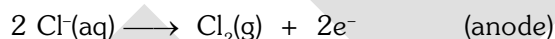
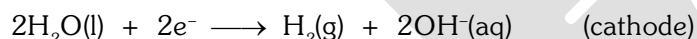
The possible anode half-reactions are



Under the standard-state conditions, H_2O is to be oxidized in preference to Cl^- . But since electrode potentials, depend on concentrations it turns out that when the solution is concentrated enough in Cl^- , Cl_2 is the product; but in dilute solution, O_2 is the product.

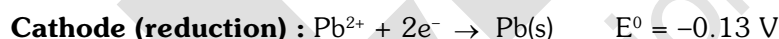
Starting with very dilute NaCl solutions, the oxidation potential of Cl^- is very negative, so H_2O is oxidised in preference to Cl^- . But as the NaCl concentration increase, the oxidation potential of Cl^- increases until eventually Cl^- is oxidized in preference to H_2O . The product changes from O_2 to Cl_2 .

The half-reactions and cell reaction for the electrolysis of aqueous sodium chloride to chlorine and hydroxide ion are as follows:



Because the electrolysis started with sodium chloride, the cation in the electrolyte solution is Na^+ . On evaporation of the electrolyte solution, sodium hydroxide NaOH is obtained.

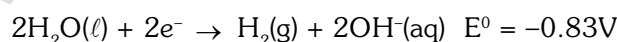
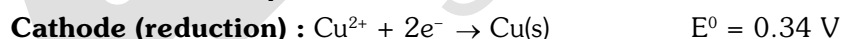
(c) **Electrolysis of Molten lead bromide using inert (pt/graphite) electrodes.**



$$E_{\text{cell}} = -0.13 - 1.09 = -1.22\text{V}$$

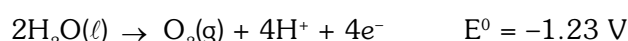
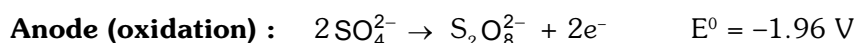
$$E_{\text{ext}} > 1.22\text{V}$$

(d) **Electrolysis of aq CuSO_4 using inert (pt/graphite) electrode**



At cathode reduction of $\text{Cu}^{2+}(\text{aq})$ will be preferred over reduction of water as its standard reduction potential is more, than that of water.

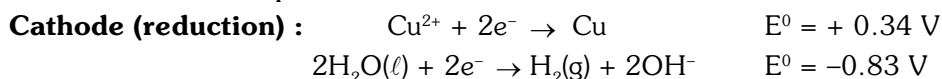
Thus reaction at cathode : $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}(\text{s})$



At anode oxidation of water is preferred over oxidation of sulphate ion as its standard oxidation potential is more, than that of sulphate ion.

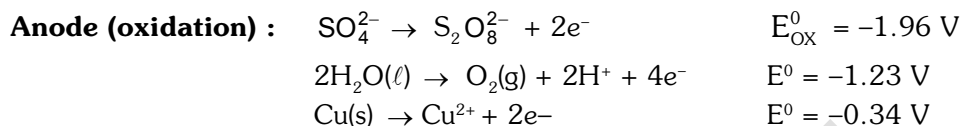
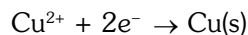
Thus reaction at anode : $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+ + 4e^-$

After electrolysis solution will become acidic ($\text{pH} < 7$)

(e) Electrolysis of aq. CuSO₄ using Cu electrode.

At cathode reduction of Cu²⁺ (aq) will be preferred over reduction of water as its standard reduction potential is more, than that of water.

Thus reaction at cathode :

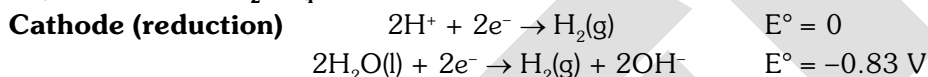


At anode oxidation of copper solid is preferred over oxidation of water and sulphate ion, as its standard oxidation potential is more than that of water and sulphate ion.

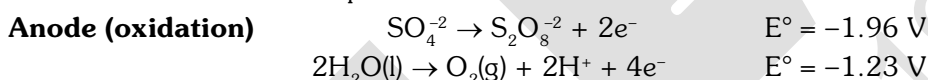
Thus reaction at anode :



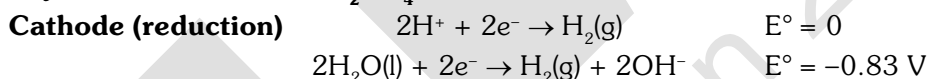
It is an example of electrolytic refining and there will be no change in the pH and concentration of electrolyte.

(f) Electrolysis of dilute H₂SO₄ :

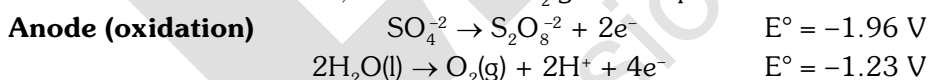
due to less availability of H⁺ ions, it will not participate in formation of H₂ gas therefore reduction of water will take place.



oxidation of water is preferred over oxidation of sulphate ion as its standard oxidation potential is more, than that of sulphate ion.

(g) Electrolysis of concentrated H₂SO₄

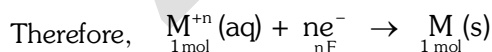
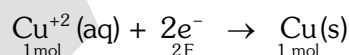
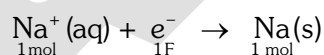
due to reduction of H⁺ ions, formation of H₂ gas takes place.



due to oxidation of SO₄²⁻ ions, formation of persulphate ion (S₂O₈²⁻) takes place.

(B) QUANTITATIVE ASPECTS OF ELECTROLYSIS

1 mole electron = N_A electron = 96500 coulomb = 1 Faraday



nF charge discharges = 1 mol of M⁺ⁿ ion

∴ 1 F charge discharges = $\frac{1}{n}$ mol of M⁺ⁿ ion

$$= \frac{1}{n} \times n = 1 \text{ g equivalent of M}^{+n} \text{ ion}$$

Hence 1 mole e⁻ = 1 F = 1 g equivalent

FARADAY'S LAWS OF ELECTROLYSIS

(a) First law of electrolysis :

Amount of substance deposited or liberated at an electrode is directly proportional to amount of charge passed through the solution.

$$W \propto Q$$

W = amount of substance deposited, Q = charge in coulomb

$$W = ZQ$$

Z = electrochemical equivalent

when $Q = 1$ coulomb, then $W = Z$

Thus, amount of substance deposited or liberated by 1 coulomb charge is called **electrochemical equivalent**.

Let I ampere current is passed till ' t ' seconds .

Then, $Q = It \therefore W = ZIt$

1 Faraday = 96500 coulomb = Charge on one mole electrons

One faraday is the charge required to liberate or deposit one gram equivalent of a substance at corresponding electrode.

Let 'E' is equivalent weight then 'E' gram will be liberated by 96500 coulomb.

$$\therefore 1 \text{ Coulomb will liberate } \frac{E}{96500} \text{ gram ;}$$

$$\text{By definition, } Z = \frac{E}{96500} \therefore W = \frac{EIt}{96500} \quad \left| \frac{W}{E} = \frac{It}{96500} = \text{number of g eq} = \text{number of faraday} \right|$$

(b) Second law of electrolysis :

When same amount of charge is passed through different electrolyte solutions connected in series then weight of substances deposited or liberated at electrodes are in ratio of their respective equivalent weights.

$$\text{i.e. } \frac{W_1}{W_2} = \frac{E_1}{E_2}$$

GOLDEN KEY POINTS

PRODUCTS OF ELECTROLYSIS OF SOME ELECTROLYTES

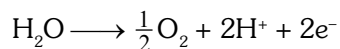
S. No.	Electrolyte	Electrode	Product obtained at anode	Product obtained at cathode
(i)	Aqueous NaCl	Pt or Graphite	Cl ₂	H ₂
(ii)	Fused NaCl	Pt or Graphite	Cl ₂	Na
(iii)	Aqueous NaOH	Pt or Graphite	O ₂	H ₂
(iv)	Fused NaOH	Pt or Graphite	O ₂	Na
(v)	Aqueous CuSO ₄	Pt or Graphite	O ₂	Cu
(vi)	Aqueous CuSO ₄	Copper	Cu oxidises to Cu ⁺² Ions	Cu
(vii)	Aqueous CuCl ₂	Copper	Cu oxidises to Cu ⁺² ions	Cu
(viii)	Aqueous HCl	Pt or Graphite	Cl ₂	H ₂
(ix)	Dilute H ₂ SO ₄	Pt or Graphite	O ₂	H ₂
(x)	Conc. H ₂ SO ₄	Pt or Graphite	S ₂ O ₈ ⁻²	H ₂
(xi)	Aqueous AgNO ₃	Pt of Graphite	O ₂	Ag
(xii)	Dilute NaCl	Pt or Graphite	O ₂	H ₂

- For electrolysis $\Delta G = +ve$, So it is non-spontaneous process.
- In electrolytic cell D.C. current is used.

Illustrations

Illustration 16 How much electric charge is required to oxidise :(a) 1 mol of H_2O to O_2 (b) 1 mol of FeO to Fe_2O_3 ?**Solution**

(a) The oxidation reaction is :



1 mol 2 mol

$$Q = 2F = 2 \times 96500 = 1.93 \times 10^5 \text{ C}$$

(b) The oxidation reaction is $\text{FeO} + \frac{1}{2} \text{H}_2\text{O} \longrightarrow \frac{1}{2} \text{Fe}_2\text{O}_3 + \text{H}^+ + \text{e}^-$

$$Q = F = 96500 \text{ C}$$

Illustration 17 One mole of electron passes through each of the solution of AgNO_3 , CuSO_4 and AlCl_3 when Ag, Cu and Al are deposited at cathode. The molar ratio of Ag, Cu and Al deposited are

(1) 1 : 1 : 1

(2) 6 : 3 : 2

(3) 6 : 3 : 1

(4) 1 : 3 : 6

Solution

Deposited equivalent will be same

$$1 \text{ eq. Ag} = 1 \text{ mol Ag}$$

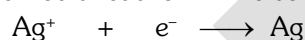
$$1 \text{ eq. Cu} = \frac{1}{2} \text{ mol Cu}$$

$$1 \text{ eq. Al} = \frac{1}{3} \text{ mol Al}$$

$$\Rightarrow \text{ratio} \quad 1 : \frac{1}{2} : \frac{1}{3} \Rightarrow 6 : 3 : 2$$

Illustration 18 Exactly 0.4 F electric charge is passed through three electrolytic cells connected in a series. First cell containing AgNO_3 , second cell containing CuSO_4 and third cell containing FeCl_3 solution. How many grams of each metal will be deposited assuming only cathodic reaction in each cell.**Solution**

The cathodic reaction in the cell are respectively,



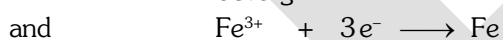
1 mol 1 mol

108 g 1 F



1 mol 1 mol

63.5 g 2 F



1 mol 1 mol

56 g 3 F

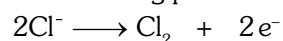
$$\text{Hence,} \quad \text{Ag deposited} = 108 \times 0.4 = 43.2 \text{ g}$$

$$\text{Cu deposited} = \frac{63.5}{2} \times 0.4 = 12.7 \text{ g}$$

$$\text{and} \quad \text{Fe deposited} = \frac{56}{3} \times 0.4 = 7.47 \text{ g}$$

Illustration 19 An electric current of 100 A is passed through a molten liquid of sodium chloride for 5 hours. Calculate the volume of chlorine gas liberated at the electrode at NTP.**Solution**

The reaction taking place at anode is

71.0 g 71.0 g $2 \times 96500 \text{ C}$

= 1 mol

$$Q = i \times t = 100 \times 5 \times 60 \times 60 \text{ C}$$

The amount of chlorine liberated by passing $100 \times 5 \times 60 \times 60 \text{ C}$ of electric charge.

$$= \frac{1}{2 \times 96500} \times 100 \times 5 \times 60 \times 60 = 9.3264 \text{ mole}$$

$$\text{Volume of } \text{Cl}_2 \text{ liberated at NTP} = 9.3264 \times 22.4 = 208.91 \text{ L}$$

Illustration 20 How long a current of 2 A has to be passed through a solution of AgNO_3 to coat a metal surface of 80 cm^2 with $5 \mu\text{m}$ thick layer? Density of silver = 10.8 g/cm^3 .

Solution

$$d = \frac{M}{V} \Rightarrow 10.8 = \frac{M}{80 \times 5 \times 10^{-4}} \Rightarrow M = 10.8 \times 400 \times 10^{-4}$$

$$W = \frac{E \times I \times t}{96500} \Rightarrow 10.8 \times 400 \times 10^{-4} = \frac{108 \times 2 \times t}{96500} \Rightarrow t = 193 \text{ s}$$

BEGINNER'S BOX-3

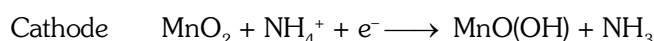
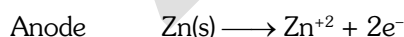
- 1 L of 1 M CuSO_4 solution is electrolysed. After passing 2 F charge, the molarity of CuSO_4 will be
(1) M/2 (2) M/4 (3) M (4) zero
- Same quantity of current is passed through molten NaCl and molten Al_2O_3 . If 4.6 g of Na was deposited in one cell, the mass of Al deposited in other cell is :-
(1) 0.9 g (2) 1.8 g (3) 2.7 g (4) 3.6 g
- A certain current liberates 0.5 g of hydrogen in 2 hr. How many grams of copper can be deposited by the same current flowing for the same time in a copper sulphate solution?
(1) 12.7 g (2) 15.9 g (3) 31.8 g (4) 63.5 g
- A quantity of electric charge that brings about the deposition of 4.5 g Al from Al^{+3} at the cathode will also produce the following volume (STP) of $\text{H}_2(\text{g})$ from H^+ at the cathode
(1) 44.8 L (2) 22.4 L (3) 11.2 L (4) 5.6 L
- A current of 9.65 A is passed through the aqueous solution of NaCl using suitable electrodes for 1000 s. The amount of NaOH formed during electrolysis is
(1) 2.0 g (2) 4.0 g (3) 6.0 g (4) 8.0 g
- If 0.224 L of H_2 gas is formed at the cathode, the volume of O_2 gas formed at the anode under identical conditions, is
(1) 0.224 L (2) 0.448 L (3) 0.112 L (4) 1.12 L

4.16 COMMERCIAL VOLTAIC CELLS

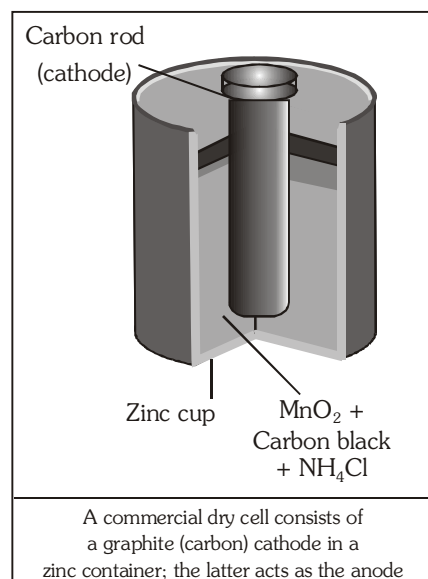
Batteries can be classified as primary and secondary. Primary batteries can not be returned to their original state by recharging, so when the reactants are consumed, the battery is "dead" and must be discarded. Secondary batteries are often called storage batteries or rechargeable batteries. The reactions in these batteries can be reversed; thus, the batteries can be recharged.

(A) Primary cells : Dry cells and mercury cells

The most familiar type of battery is the dry cell which is a compact of Leclanche cell known after its discoverer Leclanche (fig.). In this cell, the anode consists of a zinc container and the cathode is a graphite rod surrounded by powdered MnO_2 and carbon. The space between the electrodes is filled with a moist paste of NH_4Cl and ZnCl_2 . The electrode reactions are complex, but they can be written approximately as follows.



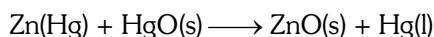
In the cathode reaction, manganese is reduced from the 4+ oxidation state to the 3+ state. Ammonia is not liberated as a gas but combines with Zn^{2+} to form $\text{Zn}(\text{NH}_3)_4^{2+}$ ion. Dry cells do not have an indefinite life as acidic NH_4Cl corrodes the zinc container even when not in use. **Dry cells have a potential of approximately 1.25 to 1.5V.**



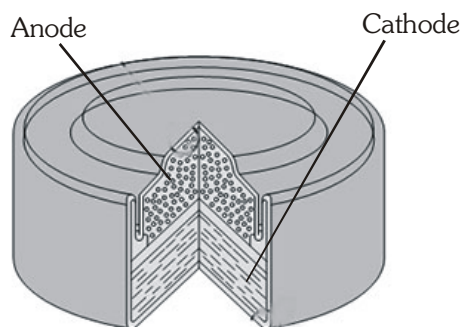
Mercury cell, suitable for low current devices like hearing aids, watches, etc. consists of zinc – mercury amalgam as anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO. The electrode reactions for the cell are given below :



The overall reaction is represented by



The cell potential is **approximately 1.35 V** and remains constant during its life as the overall reaction does not involve any ion in solution whose concentration can change during its life time.



(B) SECONDARY OR RECHARGEABLE BATTERIES

The most important secondary cell is the lead storage battery. It consists of a **lead anode** and a grid of lead packed with **lead dioxide as cathode**. A solution of sulphuric acid (38 percent by mass or having density 1.30 gm cm^{-3}) is used as an electrolyte.

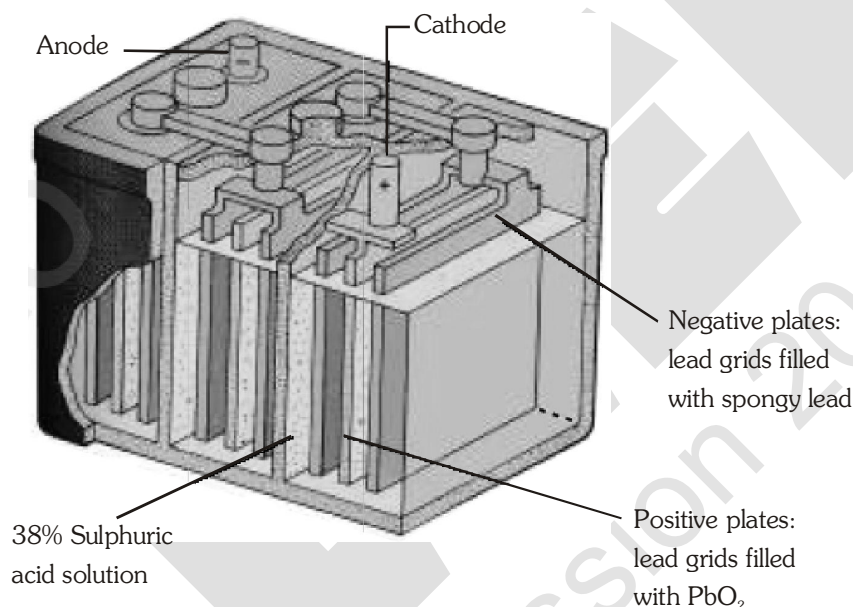
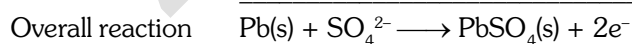


Fig : The Lead storage battery

When the cell discharges, it operates as a voltaic cell. The lead is oxidised to Pb^{+2} ion and lead plates acquire a negative charge.

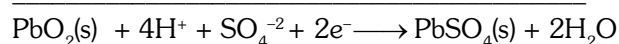
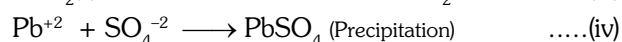
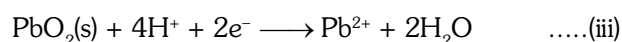


Pb^{+2} ions combine with sulphate ions of solution to form insoluble PbSO_4

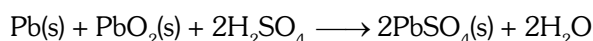
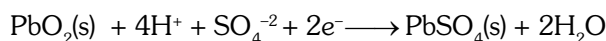
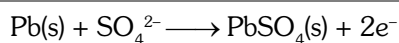


The electrons given by Pb are used at cathode (PbO_2 electrode)

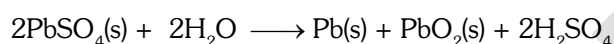
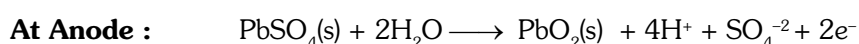
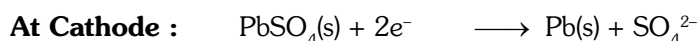
At Cathode :



Therefore the overall cell reaction during discharging



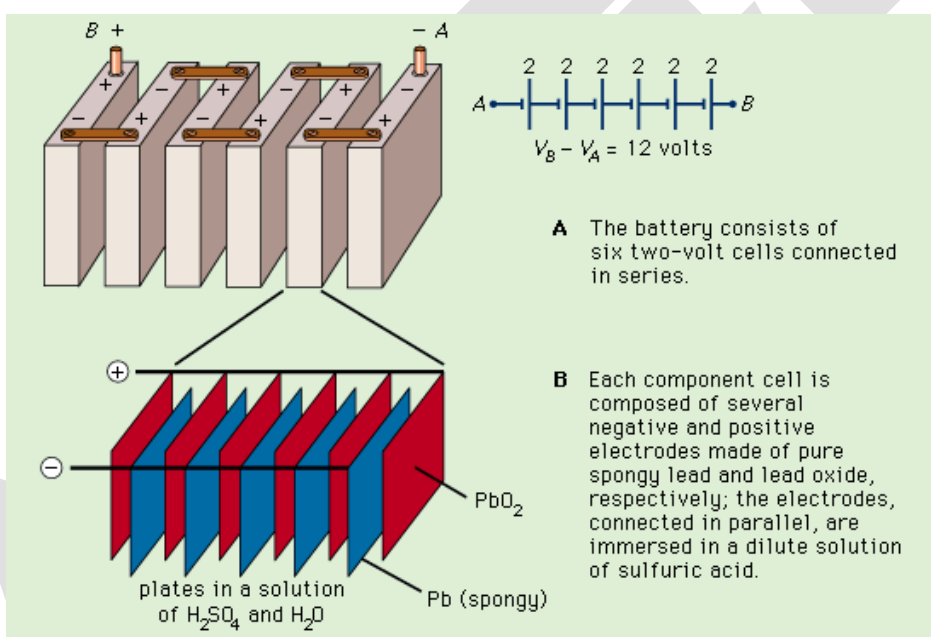
When a potential slightly greater than the potential of battery is applied, the battery can be recharged. During recharging the cell is operated like an **electrolytic cell** i.e. now electrical energy is supplied to it from an external source. The electrode reactions are reverse of those that occur during discharge :



It is clear that H_2SO_4 is used up during the discharge & produced during recharging reaction of cell.

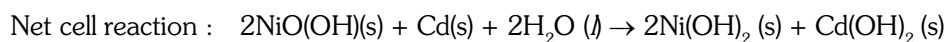
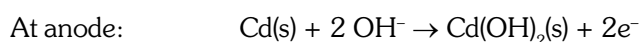
The discharging process of the storage cell is based on the principles of electrochemical cell, whereas the charging process is based upon the principles of electrolytic cells.

- n factor for H_2SO_4 during charging and discharging is equal to one.



(C) Nickel - Cadmium ("Ni - Cd") batteries :

They are used in variety of cordless appliances such as telephones, video camcorders, and cordless power tools, are lightweight and rechargeable. The chemistry of the cell utilizes the oxidation of cadmium and the reduction of nickel (III) oxide under basic conditions.

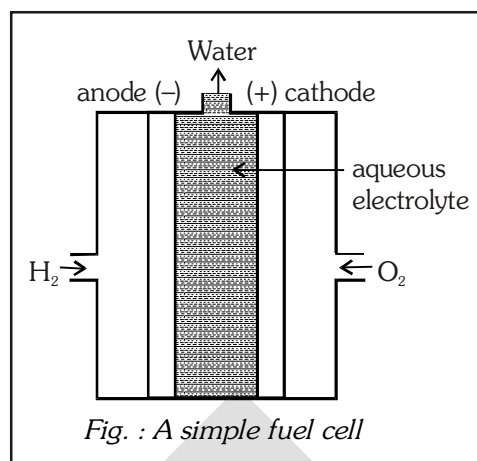


As the active masses of all reaction components are independent of their amount. The cell delivers constant EMF throughout.

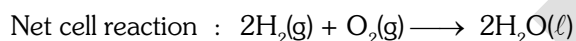
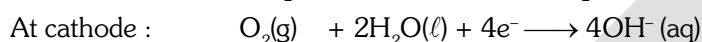
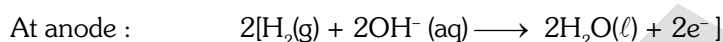
(D) Fuel Cells

It is possible to make batteries in which the reactants are fed continuously to the electrodes. Electrical cells that are designated to convert the free energy from the combustion of fuels such as hydrogen, carbon monoxide or methane directly into electrical energy are called **fuel cells**.

One of the most successful fuel cells uses the reaction of hydrogen with oxygen to form water (fig.). This cell has been used for electric power in the Appollo space programme. The water vapours produced were condensed and added to the drinking water supply for the astronauts.



In the cell shown in figure hydrogen and oxygen are bubbled through a porous carbon electrode into concentrated aqueous sodium hydroxide. Catalysts are incorporated in the electrode. The electrode reactions are :



This cell runs continuously as long as the reactants are supplied. Since fuel cells convert the energy of a fuel directly to electricity, they are potentially more efficient than the conventional method of generating electricity on a large scale by burning hydrogen, carbon fuels or by using nuclear reactor. Although **theoretically we expect 100 percent efficiency in fuel cells, so far an efficiency of only 60-70 per cent has been attained**. Since fuel cells are efficient and free from pollution, efforts are being made to produce better commercially practical fuel cells.

Efficiency of a Fuel Cell : $\eta = \frac{(\Delta G)}{(\Delta H)} \times 100$

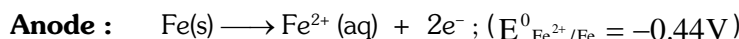
Main features of different cells.

	Name of the cell/Battery	Anode	Cathode	Electrolyte
(i)	Dry cell	Zinc	Graphite surrounded by MnO_2 & C.	$\text{NH}_4\text{Cl} + \text{ZnCl}_2$ (touching anode)
(ii)	Mercury cell	Zinc/Mercury Amalgum	Paste of HgO and carbon	Paste of KOH and ZnO
(iii)	Lead storage battery	Lead	Lead dioxide	H_2SO_4 (38%)
(iv)	Ni-Cd cell	Cadmium	Nickel dioxide	KOH sol.
(v)	H_2 - O_2 fuel cell	Porous carbon containing catalysts (H_2 passed)	Porous carbon containing catalysts (O_2 passed)	Conc. aq. NaOH sol.

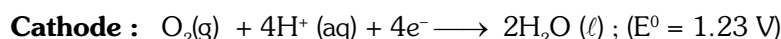
4.17 CORROSION

Corrosion is basically an electrochemical phenomenon. The rusting of iron, the tarnishing of silver, development of a green coating on copper and bronze are some of the examples of corrosion. Corrosion causes enormous damage to building, bridges, ships and to many other articles made of iron.

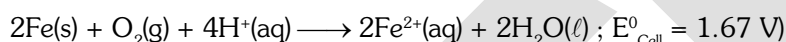
In corrosion, a metal is oxidised by loss of electrons to oxygen and forms metal oxide. Corrosion of iron (which is commonly known as rusting) occurs in presence of water and oxygen (air). Although the chemistry of corrosion is complex but it is understood that at one spot of an iron object oxidation occurs and that spot behaves as an anode.



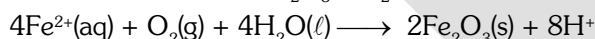
Electrons released at anode move through the metal and go to another spot on the metal and reduce the oxygen in presence of H^+ ions (which is believed to be available from H_2CO_3 formed due to dissolution of CO_2 in water). This spot works as a cathode :



Here Fe^{2+} ions move through water on the surface of the iron object. (If water present is saline, it will help more in carrying the current in the miniature cell thus formed and will enhance corrosion). The overall reaction of the miniature cell is the sum of the cathode and anode reactions as follows :



The Fe^{2+} ions are further oxidised by atmospheric oxygen to Fe^{3+} (as Fe_2O_3) and comes out as rust in the form of a hydrated iron (III) oxide expressed as $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.



H^+ ions produced in the above reaction help further in rusting.

Impurities present in iron also enhance rusting by setting a number of miniature cells. Very pure iron does not rust quickly.

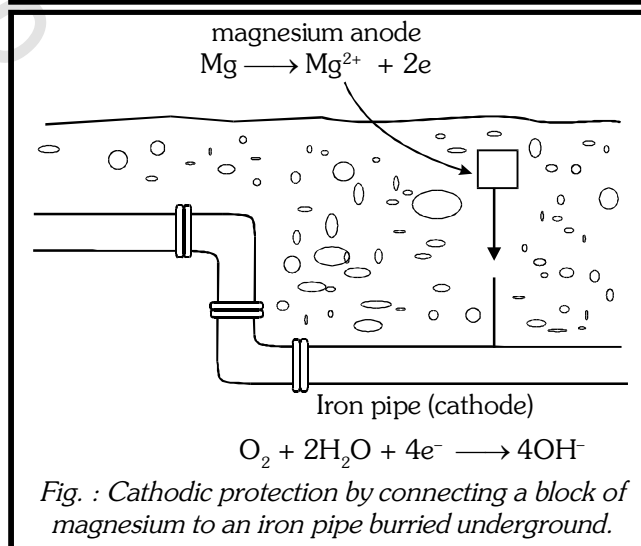
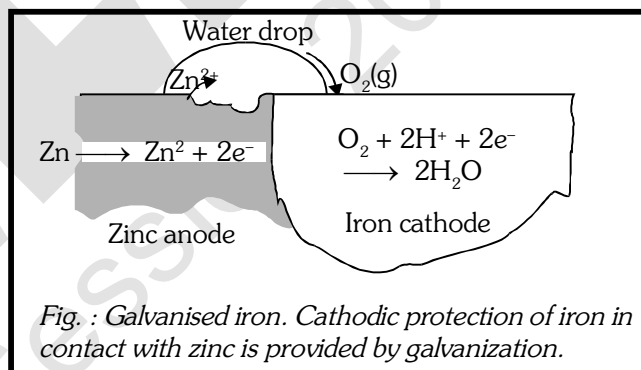
4.18 PREVENTION FROM CORROSION

Corrosion of a metal is prevented by applying protective coating (such as grease, paint, bisphenol or metal coatings) on metal surface. In case of iron this coating is done in two ways : (i) electrolysis (Cr, Ni and Cd coating) (ii) dipping iron objects in a molten metal (Zn and Sn coating). Use of zinc coating to protect iron is called Galvanisation. In galvanising iron, zinc being more reactive than iron serves as an anode and is oxidised



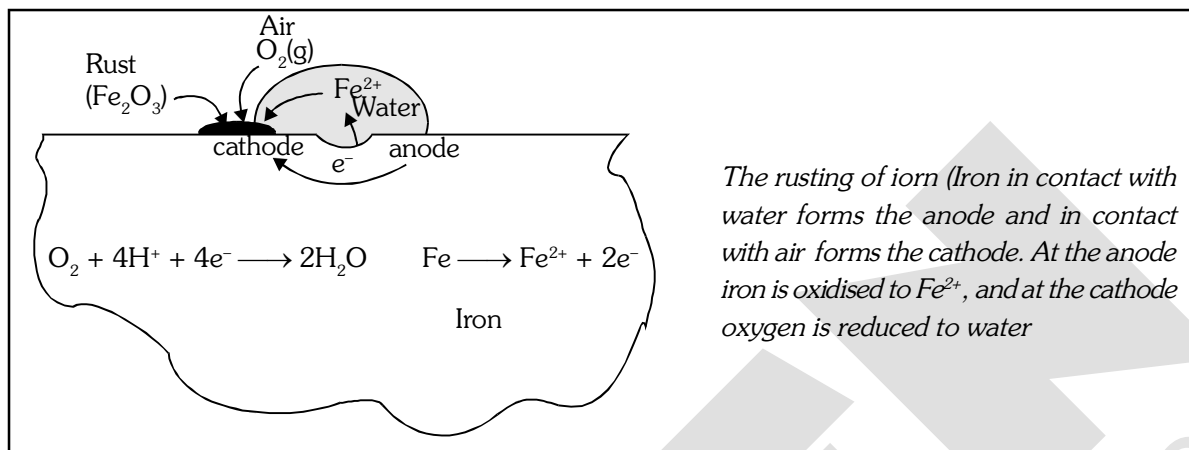
Here it is important to note that iron does not rust even after zinc coating is broken which is not true in case of tin coating over iron

($E^0_{\text{Sn}^{2+}/\text{Sn}} = -0.14\text{V}$). Now if the coating is broken, iron is exposed and iron being more active than both copper and tin, is corroded. Here iron corrodes more rapidly than it does in the absence of tin. But tin protects copper in the same way as zinc protects iron ($E^0_{\text{Cu}^{2+}/\text{Cu}} = 0.34\text{V}$).



Instead of coating more reactive metal on iron, the use of such metals is made as sacrificial anode. This method of preventing iron from corrosion is called CATHODIC PROTECTION.

In this method, a plate of reactive metal (Zn or Mg) is buried beside the iron pipe or tank and connected to it by wires. Here iron becomes **cathode** and more reactive metal becomes **anode**. The reactive metal anode is sacrificed to protect the iron. Since these reactive metal plates are oxidised quickly, they are replaced from time to time which is easy to do.



ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5					
	Ans.	1	2	4	3	4					
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6				
	Ans.	3	2	1	1	3	3				
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6				
	Ans.	4	2	2	4	2	3				

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

ELECTROLYTIC CONDUCTANCE

- Strong electrolytes are those which :
(1) dissolve readily in water
(2) conduct electricity
(3) dissociate into ions even at high concentration
(4) dissociate into ions at high dilution.
- Molten sodium chloride conducts electricity due to the presence of :
(1) free electrons
(2) free ions
(3) free molecules
(4) free atoms of Na and Cl
- Electrolytic conduction is due to the movement of :
(1) molecules (2) atoms
(3) ions (4) electrons
- Which of the following solutions of KCl has the lowest value of equivalent conductance ?
(1) 1 M (2) 0.1 M
(3) .01 M (4) .001 M
- In the equation $\Lambda = \text{sp. cond.} \times V$, If V is the volume in cc containing 1 equivalent of the electrolyte then V for a $\frac{N}{10}$ solution will be :
(1) 10 c.c. (2) 100 c.c.
(3) 1000 c.c. (4) 10,000 c.c.
- If the specific resistance of a solution of concentration $C \text{ g eq L}^{-1}$ is R , then its equivalent conductance is :
(1) $\frac{100R}{C}$ (2) $\frac{RC}{1000}$
(3) $\frac{1000}{RC}$ (4) $\frac{C}{1000R}$
- The specific conductances in $\text{ohm}^{-1} \text{ cm}^{-1}$ of four electrolytes P, Q, R and S are given in brackets :
P (5.0×10^{-5}) Q (7.0×10^{-8})
R (1.0×10^{-10}) S (9.2×10^{-3})
The one that offers highest resistance to the passage of electric current is
(1) P (2) S
(3) R (4) Q
- The specific conductance of a salt of 0.01 M concentration is $1.061 \times 10^{-4} \text{ S cm}^{-1}$. Molar conductance of the same solution will be :
(1) 1.061×10^{-4} (2) 1.061
(3) 10.61 (4) 106.1

- Which of the following solutions of NaCl will have the highest specific conductance ?
(1) 0.001 N (2) 0.1 N
(3) 0.01 N (4) 1.0 N
- The value of molar conductivity of HCl is greater than that of NaCl at a particular temperature because :
(1) Molecular mass of HCl is less than that of NaCl.
(2) Velocity of H^+ ions is more than that of Na^+ ions
(3) HCl is strongly acidic
(4) Ionisation of HCl is larger than that of NaCl
- Which statement is not correct :-
(1) Conductance of an electrolytic solution increases with dilution
(2) Conductance of an electrolytic solution decreases with dilution
(3) Specific conductance of an electrolytic solution decreases with dilution
(4) Equivalent conductance of an electrolytic solution increases with dilution.
- The resistance of 0.01 N solution of an electrolyte was found to be 210 ohm at 298 K using a conductivity cell of cell constant 0.66 cm^{-1} . The equivalent conductance of solution is :-
(1) $314.28 \text{ mho cm}^2 \text{ eq}^{-1}$
(2) $3.14 \text{ mho cm}^2 \text{ eq}^{-1}$
(3) $314.28 \text{ mho}^{-1} \text{ cm}^2 \text{ eq}^{-1}$
(4) $3.14 \text{ mho}^{-1} \text{ cm}^2 \text{ eq}^{-1}$
- Electrolytic conduction differs from metallic conduction from the fact that in the former
(1) The resistance increases with increasing temperature
(2) The resistance decreases with increasing temperature
(3) The resistance remains constant with increasing temperature
(4) The resistance is independent of the length of the conductor
- The specific conductance of a 0.01 M solution of KCl is $0.0014 \text{ ohm}^{-1} \text{ cm}^{-1}$ at 25°C . Its equivalent conductance ($\text{cm}^2 \text{ ohm}^{-1} \text{ eq}^{-1}$) is :-
(1) 140 (2) 14
(3) 1.4 (4) 0.14

15. Which one of the following is wrong :-
 (1) Specific conductance increases on dilution.
 (2) Specific conductance decreases on dilution.
 (3) Equivalent conductance increases on dilution.
 (4) Molar conductance increases on dilution.

KOHLRAUSCH LAW

16. At infinite dilution, the equivalent conductances of CH_3COONa , HCl and CH_3COOH are 91, 426 and 391 $\text{mho cm}^2 \text{eq}^{-1}$ respectively at 25 °C. The eq. conductance of NaCl at infinite dilution will be :
 (1) 126 (2) 209
 (3) 391 (4) 908
17. The equivalent conductivity of 0.1 N CH_3COOH at 25 °C is 80 and at infinite dilution it is 400. The degree of dissociation of CH_3COOH is :
 (1) 1 (2) 0.2
 (3) 0.1 (4) 0.5
18. The limiting molar conductivities Λ^0 for NaCl , KBr and KCl are 126, 152 and 150 $\text{S cm}^2 \text{mol}^{-1}$ respectively. The Λ^0 for NaBr is :
 (1) 278 $\text{S cm}^2 \text{mol}^{-1}$ (2) 176 $\text{S cm}^2 \text{mol}^{-1}$
 (3) 128 $\text{S cm}^2 \text{mol}^{-1}$ (4) 302 $\text{S cm}^2 \text{mol}^{-1}$
19. For HCl solution at 25 °C, equivalent conductance at infinite dilution, is 425 $\text{ohm}^{-1} \text{cm}^2 \text{eq}^{-1}$. The specific conductance of a solution of HCl is 3.825 $\text{ohm}^{-1} \text{cm}^{-1}$. If the degree of dissociation is 90%. The normality of the solution is :-
 (1) 0.90 N (2) 1.0 N
 (3) 10 N (4) 1.2 N
20. The molar conductivities Λ_{NaOAc}^0 and Λ_{HCl}^0 at infinite dilution in water at 25°C are 91.0 and 426.2 $\text{S cm}^2 \text{mol}^{-1}$ respectively. To calculate Λ_{HOAc}^0 the additional value required is :
 (1) Λ_{NaCl}^0 (2) $\Lambda_{\text{H}_2\text{O}}^0$
 (3) Λ_{KCl}^0 (4) Λ_{NaOH}^0
21. The molar conductance at infinite dilution of AgNO_3 , AgCl and NaCl are 116.5, 121.6 and 110.3 $\text{S cm}^2 \text{mol}^{-1}$ respectively. The molar conductance of NaNO_3 is :
 (1) 111.4 $\text{S cm}^2 \text{mol}^{-1}$ (2) 105.2 $\text{S cm}^2 \text{mol}^{-1}$
 (3) 130.6 $\text{S cm}^2 \text{mol}^{-1}$ (4) 150.2 $\text{S cm}^2 \text{mol}^{-1}$
22. The conductivity of a saturated solution of BaSO_4 is $3.06 \times 10^{-6} \text{ohm}^{-1} \text{cm}^{-1}$ and its molar conductance is 1.53 $\text{ohm}^{-1} \text{cm}^{-1} \text{mol}^{-1}$. The K_{sp} of BaSO_4 will be
 (1) 4×10^{-12} (2) 2.5×10^{-9}
 (3) 2.5×10^{-13} (4) 4×10^{-6}

GALVANIC CELL

23. In the galvanic cell
 $\text{Cu(s)} \mid \text{Cu}^{2+} (1 \text{ M}) \parallel \text{Ag}^+ (1 \text{ M}) \mid \text{Ag(s)}$
 the electrons will travel in the external circuit :
 (1) from Ag to Cu
 (2) from Cu to Ag
 (3) electrons do not travel in the external circuit
 (4) in any direction
24. The passage of electricity in the Daniell cell when Zn and Cu electrodes are connected is :
 (1) from Cu to Zn in the cell
 (2) from Cu to Zn out side the cell
 (3) from Zn to Cu outside the cell
 (4) in any direction in the cell
25. The equation representing the process by which standard reduction potential of zinc can be defined is
 (1) $\text{Zn}^{2+} (\text{s}) + 2\text{e}^- \longrightarrow \text{Zn} (\text{s})$
 (2) $\text{Zn} (\text{g}) \longrightarrow \text{Zn}^{2+} (\text{g}) + 2\text{e}^-$
 (3) $\text{Zn}^{2+} (\text{g}) + 2\text{e}^- \longrightarrow \text{Zn} (\text{s})$
 (4) $\text{Zn}^{2+} (\text{aq.}) + 2\text{e}^- \longrightarrow \text{Zn} (\text{s})$
26. A standard hydrogen electrode has zero electrode potential because :
 (1) Hydrogen is easiest to oxidize.
 (2) This electrode potential is assumed to be zero.
 (3) Hydrogen atom has only one electron.
 (4) Hydrogen is the lightest element.
27. Which is not true for a standard hydrogen electrode ?
 (1) The hydrogen ion concentration is 1 M.
 (2) Temperature is 25°C.
 (3) Pressure of hydrogen is 1 bar.
 (4) It contains a metallic conductor which does not adsorb hydrogen.
28. E° for the half cell
 $\text{Zn}^{2+} \mid \text{Zn}$ is -0.76 V. E.m.f. of the cell
 $\text{Zn} \mid \text{Zn}^{2+} (1\text{M}) \parallel 2\text{H}^+ (1\text{M}) \mid \text{H}_2 (1 \text{ atm})$ is :
 (1) -0.76 V (2) +0.76 V
 (3) -0.38 V (4) + 0.38 V
29. The chemical reaction,
 $2\text{AgCl(s)} + \text{H}_2(\text{g}) \longrightarrow 2\text{HCl(aq)} + 2\text{Ag(s)}$
 taking place in a galvanic cell is represented by the notation.
 (1) $\text{Pt(s)} \mid \text{H}_2(\text{g}).1\text{bar} \mid 1\text{MKCl(aq)} \mid \text{AgCl(s)} \mid \text{Ag(s)}$.
 (2) $\text{Pt(s)} \mid \text{H}_2(\text{g}).1\text{bar} \mid 1\text{MHCl(aq)} \mid 1\text{M Ag(aq)} \mid \text{Ag(s)}$.
 (3) $\text{Pt(s)} \mid \text{H}_2(\text{g}).1\text{bar} \mid 1\text{MHCl(aq)} \mid \text{AgCl(s)} \mid \text{Ag(s)}$.
 (4) $\text{Pt(s)} \mid \text{H}_2(\text{g}).1\text{bar} \mid 1\text{MHCl(aq)} \mid \text{Ag(s)} \mid \text{AgCl(s)}$.

30. $\text{Cu(s)} \mid \text{Cu}^{2+}(1 \text{ M}) \parallel \text{Zn}^{2+}(1 \text{ M}) \mid \text{Zn(s)}$
A cell represented above should have emf.
(1) Positive
(2) Negative
(3) Zero
(4) Cannot be predicted
31. Given electrode potentials :
 $\text{Fe}^{3+} + e \longrightarrow \text{Fe}^{2+}$; $E^\circ = 0.771 \text{ V}$
 $\text{I}_2 + 2e \longrightarrow 2\text{I}^-$; $E^\circ = 0.536 \text{ V}$
 E° cell for the cell reaction
 $2\text{Fe}^{3+} + 2\text{I}^- \longrightarrow 2\text{Fe}^{2+} + \text{I}_2$ is -
(1) $(2 \times 0.771 - 0.536) = 1.006 \text{ V}$
(2) $(0.771 - 0.5 \times 0.536) = 0.503 \text{ V}$
(3) $0.771 - 0.536 = 0.235 \text{ V}$
(4) $0.536 - 0.771 = -0.235 \text{ V}$
32. Which of the following is not an anodic reaction-
(1) $\text{Ag}^+ \rightarrow \text{Ag} - e^-$
(2) $\text{Cu} \rightarrow \text{Cu}^{2+} + 2e^-$
(3) $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + e^-$
(4) $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4e^-$
33. Which of the following statements is correct :-
(1) Oxidation occur at anode in both galvanic and electrolytic cell.
(2) Reduction occurs at anode in both galvanic and electrolytic cell,
(3) Reduction occur at anode in electrolytic cell where as oxidation occur at cathode in galvanic cell,
(4) Oxidation occur at anode in electrolytic cell where as reduction occur at anode in a galvanic cell,
34. Other things being equal, the life of a Daniell cell may be increased by :-
(1) Keeping low temperature
(2) Using large copper electrode
(3) Decreasing concentration of copper ions
(4) Using large zinc electrodes

ELECTROCHEMICAL SERIES

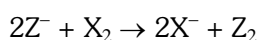
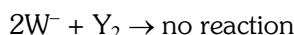
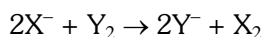
35. Zn can not displace following ions from their aqueous solution :
(1) Ag^+ (2) Cu^{2+}
(3) Fe^{2+} (4) Na^+
36. The standard reduction potentials at 25°C for the following half reactions are given against each :
 $\text{Zn}^{2+}(\text{aq}) + 2e^- \rightleftharpoons \text{Zn(s)}$, -0.762 V
 $\text{Cr}^{3+}(\text{aq}) + 3e^- \rightleftharpoons \text{Cr(s)}$, -0.740 V
 $2\text{H}^+ + 2e^- \rightleftharpoons \text{H}_2(\text{g})$, 0.00 V
 $\text{Fe}^{3+} + 2e^- \rightleftharpoons \text{Fe}^{2+}$, 0.77 V
Which is the strongest reducing agent ?
(1) Zn (2) Cr
(3) $\text{H}_2(\text{g})$ (4) $\text{Fe}^{2+}(\text{aq})$

37. Red hot carbon will remove oxygen from the oxide XO and YO but not from ZO. Y will remove oxygen from XO. Use this evidence to deduce the order of activity of the three metals X, Y and Z putting the most active first.
(1) XYZ (2) ZYX
(3) YXZ (4) ZXY
38. Adding powdered Pb and Fe to a solution containing 1.0 M in each of Pb^{2+} and Fe^{2+} ions, would result into the formation of :-
(1) More of Fe and Pb^{2+} ions
(2) More of Fe^{2+} and Pb^{2+} ions
(3) More of Pb and Fe^{2+} ions
(4) More of Fe and Pb
39. If a spoon of copper metal is placed in a solution of ferrous sulphate :
(1) Cu will precipitate out
(2) Iron will precipitate
(3) Cu and Fe will precipitate
(4) No reaction will take place
40. Which one will liberate Br_2 from KBr ?
(1) HI (2) I_2
(3) Cl_2 (4) SO_2
41. Using the standard electrode potential values given below, decide which of the statements, I, II, III and IV are correct. Choose the right answer from (1), (2), (3) and (4).
 $\text{Fe}^{2+} + 2e^- \rightleftharpoons \text{Fe}$; $E^\circ = -0.44 \text{ V}$
 $\text{Cu}^{2+} + 2e^- \rightleftharpoons \text{Cu}$; $E^\circ = +0.34 \text{ V}$
 $\text{Ag}^+ + e^- \rightleftharpoons \text{Ag}$; $E^\circ = +0.80 \text{ V}$
I. Copper can displace iron from FeSO_4 solution.
II. Iron can displace copper from CuSO_4 solution.
III. Silver can displace copper from CuSO_4 solution.
IV. Iron can displace silver from AgNO_3 solution.
(1) I and II (2) II and III
(3) II and IV (4) I and IV
42. The standard electrode potential value of the elements A, B and C are 0.68 , -2.50 and 0.50 V respectively. The order of their reducing power is :
(1) $A > B > C$ (2) $A > C > B$
(3) $C > B > A$ (4) $B > C > A$

43. The oxidation potential of Zn, Cu, Ag, H_2 and Ni are 0.76 V, -0.34 V, -0.80 V, 0 V, 0.55 V respectively. Which of the following reaction will provide maximum voltage ?

- (1) $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$
- (2) $Zn + 2Ag^+ \longrightarrow 2Ag + Zn^{2+}$
- (3) $H_2 + Cu^{2+} \longrightarrow 2H^+ + Cu$
- (4) $H_2 + Ni^{2+} \longrightarrow 2H^+ + Ni$

44. The following facts are available :-



Which of the following statements is correct :-

- (1) $E^\circ_{W^-/W_2} > E^\circ_{Y^-/Y_2} > E^\circ_{X^-/X_2} > E^\circ_{Z^-/Z_2}$
- (2) $E^\circ_{W^-/W_2} < E^\circ_{Y^-/Y_2} < E^\circ_{X^-/X_2} < E^\circ_{Z^-/Z_2}$
- (3) $E^\circ_{W^-/W_2} < E^\circ_{Y^-/Y_2} > E^\circ_{X^-/X_2} > E^\circ_{Z^-/Z_2}$
- (4) $E^\circ_{W^-/W_2} > E^\circ_{Y^-/Y_2} < E^\circ_{X^-/X_2} < E^\circ_{Z^-/Z_2}$

45. The standard reduction potential at 25 °C of Li^+ / Li , Ba^{2+} / Ba , Na^+ / Na and Mg^{2+} / Mg are -3.05 V, -2.73 V, -2.71 V and -2.37 V respectively. Which one of the following is the strongest oxidising agent ?

- (1) Na^+
- (2) Li^+
- (3) Ba^{2+}
- (4) Mg^{2+}

46. A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y^- and 1 M Z^- at 25 °C. If the reduction potential of $Z > Y > X$ then :

- (1) Y will oxidise X and not Z
- (2) Y will oxidise Z and not X
- (3) Y will oxidise both X and Z
- (4) Y will reduce both X and Z

47. The standard electrode potential of Zn, Ag and Cu are -0.76 V, 0.80 V and 0.34 V respectively, then :

- (1) Ag can oxidise Zn and Cu
- (2) Ag can reduce Zn^{2+} and Cu^{2+}
- (3) Zn can reduce Ag^+ and Cu^{2+}
- (4) Cu can oxidise Zn and Ag

48. Each of the three metals x, y and z were put in turn into aqueous solution of the other two. $x + \text{salt of y}$ (or z) $\rightarrow y$ (or z) + salt of x

Which one of the following observation is incorrect?

- (1) $y + \text{salt of x} \rightarrow \text{no action observed}$
- (2) $y + \text{salt of z} \rightarrow z + \text{salt of y}$
- (3) $z + \text{salt of x} \rightarrow x + \text{salt of z}$
- (4) $z + \text{salt of y} \rightarrow \text{no action observed}$

49. A standard reduction electrode potentials of four elements are

$$A = -0.250 \text{ V}, \quad B = -0.140 \text{ V}$$

$$C = -0.126 \text{ V}, \quad D = -0.402 \text{ V}$$

The element that displaces A from its compounds aqueous solution is :-

- (1) B
- (2) C
- (3) D
- (4) None of the above

50. The following four colourless salt solutions are placed in separate test tubes and a strip of Cu is placed in each solution which finally turns blue :-

- (1) $Zn(NO_3)_2$
- (2) $Mg(NO_3)_2$
- (3) KNO_3
- (4) $AgNO_3$

51. Which of the following displacement does not occur

- (1) $Zn + 2H^+ \rightarrow Zn^{2+} + H_2 \uparrow$
- (2) $Fe + 2Ag^+ \rightarrow Fe^{2+} + Ag \downarrow$
- (3) $Cu + Fe^{2+} \rightarrow Cu^{2+} + Fe \downarrow$
- (4) $Zn + Pb^{2+} \rightarrow Zn^{2+} + Pb \downarrow$

NERNST EQUATION

52. $E^\circ(Ni^{2+}/Ni) = -0.25 \text{ V}$

$$E^\circ(Au^{3+}/Au) = 1.50 \text{ V}$$

The emf of the voltaic cell.

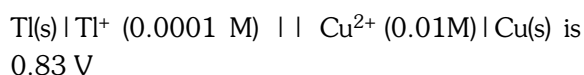


- (1) 1.25 V
- (2) -1.75 V
- (3) 1.75 V
- (4) 4.0 V

53. The emf of the cell in which the following reaction $Zn(s) + Ni^{2+} (a = 0.1) \rightleftharpoons Zn^{2+} (a = 1.0) + Ni(s)$ occurs, is found to be 0.5105 V at 298 K. The standard e.m.f. of the cell is :-

- (1) -0.5105 V
- (2) 0.5400 V
- (3) 0.4810 V
- (4) 0.5696 V

54. The emf of the cell



The emf of this cell will be increased by :-

- (1) Increasing the concentration of Cu^{2+} ions
- (2) Decreasing the concentration of Tl^+
- (3) Increasing the concentration of both
- (4) (1) & (2) both

55. The potential of hydrogen electrode

$$(P_{H_2} = 1 \text{ atm}; C_{H^+} = 0.1 \text{ M}) \text{ at } 25^\circ\text{C} \text{ will be -}$$

- (1) 0.00 V
- (2) -0.059 V
- (3) 0.118 V
- (4) 0.059 V

56. Which of the following represents the potential of silver wire dipped into 0.1 M AgNO_3 solution at 25°C ?
 (1) E°_{red} (2) $(E^\circ_{\text{red}} + 0.059)$
 (3) $(E^\circ_{\text{ox}} - 0.059)$ (4) $(E^\circ_{\text{red}} - 0.059)$
57. The potential of a hydrogen electrode at $\text{pH} = 1$ is
 (1) 0.059 V (2) 0.00 V
 (3) -0.059 V (4) 0.59 V
58. Consider the reaction
 $\text{Cl}_2(\text{g}) + 2\text{Br}^-(\text{aq}) \longrightarrow 2\text{Cl}^-(\text{aq}) + \text{Br}_2$
 The emf of the cell when
 $[\text{Cl}^-] = [\text{Br}_2] = [\text{Br}^-] = 0.01 \text{ M}$ and Cl_2 gas at 1 atm pressure will be (E° for the above reaction is = 0.29 V)
 (1) 0.54 V (2) 0.35 V
 (3) 0.24 V (4) -0.29 V
59. The standard emf for the cell reaction
 $\text{Zn} + \text{Cu}^{2+} \longrightarrow \text{Zn}^{2+} + \text{Cu}$ is 1.10 V at 25°C . The emf for the cell reaction when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solution are used at 25°C is :
 (1) 1.10 V (2) 0.110 V
 (3) -1.10 V (4) -0.110 V
60. E° for $\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$ is 2.8 V,
 E° for $\frac{1}{2}\text{F}_2 + \text{e}^- \rightarrow \text{F}^-$ is ?
 (1) 2.8 V (2) 1.4 V (3) -2.8 V (4) -1.4 V
61. ΔG° of the cell reaction
 $\text{AgCl}(\text{s}) + \frac{1}{2}\text{H}_2(\text{g}) \rightarrow \text{Ag}(\text{s}) + \text{H}^+ + \text{Cl}^-$ is -21.52 kJ
 ΔG° of $2\text{AgCl}(\text{s}) + \text{H}_2(\text{g}) \rightarrow 2\text{Ag}(\text{s}) + 2\text{H}^+ + 2\text{Cl}^-$ is :
 (1) -21.52 kJ (2) -10.76 kJ
 (3) -43.04 kJ (4) 43.04 kJ
62. How much will the potential of $\text{Zn} / \text{Zn}^{2+}$ change if the solution of Zn^{2+} is diluted 10 times
 (1) increases by 0.03 V
 (2) decreases by 0.03 V
 (3) increases by 0.059 V
 (4) decreases by 0.059 V
63. How much will the potential of a hydrogen electrode change when its solution initially at $\text{pH} = 0$ is neutralised to $\text{pH} = 7$?
 (1) increases by 0.059 V
 (2) decreases by 0.059 V
 (3) increases by 0.41 V
 (4) decreases by 0.41 V
64. Which of the following will increase the voltage of the cell with following cell reaction
 $\text{Sn}_{(\text{s})} + 2\text{Ag}^+_{(\text{aq})} \rightarrow \text{Sn}^{2+}_{(\text{aq})} + 2\text{Ag}_{(\text{s})}$
 (1) Increase in the size of silver rod
 (2) Increase in the concentration of Sn^{2+} ions
 (3) Increase in the concentration of Ag^+ ions
 (4) Decrease in the concentration of Ag^+ ions
65. E° for the reaction $\text{Fe} + \text{Zn}^{2+} \rightarrow \text{Zn} + \text{Fe}^{2+}$ is -0.35 V. The given cell reaction is :
 (1) feasible (2) not feasible
 (3) in equilibrium (4) can't say anything
66. For a reaction - $\text{A}(\text{s}) + 2\text{B}^+ \rightarrow \text{A}^{2+} + 2\text{B}(\text{s})$
 K_c has been found to be 10^{12} . The E°_{cell} is:
 (1) 0.354 V (2) 0.708 V
 (3) 0.0098 V (4) 1.36 V
67. The standard electrode potential (E°) for $\text{OCl}^- / \text{Cl}^-$ and $\text{Cl}^- / \frac{1}{2}\text{Cl}_2$ respectively are 0.94 V and -1.36 V. The E° value of $\text{OCl}^- / \frac{1}{2}\text{Cl}_2$ will be :
 (1) -2.20 V (2) -0.42 V
 (3) 0.52 V (4) 1.04 V
68. The standard reduction potentials of $\text{Cu}^{2+} / \text{Cu}$ and $\text{Cu}^{2+} / \text{Cu}^+$ are 0.337 V and 0.153 V respectively. The standard electrode potential of Cu^+ / Cu half cell is :
 (1) 0.184 V (2) 0.827 V
 (3) 0.521 V (4) 0.490 V
69. The hydrogen electrode is dipped in a solution of $\text{pH} = 3$ at 25°C . The potential of the cell would be :
 (1) 0.177 V (2) -0.177 V
 (3) 0.087 V (4) 0.059 V
70. What is the potential of the cell containing two hydrogen electrodes as represented below
 $\text{Pt}; \text{H}_2(\text{g}) | \text{H}^+(10^{-8}\text{M}) || \text{H}^+(0.001 \text{ M}) | \text{H}_2(\text{g}); \text{Pt}$;
 (1) -0.295 V (2) -0.0591 V
 (3) 0.295 V (4) 0.0591 V
71. Consider the cell $\text{Cu} / \text{Cu}^{2+} || \text{Ag}^+ / \text{Ag}$. If the concentration of Cu^{2+} and Ag^+ ions becomes ten times then the emf of the cell will :-
 (1) Becomes 10 times
 (2) Remains same
 (3) Increases by 0.0295 V
 (4) Decreases by 0.0295 V

72. The emf of the cell
 $\text{Ni} | \text{Ni}^{+2} (1.0 \text{ M}) || \text{Au}^{+3} (0.1 \text{ M}) | \text{Au}$
 $[E^\circ \text{ for } \text{Ni}^{+2}/\text{Ni} = -0.25 \text{ V}, E^\circ \text{ for } \text{Au}^{+3}/\text{Au} = 1.50 \text{ V}]$
 is given as:-
 (1) 1.25 V (2) -1.75 V
 (3) 1.75 V (4) 1.73 V

ELECTROLYSIS

73. When an electric current is passed through acidified water, 112 mL of hydrogen gas at STP collects at the cathode in 965 s. The current passed, in ampere is :
 (1) 1.0 (2) 0.5
 (3) 0.1 (4) 2.0
74. Two electrolytic cells one containing acidified ferrous chloride and another acidified ferric chloride are connected in series. The ratio of iron deposited at cathodes in the two cells when electricity is passed through the cells will be :
 (1) 3 : 1 (2) 2 : 1 (3) 1 : 1 (4) 3 : 2
75. A current of 9.65 A flowing for 10 minute deposits 3.0 g of a metal. The equivalent weight of the metal is :
 (1) 10 (2) 30 (3) 50 (4) 96.5
76. The same amount of electricity was passed through two separate electrolytic cells containing solutions of nickel nitrate $[\text{Ni}(\text{NO}_3)_2]$ and chromium nitrate $[\text{Cr}(\text{NO}_3)_3]$ respectively. If 0.3 g of nickel was deposited in the first cell, the amount of chromium deposited in the other cell is :-
 (at. wt. of Ni = 59, at. wt. of Cr = 52)
 (1) 0.1 g (2) 0.17 g
 (3) 0.3 g (4) 0.6 g
77. How many coulombs of electricity are required for the oxidation of 1 mol of H_2O to O_2 ?
 (1) $9.65 \times 10^4 \text{ C}$ (2) $4.825 \times 10^5 \text{ C}$
 (3) $1.93 \times 10^5 \text{ C}$ (4) $1.93 \times 10^4 \text{ C}$
78. When a quantity of electricity equal to that required to liberate 2.24 L of hydrogen at STP from 0.1 M aqueous H_2SO_4 is passed (At. mass of Cu = 63.5) then the mass of copper that will be deposited at cathode in electrolysis of 0.2 M solution of copper sulphate will be :
 (1) 1.59 g (2) 3.18 g
 (3) 6.35 g (4) 12.70 g
79. 10800 C of electricity through the electrolyte deposited 2.977 g of metal with atomic mass 106.4 g mol⁻¹. The charge on the metal cation is -
 (1) +4 (2) +3 (3) +2 (4) +1
80. On passing electricity through dilute H_2SO_4 solution the amount of substance liberated at the cathode and anode are in the ratio :
 (1) 1 : 8 (2) 8 : 1
 (3) 16 : 1 (4) 1 : 16
81. During electrolysis of fused calcium hydride, the hydrogen is produced at :
 (1) Cathode
 (2) Anode
 (3) Hydrogen is not liberated at all
 (4) H_2 produced reacts with oxygen to form water
82. A silver cup is plated with silver by passing 965 A current for one second, the mass of Ag deposited is :- (At. wt. of Ag = 107.87)
 (1) 9.89 g. (2) 107.87 g.
 (3) 1.0787 g. (4) 100.2 g.
83. When electricity is passed through a solution of AlCl_3 , 13.5 g Al is deposited. The number of Faradays must be :-
 (1) 5.0 (2) 1.0 (3) 1.5 (4) 3.0
84. A solution of sodium sulphate in water is electrolysed using inert electrodes. The product at the cathode and anode are respectively :-
 (1) H_2 , SO_2 (2) O_2 , H_2
 (3) O_2 , Na (4) H_2 , O_2
85. One Faraday of electricity will liberate one mole of the metal from the solution of
 (1) Auric chloride (2) Silver nitrate
 (3) Calcium chloride (4) Copper sulphate
86. When 96500 C of electricity are passed through barium chloride solution, the amount of barium deposited will be :-
 (1) 0.5 mol (2) 1.0 mol
 (3) 1.5 mol (4) 2.0 mol
87. A factory produces 40 kg of calcium in two hours by electrolysis. How much aluminium can be produced by the same current in two hours :-
 (At wt. of Ca = 40, Al = 27)
 (1) 22 kg (2) 18 kg
 (3) 9 kg (4) 27 kg

88. What would be the ratio of moles each of Ag^+ , Cu^{+2} , Fe^{+3} ions would be deposited by passage of same quantity of electricity through solutions of their salts :-

(1) 1 : 1 : 1

(2) $1 : \frac{1}{2} : \frac{1}{3}$

(3) $\frac{1}{3} : \frac{1}{2} : 1$

(4) 1 : 2 : 3

89. Electrolysis of aq. CuSO_4 causes :-

(1) An increase in pH

(2) A decrease in pH

(3) Either decrease or increase

(4) None

90. The passage of current liberates H_2 at cathode and Cl_2 at anode the solution is :-

(1) CuSO_4 (aq)

(2) CuCl_2 (aq.)

(3) NaCl (aq.)

(4) Water

COMMERCIAL CELLS

91. When lead accumulator is charged it is :

(1) an electrolytic cell

(2) a galvanic cell

(3) a Daniel cell

(4) none of the above

92. When a lead storage battery is charged :

(1) PbO_2 dissolves

(2) The lead electrode becomes coated with lead sulphate

(3) Sulphuric acid is regenerated

(4) The amount of acid decreases

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	3	1	4	3	3	3	4	2	2	1	2	1	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	3	3	1	2	4	2	2	4	2	4	2	3	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	1	1	4	4	1	2	3	4	3	3	4	2	2	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	3	3	3	4	3	3	2	4	2	4	3	2	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	1	4	3	2	1	3	3	2	3	3	4	1	4	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	3	3	1	1	2	3	3	4	2	1	2	2	2	3
Que.	91	92													
Ans.	1	3													

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

1. If $E_{\text{Fe}^{+2}/\text{Fe}}^0 = -0.441\text{V}$ and $E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^0 = 0.771\text{V}$ the standard EMF of the reaction $\text{Fe} + 2\text{Fe}^{+3} \rightarrow 3\text{Fe}^{+2}$ will be :
- (1) 0.330 V
 - (2) 1.653 V
 - (3) 1.212 V
 - (4) 0.111 V

2. A hypothetical electrochemical cell is shown



The emf measured is +0.20 V

The cell reaction is :

- (1) $\text{A}^+ + \text{B} \longrightarrow \text{A} + \text{B}^+$
- (2) $\text{A}^+ + \text{e}^- \longrightarrow \text{A}$, $\text{B}^+ + \text{e}^- \longrightarrow \text{B}$
- (3) The cell reaction cannot be predicted.
- (4) $\text{A} + \text{B}^+ \longrightarrow \text{A}^+ + \text{B}$

AIIMS 2006

3. The charge required for the reduction of 1 mol of MnO_4^- to MnO_2 is :
- (1) 1F
 - (2) 3F
 - (3) 5F
 - (4) 7F
4. The products formed when an aqueous solution of NaBr is electrolyzed in a cell having inert electrodes are :-
- (1) Na and Br_2
 - (2) Na and O_2
 - (3) H_2 , Br_2 and NaOH
 - (4) H_2 and O_2

AIPMT 2007

5. The equilibrium constant of the reaction : $\text{Cu(s)} + 2\text{Ag}^+(\text{aq.}) \longrightarrow \text{Cu}^{2+}(\text{aq.}) + 2\text{Ag(s)}$ $E^\circ = 0.46\text{V}$ at 298 K is :-
- (1) 2.0×10^{10}
 - (2) 4.0×10^{10}
 - (3) 4.0×10^{15}
 - (4) 2.4×10^{10}

AIPMT 2008

6. On the basis of the following E° values, the strongest oxidizing agent is :-
- $$[\text{Fe}(\text{CN})_6]^{4-} \rightarrow [\text{Fe}(\text{CN})_6]^{3-} + \text{e}^-, E^\circ = -0.35\text{V}$$
- $$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-, E^\circ = -0.77\text{V}$$
- (1) Fe^{3+}
 - (2) $[\text{Fe}(\text{CN})_6]^{3-}$
 - (3) $[\text{Fe}(\text{CN})_6]^{4-}$
 - (4) Fe^{2+}

7. Kohlrausch's law states that at :-
- (1) Infinite dilution, each ion makes definite contribution to conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.
 - (2) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.
 - (3) Finite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte whatever be the nature of the other ion of the electrolyte.
 - (4) Infinite dilution, each ion makes definite contribution to equivalent conductance of an electrolyte depending on the nature of the other ion of electrolyte.

AIPMT 2009

8. Given :
- (i) $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$, $E^\circ = 0.337\text{V}$
 - (ii) $\text{Cu}^{2+} + \text{e}^- \rightarrow \text{Cu}^+$, $E^\circ = 0.153\text{V}$
- Electrode potential, E° for the reaction, $\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$, will be :-
- (1) 0.38 V
 - (2) 0.52 V
 - (3) 0.90 V
 - (4) 0.30 V
9. The equivalent conductance of $\frac{M}{32}$ solution of a weak monobasic acid is $8.0\text{ mho cm}^2 \text{eq}^{-1}$ and at infinite dilution is $400\text{ mho cm}^2 \text{eq}^{-1}$. The dissociation constant of this acid is :-
- (1) 1.25×10^{-4}
 - (2) 1.25×10^{-5}
 - (3) 1.25×10^{-6}
 - (4) 6.25×10^{-4}
10. Al_2O_3 is reduced by electrolysis at low potential and high current. If $4.0 \times 10^4\text{A}$ of current is passed through molten Al_2O_3 for 6 hours, what mass of aluminium is produced ? (Assume 100% current efficiency, At. mass of Al = 27 g mol^{-1})
- (1) $1.3 \times 10^4\text{g}$
 - (2) $9.0 \times 10^3\text{g}$
 - (3) $8.1 \times 10^4\text{g}$
 - (4) $2.4 \times 10^5\text{g}$

AIPMT 2010

11. An increase in equivalent conductance of a strong electrolyte with dilution is mainly due to:-
- (1) Increase in number of ions.
 - (2) Increase in ionic mobility of ions.
 - (3) 100% ionisation of electrolyte at normal dilution.
 - (4) Increase in both i.e. number of ions and ionic mobility of ions.

12. Consider the following relations for emf of a electrochemical cell :
- emf of cell = (Oxidation potential of anode) – (Reduction potential of cathode)
 - emf of cell = (Oxidation potential of anode) + (Reduction potential of cathode)
 - emf of cell = (Reduction potential of anode) + (Reduction potential of cathode)
 - emf of cell = (Oxidation potential of anode) – (Oxidation potential of cathode)

Which of the above relations are correct :

- (a) and (b)
 - (c) and (d)
 - (b) and (d)
 - (c) and (a)
13. Which of the following expressions correctly represents the equivalent conductance at infinite dilution of $\text{Al}_2(\text{SO}_4)_3$. Given that $\Lambda^\circ_{\text{Al}^{3+}}$ and $\Lambda^\circ_{\text{SO}_4^{2-}}$ are the equivalent conductances at infinite dilution of the respective ions :-

- $\Lambda^\circ_{\text{Al}^{3+}} + \Lambda^\circ_{\text{SO}_4^{2-}}$
- $(\Lambda^\circ_{\text{Al}^{3+}} + \Lambda^\circ_{\text{SO}_4^{2-}}) \times 6$
- $\frac{1}{3} \Lambda^\circ_{\text{Al}^{3+}} + \frac{1}{2} \Lambda^\circ_{\text{SO}_4^{2-}}$
- $2\Lambda^\circ_{\text{Al}^{3+}} + 3\Lambda^\circ_{\text{SO}_4^{2-}}$

14. For the reduction of silver ions with copper metals, the standard cell potential was found to be +0.46 V at 25 °C. The value of standard Gibbs energy, ΔG° will be ($F = 96500 \text{ C mol}^{-1}$)
- 98.0 kJ
 - 89.0 kJ
 - 89.0 J
 - 44.5 kJ

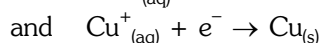
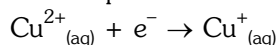
AIIMS 2010

15. Limiting molar conductance of H^+ and CH_3COO^- ions are $344 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $40 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Molar conductance of 0.008 M CH_3COOH is 48. What will be the value of K_a for CH_3COOH :-
- 1.4×10^{-5}
 - 1.2×10^{-5}
 - 1.4×10^{-4}
 - 1×10^{-5}
16. E value for the cell reaction $\text{Cu} | \text{Cu}^{2+} (0.001 \text{ M}) || \text{Cu}^{2+} (0.1 \text{ M}) | \text{Cu}$ is :-
- $-\frac{RT}{F} \ln(0.01)$
 - $-\frac{RT}{F} \ln(0.1)$
 - $+\frac{RT}{F} \ln(0.01)$
 - $+\frac{RT}{F} \ln(0.1)$

AIPMT Pre. 2011

17. Standard electrode potential of three metals X, Y and Z are -1.2 V, +0.5 V and -3.0 V respectively. The reducing power of these metals will be :-
- $Y > Z > X$
 - $Y > X > Z$
 - $Z > X > Y$
 - $X > Y > Z$

18. The electrode potentials for



are +0.15 V and +0.50 V respectively. The value of $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$ will be :-

- 0.500 V
- 0.325 V
- 0.650 V
- 0.150 V

19. Standard electrode potential for $\text{Sn}^{4+}/\text{Sn}^{2+}$ couple is +0.15 V and that for the Cr^{3+}/Cr couples is -0.74 V. These two couples in their standard state are connected to make a cell. The cell potential will be :-

- +1.19 V
- +0.89 V
- +0.18 V
- +1.83 V

20. If the E°_{cell} for a given reaction has a negative value, then which of the following gives the correct relationship for the values of ΔG° and K_{eq} ?

- $\Delta G^\circ > 0$; $K_{\text{eq}} > 1$
- $\Delta G^\circ < 0$; $K_{\text{eq}} > 1$
- $\Delta G^\circ < 0$; $K_{\text{eq}} < 1$
- $\Delta G^\circ > 0$; $K_{\text{eq}} < 1$

AIPMT Mains 2011

21. A solution contains Fe^{2+} , Fe^{3+} and I^- ions. This solution was treated with iodine at 35 °C. E° for $\text{Fe}^{3+} / \text{Fe}^{2+}$ is +0.77 V and E° for $\text{I}_2/2\text{I}^- = 0.536 \text{ V}$. The favourable redox reaction is :-
- Fe^{2+} will be oxidised to Fe^{3+}
 - I_2 will be reduced to I^-
 - There will be no redox reaction
 - I^- will be oxidised to I_2

AIPMT Pre. 2012

22. Limiting molar conductivity of NH_4OH

(i.e. $\Lambda^\circ_{\text{m}(\text{NH}_4\text{OH})}$) is equal to:-

- $\Lambda^\circ_{\text{m}(\text{NH}_4\text{OH})} + \Lambda^\circ_{\text{m}(\text{NH}_4\text{Cl})} - \Lambda^\circ_{\text{m}(\text{HCl})}$
- $\Lambda^\circ_{\text{m}(\text{NH}_4\text{Cl})} + \Lambda^\circ_{\text{m}(\text{NaOH})} - \Lambda^\circ_{\text{m}(\text{NaCl})}$
- $\Lambda^\circ_{\text{m}(\text{NH}_4\text{Cl})} + \Lambda^\circ_{\text{m}(\text{NaCl})} - \Lambda^\circ_{\text{m}(\text{NaOH})}$
- $\Lambda^\circ_{\text{m}(\text{NaOH})} + \Lambda^\circ_{\text{m}(\text{NaCl})} - \Lambda^\circ_{\text{m}(\text{NH}_4\text{Cl})}$

AIPMT Mains 2012

- 23.** Molar conductivities (Λ_m°) at infinite dilution of NaCl, HCl and CH_3COONa are 126.4, 425.9 and $91.0 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. Λ_m° for CH_3COOH will be :-
- (1) $290.8 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $390.5 \text{ S cm}^2 \text{ mol}^{-1}$
(3) $425.5 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $180.5 \text{ S cm}^2 \text{ mol}^{-1}$

AIIMS 2012

- 24.** For $\text{Zn}^{+2} \mid \text{Zn}$ $E^\circ = -0.76 \text{ V}$ then EMF of this cell $\text{Zn} \mid \text{Zn}^{+2}(1\text{M}) \parallel 2\text{H}^+(1\text{M}) \mid \text{H}_2(1 \text{ atm})$ will be :-
- (1) -0.76 V (2) 0.76 V
(3) 0.38 V (4) -0.38 V

NEET-UG 2013

- 25.** At 25 °C molar conductance of 0.1 molar aqueous solution of ammonium hydroxide is $9.54 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and at infinite dilution its molar conductance is $238 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$. The degree of ionisation of ammonium hydroxide at the same concentration and temperature is :-
- (1) 40.800 % (2) 2.080 %
(3) 20.800 % (4) 4.008 %
- 26.** A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at 1 atm pressure. The oxidation potential of electrode would be ?
- (1) 1.18 V (2) 0.059 V
(3) 0.59 V (4) 0.118 V

- 27.** A button cell used in watches function as following
- $$\text{Zn(s)} + \text{Ag}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightleftharpoons 2\text{Ag(s)} + \text{Zn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$$
- If half cell potentials are
- $$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn(s)}; E^\circ = -0.76 \text{ V}$$
- $$\text{Ag}_2\text{O(s)} + \text{H}_2\text{O(l)} + 2\text{e}^- \rightarrow 2\text{Ag(s)} + 2\text{OH}^-(\text{aq}); E^\circ = 0.34 \text{ V}$$
- The cell potential will be :-
- (1) 1.34 V (2) 1.10 V
(3) 0.42 V (4) 0.84 V

AIIMS 2013

- 28.** Electrode potential of hydrogen electrode is 18 mV then $[H^+]$ is :-
 (1) 0.2 (2) 1 (3) 2 (4) 5

AIPMT 2014

- 29.** When 0.1 mol MnO_4^{2-} is oxidised the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is :-
- (1) 96500 C (2) 2×96500 C
(3) 9650 C (4) 96.50 C
- 30.** The weight of silver (at wt. = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be :-
- (1) 5.4 g (2) 10.8 g
(3) 54.0 g (4) 108.0 g

AIIMS 2014

- 31.** Amount of charge is required to convert 17 g H_2O_2 into O_2 :-
- (1) 1F (2) 2F
(3) 6F (4) None of these

AIPMT 2015

- 32.** A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as :-
- (1) Electrolytic cell (2) Dynamo
(3) Ni-Cd cell (4) Fuel Cell

AIIMS 2015

- 33.** For the cell reaction
- $$\text{Mg} + 2\text{Ag}^+ (0.0001\text{M}) \rightarrow \text{Mg}^{2+} (0.01\text{M}) + 2\text{Ag}$$
- $E_{\text{cell}}^0 = 3.177 \text{ V}$ then what will be the value of E_{cell} for the above reaction ?
- (1) 2.5 (2) 3.0 (3) 3.2 (4) 3.8
- 34.** If electrode potential of $0.01 \text{ M Cu}^{2+} / \text{Cu}$ is 0.32 V then what will be its E_{RP}^0 -
- (1) 0.18 V (2) 0.28 V
(3) 0.74 V (4) 0.38 V

NEET-I 2016

- 35.** The pressure of H_2 required to make the potential of H_2 -electrode zero in pure water at 298 K is :-
- (1) 10^{-14} atm (2) 10^{-12} atm
(3) 10^{-10} atm (4) 10^{-4} atm

NEET-II 2016

- 36.** The molar conductivity of a 0.5 mol/dm^3 solution of AgNO_3 with electrolytic conductivity of $5.76 \times 10^{-3} \text{ S cm}^{-1}$ at 298 K is
- (1) $0.086 \text{ S cm}^2/\text{mol}$ (2) $28.8 \text{ S cm}^2/\text{mol}$
(3) $2.88 \text{ S cm}^2/\text{mol}$ (4) $11.52 \text{ S cm}^2/\text{mol}$

37. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 A is

- (1) 220 minutes (2) 330 minutes
(3) 55 minutes (4) 110 minutes

38. If the E_{cell}° for a given reaction has a negative value, which of the following gives the **correct** relationships for the values of ΔG° and K_{eq} ?

- (1) $\Delta G^{\circ} < 0$; $K_{\text{eq}} > 1$ (2) $\Delta G^{\circ} < 0$; $K_{\text{eq}} < 1$
(3) $\Delta G^{\circ} > 0$; $K_{\text{eq}} < 1$ (4) $\Delta G^{\circ} > 0$; $K_{\text{eq}} > 1$

39. The number of electrons delivered at the cathode during electrolysis by a current of 1 A in 60 s is (charge on electron = 1.60×10^{-19} C)

- (1) 3.75×10^{20} (2) 7.48×10^{23}
(3) 6×10^{23} (4) 6×10^{20}

AIIMS 2016

40. How many faraday charge is required for $\text{KMnO}_4/\text{Mn}^{2+}$:-

- (1) 3F (2) 2F (3) 5F (4) 1F

NEET(UG) 2017

41. In the electrochemical cell :-
 $\text{Zn} | \text{ZnSO}_4(0.01\text{M}) || \text{CuSO}_4(1.0\text{M}) | \text{Cu}$, the emf of this Daniel cell is E_1 . When the concentration of ZnSO_4 is changed to 1.0M and that of CuSO_4 changed to 0.01M, the emf changes to E_2 . Which one of the relationship is correct between E_1 and E_2 ?

(Given, $\frac{RT}{F} = 0.059$)

- (1) $E_1 < E_2$ (2) $E_1 > E_2$
(3) $E_2 = 0 \neq E_1$ (4) $E_1 = E_2$

AIIMS 2017

42. If limiting molar conductances of Ca^{+2} and Cl^- ions are 119 and $76.3 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The limiting molar conductivity of CaCl_2 is :-

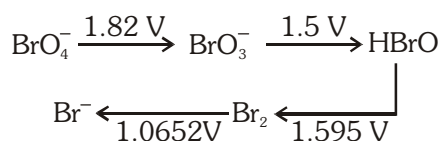
- (1) $347 \text{ S cm}^2 \text{ mol}^{-1}$ (2) $271.6 \text{ S cm}^2 \text{ mol}^{-1}$
(3) $42.7 \text{ S cm}^2 \text{ mol}^{-1}$ (4) $195.3 \text{ S cm}^2 \text{ mol}^{-1}$

EXERCISE-II (Previous Year Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	2	3	3	1	2	2	2	3	2	3	1	2	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	2	2	4	4	2	2	2	4	3	2	3	3	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	4	2	4	1	4	4	3	1	3	2	2	4	2	2
Que.	46	47													
Ans.	4	3													

NEET(UG) 2018

43. Consider the change in oxidation state of Bromine corresponding to different EMF values as shown in the diagram below:

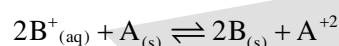


Then the species undergoing disproportionation is:-

- (1) BrO_3^- (2) BrO_4^-
(3) Br_2 (4) HBrO

AIIMS 2018

44. For given reaction



If K_{eq} of reaction is $10^{15.2}$

and $E^{\circ}(\text{A}^{+2}/\text{A}) = 0.34\text{V}$

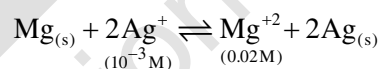
Then calculate $E^{\circ}(\text{B}^{+}/\text{B})$ will be :-

- (1) -0.80 (2) 0.80
(3) 1.27 (4) 1.60

45. For an electrolytic solution having concentration 0.1M and resistance is $1\text{K}\Omega$. If distance between electrode is 100 cm and area is 1 cm^2 then calculate λ_{m} in $\text{S cm}^2 \text{ mol}^{-1}$:-

- (1) 100 (2) 1000 (3) 500 (4) 50

46. What will be value of E_{cell} for reaction



Given E_{cell}° is 3.17 V

- (1) 2.67 (2) 2.13
(3) 3.27 (4) 3.041

47. 36 g H_2O is electrolysed by passing 3 ampere current. What will be time required for complete decomposition (in hour) :-

- (1) 17.87 hour (2) 15.84 hour
(3) 35.74 hour (4) 25.35 hour

ANSWER KEY

EXERCISE-III (Analytical Questions)

Check Your Understanding

- Specific conductance of 0.1 M Nitric acid is $6.3 \times 10^{-2} \text{ ohm}^{-1} \text{ cm}^{-1}$. The molar conductance of the solution is :
 - $630 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 - $315 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 - $100 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 - $6.300 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
 - $63.0 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- The highest electrical conductivity of the following aqueous solution is of
 - 0.1 M fluoroacetic acid
 - 0.1 M difluoroacetic acid
 - 0.1 M acetic acid
 - 0.1 M chloroacetic acid
- Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1 M is 100Ω . The conductivity of this solution is 1.29 S m^{-1} . Resistance of the same cell when filled with 0.02 M of the same solution is 520Ω . The molar conductivity of 0.02 M solution of the electrolyte will be.
 - $124 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 - $1240 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 - $1.24 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 - $12.4 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
- Resistance of 0.2 M solution of an electrolyte is 50Ω . The specific conductance of the solution is 1.4 S m^{-1} . The resistance of 0.5 M solution of the same electrolyte is 280Ω . The molar conductivity of 0.5 M solution of the electrolyte in $\text{S m}^2 \text{ mol}^{-1}$ is :
 - 5×10^3
 - 5×10^2
 - 5×10^{-4}
 - 5×10^{-3}
- Resistance of 0.2 M solution of an electrolyte is 50Ω . The specific conductance of the solution is 1.3 S m^{-1} . If resistance of the 0.4 M solution of the same electrolyte is 260Ω then its molar conductivity is:-
 - $6250 \text{ S m}^2 \text{ mol}^{-1}$
 - $6.25 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 - $625 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 - $62.5 \text{ S m}^2 \text{ mol}^{-1}$
- The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ_∞ , respectively. The correct relationship between λ_C and λ_∞ is given by :
(where the constant B is positive)
 - $\lambda_C = \lambda_\infty - (B)\sqrt{C}$
 - $\lambda_C = \lambda_\infty + (B)\sqrt{C}$
 - $\lambda_C = \lambda_\infty + (B) C$
 - $\lambda_C = \lambda_\infty - (B) C$

Electrolyte	KCl	KNO ₃	HCl	NaOAc	NaCl
Λ^∞ (S cm ² mol ⁻¹)	149.9	145.0	426.2	91.0	126.5

- Calculate $\Lambda_{\text{HOAc}}^{\infty}$ using appropriate molar conductances of the electrolytes listed above at infinite dilution in H_2O at 25°C
 (1) 390.7 (2) 217.5 (3) 517.2 (4) 552.7
- 8.** Which of the following statements is TRUE for the electrochemical Daniel cell :
- (1) Electrons flow from copper electrode to zinc electrode.
 (2) Current flows from zinc electrode to copper electrode.
 (3) Cations move toward copper electrode.
 (4) Cations move toward zinc electrode.
- 9.** Aluminium displaces hydrogen from dilute HCl whereas silver does not, the E.M.F. of a cell prepared by combining Al/Al^{+3} and Ag/Ag^+ is 2.46 V. The reduction potential of silver electrode is +0.80 V. The reduction potential of aluminium electrode is :
- (1) - 3.26 V (2) +1.66 V
 (3) -1.66 V (4) 3.26 V
- 10.** Standard reduction electrode potential of three metals A, B and C are respectively + 0.5 V, - 3.0 V and -1.2 V. The reducing power of these metals are :
- (1) $\text{C} > \text{B} > \text{A}$ (2) $\text{A} > \text{C} > \text{B}$
 (3) $\text{B} > \text{C} > \text{A}$ (4) $\text{A} > \text{B} > \text{C}$
- 11.** Consider the following E^0 values
- $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = +0.77 \text{ V}$
 $E_{\text{Sn}^{2+}/\text{Sn}}^0 = -0.14 \text{ V}$
- Under standard conditions the potential for the reaction
- $\text{Sn(s)} + 2\text{Fe}^{3+}(\text{aq}) \longrightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{Sn}^{2+}(\text{aq})$ is
- (1) 0.91 V (2) 1.40 V
 (3) 1.68 V (4) 0.63 V
- 12.** For the redox reaction :
- $\text{Zn(s)} + \text{Cu}^{2+} (0.1\text{M}) \rightarrow \text{Zn}^{2+} (1\text{M}) + \text{Cu(s)}$ taking place in a cell,
 E_{Cell}^0 is 1.10 V E_{Cell} for the cell will be
- $\left(2.303 \frac{RT}{F} = 0.0591 \right)$
- (1) 1.07 V (2) 0.82 V
 (3) 2.14 V (4) 1.80 V

- 13.** In a cell that utilises the reaction
- $$\text{Zn(s)} + 2\text{H}^+(\text{aq.}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$$
- addition of H_2SO_4 to cathode compartment, will
- (1) increase the E_{cell} and shift equilibrium to the right
 - (2) lower the E_{cell} and shift equilibrium to the right
 - (3) lower the E_{cell} and shift equilibrium to the left
 - (4) increase the E_{cell} and shift equilibrium to the left
- 14.** The e.m.f. of a Daniell cell at 298 K is E_1 .
 $\text{Zn} | \text{ZnSO}_4 | | \text{CuSO}_4 | \text{Cu}$
 (0.01M) (1.0M)
 When the concentration of ZnSO_4 is 1.0 M and that of CuSO_4 is 0.01 M, the e.m.f. changed to E_2 . What is the relationship between E_1 and E_2
- (1) $E_1 > E_2$
 - (2) $E_1 < E_2$
 - (3) $E_1 = E_2$
 - (4) $E_2 = 0 \neq E_1$
- 15.** The reduction potential of hydrogen half-cell will be negative if :-
- (1) $p(\text{H}_2) = 2 \text{ atm}$ and $[\text{H}^+] = 1.0 \text{ M}$
 - (2) $p(\text{H}_2) = 2 \text{ atm}$ and $[\text{H}^+] = 2.0 \text{ M}$
 - (3) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^+] = 2.0 \text{ M}$
 - (4) $p(\text{H}_2) = 1 \text{ atm}$ and $[\text{H}^+] = 1.0 \text{ M}$
- 16.** For a cell reaction involving two-electron change, the standard e.m.f. of the cell is found to be 0.295 V at 25° C. The equilibrium constant of the reaction at 25 °C will be :
- (1) 10
 - (2) 1×10^{10}
 - (3) 1×10^{-10}
 - (4) 29.5×10^{-2}
- 17.** For a spontaneous reaction the ΔG , equilibrium constant (K) and E_{Cell}^0 will be respectively
- (1) -ve, < 1, -ve
 - (2) -ve, > 1, -ve
 - (3) -ve, > 1, +ve
 - (4) +ve, > 1, -ve
- 18.** On the basis of the information available from the reaction
- $$\frac{4}{3}\text{Al} + \text{O}_2 \rightarrow \frac{2}{3}\text{Al}_2\text{O}_3,$$
- $\Delta G = -827 \text{ KJ mol}^{-1}$ of O_2 , the minimum e.m.f. required to carry out an electrolysis of Al_2O_3 is :-
 (F = 96500 C mol⁻¹)
- (1) 2.14 V
 - (2) 4.28 V
 - (3) 6.42 V
 - (4) 8.56 V

- 19.** Given below are the half-cell reactions :-
- $$\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}; \quad E^\circ = -1.18 \text{ V}$$
- $$2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+}); \quad E^\circ = +1.51 \text{ V}$$
- The E° for $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ will be :
- (1) -0.33 V; the reaction will not occur
 - (2) -0.33 V; the reaction will occur
 - (3) -2.69 V; the reaction will not occur
 - (4) -2.69 V; the reaction will occur
- 20.** Given the data at 25°C,
- $$\text{Ag} + \text{I}^- \rightarrow \text{AgI} + \text{e}^-, \quad E^\circ = 0.152 \text{ V}$$
- $$\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-, \quad E^\circ = -0.800 \text{ V}$$
- What is the value of $\log K_{\text{sp}}$ for AgI ?
- $$\left(2.303 \frac{RT}{F} = 0.059 \text{ V} \right)$$
- (1) -8.12
 - (2) +8.612
 - (3) -37.83
 - (4) -16.13
- 21.** The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is:
- (1) Cu
 - (2) Cr
 - (3) Ag
 - (4) Ca
- 22.** When during electrolysis of a solution of AgNO_3 9650 C of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be :
- (1) 21.6 g
 - (2) 108 g
 - (3) 1.08 g
 - (4) 10.8 g
- 23.** Aluminium oxide may be electrolysed at 1000 °C to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96500 C). The cathode reaction is $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
 To prepare 5.12 kg of aluminium metal by this method would require.
- (1) $5.49 \times 10^4 \text{ C}$ of electric charge
 - (2) $5.49 \times 10^1 \text{ C}$ of electric charge
 - (3) $5.49 \times 10^7 \text{ C}$ of electric charge
 - (4) $1.83 \times 10^7 \text{ C}$ of electric charge
- 24.** A current of 'i' A was passed for 't' s. through three cells P, Q and R connected in a series. These contain respectively silver nitrate, mercuric nitrate and mercurous nitrate. At the cathode of the cell P, 0.216 g of Ag was deposited. The weights of mercury deposited in the cathode of Q and R respectively are : (at. wt. of Hg = 200.59)
- (1) 0.4012 g and 0.8024 g
 - (2) 0.4012 g and 0.2006 g
 - (3) 0.2006 g and 0.4012 g
 - (4) 0.1003 g and 0.2006 g

25. A galvanic cell is set up from a zinc bar weighing 100 g and 1.0 L of 1.0 M CuSO_4 solution. How long would the cell will run if it is assumed to deliver a steady current of 1.0 A. (Atomic mass of Zn = 65)
- (1) 1.1 hr. (2) 46 hr.
(3) 53.6 hr. (4) 24.00 hr.
26. The electrochemical equivalent of silver is 0.0011180 g. When an electric current of 0.5 A is passed through an aqueous silver nitrate solution for 200 s, the amount of silver deposited is :
- (1) 1.1180 g (2) 0.11180 g
(3) 5.590 g (4) 0.5590 g
27. A current of 9.65 A is passed through an aqueous solution of NaCl using suitable electrodes for 1000 s. Given that 1 faraday equals 96500 C the amount of NaOH (mol wt. 40.00) formed on electrolysis is :
- (1) 2.0 g (2) 8.0 g
(3) 4.0 g (4) 1.0 g.
28. An electric current is passed through silver voltameter connected to a water voltameter. The cathode of the silver voltameter weighed 0.108 g more at the end of the electrolysis. The volume of oxygen evolved at STP is :
- (1) 56 cm^3 (2) 550 cm^3
(3) 5.6 cm^3 (4) 11.2 cm^3
(5) 22.4 cm^3
29. 4.5 g of aluminium (at. mass 27 amu) is deposited at cathode from Al^{3+} solution by a certain quantity of electric charge. The volume of hydrogen produced at STP from H^+ ions in solution by the same quantity of electric charge will be –
- (1) 44.8 L (2) 11.2 L
(3) 22.4 L (4) 5.6 L
30. The thermodynamic efficiency of cell is given by -
- (1) $\frac{\Delta H}{\Delta G}$ (2) $\frac{nFE}{\Delta G}$
(3) $-\frac{nFE}{\Delta H}$ (4) Zero

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	1	3	2	1	1	3	3	3	1	1	1	1	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	1	3	4	4	4	3	3	3	2	3	3	4	3

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** Electrolysis of molten calcium hydride produces hydrogen gas at anode.
Reason :- Hydrogen in calcium hydride is present as H^- ion.
 (1) A (2) B (3) C (4) D
- Assertion :-** Molar conductance of 0.1 M NH_4OH solution is less than that of 0.001M NH_4OH solution.
Reason :- Dilution increases the degree of ionisation of NH_4OH .
 (1) A (2) B (3) C (4) D
- Assertion :-** Salt bridge is generally used in the electrochemical cells.
Reason :- The ions of the electrolyte used in the salt bridge should have nearly same transport number.
 (1) A (2) B (3) C (4) D
- Assertion :-** The electrode potential of SHE is zero.
Reason :- SHE is a standard reference electrode.
 (1) A (2) B (3) C (4) D
- Assertion :-** Copper sulphate solution cannot be stored in iron-vessel.
Reason :- Copper atoms are oxidised by iron ions.
 (1) A (2) B (3) C (4) D
- Assertion :-** Absolute value of E_{red}^0 of an electrode cannot be determined.
Reason :- Neither oxidation nor reduction can take place alone.
 (1) A (2) B (3) C (4) D
- Assertion :-** A dry cell became dead after long time even if it has not been used.
Reason :- The NH_4Cl slowly and gradually corrodes the zinc container.
 (1) A (2) B (3) C (4) D
- Assertion :-** The molar conductance of weak electrolytes is low as compared to that of strong electrolytes at moderate concentrations.
Reason :- Weak electrolytes at moderate concentration dissociates to a much greater extent as compared to strong electrolytes.
 (1) A (2) B (3) C (4) D
- Assertion :-** Galvanised iron does not rust.
Reason :- Zinc has a more negative electrode potential than iron.
 (1) A (2) B (3) C (4) D
- Assertion :-** Increase in the concentration of copper half cell in Daniel cell increases the emf of the cell.
Reason :- According to the nernst equation

$$emf \text{ of Daniel cell} = E_{cell}^0 + \frac{0.059}{2} \log \frac{[Cu^{+2}]}{[Zn^{+2}]}$$
 (1) A (2) B (3) C (4) D
- Assertion :-** Sodium ions are discharged at the Hg electrode during electrolysis in preference to H^+ ions.
Reason :- The nature of electrode also affects the order of discharge of cations.
 (1) A (2) B (3) C (4) D
- Assertion :-** The cell constant of a cell depends upon the nature of the material of the electrodes.
Reason :- The observed conductance depends upon the nature of the electrolyte and the concentration of the solution.
 (1) A (2) B (3) C (4) D
- Assertion :-** At the end of electrolysis using platinum electrodes, an aqueous solution of copper sulphate turns colourless.
Reason :- Copper in $CuSO_4$ is converted to $Cu(OH)_2$ during the electrolysis.
 (1) A (2) B (3) C (4) D

- 14. Assertion :-** In electrolysis, the quantity of electricity needed for depositing 1 mol of silver is different from that required for 1 mol of copper.
Reason :- The atomic weight of silver and copper are different.
 (1) A (2) B (3) C (4) D
- 15. Assertion :-** pH of solution increases during electrolysis of aqueous solution of NaCl.
Reason :- Electrolysis is non-spontaneous process.
 (1) A (2) B (3) C (4) D
- 16. Assertion :-** A reaction is spontaneous if $E_{\text{cell}} = +ve$.
Reason :- For $E_{\text{cell}} = +ve$, (ΔG) is always $-ve$.
 (1) A (2) B (3) C (4) D
- 17. Assertion :-** Cu is stronger reducing agent than H_2 .
Reason :- E° of Cu^{+2}/Cu is negative.
 (1) A (2) B (3) C (4) D
- 18. Assertion :-** For cell reaction; at 298 K
 $Zn(s) + Cu^{+2}(1M) \rightleftharpoons Zn^{+2}(1M) + Cu(s)$
 $E_{\text{cell}}^0 = 0$.
Reason :- At standard condition ; cell potential is always zero.
 (1) A (2) B (3) C (4) D
- 19. Assertion :-** Molar conductivity of an electrolyte solution increases on dilution.
Reason :- Decrease in conductivity is overcome by increase in volume.
 (1) A (2) B (3) C (4) D
- 20. Assertion :-** In electrolytic refining, electrolytic cell consists of pure and impure Cu electrodes dipped in $CuSO_4(aq)$.
Reason :- Pure Cu is obtained at anode.
 (1) A (2) B (3) C (4) D
- 21. Assertion :-** Molar conductivity of CH_3COOH increases with dilution.
Reason :- The graph between λ_m v/s $(C)^{1/2}$ is a straight line for CH_3COOH .
 (1) A (2) B (3) C (4) D
- 22. Assertion :-** On electrolysis of aq. NaCl, Na is obtained.
Reason :- Na^+ is reduced at cathode.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	2	2	3	1	1	3	1	1	1	4	3	2	2
Que.	16	17	18	19	20	21	22								
Ans.	1	4	4	1	3	3	4								

IMPORTANT NOTES

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NEET SYLLABUS

Surface Chemistry : Adsorption-physisorption and chemisorption; factors affecting adsorption of gases on solids, colloidal state: distinction between true solutions, colloids and suspensions; lyophilic, lyophobic multimolecular and macromolecular colloids; properties of colloids; Tyndall effect, Brownian movement, electrophoresis, coagulation; emulsions- types of emulsions, catalysis, homogeneous and heterogeneous, activity and selectivity: enzyme catalysis.

OBJECTIVES

After studying this unit, we will be able to :

- *define adsorption and classification on the basis of nature of forces.*
- *explain factors affecting rate of adsorption*
- *enumerate the nature of colloidal state.*
- *describe preparation, properties and purification of colloids.*
- *application of colloids*
- *appreciate the role of catalysts in industry.*

"I would like to start by emphasizing the importance of surfaces. It is at a surface where many of our most interesting and useful phenomena occur. Much of chemistry is concerned with a surface."

Walter Houser Brattain

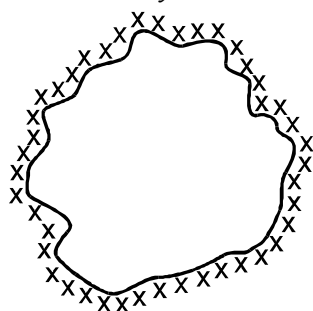
Surface Chemistry

LECTURE # 1

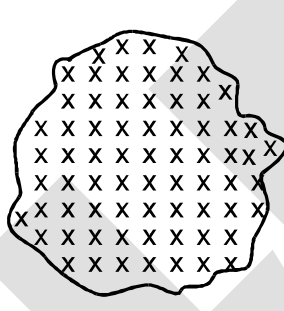
5.0 INTRODUCTION

5.1 DEFINITION OF ADSORPTION AND RELATED TERMS :

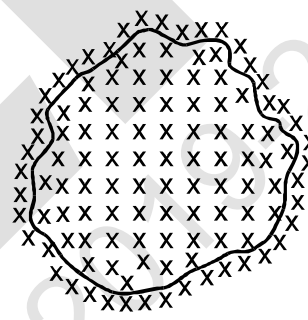
- (i) **Adsorption** : The phenomenon of increase in concentration at the surface due to molecular surface force is known as adsorption.
- (ii) **Adsorbate** : The substances solids, gases or liquids which are adsorbed on the surface of adsorbent are called adsorbate.
- (iii) **Adsorbent** : The solid or liquid substance on the surface of which adsorption takes place is called adsorbent. Examples of adsorbents are **activated charcoal, Pt, Pd, Ni, Silica gel, gelatin, Al_2O_3 , starch.**
- (iv) **Absorption** : When a substance is uniformly distributed throughout the body of a solid or liquid, the phenomenon is called absorption.
- (v) **Sorption** : It may be defined as the process in which both adsorption and absorption take place simultaneously.



Adsorption



Absorption



Sorption

Adsorption

1. It involves higher concentration of the gas or liquid at the surface of a substance.
2. It is rapid in the beginning and slows down near the equilibrium.
3. It is a surface phenomenon.
4. It is a fast process.

Absorption

1. It involves uniform distribution of the molecular species throughout the bulk
2. It occurs at a uniform rate.
3. It occurs throughout the body of the material.
4. It is a slow process.

Mechanism of adsorption

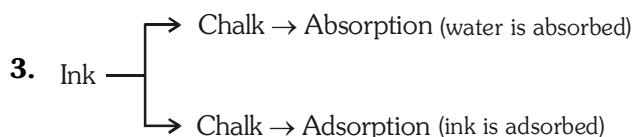
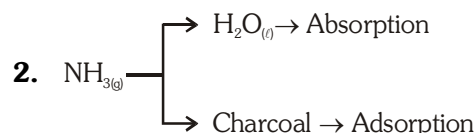
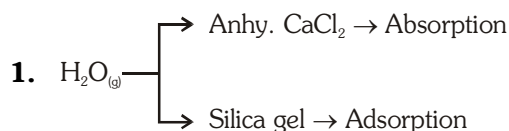
Adsorption is due to the fact that the surface particles of the adsorbent are in different state than the particles inside the bulk. Inside the adsorbent all the forces acting between the particles are mutually balanced but on the surface, particles are not surrounded by atoms or molecules on all sides and hence they possess unbalanced or residual attractive forces.

- (i) Adsorption is an exothermic process
- (ii) Freedom of movement of gas molecules decreases
- (iii) Adsorption is a spontaneous process

$$\text{so } \Delta H = -ve$$

$$\text{so } \Delta S_{\text{gas}} = -ve$$

$$\text{so } \Delta G = -ve \text{ but } \Delta S_{\text{total}} > 0$$

Examples.

- When Charcoal is mixed with Methylene blue then its blue colour is adsorbed (adsorption) by charcoal and the dye becomes colourless.
- When the Magneson reagent is mixed with white precipitate of $\text{Mg}(\text{OH})_2$ blue colour is adsorbed by white precipitate (adsorption) of $\text{Mg}(\text{OH})_2$
- When charcoal is mixed in yellow sugar solution, then yellow colour is adsorbed by charcoal and white coloured sugar solution is obtained (adsorption)

5.2 TYPES OF ADSORPTION

Depending upon the nature of forces between molecules of adsorbate and adsorbent, adsorption is of two types.

Table : Comparison of Physi-sorption and Chemi-sorption

	Physical Adsorption	Chemical Adsorption
1.	It is caused by intermolecular van der Waal's forces.	It is caused by chemical bond formation.
2.	It is not specific.	It is highly specific.
3.	It is reversible.	It is irreversible.
4.	Heat of adsorption is low (-20 to -40 kJ mol^{-1})	Heat of adsorption is high (-80 to -240 kJ mol^{-1})
5.	No appreciable activation energy is involved.	High activation energy is involved.
6.	It forms multimolecular layers on adsorbent surface under high pressure.	It forms unimolecular layer.
7.	It depends on the nature of gas. More easily liquefiable gases are adsorbed readily.	It also depends on the nature of gas. Gases which can react with the adsorbent show chemisorption.
8.	It decreases with increases of temperature.	It first increases then decreases with temperature.

5.3 FACTORS AFFECTING EXTENT OF ADSORPTION (x/m)**(i) Nature of adsorbent :**

For Solids : Adsorption $\propto$ Surface area of adsorbent or free valency of the solid.

(ii) Nature of adsorbate:

For gases : Adsorption $\propto$ Critical temperature of gas (T_c)

Volume of gases at N.T.P. adsorbed by 1 g of Charcoal.

Gas	H_2	N_2	CH_4	CO_2	HCl	NH_3	SO_2
Critical Temperature, T_c (K)	33	126	190	304	324	406	430
Volume adsorbed (mL)	4.7	8	16.2	48	72	181	380

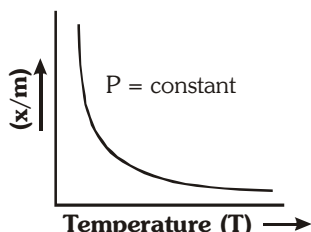
Critical temperature increases
Ease of liquification increases
Extent of adsorption increases



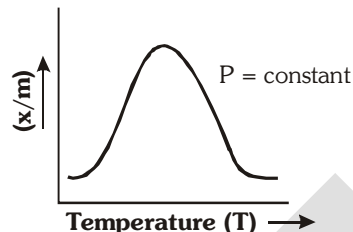
(iii) Effect of temperature : $\text{Adsorbate} + \text{Adsorbent} \rightleftharpoons \text{Adsorption}$, $\Delta H = -ve$, exothermic process.

Normally adsorption decreases with temperature because adsorption is exothermic process. Physical adsorption shows regular decrease with temperature, but chemical adsorption first increases then decreases with temperature because it is specific, requires activation energy and in this process heat supplied may be used as its energy of activation.

(Physical adsorption)



(Chemical adsorption)



(iv) Effect of Pressure : (Adsorption Isotherm)

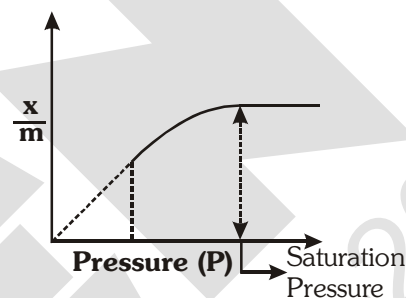
The extent of the adsorption of a gas on a solid is generally

expressed as $\frac{x}{m}$

Where :

x – Mass of adsorbate,
 m – Mass of adsorbent

Theoretically $\frac{x}{m} \propto P$



5.4 TYPES OF ADSORPTION ISOTHERM

(a) Freundlich adsorption isotherm : (Physical adsorption)

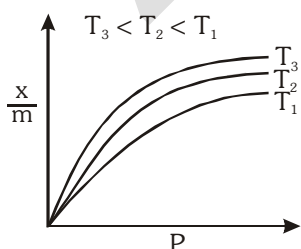
- (i) Freundlich adsorption isotherm is obeyed by the adsorption where the adsorbate forms single layer on the surface of the adsorbent.

$$\frac{x}{m} = kP^{\frac{1}{n}} \quad \text{Freundlich adsorption isotherm equation; } n \geq 1$$

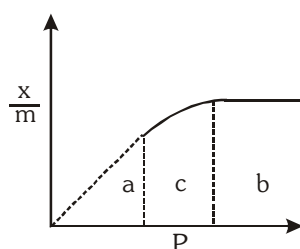
and in logarithmic form $\log \frac{x}{m} = \log k + \frac{1}{n} \log P$; $0 \leq \frac{1}{n} \leq 1$; in general $\frac{1}{n}$ is 0.1 to 0.5.

where x is the weight of the gas adsorbed by m g of the adsorbent at a pressure P , thus x/m represents the amount of gas adsorbed on the unit mass of adsorbent, k and n are constant at a particular temperature and for a particular adsorbent and adsorbate (gas), n is always greater than one, indicating that the amount of the gas adsorbed does not increase as rapidly as the pressure.

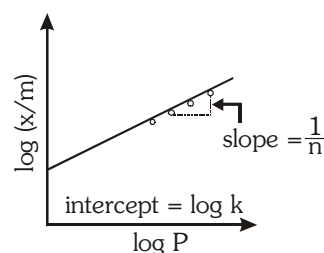
- (ii) At low pressure, the extent of adsorption varies linearly with pressure $\frac{x}{m} \propto P$... (a)
- (iii) At high pressure, it becomes independent of pressure. $\frac{x}{m} \propto P^0$... (b)
- (iv) At moderate pressure $\frac{x}{m}$ depends upon pressure raised to powers $\frac{x}{m} \propto P^{\frac{1}{n}}$... (c)



Freundlich adsorption isotherm plot of x/m against P



Freundlich adsorption isotherm plot of x/m against P



Plot of $\log x/m$ against $\log P$ for the adsorption of a gas on a solid

Adsorption in solutions (Liquid + Solid)

Ex. Dyes solution + Charcoal

$$\frac{x}{m} = kC^{\frac{1}{n}} \quad [C = \text{concentration of liquid}]$$

(b) The Langmuir adsorption isotherm :

- (i) One of the drawbacks of Freundlich adsorption isotherm is that it fails at high pressure of the gas. Irving Langmuir in 1916 derived a simple adsorption isotherm, on theoretical considerations based on kinetic theory of gases. This is named as Langmuir adsorption isotherm.
- (ii) Adsorption takes place on the surface of the solid only till the whole of the surface is completely covered with a unimolecular layer of the adsorbed gas.
- (iii) Adsorption consists of two opposing processes, namely condensation of the gas molecules on the solid surface and evaporation (desorption) of the gas molecules from the surface back into the gaseous phase.
- (iv) The rate of condensation depends upon the uncovered (bare) surface of the adsorbent available for condensation. Naturally, at start, when whole of the surface is uncovered the rate of condensation is very high and as the surface is covered more and more, the rate of condensation progressively decreases. On the contrary, the rate of evaporation depends upon the covered surface and hence increases as more of the surface is covered ultimately an equilibrium will be set up at a stage when the rate of condensation becomes equal to the rate of evaporation (adsorption equilibrium).
- (v) The rate of condensation also depends upon the pressure of the gas since according to the kinetic theory of gases, the number of molecules striking per unit area is proportional to the pressure.

Mathematically, $\frac{x}{m} = \frac{ap}{1 + bp}$, where a and b are constants and their values depend upon the nature of gas (adsorbate), nature of the solid adsorbent and the temperature. Their values can be determined from the experimental data.

- (vi) Adjacent adsorbed molecules do not interact but adsorbate molecules have equal ability to interact.

APPLICATIONS OF ADSORPTION

The phenomenon of adsorption finds a number of applications. Important ones are listed here:

- (i) **Gas masks :** Gas mask (a device which consists of activated charcoal or mixture of adsorbents) is usually used for breathing in coal mines to adsorb poisonous gases.
- (ii) **Removal of colouring matter from solutions :** Animal charcoal removes colours of solutions by adsorbing coloured impurities.
- (iii) **Separation of inert gases :** Due to the difference in degree of adsorption of gases by charcoal, a mixture of noble gases can be separated by adsorption on coconut charcoal at different temperatures.

- Generally adsorption is exothermic process but an example of chemical adsorption is endothermic.

Ex. $\text{Glass} + \text{H}_2(\text{g}) \rightleftharpoons \text{Glass}/\text{H}_2(\text{g}); \Delta H = \text{Positive}$ [Exception]

- Freundlich adsorption isotherm explains physical adsorption upto moderate pressure but fails at high pressure.
- Occlusion $\rightarrow$ Adsorption of gases on metal surface voids is called occlusion.

GOLDEN KEY POINTS

- Generally adsorption is exothermic process but an example of chemical adsorption is endothermic.
Ex. $\text{Glass} + \text{H}_2(\text{g}) \rightleftharpoons \text{Glass}/\text{H}_2(\text{g}); \Delta H = \text{Positive}$ [Exception]
- Freundlich adsorption isotherm explains physical adsorption upto moderate pressure but fails at high pressure.
- Occulusion $\rightarrow$ Adsorption of gases on metal surface voids is called occulusion.

BEGINNER'S BOX-1

- Which of the following is used to absorb water?
(1) Silica gel (2) Calcium acetate (3) Hair gel (4) Anhydrous CaCl_2
- The heat evolved in physisorption lies in the range (kJ/mol) of
(1) 20-40 (2) 40-100 (3) 100-200 (4) 200-400
- Adsorbed acetic acid on activated carbon is
(1) Adsorber (2) Absorber (3) Adsorbent (4) Adsorbate
- A plot of $\log \left(\frac{x}{m} \right)$ v/s $\log P$ for the adsorption of a gas on a solid gives a straight line with slope equal to
(1) $\frac{1}{n}$ (2) n (3) $\log k$ (4) k
- The graph between $\log \frac{x}{m}$ v/s $\log P$ is straight line inclined at an angle 45° with intercept 0.30. What will be rate of adsorption at pressure 0.4 atm.
(1) 0.4 (2) 0.6 (3) 0.8 (4) 0.9
- 1 g of charcoal having surface area $3.02 \times 10^2 \text{ m}^2$ is mixed in 100 ml of 0.5 M acetic acid solution; after few times concentrations of solution becomes 0.49 M what will be surface area occupied one molecule of acetic acid ?
(1) $2 \times 10^{-19} \text{ m}^2$ (2) $5 \times 10^{-19} \text{ m}^2$ (3) $6 \times 10^{20} \text{ m}^2$ (4) 10^{-3} m^2

5.5 COLLOIDAL SOLUTION

Thomas Graham (1861) studied the process of diffusion of dissolved substances through a parchment paper or an animal membrane and divided the substances into two classes :-

- (1) Crystalloid (2) Colloid

But this classification soon proved to be wrong since a crystalloid could behave as a colloid under different conditions and vice-versa.

For example: NaCl behaves as a crystalloid in aqueous medium and behaves as a colloid in benzene medium, whereas soap behaves as a typical colloid in water and behaves as a crystalloid in alcohol.

So new classification was given based on the size of solute particles.

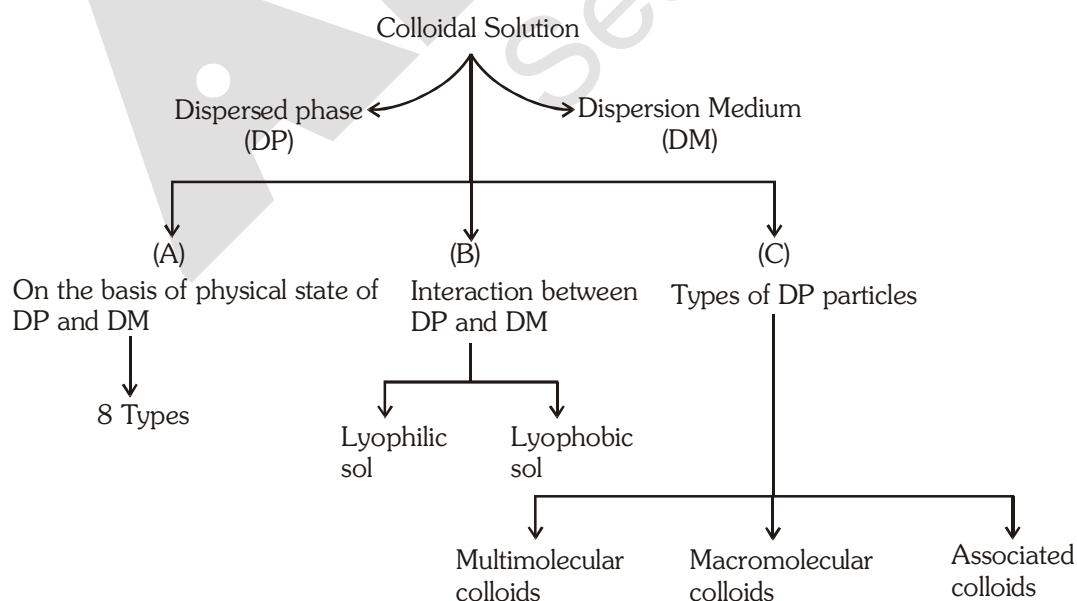
Table : Comparison of true solutions, colloids and suspension

S.No.	Property	True Solution or Crystalloid	Colloid	Suspension
1.	Particle size	< 1 nm	1 nm to 1000 nm	> 1000 nm
2.	Visibility	Not visible with any of the optical means	May be visible with ultramicroscope	Visible with naked eye
3.	Separation (a) With filter paper (b) With membranes (ultrafiltration)	Not possible Not possible	Not Possible Possible	Possible Possible
4.	Diffusion	Diffuses rapidly	Diffuses very slowly	Does not diffuse
5.	Settling	Does not settle	Does not settle due to gravity but it may settle under Centrifuge	Settle due to gravity
6.	Nature	Homogeneous	Heterogeneous	Heterogeneous
7.	Tyndall effect & Brownian moment	does not show	shows	May or may not show

5.6 TYPES OF COLLOIDAL SOLUTIONS

A colloidal system is made of two phases. The substance distributed as the colloidal particles is called **Dispersed phase** or *internal phase* or the *discontinuous phase*. The continuous phase in which the colloidal particles are dispersed is called **dispersion medium**. For example, for a colloidal solution of copper in water, copper particles constitute the dispersed phase and water the dispersion medium.

Depending on the dispersed phase or dispersion medium, colloidal solutions can be classified by 3 ways-



A. CLASSIFICATION OF COLLOIDAL SYSTEMS BASED ON DISPERSED PHASE AND DISPERSION MEDIUM

Dispersed phase	Dispersion medium	Type of Colloidal Solution	Example
Solid	Solid	Solid sol	Ruby glass (Gold dispersed in glass), Gem stones, Minerals
Solid	Liquid	Sol	Ink, Colloidal gold, Paints, Cell fluids, Silt
Solid	gas	Aerosol	Dust, Soot in air, smoke
liquid	Solid	Gel	Cheese, curd, Jellies, Boot polish, Opal, Butter
liquid	Liquid	Emulsion	Milk, Hair cream, butter , Cold Cream
liquid	gas	Aerosol	Fog, Mist, Clouds, Insecticide sprays
gas	Solid	Solid Sol	Cork, Pumice stone, Foam rubber, Bread, Cake
gas	Liquid	Foam	Whipped Cream, Shaving Cream, Soda water, Soap lather

A colloidal dispersion of one gas in another is not possible since the two gases would give a homogeneous molecular structure but colloidal solution is heterogeneous in nature.

Sol : If the colloidal system has the appearance of a fluid, then it is called a **Sol**. On the basis of dispersion medium different types of sols are possible.

S.N.	Dispersion medium	Name of solution
i.	Water	Hydrosol or aquasol
ii.	Benzene	Benzosol
iii.	Alcohol	Alcosol
iv.	Air	Aerosol

EMULSIONS :

Emulsions are liquid-liquid colloidal system. They are of two types

- (i) Oil dispersed in water (O/W type) (ii) Water dispersed in oil (W/O type)

In the first type water acts as a dispersion medium. Examples of this type of emulsions are **milk** and **vanishing cream**. In milk, liquid fat is dispersed in water.

In the second system, oil acts as dispersion medium. Common examples of this type are **butter**, **cold cream**, and **cod liver oil** etc.

Emulsions of oil and water are unstable and sometimes they separate into two layers on standing. For stabilization of an emulsion, a third component called emulsifying agent is usually added. The emulsifying agent forms an interfacial film between suspended particles and the medium. The principal emulsifying agents for o/w emulsions are **proteins**, **gums**, **soaps**, etc. For w/o emulsion the principal emulsifying agents are **heavy metal salts of fatty acids**, **long chain alcohol**, **lamp black**.

GELS :

A gel is a jelly like colloidal system in which a liquid is dispersed in a solid medium. Gelatin can behave both as sol and gel. **Gels may be classified into two types -**

- (a) **Elastic gels** : These are those which possess the property of elasticity. They change their shape on applying force and return to original shape when the force is removed. Gelatin, starch and soaps are examples of substances which form elastic gels.
- (b) **Non - elastic gels** : These are the gels which are rigid like Silica gel. These are prepared by appropriate chemical action. Thus silica gel is produced by adding concentrated hydrochloric acid to sodium silicate solution of the correct concentration.

(B) CLASSIFICATION BASED ON INTERACTION BETWEEN DISPERSED PHASE AND DISPERSION MEDIUM :**LYOPHILIC SOL AND LYOPHOBIC SOL**

Colloidal solutions in which the dispersed phase has considerable affinity for the dispersion medium, are called lyophilic sols (Solvent - loving). For example - dispersion of gelatin, starch, gum and proteins in water.

Colloidal solutions in which the dispersed phase has no affinity or attraction for the medium or for the solvent are called Lyophobic colloidal (Solvent hating) solutions. Eg. : Metals and their sulphides

Comparison of Lyophilic and Lyophobic sols

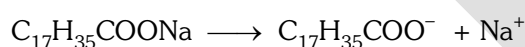
S.No	Property	Lyophilic sol (Emulsoid)	Lyophobic sol (suspensoid)
1	Preparation	Can be easily prepared by shaking or warming the substance with solvent	Can not be prepared easily Special methods are required
2	Stability	Are more stable	Are less stable
3	Reversibility	Are reversible	Are irreversible
4	Viscosity	Viscosity is much higher than that of solvent	Viscosity is nearly same as that of the solvent
5	Surface tension	Surface tension is usually low	Surface tension is almost same as that of solvent
6	Hydration or solvation	These are highly solvated as the particle have great affinity for solvent	These are less solvated as the particles have less affinity for the solvent
7	Charge	The particles have little charge or no charge at all	The particles carry a characteristic charge either positive or negative
8	Visibility	Particles can not be seen under microscope	Particles can be seen under microscope
9	Coagulation or precipitation	Precipitated by high concentration of electrolytes	Precipitated by low concentration of electrolytes
10	Tyndall effect	Less Scattering	More Scattering
11	Migration in electric field	May or may not migrate as they may or may not carry charge	Migrate towards anode or cathode as these particles carry charge
12	General Example	Mostly of organic nature Example : Gelatin, Starch, Gum, Albumin & Cellulose Solution	Mostly of Inorganic nature Example : Transition metal salt like Gold, As etc.

(C) ON THE BASIS OF TYPE OF DISPERSED PHASE

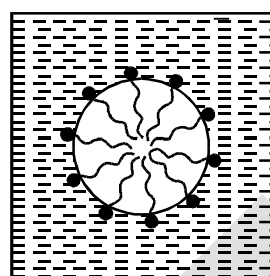
Multi Molecular	Macro Molecular	Associated colloids
Formation by aggregation of a large number of atoms or smaller molecules of substance. Ex. → Gold Sol (Au) Sulphur sol (S₈)	In suitable solvents, macromolecules form solutions in which the size of particles may be in colloidal range. Ex. → Starch, Cellulose, Protein etc.	These are the substances which behave as normal electrolytes at low concentration but get associated at higher concentration and behave as colloidal solutions. These associated particles are also called micelles. Ex. → Soap & Detergent

ASSOCIATED COLLOIDS [MICELLES]

Substances whose molecules aggregate to form particles of colloidal dimensions are called associated colloids. The molecules of soaps and detergents are usually smaller than the colloidal particles. However, in concentrated solutions, these molecules associate and form aggregates of colloidal size. These aggregates of soaps or detergent molecules are called Micelles. Soaps and detergents are strong electrolytes and give ions when dissolved in water



Soap



SOLID CIRCLE (●) REPRESENTS POLAR GROUP (–COO[–] Na⁺)
WAVE LINE (~~~~) REPRESENTS NON-POLAR GROUP (–R)

The negative ions aggregate to form a micelle of colloidal size. The negative ion has a long hydrocarbon chain and a polar group (–COO[–]) at one end. In micelle formation, the long hydrocarbon chain (tail) which is insoluble in water is directed towards the centre while the soluble polar head is on the surface in contact with water. The charge on the micelle is responsible for the stability of this system.

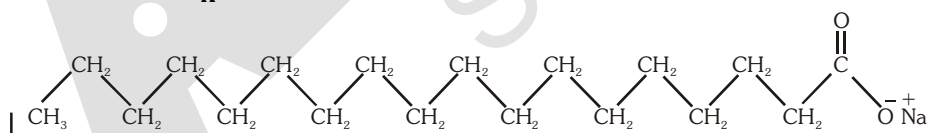
Critical Micelle concentration (CMC) :

The concentration above which associated colloids or Micelles are formed is called CMC.

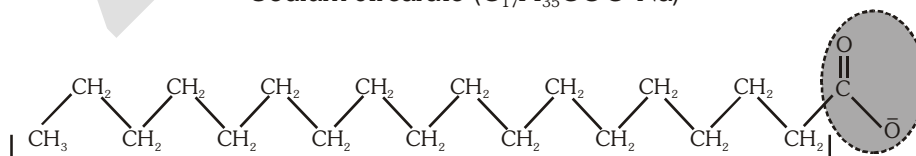
CMC of soap is 10^{–4} to 10^{–3} mol lit^{–1}.

These colloids have both lyophobic and lyophilic parts. Micelles may contain as many as 100 molecules or more.

Kraft Temperature (T_K) : Temperature above which Micelles are formed.



Sodium stearate (C₁₇H₃₅COO[–] Na⁺)



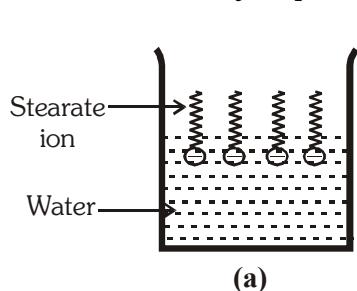
Hydrophobic tail Stearate ion



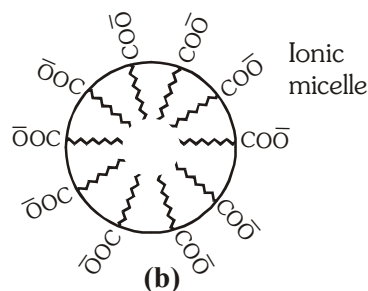
Hydrophobic tail

Hydrophilic head

Hydrophobic and hydrophilic parts of stearate ion



(a) Arrangement of stearate ions on the surface of water at low concentrations of soap



(b) Arrangement of stearate ions inside the bulk of water (ionic micelle) at critical micelle concentrations of soap

NECESSARY AND SUFFICIENT CONDITION FOR MICELLE FORMATION ARE:-

- (a) Compounds must have hydrophobic part means long chain alkyl group.
- (b) Compounds must have hydrophilic part means $-\text{SO}_3^-$, $-\text{COO}^-$, $-\ddot{\text{O}}-$, $-\text{NR}_3^+$ etc.

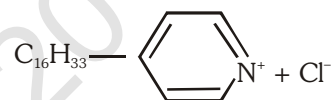
All these compounds which form micelle are also known as surfactants. They are surface active agents which reduce surface tension.

SURFACTANTS :

Surfactants are those substances which are preferentially adsorbed at the interfaces like air-water, oil-water and solid-water interfaces. Thus, it is the surfactant which is responsible for **micellisation** and emulsification. Surfactants are divided into three categories:

(i) Cationic surfactants :

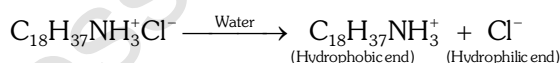
Such substances on ionisation give a cation having hydrophobic and hydrophilic group, e.g., cetyl pyridinium chloride.



Cetyltrimethyl ammonium chloride, $\text{C}_{16}\text{H}_{33}(\text{CH}_3)_3\text{N}^+\text{Cl}^-$

Octadecyl ammonium chloride $\text{C}_{18}\text{H}_{37}\text{NH}_3^+\text{Cl}^-$

Ionisation of these compounds may be represented as



(ii) Anionic surfactants : Such substances give anion which act as surfactant

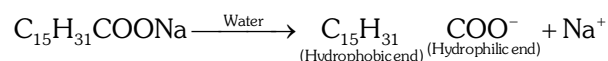
e.g. :-

Sodium Palmitate $\text{C}_{15}\text{H}_{31}\text{COONa}$

Sodium Oleate $\text{C}_{17}\text{H}_{33}\text{COONa}$

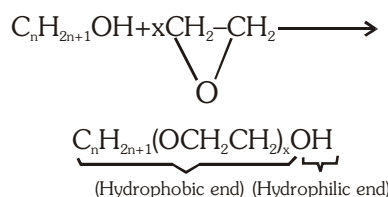
and salts of sulphonic acid having molecular formula

$\text{C}_n\text{H}_{2n+1}\text{SO}_3\text{M}$ where, $\text{M}^+ = \text{Na}, \text{K}^+, \text{NH}_4^+$, etc.



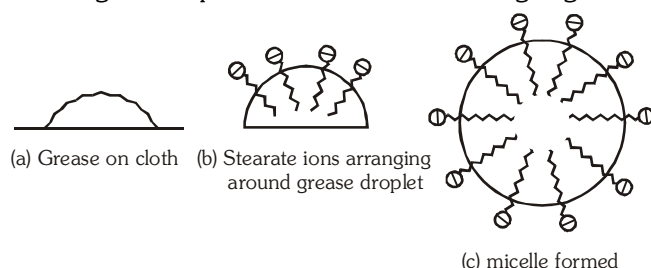
(iii) Non-ionic surfactants : These surfactants do not ionise or dissociate in aqueous medium, but these molecules also have hydrophobic and hydrophilic end.

High molecular mass alcohol adds to several molecules of ethylene oxide to form hydroxy surfactant.



CLEANSING ACTION OF SOAPS

The cleansing action of soap is due to fact that soap molecules form micelle around the oil droplet in such a way that hydrophobic part of stearate ions in the oil droplet and hydrophilic part projects out of grease droplet like bristles. Thus soap helps in emulsification and washing away of oils and fats. The negatively charged sheath around the globules prevents them from coming together and forming aggregates.



5.7 METHODS OF PREPARATIONS

Lyophilic sols may be prepared by simply warming the solid with liquid dispersion medium eg. starch with water. On the other hand lyophobic sols have to be prepared by special methods. These methods fall into two categories -

(A) Dispersion Methods : In this larger macro sized particles are broken down to colloidal size

(B) Condensation methods : In this colloidal sized particles are built up by aggregating single ions or molecules. This method is known as condensation method.

S.No	(A) Dispersion methods	S.No	(B) Aggregation or condensation methods
1	Mechanical dispersion	1	Physical methods
2	Electro-dispersion	a	Exchange of solvent
3	Peptization	b	By Excessive Cooling
		2	Chemical methods
		(a)	Double decomposition
		(b)	Oxidation
		(c)	Reduction
		(d)	Hydrolysis

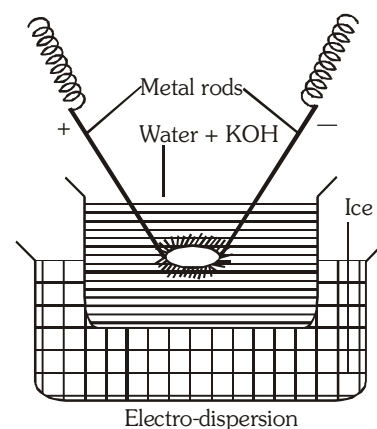
(A) Dispersion Methods.

(i) Mechanical dispersion : The substance to be dispersed is grinded as finely as possible and then shaken with the dispersion medium to form a suspension. This suspension is passed through colloidal mill where the suspended particles are broken to produce particles of colloidal size. Ex.: Ink, paints.

(ii) Electro-dispersion (Bredig's arc method):

This method is suitable for the preparation of colloidal solutions of metals like gold, silver, platinum etc. An arc is struck between the metal electrodes under the surface of water containing some stabilizing agent such as a trace of KOH. The water is cooled by immersing the container in a ice bath. The intense heat of the arc vaporizes some of the metal which condenses under cold water.

Ex. : Pt, Ag, Cu, Au, Pb Sol are prepared by this method.



(iii) Peptization (Peptization is a opposite process of coagulation)

The process of converting a precipitate into colloidal solution by shaking it with dispersion medium in the presence of small amount of electrolyte. The electrolyte used is called a **Peptizing agent**.

FEW EXAMPLES OF SOLS OBTAINED BY PEPTIZATION :

- (i) $\text{AgI} \downarrow + \text{AgNO}_3 \rightarrow [\text{AgI}] \text{Ag}^+ : \text{NO}_3^-$ (ii) $\text{AgI} \downarrow + \text{KI} \rightarrow [\text{AgI}] \text{I}^- : \text{K}^+$
 (iii) $\text{Fe(OH)}_3 \downarrow + \text{FeCl}_3 \rightarrow [\text{Fe(OH)}_3] \text{Fe}^{3+} : 3\text{Cl}^-$ (iv) $\text{Fe(OH)}_3 \downarrow + \text{NaOH} \rightarrow [\text{Fe(OH)}_3] \text{OH}^- : \text{Na}^+$
 (v) $\text{CdS} \downarrow + \text{H}_2\text{S} \rightarrow [\text{CdS}] \text{S}^{2-} : 2\text{H}^+$

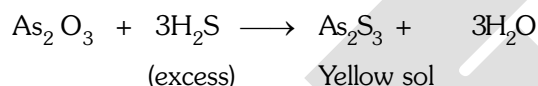
The process of peptization thus involves the adsorption of suitable ions (supplied by the electrolyte added-particularly a common ion) and electrically charged particles then split from the precipitate as colloidal particles.

(B) Condensation methods

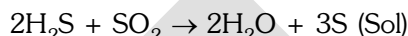
- (1) **Chemical methods :** The chemical methods involve chemical reactions in a medium in which the dispersed phase is sparingly soluble. A condition of super-saturation is produced but the actual precipitation is avoided. **Some familiar reactions used are :**

(Optional)

- (a) **Double decomposition :** A 1% solution of arsenic oxide is prepared in hot water. The solution is cooled, filtered and is then gradually added to water saturated with hydrogen sulphide, then a stream of H_2S is passed through the solution. This is continued till an intense yellow coloured solution is obtained.

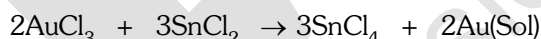


- (b) **Oxidation :** A colloidal solution of sulphur is obtained by passing H_2S into a solution of sulphur dioxide or oxidising agent Ex.: HNO_3 , Br_2 , Etc.

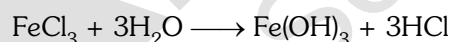


Sulphur sol can also be obtained when H_2S is bubbled through an oxidising agent (bromine water or nitric acid)

- (c) **Reduction :** Colloidal solutions of metals like gold, silver, platinum, lead etc. can be obtained when their salts solutions are acted upon by reducing agents.



- (d) **Hydrolysis :** Colloidal solution of many metal hydroxides are prepared by this method. Ferric hydroxide sol is prepared by the hydrolysis of ferric chloride with boiling water.



The hydroxide sol of Cr and Al can also be prepared by this method.

(2) Physical Methods :

- (a) **By exchange of solvent :** when a true solution is mixed with an excess of some other solvent in which the solute is insoluble but solvent is soluble, a colloidal sol is formed. For eg. when a solution of sulphur in alcohol is mixed in excess of water, a colloidal sol of sulphur is formed.
 (b) **By excessive cooling :** Molecules of certain substances condense together to form particles of colloidal size.

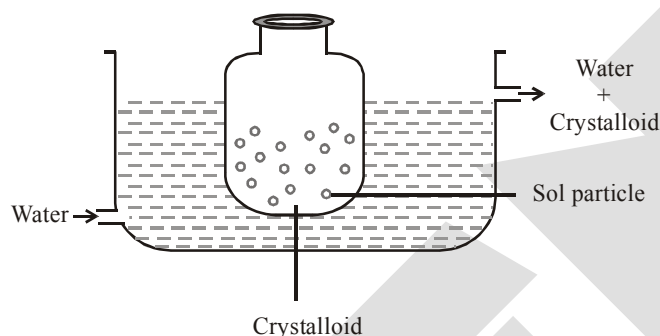
For Example : Colloidal sol of ice in an organic solvent like ether or chloroform may be prepared by freezing the mixture of water and the solvent.

BEGINNER'S BOX-2

- The nature of colloidal solution obtained when excess FeCl_3 is added in precipitate of $\text{Fe}(\text{OH})_3$?
 (1) (+) vely charged (2) (-) vely charged (3) No charge (4) All of above
- Which of the following will form negatively charged colloidal solution ?
 (1) 100 ml 0.1 M AgNO_3 + 100 ml 0.1 M KI (2) 100 ml 0.2 M AgNO_3 + 100 ml 0.1 M KI
 (3) 100 ml 0.1 M AgNO_3 + 100 ml 0.2 M KI (4) 100 ml 0.2 M AgNO_3 + 200 ml 0.1 M KI

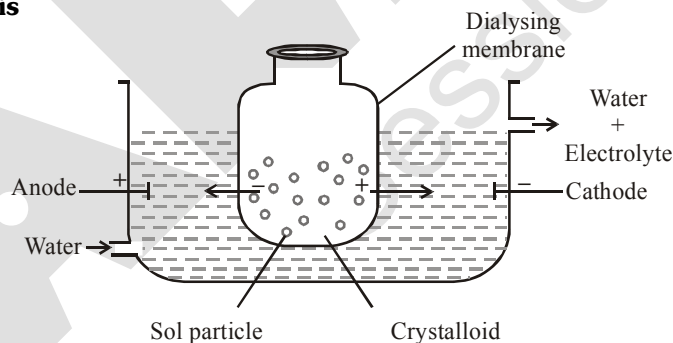
5.8 PURIFICATION OF COLLOIDAL SOLUTION :

(a) Dialysis :



- The process of separating the particles of colloid from those of crystalloid, by means of diffusion through a suitable membrane is called dialysis.
- Its principle is based upon the fact that colloidal particles can not pass through a parchment or cellophane membrane while the ions of the electrolyte can pass through it.
- The impurities slowly diffuse out of the bag leaving behind pure colloidal solution
- The distilled water is changed frequently to avoid accumulation of the crystalloids otherwise they may start diffusing back into the bag.
- Dialysis can be used for removing HCl from the ferric hydroxide sol.
- Prolonged dialysis can cause the colloid to coagulate.

(b) Electrodialysis



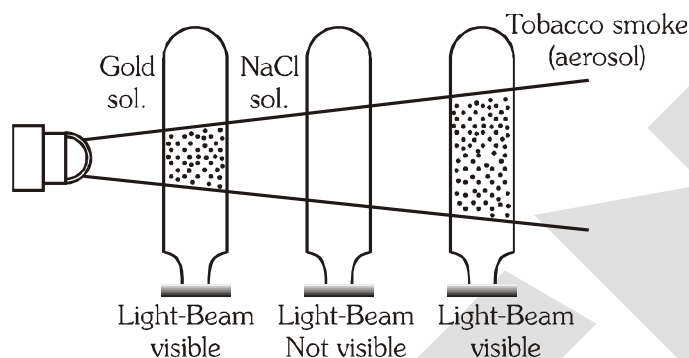
Process of dialysis can be made faster by applying an electric field if the dissolved substances in the impure colloidal solution is only an electrolyte.

(c) Ultrafiltration :

- Sol particles directly pass through ordinary filter paper because their pores are larger than the size of sol particles.
- If the pores of the ordinary filter paper are made smaller by soaking the filter paper in a solution of gelatin or collodion and subsequently hardened by soaking in formaldehyde, the treated filter paper may retain colloidal particles and allow the true solution particles to escape. Such filter paper is known as ultrafilter and the process of separating colloids by using ultra-filters is known as ultrafiltration.
- Collodion is a 4% solution of nitro cellulose in mixture of alcohol and ether.

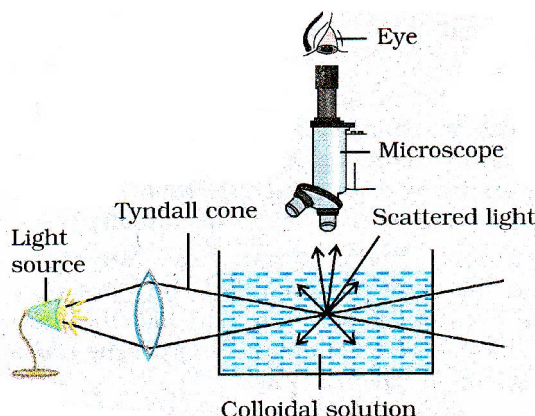
5.9 PROPERTIES OF COLLOIDAL SOLUTIONS

- (a) **Heterogenous** : Colloidal particles in a solution differ in sizes and are not homogeneously distributed through out the solution.
- (b) **Visibility** : Colloidal particles can not be seen with naked eyes or with the help of microscope. Colloidal particles, however scatter light and become visible as bright spots in a dark background when seen through a ultramicroscope. In 1903 Zsigmondy devised a microscope which is based on the principle of scattering of light by sol particles. A strong beam of light is focussed by a series of lenses in a vessel containing the sol. The colloidal solution is observed at right angle to the beam of light.



- (c) **Filterability** : Colloidal particles pass through an ordinary filter paper but do not pass through parchment and other fine membranes.
- (d) **Surface tension and viscosity** : For Lyophobic sols, surface tension and viscosity are not very different from those of the medium, as there is very slight interaction between the suspended particles and the medium. On the other hand Lyophilic sols show a high degree of solvation of the particles and therefore, the properties of the medium are modified. Thus, the viscosity is much higher for the sol than for the medium. Furthermore the surface tension of the sol is lower than that of pure medium.
- (e) **Colour** : The colour of colloidal solution depends on the wavelength of the light scattered by the dispersed particles. The wavelength of the scattered light again depends on the size and the nature of particles. For example the colour of silver sol changes with the particle (suspended) diameter in solution the colour of colloidal solution also change with the manner in which the observer receives the light. For ex. mixture of milk and water appears blue when viewed by reflected light but red when viewed by transmitted light. finest gold sol is red in colour as size of particle increases it appears purple then blue and finally golden.
- (f) **Colligative properties** : These properties depend on the number of solute particles in solution. In case of colloidal solutions, colloidal particles are the aggregates of ions or molecules and when compared to true solutions or normal solutions, the total no. of particles of solute in solution are very less due to large size and hence these solutions exhibit colligative properties to lesser extent.
- (g) **Optical properties (Sols exhibits Tyndall effect)** : When a beam of light is passed through a sol and viewed at right angles, the path of the light shows up as a hazy beam of cone (illuminated path of bluish light). This was first observed by Faraday and later by Tyndall and is known as Tyndall effect. The same phenomenon is observed when a beam of sunlight enters a dark room through a small slit. When light is focussed on a sol and observed perpendicular to the beam of light rays, a cone is seen which is called Tyndall cone.
Tyndall effect can be observed only when following two conditions can be satisfied.
- (i) *Diameter of dispersed particle is not much smaller than wavelength of light used.*

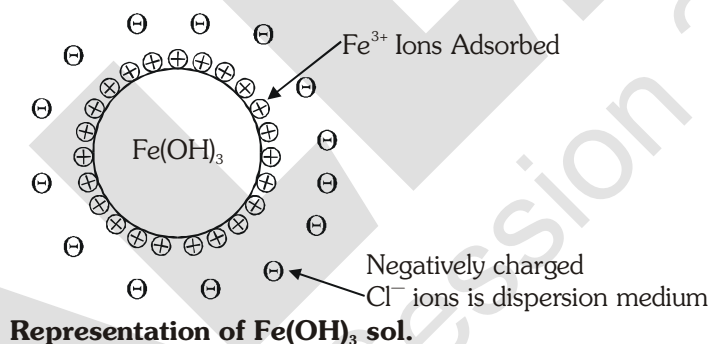
- (ii) The refractive indices of the dispersed phase and dispersion medium differ greatly in magnitude.



- (h) **Kinetic properties** : When a sol is examined with an ultramicroscope, the suspended particles are seen as shining of light. By following an individual particle, it is observed that the particle is in a state of continuous motion in zigzag path. The continuous rapid zigzag motion of a colloidal particle in the dispersion medium is called "Brownian movement or motion" (first observed by British botanist Robert Brown).

The Brownian movement has been explained to be due to the unbalanced bombardments of the colloidal particles by the molecules of dispersion medium.

- (i) **Charge on colloidal particles** : Colloidal particles always carry an electric charge. The mutual forces of repulsion between similarly charged particles prevent them from aggregating and settling under the action of gravity. This gives stability to the solution.

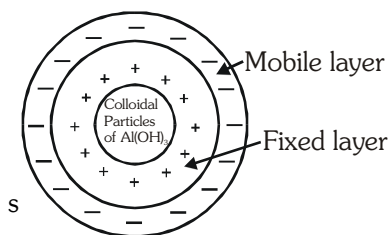


	Positively charged		Negatively charged
1.	Metallic hydroxides $\text{Cr}(\text{OH})_3$, $\text{Al}(\text{OH})_3$ & $\text{Fe}(\text{OH})_3$ and hydrated metallic oxides $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, $\text{CrO}_3 \cdot x\text{H}_2\text{O}$	1.	Metallic sulphides like As_2S_3 , Sb_2S_3 , CdS
2.	Basic dye in acidic medium stuff like:- Methylene blue sol	2.	Acid dye in basic medium stuff like:- Congo red sols
3.	Proteins in (acidic medium)	3.	Metal like – Au, Ag, Cu, Pt etc.
4.	Haemoglobin (Hb)	4.	Proteins in (basic medium), Sols of starch, gum, gelatin, clay & charcoal

NOTE: Blood is a naturally occurring negatively charged colloid but Haemoglobin (Hb) is a positively charged colloid.

ELECTRICAL DOUBLE LAYER

The combination of the two layers of +ve and -ve charges around the sol particle is called **Helmholtz double layer**. According to modern view, the first layer of ions is firmly held and is termed as **Fixed layer** or **Compact layer** while the second layer is mobile which is termed as **Diffused layer**



The diffused layer is only loosely attached to the particle surface and moves in the opposite direction under an applied electric field. The potential difference between the fixed layer and the diffused layer of opposite charge is called **Electrokinetic Potential** or **Zeta Potential**.

- (j) **Electrophoresis** : If electric potential is applied across two platinum electrodes dipped in a colloidal solution, the colloidal particles move towards one or the other electrode, due to charge on them. The movement of sol particles under an applied electric potential is called "Electrophoresis". Depending upon the direction of movement of particles towards cathode or anode electrophoresis can be called "Cataphoresis" or 'Anaphoresis'. **Electrophoresis provides an experimental proof to show that the colloidal particles are charged particles.**

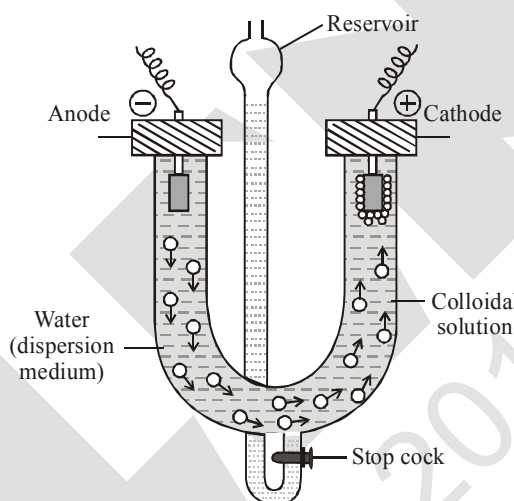


Fig. : Electrophoresis

- (k) **Electro osmosis** : The medium will move in opposite direction to the dispersed phase under the influence of applied electric potential. The movement of dispersion medium under the influence of applied potential is known as 'Electro-osmosis'. (In presence of plasma membrane)
- (l) **Coagulation or Precipitation** : We know that the stability of a lyophobic sol is due to the adsorption of positive or negative ions by the dispersed particles. The repulsion forces between the charged particles do not allow them to settle. If somehow, the charge is removed there is nothing to keep the particles apart from each other. In such cases they aggregate or flocculate and settle down under the action of gravity. The flocculation and settling down of the dispersed phase particles is called coagulation or precipitation. The precipitation can be brought about in five ways -

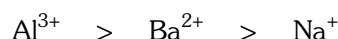
Coagulation of lyophobic Sol :

- (i) By addition of electrolyte.
 - (ii) By electrophoresis.
 - (iii) By mixing two oppositely charged sols.
 - (iv) By boiling.
 - (v) By repeated dialysis
- (i) **By addition of electrolytes** : When an electrolyte is added in excess to a sol, then the electrolyte furnishes both the type of ions in solution. The oppositely charged ions get adsorbed on the surface

of colloidal particles. This causes neutralization and the size mass of colloidal particle increases and it becomes a suspended particle. Due to greater volume and greater mass these suspended particles settle down and coagulate. The ion responsible for neutralization of charge on the particle is called the flocculating ion.

Hardy Schulze Rule : This rule states that the precipitating effect of an ion on dispersed phase of opposite charge increases with the valency of the ion.

The higher the valency of the flocculating ion, the greater is its coagulating power. Thus for the precipitation of As_2S_3 sol (-ve) the precipitating power of Al^{3+} , Ba^{2+} , and Na^+ ions is in the order



Similarly for precipitating $\text{Fe}(\text{OH})_3$ sol (positive) the precipitating power of $[\text{Fe}(\text{CN})_6]^{3-}$, SO_4^{2-} and Cl^- ions is in the order



The minimum concentration of an electrolyte in milli moles required to cause precipitation of 1 litre sol in two hours is called FLOCCULATION VALUE. The smaller the flocculating value, the higher will be the coagulating power of the ion.

- (ii) **By electrophoresis** - During electrophoresis the charged sol particles migrate towards the electrode of opposite sign where they deposit their charge and then get coagulated (As neutral particles can aggregate and change to suspension particles.)
- (iii) **By mixing two oppositely charged sols** - The coagulation of two sols of opposite charge can be effected by mixing them eg. $\text{Fe}(\text{OH})_3$ (positive sol) and Arsenic sulphide (negative sol) when mixed neutralize each other, join and coagulate.
- (iv) **By boiling** - Sols such as sulphur and silver halides dispersed in water get coagulated when boiled due to increased collisions between sol particles and water molecules, which removes the adsorbed charged layer from the sol and therefore the sol particles settle down.
- (v) **By repeated dialysis** - When dialysis is repeated again and again then colloidal solution gets coagulated.

COAGULATION OF LYOPHILIC SOLS

Lyophilic sols are stable due to less charge and high solvation of the colloidal particles. When these two factors are removed, a lyophilic sol can be coagulated.

This is done (i) By adding electrolyte (ii) By adding suitable solvent.

When solvents such as alcohol and acetone are added to hydrophilic sol, the dehydration of dispersed phase occurs. Under this condition a small quantity of electrolyte can bring about coagulation.

BEGINNER'S BOX-3

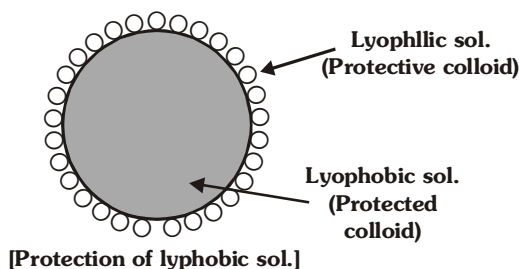
1. $[\text{Fe}(\text{OH})_3]\text{Fe}^{3+}$: can be coagulated easily by -

(1) Cl^-	(2) SO_4^{2-}	(3) PO_4^{3-}	(4) $[\text{Fe}(\text{CN})_6]^{4-}$
-------------------	------------------------	------------------------	-------------------------------------
2. To coagulate $[\text{Ag}]\text{Ag}^+; \text{NO}_3^-$ Sol; The decreasing order of coagulation power in following is -

(a) NaCl	(b) K_2SO_4	(c) $\text{Ca}_3(\text{PO}_4)_2$	(d) $\text{K}_4[\text{Fe}(\text{CN})_6]$
(1) (a) > (b) > (c) > (d)	(2) (d) > (c) > (b) > (a)		
(3) (a) > (d) > (c) > (b)	(4) (d) > (c) > (a) > (b)		

5.10 PROTECTION OR PROTECTIVE ACTION :

Lyophobic sols are readily precipitated by adding small amounts of electrolytes. However, these sols are often stabilized by the addition of lyophilic sols.



The property of lyophilic sols to prevent the precipitation or coagulation of a lyophobic sol is called protection. The lyophilic sol used to protect a lyophobic sol from precipitation is referred to as a protective colloid. lyophilic sols form a thin layer around lyophobic sol or around the ions furnished by electrolyte and therefore the coagulation can not take place (as the size does not increase much). **Gelatin, Albumin, Gum Arabia, Potato Starch** are some of the examples of **Protective colloids**. The lyophilic colloids differ in their protective power. The protective power is measured in terms of gold number.

"**Gold number**" introduced by Zsigmondy. The number of **milligrams** of a hydrophilic colloid that will just prevent the precipitation of 10 ml of standard gold sol on addition of 1 ml of 10% NaCl solution is known as **Gold number** of that protector (Lyophilic colloid).

The precipitation of the gold sol is indicated by a colour change from red to blue when the particle size just increases.

The **smaller the gold number** of a protective lyophilic colloid, **greater is its protection power**.

Gold Number of some hydrophilic colloids

Lyophilic colloid	Gold Number
Gelatin	0.005 - 0.01
Egg albumen	0.08 - 0.1
Gum arabic	0.10 - 0.15
Potato - starch	25

$$\text{Gold number} = 10 \times \frac{\text{Mass of lyophilic sol (mg)}}{\text{Volume of lyophobic sol (mL)}}$$

$$\text{Protection Capacity} \propto \frac{1}{(\text{Gold number})}$$

Gelatin and starch have the maximum and minimum protective powers respectively.

BEGINNER'S BOX-4

- On adding 1mL solution of 10% NaCl to 10 mL gold sol in presence of 0.025 g starch so that coagulation is just prevented what will be gold number of starch.
 (1) 0.25 (2) 2.5 (3) 25 (4) 0.025
- Which one is a natural colloid?
 (1) NaCl (2) Sugar (3) RCOONa (4) Blood

3. Colloidal solution of silver can be prepared by

(1) Colloidal mill	(2) Double decomposition method
(3) Bredig's Arc method	(4) Electrodialysis
4. Medicines are more effective if they are used in

(1) Colloidal state	(2) Solid state	(3) Solution state	(4) None of these
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5. Colloidal solution commonly used in treatment of skin diseases is

(1) Colloidal sulphur	(2) Colloidal silver	(3) Colloidal gold	(4) Colloidal antimony
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5.11 APPLICATIONS OF COLLOIDS

Colloids are widely used in the industry. Following are some examples:

- (i) **Electrical precipitation of smoke :** Smoke is a colloidal solution of solid particles such as carbon, arsenic compounds, dust, etc., in air. The smoke, before it comes out from the chimney, is led through a chamber containing plates having a charge opposite to that carried by smoke particles. The particles on coming in contact with these plates lose their charge and get precipitated. The particles thus settle down on the floor of the chamber. The precipitator is called Cottrell precipitator.

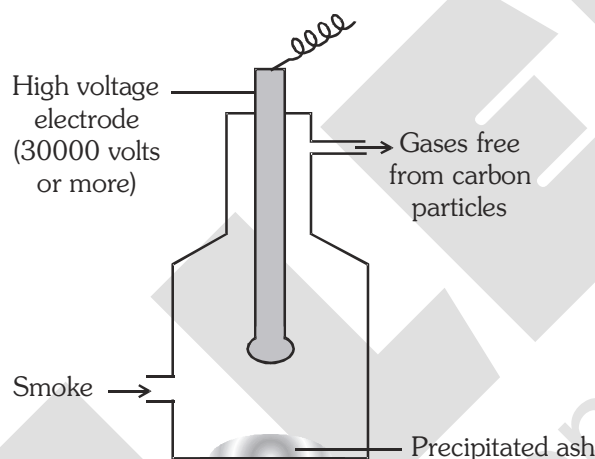


Fig : Cottrell smoke precipitator

- (ii) **Purification of drinking water:** The water obtained from natural sources often contains suspended impurities. Alum is added to such water to coagulate the suspended impurities and make water fit for drinking purposes.
- (iii) **Medicines :** Most of the medicines are colloidal in nature. For example, argyrol is a silver sol used as an eye lotion. Colloidal antimony is used in curing kalaazar. Colloidal gold is used for intramuscular injection. Milk of magnesia, an emulsion, is used for stomach disorders. Colloidal medicines are more effective because they have large surface area and are therefore easily assimilated.
- (iv) **Tanning :** Animal hides are colloidal in nature. When a hide, which has positively charged particles, is soaked in tannin, which contains negatively charged colloidal particles, mutual coagulation takes place. This results in the hardening of leather. This process is termed as tanning. Chromium salts are also used in place of tannin.
- (v) **Cleansing action of soaps and detergents :** This has already been described.

GOLDEN KEY POINTS

- Colloidal solution of graphite in water is called "**Aquadag**" while that in **Oil** is called **Oildag**.
- Gold solution in water is called **Purple of cassius**.

5.12 CATALYST AND CATALYSIS :

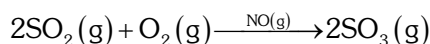
Substances which accelerate the rate of a chemical reaction and themselves remain chemically and quantitatively unchanged after the reaction are known as catalyst and the phenomenon is known as catalysis.

Catalysis can be broadly divided into two groups

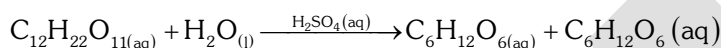
(A) Homogeneous catalysis : When the reactants and the catalyst are in the same phase (i.e. liquid or gas), the process is said to be homogeneous catalysis.

Example :

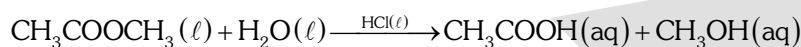
- (i) Lead chamber process : Oxidation of sulphur dioxide into sulphur trioxide with dioxygen in the presence of oxides of nitrogen as the catalyst in the lead chamber process



- (ii) Inversion of cane sugar : In aqueous solution, it is catalysed by dilute acid (hydrogen ions)



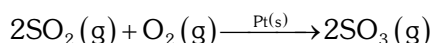
- (iii) Hydrolysis of methyl acetate is catalysed by H^+ ion furnished by hydrochloric acid



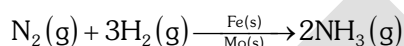
(B) Surface catalysis or heterogeneous catalysis. The catalytic process in which the reactants and the catalyst are in different phases is known as heterogeneous catalysis.

Example :

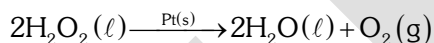
- (i) Oxidation of sulphur dioxide into sulphur trioxide in the presence of Pt.



- (ii) Haber process for NH_3



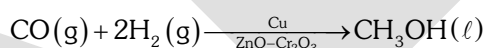
- (iii) Decomposition of H_2O_2



- (iv) Hydrogenation of vegetable oils in the presence of finely divided nickel as catalyst.



- (v) Synthesis of CH_3OH

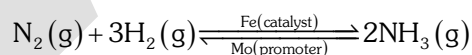


SOME OTHER RELATED TERMS

(a) Promoters : Those substances which do not themselves act as catalysts but their presence increases the activity of catalyst are called catalytic promoters.

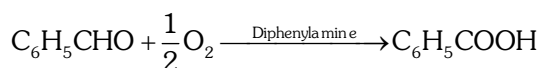
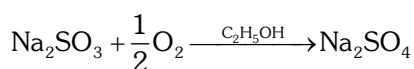
Example :

In Haber process for the synthesis of ammonia, Fe is catalyst while molybdenum (Mo) acts as a promoter



(b) Inhibitors : The substance which decreases the rate of chemical reaction is called inhibitor. It increases the activation energy for the reaction.

Example :



* H_3PO_4 , glycerol and acetamide decrease the rate of decomposition of hydrogen peroxide.

- (c) **Catalytic poison** : The substance whose presence decreases or destroys the activity of a catalyst is called catalytic poison.

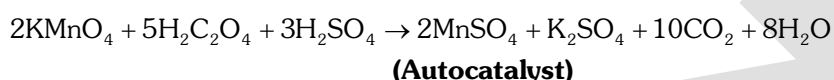
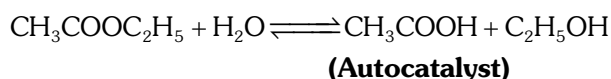
Example :

Carbon monoxide or H_2S act as poison for Fe catalyst in Haber process for manufacture of NH_3 .
 As_2O_3 acts as poison for Pt asbestos in contact process for H_2SO_4 .

- (d) **Autocatalysts** :

When one of the product of the reaction begin to act as a catalyst, it is called auto catalyst.

Example :



In the initial stage the reaction is slow but as soon as the product come into existence the reaction rate increases.

- (e) **Induced catalyst** :

When a chemical reaction enhance, the rate of another chemical reaction it is called induced catalysis.

Example :

Sodium arsenite solution is not oxidised by air if however, air is passed through a mixture of Na_3AsO_3 & Na_2SO_3 , both of them undergo simultaneous oxidation. The oxidation of sodium sulphite, thus influences the oxidation of sodium arsenite.



- (f) **General characteristics of catalysts** :

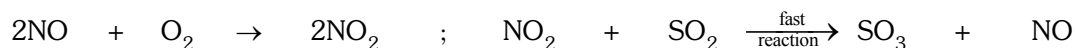
- (i) A catalyst remains unchanged in mass and chemical composition but can change their physical state.
- (ii) Only a very small amount of catalyst is sufficient to catalyse a reaction.
- (iii) A catalyst does not initiate a reaction.
- (iv) When a catalyst is a solid, it is usually more efficient when used in finely divided form.
- (v) Generally catalyst does not change the nature of products.
- (vi) A catalyst does not change the equilibrium state of a reversible reaction but helps to decrease time to achieve the equilibrium state or position of equilibrium.
- (vii) Catalysts are generally specific in nature.
- (viii) Catalyst can change rate constant of the reaction.
- (ix) Catalysts participate in mechanism of reaction.
- (x) Catalyst does not change enthalpy of reaction, free energy and entropy because they are state function.

- (g) **Theories of catalysis**

- (I) **Intermediate compound formation theory** : This theory explains homogeneous catalysis mainly. According to this theory, the catalyst combines with one of the reactants to give an intermediate compound. This intermediate compound reacts with the other reactants and gives the product and regenerates the catalyst in its original form.

Thus the reactants do not directly combine with each other, instead they react through the catalyst which provides an alternative pathway which involves lesser energy of activation.

For example: The function of nitric oxide [NO] as a catalyst in the formation of SO_3 is explained as follows.



(Catalyst) (reactant) (Intermediate) (intermediate) (reactant) (product) (catalyst regenerated)

(II) Adsorption theory : This theory explains the heterogeneous catalysis. The role of a solid catalyst in enhancing the reaction rate is explained on the basis of this theory in the following steps.

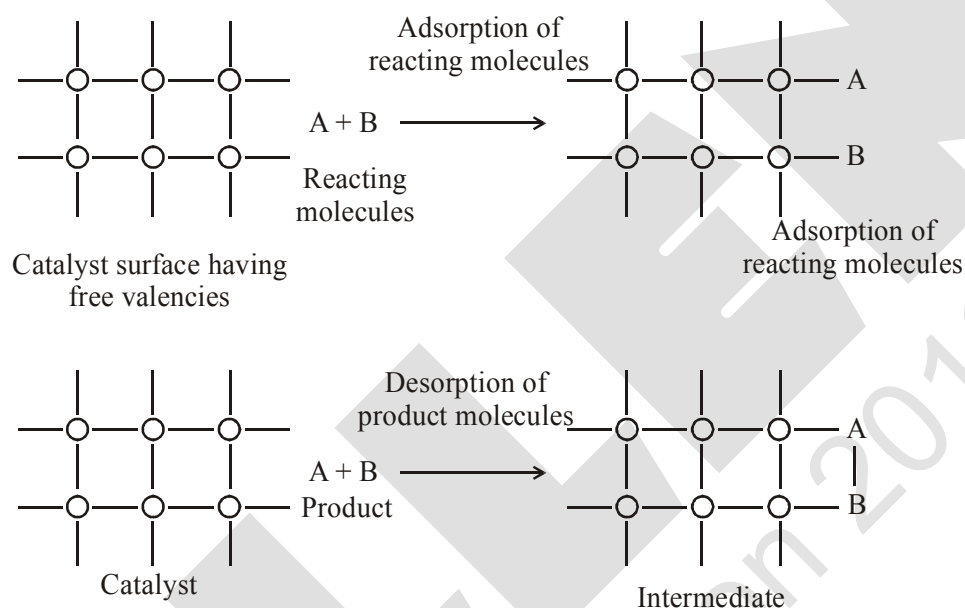


Fig. : Adsorption of reacting molecules, formation of intermediate and desorption of products

- (i) The reactant molecules are adsorbed on the surface of the catalyst at adjacent points. Adsorption leads to higher concentration of the adsorbed reactant on the surface of a catalyst.
- (ii) As adsorption is an exothermic process, the heat of adsorption provides the necessary activation energy for the chemical reaction to proceed.
- (iii) The adsorbed reactant molecules are tied on the solid surface of the catalyst. The bonds between the atoms of chemisorbed reactant molecules are weakened. The reactant molecules of sufficient energy combine together and with the surface of the catalyst to form surface activated complex. This adsorbed activated complex is decomposed to form products at a definite faster rate.
- (iv) The product molecules rapidly leave the catalyst surface to make room for the other reactant molecules to get adsorbed. Thus the chemical combination between reactant molecules goes on at the surface of the catalyst at a much faster rate.

(h) **Catalysts in Industry** : Some of the important processes and their catalyst are given in below.

Industrial process	Catalyst used
Perparation of O_2 from $KClO_3$ (Homogeneous)	MnO_2
Haber's process for ammonia (Heterogeneous)	Fe: Mo as promotor
Ostwald's process for HNO_3 from NH_3 (Heterogeneous)	Pt Gauge
Chamber process for H_2SO_4 (Homogeneous)	Nitric oxide (NO) gas
Contact process for H_2SO_4 (Heterogeneous)	V_2O_5
Deacon's process for manufacture of Cl_2 (Heterogeneous)	$CuCl_2$
Bosch process for H_2 (Heterogeneous)	$Fe_2O_3 + Cr_2O_3$ (promotor)
Manufacture of vegetable ghee (Heterogeneous)	Ni, Pt, Pd
Hydrolysis of ester (Homogeneous)	Aq. H_2SO_4/H^+
Aldol condensation of aldehyde (Homogeneous)	OH^-
Enolisation of ketone (Homogeneous)	H^+ or OH^-
Cracking of hydrocarbon (Heterogeneous)	Zeolites
Decomposition of O_3	'Cl' atom
$2CO + O_2 \rightarrow 2CO_2$	'NO'

(i) **Zeolites (Shape selective catalyst) :**

- (i) The catalytic reaction that depends upon the pore structure of the catalyst & the size of the reactant is called shape selective catalysis. The pore size is generally in range 260-740 pm.

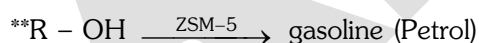
Ex. $\rightarrow$ Zeolite (honey comb - like structure).

Formula of Zeolite is $Na_{x/n} [(AlO_2)_x (SiO_2)_y] z H_2O$

- (ii) Zeolite is also called sodium aluminosilicate, water softner & cationic exchanger (Exchange only cation from hard water).
- (iii) Zeolite is 3D silicate.
- (iv) Zeolite, before using as catalyst are heated in vacuum so that the water of hydration is lost and as a result Zeolite becomes porous.

***Application of Zeolite :**

Normally used in petro chemical industries for cracking of hydrocarbon & isomerisation.



5.13 ENZYME CATALYSIS

- (i) Enzymes are complex nitrogenous organic compounds which are produced from living plants and animals.
- (ii) They are actually protein molecules of high molecular mass and form colloidal solutions in water.
- (iii) They catalyse biochemical reactions.

Reactions catalyzed by enzymes :

Enzyme	Source	Reaction
(i) Invertase	Yeast	Sucrose $\rightarrow$ Glucose and Fructose
(ii) Zymase	Yeast	Glucose $\rightarrow$ Ethyl alcohol and carbon dioxide
(iii) Diastase	Malt	Starch $\rightarrow$ Maltose
(iv) Maltase	Yeast	Maltose $\rightarrow$ Glucose
(v) Urease	Soyabean	Urea $\rightarrow$ Ammonia and CO_2
(vi) Pepsin	Stomach	Proteins $\rightarrow$ Amino acid

Characteristics of enzyme catalysis :

- Highly efficient :** One molecule of an enzyme may transform one million molecules of the reactant per minute.
- Highly specific nature :** One catalyst cannot catalyse more than one reaction.
- Highly active under optimum temperature (298 K to 310 K)
- Highly active under optimum pH (5 to 7)
- Increasing activity in presence of cofactor (Na^+ , Mn^{2+} , Co^{2+} , Cu^{2+} etc.) and co-enzymes

Mechanism of enzyme catalysis (key lock theory) : There are number of active centres of definite shape present on the surface of colloidal particles of enzymes. The molecules of the reactant (substrate), which have complementary shape fit into these cavities just like a key fits into a lock. On account of the presence of active groups an activated complex is formed which then decomposes to yield the product.

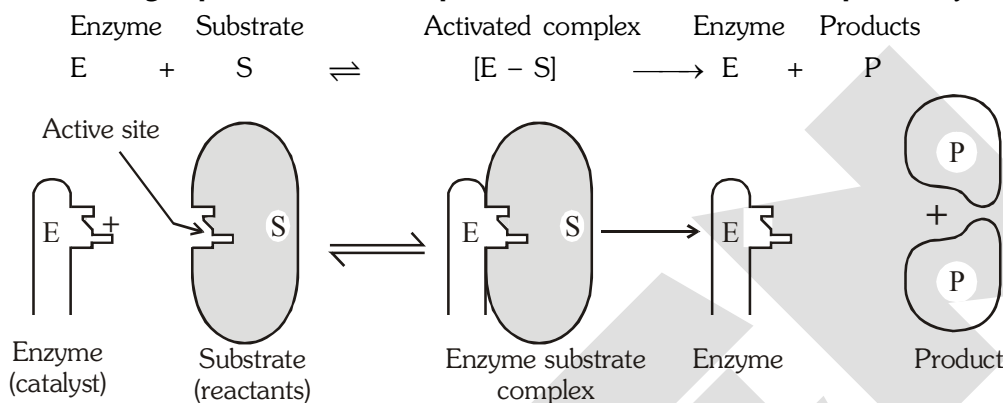


Fig : Mechanism of enzyme catalysed reaction

GOLDEN KEY POINTS

- The difference between inhibitors and poisons is that the catalytic inhibitor retards rate of reaction by itself while catalytic poison retards the rate of reaction by reducing the activity of catalyst.
- d-block metals are good catalysts, as they have higher surface area and unpaired d-electrons.

BEGINNER'S BOX-5

- Catalytic poisons are
 - Poison for catalyst
 - Poison for reactant
 - Poison for product
 - None of these
- A catalyst is more effective in
 - Finely powdered state
 - Colloidal state
 - Rough surface
 - All of these
- Zeolites are
 - Water softener
 - Catalyst
 - Both (1) and (2)
 - Colloids

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6				
	Ans.	4	1	4	1	3	2				
BEGINNER'S BOX-2	Que.	1	2								
	Ans.	1	3								
BEGINNER'S BOX-3	Que.	1	2								
	Ans.	4	2								
BEGINNER'S BOX-4	Que.	1	2	3	4	5					
	Ans.	3	4	3	1	1					
BEGINNER'S BOX-5	Que.	1	2	3							
	Ans.	1	4	3							

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

ADSORPTION

- Physical adsorption is appreciable at :-
(1) Higher temperature (2) Lower temperature
(3) At room temperature (4) 100° C
- The rate of physical adsorption:-
(1) Decreases with increase of pressure
(2) Is independent at high pressure
(3) Is maximum at one atmospheric pressure
(4) Always increases with increase of pressure
- Which of the following is not a characteristic of chemi-sorption :-
(1) Adsorption is irreversible
(2) ΔH is of the order of 40 kJ
(3) Adsorption is specific
(4) Adsorption increases with increase of surface area
- Which one of the following is not a correct statement ?
(1) Physical adsorption is reversible in nature
(2) Physical adsorption involves vander waals forces
(3) Rate of physical adsorption increases with increase of pressure on the adsorbate
(4) High activation energy is involved for physical adsorption
- The amount of gas adsorbed on charcoal increases with :-
(1) Temperature & pressure
(2) Temperature & decreases with pressure
(3) Pressure & decreases with temperature
(4) None
- Pd can adsorb 900 times its volume of hydrogen. This is called :-
(1) Absorption (2) Adsorption
(3) Occlusion (4) 2 & 3 both
- Which is correct :-
(1) Langmuir adsorption is highly specific
(2) vander-Waal's adsorption is reversible
(3) Both 1 & 2 are exothermic
(4) All are correct
- Adsorption is accompanied by :-
(1) Decrease in entropy of the system
(2) Decrease in enthalpy of the system
(3) $T\Delta S$ for the process is negative
(4) All

- Which characteristic of adsorption is wrong :-
(1) Physical adsorption in general decreases with temperature
(2) Physical adsorption in general increases with temperature
(3) Physical Adsorption is a reversible process
(4) Adsorption is limited to the surface only
- Graph between $\log \left(\frac{x}{m} \right)$ and $\log p$ is a straight line at an angle 45° with intercept on y-axis 0.3010. Calculate the amount of gas adsorbed in gram per gram of the adsorbent when pressure is 0.2 atm.
(1) 0.4 (2) 0.6 (3) 0.8 (4) 0.2
- Sorption is the term used when :
(1) Adsorption takes place
(2) Absorption takes place
(3) Both takes place
(4) Desorption takes place
- The volume of gases H_2 , CH_4 , CO_2 and NH_3 adsorbed by 1 g of charcoal at 288K are in the order :
(1) $H_2 > CH_4 > CO_2 > NH_3$
(2) $CH_4 > CO_2 > NH_3 > H_2$
(3) $CO_2 > NH_3 > H_2 > CH_4$
(4) $NH_3 > CO_2 > CH_4 > H_2$
- The extent of adsorption of a gas on a solid depends on :
(1) The nature of gas
(2) Pressure of gas
(3) Temperature of the system
(4) All
- Which forms multi molecular layers during adsorption :
(1) Physical adsorption
(2) vander Waal's adsorption
(3) Freundlich adsorption
(4) All

COLLOIDAL SOLUTION

- The number of phases present in colloidal solution is :-
(1) 2 (2) 4
(3) 3 (4) 1

- 16.** Butter is a colloid formed when :-
 (1) Fat is dispersed in fat
 (2) Fat is dispersed in water
 (3) Water is dispersed in fat
 (4) Suspension of casein in water
- 17.** Lyophobic colloids are :-
 (1) Reversible (2) Irreversible
 (3) Water loving (4) Solvent loving
- 18.** When freshly precipitated $\text{Fe}(\text{OH})_3$ is boiled with water in the presence of few drops of dilute HCl, a hydrated ferric hydroxide sol is obtained. This method is termed as :-
 (1) Dialysis (2) Peptization
 (3) Ultrafiltration (4) Electrodipersion
- 19.** Greater the valency, the higher is the coagulating power of ion. This rule was introduced by :-
 (1) Hardy-Schulze (2) Graham
 (3) Kossel & Lewis (4) Faraday
- 20.** The capacity of an ion to coagulate a colloidal solution depends on :-
 (1) Its shape
 (2) The amount of its charge
 (3) The sign of the charge
 (4) Both, the amount and the sign of the charge
- 21.** All colloidal solutions show :-
 (1) Very high osmotic pressure
 (2) High osmotic pressure
 (3) Low osmotic pressure
 (4) No osmotic pressure
- 22.** The charge of As_2S_3 sol is due to the absorbed :-
 (1) H^+ (2) OH^-
 (3) O^{2-} (4) S^{2-}
- 23.** Brownian motion shown by colloidal particles is its ----- property :-
 (1) Optical (2) Electrical
 (3) Kinetic (4) Chemical
- 24.** A freshly prepared $\text{Fe}(\text{OH})_3$ precipitate is peptized by adding FeCl_3 solution. The charge on the colloidal particles is due to preferential adsorption of :-
 (1) Cl^- ions (2) Fe^{+++} ions
 (3) OH^- ions (4) None
- 25.** In both dialysis and osmosis which particle do not pass through SPM :
 (1) Water (2) Small molecules
 (3) Colloids (4) All
- 26.** The correct statement in case of milk :-
 (1) Milk is an emulsion of fat in water
 (2) Milk is an emulsion of protein in water
 (3) Milk is unstabilized by protein
 (4) Milk is unstabilized by fat
- 27.** A colloidal system involves :-
 (1) A state of dissolution
 (2) A state of dispersion
 (3) A state of suspension
 (4) None
- 28.** In electrophoresis :-
 (1) Sol particles move towards opposite electrodes
 (2) Medium moves towards opposite electrodes
 (3) Neither (1) nor (2)
 (4) Both (1) & (2)
- 29.** Detergent action of synthetic detergents is due to:-
 (1) Interfacial area
 (2) High molecular weight
 (3) Ionisation
 (4) Emulsifying properties
- 30.** Which is not shown by sols :-
 (1) Adsorption (2) Tyndall effect
 (3) Flocculation (4) Paramagnetism
- 31.** Which of the following is an emulsifier ?
 (1) Soap (2) Water
 (3) Oil (4) NaCl
- 32.** Emulsifiers are generally :-
 (1) Soaps (2) Synthetic detergents
 (3) Lyophilic sols (4) All of the above
- 33.** Which of the following is most effective in causing the coagulation of ferric hydroxide sol :-
 (1) KCl (2) KNO_3
 (3) K_2SO_4 (4) $\text{K}_3[\text{Fe}(\text{CN})_6]$
- 34.** The colloidal sol of SnCl_4 prefers to adsorb _____ in excess of HCl :
 (1) Sn^{+4} (2) K^+
 (3) H^+ (4) Cl^-
- 35.** On adding AgNO_3 solution into KI solution, a negatively charged colloidal sol is obtained when they are mixed as :
 (1) 100 mL of 0.1 M AgNO_3 + 100 mL of 0.1 M KI
 (2) 100 mL of 0.1 M AgNO_3 + 50 mL of 0.2 M KI
 (3) 200 mL of 0.1 M AgNO_3 + 200 mL of 0.1 M KI
 (4) 100 mL of 0.1 M AgNO_3 + 100 mL of 0.15 M KI

- 36.** Micelles are :
 (1) Ideal solution (2) Associated colloids
 (3) Adsorbed surfaces (4) Adsorbent solutes
- 37.** Micelles have :
 (1) higher colligative properties as compared to common colloidal sols
 (2) lower colligative properties
 (3) same colligative properties
 (4) All of the above
- 38.** Which of the following sol is formed due to following reaction :- $\text{SnO}_2 + \text{HCl}$ (Excess) :-
 (1) $[\text{SnCl}_4] \text{Cl}^-$ (2) $[\text{SnCl}_4] \text{O}^{2-}$
 (3) $[\text{SnCl}_4] \text{H}^+$ (4) None
- 39.** Which of followig ion has minimum flocculation value :
 (1) Cl^- (2) SO_4^{2-}
 (3) PO_4^{3-} (4) $[\text{Fe}(\text{CN})_6]^{4-}$
- 40.** A negatively charged suspension of clay in water needs for precipitation the minimum amount of:
 (1) Aluminium chloride (2) Potassium sulphate
 (3) Sodium hydroxide (4) Hydrochloric acid
- 41.** Which is not a colloidal solution :
 (1) Smoke (2) Ink
 (3) Air (4) Blood
- 42.** Which one is natural colloid :
 (1) NaCl (2) Blood
 (3) RCOONa (4) Sugar
- 43.** Medicines are more effective if they are used in :
 (1) Colloidal state (2) Solid state
 (3) Granular state (4) All of the above
- 44.** Egg albumin is :
 (1) Reversible colloid (2) Lyophilic colloid
 (3) Protective colloid (4) All
- 45.** Gelatin protects :-
 (1) Gold sol (2) As_2S_3 sol
 (3) $\text{Fe}(\text{OH})_3$ sol (4) All
- 46.** The coagulating power of an effective ion carrying the charge opposite to the sol particles has been illustrated by :-
 (1) Brownian movement
 (2) Gold number
 (3) Tyndall effect
 (4) Hardy-schulze rule
- 47.** Hardy-Schulze rule states that :-
 (1) Non-electrolytes have better coagulating action on colloids than electrolytes
 (2) Sols are coagulated by effective ions whose charge is opposite to that of sol and the ions of higher charge are much more effective than the ions of lower charge
 (3) Charge of the ions has no effect on the coagulation of a sol
 (4) Sols are coagulated only by those ions whose charges is similar to that of the sol
- 48.** An example of micelle is :-
 (1) As_2O_3 sol
 (2) Ruby glass
 (3) Na_2CO_3 solution
 (4) Sodium stearate concentrated solution
- 49.** To coagulate gelatin sol, which of the following is most effective :-
 (1) NaCl (2) Na_3PO_4
 (3) AlCl_3 (4) Alcohol
- 50.** Which of the following has minimum flocculation value:-
 (1) Pb^{2+} (2) Pb^{4+}
 (3) Sr^{2+} (4) Na^+
- 51.** The gold numbers of A, B, C & D are 0.04, 0.002, 10 and 25 respectively. The protective powers of A, B, C and D are in the order :-
 (1) $A > B > C > D$ (2) $B > A > C > D$
 (3) $D > C > B > A$ (4) $C > A > B > D$
- 52.** On addition of one mL solution of 10% NaCl to 10 mL gold sol in presence of 0.025 g of starch, the coagulation is just prevented. The gold number of starch is :-
 (1) 25 (2) 2.5
 (3) 0.25 (4) 0.025
- 53.** Gold number is a measure of :-
 (1) The amount of gold present in the colloidal solution.
 (2) The amount of gold required to break the colloid.
 (3) The amount of gold required to protect the colloid.
 (4) None of the above

54. Following are various types of colloids. Match column X with column Y.

X (Colloids)		Y (Classification)	
--------------	--	--------------------	--

I	Rain cloud	A	Sol
II	Gelatin	B	Aerosol
III	Soap lather	C	Gel
IV	Boot polish	D	Foam

Correct matching is :

	I	II	III	IV
(1)	A	B	C	D
(2)	A	C	B	D
(3)	B	A	D	C
(4)	B	A	C	D

55. Silt in water is :

(1) Gel (2) Sol (3) Aerosol (4) Foam

56. Opal (mineral with liquid inclusions) is a:

(1) Gel (liquid dispersed in solid phase)
 (2) Solid sol (solid dispersed in solid phase)
 (3) Sol (solid dispersed in liquid)
 (4) Foam (gas dispersed in liquid)

57. Some of the following are true solutions :

I : Air II : Sea water
 III : Glucose solution IV : Gem stone
 V : Pearl VI : Blood

Select true solutions :

(1) I, II, III (2) II, III, IV, V
 (3) I, IV, V, VI (4) II, IV, VI

58. In multimolecular colloidal sols, atoms or molecules are held together by :

(1) H-bonding
 (2) vander-Waals forces
 (3) Ionic bonding
 (4) Polar covalent bonding

59. Tyndall effect is not observed in :

(1) Suspension (2) Starch sol
 (3) Gold sol (4) NaCl solution

60. Which is kinetic phenomenon?

(1) Brownian motion (2) Tyndall effect
 (3) Both (1) and (2) (4) None of these

61. Which pair is correctly matched?

(1) $[\text{Fe}(\text{OH})_3]$: Fe^{3+} : Cl^-
 (2) $[\text{As}_2\text{S}_3]$: As^{3+}
 (3) $[\text{SnO}_2]$: SnO_3^{2-} in acidic medium
 (4) $[\text{AgI}]$: I⁻ in excess of AgNO_3

62. Gold number of haemoglobin is 0.03. Hence, 100mL of gold sol will require haemoglobin so that gold is not coagulated by 10 mL of 10% NaCl solution.

(1) 0.03 mg (2) 30 mg
 (3) 0.30mg (4) 3 mg

63. Which is not a correct matching of emulsions:

(1) Milk : O/W
 (2) Cold cream : W/O
 (3) Butter : O/W
 (4) Vanishing cream: O/W

CATALYST

64. Which of the following statement is correct :-

(1) Catalyst accelerates the rate of a chemical reaction
 (2) A catalyst can retard the rate of a chemical reaction
 (3) A catalyst can control the speed of a reaction
 (4) A catalyst does not alters the speed of a reaction

65. Which one of the following is not the example of homogeneous catalysis :-

(1) Formation of SO_3 in the chamber process
 (2) Formation of SO_3 in the contact process
 (3) Hydrolysis of an ester in presence of acid
 (4) Decomposition of KClO_3 in presence of MnO_2

66. The decomposition of hydrogen peroxide can be slowed down by the addition of a small amount of acetanilide. The later act as :-

(1) Inhibitor (2) Promoter
 (3) Moderator (4) Poison

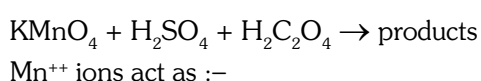
67. Efficiency of the catalyst depends on its :-

(1) Molecular weight
 (2) Number of free valencies
 (3) Physical state
 (4) Amount of reactant used

68. Which of the following types of metals make the most efficient catalysts :-

(1) Transition metals (2) Alkali metals
 (3) Alkaline earth metals (4) Radioactive metals

69. In the reaction



Mn^{++} ions act as :-

(1) Positive catalyst (2) Negative catalyst
 (3) Auto catalyst (4) Enzyme catalyst

- 70.** In the Haber's process of synthesis of NH_3 :-
 (1) Mo acts as a catalyst and Fe as a promoter
 (2) Fe acts as a catalyst and Mo as a promoter
 (3) Fe acts as inhibitor and Mo as a catalyst
 (4) Fe acts as promoter and Mo as auto-catalyst
- 71.** Which of the following statement is incorrect :-
 (1) Enzymes exist in colloidal state
 (2) Enzymes are catalysts
 (3) Enzymes can catalyse any reaction
 (4) Urease is an enzyme
- 72.** Platinized asbestos is used as a catalyst in the manufacture of H_2SO_4 . It is an example of :-
 (1) Homogeneous catalyst
 (2) Heterogeneous catalyst
 (3) Auto-catalyst
 (4) Induced catalyst
- 73.** In the Ostwald's process for the manufacturing of HNO_3 , the catalyst used is :-
 (1) Fe (2) Pt (3) V_2O_5 (4) Mo
- 74.** In a reversible reaction a catalyst :-
 (1) Increases the rate of forward reaction only
 (2) Increases the rate of forward reaction to a greater extent than that of the backward reaction
 (3) Increases the rate of forward reaction and decreases that of the backward reaction
 (4) Increases the rate of forward and backward reaction equally
- 75.** Which is false for catalyst :-
 (1) A catalyst can initiate a reaction
 (2) It does not alter the position of equilibrium in a reversible reaction
 (3) A catalyst remains unchanged in quantity and composition at the end of reaction
 (4) Catalysts are sometimes very specific in respect of a reaction
- 76.** Which acts as a catalyst in the hydrogenation of vegetable oils :-
 (1) Cu (2) Mo (3) Fe (4) Pt
- 77.** Air can oxidize sodium sulphite in aq. solution but cannot do so in the case of sodium arsenite. If however, air is passed through a solution containing both sodium sulphite & sodium arsenite then both are oxidized. This is an example of :-
 (1) Positive catalysis (2) Negative catalysis
 (3) Induced catalysis (4) Auto catalysis
- 78.** Zeolites are :-
 (1) Water softner (2) Catalyst
 (3) cation exchanger (4) All of these
- 79.** Zeolites :-
 (1) Are microporous aluminosilicates
 (2) Have general formula $\text{M}_{x/n}[(\text{AlO}_2)_x(\text{SiO}_2)_y] \cdot m\text{H}_2\text{O}$
 (3) Have pore sizes between 260 pm to 740 pm
 (4) All
- 80.** Zeolites are used as catalyst in :
 (1) Petrochemical industries during cracking
 (2) In the preparation of H_2SO_4
 (3) In the hydrolysis of ester
 (4) All
- 81.** Which is not the correct statement for a catalyst :
 (1) It does not alter activation energy.
 (2) It provides an alternate mechanism with a lower energy of activation
 (3) Catalyst may form intermediates with the reactants
 (4) Action of enzyme catalyst is always specific
- 82.** Shape selective catalysts are so called because of :
 (1) The shape of the catalysts
 (2) The specificity of the catalysts
 (3) The size of the pores of the catalysts which can trap only selective molecules
 (4) Their use for only some selected reactions

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	2	2	4	3	4	4	4	2	1	3	4	4	4	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	2	1	4	3	4	3	2	3	1	2	1	4	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	4	4	4	4	2	2	1	4	1	3	2	1	4	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	2	4	4	2	2	1	4	3	2	1	1	2	4	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	3	3	1	2	1	2	1	3	2	3	2	2	4	1
Que.	76	77	78	79	80	81	82								
Ans.	4	3	4	4	1	1	3								

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2007**

1. The Langmuir adsorption isotherm is deduced using the assumption.
- (1) The adsorption sites are equivalent in their ability to adsorb the particles.
 - (2) The heat of adsorption varies with coverage.
 - (3) The adsorption molecules interact with each other.
 - (4) The adsorption takes place in multilayers.

AIPMT 2010

2. Butter is an example of :-
- (1) Water oil emulsion
 - (2) Gas-liquid colloidal system
 - (3) Oil water emulsion
 - (4) Solid-solid colloidal system

AIPMT Pre. 2011

3. If x is amount of adsorbate and m is amount of adsorbent, which of the following relations is not related to adsorption process ?
- (1) $x/m = f(p)$ at constant T
 - (2) $x/m = f(T)$ at constant p
 - (3) $p = f(T)$ at constant (x/m)
 - (4) $\frac{x}{m} = p \times T$

AIIMS 2011

4. Which of the following is incorrect for physisorption :
- (1) Reversible
 - (2) Increases with increase in temperature
 - (3) Low heat of adsorption
 - (4) Increases with increase in surface area

AIPMT Pre. 2012

5. In Freundlich Adsorption isotherm, the value of $1/n$ is :
- (1) 1 in case of physical adsorption
 - (2) 1 in case of chemisorption
 - (3) between 0 and 1 in all cases
 - (4) between 2 and 4 in all cases

6. Which one of the following statements is **incorrect** about enzyme catalysis?
- (1) Enzymes are denaturated by ultraviolet rays and at high temperature
 - (2) Enzymes are least reactive at optimum temperature
 - (3) Enzymes are mostly proteinous in nature
 - (4) Enzyme action is specific
7. The protecting power of lyophilic colloidal sol is expressed in terms of:
- (1) Critical miscelle concentration
 - (2) Oxidation number
 - (3) Coagulation value
 - (4) Gold number

AIIMS 2012

8. A colloidal solution is kept in dark and is illuminated by a beam of light then brightness appears at the right angle of direction of light. This effect is called:-
- (1) Tyndall effect
 - (2) Brownian effect
 - (3) Hardy schulze effect
 - (4) None of these

AIIMS 2013

9. Hair cream is :-
- (1) Gel
 - (2) Emulsion
 - (3) Solid sol
 - (4) Sol
10. According to Hardy Schulze law, the flocculating power of an ion increases with :-
- (1) Decreases in size
 - (2) Increase in size
 - (3) Decrease in charge
 - (4) Increase in charge

AIPMT 2014

11. Which property of colloids is **not** dependent on the charge on colloidal particles ?
- (1) Coagulation
 - (2) Electrophoresis
 - (3) Electro - osmosis
 - (4) Tyndall effect

12. Which of the following statements is correct for the spontaneous adsorption of a gas ?

- (1) ΔS is negative and, therefore, ΔH should be highly positive
- (2) ΔS is negative and therefore, ΔH should be highly negative
- (3) ΔS is positive and, therefore, ΔH should be negative
- (4) ΔS is positive and, therefore, ΔH should also be highly positive

AIIMS 2014

13. Which of the following process is used in the purification of colloidal solution :-

- (1) Dialysis
- (2) Coagulation
- (3) Peptization
- (4) Electrophoresis

AIIMS 2015

14. FeCl_3 sol in hot water mixed with conc. NaOH is

- (1) Positively charged colloid
- (2) Negatively charged colloid
- (3) Both (+) and (-) charge
- (4) All of these

NEET-I 2016

15. Which one of the following characteristics is associated with adsorption ?

- (1) ΔG is negative but ΔH and ΔS are positive
- (2) ΔG , ΔH and ΔS all are negative
- (3) ΔG and ΔH are negative but ΔS is positive
- (4) ΔG and ΔS are negative but ΔH is positive

16. Fog is colloidal solution of :-

- (1) Liquid in gas
- (2) Gas in liquid
- (3) Solid in gas
- (4) Gas in gas

NEET-II 2016

17. The coagulation values in millimoles per litre of the electrolytes used for the coagulation of As_2S_3 are given below :

- I. $(\text{NaCl}) = 52$,
- II. $(\text{BaCl}_2) = 0.69$,
- III. $(\text{MgSO}_4) = 0.22$

The **correct** order of their coagulating power is

- (1) III > II > I
- (2) III > I > II
- (3) I > II > III
- (4) II > I > III

AIIMS 2016

18. The nature of colloidal solution of methylene blue and haemoglobin is _____ and _____ respectively?

- (1) positive, positive
- (2) negative, negative
- (3) positive, negative
- (4) negative, positive

19. Which of the following is correct example of sol and emulsion respectively?

- (1) Paints, milk
- (2) Ink, paints
- (3) Fog, mist
- (4) Smoke, colloision

AIIMS 2017

20. Milk and cheese are respectively :-

- (1) solid sol and emulsion
- (2) gel and emulsion
- (3) solid sol and gel
- (4) emulsion and gel

21. Which of the following cannot be used to coagulate hydrophobic colloids :-

- (1) Persistent dialysis
- (2) Electrophoresis
- (3) On addition of oppositely charged electrolyte
- (4) Dilution

NEET(UG) 2018

22. On which of the following properties does coagulating power of an ion depend ?

- (1) The magnitude of the charge on the alone
- (2) Size of the ion alone
- (3) Both magnitude and sign of the charge the ion
- (4) The sign of charge on the ion alone

AIIMS 2018

23. Which of the following is incorrect statement :-

- (1) Colloidal solution has large surface area so they are used in medicine
- (2) $\text{AgNO}_3 + \text{KI}_{\text{excess}}$ forms negative charge colloidal in presence of excess KI
- (3) Persistent dialysis of colloidal stabilises the sol.
- (4) On adding alum to a colloidal solution is called coagulation

24. By which method we can convert ppt to colloidal solution :-

- (1) Peptization
- (2) Homogenisation
- (3) Dispersion
- (4) Solvation

25. Which of the following is an example of emulsifying agent for w/o type emulsion

- (1) Soap
- (2) Heavy metal salts of fatty acid
- (3) Starch
- (4) Gum

26. Movement of dispersion medium in presence of electric field is known as :-

- (1) Electro osmosis
- (2) Electro dialysis
- (3) Electro dispersion
- (4) Electro phoresis

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	4	2	3	2	4	1	2	4	4	2	1	2	2
Que.	16	17	18	19	20	21	22	23	24	25	26				
Ans.	1	1	1	1	4	4	3	3	1	2	1				

EXERCISE-III (Analytical Questions)

Check Your Understanding

- Lapis lazuli is :
(1) Ferrous sulphate
(2) Copper sulphate
(3) Sodium aluminosilicate
(4) Zinc sulphate
- According to Freundlich adsorption isotherm, which of the following is correct ?
(1) $\frac{x}{m} \propto p^0$ (2) $\frac{x}{m} \propto p^1$
(3) $\frac{x}{m} \propto p^{1/n}$ (4) All of the above
- Which of the following forms cationic micelles above certain concentration :
(1) Cetyltrimethylammonium bromide
(2) Sodium dodecyl sulphate
(3) Sodium acetate
(4) Urea
- Which one of the following statements about the zeolites is false :
(1) Some of the SiO_4^{4-} units are replaced by AlO_4^{5-} and AlO_6^{9-} ions in zeolites
(2) They are used as cation exchangers
(3) They have open structure which enables them to take up small molecules
(4) Zeolites are aluminosilicates having three dimensional network

- Which one of the following forms micelles in aqueous solution above certain concentration :
(1) Glucose
(2) Urea
(3) Dodecyltrimethyl ammonium chloride
(4) Pyridinium chloride
- Fog is a colloidal system of :-
(1) gas in liquid (2) liquid in gas
(3) gas in gas (4) gas in solid
- The coagulating power of electrolytes having ions Na^+ , Al^{3+} and Ba^{2+} for arsenic sulphide sol increases in the order :-
(1) $\text{Al}^{3+} < \text{Ba}^{2+} < \text{Na}^+$
(2) $\text{Na}^+ < \text{Ba}^{2+} < \text{Al}^{3+}$
(3) $\text{Ba}^{2+} < \text{Na}^+ < \text{Al}^{3+}$
(4) $\text{Al}^{3+} < \text{Na}^+ < \text{Ba}^{2+}$
- The volume of a colloidal particle, V_c as compared to the volume of a solute particle in a true solution V_s , could be :-

- (1) $\frac{V_c}{V_s} \approx 1$ (2) $\frac{V_c}{V_s} \approx 10^{23}$
(3) $\frac{V_c}{V_s} \approx 10^{-3}$ (4) $\frac{V_c}{V_s} \approx 10^3$

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	3	4	1	1	3	2	2	4

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion :-** Langmuir adsorption is a single layer phenomenon.

Reason :- It is due to Vanderwaal's forces.

- (1) A (2) B (3) C (4) D

2. **Assertion :-** A sol of As_2S_3 prepared by the action of H_2S on As_2O_3 is negatively charged.

Reason :- Due to adsorption of S^{2-} ion on surface of As_2S_3 sol.

- (1) A (2) B (3) C (4) D

3. **Assertion :-** Physical adsorption of molecule on the surface requires activation energy.

Reason :- Because the bonds of adsorbed molecules are broken.

- (1) A (2) B (3) C (4) D

4. **Assertion :-** In case of chemical adsorption, $\frac{x}{m}$ first increase then decrease with temperature.

Reason :- The initial increase is due to the fact that heat supplied acts as activation energy. The decrease afterwards is due to the exothermic nature of adsorption at equilibrium.

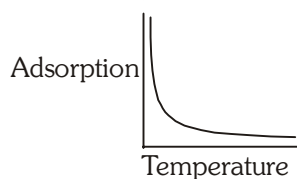
- (1) A (2) B (3) C (4) D

5. **Assertion :-** Physical adsorption forms multimolecular layer.

Reason :- It contains covalent bond.

- (1) A (2) B (3) C (4) D

6. **Assertion :-** The graphical representation of physical adsorption with temperature is



Reason :- This is weak adsorption and adsorption is exothermic process.

- (1) A (2) B (3) C (4) D

7. **Assertion :-** During physical adsorption multimolecular layers are formed.

Reason :- For physical adsorption value of $\Delta H = -20$ to -40 kJ mol^{-1} .

- (1) A (2) B (3) C (4) D

8. **Assertion :-** The micelles formed by sodium stearate in water has $-\text{COO}^-$ groups at the surface.

Reason :- Surface tension of water is reduced by the addition of stearate.

- (1) A (2) B (3) C (4) D

9. **Assertion :-** Alcohols are dehydrated to hydrocarbons in the presence of acidic zeolites.

Reason :- Zeolites are porous catalysts.

- (1) A (2) B (3) C (4) D

10. **Assertion :-** Fe^{3+} can be used for coagulation of As_2S_3 sol.

Reason :- Fe^{3+} reacts with As_2S_3 to give Fe_2S_3 .

- (1) A (2) B (3) C (4) D

11. **Assertion :-** Catalyst is more effective in powdered form.

Reason :- Number of free valencies increases with increase in surface area.

- (1) A (2) B (3) C (4) D

12. **Assertion :-** A gas mixed with another gas forms a colloidal solution.

Reason :- Mixture of gases is homogenous system.

- (1) A (2) B (3) C (4) D

13. **Assertion :-** Filtration of colloidal particles is not possible by filter paper.

Reason :- Size of colloidal particles is less than filter paper pores.

- (1) A (2) B (3) C (4) D

14. Assertion :- Lyophilic sol is more stable than lyophobic sol.

Reason :- Lyophilic sol have greater solvation energy than lyophobic sol.

(1) A (2) B (3) C (4) D

15. Assertion :- Associated colloids are formed by aggregation of large number of ions in concentrated solution of soap.

Reason :- Because above CMC they form micelles where many particles are associated.

(1) A (2) B (3) C (4) D

16. Assertion :- Lyophilic sol coagulate by adding dehydrating solvent.

Reason :- Lyophilic sol are stable due to excessive hydration.

(1) A (2) B (3) C (4) D

17. Assertion :- Charge on hydrophobic sol is greater than hydrophilic sol.

Reason :- Lyophobic sol and lyophilic sol are inorganic substance.

(1) A (2) B (3) C (4) D

18. Assertion :- Aqueous gold colloidal solution is red in colour.

Reason :- The colour arises due to scattering of light by colloidal gold particles.

(1) A (2) B (3) C (4) D

19. Assertion :- Zeolite is shape selective catalyst.

Reason :- Size of the pores of the catalyst are such that they can trap only selective molecules.

(1) A (2) B (3) C (4) D

20. Assertion : Reaction of SO_2 and H_2S in the presence of Fe_2O_3 catalyst gives elemental sulphur.

Reason: In above reaction SO_2 is a reducing agent.

(1) A (2) B (3) C (4) D

21. Assertion :- Physical adsorption increases with increase in temperature.

Reason :- Physical adsorption is exothermic in nature.

(1) A (2) B (3) C (4) D

22. Assertion :- Charcoal is used to separate inert gases.

Reason :- Size of inert gas atoms are different.

(1) A (2) B (3) C (4) D

23. Assertion :- Catalyst increases the rate of reaction.

Reason :- Catalyst increases the activation energy of reaction.

(1) A (2) B (3) C (4) D

24. Assertion :- On addition of FeCl_3 , blood gets coagulated.

Reason :- Blood is an albuminoid colloid.

(1) A (2) B (3) C (4) D

25. Assertion :- Hydrated Fe_2O_3 mixed with As_2S_3 form ppt.

Reason :- They react to form insoluble salt.

(1) A (2) B (3) C (4) D

26. Assertion : Colloidal particles do not move in presence of electric field.

Reason : Colloidal particles do not contains charge.

(1) A (2) B (3) C (4) D

27. Assertion :- On increasing particle size of gold sol, colour change from red, purple, blue and finally gold.

Reason :- Extent of metallic bonding increases.

(1) A (2) B (3) C (4) D

28. Assertion : Soap is added to stabilise oil in water emulsion.

Reason : It lowers surfaces tension between oil and water.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	1	4	1	3	1	2	2	2	2	1	4	1	1	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	1	3	1	1	3	4	2	3	2	2	4	3	2		

IMPORTANT NOTES

P-BLOCK ELEMENTS

E

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NEET SYLLABUS

p-Block Elements : General Introduction to p-Block Elements. General introduction, electronic configuration, occurrence, variation of properties, oxidation states, trends in chemical reactivity, anomalous properties of first element of the Groups 13 to 18 elements.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- appreciate general trends in the chemistry of elements of groups 13 to 18;
- explain anomalous behaviour of boron and carbon;
- describe allotropic forms of carbon;
- learn the preparation, properties and uses of dinitrogen and phosphorus and some of their important compounds;
- describe the preparation, properties and uses of dioxygen and ozone and chemistry of some simple oxides;
- know allotropic forms of sulphur, chemistry of its important compounds and the structures of its oxoacids;
- describe the preparation, properties and uses of chlorine and hydrochloric acid;
- know the chemistry of interhalogens and structures of oxoacids of halogens;
- enumerate the uses of noble gases;
- appreciate the importance of these elements and their compounds in our day to day life.

p-Block elements signify common man in society on whom all socio-economic rules are applicable.

Linus Pauling

p-BLOCK ELEMENTS

IMPORTANT CONCEPTS :

1. Back bonding
2. Dimerisation
3. Inert pair effect
4. Oxyacids
5. Hydrolysis
6. Allotropes
7. Silicates and Silicones
8. Drying agent and bleaching agents
9. Reactions and compounds (Group 13-18)

1. BACK BONDING

It is formed between two covalently bonded adjacent atoms.

Conditions for back bonding :

- (i) Both bonded atoms must belong to 2nd period or one bonded atom must belong to 2nd period and the other must belong to 3rd period.
- (ii) One bonded atom must possess vacant orbital and the other bonded atom must possess lone pair.

- As a result of back bonding between the bonded atoms, bond length decreases and bond energy increases.

Back bonding :

It is a partial sideways overlap in which there is a slight electron transference from electron rich atom to electron deficient atom. As a result the electron deficiency of the deficient atom gets slightly compensated.



Types of back bonding :

Based on type of orbital : (2 types)

- (1) $p\pi-p\pi$ back bonding
- (2) $p\pi-d\pi$ back bonding

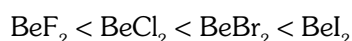
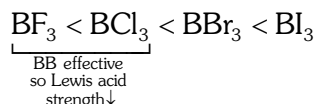
(1) $p\pi-p\pi$ back bonding :

Order of strength : $2p-2p > 2p-3p > 2p-4p \dots\dots$

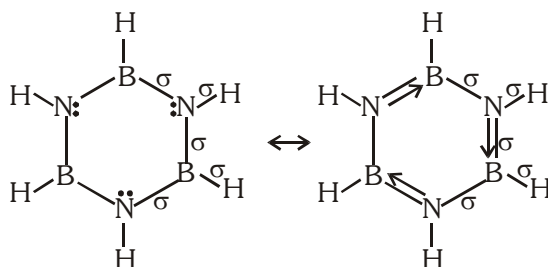
Size $\uparrow$ extent of B.B. $\downarrow$

It is used to explain following observations :-

- (a) Abnormal bond length and bond energy of B-F bond in BF_3 .
- (b) Lewis acidic order of **Boron** and **Beryllium** halides.

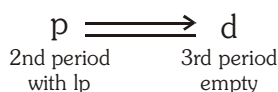


- (c) **Hybridisation** : If a lone pair participates in back bonding then it is not considered in hybridisation.
 Ex. $B_3N_3H_6$ (inorganic benzene or borazene or borazole)
- Hybridisation of B as well as N = sp^2



- Inorganic benzene is more reactive than organic benzene as in it the **bonds are polar**, although over all molecule is non polar.
- If back bonding is present then tendency to form dimer or polymer decreases.
 Ex. BF_3 , BeF_2

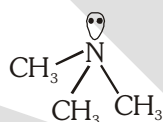
(2) **$p\pi-d\pi$ back bonding** :



it is used to explain following observations :

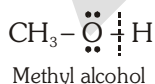
(a) **Hybridisation**

Ex. Trimethyl amine
 $(CH_3)_3N$



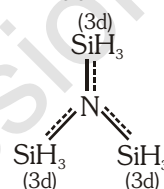
- sp^3 hybrid (N)
- trigonal pyramidal
- Lewis base (due to presence of lp)

(b) **Acidic strength**

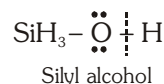


- No back bonding
- Less acidic

Trisilyl amine
 $(SiH_3)_3N$



- sp^2 hybrid (N)
- trigonal planar
- Not Lewis base
- Bond angle increases



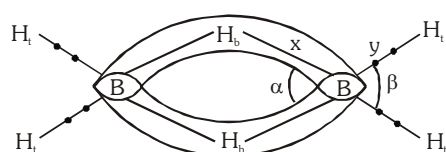
- Back bonding present in conjugate base
- More acidic

2. DIMERISATION / POLYMERISATION

Types of Dimerisation	
3C—2e	3C—4e
B ₂ H ₆	(BeCl ₂) ₂
(BeH ₂) ₂	(BeCl ₂) _n
(BeH ₂) _n	(AlCl ₃) ₂
Al ₂ (CH ₃) ₆	(ICl ₃) ₂
Ga ₂ (CH ₃) ₆	

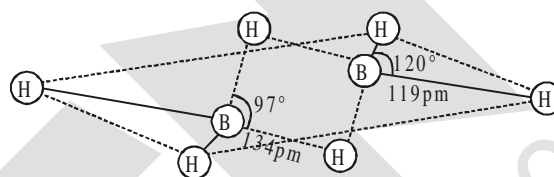
(A) By banana Bond or by 3C—2e bond or by e⁻ deficient bond

(a) B₂H₆



$$\alpha < \beta = \text{BA}$$

$$x > y = \text{BL}$$



The structure of diborane, B₂H₆

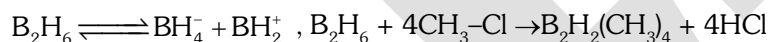
$$2\text{C} - 2e^- \text{ bonds} = 4$$

$$3\text{C} - 2e^- \text{ bonds} = 2$$

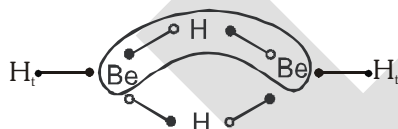
Hybridisation state = sp³

Electron deficient molecule (act as Lewis acid)

- All four terminal hydrogen and two boron atoms are present in same plane both bridging H are present in perpendicular plane.
- If substitution reaction takes place than only four terminal hydrogen atom will be substituted.



(b) (BeH₂)₂ (dimer of BeH₂ in vapour state)

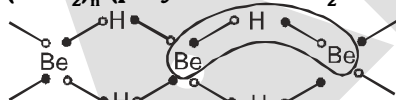


Hybridisation state = sp²

Planar

electron deficient molecule

(c) (BeH₂)_n (polymer of BeH₂ in solid state)



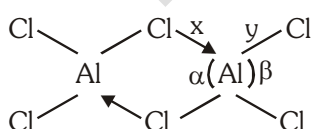
Hybridisation state = sp³

Non-planar

electron deficient molecule

(B) By-coordinate Bond / 3C-4e bond

(a) Al₂Cl₆ (dimer of AlCl₃ in liquid or solid state)



$$\text{BA} = \alpha < \beta$$

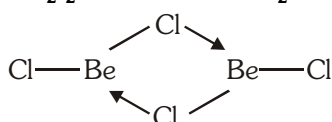
$$\text{BL} = x > y$$

Hybridisation state = sp³

Non-planar

octet complete

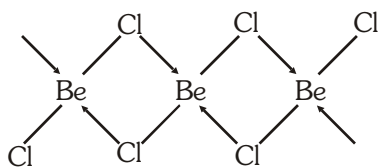
(b) (BeCl₂)₂ (dimer of BeCl₂ in vapour state)



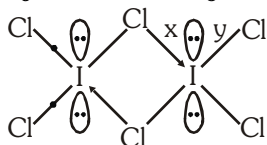
Hybridisation state = sp²

Planar

electron deficient molecule

(c) $(\text{BeCl}_2)_n$ (polymer of BeCl_2 in solid state)

Hybridisation state = sp^3
Non-planar
octet complete

(d) I_2Cl_6 (dimer of ICl_3)

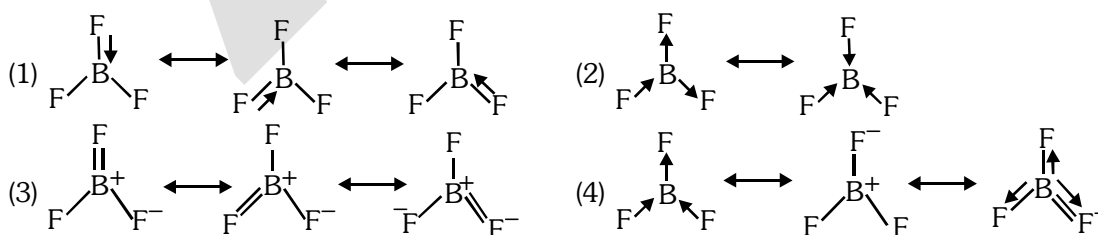
Hybridisation state = sp^3d^2
Planar
super octet complete (12 electrons)

Note :

- BCl_3 , BBr_3 & BI_3 do not form dimer due to smaller size of boron & large size of halogen (due to more steric repulsion)
- BF_3 cannot form dimer due to its back bonding.
- AlF_3 cannot form dimer due to its ionic nature.

BEGINNER'S BOX-1

- In BF_3 :
 - B-F bond has some double bond character and this bond is delocalised
 - All the B-F bonds are single covalent in nature
 - Bond energy and bond length of B-F bond indicate its single bond character
 - All the bonds are ionic
- Which of the following statements is incorrect in the context of the B-F bond in BF_3 :-
 - All the three B-F bond lengths are equal and each of them is shorter than the sum of the covalent radii of boron and fluorine.
 - The bond energy of the B-F bond is very high, higher than for any other single bond
 - The unusual shortness and strength of the B-F bond may be explained by a $\pi\pi$ - $\pi\pi$ interaction between boron and fluorine atoms.
 - The unusual shortness and strength of the bonds may be explained by a $\pi\pi$ - $d\pi$ interaction between the atoms of boron and fluorine.
- In BF_3 , the B-F bond length is $1.30 \text{ \AA}$, when BF_3 is allowed to be treated with Me_3N , it forms an adduct, $[\text{Me}_3\text{N} \rightarrow \text{BF}_3]$. The bond length of B-F in the adduct is :
 - Greater than $1.30 \text{ \AA}$
 - Smaller than $1.30 \text{ \AA}$
 - Equal to $1.30 \text{ \AA}$
 - None of these
- Which of the following structures correctly represents the boron trifluoride molecule :



- Trisilylamine $[\ddot{\text{N}}(\text{SiH}_3)_3]$ has a
 - Planar geometry
 - Tetrahedral geometry
 - Pyramidal geometry
 - None of these

6. In which of the following molecule, vacant orbitals do not participate in bonding:-
 (1) B_2H_6 (2) Al_2Cl_6 (3) $[H_3N.BF_3]$ (4) Si_2H_6
7. Which of the following is not possible due to back bonding.
 (1) State of hybridisation may change (2) Bond order increases
 (3) Bond angle always increases (4) Lewis acidic strength decreases
8. In which of the following compounds B-F bond length is shortest?
 (1) BF_4^- (2) $BF_3 \leftarrow NH_3$ (3) BF_3 (4) $BF_3 \leftarrow N(CH_3)_3$
9. In diborane –
 (1) 2 bridged hydrogen and four terminal hydrogen are present
 (2) 3 bridged and three terminal hydrogen are present
 (3) 4 bridged hydrogen atoms are not in the same plane in diborane
 (4) 1 bridged hydrogen and 1 terminal hydrogen are present
10. Which of the following statements is correct in the context of diborane (B_2H_6)?
 (1) There are 12 valence electrons-three from each of the two boron atoms and six from the six hydrogen atoms
 (2) Two of the six hydrogen atoms form two bridges between two boron atoms
 (3) The two bridging hydrogen atoms are in a plane perpendicular to the rest of the molecule and prevent rotation between the two boron atoms
 (4) All of these

3. INERT PAIR EFFECT

	Group-13	Group-14	Group-15
(group oxidation state)	$\frac{ns^2np^1}{+3}$	$\frac{ns^2np^2}{+4}$	$\frac{ns^2np^3}{+5}$

- While moving down the group the stability of lower oxidation state (2 less than group oxidation state) progressively increases, for the last element of the group the stability of lower oxidation state becomes even greater than the group oxidation state. **This is called inert pair effect.**
- For the last element group oxidation state is highly oxidising in nature.

Group-13		Group-14		Group-15	
B	+3	C	+4	N	+5
Al	+3	Si	+4	P	+5
Ga	+3 > +1	Ge	+4 > +2	As	+5 > +3
In	+3 > +1	Sn	+4 > +2	Sb	+5 > +3
Tl	+3 < +1	Pb	+4 < +2	Bi	+5 < +3

Reason : As we move down the group there is presence of d & f-orbitals in inner shells which have poor shielding effect hence Z_{eff} increases. As a result the ns^2 electron pair becomes more and more tightly held to the nucleus and becomes reluctant to participate in bonding.

Examples :

- $PbCl_2$ is more stable than $PbCl_4$.
- $TlCl$ is more stable than $TlCl_3$.
- $GaCl_3$ is more stable than $TlCl_3$.
- $SnCl_4$ is more stable than $PbCl_4$.
- Thallium (III) iodide does not exist.
- PbI_4 does not exist.
- Only BiF_5 exists but $BiCl_5$ does not exist ?

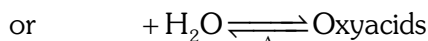


4. OXY ACID

General Formula of Oxy acid : $\text{H}_x\text{Z}_y\text{O}_z$ (Z = non metal)

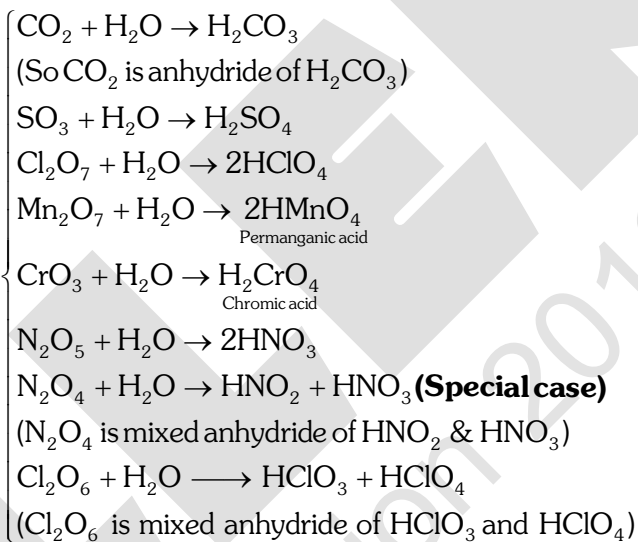
- Compounds which contain Z-OH bond are known as oxyacids.

- Acidic oxide



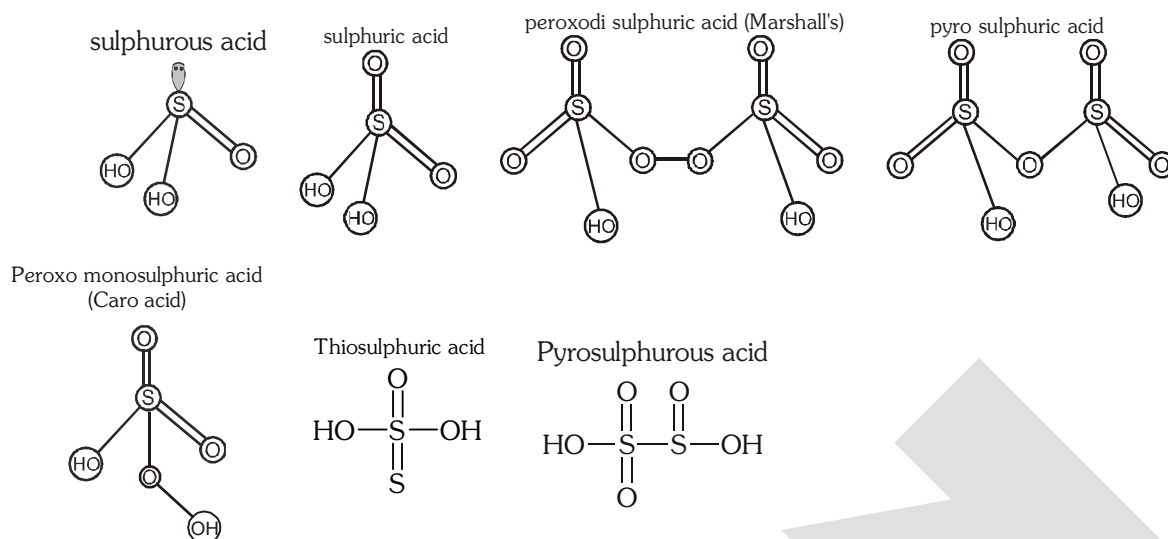
Non metallic oxide
 Anhydrides of oxyacids

Anhydrides of their
 corresponding oxyacids

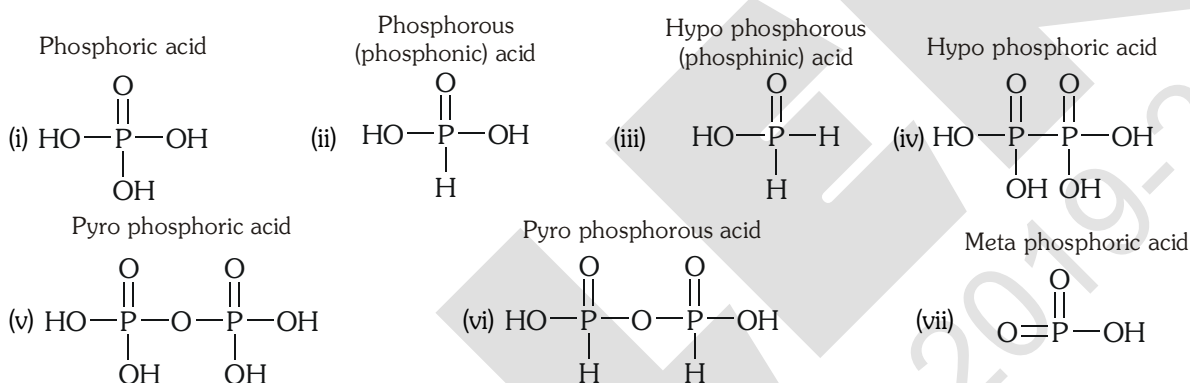


- Oxidation state of central atom does not change.

Pyro	Meta	Per oxy acid	Hypo	
2 mole oxy acid ↓ -H ₂ O ↓ pyro acid	1 mole oxy acid ↓ -H ₂ O ↓ meta acid	↓ Z-O-O-H	Ous acid ↓ -O Hypo us acid	ic acid ↓ -O Hypo ic acid
Ex. $\text{H}_2\text{S}_2\text{O}_5$ $\text{H}_2\text{S}_2\text{O}_7$ $\text{H}_4\text{P}_2\text{O}_7$	Ex. HPO_3 HBO_2	Ex. H_2SO_5 $\text{H}_2\text{S}_2\text{O}_8$ HNO_4 $\text{CH}_3\text{CO}_3\text{H}$ $\text{C}_6\text{H}_5\text{CO}_3\text{H}$	Ex. HOX H_3PO_2	Ex. $\text{H}_4\text{P}_2\text{O}_6$



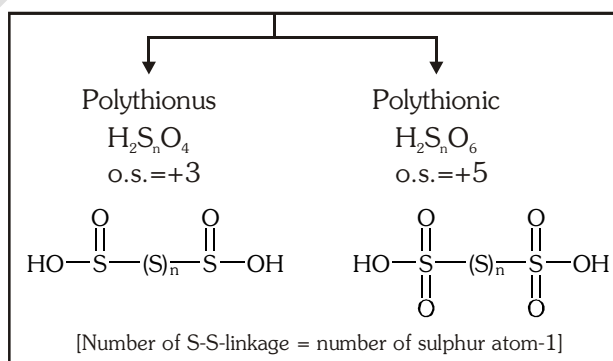
OXY ACID OF PHOSPHORUS

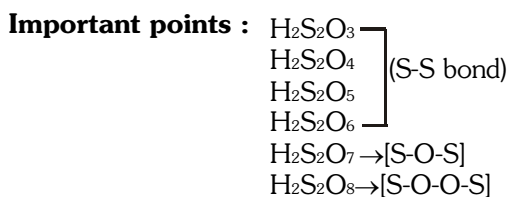


OXY ACIDS OF HALOGENS

Hypohalous	Halous	Halic	Per halic
HO—X	O=X—OH	HO—X(=O) ₂	HO—X(=O) ₃
+1 HOCl HOBr HOI	+3 HClO ₂ — —	+5 HClO ₃ HBrO ₃ HIO ₃	+7 HClO ₄ HBrO ₄ HIO ₄

OXY ACIDS CONTAINING S-S LINKAGE

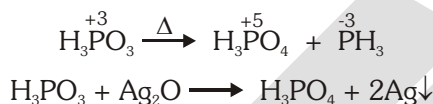
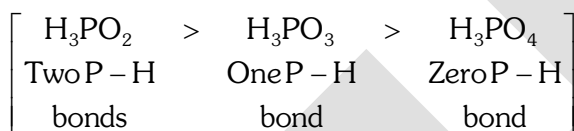


**Trick for drawing structure of oxy compound**

Oxy compounds	Group present
$\text{X}_2\text{O}_6^{-q}$	$[\text{X-X}]$
$\text{X}_2\text{O}_7^{-q}$	$[\text{X-O-X}]$
$\text{X}_2\text{O}_8^{-q}$	$[\text{X-O-O-X}]$

Some important points

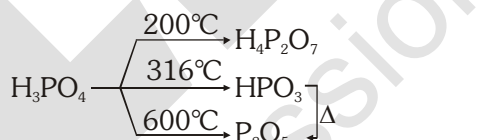
- (1) Oxy acid of phosphorus having +1 & +3 oxidation state (having P-H bond) can act as reducing agent or give disproportionation reaction on heating.

**Order of reducing nature.**

Note : Nitrogen compound having +1 & +3 oxidation state gives disproportion to reaction in acidic medium.



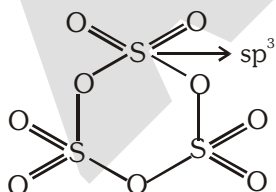
- (2) **Heating effect of phosphoric acid**



- (3) **Heating effect of boric acid**

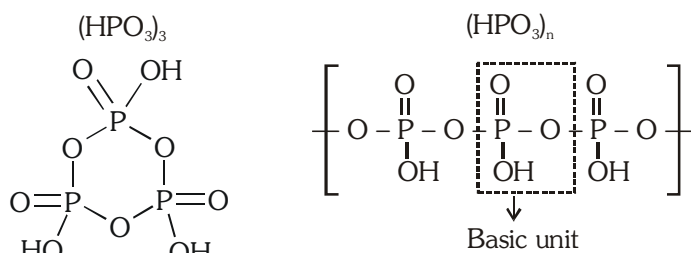


- (4) **SO_3 exists as a cyclic trimer (S_3O_9)**



6 S=O bonds
3 S-O-S bonds
 sp^3 hybrid 'S', Non planar

- (5) **Metaphosphoric acid exists as dimer, cyclic trimer or polymer**



(6) Acidic nature of oxy acids.

General concept :

$$\text{For same elements} \left(\begin{array}{l} \text{Acidic nature} \\ \text{of oxyacids} \end{array} \propto \begin{array}{l} \text{Oxidation number of} \\ \text{active element} \end{array} \right)$$

(Acidic nature of oxyacids $\propto$ EN of active element) for different elements

Exception :

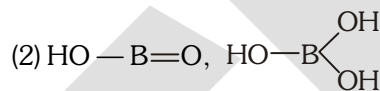
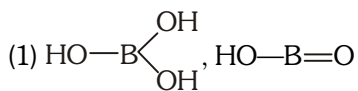
- for oxy acids of P

$$\left(\begin{array}{l} \text{Acidic nature} \propto \frac{1}{\text{Oxidation number}} \end{array} \right)$$

Example : $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_4$ (acidic order)

BEGINNER'S BOX-2

1. Structures of metaboric acid and orthoboric acid respectively are :



(3) Both the above

(4) None

2. Orthoboric acid –

(1) Donate proton to form H_2BO_3^-

(2) Accept proton of form H_4BO_3^+

(3) Donate OH^- to form H_2BO_2^+

(4) Accept OH^- to form $[\text{B}(\text{OH})_4]^-$

3. Pick out the incorrect statement :-

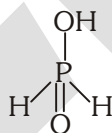
(1) Orthophosphorus acid can be obtained by reacting P_4O_6 with H_2O

(2) Orthophosphoric acid can be obtained by reacting P_4O_{10} with H_2O

(3) Pyrophosphoric acid can be obtained by heating Orthophosphorus acid

(4) Metaphosphoric acid is obtained by the dehydration of orthophosphoric acid at 316°C .

4. Hypophosphorus acid H_3PO_2 is –



(1) Tribasic acid

(2) Dibasic acid

(3) Monobasic acid

(4) Not acidic at all

5. The correct order of decreasing acid strength of oxy acids of group 15 elements is :

(1) $\text{HNO}_3 > \text{H}_3\text{SbO}_4 > \text{H}_3\text{AsO}_4 > \text{H}_3\text{PO}_4$

(2) $\text{H}_3\text{PO}_4 > \text{H}_3\text{AsO}_4 > \text{H}_3\text{SbO}_4 > \text{HNO}_3$

(3) $\text{HNO}_3 > \text{H}_3\text{PO}_4 > \text{H}_3\text{AsO}_4 > \text{H}_3\text{SbO}_4$

(4) $\text{HNO}_3 > \text{H}_3\text{AsO}_4 > \text{H}_3\text{PO}_4 > \text{H}_3\text{SbO}_4$

6. Which one of the following is a mixed anhydride :

(1) NO

(2) NO_2

(3) N_2O_3

(4) N_2O_5

7. Which of the following oxy acids of sulphur contains a sulphur-sulphur single bond :

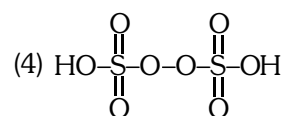
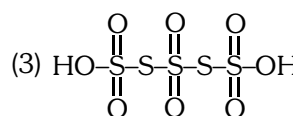
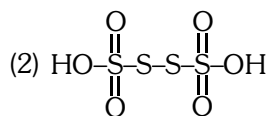
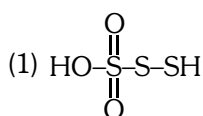
(1) $\text{H}_2\text{S}_2\text{O}_6$

(2) $\text{H}_2\text{S}_2\text{O}_7$

(3) $\text{H}_2\text{S}_2\text{O}_8$

(4) $\text{H}_2\text{S}_2\text{O}_3$

8. The structure of peroxodisulphuric acid is :



9. Number of S-S bond in $\text{H}_2\text{S}_n\text{O}_6$

(1) n

(2) (n-1)

(3) (n-2)

(4) (n+1)

10. Ga^+ acts as a reducing agent because –

(1) Ga^{3+} state is less stable than Ga^{+1}

(2) Ga^{3+} state is more stable than Ga^{+1}

(3) Ga^{3+} convert into Ga^{+1} reducing

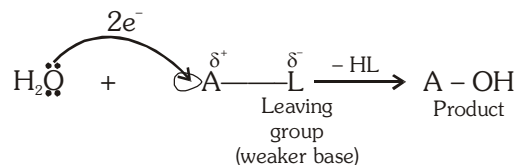
(4) None of the above

5. HYDROLYSIS

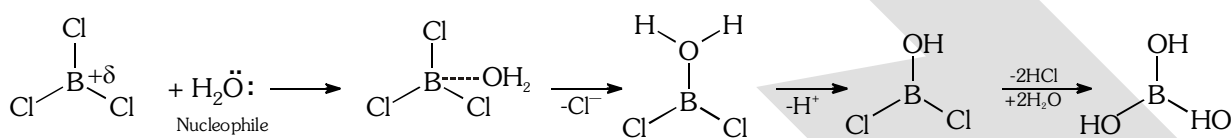
Chemical reaction of H_2O with a covalent compound.

Condition :

One atom must possess vacant orbital and positive charge.

Mechanism : (SN^2 mechanism)

- Weaker base goes out and a stronger base substitutes it.

Example

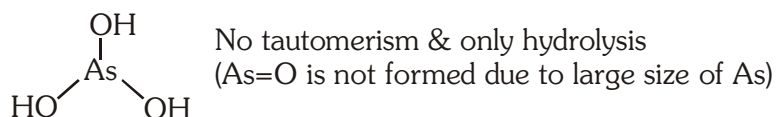
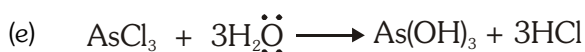
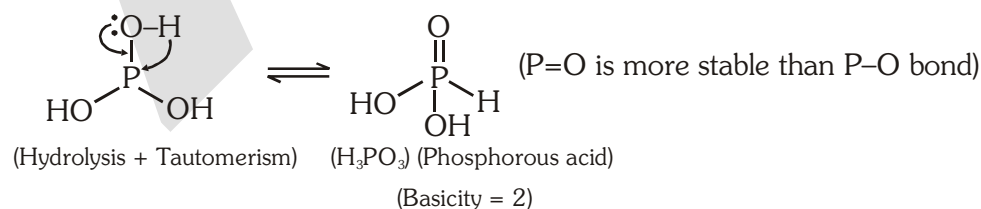
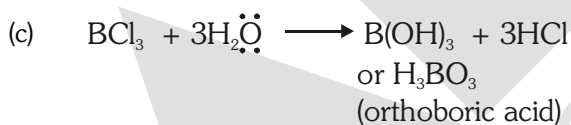
- For hybridisation of transition state

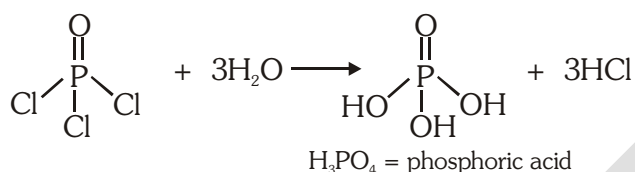
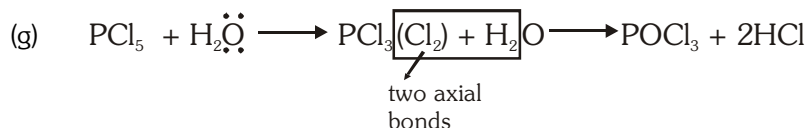
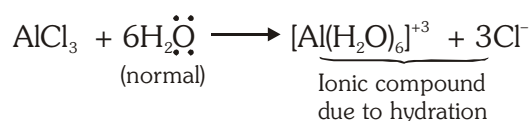
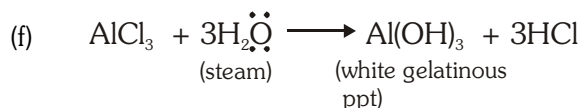
Reactant $\longrightarrow$ Transition state

Ex. : $sp \longrightarrow sp^2$

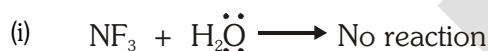
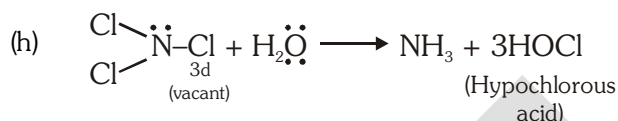
$sp^2 \longrightarrow sp^3$

$sp^3 \longrightarrow sp^3d$

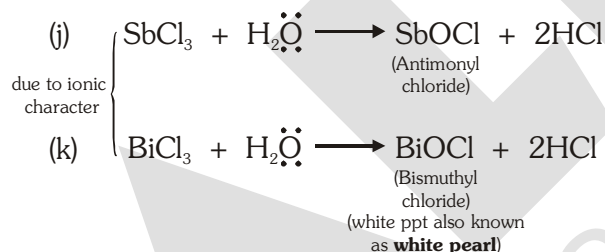
Important examples of hydrolysis**1. Hydrolysis of some important covalent molecules.**



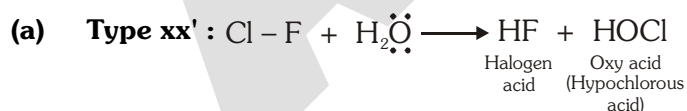
• **Hydrolysis is not a redox reaction**



• **Partial hydrolysis**

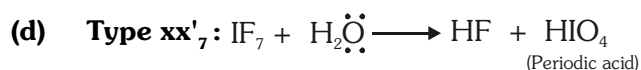
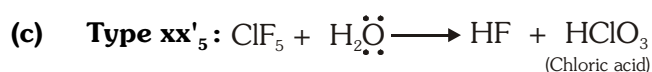
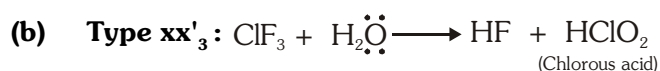


2. Hydrolysis of interhalogen compounds

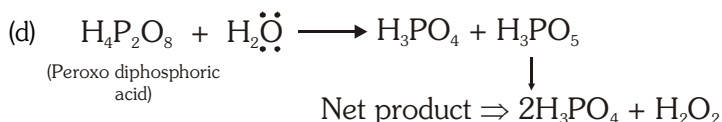
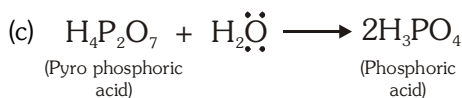
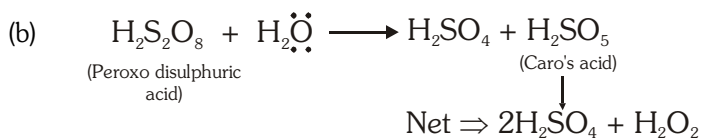
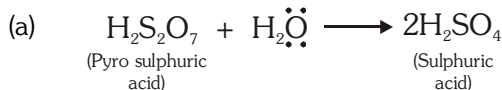


x = oxy acid (same oxidation number)

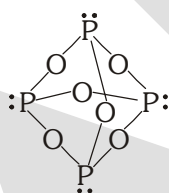
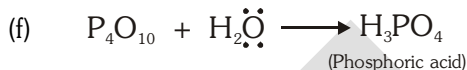
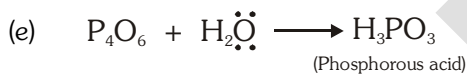
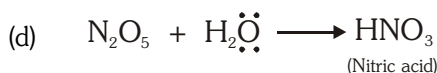
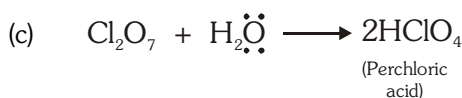
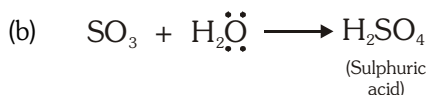
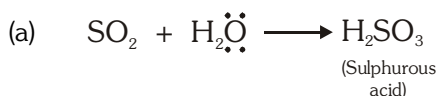
x' = Halogen acid



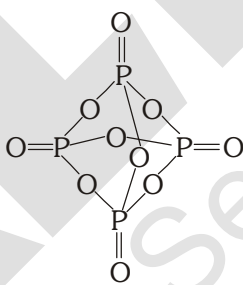
3. Hydrolysis of oxyacids



4. Hydrolysis of oxides (N, P, S, Cl)

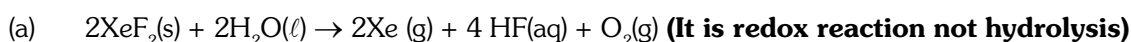


P_4O_6
 $\rightarrow sp^3$ hybrid 'P'
 $\rightarrow 6 \text{ P-O-P bonds}$
 $\rightarrow \text{lp on 'P'} = 4$

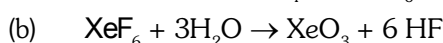


P_4O_{10}
 $\rightarrow sp^3$ hybrid 'P'
 $\rightarrow 6 \text{ P-O-P bonds}$
 $\rightarrow 4 \text{ P=O bonds}$

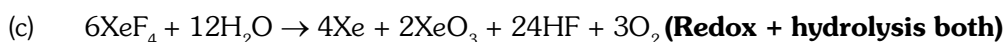
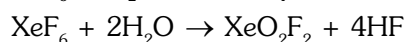
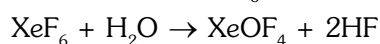
5. Hydrolysis of Xe-fluoride



Hydrolysis of XeF_4 and XeF_6 with water gives XeO_3



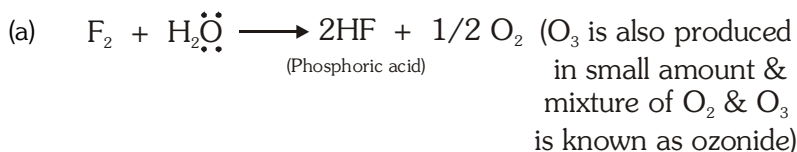
Partial hydrolysis of XeF_6 gives oxyfluorides, XeOF_4 and XeO_2F_2 .



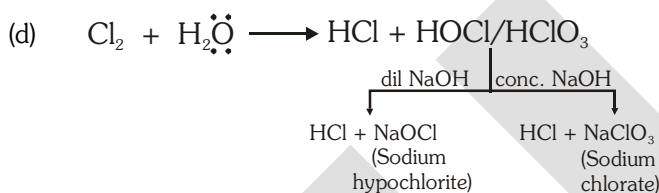
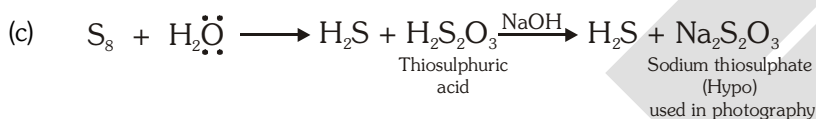
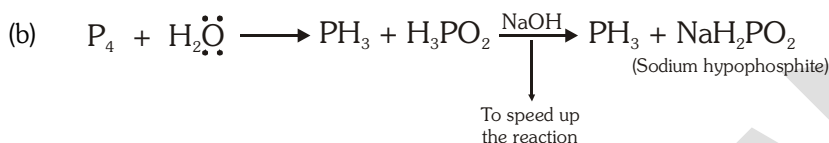
6. Hydrolysis of ionic compounds

- (a) Carbides (b) Nitrides (c) Phosphides

7. Hydrolysis of non-metals and halogen



- 'F₂' is the only non-metal that can oxidise H₂O as it is a powerful oxidising agent. Other halogens & non-metals undergo disproportionation with water in basic medium. (Redox reaction)



BEGINNER'S BOX-3

1. Which of the following statements is correct :

- (1) Aluminium carbide as well as beryllium carbide produce methane gas on treatment with water
- (2) On reacting with water, calcium carbide (CaC₂) produces acetylene while magnesium carbide (Mg₂C₃) gives propyne
- (3) Both of these
- (4) None of these

2. (a) $Al \xrightarrow{N_2} A$ (b) $Al \xrightarrow{C} B$
- High temp. High temp.

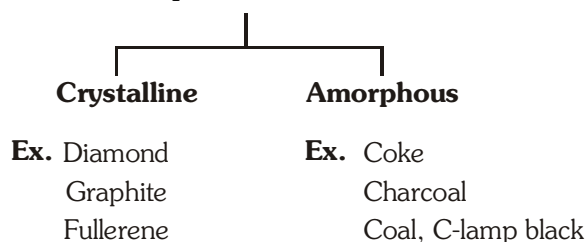
Product A and B on hydrolysis yields respectively.

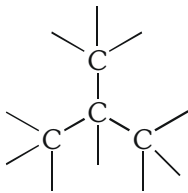
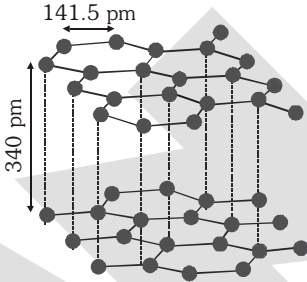
- (1) Ammonia and acetylene
- (2) Ammonia and methane
- (3) Nitric oxide and acetylene
- (4) None

3. SbCl_3 and BiCl_3 on hydrolysis gives :
- (1) Sb^{+3} and Bi^{+3} (2) Sb(OH)_3 and Bi(OH)_3
(3) SbOCl and BiOCl (4) None
4. The hydrolysis of PCl_3 , produces :
- (1) $\text{H}_3\text{PO}_3 + \text{HClO}$ (2) $\text{H}_3\text{PO}_3 + \text{HCl}$
(3) $\text{H}_3\text{PO}_4 + \text{HCl}$ (4) $\text{PH}_3 + \text{HClO}$
5. The number of molecules of water needed to convert one molecule of P_2O_5 into orthophosphoric acid is :
- (1) 2 (2) 3 (3) 4 (4) 5
6. XeF_6 on partial hydrolysis produces :-
- (1) XeF_2 (2) XeOF_4 (3) XeF_4 (4) XeO_3
7. Which of the following halides does not hydrolysed?
- (1) PbCl_4 (2) SiCl_4 (3) CCl_4 (4) SnCl_4
8. which of the following is an uncommon hydrolysis product of XeF_2 and XeF_4 ?
- (1) Xe (2) XeO_3 (3) HF (4) O_2
9. Consider the following reactions :
- (i) $\text{PCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3 + 3\text{HCl}$
(ii) $\text{SF}_4 + 3\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3 + 4\text{HF}$
(iii) $\text{BCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{BO}_3 + 3\text{HCl}$
(iv) $\text{XeF}_6 + 3\text{H}_2\text{O} \rightarrow \text{XeO}_3 + 6\text{HF}$
- Then according to given information the incorrect statement is
- (1) During the (i) reaction the hybridisation of 15th group element does not change
(2) During the (ii) reaction the hybridisation of 16th group element has been change
(3) During the (iii) reaction the hybridisation of 13th group element does not change
(4) During the (iv) reaction the hybridisation of 18th group element does not change
10. Select correct statement about hydrolysis of BCl_3 and NCl_3
- (1) NCl_3 is hydrolysed and gives HOCl but BCl_3 is not hydrolysed
(2) Both NCl_3 and BCl_3 on hydrolysis gives HCl
(3) NCl_3 on hydrolysis gives HOCl but BCl_3 gives HCl
(4) Both NCl_3 and BCl_3 on hydrolysis gives HOCl

6. ALLOTROPES

(A) Allotropes of Carbon



DIAMOND	GRAPHITE
	
Each carbon bonded with four other carbon atom sp^3 hybridisation	Each carbon bonded with three other carbon atom sp^2 hybridisation
tetrahedral structure	hexagonal layer structure
Insulator due to absence of free electrons	Conductor due to presence of delocalised electrons
Hard due to presence of strong sigma bond and 3D structure	Soft due to presence of weak van der Waals forces between two layers
Density = 3.35 gm/cm^3	Density = 2.22 gm/cm^3
High melting point (giant molecule)	Low melting point
Bond length (C—C) = $1.54 \text{ \AA}$	Bond length (C—C) = $1.41 \text{ \AA}$

Special Point :

- Thermodynamically graphite is more stable than diamond.
- Aqueous solution of graphite is known as **aquadag**.
- Graphite is also known as **plumbago (used in lead pencil)**
- Graphite use as a dry lubricants.
- Hardest allotrope** of carbon is diamond, **softest allotrope** is lamp black, **purest allotrope** is fullerene

Fullerene

- C-60 & C-70 are common fullerene.
- C-60 is also known as Buckminster fullerene (Bucky ball)

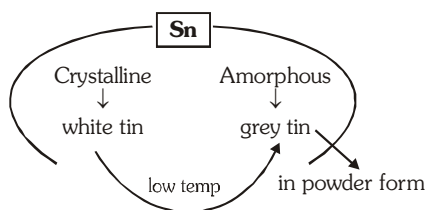
- There are 32 rings $\begin{cases} 12 \text{ pentagonal} \\ 20 \text{ hexagonal} \end{cases}$

- Each carbon atom bonded with 3 other carbon by sigma & double bond (resonance)
- sp^2 hybridisation & aromatic in nature

⚡ Dangling Bond

In diamond or graphite some surface carbon atom have free valency. These carbon atoms form new bonds with impurities. These new bonds are known as dangling bond.

(B) ALLOTROPE OF TIN



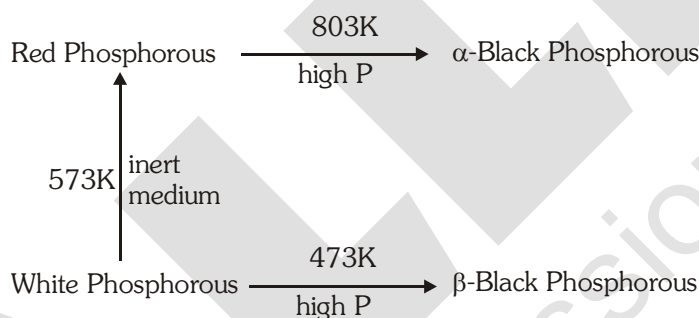
(C) ALLOTROPE OF PHOSPHOROUS

(a) white/yellow phosphorous

(b) Red phosphorous

(c) Black phosphorous

White phosphorous	Red Phosphorous
Waxy solid	Brittle powder
Poisonous	Non poisonous
Soluble in CS_2 , Insoluble in water	Insoluble in water & CS_2
Monomer of P_4	Polymer of P_4
Highly reactive due to bond angle strain	More stable than white phosphorous
It glows in dark due to slow oxidation (phosphorescence)	It does not glow in dark
It gives phosphene (PH_3) on reaction with NaOH	It give hypo phosphoric acid when on reaction with NaOH

Order of stability or MP or density $\rightarrow$ white < red < black

(D) ALLOTROPE OF SULPHUR

I.

S	
Crystalline	Amorphous
Rhombic sulphur (α -S) most stable form	Milk of sulphur Plastic sulphur (γ -S)
Monoclinic sulphur (β -S)	Colloidal sulphur
$\alpha\text{-S} \xrightleftharpoons[\text{<95}^\circ\text{,6}^\circ\text{C}]{\text{>95.6}^\circ\text{C}} \beta\text{-S}$	$\text{H}_2\text{S} + 2\text{HNO}_3 \xrightarrow{\text{Redox}} \text{S} + 2\text{NO}_2 + 2\text{H}_2\text{O}$ RA OA
95.6°C = transition Temp. both are soluble in CS_2 but insoluble in water	

- II. (a) density of $\alpha S > \beta S$
 (b) Both are puckered crown shape having S_8 units
 (c) S_2 is paramagnetic sulphur which exist in vapour form at high temperature.
 (d) S_6 is chair form of S

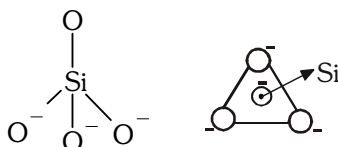
III. Effect of temperature :



7. (A) SILICATES

Basic / structural unit : SiO_4^{-4}

SiO_4^{-4} have a tendency of polymerisation



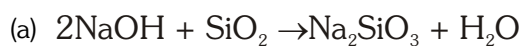
Silicate	No. of shared oxygen per unit	General formula	Structure	Example
Ortho silicate	0	SiO_4^{-4}		Zircon $ZrSiO_4$
Pyro silicate	1	$Si_2O_7^{-6}$		Hemi morphite $Zn_3Si_2O_7 \cdot Zn(OH)_2 \cdot 2H_2O$
Single chain silicate	2	$(SiO_3^{-2})_n$		$LiAlSi_2O_6$ Spodumene
Cyclic silicate	2	$(SiO_3^{-2})_n$		Beryl ($Be_3Al_2Si_6O_{18}$)
Sheet silicate	3	$(Si_2O_5^{-2})_n$		Talc $Mg_3(OH)_2(Si_2O_5)_2$
3-D silicate	4	$(SiO_2)_n$		Silica (SiO_2)

(B) Sodium Zeolite $[Na_2Al_2Si_2O_8]/[Na_2O \cdot Al_2O_3 \cdot 2SiO_2 \cdot xH_2O]$

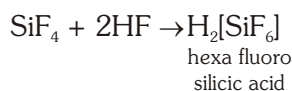
- (i) It is a 3-D silicate
 (ii) It is used in
 (a) For softening of hard water
 (b) For cracking of hydrocarbon & isomerisation

Ex. ZSM-5 (Zeolite) is used to convert ethyl alcohol into petrol.

(C) **SILICA** : Generally silica is inert but it can react with NaOH & HF.



[Na₂SiO₃ known as water glass or water soluble]



- This process is also known as **Etching of glass**.

(D) SILICONES

Silicones are organometallic polymer which contain R₂SiO as a basic unit.

Properties :

- Silicones are chemically inert due to presence of strong sigma bond.
- Silicones have water repelling nature due to presence of alkyl group.
- Silicones are insulator due to absence of free electrons.

Preparation :

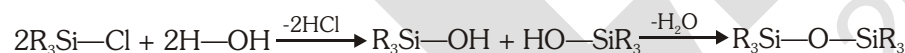
Alkyl chloro silane $\xrightarrow{\text{hydrolysis}}$ product $\xrightarrow{\text{condensation}}$ silicones

R₃SiCl : dimer (used to stop chain formation)

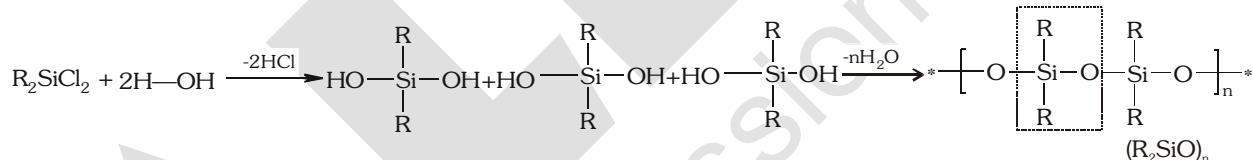
R₂SiCl₂ : linear polymer

RSiCl₃ : cross link polymer

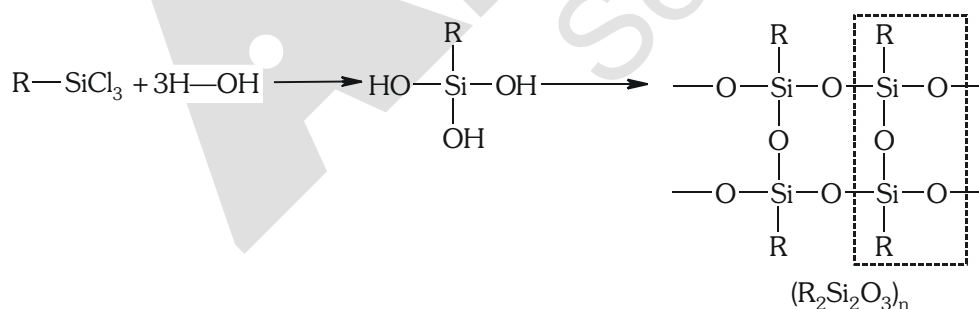
(a) Dimer silicone:



(b) Linear chain silicone



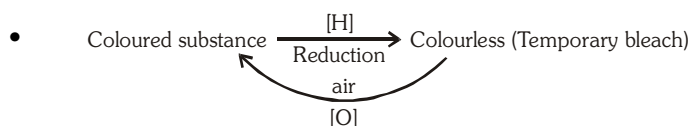
(c) Cross linked silicone



8. BLEACHING AGENTS AND DEHYDRATING AGENTS

(A) BLEACHING AGENTS

- Bleaching can be done by oxidation or reduction.
- Coloured substance $\xrightarrow[\text{Oxidation}]{[\text{O}]}$ Colourless (Permanent bleach)



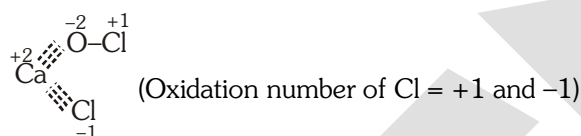
Examples :

By oxidation	By Reduction
(1) O_3 (Dry bleach)	SO_2 (with moisture)
(2) H_2O_2 (with moisture)	
(3) Cl_2 (with moisture)	

- **Bleaching powder** : Used in textile industries. Bleaches by oxidation.

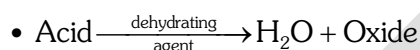
$\text{CaCl}_2 \cdot \text{Ca}(\text{OCl})_2 \cdot \text{Ca}(\text{OH})_2 \cdot 2\text{H}_2\text{O}$ (average oxidation number of Cl = 0)

or

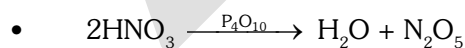
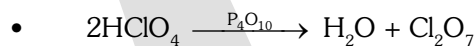
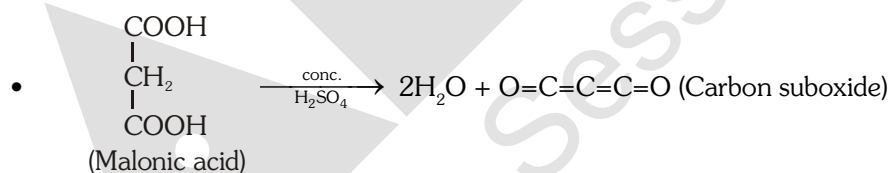
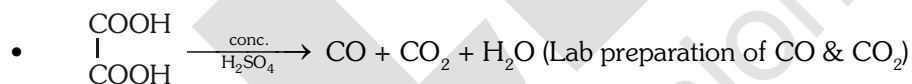
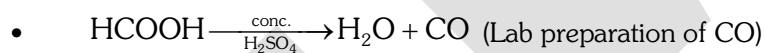


(B) DEHYDRATING AGENTS

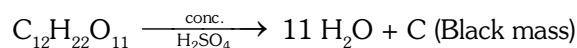
Example : $[\text{P}_4\text{O}_{10}, (\text{Conc.}) \text{H}_2\text{SO}_4, \text{CaO} (\text{Quick lime}), \text{CaCl}_2 (\text{Anhydrous})]$



Example :



- Charring of sugar (dehydration) :



Some extra points :

- NH_3 can not be dried by conc. H_2SO_4 , P_4O_{10} and CaCl_2
- Alcohols, phenols can not be dried by metal hydrides.
- $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ can not be dried by heating alone.

BEGINNER'S BOX-4

- Silicones have the general formula
 (1) SiO_4^{4-} (2) $\text{Si}_2\text{O}_7^{6-}$ (3) $(\text{R}_2\text{SiO})_n$ (4) $(\text{SiO}_3)_n^{2-}$
- Glass or silica soluble in :
 (1) HClO_4 (2) HF (3) Aqua-regia (4) H_2SO_4
- $\text{Si}_2\text{O}_7^{6-}$ anion is obtained when
 (1) no oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
 (2) one oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
 (3) two oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
 (4) three or all four oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
- Consider the following route of reactions
 $\text{R}_2\text{SiCl}_2 + \text{Water} \rightarrow (\text{A}) \xrightarrow{\text{polymerisation}} (\text{B})$
 Compound(B) in above reaction is
 (1) Dimer silicone (2) Linear silicone
 (3) cross linked silicone (4) polymerisation of (A) does not occur
- $(\text{Si}_2\text{O}_5)_n^{2n-}$ anion is obtained when
 (1) no oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
 (2) one oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
 (3) two oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
 (4) three oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron
- What is true about various allotropes of carbon ?
 (1) Diamond is the hardest and graphite is the softest
 (2) Diamond is the hardest and coke is the softest
 (3) Diamond is the hardest and lamp black is the softest
 (4) Coke is the hardest and diamond is the softest
- Different layers in graphite are held together by :
 (1) Ionic bonding (2) Metallic bonding
 (3) Covalent bonding (4) Vander Waals forces
- Bucky ball or buckminster fullerene is :
 (1) An allotrope of carbon
 (2) It is referred as C-60
 (3) It has sp^2 -hybridised nature and resembles with soccer ball
 (4) all of these
- Which of the following statement is correct for allotrops of P :-
 (1) White P is soluble in CS_2 but red P is insoluble in CS_2
 (2) P-P-P bond angle is 60° in red P
 (3) On heating in air, white P change into red P
 (4) White P change in black P at ordinary temp.
- Which of the following structural features of graphite best accounts for its use as a lubricant?
 (1) delocalized electrons
 (2) strong covalent bonds between carbon atoms
 (3) van der Waal's forces between layers
 (4) limited three covalency of carbon

BORON FAMILY (GROUP 13)

INTRODUCTION

The p-block elements are those elements of the periodic table in which the differentiating electron (or the last electron) enters the p-subshell of the last (outermost) shell. Since the maximum capacity of any atomic orbital is of two electrons with opposite spins (Pauli's principle), the maximum number of electrons that can be accommodated in any p-subshell is six. As a result, the p-block of the periodic table consists of six groups: 13 (III A), 14 (IV A), 15 (V A), 16 (VI A), 17 (VIIA) and 18 (zero group). On the basis of electronic configuration, the element He is to be excluded from the p-block as its last electron enters the s-subshell of the last shell.

BORON FAMILY

Boron is a typical non-metal, aluminium is a metal but shows many chemical similarities to boron, and gallium, indium and thallium are almost exclusively metallic in character,

GENERAL CHARACTERISTICS

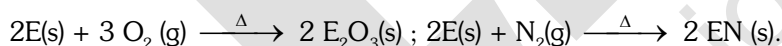
Physical properties :

- Boron is non-metallic in nature.
- It is extremely hard and black coloured solid. It exists in many allotropic forms.
- Due to very strong crystalline lattice, boron has unusually high melting point.
- Rest of the member are soft metals with low melting point and high electrical conductivity. Gallium with low melting point (303 K), could exist in liquid state during summer. Its **high boiling point (2676 K)** makes it a useful material for measuring high temperatures.

Chemical properties

Reactivity towards air

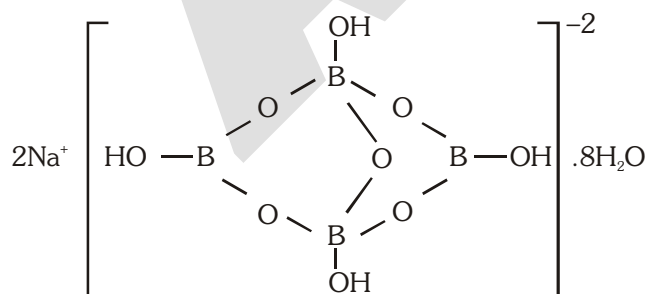
- Boron is unreactive in crystalline form.
- Aluminium forms a very thin oxide layer on the surface which protects the metal from further attack.
- Amorphous boron and aluminium metal on heating in air form B_2O_3 and Al_2O_3 respectively. With dinitrogen at high temperature they form nitrides.



The nature of these oxides varies down the group. Boron trioxide is acidic and reacts with basic (metallic) oxides forming metal borates. Aluminium and gallium oxides are amphoteric and those of indium and thallium are basic in their properties.

COMPOUNDS OF BORON

(I) Borax ($Na_2B_4O_7 \cdot 10H_2O$)

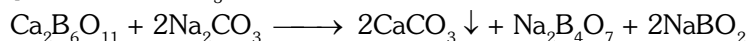


Important points :

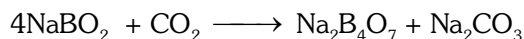
- B-O-B linkage = 5
- Total B-O bonds = 14
- Two Boron- sp^2 hybridised
- Two Boron- sp^3 hybridised

(i) From Colemanite.

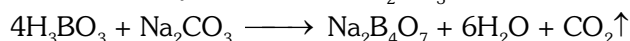
When colemanite powder is heated with Na_2CO_3 solution, the following reaction occurs with the precipitation of CaCO_3 .



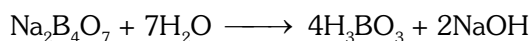
The filtrate is cooled when white crystals of borax are precipitated. The mother liquor on treatment with CO_2 converts NaBO_2 to $\text{Na}_2\text{B}_4\text{O}_7$ which precipitates out on crystallization.

**(ii) From orthoboric acid.**

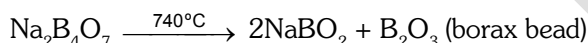
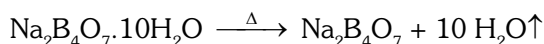
Borax is obtained by the action of Na_2CO_3 on orthoboric acid.

**Properties :**

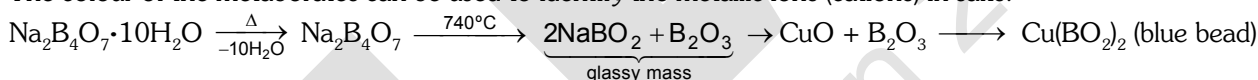
- (i) Borax is a white powder, less soluble in cold water, more soluble in hot water.
- (ii) Its aqueous solution is alkaline because of its hydrolysis to weak acid H_3BO_3 and strong alkali NaOH .

**(iii) Action of heat.**

When borax powder is heated, it first swells due to loss of water in the form of steam but at 740°C it becomes converted into colourless transparent borax bead.

**Borax-bead test :**

Borax reacts with certain metal salts such as, Ni^{2+} , Co^{2+} , Cr^{3+} , Cu^{2+} , Mn^{2+} etc. to form coloured metaborates. The colour of the metaborates can be used to identify the metallic ions (cations) in salts.

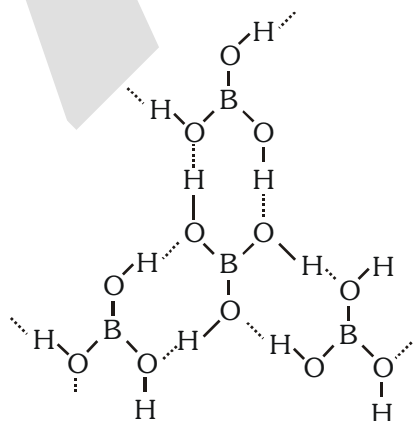


Uses : Borax is used

1. in borax bead test
2. in purifying gold
3. as flux during welding of metals and
4. in production of glass.

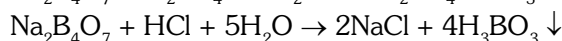
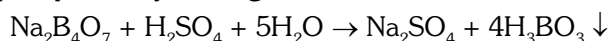
(II) ORTHO BORIC ACID [$\text{H}_3\text{BO}_3/\text{B}(\text{OH})_3$]

Orthoboric acid is greasy to touch less soluble in cold water but more soluble in hot water. In the solid state, the $\text{B}(\text{OH})_3$ units are hydrogen bonded together into two dimensional sheets with almost hexagonal symmetry.

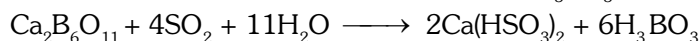


(A) Preparation :

- (i) It is precipitated by treating a concentrated solution of borax with mineral acid.

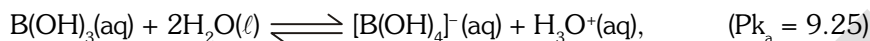


- (ii) From Colemanite: Powdered colemanite is suspended in water and excess SO_2 is passed through it. On filtering and cooling the filtrate, white crystals of H_3BO_3 are obtained.



(B) Properties:

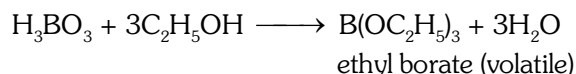
It is a weak monobasic acid soluble in water and in aqueous solution the boron atom completes its octet by accepting OH^- from water molecules:



It, therefore, functions as a Lewis acid and not as a proton donor like most acids.

(C) Test for Borate radical :

When boric acid is heated with ethyl alcohol, the evolved gas is burned forming a green edged flame.



Uses :

1. It is an antiseptic and its water solution is used as an eyewash.
2. It is also used in glass, enamel and pottery industry.

(III) DIBORANE (B_2H_6)

Binary compounds of boron with hydrogen are called boron hydrides or boranes.

Preparation :

- (i) $4\text{BF}_3 + 3\text{LiAlH}_4 \xrightarrow{\text{ether}} 2\text{B}_2\text{H}_6 + 3\text{LiF} + 3\text{AlF}_3$
- (ii) $2\text{NaBH}_4 + \text{I}_2 \xrightarrow{\text{ether}} \text{B}_2\text{H}_6 + 2\text{NaI} + \text{H}_2$
- (iii) $2\text{BF}_3 + 6\text{NaH} \xrightarrow{453\text{ K}} \text{B}_2\text{H}_6 + 6\text{NaF}$ (Industrial method)

Properties :

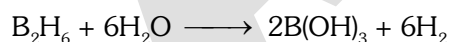
- (i) B_2H_6 is colourless gas and highly reactive (boiling point 183 K).
- (ii) It catches fire spontaneously in air and explodes with O_2 . Reaction with oxygen is extremely exothermic.



Mixtures of diborane with air or oxygen in flame spontaneously producing large amount of heat. Diborane has a higher heat of combustion per unit weight of fuel than most other fuels. It is therefore used as a rocket fuel.

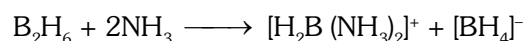
At red-heat the boranes decomposes to boron and hydrogen.

- (iii) Reaction with water is instantaneous.

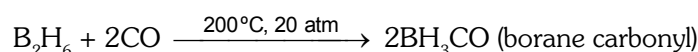


- (iv) The electron deficient 3c-2e B-H-B bridges are sites of nucleophilic attack.

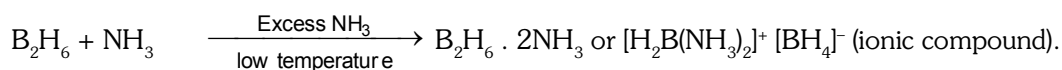
Small amines such as NH_3 , CH_3NH_2 and $(\text{CH}_3)_2\text{NH}$ give unsymmetrical cleavage of diborane.



Large amines such as $(\text{CH}_3)_3\text{N}$ and pyridine give symmetrical cleavage of diborane.



- (v) The reaction with ammonia depends on conditions.

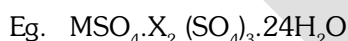


Borazole is much more reactive than benzene.

COMPOUNDS OF ALUMINIUM :

ALUM

- (a) Alums are double sulphates with their general formula $\text{M}_2\text{SO}_4 \cdot \text{M}'_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
where M = monovalent radical like Na^+ , K^+ , NH_4^+ and M' = Trivalent radical like Al^{+3} , Cr^{+3} , Fe^{+3} .
- (b) The different alums are -
- (i) Potash alum - $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 - (ii) Chrome alum - $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$
 - (iii) Iron alum - $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Fe}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 - (iv) Ammonium alum - $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
- (c) In alums each metal ion is surrounded by six water molecules.
- (d) Lithium does not form alum because Li ion is too small to have a coordination number of six.
- (e) **Pseudo alums** : Double sulphates of divalent ions and trivalent ions with 24 water molecules in their crystals are known as **pseudo alums**.



M = divalent or Bivalent ion

X = Trivalent metal ion

Properties :

- (a) It is a white crystalline solid highly soluble in water.
- (b) On heating it undergoes dehydration and swells up.
- (c) It is highly acidic in aqueous solution due to the hydrolysis of aluminium sulphate to sulphuric acid
- (d) It is a double salt and its aqueous solution gives reaction of all the constituents ions K^+ , Al^{+3} , SO_4^{-2} .

Uses :

- (i) Alum is used to stop bleeding.
- (ii) It is used for purification of water.
- (iii) It is used as a mordant in dyeing industry.
- (iv) Alum is used for tanning of leather.

BEGINNER'S BOX-5

- On the addition of mineral acid to an aqueous solution of borax, the compound formed is –
 - (1) Orthoboric acid
 - (2) Boron hydride
 - (3) Metaboric acid
 - (4) Pyroboric acid
- Which of the following can be detected by the borax-bead test ?
 - (1) Ni^{2+}
 - (2) Co^{2+}
 - (3) Pb^{+2}
 - (4) Both (1) & (2)
- A mixture of boric acid with ethyl alcohol burns with green edged flame due to the formation of –
 - (1) Ethyl borax
 - (2) Ethyl borate
 - (3) Methyl borax
 - (4) Methyl borate
- The hydrolysis of borax produces –
 - (1) An acidic medium
 - (2) A basic medium
 - (3) A neutral medium
 - (4) An acidic or an neutral medium
- In alums, each metal ion is surrounded by–
 - (1) Two water molecules
 - (2) Four water molecules
 - (3) Six water molecules
 - (4) Eight water molecules
- Which of the following is false statements :-
 - (1) Boranes are easily hydrolysed
 - (2) LiAlH_4 reduces BCl_3 to borane
 - (3) BH_3 is a Lewis acid
 - (4) All the B–H distances in diborane (B_2H_6) are equal
- Inorganic graphite is :-
 - (1) $\text{B}_3\text{N}_3\text{H}_6$
 - (2) B_2H_6
 - (3) BN
 - (4) BF_3
- By which of the following reactions is borazine prepared ?
 - (1) $\text{B}_2\text{H}_6 + \text{NH}_3 \xrightarrow{\text{excess, low temperature}}$
 - (2) $\text{B}_2\text{H}_6 + \text{NH}_3 \xrightarrow{\text{excess, high temperature}}$
 - (3) $\text{B}_2\text{H}_6 + \text{NH}_3 \xrightarrow[\text{high temperature}]{\text{(ratio } 2\text{NH}_3 : 1\text{B}_2\text{H}_6 \text{)}}$
 - (4) None of the above
- When an inorganic compound (X) having 3c–2e as well as 2c–2e bonds reacts with ammonia gas at a certain temperature, gives a compound (Y), isostructural with benzene. Compound (X) with ammonia at a high temperature produces a substance (Z) :
 - (1) (X) is B_2H_6
 - (2) (Z) is known as inorganic graphite
 - (3) (Y) is $\text{B}_3\text{N}_3\text{H}_6$
 - (4) All of these
- Which one of the following has the lowest m.p.?
 - (1) B
 - (2) Al
 - (3) Ga
 - (4) Tl

CARBON FAMILY (GROUP 14)

PHYSICAL PROPERTIES

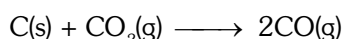
- All group 14 members are solids.
- Carbon and silicon are non-metals, germanium is metalloid whereas tin and lead are soft metals with low melting points.
- Melting points and boiling points of group 14 elements are much higher than those of corresponding elements of group 13 due to stronger metallic bonding.

COMPOUNDS OF CARBON

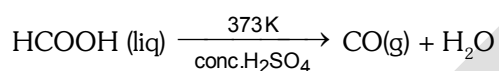
(A) Carbon Monoxide (CO)

(I) Preparation

- (i) It is formed together with CO_2 , when carbon or carbonaceous matter is oxidized by air or oxygen. It is also produced when CO_2 is reduced by red-hot carbon; this reaction is of importance in metal extractions.



- (ii) In the laboratory it can be prepared by dehydrating methanoic acid with concentrated sulphuric acid.



(II) Physical Properties

- Carbon monoxide is a colourless, odourless gas which burns in air with a blue flame, forming CO_2 .
- It is sparingly soluble in water and is a neutral oxide.
- CO is toxic, because it forms a complex with haemoglobin in the blood and this complex is more stable than oxy-haemoglobin. This prevents the haemoglobin in the red blood corpuscles from carrying oxygen round the body. This causes oxygen deficiency, leading to unconsciousness and then death.
- It acts as good reducing agents for all metal oxide.

(B) CARBON DIOXIDE (CO_2)

Preparation :

- Complete combustion of carbon containing compounds.

$$\text{CH}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$$
- In the laboratory it can be conveniently made by the action of dilute hydrochloric acid on marble chips :

$$\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$$
- Industrially it is produced as a by-product during the manufacture of quicklime and in fermentation processes:

$$\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g}) ; \quad \text{C}_6\text{H}_{12}\text{O}_6(\text{aq})\{\text{glucose}\} \longrightarrow 2\text{C}_2\text{H}_5\text{OH}(\text{aq}) + 2\text{CO}_2(\text{g})$$

Properties :

- It is a colourless, odourless and heavy gas.
- Carbon dioxide is the acid anhydride of carbonic acid, which is a weak dibasic acid and ionises in two steps as follows :

$$\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$$

$$\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$$

$$\text{H}_2\text{CO}_3 / \text{HCO}_3^- \text{ buffer system helps to maintain pH of blood between 7.26 to 7.42.}$$

A solution of carbonic acid in water will slowly turn blue litmus red and when the solution is boiled, all the CO_2 is evolved.
- Carbon dioxide readily reacts with alkalis forming the carbonate and, if CO_2 is in excess, the hydrogen carbonate. This is the basis of the lime-water test for CO_2 gas.

$$\text{Ca(OH)}_2(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{liq}) ; \quad \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{liq}) + \text{CO}_2(\text{g}) \rightarrow \text{Ca(HCO}_3)_2(\text{aq})$$

The above reaction accounts for the formation of temporarily hard water.
- Gaseous CO_2 is extensively used to carbonate soft drinks. Being heavy and non-supporter of combustion it is used as fire extinguisher. A substantial amount of CO_2 is used to manufacture urea.

(C) Carbides :

On the basis of type of bonding carbides are of three types :-

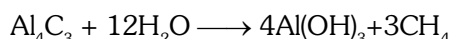
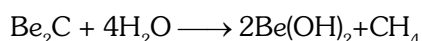
- (I) Salt like carbides (ionic carbides)
- (II) Covalent carbides (giant molecular carbides)
- (III) Interstitial carbides.

(I) Salt like carbides (ionic carbides) :

- (a) These are formed by strong electropositive elements of groups 1, 2 and 13
- (b) These are transparent crystalline substances and do not conduct electric current in the solid state.
- (c) These are easily hydrolysed by water or dilute acids to give aliphatic hydrocarbons,

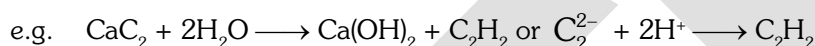
(i) **Methanides (Methides)** – Methanides are the carbides which give methane on hydrolysis.

They contain C^{4-} ions.



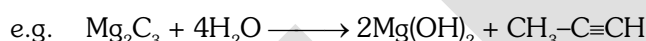
(ii) **Acetylides** – These are the carbides which yield acetylene on hydrolysis.

They contain the ion C_2^{2-} .



(iii) **Allylides** – These carbides give methylacetylene or allylene ($CH_3-C\equiv CH$) on hydrolysis.

They contain C_3^{4-} ions.



(II) Covalent carbides (giant molecular carbides)

- (a) Covalent carbides are the compounds of carbon with elements having similar electronegativity.
- (b) These are covalent, polymeric, thermally stable, extremely hard, high melting and technically inert solid.
- (c) B_4C and SiC are two main covalent carbides.

(III) Interstitial carbides.

Transition metals like Ti, Zr, Hf, V, Nb, Ta and Mo form these carbides. The metals retain their **close packed lattice** and carbons are incorporated in their interstitial spaces. These carbides are extremely hard and have very high melting points. The carbides of V and Ta are very hard hence they are used for manufacture of high speed cutting tools.

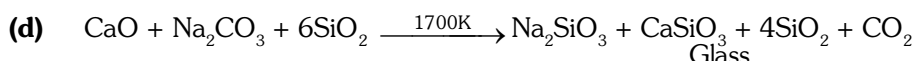
GLASS

- (a) Glass is a hard, brittle transparent, amorphous solid or supercooled liquid made up of silicates of Na, Ca and other metals like Pb, Zn etc.
- (b) Being amorphous solid (not a true solid) it has no sharp melting and melts at high temperature.
- (c) Glass has no definite chemical formula, however it may be represented as :



where M = monovalent metals (eg : Na, K)

M' = bivalent metals Ca, Pb, Zn etc. x and y are integers.



Different Types of Glass

- (i) **Soda or Soft glass or Sodaline glass** : It is common glass. Contains Na & Ca metal carbonate. Used for formation of tube, window.
- (ii) **Potash or Hard glass** : It is a mixture of potassium and calcium silicates. It fuses with difficulty. It is used for making hard glass apparatus.

- (iii) **Flint glass** : It is mixture of potassium and lead silicates. It has a very high refractive index and is used for making electric bulbs and optical instruments.
- (iv) **Crooke's glass** : It is a optical glass containing CeO_2 which prevents the entry of UV rays.
- (v) **Pyrex glass or borasil or borosilicate glass** : It is used to make lab appliances as it is resistant to heat, shock and common reagents. It is a mixture of zinc and barium boro silicates and alumina silicate.

Small part of SiO_2 is replaced by boric oxide/borax.

Low coefficient and thermal expansion, can with stand sudden change in temperature so used for making lab glass wares.

SOME GASEOUS FUELS

- | | |
|---|---|
| (a) Water gas or syn gas = $\text{CO} + \text{H}_2$ | (b) Producer gas = $\text{CO} + \text{N}_2$ |
| (c) Semi water gas = $\text{CO} + \text{N}_2 + \text{H}_2$ | (d) Natural gas = CH_4 |
| (e) Coal gas = $[\text{CO}_2 + \text{CO} + \text{H}_2 + \text{CH}_4]$ | |

BEGINNER'S BOX-6

- Carbon has no tendency to form complex compounds because of :
 - (1) Its small size
 - (2) The availability of vacant d-orbitals
 - (3) Non availability of vacant d-orbitals
 - (4) No tendency to form covalent bonds
- Which of the following statement is false :
 - (1) Dry ice is solid CO_2
 - (2) CO_2 is weakly acidic
 - (3) CO_2 and SiO_2 are linear molecules
 - (4) CO_2 is a gas while SiO_2 is solid
- CO is :
 - (1) Reductant
 - (2) Poisonous gas
 - (3) Neutral to litmus
 - (4) all
- Which is/are true statements :
 - (1) Diamond is unaffected by conc. acids but graphite reacts with hot conc. HNO_3 forming mellitic acid, $\text{C}_6(\text{COOH})_6$
 - (2) CO is toxic because it forms a complex with haemoglobin in the blood
 - (3) In C_3O_2 , all carbons are sp hybridised
 - (4) All are true statements

NITROGEN FAMILY (GROUP 15)

Group 15 includes nitrogen phosphorus, arsenic, antimony and bismuth.

PHYSICAL PROPERTIES:

- Dinitrogen is a diatomic gas while all others are solids.
- Except nitrogen, all the elements show allotropy.
- As we go down the group, there is a shift from non-metallic to metallic through metalloid character. Nitrogen and phosphorus are non-metal, arsenic and antimony are metalloid and bismuth is a typical metal.

CHEMICAL PROPERTIES

Oxidation States and trends in a chemical reactivity :

- The common oxidation states of these elements are -3 , $+3$ and $+5$. The tendency to exhibit -3 oxidation state decreases down the group, bismuth hardly forms any compound in -3 oxidation state.
- Nitrogen also exhibits $+1$, $+2$, $+4$ oxidation states when it reacts with oxygen. Phosphorus also shows $+1$ and $+4$ oxidation states in some oxyacids.
- In the case of nitrogen, all oxidation states from $+1$ to $+4$ tend to disproportionate in acid solution.
For example, $3 \text{HNO}_2 \longrightarrow \text{HNO}_3 + \text{H}_2\text{O} + 2 \text{NO}$
- Similarly, in case of phosphorus nearly all intermediate oxidation states disproportionate into $+5$ and -3 both in alkali and acid. However $+3$ oxidation state in case of arsenic, antimony and bismuth become increasingly stable with respect to disproportionation.

Reactivity towards hydrogen :

- All the elements of Group 15 form hydrides of the type EH_3 where $\text{E} = \text{N}, \text{P}, \text{As}, \text{Sb}$ or Bi .
- The stability of hydrides decreases from NH_3 to BiH_3 . Consequently the reducing character of the hydrides increases. Ammonia is only a mild reducing agent while BiH_3 is the strongest reducing agent amongst all the hydrides. Due to weaker Bi-H bond.

Reactivity towards oxygen :

- All these elements form two types of oxides : E_2O_3 and E_2O_5 . The oxides of the type E_2O_3 of nitrogen and phosphorus are purely acidic, that of arsenic and antimony amphoteric and those of bismuth is predominantly basic.

Reactivity towards halogens :

- These elements react to form two series of halides : EX_3 and EX_5 .
- All the trihalides of these elements except those of nitrogen are stable. In case of nitrogen, only NF_3 is known to be stable. Trihalides except BiF_3 are predominantly covalent in nature.

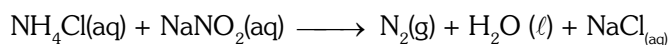
COMPOUND OF NITROGEN

1. NITROGEN GAS (N_2) :

Nitrogen is an important and essential constituent of proteins and amino acids. Nitrates and other nitrogen compounds are extensively used in fertilizers and explosive.

(a) Preparation :

(i) Laboratory method of preparation :

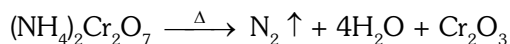


N_2 is collected by the downward displacement of water.

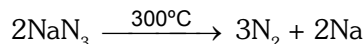
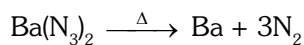
This reaction takes place in two steps as given below :



- (ii)
- By heating ammonium dichromate :**



- (iii)
- Very pure nitrogen can be obtained by heating sodium or barium azide.**



- (iv)
- Industrial methods of preparation :**

From liquified air by fractional distillation : The boiling point of N_2 is -196°C and that of oxygen is -183°C and hence they can be separated by fractional distillation of air.

(b) Properties :

- (i) N_2 is a colourless, odourless gas very less soluble in water. It is neither combustible nor a supporter of combustion.
- (ii) **Reaction with oxygen:** Dinitrogen combines with dioxygen only at very high temperature (at about 2000K) to form nitric oxide.

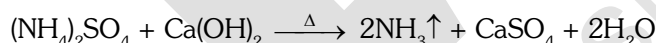
This reaction is endothermic. $\text{N}_2 + \text{O}_2 \longrightarrow 2\text{NO}$

(c) Uses :

- (i) For providing an inert atmosphere during many industrial processes where presence of air or O_2 is to be avoided.
- (ii) For manufacture of NH_3 by the Haber's process.
- (iii) Liquid dinitrogen is used as a refrigerant to preserve biological materials, food items & cryosurgery.

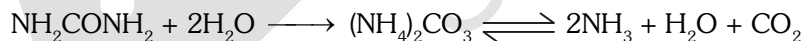
2. AMMONIA (NH_3)**(a) Preparation :**

- (i) By the action of any base or alkali on any ammonium salt :

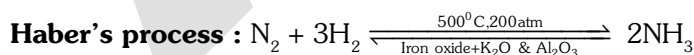


This is a general method and is used as a test for ammonium salts.

Ammonia is present in small quantities in air and soil where it is formed by the decay of nitrogenous organic matter e.g., urea.



- (ii)
- Industrial methods of preparation :**

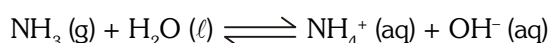


In accordance with Le Chatelier's principle, high pressure would favour the formation of ammonia. The optimum conditions for the production of ammonia are a pressure of $200 \times 10^5 \text{ Pa}$ (about 200 atm), a temperature of $\sim 700 \text{ K}$ and the use of a catalyst such as iron oxide with small amounts of K_2O and Al_2O_3 to increase the rate of attainment of equilibrium.

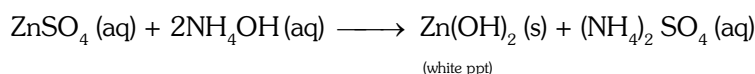
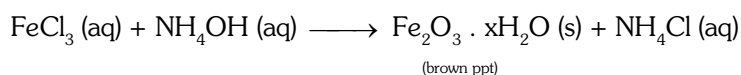
- (b) Physical properties :**
- Ammonia is a colourless gas with a pungent odour.

(c) Chemical properties :

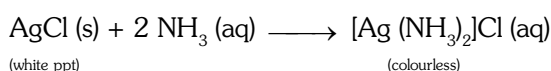
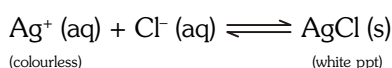
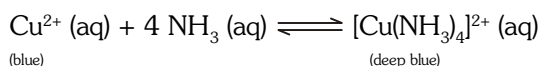
- (i) Its aqueous solution is weakly basic due to the formation of
- OH^-
- ions.



- (ii) It forms ammonium salts with acids, e.g., NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$ etc. As a weak base, it precipitates the hydroxides of many metals from their salt solutions. For example ,



- (iii) The presence of lone pair of electrons on the nitrogen atoms of the ammonia molecule makes it a Lewis base. It donates the electrons pair and forms linkage with metal ions and the formation of such complex compounds finds applications in detection of metal ions such as Cu^{2+} , Ag^+

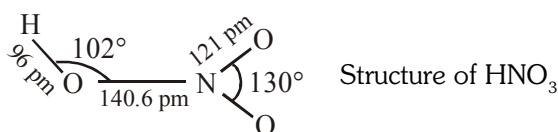


3. OXIDES OF NITROGEN :

Nitrogen forms a number of oxides, N_2O , NO , N_2O_3 , NO_2 or N_2O_4 and N_2O_5 . All these oxides of nitrogen exhibit $\pi\pi$ - $\pi\pi$ multiple bonding between nitrogen and oxygen.

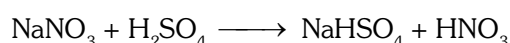
Name	Formula	Oxidation state of nitrogen	Common methods of preparation	Physical appearance and chemical nature
Dinitrogen oxide [Nitrogen(I) oxide]	N_2O	+ 1	$\text{NH}_4\text{NO}_3 \xrightarrow{\text{Heat}} \text{N}_2\text{O} + 2\text{H}_2\text{O}$	colourless gas , neutral
Nitrogen monoxide [Nitrogen(II) oxide]	NO	+ 2	$2\text{NaNO}_2 + 2\text{FeSO}_4 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 2\text{NaHSO}_4 + 2\text{H}_2\text{O} + 2\text{NO}$	colourless gas , neutral
Dinitrogen trioxide [Nitrogen(III) oxide]	N_2O_3	+ 3	$2\text{NO} + \text{N}_2\text{O}_4 \xrightarrow{250\text{ K}} 2\text{N}_2\text{O}_3$	blue solid , acidic
Nitrogen dioxide [Nitrogen(IV) oxide]	NO_2	+ 4	$2\text{Pb}(\text{NO}_3)_2 \xrightarrow{673\text{ K}} 4\text{NO}_2 + 2\text{PbO} + \text{O}_2$	brown gas, acidic
Dinitrogen tetroxide [Nitrogen(IV) oxide]	N_2O_4	+ 4	$2\text{NO}_2 \xrightleftharpoons[\text{Heat}]{\text{cool}} \text{N}_2\text{O}_4$	colourless solid / liquid , acidic
Dinitrogen pentoxide [Nitrogen(V) oxide]	N_2O_5	+ 5	$4\text{HNO}_3 + \text{P}_4\text{O}_{10} \longrightarrow 4\text{HPO}_3 + 2\text{N}_2\text{O}_5$	colourless solid, acidic

4. NITRIC ACID (HNO_3)



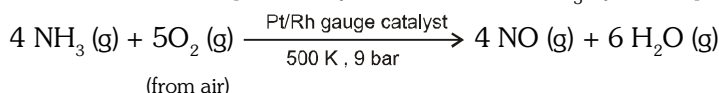
(a) Preparation :

In the laboratory, nitric acid is prepared by heating KNO_3 or NaNO_3 and concentrated H_2SO_4 in a glass retort.

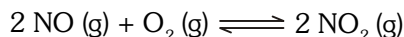


On a large scale it is prepared mainly by **Ostwald's process**.

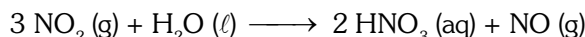
This method is based upon catalytic oxidation of NH_3 by atmospheric oxygen.



Nitric oxide thus formed combines with oxygen giving NO_2 .



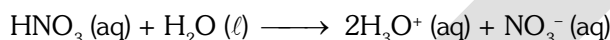
Nitrogen dioxide so formed, dissolves in water to give HNO_3 .



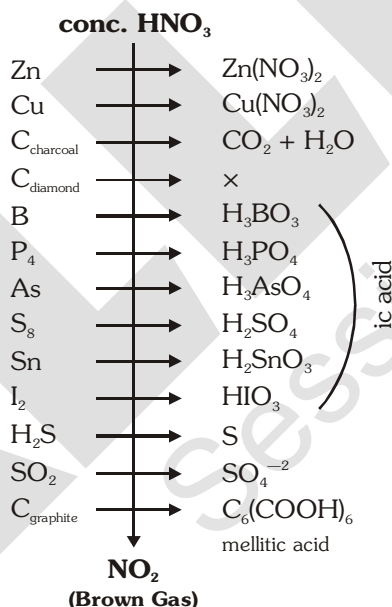
NO thus formed is recycled and the aqueous HNO_3 can be concentrated by distillation upto ~ 68% by mass. Further concentration to 98% can be achieved by dehydration with concentrated H_2SO_4 .

(b) Physical properties :

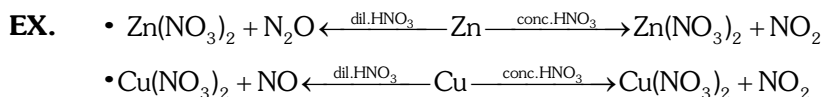
- (i) It is a colourless liquid. Freezing point is 231.4 K and boiling point is 355.6 K. Laboratory grade nitric acid contains ~ 68% of the HNO_3 by mass and has a specific gravity of 1.504.
- (ii) In the gaseous state, HNO_3 exists as a planar molecule.
- (iii) In aqueous solution, nitric acid behaves as a strong acid giving hydronium and nitrate ions.



- (iv) Concentrate nitric acid is a strong oxidising agent and attacks most metals except Au & Pt. The product of oxidation depend upon the concentration of the acid, temperature and nature of metal.
- (v) Concentrate HNO_3 also oxidises non-metals and their compounds.

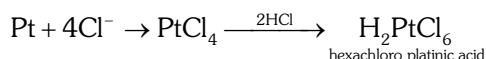
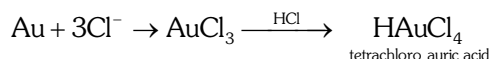


Types of Metal	conc. HNO_3	dil. HNO_3	very dil. HNO_3
negative SRP metal	Metal nitrate + NO_2	Metal nitrate + N_2O	Metal nitrate + $\text{NH}_4\text{NO}_3/(\text{NH}_3)$
positive SRP metal & Pb	Metal nitrate + NO_2	Metal nitrate + NO	×
Metalloids, Non-metals & Sn	ic-acid + NO_2	×	×



SOME EXCEPTIONAL POINTS

- **Mg and Mn form H_2 gas on reaction with very dilute HNO_3**
- **Be, Al, Fe, Cr do not react with conc. HNO_3 due to formation of protective oxide layer.**
- **Au and Pt (Noble metals) do not react with conc. HNO_3**
- **Sn reacts with conc. HNO_3 and forms H_2SnO_3 (metastannic acid)**
- **Au and Pt react only with aqua regia (3 part of conc. HCl and 1 part of conc. HNO_3)**



conc. HNO_3 + skin $\rightarrow$ yellow spot (protein $\rightarrow$ xanthoprotein)

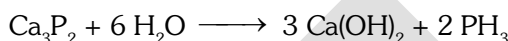
- **The major use of nitric acid is in the manufacture of ammonium nitrate for fertilizers and other nitrates for use in explosives and pyrotechnics. It is also used for the preparation of nitroglycerin, trinitrotoluene and other organic nitro compounds. Other major uses are in the pickling of stainless steel, etching of metals and as an oxidiser in rocket fuels.**

COMPOUNDS OF PHOSPHORUS :

1. Phosphine (PH_3) :

(a) Preparation :

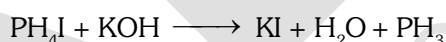
- (i) Phosphine is prepared by the reaction of calcium phosphide with water.



- (ii) In the laboratory, it is prepared by heating white phosphorus with concentrated NaOH solution in an inert atmosphere of CO_2 .

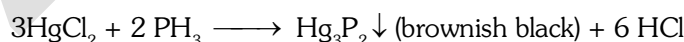
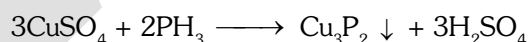


- (iii) When pure, it is non inflammable but becomes inflammable owing to the presence of P_2H_4 or P_4 vapours. To purify it from the impurities, it is absorbed in HI to form phosphonium iodide (PH_4I) which on treating with KOH gives off phosphine.



(b) Properties :

- (i) It is a colourless gas with rotten fish smell and is highly poisonous. It is slightly soluble in water. The solution of PH_3 in water decomposes in presence of light giving red phosphorus and H_2 .
- (ii) When absorbed in copper sulphate or mercuric chloride, the corresponding phosphides are obtained.



- (iii) Phosphine is weakly basic and like ammonia, gives phosphonium compounds with acids



- **Uses :** The spontaneous combustion of phosphine is technically used in **Holme's signals**. Containers containing calcium carbide and calcium phosphide are pierced and thrown in the sea when the gases evolved burn and serve as a signal.
- It is also used in the production of smoke screens. Calcium phosphide reacts with water producing phosphine which burns in air to give clouds of phosphorus pentaoxide and that acts as smoke screens.

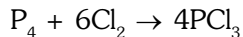
2. PHOSPHOROUS HALIDES

Phosphorous forms two types of halides, PX_3 ($X = F, Cl, Br, I$) and PX_5 ($X = F, Cl, Br$).

Phosphorous Trichloride (PCl_3)

(a) Method of preparation

(i) It is obtained by passing dry chlorine over heated white phosphorus.

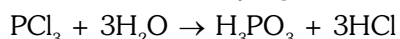


(ii) It is also obtained by the action of thionyl chloride with white phosphorus.

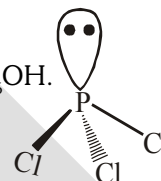
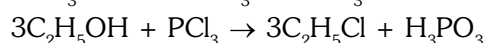
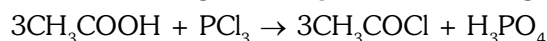


(b) Properties

(i) It is a colourless oily liquid and hydrolyses in the presence of moisture.

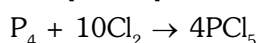


(ii) It reacts with organic compounds containing $-OH$ group such as CH_3COOH , C_2H_5OH .

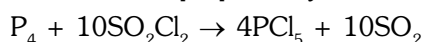
PHOSPHORUS PENTACHLORIDE (PCl_5)

(a) Method of preparation

(i) Phosphorus pentachloride is prepared by the reaction of white phosphorus with excess of dry chlorine.

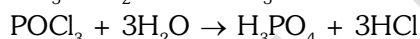
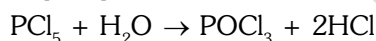


(ii) It can also be prepared by the action of SO_2Cl_2 on phosphorus.

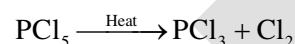


(b) Properties

(i) PCl_5 is a yellowish white powder and in moist air, it hydrolyses to $POCl_3$ and finally gets converted to phosphoric acid.



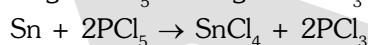
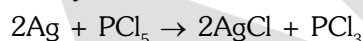
(ii) When heated, it sublimes but decomposes on stronger heating.



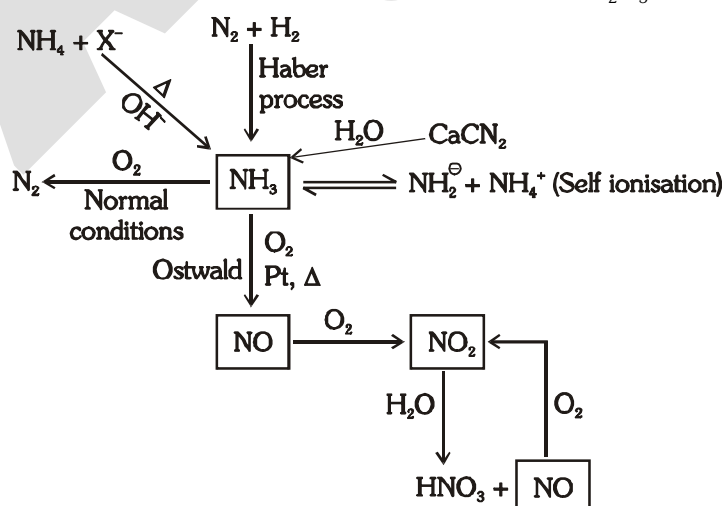
(iii) It reacts with organic compounds containing $-OH$ group converting them to chloro derivatives.



(iv) Finally divided metals on heating with PCl_5 give corresponding chlorides.



(v) It is used in the synthesis of some organic compounds, e.g., C_2H_5Cl , CH_3COCl .



NCERT QUESTIONS (REASONING)

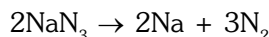
Q.1 Why BiH_3 is the strongest reducing agent amongst all the hydrides of Group 15 elements ?

Ans. In hydrides of nitrogen family on moving down the group M-H bond length increases so bond strength decreases hence tendency to release hydrogen increases and reducing nature increases.

Increasing order of reducing nature is $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$

Q.2 Write the reaction of thermal decomposition of sodium azide.

Ans. Thermal decomposition of sodium azide gives dinitrogen gas.



Q.3 Why N_2 is less reactive at room temperature?

Ans. N_2 is less reactive at room temperature because of the high bond enthalpy of $\text{N} \equiv \text{N}$ bond.

Q.4 Why does $\text{R}_3\text{P} = \text{O}$ exist but $\text{R}_3\text{N} = \text{O}$ does not (R = alkyl group)?

Ans. Due to presence of vacant d orbital phosphorous can form five covalent bond, while nitrogen restricts its covalency to four due to absence of vacant d orbitals.

Q.5 Why does nitrogen show catenation properties less than phosphorus?

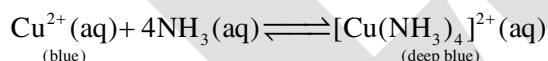
Ans. N-N bond is weaker than the single P-P bond due to high interelectronic repulsion of non-bonding electrons as a result the catenation tendency is weaker in nitrogen.

Q.6 Mention the conditions required to maximise the yield of ammonia.

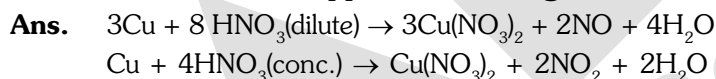
Ans. In accordance with Le Chatelier's principle, high pressure would favour the formation of ammonia. The optimum conditions for the production of ammonia are a pressure of 200×10^5 Pa (about 200 atm), a temperature of ~ 700 K and the use of a catalyst such as iron oxide with small amounts of K_2O and Al_2O_3 to increase the rate of attainment of equilibrium.

Q.7 How does ammonia react with a solution of Cu^{2+} ?

Ans. The presence of a lone pair of electrons on the nitrogen atom of the ammonia molecule makes it a Lewis base. It donates the electron pair and forms linkage with metal ions and the formation of such complex compounds finds applications in detection of Cu^{2+} .



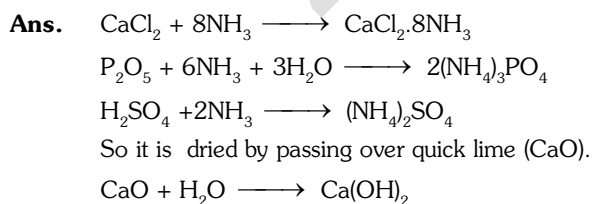
Q.8 Illustrate how copper metal can give different products on reaction with HNO_3 .



Q.9 Why is nitrogen di-oxide paramagnetic in gaseous state but the solid obtained on cooling it is diamagnetic.

Ans. NO_2 contains odd number of electrons in its valence shell. On cooling it forms dimer and converted to stable N_2O_4 which is a colourless solid and diamagnetic in nature.

Q.10 Why NH_3 gas cannot be dried by passing over P_2O_5 , CaCl_2 and H_2SO_4 ?

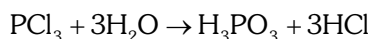


Q.11 Why inert atmosphere of CO_2 is taken in the formation of PH_3 by the reaction of white phosphorous with conc. NaOH solution.

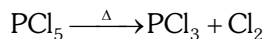
Ans. To decrease the partial pressure of O_2 in atmosphere.

Q.12 Why does PCl_3 fumes in moisture?

Ans. PCl_3 hydrolyses in the presence of moisture giving fumes of HCl

**Q.13 What happens when PCl_5 is heated?**

Ans. When heated, it sublimes but decomposes on stronger heating

**Q.14 Can PCl_5 act as an oxidising as well as reducing agent? Justify.**

Ans. It can act as an oxidising as well as reducing agent due to oxidising nature of P(V) and reducing nature of Cl⁻

BEGINNER'S BOX-7

- Which of the following halide of nitrogen is stable?
 (1) NF_3 (2) NCl_3 (3) NBr_3 (4) NI_3
- The nitrogen oxide(s) that does not contain N-N bond(s) is
 (1) N_2O (2) N_2O_3 (3) N_2O_4 (4) N_2O_5
- What is false about N_2O_5 ?
 (1) It is anhydride of HNO_3 (2) It is a powerful oxidizing agent
 (3) Solid N_2O_5 is called nitronium nitrate (4) Structure of N_2O_5 contains no $[\text{N} \rightarrow \text{O}]$ bond
- Conc. HNO_3 is yellow coloured liquid due to –
 (1) Dissolution of NO in conc. HNO_3 (2) Dissolution of NO_2 in conc. HNO_3
 (3) Dissolution of N_2O in conc. HNO_3 (4) Dissolution of N_2O_3 in conc. HNO_3
- Which of the following process is not involved in Ostwald's process for the manufacture of HNO_3 ?
 (1) $4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \xrightarrow[500\text{K}, 9\text{bar}]{\text{Pt}} 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
 (2) $2\text{NO}(\text{g}) + \text{O}_2 \rightleftharpoons 2\text{NO}_2(\text{g})$
 (3) $3\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{HNO}_3(\text{aq}) + \text{NO}(\text{g})$
 (4) None
- Which of the following salts give NH_3 in alkaline medium?
 (1) $(\text{NH}_4)_2\text{CO}_3$ (2) $(\text{NH}_4)_2\text{SO}_4$ (3) NH_4Cl (4) All of the above
- Which of the following reaction is suitable for obtaining very pure nitrogen?
 (1) $\text{NH}_4\text{Cl}(\text{aq}) + \text{NaNO}_2(\text{aq}) \longrightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + \text{NaCl}(\text{aq})$
 (2) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta} \text{N}_2 + 4\text{H}_2\text{O} + \text{Cr}_2\text{O}_3$
 (3) $\text{Ba}(\text{N}_3)_2 \xrightarrow{\Delta} \text{Ba} + 3\text{N}_2$
 (4) All of the above
- In warfare smoke screens are prepared from
 (1) CaC_2 (2) PH_3 (3) P_2O_5 (4) COCl_2
- White phosphorous on reaction with NaOH gives PH_3 as one of the products. This is a
 (1) dimerisation reaction (2) disproportionation reaction
 (3) condensation reaction (4) precipitation reaction
- Which of the following is not correctly matched?
 (1) $\text{P}_4\text{O}_{10} + \text{H}_2\text{O} \longrightarrow$ reactants involved in formation of H_3PO_4
 (2) $\text{CaC}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_2\text{H}_2 + \text{Ca}(\text{OH})_2$;
 $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \longrightarrow \text{PH}_3 + \text{Ca}(\text{OH})_2$ reactions involved in Holmes signal
 (3) $\text{PH}_3 + \text{HI} \longrightarrow \text{PH}_4\text{I} \xrightarrow{\text{KOH}} \text{KI} + \text{H}_2\text{O} + \text{PH}_3$; purification of PH_3
 (4) $\text{PH}_3 + \text{HI} \longrightarrow \text{PH}_4\text{I}$; shows Lewis acidic nature of PH_3

OXYGEN FAMILY (GROUP 16)

INTRODUCTION

Oxygen, sulphur, selenium, tellurium and polonium constitute group 16 of the periodic table. This is sometimes known as group of chalcogens (ore forming elements) because a large number of metals ores are oxides or sulphides.

Physical Properties :

- Oxygen and sulphur are non-metal, selenium and tellurium metalloids, whereas polonium is a metal. Polonium is radioactive and is short lived (Half-life 13.8 days).
- The melting and boiling points increase with an increase in atomic number down the group. The larger difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of their atomicity; oxygen exist as diatomic molecules (O_2) whereas sulphur exists as polyatomic molecule (S_8).

CATENATION

Tendency for catenation decreases down the group. This property is prominently displayed by sulphur (S_8). The S—S bond is important in biological system and is found in some proteins and enzymes such as cysteine.

Chemical Properties

Oxidation states and trends in chemical reactivity :

The elements of group 16 exhibit a number of oxidation states. The stability of -2 oxidation state decreases down the group. Except oxygen other elements of the group exhibit + 2, + 4, + 6 oxidation states but + 4 and + 6 are more common. Sulphur, selenium and tellurium usually show + 4 oxidation in their compounds with oxygen and +6 oxidation state with fluorine. The stability of +6 oxidation state decreases down the group and stability of + 4 oxidation state increases (inert pair effect). Bonding in + 4 and + 6 oxidation states are primarily covalent.

(i) Reactivity with hydrogen :

All the elements of group 16 form hydrides of the type H_2E ($E = O, S, Se, Te, Po$).

(ii) Reactivity with oxygen :

All these elements form oxides of the EO_2 and EO_3 types where $E = S, Se, Te$ or Po . Ozone (O_3) and sulphur dioxide (SO_2) are gases while selenium dioxide (SeO_2) is solid. Reducing property of dioxide decreases from SO_2 to TeO_2 ; SO_2 is reducing while TeO_2 is an oxidising agent. Besides EO_2 type sulphur, selenium and tellurium also form EO_3 type oxides (SO_3, SeO_3, TeO_3). Both types of oxides are acidic in nature.

(iii) Reactivity toward the halogens :

Elements of group 16 form a larger number of halides of the type EX_6 , EX_4 and EX_2 where E is an element of the group -16 and X is an halogen. The stabilities of the halides decrease in the order $F > Cl > Br > I$. Amongst hexahalides, hexafluorides are the only stable halides. All hexafluorides are gaseous in nature. They have octahedral structure. Sulphur hexafluoride SF_6 is exceptionally stable for steric reasons.

Amongst tetrafluorides, SF_4 is a gas, SeF_4 liquid and TeF_4 a solid

All elements except selenium form dichlorides and dibromides. The well known monohalides are dimeric in nature, Examples are $S_2F_2, S_2Cl_2, S_2Br_2, Se_2Cl_2$ and Se_2Br_2 . These dimeric halides undergo disproportionation

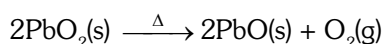
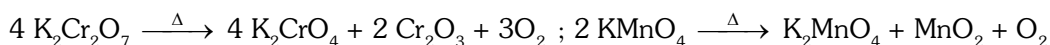
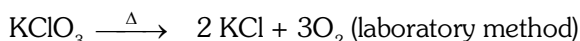
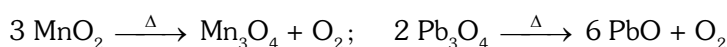
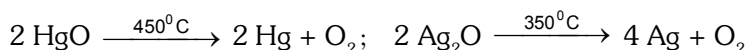
as given $2Se_2Cl_2 \longrightarrow SeCl_4 + 3Se$.

COMPOUNDS OF OXYGEN

1. DIOXYGEN (O_2)

(a) Preparation of DiOxygen (O_2) :

By thermal decomposition of oxides of metals.

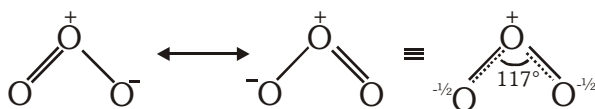


Properties:

- Dioxygen is a colourless and odourless gas.
- Oxygen atom has three stable isotopes: ^{16}O , ^{17}O and ^{18}O . Molecular oxygen.
- O_2 is paramagnetic (by MOT)
- Dioxygen directly reacts with nearly all metals and non-metals except some metals (e.g., Au, Pt) and some noble gases.

2. OZONE (O_3) :

Preparation : It is prepared by passing silent electric discharge through pure and dry oxygen.



Mixture obtained contains 5-10% ozone by volume and this mixture is called ozonised oxygen.

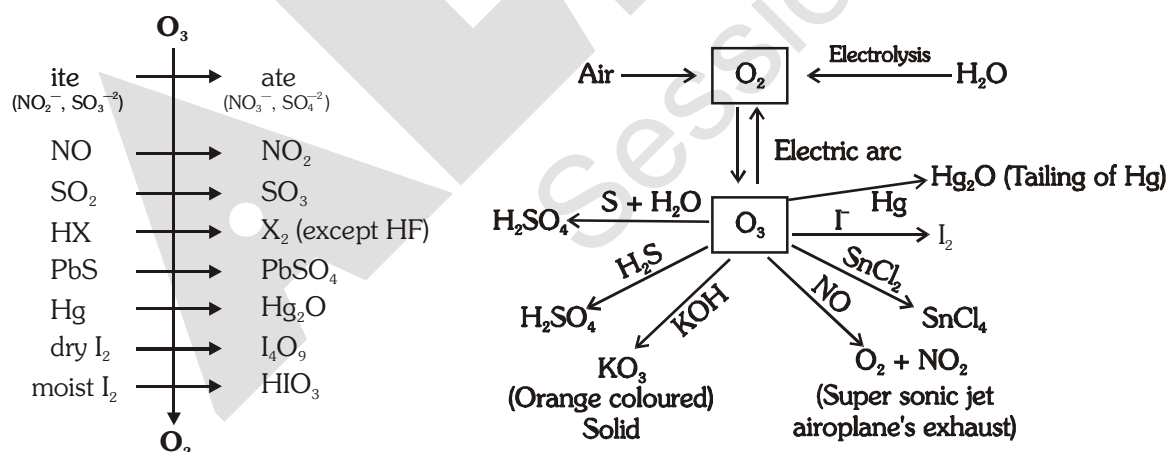
Since the formation of ozone from oxygen is an endothermic process, it is necessary to use a silent electrical discharge in its preparation to prevent its decomposition. If concentrations of ozone greater than 10 per cent are required, a battery of ozonisers can be used, and pure ozone (b.p. 385 K) can be condensed in a vessel surrounded by liquid oxygen.

Properties :

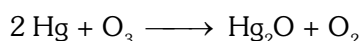
- Pure ozone is a pale blue gas, dark blue liquid and violet-black solid.
- Ozone is thermodynamically unstable with respect to oxygen since its decomposition into oxygen results in the liberation of heat (ΔH is negative) and an increase in entropy (ΔS is positive). These two effects reinforce each other, resulting in large negative Gibbs energy change (ΔG) for its conversion into oxygen. Therefore, high concentrations of ozone can be dangerously explosive.

Oxidising behaviour of ozone :

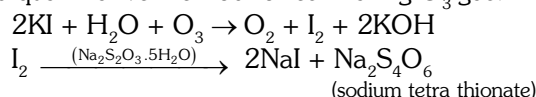
Due to the ease with which it liberates atoms of nascent oxygen ($\text{O}_3 \rightarrow \text{O}_2 + \text{O}$), it acts as a powerful oxidising agent. For example, it oxidises lead sulphide to lead sulphate and iodide ions to iodine.

**Tests for Ozone**

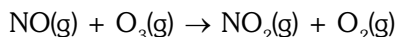
Tailing of mercury : Pure mercury is a mobile liquid but when brought in contact with O_3 its mobility decreases and it starts sticking to glass surface forming a type of tail due to the dissolution of Hg_2O (mercury sub-oxide) in Hg.



Estimation of Ozone : When ozone reacts with an excess of potassium iodide solution buffered with a borate buffer (pH 9.2), iodine is liberated which can be titrated against a standard solution of sodium thiosulphate. This is a quantitative method for estimating O_3 gas.



Depletion of ozone layer : Experiments have shown that nitrogen oxides (particularly nitric oxide) combine very rapidly with ozone and there is, thus, the possibility that nitrogen oxides emitted from the exhaust systems of supersonic jet aeroplanes might be slowly depleting the concentration of the ozone layer in the upper atmosphere.

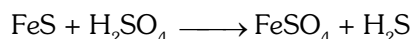


Another threat to this ozone layer is probably posed by the use of freons which are used in aerosol sprays and as refrigerants.

COMPOUNDS OF SULPHUR :

1. Hydrogen Sulphide (H_2S)

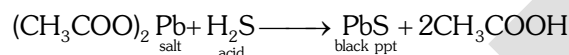
Preparation :



It is prepared in kipp's apparatus

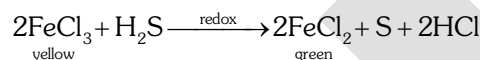
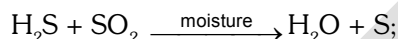
Properties :

- (i) Colourless gas with rotten egg smell .
- (ii) Moderately soluble in water but solubility decreases with increasing temperature.
- (iii) It gives black ppt with lead acetate



Reducing behaviour :

Acts as a strong reducing agent as it decomposes evolving hydrogen.



2. SO_2 (Sulphur Dioxide)

Preparation :

- (i) $S + O_2$ or air $\xrightarrow{\text{Burn}} SO_2$
- (ii) By reaction of metal sulphites with dilute HCl (Laboratory method)

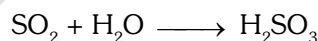
$$Na_2SO_3 + 2HCl \longrightarrow 2NaCl + SO_2 + H_2O$$
 Similarly bisulphites also give SO_2 with dilute HCl

$$NaHSO_3 + HCl \longrightarrow NaCl + SO_2 + H_2O$$
- (iii) By heating sulphides (metal sulphide ores) in excess of air.

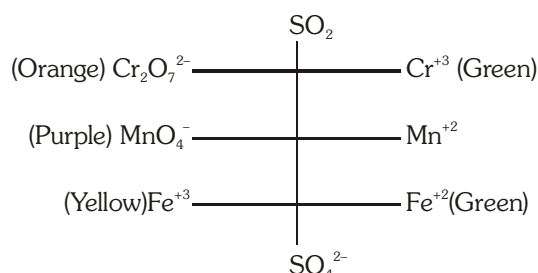
$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$
 By this method SO_2 is obtained in large scale

Properties :

- (i) Colourless gas with pungent smell.
- (ii) It is heavier than air and is highly soluble in water.
- (iii) Acidic Nature : Acidic oxide and thus dissolve in water forming sulphurous acid.

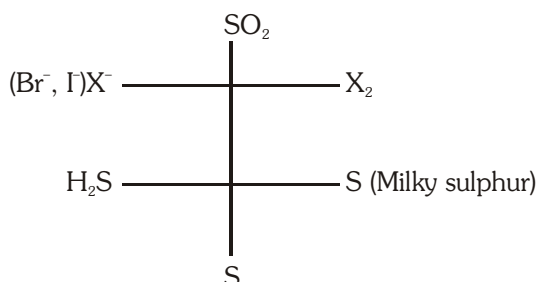


Reducing nature :



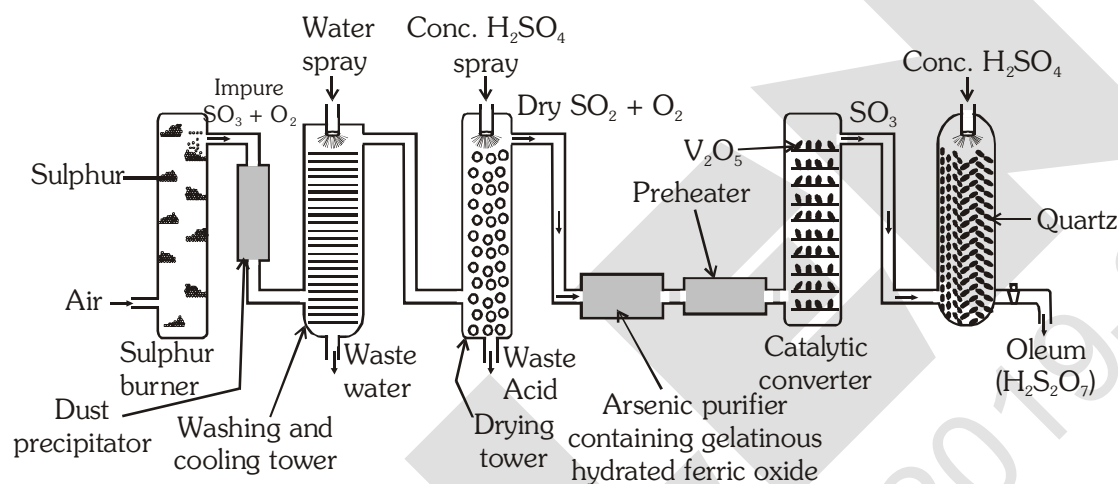
The above reactions are not given by CO_2 , so they are used to distinguish between SO_2 and CO_2

Oxidising nature : Acts as oxidising agent with strong reducing agent



3. SULPHURIC ACID (H_2SO_4) :

Manufacture by contacts process :



Flow diagram for the manufacture of sulphuric acid

Sulphuric acid is one of the most important industrial chemicals worldwide.

Sulphuric acid is manufactured by the contact process which involves three steps :

- burning of sulphur or sulphide ores in air to generate SO_2
- Conversion of SO_2 to SO_3 by the reaction with oxygen in the presence of a catalyst (V_2O_5), and
- Absorption of SO_3 in H_2SO_4 to give Oleum ($\text{H}_2\text{S}_2\text{O}_7$)

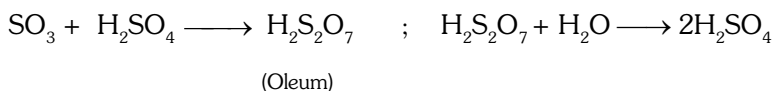
The SO_2 produced is purified by removing dust and other impurities such as arsenic compounds.

The key step in the manufacture of H_2SO_4 is the catalytic oxidation of SO_2 with O_2 to give SO_3 in the presence of V_2O_5 (catalyst).



The reaction is exothermic reversible and the forward reaction leads to a decrease in volume. Therefore, low temperature and high pressure are the favourable conditions for maximum yield. But the temperature should not be very low otherwise rate of reaction will become slow.

Dilution of oleum with water gives H_2SO_4 of the desired concentration. In the industry two steps are carried out simultaneously to make the process a continuous one and also to reduce the cost.

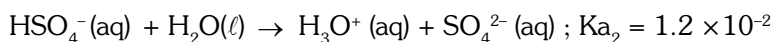
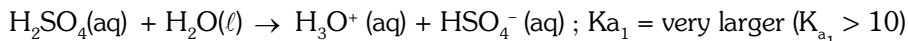


The sulphuric acid obtained by Contact process is 96-98% pure.

Properties : Sulphuric acid is a colourless, dense, oily liquid with a specific gravity of 1.84 at 298 K. The acid freezes at 283 K and boils at 611 K. It dissolves in water with the evolution of a larger quantity of heat.

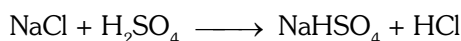
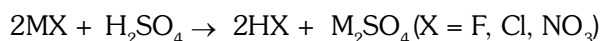
The chemical reaction of sulphuric acid are as a result of the following characteristics : (a) low volatility (b) strong acidic character (c) strong affinity for water and (d) ability to act as an oxidising agent in aqueous solution,

- (i) Sulphuric acid ionises in two steps.

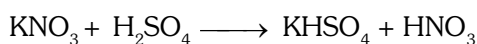


The larger value of K_{a1} ($K_{a1} > 10$) means that H_2SO_4 is largely dissociated into H^+ and HSO_4^- . Greater the value of dissociation constant (K_a) the stronger is the acid.

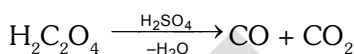
- (ii) **The acid forms two series of salts :** Normal sulphates (such as sodium sulphate and copper sulphate and acid sulphate (e.g., sodium hydrogen sulphate)
- (iii) Sulphuric acid, because of its low volatility can be used to manufacture more volatile acid from their corresponding salts.



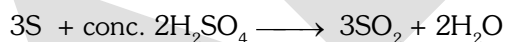
(M = Metal)



- (iv) Concentrated sulphuric acid is a strong dehydrating agent. Many wet gases can be dried by passing them through sulphuric acid, provided the gases do not react with the acid. Sulphuric acid removes water from organic compound; it is evident by its charring action on carbohydrates.



- (v) Hot concentrated sulphuric acid is moderately strong oxidising agent. In this respect it is intermediate between phosphoric and nitric acids. Both metals and non-metals are oxidised by concentrated sulphuric acid, which is reduced to SO_2 .



NCERT QUESTIONS (REASONING)

Q.1 H_2S is less acidic than H_2Te . Why?

Ans. Due to the decrease in bond (E–H) dissociation enthalpy down the group, acidic character increases.

Q.2 Why is H_2O a liquid and H_2S a gas?

Ans. Hydrogen bonds are present between H_2O molecules while between H_2S molecules, vander Waal's forces are present.

Q.3 Why is dioxygen a gas but sulphur a solid?

Ans. Oxygen exist as a O_2 molecule while sulphur exist as a S_8 molecule due to more molecular mass sulphur is solid.

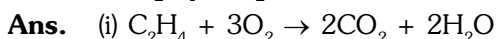
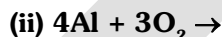
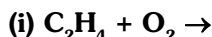
Q.4 Knowing the electron gain enthalpy values for $\text{O} \rightarrow \text{O}^-$ and $\text{O} \rightarrow \text{O}^{2-}$ as -141 and 702 kJ mol^{-1} respectively, how can you account for the formation of a large number of oxides having O^{2-} species and not O^- ?

Ans. Consider lattice energy factor in the formation of compounds.

Q.5 Which of the following does not react with oxygen directly? Zn, Ti, Pt, Fe

Ans. Pt is a noble metal which does not react directly with oxygen.

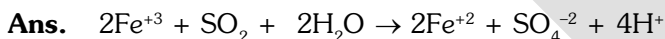
Q.6 Complete the following reactions:



Q.7 How is O_3 estimated quantitatively?

Ans. When ozone reacts with an excess of potassium iodide solution buffered with a borate buffer (pH 9.2), iodine is liberated which can be titrated against a standard solution of sodium thiosulphate. This is a quantitative method for estimating O_3 gas.

Q.8 What happens when sulphur dioxide is passed through an aqueous solution of Fe(III) salt?



Q.9 How is the presence of SO_2 detected?

Ans. It has colourless gas with pungent smell and decolourise acidified KMnO_4 solution.

Q.10 Write the conditions to maximise the yield of H_2SO_4 by Contact process.

Ans. The reaction is exothermic, reversible and the forward reaction leads to a decrease in volume. Therefore, low temperature and high pressure are the favourable conditions for maximum yield. But the temperature should not be very low otherwise rate of reaction will become slow.

Q.11 Why is $K_{a2} \ll K_{a1}$ for H_2SO_4 in water?

Ans. It is difficult to remove H^+ ion from HSO_4^- ion.

BEGINNER'S BOX-8

- Which of the following is not oxidised by O_3 ?
 (1) KI (2) FeSO_4 (3) KMnO_4 (4) K_2MnO_4
- In which of the following reaction conc. H_2SO_4 is not used as an oxidising agent?
 (1) $\text{Cu} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$ (2) $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
 (3) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + 2\text{HF}$ (4) None
- Hot conc. H_2SO_4 acts as strong oxidising agent which of the following element is oxidised by conc. H_2SO_4 into two gaseous products?
 (1) Cu (2) S (3) C (4) Zn
- HCOOH reacts with conc. H_2SO_4 to produce
 (1) CO (2) CO_2 (3) NO (4) NO_2

HALOGEN FAMILY (GROUP-17)

(A) PHYSICAL PROPERTIES

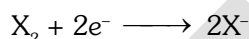
- (i) Fluorine and chlorine are gases, bromine is a liquid whereas iodine is a solid.
- (ii) Their melting and boiling points steadily increase with atomic number.
- (iii) All halogens are coloured. This is due to absorption of radiations in visible region which results in the excitation of outer electrons to higher energy level. By absorbing different quanta of radiation, they display different colours.
For example, F_2 , has yellow, Cl_2 , greenish yellow, Br_2 , red and I_2 , violet colour.
- (iv) Fluorine and chlorine react with water. Bromine and iodine are only sparingly soluble in water. But are soluble in organic solvents such as chloroform, carbon tetrachloride, carbon disulphide and hydrocarbons to give coloured solutions.
- (v) Except the smaller enthalpy of dissociation of F_2 compared to that of Cl_2 . **The X-X bond disassociation enthalpies from chlorine onwards show the expected trend : $Cl - Cl > Br - Br > F - F > I - I$. The reason for the smaller enthalpy of dissociation of F_2 is the relatively larger electrons-electrons repulsion among the lone pairs in F_2 molecule where they are much closer to each other than in case of Cl_2 .**

(B) CHEMICAL PROPERTIES

(i) Oxidation states and trends in chemical reactivity

- All the halogens exhibit -1 oxidation state. However, chlorine, bromine and iodine exhibit +1, +3, +5 and +7 oxidation states also. The higher oxidation states of chlorine, bromine and iodine are realised mainly when the halogens are in combination with the small and highly electronegative fluorine and oxygen atoms e.g., in interhalogens, oxides and oxoacids.
- The fluorine atom has no d orbitals in its valence shell and therefore cannot expand its octet. Being the most electronegative, it exhibits only -1 oxidation state.
- All the halogens are highly reactive. They react with metals and non-metals to form halides. The reactivity of the halogens decreases down the group.

Standard Reduction Potential (SRP)



More the value of the SRP, more powerful is the oxidising agent.

Hence the order of oxidising power is $F_2 > Cl_2 > Br_2 > I_2$

Order of reducing behaviour is $I^- > Br^- > Cl^- > F^-$

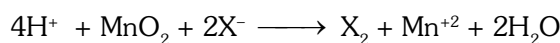
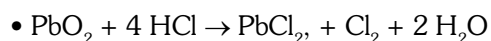
(ii) Halogen oxides :

- Fluorine forms two oxides OF_2 and O_2F_2 . However, only OF_2 is the thermally stable at 298 K. These oxides are essentially oxygen fluorides because of the higher electronegativity of fluorine than oxygen. Both are strong fluorinating agents.
- O_2F_2 oxidises plutonium to PuF_6 and the reaction is used in removing plutonium as PuF_6 from spent nuclear fuel.
- Chlorine, bromine and iodine form oxides in which the oxidation states of these halogens vary from +1 to +7. A combination of kinetic and thermodynamic factors lead to the generally decreasing order of stability of oxides formed by halogens, $I > Cl > Br$. The higher oxides of halogens tend to be more stable than the lower ones. This is called middle row anomaly.
- Chlorine oxides, Cl_2O , ClO_2 , Cl_2O_6 and Cl_2O_7 are highly reactive oxidising agents and tend to explode. ClO_2 is used as a bleaching agent for paper pulp and textiles and in water treatment.
- The bromine oxides, Br_2O , BrO_2 , BrO_3 are the least stable halogen oxides and exist only at low temperature. They are very powerful oxidising agents.
- The iodine oxides, I_2O_4 , I_2O_5 , I_2O_7 are insoluble solids and decompose on heating. I_2O_5 is very good oxidising agent and is used in the estimation of carbon monoxide.

COMPOUNDS OF HALOGEN

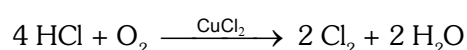
1. CHLORINE GAS (Cl_2)

(a) Preparation :

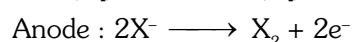
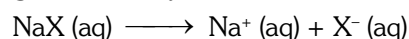
(i) By heating chloride with concentrated H_2SO_4 in presence of MnO_2 .Bromides and iodides also liberate Br_2 and I_2 respectively with concentrated H_2SO_4 and MnO_2 .(ii) • $\text{CaOCl}_2 + 2\text{HCl} \longrightarrow \text{CaCl}_2 + \text{Cl}_2 + \text{H}_2\text{O}$ 

(iii) Manufacture of chlorine :

- **Deacon's process** : By oxidation of hydrogen chloride gas by atmospheric oxygen in the presence of CuCl_2 (catalyst) at 723 K.



- **Electrolytic process** : Chlorine is obtained by the electrolysis of brine (concentrated NaCl solution). Chlorine is liberated at anode. It is obtained as a by-product in many chemical industries e.g.; in manufacturing of sodium hydroxide.

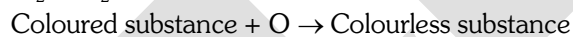


(b) Properties :

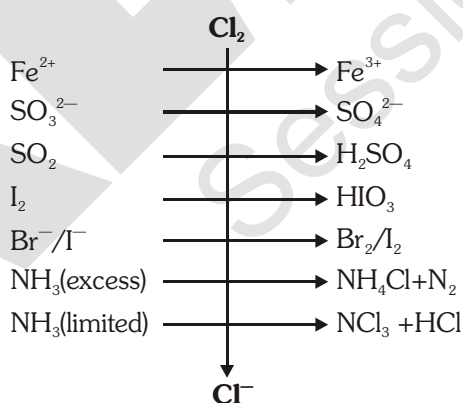
(i) It is a greenish-yellow gas with pungent and suffocating odour. It is about 2–5 times heavier than air. It can be liquefied into greenish-yellow liquid which boils at 239 K. It is soluble in water.

(ii) **Oxidising & bleaching properties** : Chlorine dissolves in water giving HCl and HOCl . Hypochlorous acid (HOCl) so formed, gives nascent oxygen which is responsible for oxidising and bleaching properties of chlorine.

It is a powerful bleaching agent ; bleaching action is due to oxidation.

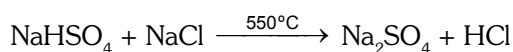
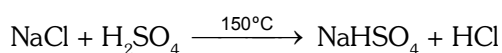


It bleaches vegetable or organic matter in the presence of moisture. Bleaching effect of chlorine is permanent.

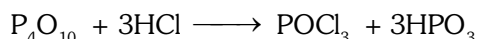
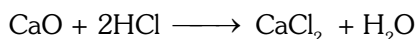
2. HALOGEN ACIDS (HCl)

(a) Preparation :

(i) By heating a halide with concentrated acid :



- Above method is called as salt cake method as it involves the formation of NaHSO_4 (salt cake).
- HCl cannot be dried over P_2O_5 (P_4O_{10}) or quick lime since they react with gas chemically.



- HCl is, hence dried by passing through concentrated H_2SO_4 .

(b) Properties :

- This is colourless, pungent smelling gas with acidic taste.
- This is neither combustible nor supporter of combustion.
- When perfectly dry, HX have no action on litmus, but in presence of moisture, they turn blue litmus red, showing acidic nature. Among HX , HI is the strongest and HF is the weakest acid.

- 3. Pseudo halogens and pseudo halides :** Some inorganic compounds consisting of two or more atoms of which at least one is N have been found to behave like halogens & they are known as pseudo halogen solids,

Pseudo halogens	Pseudo halides
(i) $(\text{CN})_2$ cyanogen	(i) $(\text{CN})^-$ cyanide ion
(ii) $(\text{SCN})_2$ thiocyanogen	(ii) $(\text{SCN})^-$ thiocyanate ion
(iii) $(\text{SeCN})_2$ selenocyanogen	(iii) $(\text{SeCN})^-$ selenocyanate ion
(iv) $(\text{SCSN}_3)_2$ azidocarbonyl disulphide	(iv) $(\text{OCN})^-$ cyanate ion
	(v) $(\text{NCN})^{2-}$ cyanamide ion
	(vi) $(\text{N}_3)^-$ azide ion etc.

4. INTERHALOGEN COMPOUNDS :

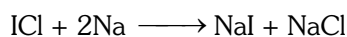
We know that halogen atoms have different electronegativity. Due to this difference in electronegativity the halogen atoms combine with each other and give rise to the formation of binary covalent compounds, which are called interhalogen compounds. These are of four types.

AB	AB_3	AB_5	AB_7^*
ClF	ClF_3	ClF_5	IF_7
BrF	BrF_3	BrF_5	
ICl	ICl_3	IF_5	
IF	IF_3		

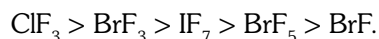
* Where A - halogen of smaller size and A is more electro positive than B.

(a) Properties :

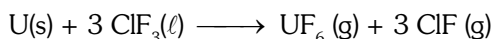
- These compounds may be gases, liquids or solids.
Gases : ClF , BrF , ClF_3 , IF_7 ; Liquids : BrF_3 , BrF_5 ; Solids : ICl , IBr , IF_3 , ICl_3 .
- Interhalogens containing fluorine are generally colourless but inter halogens consisting of heavier halogens are coloured. The intensity of colour increases with increase in the molecular weight of the compounds.
- All interhalogens are covalent molecules and are diamagnetic in nature since all the valence electrons present as bonding or non-bonding electrons are paired.
- Interhalogen compounds are more reactive than the parent halogens but less reactive than F_2 .



The order of reactivity of some interhalogens is as follows :



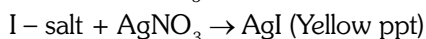
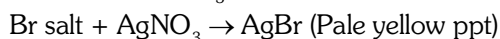
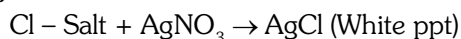
- (v) These compounds can be used as non aqueous solvents. Interhalogen compounds are very useful fluorinating agents. ClF_3 and BrF_3 are used for the production of UF_6 in the enrichment of ^{235}U .



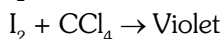
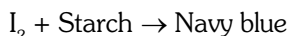
TEST OF HALOGENS

1. F^- = Itching of glass

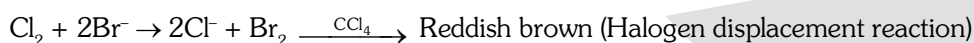
2. **AgNO_3 test** (Cl^- , Br^- , I^-)



3. **Test of I_2**



4. **Layer test (Br^- , I^-)**



(Note : In case of I^- violet colour is obtained.)

5. **Chromyl chloride test** (Test of Cl^-) : see d-block

NCERT QUESTIONS (REASONING)

Q.1 Covalent radius of fluorine is 64 pm but the bond length is not equal to 128 pm and that is 143 pm and bond energy is found to be comparable to I_2 .

Ans. This may be attributed to $\ell.p - \ell.p$ repulsions due to small size of F atom.

Q.2 Electron affinity of chlorine is more than F. Inspite of this F_2 is the better oxidising agent. Why ?

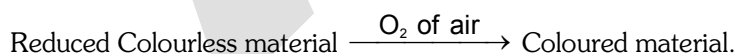
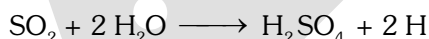
Ans. SRP of F_2 is much higher than that of Cl_2 on account of smaller bond dissociation energy and high hydration energy of F^- ion.

Q.3 Layer test of Br^- is successful with Cl_2 but not with I_2 . Explain?

Ans. Br^- is oxidised by Cl_2 but not by I_2

Q.4 What is the difference between bleaching action of SO_2 and Cl_2

Ans. The bleaching action of SO_2 is temporary because it takes place through reduction.



The bleaching action of Cl_2 is permanent because it takes place through oxidation

Q.5 (a) When HCl reacts with finely powdered iron, it forms ferrous chloride and not ferric chloride. why ?

(b) Chlorine water turns blue litmus red but solution becomes colourless after sometime.

Ans. (a) It forms H_2 gas. $\text{Fe} + 2 \text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2$.

Liberation of hydrogen prevents the formation of ferric chloride.

(b) Blue litmus change into red due to acidic nature ($\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{HCl}$) but it is bleaching agent, therefore, it decolourises the red litmus.

ZERO GROUP (GROUP-18)

INTRODUCTION

Group 18 consists of six elements: helium, neon, argon, krypton, xenon and radon. All these are gases and chemically unreactive. They form very few compounds. Because of this they are termed noble gases.

(A) PHYSICAL PROPERTIES

- (i) All the noble gases are mono-atomic.
- (ii) They are colourless, and tasteless. They are sparingly soluble in water.
- (iii) They have very low melting and boiling points because the only type of interatomic interaction in these elements is weak dispersion forces.
- (iv) Helium has the lowest boiling point (4.2K) of any known substance. It has a unusual property of diffusing through most commonly used laboratory materials such as rubber, glass or plastics.

(B) CHEMICAL PROPERTIES

In general, noble gases are least reactive. Their inertness to chemical reactivity is attributed to the following reasons:

- (i) The noble gases except helium ($1s^2$) have completely filled $ns^2 np^6$ electronic configuration in their valence shell.
- (ii) They have high ionisation enthalpy and more positive electron gain enthalpy.

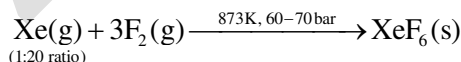
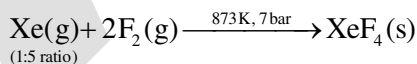
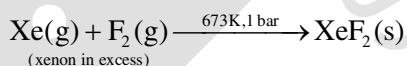
Neil Bartlett, observed the reaction of a noble gas. First, he prepared a red compound which is formulated as $O_2^+ [PtF_6]^-$. He, then realised that the first ionisation enthalpy of molecular oxygen (1175 kJ mol^{-1}) was almost identical with that xenon (1170 kJ mol^{-1}). He made efforts to prepare same type of compound with $Xe^+ [PtF_6]^-$ by mixing PtF_6 and Xenon. After this discovery, a number of xenon compounds mainly with most electronegative elements like fluorine and oxygen, have been synthesised.

The compounds of krypton are fewer. Only the difluoride (KrF_2) has been studied in detail. Compounds of radon have not been isolated but only identified (e.g., RnF_2) by radiotracer technique. No true compounds of Ar, Ne or He are yet known.

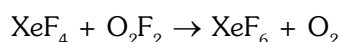
COMPOUNDS OF XENON

Xenon-fluorine compounds

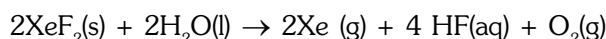
Xenon forms three binary fluorides, XeF_2 , XeF_4 and XeF_6 by the direct reaction of elements under appropriate experimental conditions.



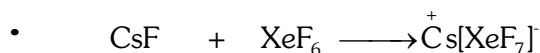
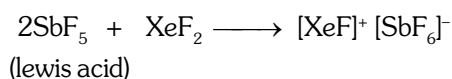
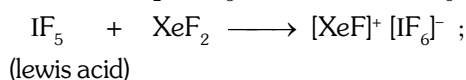
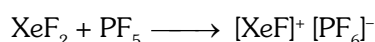
XeF_6 can also be prepared by the interaction of XeF_4 and O_2F_2 at 143K.



XeF_2 , XeF_4 and XeF_6 are colourless crystalline solids and sublime readily at 298 K. They are powerful fluorinating agents. They are readily hydrolysed even by traces of water. For example, XeF_2 is hydrolysed to give Xe, HF and O_2 .



Formation of addition compounds : XeF_2 reacts with fluoride ion acceptors to form cationic species and fluoride ion donors to form fluoroanions.



[With s-block fluorides and alkyl fluorides it is F^- acceptor with others it is F^- donor]

USES OF INERT GASES :

- (1) He is non-inflammable and light gas, so it is used in filling balloons for meteorological observations.
- (2) He is used in gas cooled nuclear reactors.
- (3) Liquid He is used as cryogenic agent.
- (4) He is used to produce powerful superconducting magnets.
- (5) Ne is used in discharge tubes.
- (6) Ar is used as inert atmosphere in metallurgical process.
- (7) Xenon and Krypton are used in light bulbs designed for special purposes.

IMPORTANT ORDER :

- (1) $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$
- (2) $\text{He} > \text{Ne} > \text{Ar} > \text{Kr} > \text{Xe}$
- (3) $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe} < \text{Rn}$
- (4) $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe} < \text{Rn}$

Atomic radius

Ionisation energy

Density

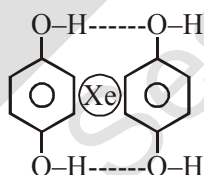
mpt. bpt

[Atomic mass / size / polarisability / London force] $\uparrow$

- (5) $\text{He} < \text{Ne} < \text{Ar} < \text{Xe} < \text{Rn}$ [solubility in water because of dipole-induced dipole attraction]

Clathrate Compounds : Larger inert gas atoms are trapped into cavity or crystal structure of some organic compounds and form clathrate compounds. For eg xenon diquinoxaline.

- Force of attraction between noble gas and organic compound is **dipole-induced dipole** attraction.



He, Ne do not form clathrate compounds due to smaller size

BEGINNER'S BOX-9

1. Which is correct
 - (1) $\text{NaCl} \xrightarrow{\text{conc. H}_2\text{SO}_4} \text{yellow green gas}$
 - (2) $\text{NaBr} \xrightarrow{\text{conc. H}_2\text{SO}_4} \text{red brown vapour}$
 - (3) $\text{NaF} + \text{Cl}_2 \longrightarrow \text{NaCl} + \text{F}_2$
 - (4) All
2. "Middle row anomaly" is related with
 - (1) stability of transition metal oxides
 - (2) stability of alkali metal oxides
 - (3) stability of halogen oxides
 - (4) stability of alkaline earth metal oxides
3. True statement about I^- will be
 - (1) weak base
 - (2) strong nucleophile
 - (3) strong reducing agent
 - (4) all

4. Of the following statements :
 (a) Cl_2 gas is dried by using conc. H_2SO_4
 (b) Fluorine have highest oxidising power
 (c) Oxidising power of halogens follow the order $\text{I}_2 > \text{Br}_2 > \text{Cl}_2$
 (d) HI is the strongest acid among HI, HBr, HCl
 (1) a, b and d are correct (2) a, c are correct (3) b, c are correct (4) c, d are correct
5. Chromyl chloride test is performed for the confirmation of the presence of the following in mixture:
 (1) SO_4^{2-} (2) Cr^{+++} (3) Cl^- (4) Cr^{+++} and Cl^-
6. The ion that cannot undergo disproportionation is
 (1) ClO_4^- (2) ClO_3^- (3) ClO_2^- (4) ClO^-
7. A greenish yellow gas reacts with an alkali metal hydroxide to form a halate which can be used in fire works and safety matches. The gas and halate respectively are :
 (1) Br_2 , KBrO_3 (2) Cl_2 , KClO_3 (3) I_2 , NaIO_3 (4) Cl_2 , NaClO
8. In the clathrates of xenon with water, the nature of bonding between xenon and water molecule is :
 (1) covalent (2) hydrogen bonding (3) co-ordinate (4) dipole-induced dipole interaction
9. XeF_2 reacts with PF_5 to give :
 (1) XeF_6 (2) $[\text{XeF}]^+ [\text{PF}_6]^-$ (3) XeF_4 (4) $[\text{PF}_4]^+ [\text{XeF}_3]^-$
10. The first compound of noble gases prepared by Neil-Bartlett was :-
 (1) $\text{Xe}^+ [\text{PtF}_6]^-$ (2) XeF_4 (3) XeF_6 (4) XeOF_4

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	1	4	1	1	1	4	3	3	1	4
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	4	3	3	3	2	1	4	2	2
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	2	3	2	2	2	3	2	4	3
BEGINNER'S BOX-4	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	2	2	2	4	3	4	4	1	3
BEGINNER'S BOX-5	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	1	4	2	2	3	4	3	3	4	3
BEGINNER'S BOX-6	Que.	1	2	3	4						
	Ans.	3	3	4	4						
BEGINNER'S BOX-7	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	1	4	4	2	4	4	3	2	2	4
BEGINNER'S BOX-8	Que.	1	2	3	4						
	Ans.	3	3	3	1						
BEGINNER'S BOX-9	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	3	4	1	3	1	2	4	2	1

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

IMPORTANT CONCEPTS

- "Hybridisation of central atom does not always change due to back bonding". This statement is valid for which of the following compounds?
 (i) CCl_3^- (ii) CCl_2
 (iii) CF_2 (iv) $\text{N}(\text{SiH}_3)_3$
 (1) (i), (ii) (2) (i), (iii)
 (3) (ii), (iii) (4) All
- The geometry with respect to the central atom of the following molecules are $\text{N}(\text{SiH}_3)_3$, Me_3N , $(\text{SiH}_3)_3\text{P}$
 (1) planar, pyramidal, planar
 (2) planar, pyramidal, pyramidal
 (3) pyramidal, pyramidal, pyramidal
 (4) pyramidal, planar, pyramidal
- In which of the following compounds, observed bond angle is found to be greater than expected, but not due to back bonding
 (1) $\text{N}(\text{SiH}_3)_3$ (2) $\text{O}(\text{CH}_3)_2$
 (3) $\text{O}(\text{SiH}_3)_2$ (4) All of these
- According to following reactions,

$$\text{CHF}_3 \xrightarrow{K_\alpha} \text{CF}_3^- + \text{H}^+$$

$$\text{CHCl}_3 \xrightarrow{K'_\alpha} \text{CCl}_3^- + \text{H}^+$$
 correct statement(s) is :
 (1) $K_\alpha > K'_\alpha$
 (2) CHF_3 act as a stronger bronsted acid than CHCl_3
 (3) CCl_3^- is more stable than CF_3^-
 (4) None of these
- The product formed in the reaction,
 $\text{BCl}_3 + \text{H}_2\text{O} \longrightarrow \text{Product is -}$
 (1) $\text{H}_3\text{BO}_3 + \text{HCl}$ (2) $\text{B}_2\text{O}_3 + \text{HOCl}$
 (3) $\text{B}_2\text{H}_6 + \text{HCl}$ (4) No reaction
- Which of the following statements is incorrect in relation to the structure of diborane
 (1) All the terminal B-H bond length are equal
 (2) The terminal B-H bond is a 2-centre 3-electron bond
 (3) The terminal B-H bond is a 2-centre 2-electron bond
 (4) The bridge $\begin{array}{c} \text{H} \\ \diagup \quad \diagdown \\ \text{B} \quad \text{B} \end{array}$ is a 3-centre 2-electron bond
- The type of hybridization of boron in diborane is -
 (1) sp (2) sp^2
 (3) sp^3 (4) sp^3d^2

- Which is less hydrolysed :
 (1) PCl_3 (2) NCl_3
 (3) AsCl_3 (4) SbCl_3
- Which of the following halides cannot be hydrolysed?
 (i) TeF_6 (ii) SF_6 (iii) NCl_3 (iv) NF_3
 Choose the correct code
 (1) iii and iv (2) i, ii and iii
 (3) i, ii and iv (4) ii and iv
- Which of the following is an uncommon hydrolysis product of XeF_2 and XeF_4 ?
 (1) Xe (2) XeO_3 (3) HF (4) O_2
- In which of the following dimer empty atomic orbital of central atom of monomer does not involve in hybridisation?
 (1) Ga_2H_6 (2) Al_2Br_6 (3) Be_2H_4 (4) Cl_2O_6
- Which of the following molecule is having complete octet
 (1) BeCl_2 (dimer) (2) BeH_2 (dimer)
 (3) BeH_2 (s) (4) BeCl_2 (s)
- Which one of the following oxy acid of fluorine exists ?
 (1) HOF (2) HFO_3
 (3) HFO_4 (4) HFO_2
- Which of the following statements is correct
 (1) All form HOXO_3 type oxy acid
 (2) Only chlorine and bromine form oxyacids
 (3) All halogens form oxyacids
 (4) Only iodine forms oxyacid
- Consider the oxy acids HClO_n series, here value of n is 1 to 4. Then incorrect statement regarding these oxy acids is
 (1) acidic character of oxy acids increases with increasing value of n.
 (2) oxidising power of oxy acids increases with decreasing value of n.
 (3) thermal stability of oxy acids decreases with increasing value of n.
 (4) Cl-O bond order decreases with decreasing value of n.
- $2\text{P} \xrightarrow{-\text{H}_2\text{O}} \text{Q} \xrightarrow{-[\text{O}]} \text{R}$
 If P is parent phosphoric acid then according to given information the incorrect statement is
 (1) Q is pyro form and R is hypo form of given present oxy acid P
 (2) Number of H-atoms present in each given oxy acid is equal to its basicity
 (3) In P, Q, R oxy acids, oxidation state of central atom remains same
 (4) All given oxy acids have $\text{p}\pi\text{-d}\pi$ bond(s) in their structure

17. Silicate having one monovalent corner oxygen atom in each tetrahedron unit is
 (1) sheet silicate (2) cyclic silicate
 (3) single chain silicate (4) double chain silicate
18. The silicate anion in the mineral kinoite is a chain of three SiO_4^{4-} tetrahedra, that share corners with adjacent tetrahedra. The charge of the silicate anion is
 (1) -4 (2) -8
 (3) -6 (4) -2
19. Which of the following is an organo silicon polymer?
 (1) silica (2) silicon
 (3) silicon carbide (4) silicic acid
20. Which reacts rapidly with oxygen in the air at ordinary temperature :
 (1) White P (2) Red P
 (3) N_2 (4) C
21. Red and yellow phosphorus are :
 (1) Allotropes (2) Isobars
 (3) Isomers (4) Isotopes
22. Phosphorus is kept in :
 (1) Kerosene oil (2) Alcohol
 (3) Water (4) Ammonia
23. Graphite conducts electricity because of the –
 (1) Highly polarized nature of π -electrons.
 (2) Highly delocalized nature of π -electrons
 (3) Highly localized nature of π -electrons
 (4) None of these

BORON AND CARBON FAMILY

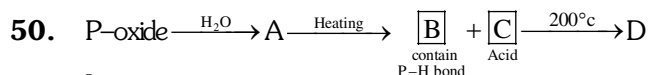
24. Melting point is higher for –
 (1) B (2) Al (3) Ga (4) In
25. Alane is chemically –
 (1) AlH_3 (2) $(\text{AlH}_3)_n$ (3) LiAlH_4 (4) None
26. Aluminium is not acted upon by pure water as –
 (1) Impurities in water are essential for the reaction to occur
 (2) It is light metal
 (3) It is protected by a film of aluminium oxide
 (4) It is not a reactive metal
27. The borax bead test is based upon the formation of
 (1) Boron oxide (2) Boric acid
 (3) Meta borates (4) Elemental boron
28. Boric acid polymerizes due to –
 (1) The presence of hydrogen bonds
 (2) Its acidic nature
 (3) Its geometry
 (4) Its monobasic nature

29. Alum is found to contain hydrated monovalent cation $[\text{M}(\text{H}_2\text{O})_6]^+$, trivalent cation $[\text{M}'(\text{H}_2\text{O})_6]^{+3}$ and SO_4^{2-} in the ratio of :
 (1) 1 : 1 : 1 (2) 1 : 1 : 2
 (3) 1 : 2 : 2 (4) 1 : 2 : 3
30. Borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$ is actually :-
 (1) $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$
 (2) $\text{Na}_2[\text{B}_4\text{O}_4(\text{OH})_6] \cdot 7\text{H}_2\text{O}$
 (3) $\text{Na}_2[\text{B}_4\text{O}_3(\text{OH})_8] \cdot 6\text{H}_2\text{O}$
 (4) $\text{Na}_2[\text{B}_4\text{O}_2(\text{OH})_{10}] \cdot 5\text{H}_2\text{O}$
31. Diborane is a Lewis acid forming addition compound $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ with NH_3 , a Lewis base. This
 (1) Is ionic and exists as $[\text{BH}_2(\text{NH}_3)_2]^+$ and $[\text{BH}_4]^-$ ions
 (2) On heating, is converted into borazine, $\text{B}_3\text{N}_3\text{H}_6$
 (3) Both are correct
 (4) None is correct
32. Which alum is a double salt made up of two salts :
 (1) Salt of a (SA + WB) + Salt of a (WA + WB)
 (2) Salt of a (SA + SB) + Salt of a (SA + WB)
 (3) Salt of a (SA + SB) + Salt of a (WA + WB)
 (4) Salt of a (SA + WB) + Salt of a (WA + WB)
33. From B_2H_6 all the following can be prepared except :
 (1) H_3BO_3 (2) $\text{B}_2(\text{CH}_3)_4\text{H}_2$
 (3) $\text{B}_2(\text{CH}_3)_6$ (4) NaBH_4
34. Borax on heating with cobalt oxide forms a blue bead of
 (1) $\text{Co}(\text{BO}_2)_2$ (2) CoBO_2
 (3) $\text{Co}_3(\text{BO}_3)_2$ (4) $\text{Na}_3\text{Co}(\text{BO}_3)_2$
35. The hydrides of group 14 elements are :
 (1) Ionic (2) Oxidising
 (3) Covalent (4) None of these
36. Which gas is responsible for green house effect :
 (1) CO_2 (2) SO_2 (3) CO (4) SO_3
37. Artificial gem used for cutting glass is :
 (1) Graphite (2) Diamond
 (3) SiC (4) CaCN_2

NITROGEN FAMILY

38. Nitrogen reacts with calcium carbide to give –
 (1) Calcium nitride (2) Calcium cyanide
 (3) Calcium cyanamide (4) Calcium nitrate
39. P_2O_5 is used extensively as a :
 (1) Dehydrating agent (2) Catalytic agent
 (3) Reducing agent (4) Preservative
40. PH_3 produces smoky rings when it comes in contact with air. This is because :
 (1) It is inflammable
 (2) It combines with water vapours
 (3) It combines with nitrogen
 (4) It contains impurity of P_2H_4

41. Which of the following is the correct statement for PH_3
- (1) It is less basic than NH_3
 - (2) It is less poisonous than NH_3
 - (3) Bond angle of $\text{PH}_3 > \text{NH}_3$
 - (4) It does not show reducing properties
42. Ammonia reacts with excess of chlorine to form:
- (1) N_2 and NH_4Cl
 - (2) NCl_3 and HCl
 - (3) NH_4Cl and NCl_3
 - (4) N_2 and HCl
43. Concentrated nitric acid reacts with iodine to give:-
- (1) HI
 - (2) HOI
 - (3) HOIO_2
 - (4) HOIO_3
44. Each of the following is true of white and red phosphorus except that they
- (1) Are both soluble in CS_2
 - (2) Can be oxidised by heating in air
 - (3) Consist of the same kind of atoms
 - (4) Can be converted into one another
45. A gas which is used as anaesthetic in dental surgery is :
- (1) N_2
 - (2) CO
 - (3) N_2O
 - (4) NH_3
46. The wrong statement about NH_3 is :
- (1) It is oxidised with oxygen at 700°C in the presence of platinum
 - (2) It gives reddish brown ppt with nessler's reagent
 - (3) It can be dried by P_2O_5 , H_2SO_4 and CaCl_2
 - (4) It gives white fumes with HCl
47. On heating, ammonium dichromate and barium azide separately, we get :-
- (1) N_2 with ammonium dichromate and NO with barium azide
 - (2) N_2O with ammonium dichromate and NO_2 with barium azide
 - (3) N_2O with ammonium dichromate and NO with barium azide
 - (4) N_2 in both cases
48. Which one of the following acid possesses oxidising, reducing and complex forming properties?
- (1) HNO_3
 - (2) HCl
 - (3) H_2SO_4
 - (4) HNO_2
49. NO_2 is formed when
- (1) Cu reacts with conc. HNO_3
 - (2) Zn reacts with conc. HNO_3
 - (3) $\text{Pb}(\text{NO}_3)_2$ is heated
 - (4) All



Incorrect statement is :-

- (1) Gas B is PH_3
 - (2) D is pyrophosphoric acid
 - (3) A is H_3PO_3
 - (4) D on strong heating gives P_2O_3
51. Which of the following does not produce NO_2 gas with conc. HNO_3 ?
- (1) Cu
 - (2) I_2
 - (3) Ag
 - (4) Au

OXYGEN FAMILY

52. SO_2 can act as -
- (1) Reducing agent
 - (2) Oxidising agent
 - (3) Bleaching agent
 - (4) All
53. Which reaction cannot be used for the preparation of the halogen acid-
- (1) $2\text{KBr} + \text{H}_2\text{SO}_4 \xrightarrow{\text{Conc.}} \text{K}_2\text{SO}_4 + 2\text{HBr}$
 - (2) $\text{NaCl} + \text{H}_2\text{SO}_4 \xrightarrow{\text{conc.}} \text{NaHSO}_4 + \text{HCl}$
 - (3) $\text{NaHSO}_4 + \text{NaCl} \xrightarrow{\text{conc.}} \text{Na}_2\text{SO}_4 + \text{HCl}$
 - (4) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \xrightarrow{\text{conc.}} \text{CaSO}_4 + 2\text{HF}$
54. Ozone acts as
- (1) Oxidising agent
 - (2) reducing agent
 - (3) bleaching agent
 - (4) all
55. A black sulphide when treated with ozone becomes white. The white compound is :
- (1) ZnSO_4
 - (2) CaSO_4
 - (3) BaSO_4
 - (4) PbSO_4
56. H_2S gas changes a filter paper dipped in lead acetate solution into :
- (1) Black
 - (2) Red
 - (3) Green
 - (4) Yellow
57. The number of S-S bonds in sulphur trioxide trimer (S_3O_9) is :
- (1) Three
 - (2) Two
 - (3) One
 - (4) Zero
58. Dry bleaching is done by :
- (1) Cl_2
 - (2) SO_2
 - (3) O_3
 - (4) None
59. Compounds A and B are treated with dilute HCl separately. The gases liberated are Y and Z respectively. Y turns acidified dichromate paper green while Z turns lead acetate paper black. So A and B compounds are respectively :-
- (1) Na_2SO_3 , Na_2S
 - (2) NaCl , Na_2CO_3
 - (3) Na_2S , Na_2SO_3
 - (4) Na_2SO_3 , K_2SO_4

60. When KBr is treated with conc. H_2SO_4 reddish brown gas is evolved. The gas is :
 (1) Br_2 (2) HOBr (3) NO_2 (4) H_2O_2
61. One gas bleaches the colour of the flowers by reduction while the other by oxidation. The gases are :
 (1) CO and CO_2 (2) H_2S and Br_2
 (3) SO_2 and Cl_2 (4) NH_3 and SO_3
62. On addition of conc. H_2SO_4 to a chloride salt, colourless fumes are evolved but in case of iodide salt, violet fumes come out. This is because:-
 (1) H_2SO_4 reduces HI to I_2
 (2) HI is of violet colour
 (3) HI gets oxidised to I_2
 (4) HI changes to HIO_3
63. Which of the following is responsible for turning starch-iodide paper blue when it is brought in contact with O_3 ?
 (1) Liberation of iodine
 (2) Liberation of oxygen
 (3) Formation of alkali
 (4) Reaction of ozone with litmus paper.
64. Which one of the following property is not correct for ozone ?
 (1) It oxidises lead sulphide
 (2) It oxidises potassium iodide
 (3) It oxidises mercury
 (4) It cannot act as bleaching agent in dry state.
65. By which of the following SO_2 is formed ?
 (1) Reaction of dilute H_2SO_4 with O_2 .
 (2) Heating $\text{Fe}_2(\text{SO}_4)_3$.
 (3) Reaction of concentrated H_2SO_4 with Cu .
 (4) None.
66. Pick out the statement/s that is/are wrong :-
 (1) Oxygen is paramagnetic in all the three states of matter
 (2) Ozone is diamagnetic
 (3) Ozone is a linear molecule
 (4) The O-O bonds in ozone have considerable double bond character
67. **Column-I** **Column-II**
 (A) Caro's acid (p) - S - S - bond
 (B) Polythionic acid (q) - S = S bond
 (C) Pyrosulphuric acid (r) - S - O - S - bond
 (D) Thiosulphuric acid (s) - S - O - O - bond
 (1) A-p, B-q, C-r, D-s (2) A-p, B-q, C-s, D-r
 (3) A-s, B-p, C-r, D-q (4) A-q, B-s, C-r, D-p

HALOGEN FAMILY AND INERT GASES

68. Select the correct statement (s) from the following-
 (1) Fluorine displaces other halogens from the corresponding halides
 (2) Fluorine reacts slowly with halogens
 (3) Fluorine does not decompose water
 (4) Except fluorine, other halogens directly combine with carbon
69. Which of the following is a false statement
 (1) Halogens are oxidizing agent
 (2) Halogens show only (-1) oxidation state
 (3) HF molecules form intermolecular H-bonds
 (4) Fluorine is highly reactive
70. The halide which does not give a precipitate with AgNO_3 is-
 (1) F^- (2) Cl^- (3) Br^- (4) I^-
71. Volatile nature of halogen is because :
 (1) Halogen molecules are bonded by strong forces
 (2) Halogen molecules are bonded by electrostatic forces
 (3) The forces existing between the discrete molecule are only weak vander Waals force.
 (4) Halogen molecules are more reactive
72. Iodine gas turns starch iodide paper :
 (1) Blue (2) Red
 (3) Colourless (4) Yellow
73. BrF_5 is a :
 (1) Interhalogen compound
 (2) Pseudohalogen compound
 (3) Both the above
 (4) None of the above
74. Which of the following statements is not correct when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 :
 (1) A deep red vapour is evolved
 (2) The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
 (3) Chlorine gas is evolved
 (4) Chromyl chloride is formed
75. Which of the following does not decolourise iodine?
 (1) Na_2SO_3 (2) $\text{Na}_2\text{S}_2\text{O}_3$
 (3) NaCl (4) NaOH
76. Which one of the following halogen liberates oxygen on reacting with H_2O
 (1) I_2 (2) Cl_2 (3) Br_2 (4) F_2
77. Which of the following gases can be dried by concentrated H_2SO_4 ?
 (1) HCl (2) HBr
 (3) HI (4) H_2S

- 78.** Helium is added to oxygen used by deep sea divers because :
 (1) It is less soluble in blood than nitrogen under high pressure
 (2) It is lighter than nitrogen
 (3) It is readily miscible with oxygen
 (4) It is less poisonous than nitrogen
- 79.** Which of the following is not correct :
 (1) XeO_3 has four σ and four π bonds
 (2) The hybridization of Xe in XeF_4 is sp^3d^2
 (3) Among noble gases, the occurrence of argon is highest in air
 (4) Liquid helium is used as cryogenic liquid
- 80.** XeF_2 reacts with SbF_5 to form :
 (1) $[\text{XeF}]^+ [\text{SbF}_6]^-$ (2) $[\text{XeF}_3]^+ [\text{SbF}_4]^-$
 (3) $\text{Xe}^- [\text{PtF}_6]^+$ (4) XeF_4
- 81.** Which factor is most responsible for the increase in boiling points of noble gases from He to Xe?
 (1) decrease in I.E.
 (2) Monoatomic nature
 (3) decrease in polarisability
 (4) increase in polarisability
- 82.** The compound that cannot be formed by xenon is
 (1) XeO_3 (2) XeF_4
 (3) XeCl_4 (4) XeO_2F_2
- 83.** XeF_6 dissolves in anhydrous HF to give a good conducting solution which contains
 (1) H^+ and XeF_7^- ion
 (2) HF_2^- and XeF_5^+ ions
 (3) HXeF_6^+ and F^- ions
 (4) none of these
- 84.** SbF_5 reacts with XeF_4 to form an adduct. The shapes of cation and anion in the adduct are respectively
 (1) square planar, trigonal bipyramidal
 (2) T-shaped, octahedral
 (3) square pyramidal, octahedral
 (4) square planar, octahedral
- 85.** Which of the following noble gas does not form clathrate compound?
 (1) Kr (2) Ne (3) Xe (4) Ar
- 86.** An inorganic salt when heated with concentrated H_2SO_4 evolves a colourless pungent smelling gas but with concentrated H_2SO_4 and MnO_2 evolves a coloured pungent smelling gas which bleaches moist litmus paper. The coloured gas is :-
 (1) NO_2 (2) Cl_2 (3) Br_2 (4) I_2
- 87.** Identify the incorrect statement (Structure is including the location of lone pair of electrons) :-
 (1) XeO_2F_2 has 7 bonded electron pairs and one valence shell lone pair of electron. Its structure is square pyramidal with one pi bond.
 (2) XeO_2F_2 has 6 bonded electron pairs and one valence shell lone pair of electrons. Its structure is trigonal bipyramidal with two pi bonds.
 (3) XeF_6 has 6 bonded electron pairs and one valence shell lone pair of electrons. Its structure is pentagonal bipyramidal.
 (4) XeF_2 has 5 electron pairs (bonded and valence shell lone pairs both) and its structure is trigonal bipyramidal.

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	3	1	2	3	4	4	2	4	4	1	3	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	2	2	1	1	3	2	1	2	3	3	1	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	2	3	1	3	1	3	3	1	4	1	2	3	1	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	4	4	4	4	4	4	1	4	4	1	4	3	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	3	1	4	3	3	3	1	2	1	3	1	1	3	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87			
Ans.	4	1	1	1	1	4	3	2	2	2	2	1			

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

1. Which of the following is **not** a correct statement?
 (1) Multiple bonds are always shorter than corresponding single bonds
 (2) The electron-deficient molecules can act as Lewis acids
 (3) The canonical structures have no real existence
 (4) Every AB_5 molecule does in fact have square pyramid structure.

AIIMS 2006

2. The colour imparted by Co(II) compounds to glass is –
 (1) Green (2) Deep-Blue
 (3) Yellow (4) Red
3. Tincture of iodine is –
 (1) Aqueous solution of I_2
 (2) Solution of I_2 in aqueous KI
 (3) Alcoholic solution of KI with I_2
 (4) Aqueous solution of KI
4. Which two of the following salts are used for preparing iodized salt–
 (i) KIO_3 (ii) KI
 (iii) I_2 (iv) HI
 (1) (i) and (ii) (2) (i) and (iii)
 (3) (ii) and (iv) (4) (iii) and (iv)
5. Borax is used as cleansing agent because on dissolving in water it gives.
 (1) Alkaline solution (2) Acidic solution
 (3) Bleaching solution (4) Colloidal solution

AIPMT 2007

6. Which one of the following anions is present in the chain structure of silicates :
 (1) SiO_4^{4-} (2) $Si_2O_7^{6-}$
 (3) $(Si_2O_5^{2-})_n$ (4) $(SiO_3^{2-})_n$

AIPMT 2009

7. The straight chain polymer is formed by :-
 (1) Hydrolysis of $(CH_3)_2SiCl_2$ followed by condensation polymerisation
 (2) Hydrolysis of $(CH_3)_3SiCl$ followed by condensation polymerisation
 (3) Hydrolysis of CH_3SiCl_3 followed by condensation polymerisation
 (4) Hydrolysis of $(CH_3)_4Si$ by addition polymerisation

AIPMT Pre. 2010

8. Which one of the following molecular hydrides acts as a Lewis acid ?
 (1) CH_4 (2) NH_3
 (3) H_2O (4) B_2H_6

9. Oxidation states of P in $H_4P_2O_5$, $H_4P_2O_6$, $H_4P_2O_7$, are respectively :-
 (1) +3, +4, +5 (2) +3, +5, +4
 (3) +5, +3, +4 (4) +5, +4, +3
10. How many bridging oxygen atoms are present in P_4O_{10} :-
 (1) 4 (2) 2
 (3) 5 (4) 6

AIIMS 2010

11. The product obtained when one mole of diborane reacts with two mole of NH_3 at high temperature
 (1) $B_2H_6 \cdot 2NH_3$ (2) $B_3N_3H_6$
 (3) $(BN)_x$ (4) $[BH_2(NH_3)_2]^+ [BH_4]^-$

AIPMT Pre. 2011

12. Name the type of the structure of silicate in which one oxygen atom of $[SiO_4]^{4-}$ is shared ?
 (1) Linear chain silicate
 (2) Sheet silicate
 (3) Pyrosilicate
 (4) Three dimensional

AIPMT Mains 2010

13. Which of the following oxide is amphoteric :-
 (1) CO_2 (2) SnO_2 (3) CaO (4) SiO_2

AIIMS 2011

14. The wrong statement about fullerene is–
 (1) It has 5-membered carbon ring
 (2) It has 6-membered carbon ring
 (3) It has sp^2 hybridization
 (4) It has no. of 5-membered ring more than 6-membered ring
15. First compound for Xe synthesized was–
 (1) $[XeF^+][XePtF_5]^-$ (2) $[XeO_2]$
 (3) $Xe[PtF_6]$ (4) $O_2[XeF_6]$

16. Iodine oxidises sodium borohydride to gives:
 (1) B_2H_6 (2) Sodium hydride
 (3) HI (4) I_3^-

AIPMT Pre. 2012

17. Which of the following statements is not valid for oxyacids of phosphorus?
 (1) All oxyacids contain tetrahedral four coordinated phosphorus
 (2) All oxyacids contain atleast one $P=O$ unit and one $P-OH$ group
 (3) Orthophosphoric acid is used in the manufacture of triple superphosphate
 (4) Hypophosphorous acid is a diprotic acid

18. Sulphur trioxide can be obtained by which of the following reaction:

- (1) $S + H_2SO_4 \xrightarrow{\Delta}$
 (2) $H_2SO_4 + PCl_5 \xrightarrow{\Delta}$
 (3) $CaSO_4 + C \xrightarrow{\Delta}$
 (4) $Fe_2(SO_4)_3 \xrightarrow{\Delta}$

AIIMS 2012

19. How many P = O bond present in $(HPO_3)_3$?

- (1) 0 (2) 3 (3) 6 (4) 9

20. N-N bond length minimum in :-

- (1) N_2O (2) N_2O_3 (3) N_2O_4 (4) N_2O_5

21. $S_2O_8^{2-}$ have :-

- (1) S-S bond
 (2) S-O bridge
 (3) O-O bridge
 (4) All S-O bond length same

NEET-UG 2013

22. The basic structural unit of silicates is :-

- (1) SiO_4^{2-} (2) SiO^- (3) SiO_4^{4-} (4) SiO_3^{2-}

23. Which of these is not a monomer for a high molecular mass silicone polymer ?

- (1) $PhSiCl_3$ (2) $MeSiCl_3$
 (3) Me_2SiCl_2 (4) Me_3SiCl

24. Which of the following does not give oxygen on heating ?

- (1) $(NH_4)_2Cr_2O_7$ (2) $KClO_3$
 (3) $Zn(ClO_3)_2$ (4) $K_2Cr_2O_7$

25. Roasting of sulphides gives the gas X as a by product. This is colourless gas with choking smell of burnt sulphur and causes great damage to respiratory organs as a result of acid rain. Its aqueous solution is acidic, acts as reducing agent and its acid has never been isolated. The gas X is :-

- (1) SO_3 (2) H_2S (3) SO_2 (4) CO_2

AIIMS 2013

26. Bleaching powder does not contain :-

- (1) $CaCl_2$ (2) $Ca(OH)_2$
 (3) $Ca(OC)_2$ (4) $Ca(ClO)_2$

AIIMS 2014

27. sp^2 carbon is not present in :-

- (1) Fullerene (2) Graphite
 (3) Carbonic acid (4) Dry ice

28. Which of the following halogen form only one oxyacids (HOX)

- (1) F (2) Br (3) I (4) Cl

29. Rhombic sulphur is not soluble in :-

- (1) CS_2 (2) Benzene (3) Water (4) Ether

30. Which compound is used for enrichment of U_{235} concentration in nuclear reactions ?

- (1) ClF_3 (2) IF_7
 (3) U_3O_8 (4) $XePtO_4$

AIPMT 2015

31. Nitrogen dioxide and sulphur dioxide have some properties in common. Which property is shown by one of these compounds, but not by the other?

- (1) is a reducing agent
 (2) is soluble in water
 (3) is used as a food-preservative
 (4) forms 'acid-rain'

Re-AIPMT 2015

32. Strong reducing behaviour of H_3PO_2 is due to :

- (1) High oxidation state of phosphorus
 (2) Presence of two -OH groups and one P-H bond
 (3) Presence of one -OH group and two P-H bonds
 (4) High electron gain enthalpy of phosphorus

33. Which of the statements given below is incorrect?

- (1) ONF is isoelectronic with O_2N^-
 (2) OF_2 is an oxide of fluorine
 (3) Cl_2O_7 is an anhydride of perchloric acid
 (4) O_3 molecule is bent

34. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence :

- (1) $Tl < In < Ga < Al$
 (2) $In < Tl < Ga < Al$
 (3) $Ga < In < Al < Tl$
 (4) $Al < Ga < In < Tl$

AIIMS 2015

35. Which of the following statement is correct about solid state of boric acid.

- (1) H-bond is present
 (2) dissolves in water to yield H^+
 (3) shape of B is tetrahedral
 (4) None of these

36. Which of the following allotrope of phosphorous has tetrahedral discrete unit?

- (1) Red black P
- (2) White P
- (3) Red P
- (4) Red, black & white P

37. Chain growth can be controlled by

- (1) R_2SiCl_2
- (2) R_3SiCl
- (3) $RSiCl_3$
- (4) $R_4Si/SiCl_4$

38. Lowest oxidation state of phosphorous is in

- (1) H_3PO_2
- (2) H_3PO_4
- (3) $H_4P_2O_7$
- (4) H_3PO_3

39. Which one of the following element can shown +7 oxidation state with fluorine

- (1) I
- (2) Mn
- (3) F
- (4) Br

40. S-S linkage absent in

- (1) $H_2S_2O_7$
- (2) $H_2S_2O_3$
- (3) $H_2S_2O_5$
- (4) $H_2S_2O_6$

41. Which of the following gives B_2H_6 on reaction with $NaBH_4$

- (1) $LiAlH_4$
- (2) $NaOH$
- (3) I_2
- (4) None

NEET-I 2016

42. Among the following, the correct order of acidity is

- (1) $HClO_3 < HClO_4 < HClO_2 < HClO$
- (2) $HClO < HClO_2 < HClO_3 < HClO_4$
- (3) $HClO_2 < HClO < HClO_3 < HClO_4$
- (4) $HClO_4 < HClO_2 < HClO < HClO_3$

43. When copper is heated with conc. HNO_3 it produces

- (1) $Cu(NO_3)_2$ and NO_2
- (2) $Cu(NO_3)_2$ and NO
- (3) $Cu(NO_3)_2$, NO and NO_2
- (4) $Cu(NO_3)_2$ and N_2O

44. Which is the **correct** statement for the given acids?

- (1) Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid
- (2) Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid
- (3) Both are triprotic acids
- (4) Both are diprotic acids

NEET-II 2016

45. Boric acid is an acid because its molecule

- (1) accepts OH^- from water releasing proton
- (2) combines with proton from water molecule
- (3) contains replaceable H^+ ion
- (4) gives up a proton

46. AlF_3 is soluble in HF only in presence of KF . It is due to the formation of

- (1) AlH_3
- (2) $K[AlF_3H]$
- (3) $K_3[AlF_3H_3]$
- (4) $K_3[AlF_6]$

AIIMS 2016

47. At $1000^\circ C$ air is passed on coal then which product will be form :-

- (1) producer gas
- (2) water gas
- (3) CH_4
- (4) none of these

48. Which one of the following statement is not true regarding B_2H_6 :-

- (1) It react with NaH to form $NaBH_4$
- (2) It is non inflammable gas
- (3) It contain four equal B-H bonds
- (4) It react with lewis base to form always adduct

49. Which of the following is correct :-

- (1) N_2O is a coloured gas
- (2) CO is an acidic oxide
- (3) CO_2 is not absorbed in $CsOH$
- (4) N_2O is a neutral oxide

50. Which of the following has least tendency in +3 oxidation state to show disproportionation reaction.

- (1) N
- (2) P
- (3) Bi
- (4) Sb

NEET(UG) 2017

51. In which pair of ions both the species contain S-S bond?

- (1) $S_4O_6^{2-}$, $S_2O_3^{2-}$
- (2) $S_2O_7^{2-}$, $S_2O_8^{2-}$
- (3) $S_4O_6^{2-}$, $S_2O_7^{2-}$
- (4) $S_2O_7^{2-}$, $S_2O_3^{2-}$

52. It is because of inability of ns^2 electrons of the valence shell to participate in bonding that:-

- (1) Sn^{2+} is oxidising while Pb^{4+} is reducing
- (2) Sn^{2+} and Pb^{2+} are both oxidising and reducing
- (3) Sn^{4+} is reducing while Pb^{4+} is oxidising
- (4) Sn^{2+} is reducing while Pb^{4+} is oxidising

AIIMS 2017

53. Which of the following statement is incorrect about silicones ?
 (1) R_2SiCl_2 & $RSiCl_3$ are monomers of silicones
 (2) Silicones are hydrophobic
 (3) R_3SiCl is used to close polymeric chain
 (4) Polymers of $(R_3SiO)_2$
54. Boric acid in water produces ?
 (1) $H_2BO_3^-$ (2) BO_3^{3-}
 (3) $(H_4BO_4)^-$ (4) $[B(OH)]^-$
55. Which of the following is not completely soluble in conc. HNO_3 .
 (1) S_8 (2) I_2
 (3) Al (4) Zn
56. Which of the following pair of compounds gives white fumes, on exposing to moist air
 (1) $AlCl_3$ & $NiCl_2$ (2) $AlCl_3$ & $CrCl_3$
 (3) $TiCl_4$ & $SiCl_4$ (4) $SiCl_4$ & $SnCl_4$
57. Which of the following covalent hydride having highest lewis acidic character ?
 (1) $Al(CH_3)_3$ (2) $NaBH_4$
 (3) B_2H_6 (4) AsH_3
58. The structure of graphite belongs to :-
 (1) Monoclinic (2) Triclinic
 (3) Hexagonal (4) Rhombohedral

NEET(UG) 2018

59. Which of the following statements is **not** true for halogens ?
 (1) All form monobasic oxyacids.
 (2) All are oxidizing agents.
 (3) All but fluorine show positive oxidation states.
 (4) Chlorine has the highest electron-gain enthalpy.
60. The correct order of N-compounds in its decreasing order of oxidation states is
 (1) HNO_3 , NO, N_2 , NH_4Cl
 (2) HNO_3 , NO, NH_4Cl , N_2
 (3) HNO_3 , NH_4Cl , NO, N_2
 (4) NH_4Cl , N_2 , NO, HNO_3

AIIMS 2018

61. Halides of which of the following element always have only one lone pair on central atom :-
 (1) Xe (2) Cl
 (3) Se (4) N
62. Which of the following statement is not correct :-
 (1) Black phosphorus is more stable than white & red phosphorus
 (2) Red phosphorus is highly toxic
 (3) White phosphorus is highly soluble in CS_2
 (4) Red phosphorus converts in black on heating
63. The best compound which convert nitrite in N_2 is:-
 (1) Urea
 (2) Thiourea
 (3) Sulphamic acid
 (4) Ammonium chloride
64. Which pair not shows same shape :-
 (1) I_3^- , $BeCl_2$ (2) BF_3 , ClO_3^-
 (3) SO_2 , H_2O (4) XeF_4 , BrF_4^-
65. Which one is wrong :-
 (1) Pyroxene is a linear silicate
 (2) Zeolite is a 2 dimensional double strand silicate
 (3) Beryl is a cyclic silicate
 (4) Pyrex is a man-made silicate
66. When cyanogen gas is passed through strong alkali then product formed is ?
 (1) OCN^- (2) N_2
 (3) NCN^- (4) NH_2CH
67. Which of the following insoluble in dil HCl :-
 (1) $SnCl_2$ (2) $PbCl_2$
 (3) $CdCl_2$ (4) $BiOCl$
68. Which gas will diffuse fastest in Rubber
 (1) Xenon (2) Fluorine
 (3) Oxygen (4) Helium
69. Which will give Blue colour with SO_2
 (1) I_2 + Cellulose (2) Iodate + Starch
 (3) $CuSO_4$ + KI (4) I_2 solution
70. Which of the following will not convert N_2H_4 into N_2
 (1) Oxygen (2) $CuSO_4$
 (3) Ozone (4) Zn + HCl

71. Which change will not take place when NaCl and MnO_2 is placed in conc. H_2SO_4
- (1) Oxidation state of sulphur will change
 - (2) Cl^- will be oxidised to Cl_2
 - (3) Greenish-yellow coloured vapour will be of Cl_2
 - (4) Oxidation state of Mn^{+4} is change to Mn^{+2}

72. Which is incorrect?
- (1) ΔG_f° of Br_2 is maximum in halogen.
 - (2) Bond dissociation energy of Br_2 is more than F_2 .
 - (3) Hydration energy of F^- is more than I^-
 - (4) I_2 is more soluble in water than Br_2

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	2	3	1	1	4	1	4	1	4	2	3	2	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	4	4	2	1	3	3	4	1	3	4	4	1	3	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	3	2	4	1	2	2	1	1	1	3	2	1	2	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	1	2	4	3	1	4	4	3	3	4	3	3	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72			
Ans.	4	2	3	2	2	1	2	4	2	4	1	1			

EXERCISE-III (Analytical Questions)

Check Your Understanding

- One mole of magnesium nitride on the reaction with an excess of water gives
(1) Two moles of ammonia (2) One mole of nitric acid
(3) One mole of ammonia (4) Two moles of nitric acid
- Which of the following oxides of nitrogen is solid :-
(1) NO_2 (2) N_2O
(3) NO (4) N_2O_5
- Which has no S-S-bond.
(1) $\text{S}_2\text{O}_4^{-2}$ (2) $\text{S}_2\text{O}_5^{-2}$
(3) $\text{S}_2\text{O}_3^{-2}$ (4) $\text{S}_2\text{O}_7^{-2}$
- The products obtained on heating LiNO_3 will be:-
(1) $\text{LiNO}_2 + \text{O}_2$
(2) $\text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$
(3) $\text{Li}_3\text{N} + \text{O}_2$
(4) $\text{Li}_2\text{O} + \text{NO} + \text{O}_2$
- Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ?
(1) NH_4NO_3 (2) NaNO_3
(3) KClO_3 (4) CaCO_3
- An example of a cyclic silicate is :-
(1) Beryl (2) Zeolite
(3) Talc (4) Zircon
- Which one of the following reacts with glass ?
(1) H_2SO_4 (2) HF
(3) HNO_3 (4) $\text{K}_2\text{Cr}_2\text{O}_7$
- When I_2 is passed through KCl , KF , KBr solutions :
(1) Cl_2 and Br_2 are evolved
(2) Cl_2 is evolved
(3) Cl_2 , Br_2 , F_2 are evolved
(4) None of these
- Ammonia can be dried by
(1) Conc. H_2SO_4 (2) P_4O_{10}
(3) CaO (4) Anhydrous CaCl_2
- Which of the following product is not formed by the reaction of PH_4I and KOH :-
(1) KI (2) PH_3
(3) H_2O (4) P_2O_3
- Out of the following which does not liberate N_2 gas on heating :-
(1) $\text{NaNO}_2 + \text{NH}_4\text{Cl}$ (2) $\text{Ba}(\text{N}_3)_2$
(3) $\text{O}_3 + \text{NO}$ (4) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- Which of the following is incorrect ?
(1) O_2 is weaker oxidant than O_3
(2) O_2 has small bond length than O_3
(3) Both O_2 and O_3 are paramagnetic
(4) O_3 is angular in shape
- Which of the following statements regarding sulphur is incorrect ?
(1) At 600°C the gas mainly consists of S_2 molecules
(2) The oxidation state of sulphur is never less than +4 in its compounds
(3) S_2 molecule is paramagnetic
(4) The vapour at 200°C consists mostly of S_8 rings
- The correct order of acidic strength is :
(1) $\text{Cl}_2\text{O}_7 > \text{SO}_3 > \text{P}_4\text{O}_{10}$
(2) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
(3) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
(4) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
- Which of the following statements is correct ?
(1) BCl_3 and AlCl_3 are both Lewis acids and BCl_3 is stronger than AlCl_3
(2) BCl_3 and AlCl_3 both Lewis acids and AlCl_3 is stronger than BCl_3
(3) BCl_3 and AlCl_3 are both equally strong Lewis acids
(4) Both BCl_3 and AlCl_3 are not Lewis acids.
- Which of the following statements about H_3BO_3 is not correct
(1) It is a strong tribasic acid
(2) It is prepared by acidifying an aqueous solution of borax
(3) It has a layer structure in which planar H_3BO_3 units are joined by hydrogen bonds
(4) It does not act as proton donor but acts as a Lewis acid by accepting hydroxyl ion
- Which of the following property is not related with PH_3
(1) It is a colorless gas having rotten fish smell
(2) it is non poisonous
(3) it is slightly soluble in water
(4) it is a weak Lewis base

18. Which of the following is a mixed anhydride

- (1) P_4O_{10} (2) SO_3
(3) Cl_2O_6 (4) SO_2

19. In which of the following phosphorous atoms are at the corner of tetrahedron.

- (1) P_4 (2) P_4O_6
(3) P_4O_{10} (4) All of these

20. In which of the following option product gas X and Y (other than water vapour) are same ?

- (1) $Mg_2C_3 + H_2O \rightarrow X$; $Al_4C_3 + H_2O \rightarrow Y$
(2) $NH_4NO_3 \xrightarrow{\Delta} X$; $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Y$
(3) $NH_4Cl \xrightarrow{\Delta} X$; $UREA \xrightarrow{H_2O} Y$
(4) $Zn + \text{dil. } HNO_3 \rightarrow X$; $Ag + \text{dil. } HNO_3 \rightarrow Y$

21. When con. H_2SO_4 is added to charcoal :

- (1) There is no reaction
(2) Water gas is formed
(3) SO_2 and CO_2 are evolved
(4) CO and SO_2 are evolved

22. Which of the following statements are correct for SO_2 gas?

- (1) It acts as bleaching agent in moist conditions.
(2) It's molecule has linear geometry.
(3) It's dilute solution is used as lubricant
(4) It can be prepared by the reaction of dilute H_2SO_4 with metal sulphide

23. Iodine is placed between two liquids C_6H_6 and water then :

- (1) It dissolves more in C_6H_6
(2) It dissolves more in water
(3) It dissolves equally in both
(4) Does not dissolve in both

24. Chromyl chloride vapours are dissolved in water and acetic acid and barium acetate solution is added, then :-

- (1) The solution will remain colourless
(2) The solution will become dark green
(3) A yellow solution will be obtained
(4) A yellow precipitate will be obtained

25. The correct order of sulphur-oxygen bond length is :-

- (1) $SOF_2 > SOCl_2 > SOBr_2$
(2) $SOCl_2 > SOBr_2 > SOF_2$
(3) $SOF_2 = SOCl_2 = SOBr_2$
(4) $SOBr_2 > SOCl_2 > SOF_2$

26. Which is/are wrong about P_4O_{10} molecule :-

- (1) Each 'P' atom can be considered to be sp^3 hybridised
(2) There are six P-O-P bonds in the molecule
(3) There are two types of P-O bond lengths
(4) P-O-P angle is 180°

27. Which of the following order(s) is/are incorrect ?

- (1) $H_3PO_4 > H_3PO_3 > H_3PO_2$ (reducing character)
(2) $N_2O < NO < N_2O_3 < N_2O_5$ (oxidation state on nitrogen atom)
(3) $NH_3 > PH_3 > AsH_3 > SbH_3$ (basicity)
(4) $HClO < HClO_2 < HClO_3 < HClO_4$ (Acidic nature)

28. Which of the following is correct :-

- (1) H_3BO_3 can be considered to be a lewis acid
(2) The correct order of lewis acid strength order is $BF_3 > BCl_3 > BBr_3$
(3) One mole borax in aqueous solution will require one mole HCl for titration
(4) B_2H_6 can be methylated completely to give $B_2(CH_3)_6$

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	4	4	2	4	1	2	4	3	4	3	3	2	1	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	1	2	3	4	3	3	1	1	4	4	4	1	1		

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion** : CCl_4 is not hydrolysed by water
Reason : Carbon in CCl_4 is sp^3 hybridised
 (1) A (2) B (3) C (4) D
- Assertion** : Between SiCl_4 and CCl_4 only SiCl_4 reacts with water.
Reason : SiCl_4 is ionic and CCl_4 is covalent.
 (1) A (2) B (3) C (4) D
- Assertion** : PbI_4 is a stable compound.
Reason : Iodine stabilizes higher oxidation state.
 (1) A (2) B (3) C (4) D
- Assertion** : The atoms in a covalent molecule are said to share electrons, yet some covalent molecules are polar.
Reason : In polar covalent molecules, the shared electrons spend more time on the average near one of the atoms due to high EN.
 (1) A (2) B (3) C (4) D
- Assertion** : Nitrogen is unreactive at room temperature but becomes reactive at elevated temperature (on heating or in the presence of catalysts).
Reason : In nitrogen molecule, there is extensive delocalization of electrons.
 (1) A (2) B (3) C (4) D
- Assertion** :- H_3PO_2 can act as strong reducing agent and gives disproportionation reaction on heating.
Reason :- Due to presence of one $-\text{OH}$ group and two $\text{P}-\text{H}$ group, H_3PO_2 is a monobasic bronsted acid.
 (1) A (2) B (3) C (4) D
- Assertion** :- NO_2 is paramagnetic in gaseous state but become diamagnetic solid on cooling.
Reason :- In gaseous state, NO_2 exists as monomer odd electron species but on cooling it dimerise to N_2O_4 .
 (1) A (2) B (3) C (4) D
- Assertion** :- For drying H_2 , conc. H_2SO_4 cannot be used.
Reason :- conc. H_2SO_4 on absorbing H_2O form moist H_2 along with large amount of heat which burn H_2 .
 (1) A (2) B (3) C (4) D
- Assertion** :- NH_3 can be dried by quick lime.
Reason :- Quick lime is also basic in nature and no reaction take place with NH_3 .
 (1) A (2) B (3) C (4) D
- Assertion** :- $\text{Cl}_2 + \text{H}_2\text{O}$ reaction is a disproportionation reaction and HOCl is formed.
Reason :- Increase in pH will increase the yield of HOCl .
 (1) A (2) B (3) C (4) D
- Assertion** :- NO_2 and ClO_2 both being odd electron molecule dimerise.
Reason :- On dimerisation, NO_2 is converted to stable N_2O_4 molecule with even number of electrons.
 (1) A (2) B (3) C (4) D
- Assertion** :- All inter halogens are diamagnetic.
Reason :- AB type of interhalogen undergoes hydrolysis giving a halide ion derived from the smaller halogen and a hypohalite ion derived from the larger halogen.
 (1) A (2) B (3) C (4) D

- 13. Assertion :-** Anhyd. calcium chloride cannot be used to dry NH_3 .
Reason :- Anhyd. CaCl_2 forms an adduct $\text{CaCl}_2 \cdot 8\text{NH}_3$ with NH_3 .
 (1) A (2) B (3) C (4) D
- 14. Assertion :-** Surface of diamond is generally impure.
Reason :- It has dangling bonds.
 (1) A (2) B (3) C (4) D
- 15. Assertion :-** SiCl_4 is more hydrolysed than SiF_4 .
Reason :- -I effect of $\text{F} > \text{Cl}$.
 (1) A (2) B (3) C (4) D
- 16. Assertion :-** The solubility of I_2 increases in water in presence of KI .
Reason :- I_2 forms ionic polyhalide with KI .
 (1) A (2) B (3) C (4) D
- 17. Assertion :-** H_3PO_2 is monoprotic acid.
Reason :- It has H-bonding in solid state.
 (1) A (2) B (3) C (4) D
- 18. Assertion :-** O_3 is a good oxidising agent.
Reason :- O_3 very rapidly gives dioxygen.
 (1) A (2) B (3) C (4) D
- 19. Assertion :-** Sulphur is soluble in CS_2 .
Reason :- Sulphur exists in S_8 form.
 (1) A (2) B (3) C (4) D
- 20. Assertion :-** P_4O_{10} dehydrates HNO_3 .
Reason :- In P_4O_{10} oxidation state of P is +5.
 (1) A (2) B (3) C (4) D
- 21. Assertion :-** Peroxy disulphuric acid & oleum have same oxidation state.
Reason :- In oleum oxygen bridges two sulphur atoms.
 (1) A (2) B (3) C (4) D
- 22. Assertion :-** $3\text{O}_{2(g)} \longrightarrow 2\text{O}_{3(g)}$ the reaction occurs in presence of silent electric discharge.
Reason :- because at 298 K, $\Delta S = +ve$.
 (1) A (2) B (3) C (4) D
- 23. Assertion :-** Conc. H_2SO_4 on addition to HCOOH liberates CO .
Reason :- Hot conc. H_2SO_4 is a moderate oxidising agent.
 (1) A (2) B (3) C (4) D
- 24. Assertion :-** Salts of H_3PO_3 and H_3PO_4 used as fertilizers.
Reason :- Formed H_3PO_3 increases solubility of water in soil on addition of phosphate fertilizers.
 (1) A (2) B (3) C (4) D
- 25. Assertion :-** Pb_3O_4 reacts with HNO_3 forms PbO_2 .
Reason :- Pb^{+4} is most stable.
 (1) A (2) B (3) C (4) D
- 26. Assertion :-** SO_2 mixed with H_2SO_4 to make oleum.
Reason :- SO_2 in presence of water forms H_2SO_4 in air.
 (1) A (2) B (3) C (4) D
- 27. Assertion :-** Cl_2 is oxidised by hot & conc. NaOH .
Reason :- It is favoured by formation of NaClO_3 .
 (1) A (2) B (3) C (4) D
- 28. Assertion :-** XeF_4 is a good fluorinating agent.
Reason :- XeF_4 sublimes at 298 K.
 (1) A (2) B (3) C (4) D
- 29. Assertion :-** $\text{Ba}(\text{N}_3)_2$ gives highly pure N_2 .
Reason :- Atomic weight of Ba is very high.
 (1) A (2) B (3) C (4) D
- 30. Assertion :-** As temp. increases intensity of brown colour of NO_2 decreases.
Reason :- Formation of NO_2 is an endothermic process.
 (1) A (2) B (3) C (4) D
- 31. Assertion :-** Black phosphorus is oxidised easily in air.
Reason :- Black phosphorus consists of polymer tetrahedral units of P_4 .
 (1) A (2) B (3) C (4) D
- 32. Assertion :-** Fluorine oxidises H_2O to O_2 but Cl_2 does not.
Reason :- Electronegativity of fluorine is greater than oxygen but EN of chlorine is lesser than oxygen.
 (1) A (2) B (3) C (4) D
- 33. Assertion :-** SO_2 (at room temp.) and SeO_2 (at higher temp. vapour condition) have same shape.
Reason :- Covalent radius of Se is somewhat larger than sulphur.
 (1) A (2) B (3) C (4) D

34. **Assertion :-** I_2O_5 oxidises CO

Reason :- Oxidation Number of Iodine is +5 in I_2O_5 .

- (1) A (2) B (3) C (4) D

35. **Assertion :-** O_2F_2 is used for the purification of Plutonium from spent nuclear fuel.

Reason :- O_2F_2 convert Plutonium to PuF_6 .

- (1) A (2) B (3) C (4) D

36. **Assertion :-** NH_3 is easily desolved in alkaline HgI_4^{-2}

Reason :- Compound of Hg and I^- is used in detection of NH_3 .

- (1) A (2) B (3) C (4) D

37. **Assertion :-** Liquid N_2 is used as refrigerant.

Reason :- Because Liquid N_2 is not Harmful for environment and its is non reactive.

- (1) A (2) B (3) C (4) D

38. **Assertion :-** 1ppm of NaF is compulsory in drinking water.

Reason :- It helps in maintaining blood pressure.

- (1) A (2) B (3) C (4) D

39. **Assertion :-** Graphene is obtained from graphite

Reason :- Graphene and graphite both have layered structure.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	4	1	3	2	1	1	1	2	4	2	1	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	2	2	2	2	3	2	1	3	4	2	2	2	4
Que.	31	32	33	34	35	36	37	38	39						
Ans.	4	2	2	2	1	1	1	1	3						

d & f-BLOCK ELEMENTS

E

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NEET SYLLABUS

d and f Block Elements : General introduction, electronic configuration, characteristics of transition metals, general trends in properties of the first row transition metals- metallic character, ionization enthalpy, oxidation states, ionic radii, colour, catalytic property, magnetic properties, interstitial compounds, alloy formation. Preparation and properties of $K_2Cr_2O_7$ and $KMnO_4$.

Lanthanoids - electronic configuration, oxidation states, chemical reactivity, and lanthanoid contraction and its consequences.

Actinoids: Electronic configuration, oxidation states and comparison with lanthanoids.

OBJECTIVES

After studying this unit, you will be able to :

- learn the positions of the d- and f-block elements in the periodic table;
- know the electronic configurations of the transition (d-block) and the inner transition (f-block) elements;
- appreciate the relative stability of various oxidation states in terms of electrode potential values;
- describe the preparation, properties, structures and uses of some important compounds such as $K_2Cr_2O_7$ and $KMnO_4$;
- understand the general characteristics of the d- and f-block elements and the general horizontal and group trends in them;
- describe the properties of the f-block elements and give a comparative account of the lanthanoids and actinoids with respect to their electronic configurations, oxidation states and chemical behaviour.

"d & f-block elements signify special people in society, on whom no social or economic rule is applicaton."

Mitch Album

d & f-BLOCK ELEMENTS

2.0 INTRODUCTION :

- (i) d-block elements are often called as 'transition elements' because their position in periodic table is between the s-block and p-block elements.
- (ii) Their properties are transitional between the highly reactive metallic elements of s-block (which form ionic compounds) and the elements of p-block (which are largely covalent).
- (iii) Typically the transition elements have an incompletely filled d-orbital. A transition element may be defined as the element whose atom in ground state or ion in one of common oxidation states, has partly filled d-sub shell i.e. having electrons between 1 to 9.
- (iv) Group 12 (the zinc group) elements have completely filled d-orbitals in the ground state as well as in their common oxidation states. Therefore, they are not regarded as transition elements.
- (v) The general electronic configuration of d-block elements is $(n-1) d^{1-10} ns^{1-2}$, where n is the outer most shell. However, palladium does not follow this general electronic configuration. It has electron configuration $[Kr]^{36} 4d^{10} 5s^0$ in order to have stability.
- (vi) Zn, Cd and Hg are involved in transition series but they are called non transition elements due to completely filled d-orbitals.

1st Series										
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10

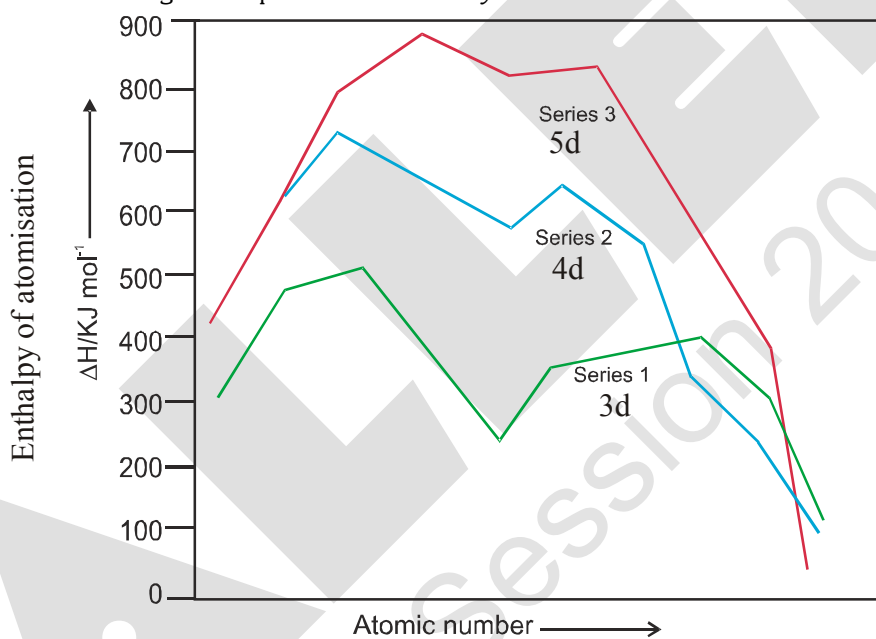
2nd Series										
	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	1	1	1	0	1	2
4d	1	2	4	5	6	7	8	10	10	10

3rd Series										
	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

2.1 GENERAL TRENDS IN THE CHEMISTRY OF TRANSITION ELEMENTS.

(A) Metallic character :

- (i) In d-block elements the last but one (i.e. the penultimate) shell of electrons is expanding. Thus they have many physical and chemical properties in common.
- (ii) Nearly all the transition elements display typical metallic properties such as high tensile strength, ductility, malleability, high thermal and electrical conductivity and metallic lustre.
- (iii) Most transition elements have ccp/hcp type structures.
- (iv) The transition elements are very much hard and have low volatility, they have high enthalpy of atomisation.
- (v) Cr, Mo and W have maximum number of unpaired electrons and therefore, these are very hard metals and have maximum enthalpies of atomisation in their respective period.
- (vi) The metals with very high enthalpy of atomisation (i.e., very high boiling point) tend to be noble in their reactions.
- (vii) The metals of the 4d and 5d series have greater enthalpies of atomisation than the corresponding elements of the 3d and this is an important factor indicating for the occurrence of much more frequent metal-metal bonding in compounds of the heavy transition metals.



Graph showing Trends in enthalpies of atomisation of transition elements

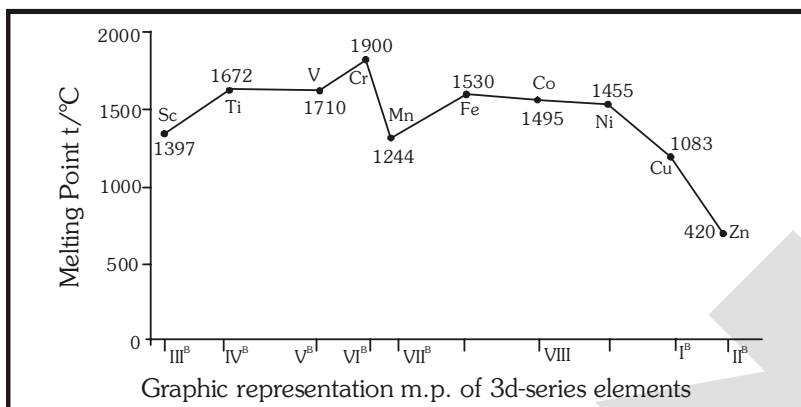
(B) MELTING AND BOILING POINTS

- (i) Melting and boiling point of d-block > s-block

Reason : Stronger metallic bond formed present by unpaired d-electrons.

- (ii) In Zn, Cd, and Hg there is no unpaired electron present in d-orbital, hence due to absence of covalent bond melting and boiling point are very low in series. (Volatile metals Zn, Cd, Hg)
- (iii) In 3d series Sc to Cr melting and boiling point increases then Mn to Zn melting and boiling point decreases
- (iv) As the number of unpaired d-electron increases, the number of covalent bond and bond energy between the atoms is expected to increase up to Cr-Mo-W family where each of the d-orbital has only unpaired electron and the opportunity for covalent sharing is greatest.

- (v) Mn and Tc have comparatively low melting point, due to weak metallic bond because of stable Half filled (d^5) configuration and high IP.
- (vi) Lowest melting point Hg (-38°C) ; Highest melting point W ($\approx 3400^\circ\text{C}$)



Illustrations

Illustration 1 : Why do the transition elements have higher boiling & melting points ?

Solution Because of having larger number of unpaired electrons in their atoms, they have stronger interatomic interaction and hence stronger bonding between atoms. Hence strong metallic bonds between the atoms of these elements attribute to their high melting and boiling points.

(C) DENSITY :

- (i) The atomic volumes of the transition elements are low compared with the elements of group 1 and 2. This is because the increased effective nuclear charge. In addition, the extra electrons added occupy inner orbitals. Consequently the densities of the transition metals are high.
- (ii) Across a period from left to right atomic volumes decrease and atomic masses increase. Hence the densities also increase across a period.

$$\text{Sc} < \text{Ti} < \text{V} < \text{Cr} < \text{Mn} < \text{Fe} < \text{Co} < \text{Ni} \approx \text{Cu} > \text{Zn}$$

(zinc is an exception, having large atomic volume and hence has lower density)

- (iii) $3d < 4d < 5d$ Series

- (iv) Maximum density : $\text{Ir} \geq \text{Os}$

(D) Ionisation energies or Ionisation enthalpies :

- (i) The first ionisation energy of Zn, Cd, and Hg are very high because of their fully filled $(n-1) d^{10} ns^2$ configuration.
- (ii) Order of second ionisation energy : $\text{Sc} < \text{Ti} < \text{V} < \text{Mn} < \text{Cr}$

(E) Oxidation states :

- (i) The transition metals exhibit a large number of oxidation states. With the exception of a few elements, most of these show variable oxidation states. These different oxidation states are related to the electronic configuration of their atoms.
- (ii) The existence of the transition elements in different oxidation states means that their atoms can lose different number of electrons. This is due to the participation of inner $(n-1) d$ -electrons in addition to outer ns -electrons because, the energies of the ns and $(n-1) d$ -sub-shells are nearly same.
- (iii) In general oxidation state of d-block element is given as minimum oxidation state = number of 's' electrons
maximum oxidation state = number of 's' electrons + unpaired 'd' electron

Different oxidation states of first transition series.

Element	Outer electronic configuration	Oxidation states
Sc	$3d^1 4s^2$	+3
Ti	$3d^2 4s^2$	+2, +3, +4
V	$3d^3 4s^2$	+2, +3, +4, +5
Cr	$3d^5 4s^1$	+2, +3, (+4), (+5), +6
Mn	$3d^5 4s^2$	+2, +3, +4, (+5), +6, +7
Fe	$3d^6 4s^2$	+2, +3, (+4), (+5), (+6)
Co	$3d^7 4s^2$	+2, +3, (+4)
Ni	$3d^8 4s^2$	+2, +3, +4
Cu	$3d^{10} 4s^1$	+1, +2
Zn	$3d^{10} 4s^2$	+2

• Oxidation states given in parenthesis are unstable.

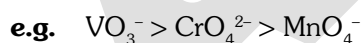
Higher oxidation state in d-block = +8 (Os, Ru)

Zero oxidation state for e.g. $[\text{Ni}(\text{CO})_4]$, $[\text{Fe}(\text{CO})_5]$

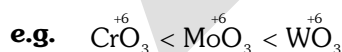
(iv) Relative stability of oxidation state :

$\text{Sc}^{+2} < \text{Sc}^{+3}$	inert gas stable configuration
$\text{Ti}^{+3} < \text{Ti}^{+4}$	
$\text{Mn}^{+6} < \text{Mn}^{+7}$	
$\text{V}^{+4} < \text{V}^{+5}$	
$\text{Mn}^{+3} < \text{Mn}^{+2}$	d^5 stable configuration
$\text{Cr}^{+2} < \text{Cr}^{+3}$	d^3 is more stable than d^4 in aq. solution
$\text{Ni}^{+2} > \text{Ni}^{+4}$	On the basis of sum of first four I.P's.
$\text{Pt}^{+2} < \text{Pt}^{+4}$	

(v) On moving in a period stability of higher oxidation state decreases

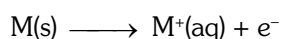


(vi) On moving down the group stability of higher oxidation state increases

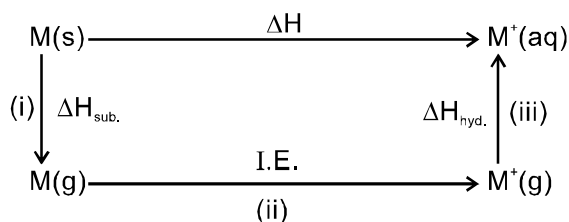


(F) ELECTRODE POTENTIALS :

In addition to ionisation enthalpy, the other factors such as enthalpy of sublimation, hydration enthalpy, ionisation enthalpy etc. determine the stability of a particular oxidation state in solution. This can be explained in terms of their electrode potential values. The oxidation potential of a metal involves the following process:



This process actually takes place in the following three steps as given in following flowchart :



The oxidation potential which gives the tendency of the overall change to occur, depends upon the net effect of these three steps. The overall energy change is $\Delta H = \Delta_{\text{sub}} H^\ominus + \Delta H + \Delta_{\text{hyd}} H$

If SOP is +ve $\rightarrow$ Means Oxidation easy

If SRP is +ve $\rightarrow$ Means reduction easy

Some important examples :

1. $E_{\text{M}^{+2}/\text{M}}^0 \Rightarrow$ +ve only for Cu among 3d elements because HE of Cu^{+2} is not enough to compensate for sublimation energy, IP_1 and IP_2 for Cu.

2. $E_{\text{M}^{+3}/\text{M}^{+2}}^0 \Rightarrow$

(a) $E_{\text{Cr}^{+3}/\text{Cr}^{+2}}^0 = -ve$

Reason : $t_2g^3 eg^0 \leftarrow d^3 > d^4$

(b) $E_{\text{Mn}^{+3}/\text{Mn}^{+2}}^0 = +ve \text{ (high)}$

Reason : $d^4 < d^5$

(c) $E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^0 = \text{less +ve (learn)}$

Reason : $d^5 > d^6$

Note : The SRP of $E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^0$ is less than expected because of extra stability of d^5 configuration of Fe^{+3} ion.

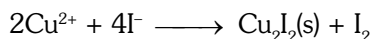
(G) TRENDS IN STABILITY OF HIGHER OXIDATION STATES

Standard electrode potential data provide valuable information about the stabilities of different oxidation states shown by an element. The highest oxidation states are shown generally among halides and oxides.

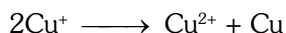
Halides of first transition series

- In metal halides.** The transition elements react with halogens at high temperatures to form transition metal halides. These reactions have very high heat of reaction. But once the reaction starts, the heat of reaction is sufficient to continue the reaction. The halogens react in the following decreasing order ; $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- In general, the elements of first transition series tend to exist in low oxidation states. Chromium to zinc form stable difluorides and the other chlorides are also known.
- Since fluorine is the most electronegative element, the transition metals show highest oxidation states with fluorine. The highest oxidation states are found in TiX_4 (tetrahalides, $\text{X} = \text{F}, \text{Cl}, \text{Br}$ and I), VF_5 and CrF_6 .

- (iv) The +7 oxidation state for Mn is not shown by simple halides. However, MnO_3F is known in which the oxidation state of Mn is +7.
- (v) The tendency of fluorine to stabilise the highest oxidation state is due to either higher lattice enthalpy as in case of CoF_3 or higher bond enthalpy due to higher covalent bonds e.g., VF_5 and CrF_6 .
- (vi) Fluorides are relatively unstable in their low oxidation states. For example, vanadium form only VX_2 ($\text{X} = \text{Cl}, \text{Br}$ or I) and copper can form CuX ($\text{X} = \text{Cl}, \text{I}$). All copper (II) halides are known except the iodide. This is because, Cu^{2+} oxidises I^- to I_2 .



It has been observed that many copper (I) compounds are unstable in aqueous solution and they undergo disproportionation to Cu(II) and Cu(0) as :



Copper in +2 oxidation state is more stable than in +1 oxidation state. This can be explained on the basis of much larger negative hydration enthalpy ($\Delta_{\text{hyd}}H^\ominus$) of $\text{Cu}^{2+}(\text{aq})$ than Cu^+ , which is much more than compensates for the large energy required to remove the second electron i.e., second ionisation enthalpy of copper.

(H) IN METAL OXIDES AND OXOCATIONS.

- (i) The ability of oxygen to stabilize the highest oxidation state is demonstrated in their oxides. The highest oxidation states in their oxides coincides with the group number. For example, the highest oxidation state of scandium of group 3 is +3 in its oxides, Sc_2O_3 whereas the highest oxidation state of manganese of group 7 is +7, in Mn_2O_7 .
- (ii) Besides the oxides, oxocation of the metals also stabilise higher oxidation states. For example, V^{V} as VO_2^+ , V^{IV} as VO^{2+} and Ti^{IV} as TiO^{2+} .
- (iii) It may be noted that the ability of oxygen to stabilise these high oxidation states exceeds that of fluorine. For example, manganese forms highest fluoride as MnF_4 whereas the highest oxide is Mn_2O_7 . This is due to the fact that oxygen has great ability to form multiple bonds to metals.
- (iv) The transition elements in the +2 and +3 oxidation states mostly form ionic bonds whereas with higher oxidation states, the bonds are essentially covalent e.g., in MnO_4^- all bonds are covalent. As the oxidation number of a metal increases, the ionic character of their oxides decrease. For example, in case of Mn, Mn_2O_7 is a covalent. In these higher oxides the acidic character is predominant. Thus CrO_3 gives H_2CrO_4 and $\text{H}_2\text{Cr}_2\text{O}_7$ and Mn_2O_7 gives HMnO_4 . **V_2O_5 is, however amphoteric though mainly acidic and with alkalis as well as acids gives VO_4^{3-} and VO_2^+ respectively.**

In general

Oxidation state	Nature of oxide/Hydroxide (d-block)
+1, +2, +3	Basic
+4	Amphoteric
+5, +6, +7, +8	Acidic

Except $\text{Cr}_2\text{O}_3 = \text{Amphoteric}$

(I) **FORMATION OF COLOURED IONS** : See coordination compound

(J) MAGNETIC PROPERTIES

It is interesting to note that when the various substances are placed in a magnetic field, they do not behave in a similar way i.e., they show different behaviour which are known as magnetic behaviour.

These are classified as :

- (i) **Paramagnetic substances.** The substances which are attracted by magnetic field are called paramagnetic substances and this character arises due to the presence of unpaired electrons in the atomic orbitals.
- (ii) **Diamagnetic substances.** The substances which are repelled by magnetic field are called diamagnetic substances and this character arises due to the presence of paired electrons in the atomic orbitals. Most of the compounds of transition elements are paramagnetic in nature and are attracted by the magnetic field.
- The transition elements involve the partial filling of d-sub-shells. Most of the transition metal ions or their compounds have unpaired electrons in d-sub-shell (from configuration d^1 to d^9) and therefore, they give rise to paramagnetic character.
- (iii) The magnetic moment arise only from the spin of the electrons. This can be calculated from the relation $\mu = \sqrt{n(n+2)}$ B.M. n = Number of unpaired electrons

Table : Calculated and Observed Magnetic Moments (BM)

Ion	Configuration	Unpaired electron(s)	Magnetic moment	
			Calculated	Observed
Sc ³⁺	3d ⁰	0	0	0
Ti ³⁺	3d ¹	1	1.73	1.75
Ti ²⁺	3d ²	2	2.84	2.76
V ²⁺	3d ³	3	3.87	3.86
Cr ²⁺	3d ⁴	4	4.90	4.80
Mn ²⁺	3d ⁵	5	5.92	5.96
Fe ²⁺	3d ⁶	4	4.90	5.3 – 5.5
Co ²⁺	3d ⁷	3	3.87	4.4 – 5.2
Ni ²⁺	3d ⁸	2	2.84	2.9 – 3.4
Cu ²⁺	3d ⁹	1	1.73	1.8 – 2.2
Zn ²⁺	3d ¹⁰	0	0	

(K) **FORMATION OF COMPLEXES :** See coordination compound

(L) **FORMATION OF INTERSTITIAL COMPOUNDS**

Transition metals form interstitial compounds with elements such as hydrogen, boron, carbon and nitrogen. The small atoms of these non-metallic elements (H, B, C, N, etc.) get trapped in vacant spaces of the lattices of the transition metal atoms as shown below.

e.g. WC, Fe₃C, TiH_{1.30 - 1.80}

Properties :

- (1) They are generally non-stoichiometric compound.
- (2) Neither typical ionic nor covalent
- (3) Harder than pure metal
- (4) High melting point than pure metal
- (5) Malleability and ductility less than pure metal
- (6) Chemically inert

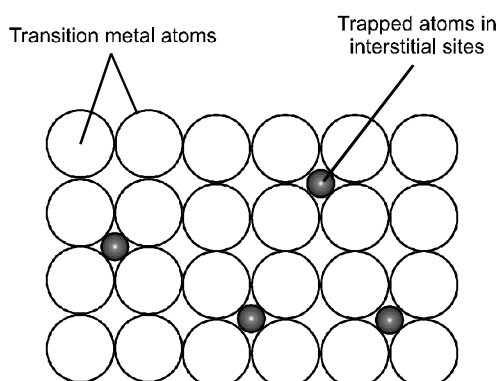


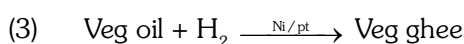
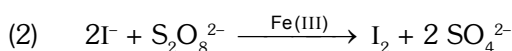
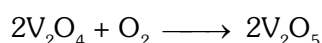
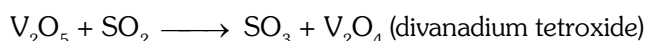
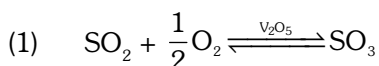
Figure showing formation of interstitial compounds

(M) CATALYTIC PROPERTIES

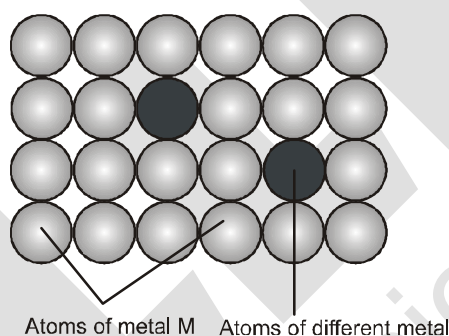
d-block metals or their compound act as catalyst in many reaction due to :

- Possibility of variable oxidation state
- Presence of free valencies over metal surface (adsorption power)
- Presence of vacant d-orbital

e.g.

**(N) ALLOY FORMATION :**

Alloys are homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other metal. The alloys are generally formed by those atoms which have metallic radii within about 15% of each other. Transition metals form a large number of alloys. The transition metals are quite similar in size and therefore the atoms of one metal can substitute the atoms of other metal in its crystal lattice.

**Alloy formation**

Note : Fe, Co, Ni do not form amalgam with Hg (Due to large difference in size)

BEGINNER'S BOX-1

- The general electronic configuration of d-block elements is $(n-1)d^{1-10} ns^{1-2}$. This generalisation has several exceptions mainly due to
 - strong shielding of $(n-1)$ d-electrons
 - very little energy difference between $(n-1)d$ and ns orbitals
 - presence of unpaired electrons
 - high energy difference between $(n-1)d$ and ns orbitals
- Select group of transition elements

(1) Pb, Sc, V	(2) Pd, Cd, La	(3) Pd, La, Fe	(4) Zn, Pd, Bi
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- Which element is having lowest melting and boiling point :

(1) Ti	(2) Cu	(3) Zn	(4) Mn
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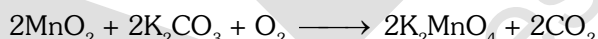
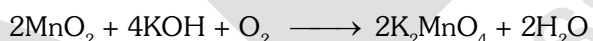
4. Density of which of the following element is highest
(1) Pt (2) Hg (3) Mn (4) Cu
5. Not more than one oxidation state is shown by :
(1) Mn (2) Cr (3) Fe (4) Sc
6. Consider the ions Fe^{2+} , Fe^{3+} , Cr^{2+} , Cr^{3+} , Mn^{2+} , Mn^{3+}
(i) Strongest oxidising ion is
(1) Fe^{2+} (2) Cr^{3+} (3) Mn^{3+} (4) Mn^{2+}
(ii) Strongest reducing ion is
(1) Fe^{2+} (2) Mn^{2+} (3) Cr^{2+} (4) Mn^{3+}
7. Which compound does exist
(1) MnO_3F (2) $\text{K}_2[\text{CuI}_4]$ (3) MnF_7 (4) CrF_7
8. In the form of dichromate, Cr(VI) is a strong oxidising agent in acidic medium but Mo(VI) in MoO_3 and W(VI) in WO_3 are not because
(1) Cr(VI) is more stable than Mo(VI) & W(VI)
(2) Mo(VI) and W(VI) are less stable than Cr(VI)
(3) H.O.S. of heavier members of group 6 of transition metals are more stable
(4) L.O.S. of heavier members of group 6 of transition metals are more stable
9. By which of the following species, a transitional metal can brought in to its highest oxidation state
(1) F_2 (2) Cl_2 (3) Br_2 (4) I_2

2.2 COMPOUNDS OF d-BLOCK ELEMENTS :

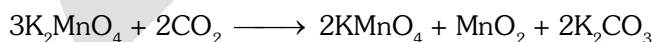
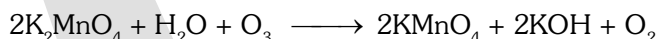
POTASSIUM PERMANGANATE (KMnO_4) :

(a) Preparation

This is the most important and well known salt of permanganic acid and is prepared from the pyrolusite ore. It is prepared by fusing pyrolusite ore either with KOH or K_2CO_3 in presence of atmospheric oxygen or any other oxidising agent such as KNO_3 . The fused mass turns green with the formation of potassium manganate, K_2MnO_4 .

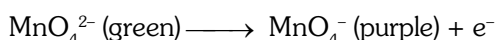


The fused mass is extracted with water and the solution is now treated with a current of chlorine or ozone or carbon dioxide to convert manganate into permanganate.

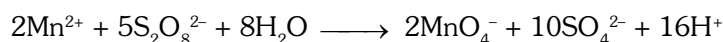


(disproportionate in acidic medium)

Commercially it is prepared by fusion of MnO_2 with KOH followed by electrolytic oxidation of manganate.



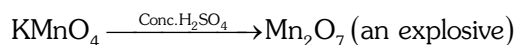
In the laboratory, a manganese(II) ion salt is oxidised by peroxodisulphate to permanganate



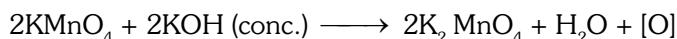
- (b) **Physical property :** It is purple coloured crystalline compound. It is moderately soluble in water at room temperature.

(c) Chemical property:

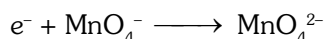
- Effect of heating** $2\text{KMnO}_4 \xrightarrow{750\text{K}} \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
- On treatment with concentrated H_2SO_4 (KMnO_4 is taken in excess), it forms manganese heptoxide via permanganyl sulphate which decomposes explosively on heating.



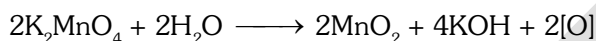
- Potassium permanganate is a powerful oxidising agent.
Potassium permanganate acts as an oxidising agent in alkaline, neutral or acidic solutions.
- In alkaline & neutral medium :
In strongly alkaline medium KMnO_4 is reduced to manganate.



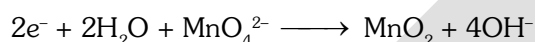
or



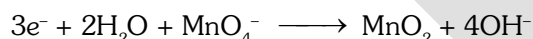
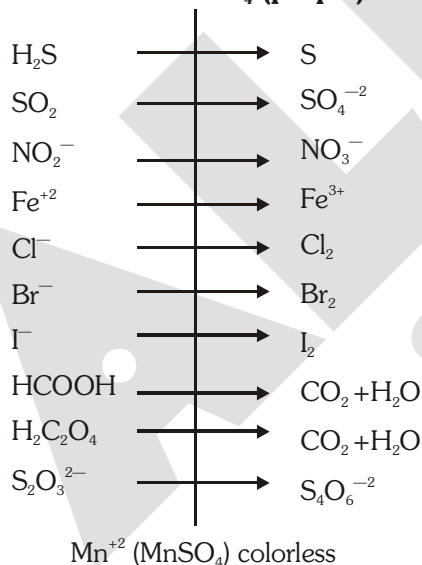
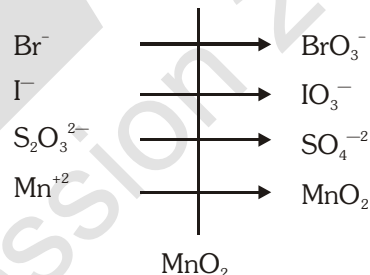
However if solution is dilute then K_2MnO_4 is converted in to MnO_2 which appears as a brownish precipitate.



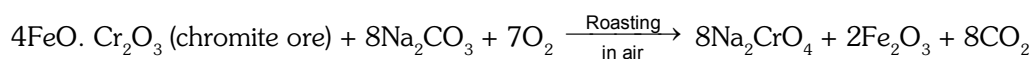
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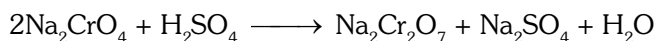
This type of behaviour is shown by KMnO_4 itself in neutral medium

**(d) Other reactions****acidified KMnO_4 (purple)****Neutral or weak alkaline KMnO_4** **POTASSIUM DICHROMATE ($\text{K}_2\text{Cr}_2\text{O}_7$) :**

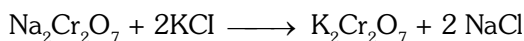
- (a) Preparation :** The chromite ore is roasted with sodium carbonate in presence of air in a reverberatory furnace



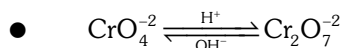
The roasted mass is extracted with water when Na_2CrO_4 goes into the solution leaving behind insoluble Fe_2O_3 . The solution is then treated with calculated amount of H_2SO_4 .



The solution is concentrated when less soluble Na_2SO_4 crystallises out. The solution is further concentrated when crystals of $\text{Na}_2\text{Cr}_2\text{O}_7$ are obtained. Hot saturated solution of $\text{Na}_2\text{Cr}_2\text{O}_7$ is then treated with KCl when orange red crystals of $\text{K}_2\text{Cr}_2\text{O}_7$ are obtained on crystallisation.



- $\text{K}_2\text{Cr}_2\text{O}_7$ is preferred over $\text{Na}_2\text{Cr}_2\text{O}_7$ as a primary standard in volumetric estimation because $\text{Na}_2\text{Cr}_2\text{O}_7$ is hygroscopic in nature but $\text{K}_2\text{Cr}_2\text{O}_7$ is not.



- Solubility of $\text{Na}_2\text{Cr}_2\text{O}_7$ is higher than $\text{K}_2\text{Cr}_2\text{O}_7$

(b) **Physical property :** It is orange-red coloured crystalline compound. It is moderately soluble in cold water but freely soluble in hot water. It melts at 398°C .

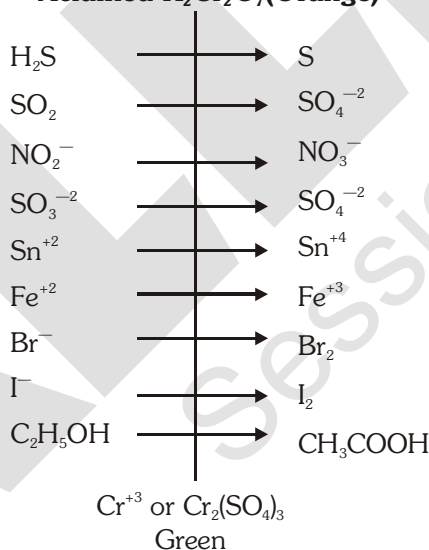
(c) **Chemical property :**

- **Effect of heating :** On strongly heating, it decomposes with liberating oxygen.

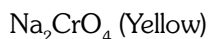
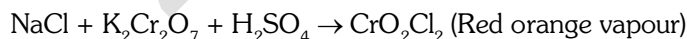


(d) **Other reactions**

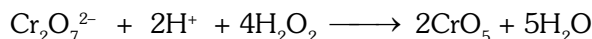
Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ (Orange)



1. **Chromyl chloride test :**



2. Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution reacts with H_2O_2 in an ethereal solution to give a deep blue solution due to the formation of CrO_5 .



Note : Reaction with $\text{C}_2\text{H}_5\text{OH}$ is used in drunken driver test.

f-BLOCK ELEMENTS

2.3 INNER TRANSITION ELEMENTS

The elements in which the additional electron enters in $(n-2)f$ orbitals are called **inner transition elements** or **f-block elements**.

Position in The Periodic Table

The lanthanoids resemble with Yttrium in most of their properties. So it became necessary to accommodate all the fifteen elements together at one place. This has been done by placing the first element, lanthanum below yttrium and placing the remaining fourteen elements separately in the lower part of the periodic table.

Lanthanoid series (Z = 58 – 71) (Ce – Lu)

Actinide series (Z = 90 – 103) (Th – Lr)

2.4 LANTHANIDS (Rare Earths or Lanthanones)

- (i) Lanthanoids are reactive elements so do not found in free state in nature.
- (ii) Most important minerals for lighter Lanthanoids are – Monazite, cerites and orthite and for heavier lanthanoids – Gadolinite and Xenotime

(a) Electronic configuration

- (i) The general configuration of lanthanoids may be given as $[\text{Xe}] 4f^{1-14} 5s^2 5p^6 5d^{0-1} 6s^2$.

Atomic No.	Element	Symbol	Outer electronic configuration	
			Atomic	+3 ion
58	Cerium	Ce	$4f^1 5d^1 6s^2$	$4f^1$
59	Praseodymium	Pr	$4f^3 6s^2$	$4f^2$
60.	Neodymium	Nd	$4f^4 6s^2$	$4f^3$
61.	Promethium	Pm	$4f^5 6s^2$	$4f^4$
62.	Samarium	Sm	$4f^6 6s^2$	$4f^5$
63.	Europium	Eu	$4f^7 6s^2$	$4f^6$
64.	Gadolinium	Gd	$4f^7 5d^1 6s^2$	$4f^7$
65.	Terbium	Tb	$4f^9 6s^2$	$4f^8$
66.	Dysprosium	Dy	$4f^{10} 6s^2$	$4f^9$
67.	Holmium	Ho	$4f^{11} 6s^2$	$4f^{10}$
68.	Erbium	Er	$4f^{12} 6s^2$	$4f^{11}$
69.	Thulium	Tm	$4f^{13} 6s^2$	$4f^{12}$
70.	Ytterbium	Yb	$4f^{14} 6s^2$	$4f^{13}$
71.	Lutecium	Lu	$4f^{14} 5d^1 6s^2$	$4f^{14}$

- (ii) It is to be noted that filling of 4f orbitals in the atoms is not regular. A 5d electron appears in gadolinium (Z = 64) with an outer electronic configuration of $4f^7 5d^1 6s^2$ (and not $4f^8 6s^2$). This is because the 4f and 5d electrons are at about the same potential energy and that the atoms have a tendency to retain stable half filled configuration.
- (iii) On the other hand, the filling of f-orbitals is regular in tripositive ions.
- (iv) After losing outer electrons, the f-orbitals shrink in size and became more stable.
- (v) **Pm** is the only synthetic radioactive lanthanoid.

(b) Oxidation states

- (i) In lanthanoids +3 oxidation state is most common.
- (ii) **Except Ce, Tb, Nd, Dy, Pr** which also exhibit +4.
- (iii) **Sm, Eu, Tm, and Yb**, also exhibit +2 state.

(c) Magnetic Properties

- (i) In tripositive lanthanoid ions the number of unpaired electrons regularly increases from lanthanum to Gadolinium (0 to 7) and then continuously decreases upto lutecium (7 to 0).
- (ii) lanthanum and lutecium ions are diamagnetic, while all other tripositive lanthanoid ions are paramagnetic. (Exception – Neodymium is the most paramagnetic lanthanoid).
- (iii) Ce^{+4} and Yb^{+2} are also diamagnetic ions.

(d) Colour

- (i) The lanthanoid ions have unpaired electrons in their 4f orbitals. Thus these ions absorb visible region of light and undergo f-f transition and hence exhibit colour.
- (ii) The colour exhibited depends on the number of unpaired electrons in the 4f orbitals.
- (iii) The ions often with $4f^n$ configuration have similar colour to those ions having $4f^{14-n}$ configuration.
- (iv) Lanthanoid ions having $4f^0$, $4f^{14}$ are colourless.

(e) Lanthanoid Contraction

- (i) In the lanthanoid series with increasing atomic number, there is a progressive decrease in the size from lanthanum to lutecium or from La^{+3} to Lu^{+3} . This contraction in size is known as lanthanoid contraction.
- (ii) The general electronic configuration of these elements is $4f^{1-14}5s^2p^6d^{0-1}6s^2$. In these elements the added electron enters the deep seated f-orbitals and therefore experiences considerable pull by the nucleus.
- (iii) Due to very poor shielding effect of (n-2)f electrons, they exert very little screening effect on the outermost $6s^2$ electrons.

Hence with increasing atomic number, the enhanced nuclear charge leads to contraction in the size of atoms and ions.

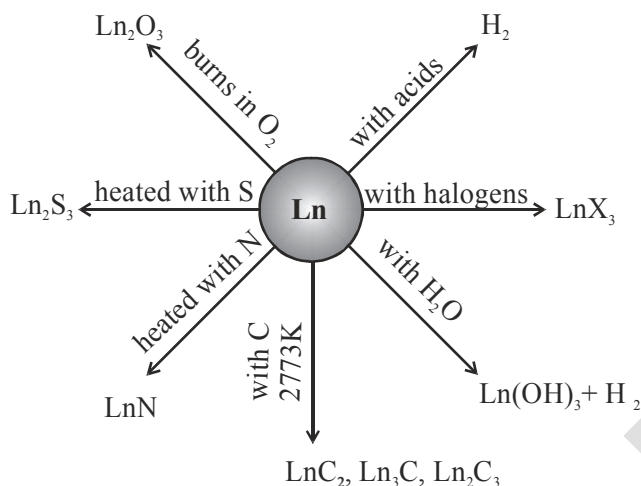
APPLICATIONS OF LANTHANOID CONTRACTION

- **Atomic size** : The ionic radii of Zr^{+4} is about 9% more than Ti^{+4} . Similar trend is not maintained on passing from the second to third transition series. The ionic radius of Hf^{+4} , instead of increasing (because of inclusion of one more electronic shell), decreases (or is virtually equal to Zr^{+4}) as a consequence of the lanthanoid contraction.

This explains the close similarities between the members of the second and third transition series than between the elements of the first and second series.

- **Basic nature of hydroxide** : The effect of lanthanoid contraction basic nature of hydroxide regularly decreases.
- **Similarity in properties of lanthanoid** : Due to small difference in size the chemical properties of lanthanoids are similar and their separation is difficult.

(e) Important reactions of Lanthanoids



Chemical reactions of the lanthanoids

Mish metal is an alloy of lanthanoids (Mainly cerium) and iron.

2.5 ACTINOIDS OR (5f - SERIES)

- The elements in which the extra electron enters 5f-orbitals of $(n - 2)$ th main shell are known as actinoids.
- The man-made eleven elements $\text{Np}_{93} - \text{Lr}_{103}$ are placed beyond uranium in the periodic table and are collectively called trans-uranium elements.
- Th, Pa and U first three actinoids are natural elements.

(a) Electronic Configuration

The general configuration of actinoids may be given as $[\text{Rn}] 5f^{1-14} 6d^{0,1,2} 7s^2$.

Atomic No.	Elements	Symbol	Outer Electronic Configuration	
			Atomic	+3 ion
90	Thorium	Th	$6d^2 7s^2$	$5f^1$
91	Proactenium	Pa	$5f^2 6d^1 7s^2$	$5f^2$
92	Uranium	U	$5f^3 6d^1 7s^2$	$5f^3$
93	Neptunium	Np	$5f^4 6d^1 7s^2$	$5f^4$
94	Plutonium	Pu	$5f^6 6d^0 7s^2$	$5f^5$
95	Americium	Am	$5f^6 6d^1 7s^2$	$5f^6$
96	Curium	Cm	$5f^7 6d^1 7s^2$	$5f^7$
97	Berkellium	Bk	$5f^9 6d^0 7s^2$	$5f^8$
98	Californium	Cf	$5f^{10} 6d^0 7s^2$	$5f^9$
99	Einsteinium	Es	$5f^{11} 6d^0 7s^2$	$5f^{10}$
100	Fermium	Fm	$5f^{12} 6d^0 7s^2$	$5f^{11}$
101	Mandeleium	Md	$5f^{13} 6d^0 7s^2$	$5f^{12}$
102	Nobellium	No	$5f^{14} 6d^0 7s^2$	$5f^{13}$
103	Lowrencium	Lr	$5f^{14} 6d^1 7s^2$	$5f^{14}$

(b) Oxidation states

- In lanthanoids and actinoids +3 oxidation is the most common for both of the series of elements.
- This oxidation state becomes increasingly more stable as the atomic number increases in the actinide series.

- (iii) Highest oxidation states in the actinoids is +7 exhibited by Np_{93} & Pu_{94} , it is unstable.
 (iv) Highest stable oxidation state is +6 shown by U_{92} .

Other Properties

- **Physical appearance** : Actinides are silvery white metals. They get tarnished when exposed to the attack of alkalis.
- **Density** : All the actinoids except **thorium** and **americium** have high densities.
- **Colour** : Actinide ions are generally coloured. The colour of actinide ions depends upon the number of 5f-electrons. The ions containing no unpaired 5f-electrons (exactly full filled f-subshell) are colourless, as expected.
- **Ionisation energies** : Ionisation energies values of actinoids are low.
- **Electropositive character** : All the known actinide metals are **highly electropositive**. They resemble lanthanoid series in this respect.
- **Melting Boiling properties** : They have **high melting and boiling points**. They do not follow regular gradation of melting or boiling points with increase in atomic number.
- **Magnetic properties** : The actinide elements are paramagnetic due to the presence of unpaired electrons.
- **Radioactive nature** : All the actinoids are radioactive in nature.
- **Actinide contraction** : The size of atom/cation decrease regularly along the actinoids series. The steady decrease in ionic radii with increase in atomic number is referred to as **actinide contraction**. This is due to poor shielding of 5f-electrons.

Comparison of Lanthanides and Actinoids

Points of Resemblance :

- Both lanthanoids and actinoids show a dominant oxidation state of +3.
- Both are electropositive and act as strong reducing agents.
- Cations with unpaired electrons in both of them are paramagnetic.
- Most of the cations of lanthanoids and actinoids are coloured.
- Both of them show a steady decrease in their ionic radii along the series. Thus, lanthanoids show **lanthanoid contraction** and actinoids show **actinide contraction**.

POINTS OF DISTINCTION

Lanthanoids	Actinoids
1. Besides the most common oxidation state of +3 lanthanoids show +2 and +4 oxidation states in case of certain elements.	Besides the most common oxidation state of +3, actinoids show +4, +5 and +6 oxidation states in case of certain elements.
2. Lanthanoids have less tendency towards complex formation.	Actinoids have a stronger tendency towards complex formation.
3. Except promethium, they are non radioactive.	All the actinoids are radioactive.
4. Oxides and hydroxide of lanthanoids are less basic.	Oxides and hydroxides of actinoids are more basic

Some important uses of actinoids are as follows –

Thorium : Thorium is used in atomic reactors as fuel rods and in the treatment of cancer.

Uranium : Uranium is used as nuclear fuel. Its salts are used in glass industry (for imparting green colour). textile industry and also in medicines.

Plutonium : Plutonium is used as fuel for atomic reactors as well as in atomic bombs.

BEGINNER'S BOX-2

1. KMnO_4 can be acidified by
 (1) Dil. HCl (2) conc. H_2SO_4 (3) conc. HNO_3 (4) dil. H_2SO_4
2. $\text{NaCl} \xrightarrow[\text{X}]{\text{conc. H}_2\text{SO}_4}$ yellow green gas, X is
 (1) $\text{K}_2\text{Cr}_2\text{O}_7$ (2) MnO_2 (3) KMnO_4 (4) both (2) & (3)
3. $\text{X} \xrightarrow[\text{K}_2\text{Cr}_2\text{O}_7]{\text{conc. H}_2\text{SO}_4}$ red brown vapours. X can be
 (1) NaI (2) NaBr (3) Both (1) & (2) (4) None of these
4. $\text{X} \xrightarrow[\text{black coloured ore}]{\text{KOH, air}} \text{Y} \xrightarrow[\text{dark green}]{\text{H}^+} \text{Z} + \text{X}$
 an oxidising agent
 True statement is
 (1) X can't be MnO_2 (2) Y can be KMnO_4
 (3) Z can be K_2MnO_4 (4) step (1) is oxidation while step (ii) is disproportionation
5. $\text{X} \xrightleftharpoons[\text{OH}]{\text{H}^+} \text{Y}; \text{Y} \xrightarrow{\text{SO}_2} \text{Z} + \text{SO}_4^{2-}$
 X and Y are oxy anion of Cr than oxidation state of Cr in X, Y, Z will be respectively
 (1) +3, +6, +6 (2) +6, +3, +6 (3) +6, +6, +6 (4) +6, +6, +3
6. When KMnO_4 solution is added to oxalic acid, solution, the decolourisation is slow in beginning but become instantaneous after some time because
 (1) CO_2 is a by product (2) Mn^{2+} acts as a catalyst
 (3) reaction is exothermic (4) MnO_4^- catalyzes the reaction
7. The elements from cerium (At.No. 58) to lutetium (At.No. 71) in which 4f energy levels are filled up are called :
 (1) lanthanides (2) rare earths (3) lanthanones (4) all the above
8. Which of the following is a lanthanide :
 (1) Ta (2) Rh (3) Th (4) Gd
9. The element with the electronic configuraton $[\text{Xe}]^{54} 4f^{14} 5d^1 6s^2$ is a
 (1) representative element (2) transition element
 (3) lanthanide (4) actinide
10. Name the three lanthanides which show +2 oxidation state also :
 (1) Sm, Tb, Gd (2) Sm, Eu, Yb (3) La, Gd, Lu (4) Yb, Pm, Sm

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	
	Ans.	2	3	3	1	4	3,3	1	3	1	
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	4	4	2	4	4	2	4	4	3	2

NCERT QUESTIONS (REASONING)

Q.1 Why do the transition elements have higher boiling & melting points ?

Ans. Because of having larger number of unpaired electrons in their atoms, they have stronger interatomic interaction and hence stronger bonding between atoms. Hence strong metallic bonds between the atoms of these elements attribute to their high melting and boiling points.

Q.2 Give the main difference in oxidation states of transition elements & p-block elements.

Ans. (i) The variation of oxidation state in transition elements is in such a way that their oxidation state differ from each other by unity e.g. V^{+2} , V^{+3} , V^{+4} , V^{+5} . While in p block elements oxidation states normally differ by two due to inert pair effect.

(ii) Stability of higher oxidation state in transition elements increase down the group

e.g. Pt^{+4} is more stable than Pt^{+2}

While in p block elements lower oxidation state become increasingly stable down the group as result of inert pair effect.

Q.3 For the first series of transition metals the E° values are

E°	V	Cr	Mn	Fe	Co	Ni	Cu
(M^{2+}/M)	-1.18	-0.91	-1.18	-0.44	-0.28	-0.25	+0.34

Explain the irregularity in the above values.

Ans. This is because of irregular variation of ionization energies ($IE_1 + IE_2$) and also the sublimation energies which are much less for manganese and vanadium

Q.4 Zinc does not show variable valency because of :

(A) complete 'd' sub-shell (B) inert pair effect (C) $4s^2$ sub-shell (D) none.

Ans. $Zn = [Ar]^{18} 3d^{10} 4s^2$. Zinc has completely filled d-sub-shell so removal of electron from completely filled 4d sub-shell would be quite difficult. Thus it does not show variable valency.

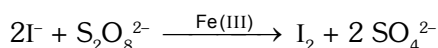
Q.5 Explain the blue colour of $CuSO_4 \cdot 5H_2O$.

Ans. Cu^{2+} ion ($3d^9$) absorbs red light from the visible region, for the promotion of 3d electrons, the ions reflect blue light and appear blue.

Q.6 How iron (III) catalyses the reaction between iodide & persulphate?

Ans. $2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2$

$2Fe^{2+} + S_2O_8^{2-} \longrightarrow 2Fe^{3+} + 2SO_4^{2-}$



EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

d-BLOCK ELEMENTS

- For a catalyst which condition is not essential :
 (1) Variable valency
 (2) High ionisation energy
 (3) Empty orbitals
 (4) Free valency on the surface
- To which of the following series the transition element from $Z = 39$ to $Z = 48$ belong :
 (1) 3d series (2) 4d series
 (3) 5d series (4) 6d series
- KMnO_4 on treatment with conc. H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively
 (1) Mn_2O_7 , MnO_2 (2) Mn_2O , Mn_2O_3
 (3) MnSO_4 , Mn_2O_3 (4) Mn_2O_3 , MnO_2
- Which pair of ions is colourless :
 (1) Mn^{3+} , Co^{3+} (2) Fe^{3+} , Cr^{3+}
 (3) Zn^{2+} , Sc^{3+} (4) Ti^{2+} , Cu^{2+}
- $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is :
 (1) Cr_2O_3 (2) CrO_2 (3) Cr_2O_4 (4) CrO_3
- $\text{Cr}_2\text{O}_7^{2-} \xrightleftharpoons[X]{Y} 2\text{CrO}_4^{2-}$, X and Y are respectively
 (1) $\text{X} = \text{OH}^-$, $\text{Y} = \text{H}^+$ (2) $\text{X} = \text{H}^+$, $\text{Y} = \text{OH}^-$
 (3) $\text{X} = \text{OH}^-$, $\text{Y} = \text{H}_2\text{O}_2$ (4) $\text{X} = \text{H}_2\text{O}_2$, $\text{Y} = \text{OH}^-$
- During estimation of oxalic acid Vs KMnO_4 , self indicator is :
 (1) KMnO_4 (2) Oxalic acid
 (3) K_2SO_4 (4) MnSO_4
- The higher oxidation states of transition elements are found to be the combination with A and B, which are:
 (1) F, O (2) O, N
 (3) O, Cl (4) F, Cl
- Magnetic moment of x^{n+} is $\sqrt{24}$ B.M. Hence No. of unpaired electron and value of 'n' respectively. (Atomic number = 26)
 (1) 4, 3 (2) 3, 5
 (3) 4, 2 (4) 4, 1

- The product of oxidation of I^- with MnO_4^- in alkaline medium is :
 (1) IO_3^- (2) I_2 (3) IO^- (4) IO_4^-
- Copper becomes green when exposed to moist air for longer period.
 (1) Because of the formation of a layer of cupric oxide on the surface of copper
 (2) Because of the formation of a layer of basic carbonate of copper on the surface of copper
 (3) Because of the formation of a layer of cupric hydroxide on the surface of copper.
 (4) (1) and (3) both
- Which of the following oxide of chromium is amphoteric in nature
 (1) CrO (2) Cr_2O_3
 (3) CrO_3 (4) CrO_5
- Compared to Cu^{2+} having $3d^9$ configuration, Cu^+ having $3d^{10}$ configuration (aq. solution) :-
 (1) Is more stable
 (2) Is equally stable
 (3) Is less stable
 (4) Stability depends upon nature of copper salt
- Pick out the wrong statement :-
 (1) $\text{K}_2\text{Cr}_2\text{O}_7$ reduces ferric sulphate to ferrous sulphate
 (2) Iron do not form amalgam
 (3) Permanent magnet is made by an alloy called Alnico
 (4) In the Lathanides ionic radius decreases from La^{+3} to Lu^{+3}
- CrO_3 is red or orange in colour. The nature of oxide is :-
 (1) Acidic (2) Basic
 (3) Amphoteric (4) Neutral
- Cl_2 gas is obtained by various reactions but not by :
 (1) $\text{KMnO}_4 + \text{conc. HCl} \xrightarrow{\Delta}$
 (2) $\text{KCl} + \text{K}_2\text{Cr}_2\text{O}_7 + \text{conc. H}_2\text{SO}_4 \xrightarrow{\Delta}$
 (3) $\text{MnO}_2 + \text{conc. HCl} \xrightarrow{\Delta}$
 (4) $\text{KCl} + \text{F}_2 \xrightarrow{\Delta}$

- 17.** Maximum magnetic moment is shown by :
 (1) d^5 (2) d^6
 (3) d^7 (4) d^8
- 18.** Disproportion can be shown by
 (1) MnO_4^{2-} in acidic medium
 (2) Cu^+ in aqueous medium
 (3) Cl_2 in alkaline medium
 (4) All of these
- 19.** The basic character of the transition metal monoxides follows the order :
 (Atomic no. Ti = 22, V = 23, Cr = 24, Fe = 26)
 (1) $TiO > FeO > VO > CrO$
 (2) $TiO > VO > CrO > FeO$
 (3) $VO > CrO > TiO > FeO$
 (4) $CrO > VO > FeO > TiO$
- 20.** Which of the following reactions is used to estimate copper volumetrically ?
 (1) $2Cu^{2+} + 4F^- \longrightarrow Cu_2F_2 + F_2$
 (2) $Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$
 (3) $2Cu^{2+} + 2CNS^- + SO_2 + 2H_2O \longrightarrow Cu_2(CNS)_2 + H_2SO_4 + 2H^+$
 (4) $2Cu^{2+} + 4I^- \longrightarrow Cu_2I_2 + I_2$
- 21.** Atomic size of gold is almost the same as that of silver. It is due to :
 (1) the same crystal structure of silver and gold
 (2) almost the same electropositive character of the two metals
 (3) transition metals contraction in a series
 (4) the effect of lanthanide contraction
- 22.** Which among the following statements is incorrect
 (1) In d-block elements oxidation state differ by unity.
 (2) In p-block metals oxidation state differ by two units.
 (3) In a group of p-block lower oxidation states are favoured by the heavier members.
 (4) In a group of d-block higher oxidation states are favoured by the lighter member.

- 23.** Which among the following order of oxidising character is correct—
 (1) $CrO_3 > MoO_3$ (2) $K_2Cr_2O_7 > KMnO_4$
 (3) $Fe(CO)_5 > Mn(CO)_5$ (4) $V_2O_3 > V_2O_5$
- 24.** Which of the following configurations of 3d series metals exhibits the largest number of oxidation states—
 (1) $[Ar] 3d^8 4s^2$ (2) $[Ar] 3d^{10} 4s^1$
 (3) $[Ar] 3d^5 4s^2$ (4) $[Ar] 3d^7 4s^2$
- 25.** Which of the following statement regarding interstitial compounds is incorrect—
 (1) They are chemically inert
 (2) They are soft and nonconductive
 (3) They retain metallic conductivity
 (4) They have high melting point.
- 26.** Which of the following pair of ions has same value of "spin-only" magnetic moment
 (1) Cu^+ , Cu^{2+} (2) Co^{3+} , Fe^{2+}
 (3) Ti^{2+} , V^{2+} (4) Sc^{2+} , Zn^{+2}

f-BLOCK ELEMENTS

- 27.** The elements from thorium (At.No. 90) to lawrencium (At.No. 103) in which 5f energy levels are filled up are called :
 (1) lanthanides (2) rare earths
 (3) actinides (4) transuranics
- 28.** Select the element in the following which does not show +4 oxidation state :
 (1) Ti (2) Zr
 (3) La (4) Pt
- 29.** With increase in atomic number the ionic radii of actinides :
 (1) contract slightly
 (2) increase gradually
 (3) show no change
 (4) change irregularly
- 30.** The general electronic configuration of lanthanide is :
 (1) $[Xe] 4f^{14} 5d^{0-1} 6s^2$
 (2) $[Xe] 4f^{0-14} 5d^{1-2} 6s^1$
 (3) $[Xe] 4f^{0-14} 5d^{0-1} 6s^{1-2}$
 (4) None of these

- 31.** Cerium can show the oxidation state of +4 because
 (1) it resemble alkali metals
 (2) it has very low value of I.E.
 (3) of its tendency to attain noble gas configuration of xenon
 (4) of its tendency to attain $4f^7$ configuration
- 32.** In aqueous solution Eu^{+2} acts as :
 (1) an oxidising agent
 (2) reducing agent
 (3) can act as redox agent
 (4) None of these
- 33.** The maximum oxidation state shown by actinides is :
 (1) +6 (2) +7
 (3) +5 (4) +4
- 34.** The outer electronic configuration of gadolinium (At. No. 64) is :
 (1) $4f^7 5d^1 6s^2$ (2) $4f^8 5d^0 6s^2$
 (3) $4f^8 5d^1 6s^1$ (4) $4f^7 5d^0 6s^2$
- 35.** The most characteristic oxidation state of lanthanides is :
 (1) +2 (2) +3
 (3) +4 (4) none of these
- 36.** The common oxidation state of actinides is :
 (1) +4 (2) +3
 (3) +5 (4) +6
- 37.** Which of the following f - block elements, will change its group on emitting α -particle (alpha particle) :
 (1) ${}_{58}\text{Ce}$ (2) ${}_{70}\text{Lu}$
 (3) ${}_{90}\text{Th}$ (4) ${}_{92}\text{U}$
 Correct answer is :-
 (1) Only 1 and 3 (2) Only 2 and 4
 (3) All (4) None
- 38.** Which of the following pair have almost similar size
 (1) Ti_{22} and Zr_{40} (2) Nb_{41} and Ta_{73}
 (3) Y_{39} and La_{57} (4) Ca_{20} and Ir_{31}
- 39.** An increase in both atomic and ionic radii with atomic number occurs in any group of the periodic table. In accordance of this the ionic radii of Ti (IV) and Zr (IV) ions are $0.68 \text{ \AA}$ and $0.74 \text{ \AA}$ respectively but for Hf (IV) ion the ionic radius is $0.75 \text{ \AA}$, which is almost the same as that for Zr (IV) ion. This is due to :-
 (1) greater degree of covalency in compounds of Hf^{4+}
 (2) Lanthanide contraction
 (3) Difference in the co-ordination number of Zr^{+4} and Hf^{+4} in their compounds
 (4) Actinide contraction

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	2	1	3	1	1	1	1	3	1	2	2	3	1	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	4	2	4	4	4	1	3	2	2	3	3	1	4
Que.	31	32	33	34	35	36	37	38	39						
Ans.	3	2	2	1	2	2	1	2	2						

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

- In which of the following pairs are both the ions coloured in aqueous solution ?
(At. no. : Sc = 21, Ti = 22, Ni = 28, Cu = 29, Co = 27)
(1) Ni^{2+} , Cu^{+} (2) Ni^{2+} , Ti^{3+}
(3) Sc^{3+} , Ti^{3+} (4) Sc^{3+} , Co^{2+}
- More number of oxidation states are exhibited by the actinoids than by the lanthanoids. The main reason for this is
(1) more active nature of the actinoids
(2) more energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals
(3) lesser energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals
(4) greater metallic character of the lanthanoids than that of the corresponding actinoids

AIPMT 2007

- Which one of the following ions is the most stable in aqueous solution
(1) V^{3+} (2) Ti^{3+} (3) Mn^{3+} (4) Cr^{3+}
- Identify the incorrect statement among the following
(1) Lanthanoid contraction is the accumulation of successive shrinkages.
(2) As a result of lanthanoid contraction, the properties of 4d series of the transition elements have no similarities with the 5d series of elements.
(3) Shielding power of 4f electrons is quite weak.
(4) there is a decrease in the radii of the atoms or ions as one proceeds from La to Lu.

AIPMT 2008

- The correct order of decreasing second ionisation enthalpy of Ti (22), V (23), Cr (24) and Mn (25) is:-
(1) $\text{Mn} > \text{Cr} > \text{Ti} > \text{V}$ (2) $\text{Ti} > \text{V} > \text{Cr} > \text{Mn}$
(3) $\text{Cr} > \text{Mn} > \text{V} > \text{Ti}$ (4) $\text{V} > \text{Mn} > \text{Cr} > \text{Ti}$

AIPMT 2009

- Which one of the elements with the following outer orbital configurations may exhibit the largest number of oxidation states?
(1) $3d^2 4s^2$ (2) $3d^3 4s^2$ (3) $3d^5 4s^1$ (4) $3d^5 4s^2$

AIPMT 2010

- Which of the following oxidation states is the most common among the lanthanoids :-
(1) 2 (2) 5 (3) 3 (4) 4
- Which of the following pairs has the same size ?
(1) Zn^{2+} , Hf^{4+} (2) Fe^{2+} , Ni^{2+}
(3) Zr^{4+} , Ti^{4+} (4) Zr^{4+} , Hf^{4+}
- Which one of the following ions has electronic configuration $[\text{Ar}]3d^6$? (At. nos. Mn = 25, Fe = 26, Co = 27, Ni = 28)
(1) Co^{3+} (2) Ni^{3+} (3) Mn^{3+} (4) Fe^{3+}
- Which of the following ions will exhibit colour in aqueous solutions ?
(1) Sc^{3+} (z = 21) (2) La^{3+} (z = 57)
(3) Ti^{3+} (z = 22) (4) Lu^{3+} (z = 71)

AIIMS 2010

- Uranium can be obtained by the electrolysis of:-
(1) UF_6 (2) UF_4 (3) UCl_6 (4) UI_6

AIPMT Pre. 2011

- For the four successive transition elements (Cr, Mn, Fe and Co), the stability of +2 oxidation state in gaseous state will be there in which of the following order ?
(1) $\text{Mn} > \text{Fe} > \text{Cr} > \text{Co}$
(2) $\text{Fe} > \text{Mn} > \text{Co} > \text{Cr}$
(3) $\text{Co} > \text{Mn} > \text{Fe} > \text{Cr}$
(4) $\text{Cr} > \text{Mn} > \text{Co} > \text{Fe}$
(At. nos. Cr = 24, Mn = 25, Fe = 26, Co = 27)

AIIMS 2011

- When KMnO_4 reacts with Br_2 in weak alkaline medium gives bromate ion. Then oxidation state of Mn changes from +7 to:
(1) +6 (2) +4 (3) +3 (4) +2

AIPMT Pre. 2012

- Which of the statements is not true?
(1) $\text{K}_2\text{Cr}_2\text{O}_7$ solution in acidic medium is orange
(2) $\text{K}_2\text{Cr}_2\text{O}_7$ solution becomes yellow on increasing the pH beyond 7
(3) On passing H_2S through acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution, a milky colour is observed
(4) $\text{Na}_2\text{Cr}_2\text{O}_7$ is preferred over $\text{K}_2\text{Cr}_2\text{O}_7$ in volumetric analysis

AIPMT Mains 2012

15. Which one of the following does not correctly represent the correct order of the property indicated against it ?
 (1) $Ti < V < Cr < Mn$: increasing melting points
 (2) $Ti < V < Mn < Cr$: increasing 2nd ionization enthalpy
 (3) $Ti < V < Cr < Mn$: increasing number of oxidation states
 (4) $Ti^{3+} < V^{3+} < Cr^{3+} < Mn^{3+}$: increasing magnetic moment
16. The catalytic activity of transition metals and their compounds is ascribed mainly to :-
 (1) their ability to adopt variable oxidation states
 (2) their chemical reactivity
 (3) their magnetic behaviour
 (4) their unfilled d-orbitals

AIIMS 2012

17. Magnetic moment of Cr^{2+} is nearest to :-
 (1) Fe^{2+} (2) Mn^{2+} (3) Co^{+2} (4) Ni^{2+}

AIIMS 2013

18. Which of the following element has lowest melting point :-
 (1) Cr (2) Fe (3) Ni (4) Cu
19. Maximum no. of unpaired electrons are present in :-
 (1) Gd^{+3} (2) Yb^{+2} (3) Tb^{+2} (4) Pm^{+3}
20. $K_2Cr_2O_7$ in acidic medium converts into :-
 (1) Cr^{2+} (2) Cr^{3+} (3) Cr^{4+} (4) Cr^{+5}
21. Which of the following metal ion forms unstable complex with CN^- :-
 (1) $Ag(I)$ (2) $Zn(II)$ (3) $Cu(II)$ (4) $Cr(II)$
22. Which of the following ion does not exist ?
 (1) $[CuI_4]^{2-}$ (2) VO_4^{3-} (3) WO_4^{2-} (4) CrO_4^{2-}

AIPMT 2015

23. Magnetic moment 2.84 B.M. is given by :-
 (At. no.), Ni = 28, Ti = 22, Cr = 24, Co = 27
 (1) Ti^{3+} (2) Cr^{2+} (3) Co^{2+} (4) Ni^{2+}
24. Which of the following processes does not involve oxidation of iron ?
 (1) Decolourization of blue $CuSO_4$ solution by iron
 (2) Formation of $Fe(CO)_5$ from Fe
 (3) Liberation of H_2 from steam by iron at high temperature
 (4) Rusting of iron sheets

25. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii ? (Numbers in the parenthesis are atomic numbers).
 (1) Zr (40) and Nb (41)
 (2) Zr (40) and Hf (72)
 (3) Zr (40) and Ta (73)
 (4) Ti (22) and Zr (40)

AIIMS 2015

26. Highly unstable species is
 (1) $Sc(II)$ (2) $Ti(II)$
 (3) $Co(II)$ (4) $V(II)$
27. Which of the following is diamagnetic
 (1) Aurum (2) Tungsten
 (3) Palladium (4) Niobium

NEET-I 2016

28. Which one of the following statements is correct when SO_2 is passed through acidified $K_2Cr_2O_7$ solution ?
 (1) The solution turns blue
 (2) The solution is decolourized
 (3) SO_2 is reduced
 (4) Green $Cr_2(SO_4)_3$ is formed
29. The electronic configurations of Eu (Atomic No 63), Gd (Atomic No 64) and Tb (Atomic No. 65) are
 (1) $[Xe]4f^7 6s^2$, $[Xe]4f^8 6s^2$ and $[Xe]4f^8 5d^1 6s^2$
 (2) $[Xe]4f^7 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 6s^2$
 (3) $[Xe]4f^6 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^8 5d^1 6s^2$
 (4) $[Xe]4f^7 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^9 6s^2$

NEET-II 2016

30. Which one of the following statements related to lanthanons is **incorrect** ?
 (1) All the lanthanons are much more reactive than aluminium
 (2) $Ce(+4)$ solutions are widely used as oxidizing agent in volumetric analysis
 (3) Europium shows +2 oxidation state.
 (4) The basicity decreases as the ionic radius decreases from Pr to Lu.

AIIMS 2016

31. Which of the following is ferromagnetic :-
 (1) U (2) Co
 (3) Cr (4) Cu
32. Aquaregia convert :-
 (1) Pt to $[\text{PtCl}_4]^{-2}$ (2) Ag to AgCl
 (3) Pt to PtCl_2 (4) Au to $[\text{AuCl}_4]^{-}$
33. Which of the following can convert Cl^- into Cl_2 :-
 (1) KMnO_4 (2) H_2SO_3
 (3) $\text{Fe}_2(\text{SO}_4)_3$ (4) HIO_3
34. Which of the following Lanthanoid show yellow colour :-
 (1) La^{+3} (2) Yb^{+3}
 (3) Lu^{+3} (4) Ce^{+4}

NEET(UG) 2017

35. Name the gas that can readily decolourise acidified KMnO_4 solution :
 (1) SO_2 (2) NO_2 (3) P_2O_5 (4) CO_2
36. HgCl_2 and I_2 both when dissolved in water containing I^- ions the pair of species formed is :
 (1) HgI_2, I^- (2) $\text{HgI}_4^{2-}, \text{I}_3^-$
 (3) $\text{Hg}_2\text{I}_2, \text{I}^-$ (4) $\text{HgI}_2, \text{I}_3^-$
37. The reason for greater range of oxidation states in actinoids is attributed to :-
 (1) actinoid contraction
 (2) 5f, 6d and 7s levels having comparable energies
 (3) 4f and 5d levels being close in energies
 (4) the radioactive nature of actinoids

AIIMS 2017

38. Gadolinium is :-
 (1) Ferromagnetic
 (2) Anti ferromagnetic
 (3) Ferrimagnetic
 (4) Diamagnetic
39. Which of the following pair contains all diamagnetic ions.
 (1) Yb^{+2} & Lu^{+3} (2) Ce^{+4} & Pr^{+4}
 (3) Dy^{+3} & Lu^{+3} (4) Lu^{+3} & Gd^{+3}

40. Which of the following is incorrect about interstitial compound.

- (1) they have high melting points in comparison to pure metals.
 (2) they are very hard
 (3) they retain metallic conductivity
 (4) they are chemically very reactive

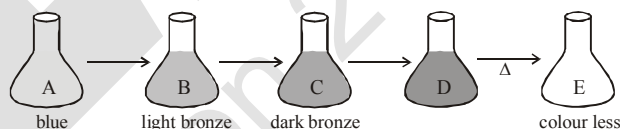
41. Which of the following does not decolourises acidic KMnO_4 solution.

- (1) KI & NaNO_2
 (2) NaNO_3 & $\text{Na}_2\text{C}_2\text{O}_4$
 (3) Na_2SO_4 & $\text{Na}_2\text{C}_2\text{O}_4$
 (4) I_2 & NaNO_3

42. Select the incorrect statement about iodide of Millon's base.

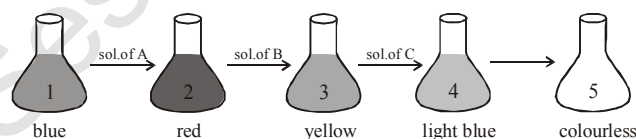
- (1) it is formed by the reaction $\text{K}_2[\text{HgI}_4] + \text{KOH} + \text{NH}_4^+$ solution
 (2) its formula $\left[\text{Hg} \begin{array}{c} \diagup \text{O} \diagdown \\ \diagdown \text{NH}_2 \diagup \end{array} \text{Hg} \right] \text{I}$
 (3) it is a brown ppt. but give test of complex cation I^- ions.
 (4) it is a Nesler's reagent of alkaines solution of $\text{K}_2[\text{HgI}_4]$

43. Which of the following will not be present in last flask during the iodometric titration of Cu^{+2}



- (1) starch (2) Cu_2I_2
 (3) I_2 (4) $\text{Na}_2\text{S}_2\text{O}_3$

44. In iodometric titration of Cu^{+2} .



Select the correct statement.

- (1) the colour of solution 2 is due to Iodine
 (2) the colour of solution 2 & solution 4 are due to the same ions.
 (3) solution 4 on strongly heating gives colourless solution 5.
 (4) None of these

NEET(UG) 2018

45. Which one of the following ions exhibits d-d transition and paramagnetism as well ?

- (1) CrO_4^{2-} (2) $\text{Cr}_2\text{O}_7^{2-}$
 (3) MnO_4^- (4) MnO_4^{2-}

46. Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the **correct** code :

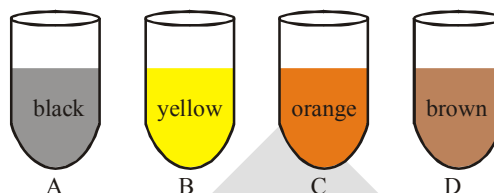
Column I**Column II**

- | | |
|---------------------|----------------------|
| a. Co^{3+} | i. $\sqrt{8}$ B.M. |
| b. Cr^{3+} | ii. $\sqrt{35}$ B.M. |
| c. Fe^{3+} | iii. $\sqrt{3}$ B.M. |
| d. Ni^{2+} | iv. $\sqrt{24}$ B.M. |
| | v. $\sqrt{15}$ B.M. |

- | | | | | |
|-----|----------|----------|----------|----------|
| | a | b | c | d |
| (1) | iv | v | ii | i |
| (2) | i | ii | iii | iv |
| (3) | iv | i | ii | iii |
| (4) | iii | v | i | ii |

AIIMS 2018

47. Which of the following elements have least difference in E^0 value of M^{+2}/M :-
 (1) V & Mn (2) Mn & Fe
 (3) Fe & Co (4) Co & Ni
48. Violet color of glass is due to :-
 (1) K(I) (2) Mn(IV) (3) I_2 (4) Cr(III)
49. In which compound oxidation state not changed in acidic & basic medium :-
 (1) MnO_4^- (2) CrO_4^{2-}
 (3) FeO_4^{2-} (4) MnO_4^{2-}
50. Which can displace Al from Alum :-
 (1) Mn (2) Tl
 (3) Sc (4) In
51. Which shows least number of oxidation states ?
 (1) Mn (2) Th
 (3) Cr (4) Np
52. Spin only magnetic moment of a divalent metal ion having total 23 electrons :-
 (1) 5.9 (2) 5.2
 (3) 4.0 (4) 5.0
53. In salts of Co, Sb, Sn & Cd, H_2S was added. The colour of the precipitates obtained were as follows



Which set is correct for its colour of its sulphide:-

- | | |
|----------|----------|
| (1) A-Co | (2) B-Sb |
| (3) C-Sn | (4) D-Cd |
54. The colour of compound formed when KMnO_4 is fused with KOH :-
 (1) Green (2) Black
 (3) Brown (4) Pale Pink
55. In which of the following conversion Blue colour is appeared.
 (1) $\text{MnO}_4^- + \text{SO}_2 \longrightarrow$
 (2) $\text{Cr}_2\text{O}_7^{2-} + \text{KI} \longrightarrow$
 (3) $\text{K}_2\text{Cr}_2\text{O}_7 + \text{NaCl} + \text{conc. H}_2\text{SO}_4 \longrightarrow$
 (4) $\text{CuSO}_4 + \text{KI} + \text{Starch} \longrightarrow$
56. Which of the following pair is soluble in excess NH_4OH
 (1) $\text{Zn}(\text{OH})_2, \text{Pb}(\text{OH})_2$
 (2) $\text{Zn}(\text{OH})_2, \text{Sn}(\text{OH})_2$
 (3) $\text{Cd}(\text{OH})_2, \text{Zn}(\text{OH})_2$
 (4) $\text{Be}(\text{OH})_2, \text{Cd}(\text{OH})_2$

EXERCISE-II (Previous Year Questions)**ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	4	2	3	4	3	4	1	3	2	1	2	4	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	4	1	2	3	1	4	2	2	1	3	4	4	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	4	1	4	1	2	2	1	1	4	4	4	3	3	4
Que.	46	47	48	49	50	51	52	53	54	55	56				
Ans.	1	1	2	2	3	2	1	1	1	4	3				

EXERCISE-III (Analytical Questions)

Check Your Understanding

- The correct order of ionic radii of Y^{3+} , La^{3+} , Eu^{3+} and Lu^{3+} is
 (1) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$
 (2) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$
 (3) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$
 (4) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$
- Among the following series of transition metal ions, the one where all metal ions have $3d^2$ electronic configuration is :-
 (1) Ti^{2+} , V^{3+} , Cr^{4+} , Mn^{5+}
 (2) Ti^{2+} , V^{2+} , Cr^{3+} , Mn^{4+}
 (3) Ti^{+} , V^{4+} , Cr^{6+} , Mn^{7+}
 (4) Ti^{4+} , V^{3+} , Cr^{2+} , Mn^{3+}
- The aqueous solution containing which one of the following ions will be colourless :-
 [Atomic n : Sc = 21, Fe = 26, Ti = 22, Mn = 25]
 (1) Fe^{2+} (2) Mn^{2+}
 (3) Ti^{3+} (4) Sc^{3+}
- The oxidation state of chromium in the final product formed by the reaction between KI and acidified potassium dichromate solution is
 (1) +4 (2) +6 (3) +2 (4) +3
- The lanthanide contraction is responsible for the fact that
 (1) Zr and Y have about the same radius
 (2) Zr and Nb have similar oxidation state
 (3) Zr and Hf have about the same radius
 (4) Zr and Zn have the same oxidation state
- Which one of the following nitrates will leave behind a metal on strong heating ?
 (1) Copper nitrate (2) Manganese nitrate
 (3) Silver nitrate (4) Ferric nitrate
- Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction :-
 (1) $Na_2S_2O_3$ is oxidised
 (2) CuI_2 is formed
 (3) Cu_2I_2 is formed
 (4) Evolved I_2 is reduced
- In which of the following oxidation state of Mn is six
 (1) K_2MnO_4 (2) MnO_2
 (3) $KMnO_4$ (4) MnO_4^{-1}
- Which has maximum magnetic moment :-
 (1) Co^{+2} (2) Mn^{+3}
 (3) Mn^{+2} (4) Cr^{+3}

- Atomic number 21, shows the oxidation states
 (1) +1 (2) +3 (3) -1 (4) +5
- Transition element makes complex compound due to -
 (1) Presence of empty d-orbital
 (2) large size
 (3) Variable valency
 (4) presence of $2e^-$
- Lanthanoid contraction is caused due to
 (1) the same effective nuclear charge from Ce to Lu
 (2) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 (3) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 (4) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
- $KMnO_4$ is a strong oxidizing agent in acid medium. To provide acid medium H_2SO_4 is used instead of HCl. This is because
 (1) H_2SO_4 is a stronger acid than HCl
 (2) HCl is oxidized by $KMnO_4$ to Cl_2
 (3) H_2SO_4 is a dibasic acid
 (4) rate is faster in the presence of H_2SO_4
- When hydrogen peroxide is added to acidified potassium dichromate, a blue colour is produced due to formation of
 (1) CrO_3 (2) Cr_2O_3
 (3) CrO_5 (4) CrO_4^{2-}
- Identify the incorrect statement among the following
 (1) 4f and 5f orbitals are equally shielded.
 (2) d-Block elements show irregular and erratic chemical properties among themselves.
 (3) La and Lu have partially filled d orbitals and no other partially filled orbitals.
 (4) The chemistry of various lanthanoids is very similar.
- The actinoids exhibit more number of oxidation states in general than the lanthanoids. This is because
 (1) due to less energy difference between 6d & 5f
 (2) the 5f orbitals are more buried than the 4f orbitals
 (3) there is a similarity between 4f and 5f orbitals in their angular part of the wave function
 (4) the actinoids are more reactive than the lanthanoids.

17. In $\text{Cr}_2\text{O}_7^{-2} \longrightarrow \text{CrO}_4^{-2}$ reaction the change in oxidation number is :-
 (1) +3 (2) -3 (3) 2 (4) 0
18. Which of the following properties of Zr and Hf can be explained due to lanthanide contraction :-
 (1) Radius of Zr and Hf is same
 (2) Their EN is same
 (3) They are f-block elements
 (4) All
19. Similar sizes of second and third transition elements can be explained on the basis of :-
 (1) Inert-pair effect
 (2) Screening effect
 (3) Lanthanide contraction
 (4) Increasing effective nuclear charge
20. Iron exhibits +2 and +3 oxidation states. Which of the following statements about iron is **incorrect**?
 (1) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.
 (2) Ferrous oxide is more basic in nature than the ferric oxide.
 (3) Ferrous compounds are relatively more ionic than the corresponding ferric compounds.
 (4) Ferrous compounds are less volatile than the corresponding ferric compounds.
21. Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest positive $E_{M^{3+}/M^{2+}}^0$ value ?
 (1) Cr(Z = 24) (2) Mn(Z = 25)
 (3) Fe (Z = 26) (4) Co(Z = 27)
22. Which of the following metal cation is reduced from its higher oxidation state (+2) to (+1) by both KI solution and excess of KCN solution ?
 (1) Zn^{2+} (2) Hg^{2+} (3) Cu^{2+} (4) None
23. A test tube containing a nitrate and another containing a bromide and MnO_2 are treated with concentrated H_2SO_4 . The reddish brown fumes evolved are passed through water. The water will be coloured by :-
 (1) The nitrate (2) The bromide
 (3) Both (4) None of the two
24. Cr^{2+} is reducing in nature because—
 (1) In Cr^{2+} , configuration changes from d^4 to d^3 to achieve half filled t_{2g} .
 (2) Cr^{2+} gain an electron to achieve d^5 configuration
 (3) Cr^{2+} give an electron to achieve d^5 configuration
 (4) In Cr^{2+} configuration changes from d^4 to d^3 to achieve half filled d-subshell.
25. Highest stable Mn fluoride is MnF_4 where as highest Mn oxide is Mn_2O_7 due to—
 (1) O atom is smaller the F.
 (2) Oxygen has ability to form multiple bonds
 (3) Mn^{7+} does not exist
 (4) F can not-stabilise higher oxidation states.
26. Which among the following order of oxidising power is correct—
 (1) $\text{CrO}_4^- < \text{MoO}_4^- < \text{WO}_4^-$
 (2) $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$
 (3) $\text{CrO}_3 < \text{MnO}_2 < \text{Fe}_2\text{O}_3$
 (4) $\text{Pb}^{4+} < \text{Sn}^{4+} < \text{C}^{4+}$
27. Cu^+ ion is not stable in aqueous solution, due to—
 (1) d^{10} configuration
 (2) disproportionation
 (3) less charge
 (4) pseudo inert gas configuration

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	4	4	3	3	2	1	3	2	1	2	2	3	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans.	1	4	1	3	1	4	3	2	1	2	2	2			

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion** : 1st ionisation potential of mercury is greater than cadmium

Reason : Hg has stable electronic configuration ($5d^{10} 6s^2$)

- (1) A (2) B (3) C (4) D

2. **Assertion** : Zr and Hf have about the same atomic radius.

Reason : Zr and Hf lies in the same group.

- (1) A (2) B (3) C (4) D

3. **Assertion** : Zn, Cd, Hg are non-transition elements while Cu, Ag, Au are transition element

Reason : In Zn, Cd, Hg ($n - 1$)d orbitals are completely filled in their atomic state where as in Cu, Ag, Au they are incomplete.

- (1) A (2) B (3) C (4) D

4. **Assertion** : Cu^+ is more stable than Cu^{+2}

Reason : ΔIP is greater than 16 eV

- (1) A (2) B (3) C (4) D

5. **Assertion** : $KMnO_4$ is dark pink coloured compound

Reason : In the $KMnO_4$ charge transfer spectrum occurs.

- (1) A (2) B (3) C (4) D

6. **Assertion** : Hg is the only metal which is liquid at $0^\circ C$.

Reason : It has very high IP and weak metallic bond

- (1) A (2) B (3) C (4) D

7. **Assertion** : Valency of transition elements is variable

Reason : Energy of ns and $(n-1)d$ orbital is almost same.

- (1) A (2) B (3) C (4) D

8. **Assertion** : Melting point of Mn less than that of Fe

Reason : Mn has less number of unpaired e^- than Fe in atomic state

- (1) A (2) B (3) C (4) D

9. **Assertion** : Solution of Na_2CrO_4 in water is intensely coloured.

Reason : Ox. state of Cr in Na_2CrO_4 is +6.

- (1) A (2) B (3) C (4) D

10. **Assertion** : Ce^{+4} acts as oxidising agent in aqueous medium

Reason : +4 is common oxidation state of lanthanides

- (1) A (2) B (3) C (4) D

11. **Assertion** : Neptunium is transuranic element.

Reason : It is heavier than uranium

- (1) A (2) B (3) C (4) D

12. **Assertion** : $La(OH)_3$ is more basic than $Lu(OH)_3$

Reason : Lanthanum is d-block element

- (1) A (2) B (3) C (4) D

13. **Assertion** : Actinides show much higher range of oxidation states

Reason : Energy difference between 5f and 6d orbitals is large

- (1) A (2) B (3) C (4) D

14. **Assertion** : All the lanthanide elements exhibits a common oxidation state of +3 in their compounds.

Reason : The atoms of the lanthanide elements contains three electron in their outermost shell.

- (1) A (2) B (3) C (4) D

15. **Assertion:-** $K_2Cr_2O_7$ is used as a primary standard in volumetric analysis.

Reason:- It has a good solubility in water.

(1) A (2) B (3) C (4) D

16. **Assertion:-** Change in colour of acidic solution of potassium dichromate by breath is used to test drunk drivers.

Reason:- Change in colour is due to the complexation of alcohol with potassium dichromate.

(1) A (2) B (3) C (4) D

17. **Assertion:-** Eu^{2+} & Yb^{2+} are reducing agents for their ions.

Reason:- Both ions have stable half filled configuration.

(1) A (2) B (3) C (4) D

18. **Assertion :-** MnO_2 is anti ferromagnetic in nature.

Reason :- In MnO_2 , equal number of domain are aligned with parallel and antiparallel spin.

(1) A (2) B (3) C (4) D

19. **Assertion :-** La_2O_3 is basic nature.

Reason :- La in aqueous solution gives $La(OH)_3$

(1) A (2) B (3) C (4) D

20. **Assertion :-** $FeCl_3$ does not affect iodometric titration of $CuSO_4$ Solution

Reason :- FeI_3 is formed.

(1) A (2) B (3) C (4) D

21. **Assertion :-** Actinoids can possess +4 O.S. more easily than lanthanoid.

Reason :- 4f, 5d, 6s have almost same energy levels.

(1) A (2) B (3) C (4) D

22. **Assertion :-** UF_6 is more covalent than UF_4 .

Reason :- Fluorine is smaller in size.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	2	3	4	1	1	1	3	2	3	1	2	3	3	3
Que.	16	17	18	19	20	21	22								
Ans.	3	3	1	1	4	2	2								

COORDINATION COMPOUND

E

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NEET SYLLABUS

Coordination Compounds : Coordination compounds: Introduction, ligands, coordination number, colour, magnetic properties and shapes, IUPAC nomenclature of mononuclear coordination compounds, isomerism (structural and stereo) bonding, Werner's theory VBT, CFT; importance of coordination compounds (in qualitative analysis, biological systems).

OBJECTIVES

After studying this unit, you will be able to :

- appreciate the postulates of Werner's theory of coordination compounds;
- know the meaning of the terms: coordination entity, central atom/ion, ligand, coordination number, coordination sphere, coordination polyhedron, oxidation number, homoleptic and heteroleptic;
- learn the rules of nomenclature of coordination compounds;
- write the formulas and names of mononuclear coordination compounds;
- define different types of isomerism in coordination compounds;
- understand the nature of bonding in coordination compounds in terms of the Valence Bond and Crystal Field theories;
- learn the stability of coordination compounds;
- appreciate the importance and applications of coordination compounds in our day to day life.

Every science consists in the coordination of facts, if the different observations were entirely isolated, there would be no science.

Auguste Comte

COORDINATION COMPOUND

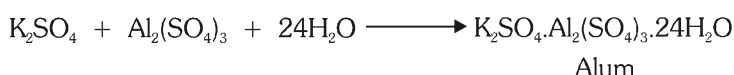
3.0 INTRODUCTION

- The concept of coordination compounds arises from the complex formation tendency of transition elements.
- These compounds play a vital role in our lives. Haemoglobin of blood and chlorophyll of plants are also coordination compounds of Fe, Mg respectively.

3.1 TYPE OF COMPOUNDS

Molecular or Addition Compounds

- When two or more simple salts are chemically combined together in fixed proportion by weight the molecular or addition compounds are formed.
- Some common examples are as follows.



- Depending upon the behavior in an aqueous solution they are of two types
 - Double salt or lattice compound
 - Coordination or complex compounds.

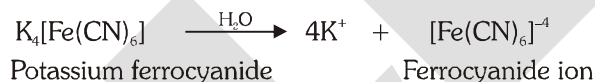
Double Salt or Lattice Compounds

The addition compounds which lose their identity in aq. solution and get completely ionised in water.

- Ex.** (i) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (Potash Alum)
 (ii) $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ (Mohr's salt)
 (iii) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (Carnallite)

Coordination or Complex Compounds

- Those molecular or addition compounds which retain their identity in aq. solution or lose the identity of constituents are called complex salt.
- Complexes retain their identity in aqueous solution. i.e., they do not dissociate into separate ion.



- On the basis of type of ligands complex compounds are divided as follows :

(i) Homoleptic complexes :

Complex in which all the ligands are identical is called homoleptic complex.



(ii) Heteroleptic complexes :

Complex in which all the ligands are not identical is known as heteroleptic complex.



TERMINOLOGY

(a) Complex Ion

An aggregate of metal ion with anions, cation or neutral molecules is called as **complex ion**.

(b) Central Metal Ion

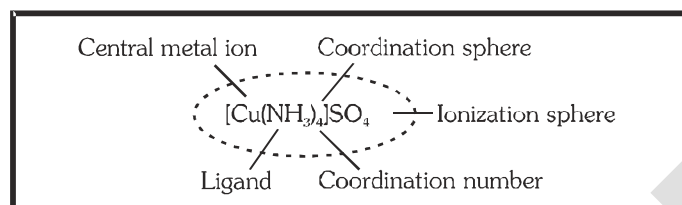
- The metal ion which forms complex ion in combination with anions, cation or neutral molecules is called as **central metal ion**.
- Central metal ion acts as an electron pair acceptor and forms coordinate covalent bond.

(c) Coordination Number

The total number of coordinate covalent bonds formed by central metal ion with ligands is called as coordination number.

(d) Coordination Sphere - (Entity)

- (i) The central metal atom and the ligands directly attached to it are collectively termed as the coordination sphere.
- (ii) Coordination sphere is written inside square bracket, for example $[\text{Co}(\text{NH}_3)_6]^{3+}$
- (iii) The part outside the bracket is called **ionisation sphere**.
- (iv) The species present in the coordination sphere are non-ionisable.
- (v) The species present in the ionization sphere are ionisable.

**(e) Oxidation State**

It is a number which represents the electrical charge on the central metal atom of a complex ion.

Ex. the oxidation number of Fe, Co and Ni in $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $\text{Ni}(\text{CO})_4$ is +2, +3 and zero respectively.

(f) Charge On the Complex Ion

It is the algebraic sum of the total charge of the ligands and central metal ion.

(g) Ligands

- (i) The anions, cation or neutral molecules which combine with central metal ion to form complex ion are called as ligands.
- (ii) They act as electron pair donor (Lewis bases), but strong crystal field ligands like CO, CN^- etc can accept electron pair from the metal ion, because these ligands are π -acids.

(h) Naming of ligands

- (i) o-suffix is provided to the name of anionic ligands.
- (ii) ium-suffix is provided to the name of cationic ligands.

Classification of ligands :

On the basis of denticity :

1. **Monodentate ligands :** Which has only one donor site.

Neutral monodentate ligands	
NH_3	ammine
H_2O	aqua
NO	nitrosyl
PH_3	phosphine
PPh_3	triphenyl phosphine
O_2	dioxygen
N_2	dinitrogen
$\text{CH}_3\text{-OH}$	methyl alcohol
$\text{CH}_3\text{-NH}_2$	methyl amine
NH_2CONH_2	urea
NH_2NH_2	hydrazine
$\text{CH}_3\text{-O-CH}_3$	dimethyl ether
CO	carbonyl
CS	thiocarbonyl

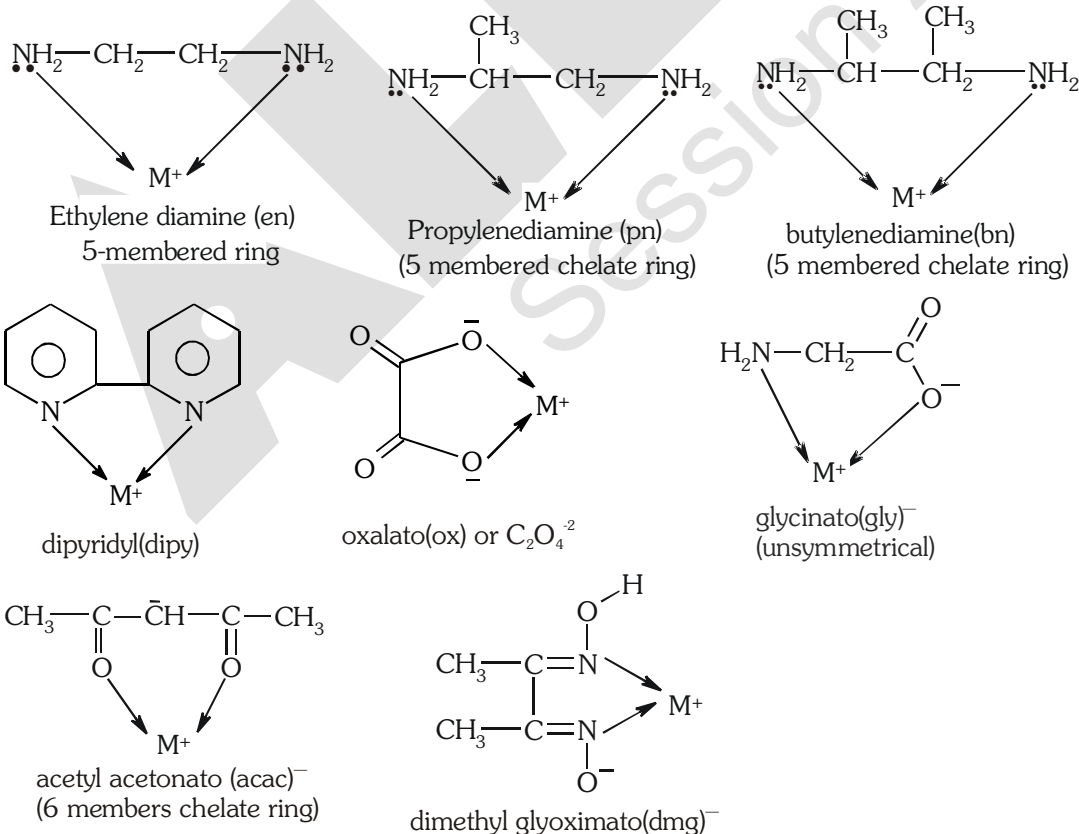
Cationic monodentate ligands :

O_2^+	oxygenium
NO^+	nitrosylium or nitrosonium
NO_2^+	nitronium
$H_2N^+-NH_3$	hydrazinium

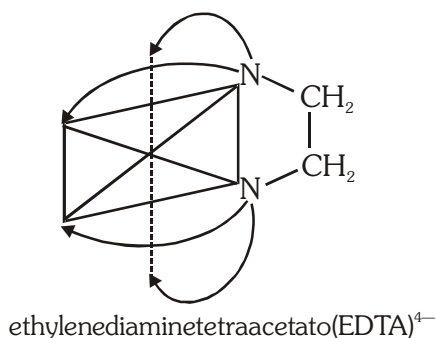
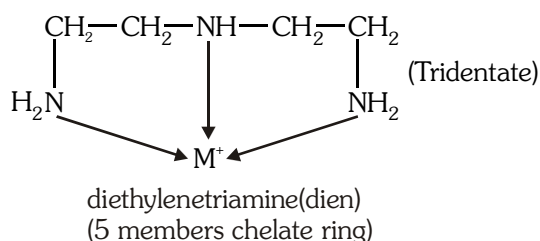
Anionic monodentate ligands :

F^-	Fluorido / fluoro
Cl^-	Chlorido / chloro
Br^-	bromido / bromo
I^-	iodido / iodo
O^{2-}	oxido/oxo
S^{2-}	sulphido/sulpho
OH^-	hydroxido / hydroxo
CH_3O^-	methoxido / methoxo
O_2^-	superoxido / superoxo
O_2^{2-}	peroxido / peroxo
N^{3-}	nitrido
N_3^-	azido
$(NH_2)^-$	amido
NH^{2-}	imido
H^-	hydrido

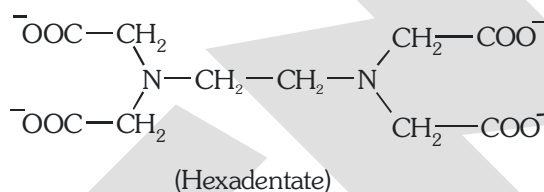
2. Bidentate ligands : Ligands which has two donor sites (simultaneously)



3. Polydentate ligands



or



4. **Ambidentate ligand** : Ligands which have two different donor atoms but at the time of coordination can donate electron pair from any one donor atoms are called ambidentate ligand.

- (i) $\text{M} \leftarrow \bar{\text{C}} \equiv \text{N}$ cyanido (CN)
 $\text{M} \leftarrow \text{N} \equiv \bar{\text{C}}$ isocyanido (NC)
- (ii) $\text{M} \leftarrow \bar{\text{O}} - \text{C} \equiv \text{N}$ cyanato / cyanato-O
 $\text{M} \leftarrow \text{N} \equiv \text{C} - \bar{\text{O}}$ isocyanato / cyanato-N
- (iii) $\text{M} \leftarrow \bar{\text{S}} - \text{C} \equiv \text{N}$ Thiocyanato / Thiocyanato-S
 $\text{M} \leftarrow \text{N} \equiv \text{C} - \bar{\text{S}}$ Isothiocyanato/thiocyanato-N
- (iv) $\text{M} \leftarrow \bar{\text{O}} - \text{N} = \text{O}$ Nitrito / Nitrito-O
 $\text{M} \leftarrow \text{N} \begin{array}{l} \text{O} \\ \text{O}^- \end{array}$ Nitro / Nitrito-N

5. **Flexidentate ligands** : Ligands which can change their denticity.

SO_4^{2-} , $\text{S}_2\text{O}_3^{2-}$, CO_3^{2-} , NO_3^- , CH_3COO^- etc are generally act as monodentate ligand but in some compound they can act as bidentate ligand

eg. (i) $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Cl}$ in this complex SO_4^{2-} act as a monodentate ligand.

(ii) $[\text{Co}(\text{en})_2(\text{SO}_4)]\text{Cl}$ in this complex SO_4^{2-} act as a bidentate ligand.

6. **Chelating ligands** : Bidentate or polydentate ligands which attach themselves with central metal ion in such a way that they form a ring, are known as chelating ligands.

No. of rings = denticity - 1

5 or 6 membered rings are more stable

e.g. (i) $[\text{Pt}(\text{trien})]\text{SO}_4$
 denticity = 4
 chelating rings = 3

(ii) $\text{Na}_2[\text{Fe}(\text{OH})(\text{EDTA})]$
 denticity of EDTA = 5
 chelating rings = 4

On the basis of nature of bonding between central metal atom and ligand.

(i) Normal or classical ligands :

Ligand which only donate electron pair to central metal ion & form coordinate (σ) bond
e.g. OH^- , NH_2^- , Cl^- , N^{3-} , O^{2-} etc.

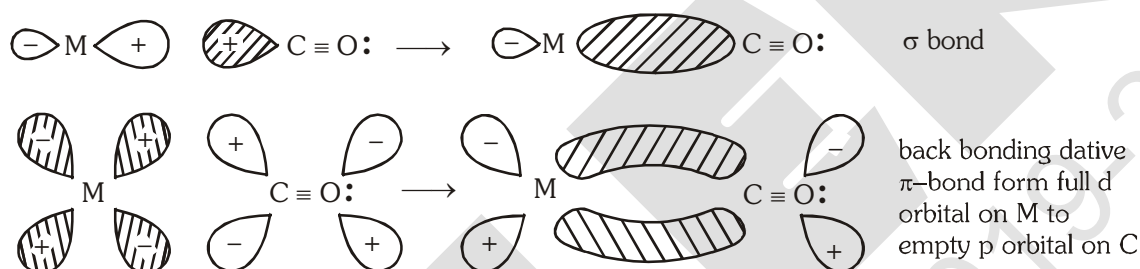
(ii) Non classical or π -acid or π -acceptor ligands :

Ligand which donate electron pair to central metal ion & form coordinate σ bond but simultaneously they accept electron pair from central metal ion through back bonding or synergic bonding.

CO , NO^+ , CN^- , R_3P , R_3As etc.

The electronic configuration of CO molecule shows that it has lone pair of electrons on carbon and oxygen atom each. Carbon atom can donate its electron pair of a transition metal atom (M), forming $\text{OC} \rightarrow \text{M}$ coordinate bond.

Since the metal atom in metal carbonyl is in zero oxidation state, the formation of $\text{M} \leftarrow \text{CO}$ σ bond accumulates a negative charge on the metal atom. The accumulation of negative charge on the metal atom can be counter balanced by transferring some negative charge from the metal atom to CO molecule (ligand). This transfer can be done by making a $\text{M} \rightarrow \text{CO}$ π bond by the overlap between an appropriate filled orbital on the metal atom and empty π_y^* or π_z^* molecular orbital on CO molecule. This type of bonding between M and CO is called **synergic bonding**.



[Schematic of orbital overlaps in metal carbonyls]

Conclusion of synergic bonding : due to synergic bonding

- $\text{M}-\text{C}$ bond strength increases; $\text{M}-\text{C}$ bond length decreases; because double bond character increases
- $\text{C}-\text{O}$ bond strength decreases; $\text{C}-\text{O}$ bond length increases; because bond order of $\text{C}-\text{O}$ decreases, electron density in ABMO of CO increases.

π -donor and π -acceptor ligands

ligands which donate π electron to central metal ion & also accept electron density from central metal ion through synergic bonding.

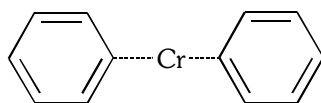
(i) $\text{H}_2\text{C}=\text{CH}_2$ (η^2 - ethylene) : it is a 2π electron donor

(ii) (η^6 -benzene) : it is a 6π electron donor

(iii) / $[\text{C}_5\text{H}_5]^-$ (η^5 -cyclopentadienyl) : it is a 6π electron donor

for example

(a) $[\text{Cr}(\eta^6-\text{C}_6\text{H}_6)_2]$



sandwich compound

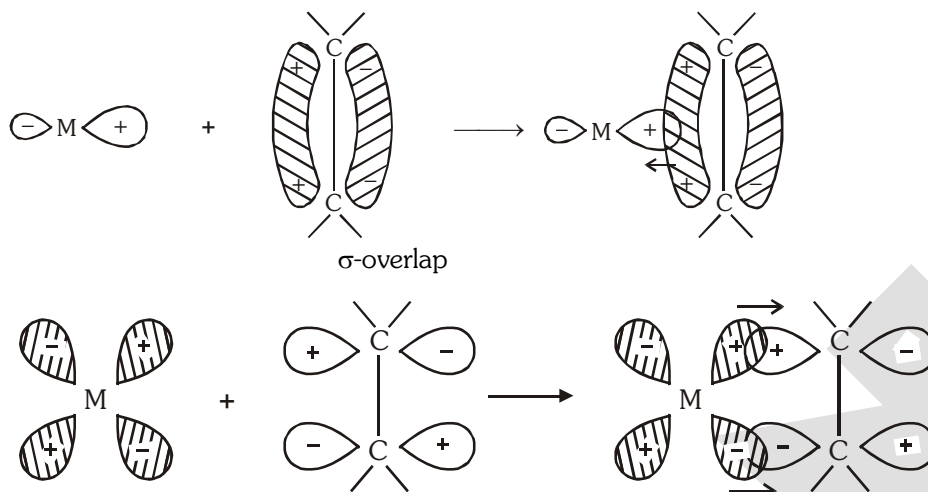
(b) $[\text{Fe}(\eta^5-\text{C}_5\text{H}_5)_2]$

ferrocene (sandwich compound)

(c) $\text{K}[\text{PtCl}_3(\eta^2-\text{C}_2\text{H}_4)]$

(Zeise's salt)

The bonding of alkenes to a transition metal to form complexes has two components. First, the π -electron density of the alkene overlaps with a σ -type vacant orbital of the metal atom. Second is the back bonding formed by the flow of electron density from a filled d-orbital on the metal into the vacant π^* -antibonding molecular orbital of the carbon atom as shown below:



BEGINNER'S BOX-1

- Which of the following is true about metal-EDTA complex (if it is a 3d metal)
 - (1) The ratio of moles of metal & EDTA in complex is 1:6
 - (2) The ratio of moles of metal & EDTA in complex is 6:1
 - (3) The ratio of moles of metal & EDTA in complex is 1:1
 - (4) 3d metal ion cannot bonded with EDTA
- CN^- is ambidentate ligand because
 - (1) it has multiple bonds
 - (2) carbon as a negative charge
 - (3) it form chelate
 - (4) both C & N can donate lone pair
- Highest C–O bond length will be in
 - (1) $[\text{Mn}(\text{CO})_6]^+$
 - (2) $[\text{Cr}(\text{CO})_6]$
 - (3) $[\text{V}(\text{CO})_6]^-$
 - (4) same in all
- The strongest –CO bond is present in
 - (1) $[\text{Cr}(\text{CO})_6]^+$
 - (2) $[\text{Fe}(\text{CO})_5]$
 - (3) $[\text{V}(\text{CO})_6]^-$
 - (4) all have equal strength
- Match the following

Ligand

- (A) ethylenediamine
- (B) hydrazene
- (C) dimethylglyoximate(dmg^-)
- (D) dien

Property

- (P) bidentate
- (Q) tridentate
- (R) form hydrazenium ion on protonation
- (S) can form dimer

[Ans. A $\rightarrow$ P, B $\rightarrow$ R, C $\rightarrow$ P, D $\rightarrow$ Q]

- Match the following

Complex

- (A) $[\text{Ni}(\text{dmg})_2]$
- (B) $[\text{Ca}(\text{EDTA})]^{2-}$
- (C) $[\text{Fe}(\text{en})_3]\text{Cl}_3$
- (D) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$

Property

- (P) number of 5 member rings = 2
- (Q) number of 6 member rings = 2
- (R) coordination number of central metal ion = 2
- (S) coordination number of central metal ion = 6
- (T) intramolecular H-bonding

[Ans. A $\rightarrow$ P, Q, T; B $\rightarrow$ S; C $\rightarrow$ S; D $\rightarrow$ R]

7. The oxidation state of metal in Iron in $[\text{Fe}(\text{CN})_6]^{3-}$ is
 (1) -6 (2) +3 (3) -3 (4) +6
8. Coordination number of Co in $[\text{CoF}_6]^{3-}$ is
 (1) 4 (2) 5 (3) 6 (4) 8
9. The neutral ligand is
 (1) Chlorido (2) Hydroxido (3) Ammine (4) Oxalato
10. $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$ serves as :
 (1) Monodentate ligand (2) Chelating ligand (3) Bridging ligand (4) Cationic ligand

3.2 IUPAC NOMENCLATURE OF COORDINATION COMPOUNDS

The present system of nomenclature derived from the suggestions of **Alfred Werner** and recommended by the Inorganic Nomenclature Committee of the I.U.P.A.C. The main rules of naming of complexes are -

- (a) Like simple salts, the positive part of the coordination compound is named first.

Ex. $\text{K}_4[\text{Fe}(\text{CN})_6]$ the naming of this complex starts with potassium.

- (b) Then after ligands of the coordination sphere are to be named.

- (c) The ligands can be neutral, anionic or cationic.

- (i) The neutral ligands are named as the molecule

Ex. $\text{C}_5\text{H}_5\text{N}$ (pyridine), $(\text{C}_6\text{H}_5)_3\text{P}$ (Triphenyl phosphine),

$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$ (ethylene diamine)

The neutral ligands which are not named as the molecule are

CO (carbonyl), NO (nitrosyl), H_2O (Aqua), NH_3 (ammine)

- (ii) Anionic ligands ending with 'ide' are named by replacing the 'ide' with suffix 'O'.

Symbol	Name as ligand	Symbol	Name as ligand
Cl^-	Chloro/Chlorido	N^{3-}	Nitrido
Br^-	Bromo/Bromido	O_2^{2-}	Peroxo, O_2^- superoxo
CN^-	Cyano	O_2H^-	Perhydroxo
O^{2-}	Oxo	S^{2-}	Sulphido
OH^-	Hydroxo	NH_2^-	Amido
H^-	Hydrido	NH^{2-}	imido

Ligands whose names end in 'ite' or 'ate' become 'ito' i.e., by replacing the ending 'e' with 'o' as follows.

CO_3^{2-}	Carbonato	SO_3^{2-}	Sulphito
$\text{C}_2\text{O}_4^{2-}$	Oxalato (Ox) ²⁻	CH_3COO^-	Acetato
SO_4^{2-}	Sulphato	NO_2^-	(bonded through oxygen) nitrite
NO_3^-	Nitrato		(bonded through nitrogen) nitro
$\text{S}_2\text{O}_3^{2-}$	Thiosulphato		

→ $\begin{bmatrix} \text{Nitrito} - \text{N} \\ \text{Nitrito} - \text{O} \end{bmatrix}$

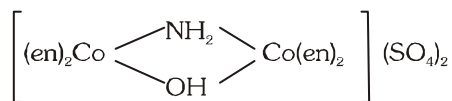
- (iii) Positive ligands naming ends in 'ium'

$\text{NH}_2-\text{NH}_3^+$ (Hydrazinium), NO_2^+ (nitronium); NO^+ (nitrosonium)

- (d) If ligands are present more than once, then their repetition is indicated by prefixes like di, tri, tetra etc. However, when the name of the ligand includes a number.

Ex. dipyriddy, ethylene diamine, then bis, tris, tetrakis are used in place of di, tri, tetra, etc.

- (e) In case of chelating ligands or ligands having di, tri, tetra, etc, in their name the prefixes bis, tris, tetrakis are used before ligands placed in paranthesis.



Bis(ethylenediamine) cobalt(III)-μ- amido-μ- hydroxobis(ethylenediamine) cobalt (III) sulphate

- (f) When more than one type of ligand is present in the complex, then the ligands are named in the alphabetical order.
- (g) After naming of ligands the central metal ion is to be named followed by its oxidation state in Roman numbers in brackets. (as per IUPAC)

If the complex is neutral or provides a cationic complex ion, then the central metal ion is to be named as it is.

If the complex provides anionic complex ion then the name of central metal ion ends in 'ate'

- (h) After the naming of central metal ion, anion which is in the outer sphere is to be named.

The naming of some of the complexes is done as follows – (as per IUPAC)

Complex Compounds	IUPAC Name
(i) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (anionic complex) so suffix ate is added with metal's name	Potassium hexacyanidoferrate (II)
(ii) $\text{K}_2[\text{Pt Cl}_6]$	Potassium hexachloridoplatinate (IV)
(iii) $[\text{Co}(\text{NH}_3)_6] \text{Cl}_3$ (Cationic complex) so metal is without any suffix	Hexamminecobalt (III) chloride
(iv) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \text{Cl}$	Tetraaquadichlorochromium (III) chloride
(v) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	Diammine tetrachloroplatinum (IV)
(vi) $[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$ (Neutral complex) So no suffix is used with metal ion	Triamminetrichlorocobalt (III)
(vii) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$	Potassium hexanitrocobaltate (III)
(viii) $\text{Na}_2[\text{Fe}(\text{CN})_5 \text{NO}]$	Sodium pentacyanonitrososulfate (II)
(ix) $[\text{NiCl}_4]^{2-}$	Tetrachloronickelate (II) ion
(x) $[\text{Ru}(\text{NH}_3)_5\text{Cl}]^{+2}$	Pentamminechlororuthenium (III) ion
(xi) $[\text{Fe}(\text{en})_3]\text{Cl}_3$	Tris (ethylenediamine) iron (III) chloride
(xii) $[\text{Ni}(\text{Gly})_2]$	Bis (glycinato) nickel (II)

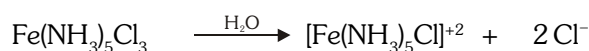
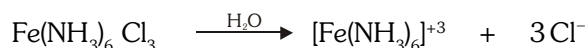
- (i) If coordination compound contains 2 coordination spheres then both spheres should have opposite charges. Name of cationic part written first following the name of anionic sphere according to the general IUPAC rule.

Ex. $[\text{Fe}(\text{NH}_3)_6]^{3+}[\text{Co}(\text{NO}_2)_6]^{3-}$ [Hexamine iron (II) hexanitro cobaltate(III)]

3.3 BONDING IN COORDINATION COMPOUNDS

(A) Werner's Theory

This was postulated by **Alfred Werner** in 1893. The postulates of Werner's theory of complexes are –



Every metal forming a complex exhibits two types of valencies - Primary valency and Secondary Valency. In the above complexes chlorine act as primary valency and NH_3 acts as secondary valency.

According to Werner central metal have 2 types of valencies :

Primary valencies	Secondary valencies
(i) Equal to oxidation state of CMA	(i) Equal to coordination number of CMA
(ii) Satisfied by anions	(ii) satisfied by ligands
(iii) Non directional	(iii) Directional
(iv) Ionisable	(iv) Non-ionisable
(v) not helpful geometry predictions	(v) helpful in geometry predictions

Shapes : According to Werner all secondary valencies has shapes arrangement according to coordination number

Experimental evidence of werner's theory

- (a) Precipitation of primary valencies on the addition of a suitable reagent.

Ex. $\text{Fe}(\text{NH}_4)_2\text{Cl}_6$ forms 3 moles of AgCl in the form of precipitate on addition of AgNO_3 solution.



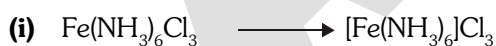
- (b) **Electrical conductance of complexes** : More the number of ions provided greater is the electrical conductance of the complex in aqueous medium. **Electrical conductivity $\propto$ no. of Ions.**

Ex. The electrical conductance of aqueous $[\text{Fe}(\text{NH}_3)_6] \text{Cl}_3$ is greater than that of aqueous solution of $[\text{Fe}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$.

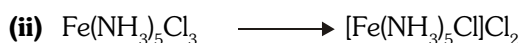
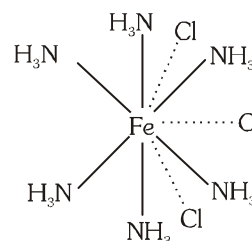


Complex	Modern formula	No. of Cl ⁻ ions precipitated	Total number of ions
PtCl ₄ ·6NH ₃	[Pt(NH ₃) ₆]Cl ₄	4	5
PtCl ₄ ·5NH ₃	[Pt(NH ₃) ₅ Cl]Cl ₃	3	4
PtCl ₄ ·4NH ₃	[Pt(NH ₃) ₄ Cl ₂]Cl ₂	2	3
PtCl ₄ ·3NH ₃	[Pt(NH ₃) ₃ Cl ₃]Cl	1	2
PtCl ₄ ·2NH ₃	[Pt(NH ₃) ₂ Cl ₄]	0	0 (non-electrolyte)

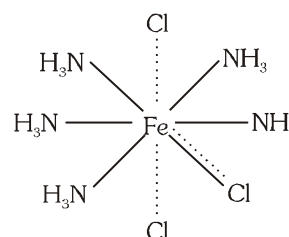
Werner's Representation of complexes

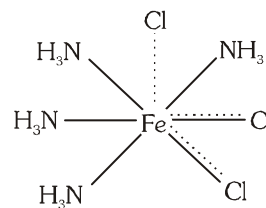
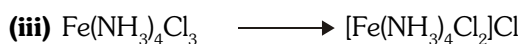


Dotted lines indicate primary valency and continuous lines indicate secondary valency of metal ion.



In this complex 'Cl' groups act as primary valencies and one of the 'Cl' acts as secondary valency also. Thus werner's representation of this complex –





In this complex 'Cl' groups act as primary valency and two of the 'Cl' group act as secondary valencies also. Thus
werner's representation of this complex –

$$\text{E.A.N.} = (\text{No. of electron in the metal ion}) + (\text{No. of electrons gained from the ligands})$$

OR

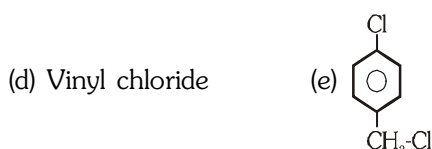
$$\text{Atomic number} - (\text{Oxidation state}) + (2 \times \text{Coordination number})$$

Example :

Complex	Metal (Oxidation State)	Atomic Number of Metal	Coordination number	Effective atomic number (E.A.N.)
$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$	+2	29	4	$(29 - 2) + (4 \times 2) = 35$
$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	+3	27	6	$(27 - 3) + (6 \times 2) = 36$ [Kr]
$\text{K}_2[\text{HgI}_4]$	+2	80	4	$(80 - 2) + (4 \times 2) = 86$ [Rn]
$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	+1	47	2	$(47 - 1) + (2 \times 2) = 50$

BEGINNER'S BOX-2

- The IUPAC name of $\text{Fe}(\text{CO})_5$ is
 (1) Pentacarbonyl ferrate (0) (2) Pentacarbonyl Ferrate (III)
 (3) Pentacarbonyl Iron (0) (4) Pentacarbonyl Iron (II)
- The compound $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$ is called
 (1) Sodium pentacyanonitrosonium ferrate (II) (2) Sodium nitroprusside
 (3) Sodium nitrosoferrocyanide (4) 1 & 2 both
- $\text{FeCl}_3 \cdot 4\text{H}_2\text{O}$ is actually :
 (1) $[\text{Fe}(\text{H}_2\text{O})_4]\text{Cl}_3$ (2) $[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (3) $[\text{Fe}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}$ (4) $[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$
- Which of the following compound does not give test of sulphate ion in aqueous solution
 (1) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ (2) $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4 \cdot \text{H}_2\text{O}$
 (3) $[\text{CoSO}_4(\text{NH}_3)_5]\text{Br}$ (4) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
- Which of the following compound is expected to yield a white precipitate with AgNO_3 solution :-
 (a) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (b) $\text{CoCl}_3 \cdot 3\text{NH}_3$ (c) Impure chloroform



(f) Carnallite

(1) a, c, f, e

(2) a, d, e

(3) a, b, c, f

(4) a, b, e, f

6. Which of the following is most likely formula of platinum complex, if $\frac{1}{4}$ of total chlorine of the compound is precipitated by adding AgNO_3 to its aqueous solution
 (1) $\text{PtCl}_4 \cdot 6\text{H}_2\text{O}$ (2) $\text{PtCl}_4 \cdot 5\text{H}_2\text{O}$ (3) $\text{PtCl}_2 \cdot 2\text{H}_2\text{O}$ (4) $\text{PtCl}_4 \cdot 3\text{H}_2\text{O}$
7. The Effective atomic number of Cr in $\text{Cr}(\text{CO})_6$ is
 (1) 36 (2) 38 (3) 28 (4) 54
8. According to the rule of effective atomic number, central atom acquires :
 (1) Inert gas configuration (2) Duplet
 (3) Pseudo inert electronic configuration (4) Quartet
9. The Effective atomic number of iron in $\text{K}_4[\text{Fe}(\text{CN})_6]$ is :
 (1) 35 (2) 34 (3) 36 (4) 38
10. The formula of the complex tris (ethylene diamine) cobalt (III) sulphate is
 (1) $[\text{Co}(\text{en})_2\text{SO}_4]$ (2) $[\text{Co}(\text{en})_3\text{SO}_4]$ (3) $[\text{Co}(\text{en})_3]\text{SO}_4$ (4) $[\text{Co}(\text{en})_{3/2}(\text{SO}_4)_3]$

(B) VALENCE BOND THEORY

This theory was mainly developed by **Linus Pauling**. The main features of this theory are -

- Every metal ion when it forms a complex compound undergoes formation of coordinate covalent bond.
- During this bond formation, the metal ion acts as electron pair acceptor. For this the metal ion provides vacant orbitals.
- The number of vacant orbitals provided is equal to the coordination number of metal ion.

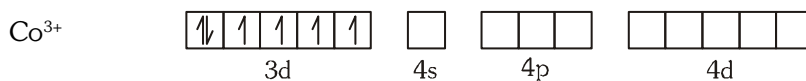
Ex. In the formation of $[\text{Fe}(\text{NH}_3)_6]^{3+}$, Fe^{+3} ion provides six vacant orbitals.

In $[\text{Cu}(\text{NH}_3)_4]^{2+}$, Cu^{+2} ion provides four vacant orbitals.

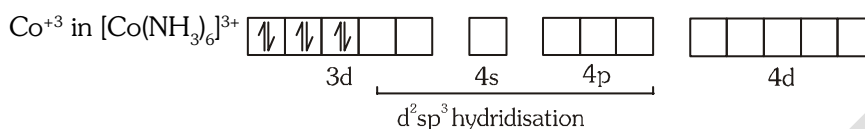
- The metal provides vacant orbitals only after the process of hybridisation, thus vacant hybrid orbitals are provided by the metal ion.
- The vacant hybrid orbitals of metal ion get overlapped by orbitals of ligands containing lone pair of electrons.
- The number of such overlappings is equal to the coordination number of metal ion.
- The empty 'd' orbitals involved in hybridisation may be inner (n-1)d or outer "nd" orbitals.
- These complexes are called as **Inner orbital complexes** and **outer orbital complexes** respectively. It is applicable for coordination number = 6 (complexes).
- If inner 'd' orbitals are involved in hybridisation, then it is through the pairing of unpaired electrons in the (n-1)d orbitals of metal ion.
- Then such type of complexes will be **diamagnetic** or **less paramagnetic** and will be called as **Low spin complexes**.
- Generally outer orbital complexes have **paramagnetic** nature and they are called as **High spin complexes**.

Applications of valence bond Theory

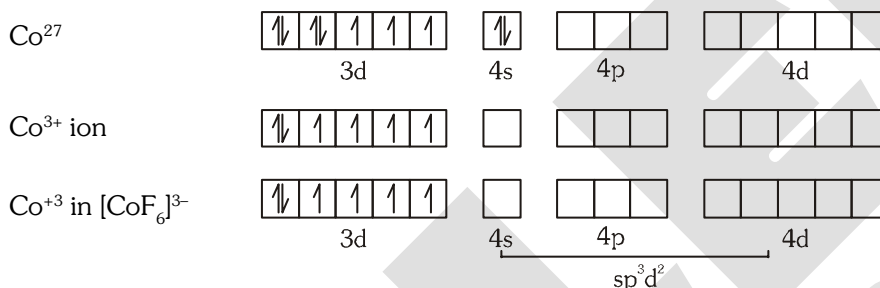
(a) To 6-coordinated complexes :

 e^- configuration of $_{27}\text{Co} = [\text{Ar}] 3d^7 4s^2$ So, $\text{Co}^{+3} = [\text{Ar}] 3d^6$ 

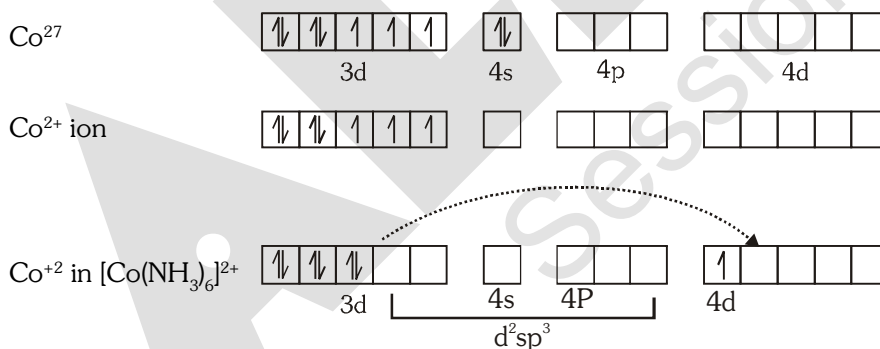
Due to presence strong ligand electrons get paired up so metal ion provides vacant '3d' orbitals for hybridisation.



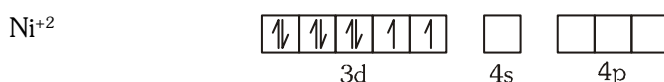
- This is a **diamagnetic complex**.
- As inner 'd' orbitals are involved in hybridisation, hence it is an **inner orbital complex**.
- It is a **low spin complex**.



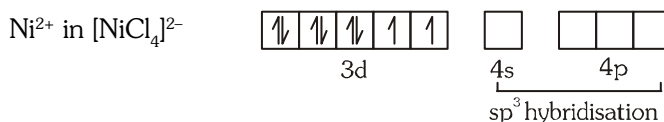
- This is a **paramagnetic complex** and **high spin complex**.
- The outer 'd' orbitals are involved in hybridisation, hence it is an **outer orbital complex**.



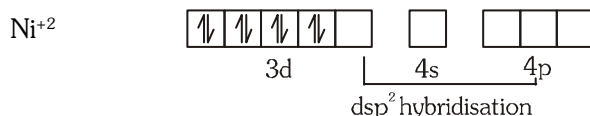
(b) 4-coordinated complexes :



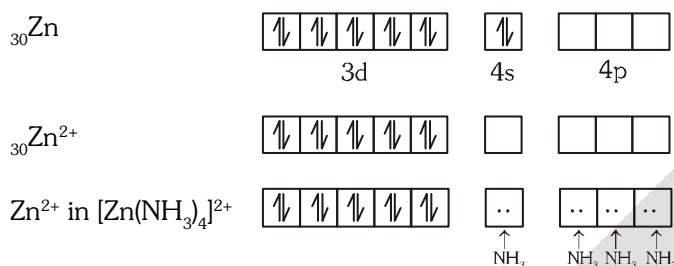
- The unpaired electrons remain as such because Cl^- is a weak ligand as it is a **paramagnetic complex**.



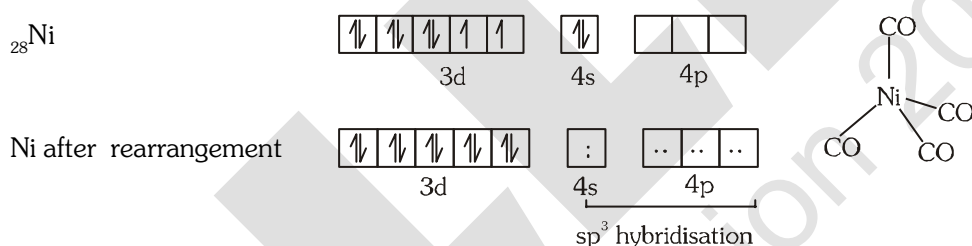
- $[\text{NiCl}_4]^{2-}$ has **tetrahedral geometry**.
- It is a **high spin complex**.



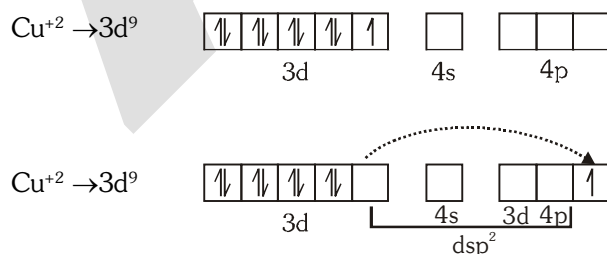
- CN^- is **strong field ligand**. Thus the complex has **square planar geometry**.
- The unpaired electrons in '3d' orbitals of Ni^{2+} get paired up, So $[\text{Ni}(\text{CN})_4]^{2-}$ is **diamagnetic** in nature.
- This Ni^{2+} complex is an **inner orbital complex** and a **low spin complex**.



- Because the complex is formed by sp^3 hybridisation, hence it is **tetrahedral**.
- Since all electrons are paired, hence it is **diamagnetic**.

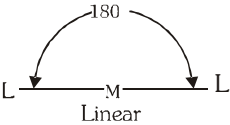
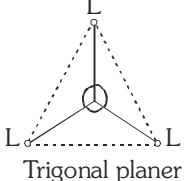
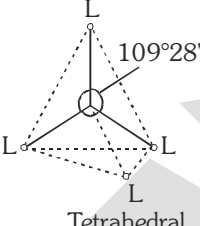
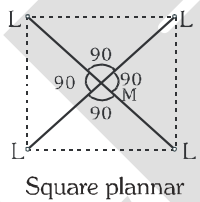
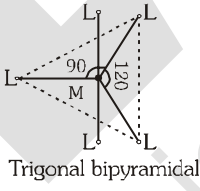
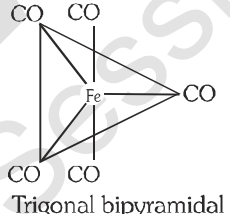
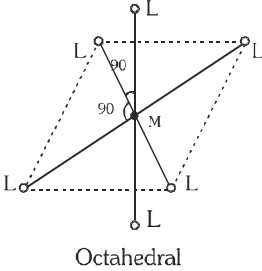


- CO is a strong ligand. So it is **low spin complex**.
- The empty one 4s and three 4p orbitals mix to form four new sp^3 hybridised orbitals.
- Each orbital accepts one electron pair from carbon monoxide molecule forming $[\text{Ni}(\text{CO})_4]$.
- It is **diamagnetic** in nature, the shape of nickel tetra carbonyl is **tetrahedral** as shown above.



Note : In case, if metal belongs to 4d & 5d series all ligand will act as a strong field ligand.

e.g. $[\text{RhF}_6]^{3-}$, $[\text{Rh}(\text{NH}_3)_6]^{3+}$

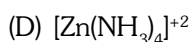
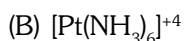
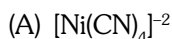
Coordination Number	Hybridised orbital	Geometrical shape of the Complex	Examples of Complex
2	sp	 Linear	$[\text{Ag}(\text{NH}_3)_2]^+$ $[\text{Ag}(\text{CN})_2]^-$
3	sp^2	 Trigonal planar	$[\text{HgI}_3]^-$
4	sp^3	 Tetrahedral	$[\text{CuCl}_4]^{2-}$ $[\text{ZnCl}_4]^{2-}$ $[\text{FeCl}_4]^-$ $[\text{Ni}(\text{CO})_4]$ $[\text{Zn}(\text{NH}_3)_4]^{+2}$
4	dsp^2 ($d_{x^2-y^2}$ orbital)	 Square planar	$[\text{PdCl}_4]^{2-}$ $[\text{Ni}(\text{CN})_4]^{2-}$ $[\text{Pt}(\text{NH}_3)_4]^{+2}$ $[\text{Cu}(\text{NH}_3)_4]^{+2}$ $[\text{PtCl}_4]^{2-}$
5	sp^3d	 Trigonal bipyramidal	$[\text{CuCl}_5]^{3-}$
5	dsp^3	 Trigonal bipyramidal	$[\text{Fe}(\text{CO})_5]$
6	d^2sp^3 When d-orbitals are (n-1)d-orbitals (Inner orbital complexes) or sp^3d^2 When d-orbitals are nd orbitals (Outer orbital complexes) In both cases d-orbitals are d_{z^2} and $d_{x^2-y^2}$ orbitals	 Octahedral	$[\text{Cr}(\text{NH}_3)_6]^{+3}$ $[\text{Ti}(\text{H}_2\text{O})_6]^{+3}$ $[\text{Fe}(\text{CN})_6]^{3-}$ $[\text{Co}(\text{NH}_3)_6]^{+3}$ $[\text{PtCl}_6]^{2-}, [\text{CoF}_6]^{3-}$

BEGINNER'S BOX-3

- Complex ion square planar geometry :-
 (1) $[\text{Ni}(\text{CN})_4]^{2-}$ (2) $[\text{Cu}(\text{CN})_4]^{-2}$ (3) $[\text{Cu}(\text{NH}_3)_4]^{+2}$ (4) All
- Which is low spin complex ?
 (1) $[\text{Fe}(\text{CN})_6]^{3-}$ (2) $[\text{Co}(\text{NO}_2)_6]^{3-}$ (3) $[\text{Mn}(\text{CN})_6]^{3-}$ (4) All
- For the complex $[\text{Fe}(\text{CO})_x]$ what is wrong :-
 (1) It is σ - π bonded organometallic compound (2) In the complex value of $x = 6$
 (3) In the complex CO is π -acid ligand (4) It is trigonal bipyramidal shape
- In the complex ion ML_6^{n+} , M^{n+} has five d-electrons and L is weak field ligand. According to crystal field theory, the magnetic properties of the complex ion corresponds to how many unpaired electrons
 (1) 0 (2) 5 (3) 2 (4) 3
- Among $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{Cl})_4]^{2-}$:
 (1) $[\text{Ni}(\text{CO})_4]$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic
 (2) $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CO})_4]$ is paramagnetic
 (3) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic
 (4) $[\text{Ni}(\text{CO})_4]$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic
- Which is correct geometry in the case of $[\text{NiCl}_4]^{2-}$ complex ?
 (1) Tetrahedral (2) Octahedral (3) Square planar (4) Pyramidal
- For $\text{K}_3[\text{CoF}_6]$, incorrect statement is :
 (1) It is high spin complex (2) Its magnetic moment is $\sqrt{24}$ BM
 (3) Primary valency of Co is six (4) Hybridisation state of CoF_6^{3-} is sp^3d^2
- Which of the following compound is square planar and does not have any unpaired electron:
 (a) $[\text{Cu}(\text{NH}_3)_4]^{+2}$ (b) XeF_4 (c) $[\text{Ni}(\text{CO})_4]$ (d) $[\text{NiCl}_4]^{2-}$ (e) $[\text{Ni}(\text{CN})_4]^{2-}$ (f) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 (1) a, e, f (2) a, b, c (3) b, e (4) b, c, f
- Which of the following complex does not have chelation
 (1) $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$ (2) $[\text{Pt}(\text{en})_2]\text{Cl}_2$ (3) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ (4) $\text{K}_2[\text{Ni}(\text{EDTA})]$
- Match list - I with list - II and choose the correct answer from the code given below

List - I (complex)

List - II (geometry)



A	B	C	D
(1) b	a	d	c
(2) a	b	c	d
(3) c	d	b	a
(4) b	c	d	a

(C) Crystal field theory

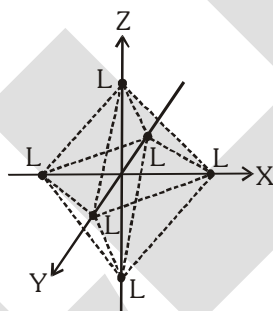
Crystal Field Theory : This is a model of electronic structure of transition-metal complexes that considers how the energies of the d-orbitals of a metal ion are affected by the electric field of the ligand. According to this theory.

- The ligands in a transition-metal complex are treated as point charges.
- A ligand anion becomes simply a point of negative charge. A neutral molecule, with its electron pair that it donates to the metal atom, is replaced by a negative charge, representing the negative end of the molecular dipole.
- In the electric field of these negative charges, the five d orbitals of the metal atom no longer have exactly same energy. Splitting of five degenerate d-orbitals of the metal ion into sets of orbitals having different energies is called crystal field splitting.
- The extent of splitting of metal d-orbitals depends upon the nature and number of ligands surrounding it and the charge on the central metal ion.
- The extent of splitting of metal d-orbitals determines the magnetic and spectroscopic properties of the complexes.

SPLITTING IN OCTAHEDRAL COMPLEXES

In an octahedral complex, the co-ordination number is 6. The metal ion is at the centre and the ligands occupy the six corners of the octahedron as shown in figure.

We know that two orbitals, $d_{x^2-y^2}$ and d_{z^2} are oriented along the axis while the remaining three orbitals, viz., d_{xy} , d_{yz} and d_{zx} are oriented in between the axis.

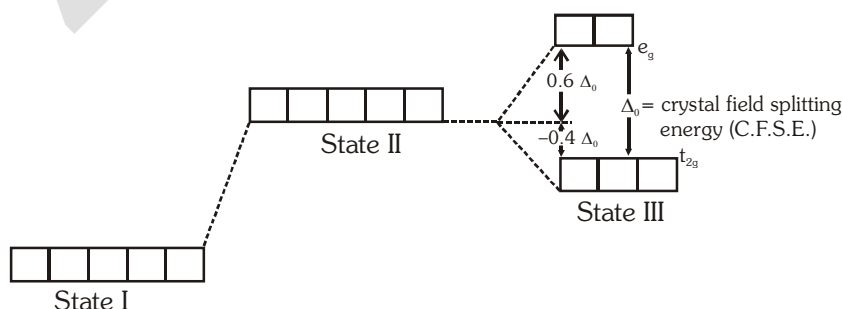


The two orbitals $d_{x^2-y^2}$ and d_{z^2} are designated as e_g orbitals while the three orbitals d_{xy} , d_{yz} and d_{zx} are designated as t_{2g} orbitals. As the six ligands approach the central ion along the axis, e_g orbitals, is repelled more by the ligand than in the t_{2g} orbitals.

In other words, the energy of the d_{z^2} and $d_{x^2-y^2}$ orbitals increases

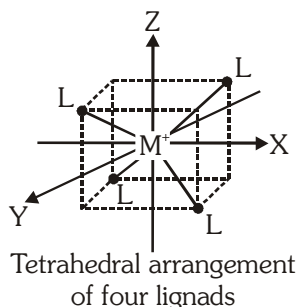
much more than the energy of the d_{xy} , d_{yz} and d_{zx} orbitals.

Thus, in octahedral complexes, the five d-orbitals split up into two sets : one set consisting of two orbitals ($d_{x^2-y^2}$ and d_{z^2}) of higher energy (e_g orbitals) and the other set consisting of three orbitals (d_{xy} , d_{yz} and d_{zx}) of lower energy (t_{2g} orbitals).



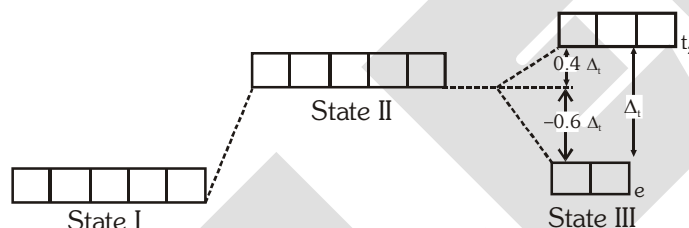
Splitting in tetrahedral complex

The co-ordination number for tetrahedral complexes is 4. The tetrahedral arrangement of four ligands surrounding a metal ion may be visualized by placing ligands at the alternate corners of a cube, as shown in figure.



It can be shown that in a tetrahedral structure, none of the d-orbitals points exactly towards the ligands. When ligand approaches it is more close d_{xy} , d_{yz} , d_{zx} in comparison of $d_{x^2-y^2}$ and d_{z^2} because d_{xy} , d_{yz} , d_{zx} are between the axis and d_{z^2} and $d_{x^2-y^2}$ are along the. So d_{xy} , d_{yz} , d_{zx} feels more repulsion as compare to d_{z^2} and $d_{x^2-y^2}$.

Thus, the d orbitals are also splitted into two groups but in a reverse order. The three orbitals, d_{xy} , d_{yz} and d_{zx} , designated as t_2 orbitals, now have higher energy than the two orbitals $d_{x^2-y^2}$ and d_{z^2} designated as e_g-orbitals.



Calculation of crystal field stablizing energy (CFSE)

- (i) For octahedral CFSE = $[-0.4(n_{t_{2g}}) + 0.6(n_{e_g})]\Delta_o$ + Pairing energy (P.E.)

where $n_{t_{2g}}$ = number of electron in t_{2g} orbitals

n_{e_g} = number of electron in e_g orbitals

Δ_o = crystal field **splitting** energy

- (ii) For tetrahedral CFSE = $[-0.6(n_e) + 0.4(n_{t_2})]\Delta_t$ + Pairing energy (P.E.)

where n_{t_2} = number of electron in t_2 orbitals

n_e = number of electron in e orbitals

Δ_t = crystal field **splitting** energy

Example

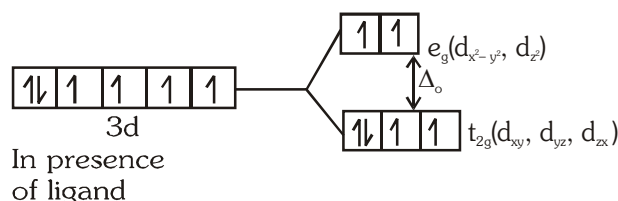
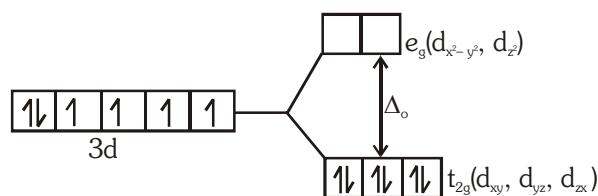
Explain the formation of $\text{Na}_4[\text{Fe}(\text{CN})_6]$ and $\text{Na}_4[\text{FeF}_6]$? Show which is low spin and which is high spin complex and also calculate the Crystal field stablisation energy (CFSE) ?

Solution

In given compounds CN^- is strong field ligands and F^- is weak field ligand and in both compounds ions is in +2 oxidation state d^6 configuration.

So in $\text{Na}_4[\text{Fe}(\text{CN})_6]$

$\text{Na}_4[\text{FeF}_6]$



For octahedral CFSE = $\left[-0.4(n_{t_{2g}}) + 0.6(n_{e_g})\right] \Delta_0 + P.E.$

where $n_{t_{2g}}$ = number of electron in t_{2g} orbitals

n_{e_g} = number of electron in e_g orbitals

$Na_4[Fe(CN)_6]$ CFSE = $-2.4 \Delta_0 + 3P$. $[n_{t_{2g}} = 6, n_{e_g} = 0]$

$Na_4[Fe(F)_6]$ CFSE = $-0.4 \Delta_0 + P$ $[n_{t_{2g}} = 4, n_{e_g} = 2]$

where P = pairing energy to pair up electron.

Factors affecting Δ

factor affecting splitting

- Strength of ligand
- Oxidation state of central metal ion
- Transition series (d-series)
- Geometry (number of ligands).

[C.F.S.E. is more in case of S.F.L. as compared to W.F.L.]

[C.F.S.E. $\propto$ oxidation state]

[C.F.S.E. , $5d > 4d > 3d$]

$[\Delta_{sq} > \Delta_0 > \Delta_t]$

$$\Delta_{sp} = \frac{4}{3} \Delta_0 \quad ; \quad \Delta_t = \frac{4}{9} \Delta_0 \quad ; \quad [\Delta_{sp} = 1.3 \Delta_0]$$

Factors affecting strength of ligands

Strength of ligand depends upon :

- good σ donor
- good π acceptor
- high negative charge
- Small in size

Spectro-chemical series for ligands

Series which shows the relative strength of ligands

I(weakest) < Br < SCN⁻ < Cl⁻ < S²⁻ > NO₃⁻ < F⁻ < OH⁻ < C₂O₄²⁻ < H₂O < NCS⁻ < edta⁴⁻ < NH₃ < en < NO₂⁻ < CN⁻ < CO(strongest)

APPLICATION OF CRYSTAL FIELD THEORY

Applications of C.F.T are

- To predict the geometry that the compound is either inner orbital or outer orbital complex.
- To calculate the magnitude of paramagnetism.
- To show the colour property.

Que.

Give the magnetism in following configuration

- | | |
|---|---|
| (i) d ⁴ high spin octahedral | (ii) d ⁴ low spin octahedral |
| (iii) d ⁵ high spin octahedral | (iv) d ⁵ tetrahedral |
| (v) d ⁶ tetrahedral | (vi) d ⁸ low spin octahedral |
| (vii) d ⁷ tetrahedral | (viii) d ⁷ high spin octahedral. |

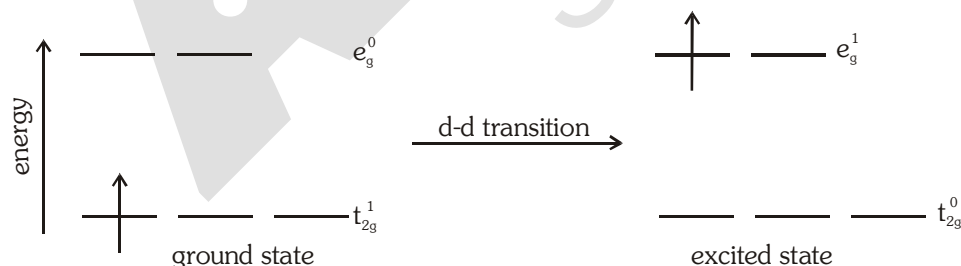
Solution

- | | |
|-------------------------------|--------------------------------|
| (i) four unpaired electron | (ii) two unpaired electron |
| (iii) five unpaired electron | (iv) five unpaired electron |
| (v) four unpaired electron | (vi) two unpaired electron |
| (vii) three unpaired electron | (viii) three unpaired electron |

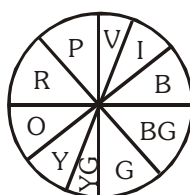
(II)

COLOUR PROPERTY :

Reason for colour in complexes is due to d-d-transition



Complementary colour wheel



For example complementary colour of red is blue green.

ALLEN**Example.** **$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is violet in colour explain using CFT.****Solution**

In $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ d-orbitals of Ti^{3+} lost their degeneracy in the presence of octahedral ligand field and produce t_{2g}^1 & e_g^0 . orbital of different energy complex absorbed visible light for excitation of electron from t_{2g}^0 to e_g^1 (d-d transition) and show complimentary violet colour.

Example.**How the complex compounds show the colour?****Solution**

When d-electrons absorb energy from visible region they will get excited. Absorbed energy is related to a particular wavelength.

$$E(\Delta)_{\text{absorbed}} = \frac{hc}{\lambda_{\text{absorbed}}}$$

when electrons fall into lower energy level it will show colour whose wavelength (λ) is the complimentary of absorbed wavelength ($\lambda_{\text{absorbed}}$).

Example.**Why violet coloured $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$ becomes colourless when heated ?****Solution**

When $[\text{Ti}(\text{H}_2\text{O})_6]\text{Cl}_3$ is heated water molecules are removed and in the absence of ligand crystal field splitting does not occur and hence the substance is colourless.

Example. **$[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ are of different colours in dilute solutions. Why ?****Solution**

It is due to difference in power of ligands.

Example.**What will be the correct order for the wavelengths of absorption in the visible region for the following : $[\text{Ni}(\text{NO}_2)_6]^{4-}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$?****Solution**

$[\text{Ni}(\text{H}_2\text{O})_6]^{2+} > [\text{Ni}(\text{NH}_3)_6]^{2+} > [\text{Ni}(\text{NO}_2)_6]^{4-}$

GOLDEN KEY POINTS

- Red colour of ruby is due to presence of small amount of Cr^{+3} in atmosphere of Al_2O_3
- Green colour of emerald is due to presence of Cr^{+3} in atmosphere of $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ [beryl]
- Complex become colourless in absence of ligand field because in absence of ligand d-d-transition does not occur.
- $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ become colourless on heating due to removal of water molecules.
- Anhydrous CuSO_4 is colourless but hydrated CuSO_4 is blue due to moisture.
- Halogens are coloured due to the HOMO-LUMO transition.
- KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, CrO_2Cl_2 are coloured due to charge transfer spectra.

3.4 STABILITY OF CO-ORDINATION COMPOUNDS IN SOLUTION

According to thermodynamic stability, the reaction between a metal ion and the ligands may be considered as a Lewis acid base reaction in solution. If the equilibrium constant is high then the complex obtained is thermodynamically stable in solution. The reaction can be written as follows : $M + nL \rightleftharpoons [ML_n]$

The stability constant K , of the complex $[ML_n]$ is given by the relation, $K = \frac{[ML_n]}{[M][L]^n}$. The greater the value of K ,

more stable is the complex.

The strength of a complex ion also depends upon –

- Higher charge of the central metal ion.
- Greater base strength of the ligand.
- Ring formation (chelation) in structure of complexes.

BEGINNER'S BOX-4

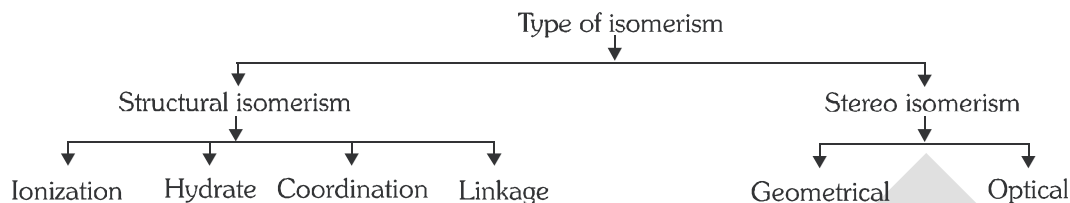
- Consider the following statements
 S_1 : $[Cr(NH_3)_6]^{3+}$ is an inner orbital complex with crystal field stabilization energy equal to $-1.2 \Delta_o$.
 S_2 : The complex formed by joining the CN^- ligands to Fe^{3+} ion has theoretical value of 'spin only' magnetic moment equal to 1.73 B.M.
 S_3 : $Na_2S + Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$,
 In reactant and product the oxidation states of iron are same and arrange in the order of true/false.
 (1) T T F (2) T T F (3) T T T (4) F F F
- For the $t_{2g}^6 e_g^2$ system, the value of magnetic moment (μ) is :
 (1) 2.83 B.M. (2) 1.73 B.M. (3) 3.87 B.M. (4) 4.92 B.M.
- Which of the following compound is not coloured :-
 (1) $Na_2[CuCl_4]$ (2) $Fe_4[Fe(CN)_6]_3$ (3) $Na_2[CdCl_4]$ (4) Na_2CrO_4
- The stability constants of the complexes formed by a metal ion M^{2+} with NH_3 , CN^- , H_2O are of the order of 10^{15} , 10^{27} , 10^{11} respectively. Then :-
 (1) NH_3 is the strongest ligand
 (2) CN^- is the strongest ligand
 (3) These values cannot predict the strength of the ligand
 (4) All the ligands are equally strong
- The value for crystal field stabilisation energy is zero for
 (1) $K_2[MnF_6]$ (2) $K_3[Fe(CN)_6]$ (3) $K_3[FeF_6]$ (4) $K_4[Fe(CN)_6]$
- Match the following

Complexes	Property
(i) $(CoCl_6)^{3-}$	(a) No. of electrons in $t_{2g} = 5$
(ii) $[Co(ox)_3]^{3-}$	(b) No. of electrons in $e_g = 0$
(iii) $[Co(H_2O)_6]^{2+}$	(c) No. of electrons in $t_{2g} = 6$
(iv) $[Fe(CN)_6]^{4-}$	(d) No. of electrons in $e_g = 2$

[Ans. i-d, ii-b, iii-a, iv-c]
- Which order is correct
 (1) $[NiCl_6]^{4-} > [NiCl_4]^{2-}$ splitting energy Δ_o
 (2) $[Cr(en)_3]Cl_3 > [Cr(H_2O)_6]Cl_3$ crystal field stabilisation energy
 (3) $[Co(CN)_6]^{4-} > [Co(CN)_6]^{3-}$ reducing behaviour
 (4) All of these
- Which is inner d-complex
 (1) $[Co(H_2O)_6]^{3+}$ (2) $[RhF_6]^{3-}$ (3) $[IrF_6]^{3-}$ (4) All
- Which stability order is incorrect
 (1) $[Fe(C_2O_4)_3]^{3-} > [Fe(H_2O)_6]^{3+}$ (2) $[Fe(EDTA)]^{2-} > [Fe(en)_3]^{3+}$
 (3) $[Ni(en)_2]^{2+} > [Ni(dmg)_2]$ (4) $[Fe(CN)_6]^{3-} > [Fe(CN)_6]^{4-}$
- The complex exhibit lowest energy absorption band is
 (1) $[NiCl_4]^{2-}$ (2) $[Ni(CO)_4]$ (3) $[Ni(CN)_4]^{2-}$ (4) $[Ni(H_2O)_6]^{2+}$

3.5 ISOMERISM IN COMPLEXES

- Compounds which have the same molecular formula, but differ in their properties due to the difference in structure are called as **Isomers**.
- Isomerism is commonly considered, to be the characteristic of only organic compounds, it is also found although less frequently among inorganic substances.
- Classification of isomerism :**



• Structural Isomerism

It arises due to the difference in the type of chemical linkages and distribution of ligands within and outside the coordination sphere.

• Ionisation Isomers

The type of isomerism which is due to the exchange of groups or ion between the coordinating sphere and the ionisation sphere.

Ex. (a) $\text{Co}(\text{NH}_3)_4\text{Br}_2\text{SO}_4$ can represent

(i) $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{SO}_4$ (red violet)

(ii) $[\text{Co}(\text{NH}_3)_4\text{SO}_4]\text{Br}_2$ (red).

These complexes give sulphate ion and bromide ion respectively

(b) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$

(c) $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_4\text{SO}_4](\text{NO}_3)_2$

• Hydrate Isomers

This type of isomerism is due to presence of different number of water molecules inside a coordination sphere.

Ex. (a) $\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_3$ has three possible structures

(i) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ violet

(ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ green

(iii) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ dark green.

These complexes differ from one another with respect to the number of water molecules acting as ligands.

Other hydrate isomers are

(i) $[\text{Co}(\text{NH}_3)_4\text{H}_2\text{OCl}]\text{Cl}_2$

(ii) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$

• Linkage or Salt Isomers

(a) This type of isomerism arises due to presence of ambidentate ligands like NO_2^- , CN^- and SCN^-

(b) These ligands have two donor atoms but at a time only one atom is directly linked to the central metal atom of the complex.

(c) Such type of isomers are distinguished by infra red (I.R.) spectroscopy.

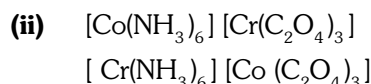
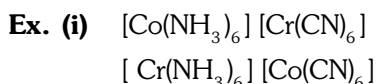
Ex. (i) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$

(ii) In NO_2^- ligand, The coordinating sites are nitrogen (i.e., NO_2^- Nitro ligand) or through oxygen (i.e. ONO Nitrito ligand)

(iii) The nitro isomer is yellow and is stable to acids whereas nitrito isomer is red and is decomposed by acids.

- Coordination Isomers**

- This type of isomerism is exhibited when the complex has two complex ions in it - '**Cationic and Anionic**'.
- This type of isomerism is caused by the interchange of ligands between the two complex ions of the same complex.



- Stereo Isomerism**

- They have same molecular formula, same constitution, they differ only with respect to the spatial orientation of ligands in space around the metal ion.
- The two stereo isomers which are possible are - Geometrical and optical.

(A) Geometrical or cis - Trans Isomers

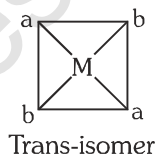
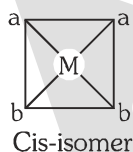
- The ligands occupy different positions around the central metal ion.
- When two identical ligands are coordinated to the metal ion from same side then it is **cis isomer**. (Latin, cis means same).
- If the two identical ligands are coordinated to the metal ion from opposite side then it is **Trans isomer** (in Latin, Trans means across).
- These geometrical isomers differ in physical as well as in chemical properties.
- Geometrical isomerism is most important in compounds with coordination numbers 4 and 6.
- 4-coordinated complexes with tetrahedral geometry do not exhibit cis- Trans isomerism, because relative position of ligands remain same for each other.
- It is exhibited by 4-coordinated complexes with square planar geometry.

Note : There are only 5 possible cases which shows geometrical isomerism in square planar complex.

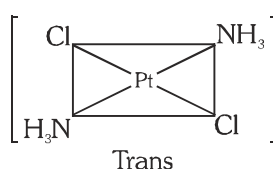
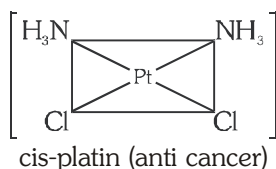
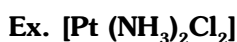
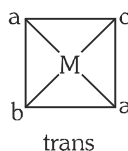
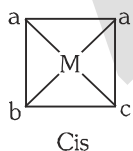
such as Ma_2b_2 , Ma_2bc , Mabcd , $\text{M}(\text{AB})_2$, $\text{M}(\text{AB})\text{cd}$

Geometrical Isomers With Coordination Number = 4 (Square Planar Complexes)

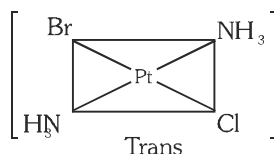
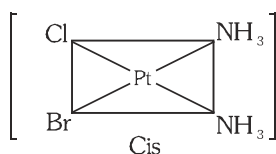
- (i)** Complexes with general formula, Ma_2b_2 (where both a and b are monodentate) can have Cis-and trans isomers.



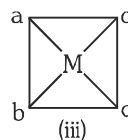
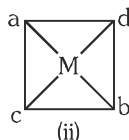
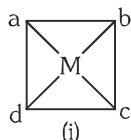
- (ii)** Complexes with general formula Ma_2bc can have Cis - and trans-isomers.



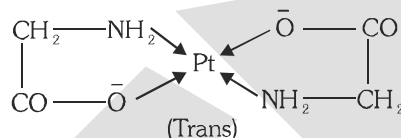
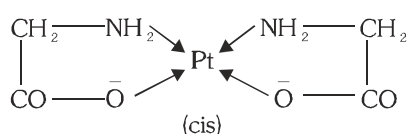
Ex. $[\text{Pt}(\text{NH}_3)_2\text{ClBr}]$



(iii) Complexes with general formula, Mabcd can have three isomers.

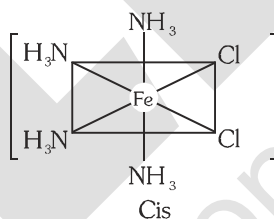
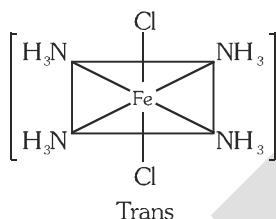


(iv) Diglycinato platinum (II) complexes

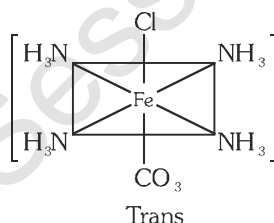
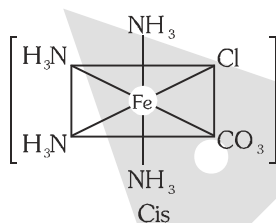


Geometrical isomers with Coordination number = 6

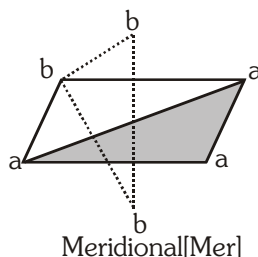
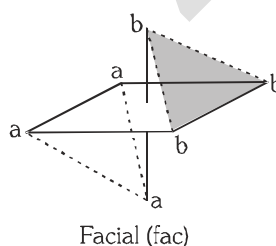
(i) $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]$



(ii) $[\text{Fe}(\text{NH}_3)_4\text{ClCO}_3]$



(iii) Facial and Meridional isomerism (Ma_3b_3)



Note : Other 6-Coordinated geometrical isomers are –

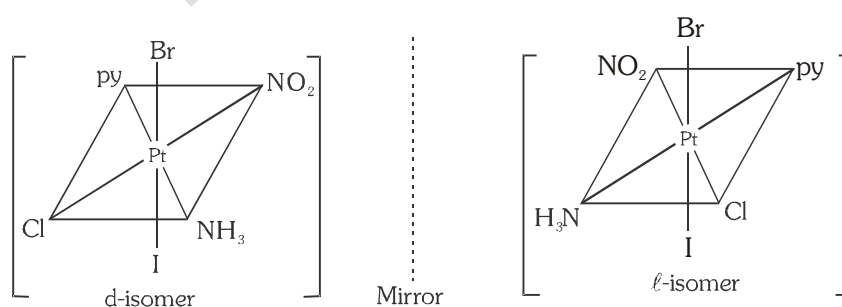
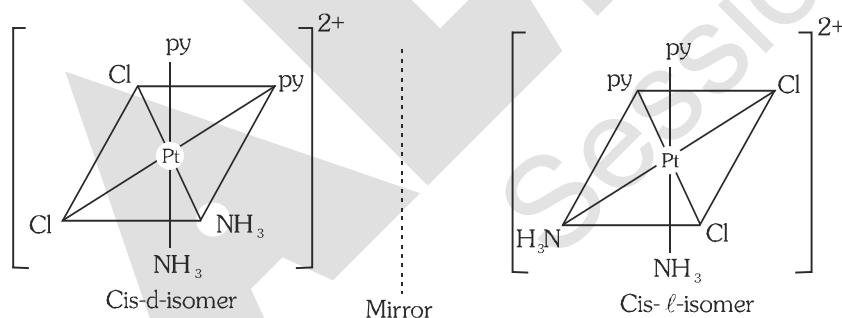
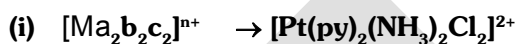
General formula	Total No. of geometrical isomers
Mabcdef	15 (30 optical isomers)
Ma ₂ bcd	9
Ma ₂ b ₂ cd	6
Ma ₂ b ₂ c ₂	5
Ma ₃ bcd	4
Ma ₃ b ₂ c	3
Ma ₃ b ₃	2
Ma ₄ bc	2
Ma ₄ b ₂	2
Ma ₅ b	Nil
Ma ₆	Nil

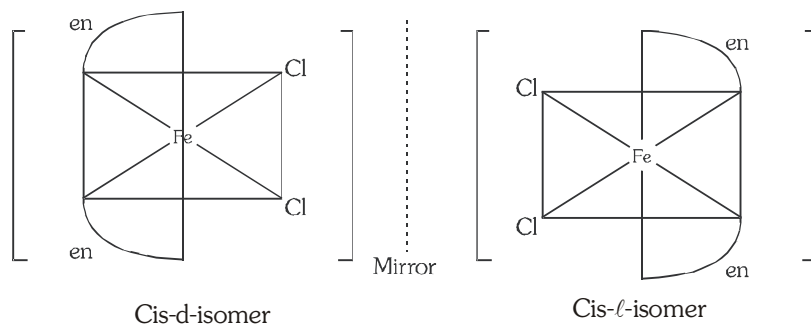
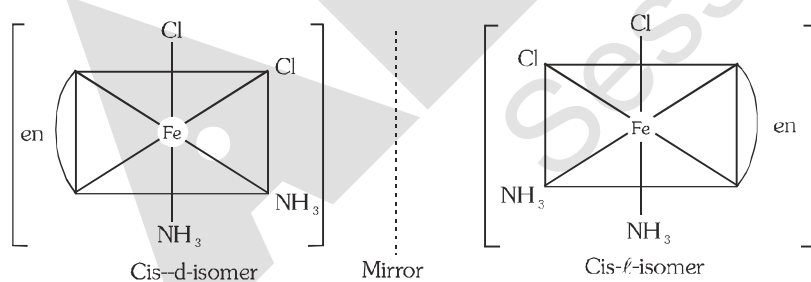
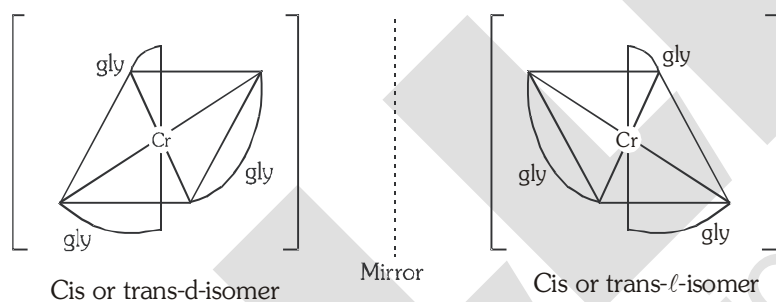
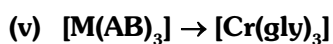
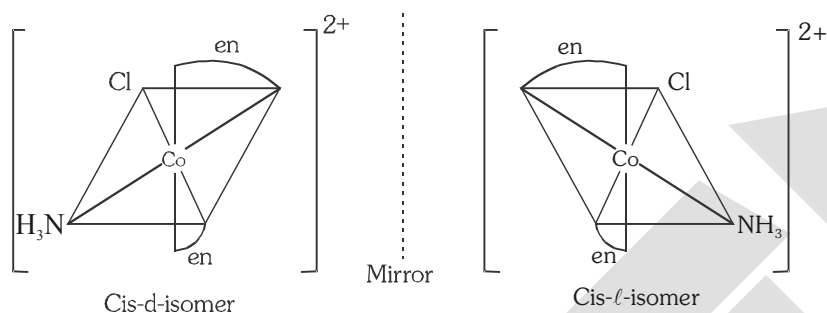
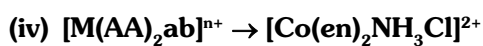
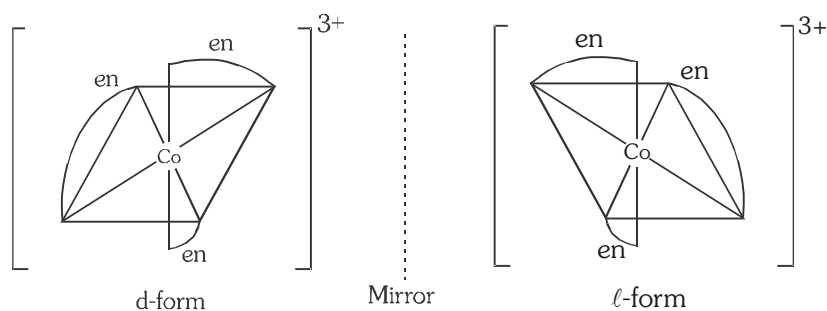
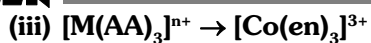
Here M = central atom ; a, b, c, d, e, f = Monodentate ligands

(B) Optical Isomers

- Optically active complexes are those which are non-superimposable over the mirror image structure.
- An optically active complex is one which is asymmetric in nature i.e., not divisible into two identical halves.
- The complex which rotates plane polarised light to left hand side is **laevo rotatory i.e. 'ℓ' or '—'** and if the complex rotates the plane polarised light to right hand side then it is **dextro rotatory 'd' or '+'**.
- Thus complexes which have same physical and chemical properties but differ in their action towards plane polarised light are called as **optical isomers**.
- The 'd' and 'ℓ' isomers of a compound are called as **Enantiomers or Enantiomorphs**.

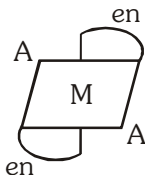
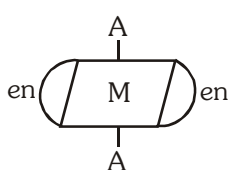
Optical isomers with Coordination number = 6





BEGINNER'S BOX-5

- Which of the following sets is/are example of co-ordination isomerism in complexes ?
 (1) $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$ and $[\text{Co}(\text{CN})_6] [\text{Cr}(\text{NH}_3)_6]$
 (2) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 (3) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Cr}(\text{NH}_3)_5\text{SO}_4]\text{Br}$
 (4) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and $[\text{Pt}(\text{NH}_3)_4] [\text{PtCl}_4]$
- Which of the following does/do not exhibit optical isomerism
 (1) Tetrahedral complexes (2) Square planar complexes
 (3) Octahedral complexes (4) Polynuclear complexes
- The number of geometrical isomers of $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$ are :
 (1) 0 (2) 2 (3) 3 (4) 4
- Which of the following coordination entity form racemic mixture when mixed in equimolar concentration
 (1) $[\text{Cr}(\text{OX})_3]^{3-}$ (2) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (3) $[\text{Co}(\text{en})\text{Cl}_4]^-$ (4) All
- The number of isomeric forms in which $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^{1+}$ ion can occur is
 (1) 2 (2) 3 (3) 4 (4) 1
- The total number of possible isomer for the complex comp. :- $[\text{Cu}(\text{NH}_3)_4] [\text{PtCl}_4]$
 (1) 3 (2) 6 (3) 5 (4) 4
- The complexes $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ and $[\text{Co}(\text{ONO})(\text{NH}_3)_5]\text{Cl}_2$ are the examples of
 (1) Coordination isomerism (2) Ionisation isomerism
 (3) Geometrical isomerism (4) Linkage isomerism
- The type of isomerism exhibited $[\text{Rh}(\text{en})_2\text{Cl}_2][\text{Rh}(\text{en})\text{Cl}_4]$ and $[\text{Rh}(\text{en})_3][\text{RhCl}_6]$ is
 (1) Linkage (2) Coordination
 (3) Ligand (4) Ionisation
- Which of the following will give maximum number of isomers :-
 (1) $[\text{Co}(\text{py})_3(\text{NH}_3)_3]^{3+}$ (2) $[\text{Ni}(\text{en})(\text{NH}_3)_4]^{2+}$
 (3) $[\text{Fe}(\text{C}_2\text{O}_4)(\text{en})_2]^{2-}$ (4) $[\text{Cr}(\text{NO}_2)_2(\text{NH}_3)_4]^+$
- The Complexes given below are :-



- (1) Geometrical isomers (2) Position isomers (3) Optical isomers (4) Identical

3.6 ORGANOMETALLIC COMPOUNDS

Compounds in which C-atom is directly bonded with less electronegative element.

(Ge, Sb, B, Si, P, As) Also form organometallic compounds.

Ex. $\begin{cases} R-Zn-R & \text{dialkyl zinc (Frankland reagent)} \\ R-Mg-X & \text{Alkyl Magnesium halide (Grignards reagent)} \end{cases}$

(b) $\begin{matrix} \text{Sodium acetate} & \text{CH}_3-\overset{\text{O}}{\parallel}{C}-\text{ONa} \\ \text{Sodium ethoxide} & \text{C}_2\text{H}_5-\text{O}-\text{Na} \\ \text{Sodium Mercaptide} & \text{H}_3\text{C}-\text{SNa} \end{matrix}$ } These are not organometallic compounds because in these compounds metal is not directly attached with carbon atom.

(c) $\text{B}(\text{OCH}_3)_3$ is not OMC $\rightarrow \begin{matrix} \text{H}_3\text{CO} \\ \text{H}_3\text{CO} \end{matrix} \text{B}-\text{OCH}_3$ and $\text{CH}_3\text{B}(\text{OCH}_3)_2$ is OMC $\rightarrow \text{CH}_3-\text{B} \begin{matrix} \text{OCH}_3 \\ \text{OCH}_3 \end{matrix}$

(d) Carbides and carbonates are not considered OMC because in these compounds metal carbon bond has ionic nature.

Classification of Organometallic Compounds

OMC can be classified into three main categories –

σ - bonded OMC : These compounds are formed mostly by non transition and metalloid elements.

Ex. $R-Mg-X$ (Grignard's reagent)

$(\text{CH}_3)_2\text{Zn}$ (Dimethyl zinc or Frankland's reagent)

R_2Cd (dialkyl cadmium)

$(\text{C}_2\text{H}_5)_4\text{Pb}$ (Tetra ethyl lead) – used as antiknocking agent in petrol.

$(\text{C}_2\text{H}_5)_3\text{Al}+\text{TiCl}_4$ (Ziegler natta catalyst) – Heterogeneous catalyst, used in polymerisation of alkene.

π - bonded OMC : These are usually formed by transition metals.

Ex. Ferrocene $[\text{Fe}(\eta^5-\text{C}_5\text{H}_5)_2]$ where η = No. of carbon atoms combined with metal atom.

Zeise's salt $[\text{PtCl}_3(\eta^2-\text{C}_2\text{H}_4)]$

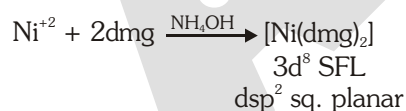
Dibenzene chromium $[\text{Cr}(\eta^6-\text{C}_6\text{H}_6)_2]$

σ and π - bonded OMC : Transition metals of gp. 6, 7, 8, 9 and 10 in their zero oxidation state form such type of OMC. The carbonyl compounds of these metals have both σ and π bonds.

Ex. $[\text{Ni}(\text{CO})_4], [\text{Fe}(\text{CO})_5], [\text{Mn}_2(\text{CO})_{10}]$
 $[\text{M}(\text{CO})_6]$ $[\text{M} = \text{Cr}, \text{Mo}, \text{W}]$

3.7 APPLICATION OF COORDINATION CHEMISTRY

(a) Test of Ni^{+2} : Ni^{+2} ion give rosy red ppt when react with dmg in weak alkaline medium.

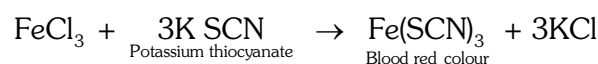


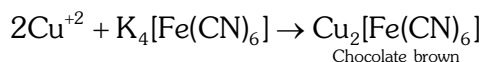
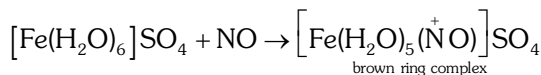
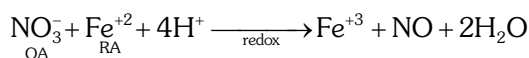
(b) Test of S^{2-} : $\text{Na}_2\text{S} + \text{Na}_2[\text{Fe}(\text{CN})_5(\text{NO})] \rightarrow \text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$
sodium nitro prousside
violet colour

(c) Test of Fe^{+2} & Fe^{+3} :

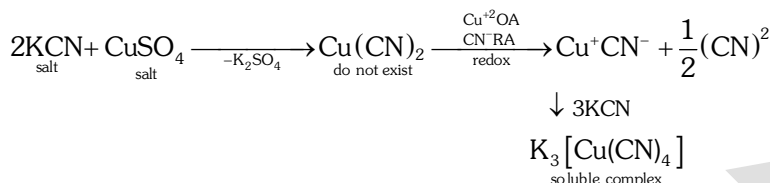


(d) Test of Fe^{+3}

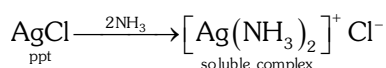


(e) Test of Cu^{+2} **(f) Brown ring test : Test for Nitrate ion**

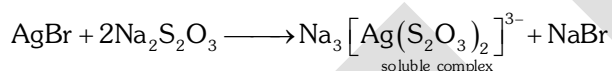
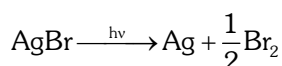
(g) If excess of KCN & KI is added in CuSO_4 insoluble CuCN & CuI is formed which later turn into soluble complex.



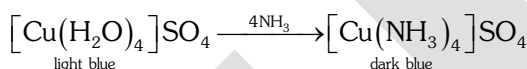
(h) AgCl or AgBr ppt (Not AgI) are soluble in ammonia & NH_4OH



(i) In photography : undecomposed AgBr is removed from photography film by using hyposolution $[\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}]$



(j) Light blue colour of aqueous solution of CuSO_4 turn into dark blue in presence of ammonia



But in the presence of acid in above there is no colour change



(k) Rusting spots of iron $[\text{Fe}_2\text{O}_3 \cdot \text{XH}_2\text{O}]$ can be removed by using oxalic acid.

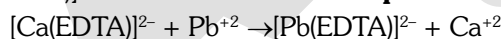
(l) cyano complex are used for electroplating of Ag or Au due to high stability.

e.g. $[\text{Ag}(\text{CN})_2]^-$

(m) Wilkinson catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ is used for hydrogenation of alkene. (Homogeneous catalysis)

(n) $[\text{EDTA}]^{4-}$ is used for **estimation of hardness of water**.

(o) $[\text{Ca}(\text{EDTA})]^{2-}$ is **used to remove Pb poison**.



(p) Biological importance

Chlorophyll (Mg)

haemoglobin (Fe^{+2})

Myoglobin (Fe^{+2})

Vit B₁₂ (Co)

Corboxypeptide (Zn)

Insulin (Zn)

JAHN TELLER EFFECT

In octahedral field if the d electrons are symmetrically arranged, they will repel all six ligands equally. Thus the structure will be a completely regular octahedron.

If the d electrons are unsymmetrically arranged, they will repel some ligands in the complex more than others. Thus the structure is distorted. The e_g orbitals point directly at the ligands. Thus unsymmetric filling of the e_g orbitals results in some ligands being repelled more than others. This causes a significant distortion of the octahedral shape. In contrast the t_{2g} orbitals do not point directly at the ligands, but point in between the ligand directions. Thus unsymmetric filling of the t_{2g} orbitals has only a very small effect on the stereochemistry. Distortion caused by unsymmetric filling of the t_{2g} orbitals is usually too small to measure.

If the d_{z^2} orbital contains one more electron than the $d_{x^2-y^2}$ orbital then the ligands approaching along +z and -z will face greater repulsion than the other four ligands. The repulsion and distortion result in elongation of the octahedron along the z-axis. This is called tetragonal distortion or tetragonal elongation.

If the $d_{x^2-y^2}$ orbital contains the extra electron, then elongation will occur along the x and y axes. This means that the ligands approach more closely along the z axis. Thus there will be four long bonds and two short bonds. This is equivalent to compressing the octahedron along the z axis, and is called tetragonal compression.

Example-I

Many Cu(+II) salts and complexes also show tetragonally distorted octahedral structures. Cu^{2+} has a d^9 configuration:



To minimize repulsion with the ligands, two electrons occupy the d_{z^2} orbital and one electron occupies the $d_{x^2-y^2}$ orbital. Thus the two ligands along +z and -z are repelled more strongly than are the other four ligands.

Therefore according to Jahn Teller effect distortion in regular octahedral geometry occurs when unsymmetrical configuration is present in e_g orbitals of central metal ion/atom.

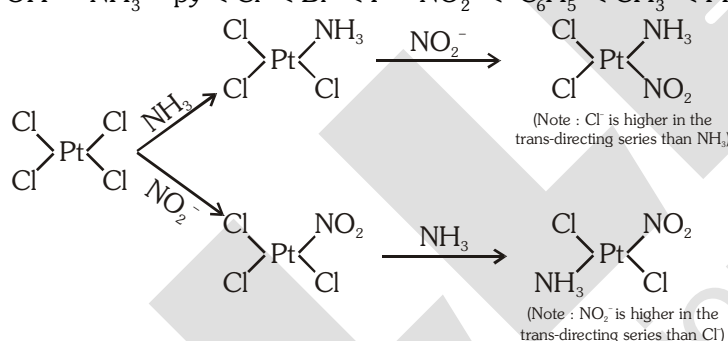
Example-II

High spin complexes in which J.T. effect is i.e. d^1 , d^2 , d^3 (negligible), d^4 , d^6 , d^7 , d^9 observed.

TRANS EFFECT (The kinetic trans effect)

The kinetic trans effect, defined as the relationship between the rate of substitution of square planar complexes and the nature of the species trans to the ligands being displaced.

A trans-directing series, such as the one shown below, can be constructed :



In a substitution reaction in which there is more than one possibility as to which chloride will be replaced, the trans directing series is used to predict which structure will be possible.

BEGINNER'S BOX-6

- $\text{K}_4[\text{Fe}(\text{CN})_6]$ reacts with FeCl_3 to form :-
(1) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (2) $\text{K}_4[\text{Fe}(\text{CN})_3\text{Cl}_3]$ (3) $\text{K}_3[\text{Fe}(\text{CN})_5\text{Cl}]$ (4) $\text{KFe}[\text{Fe}(\text{CN})_6]$
- A blue colouration is not obtained when :-
(1) NH_4OH is added to CuSO_4 (2) CuSO_4 solution reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$
(3) FeCl_3 reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$ (4) Anhydrous white CuSO_4 is dissolved in water
- A reagent used for identifying nickel ion is :-
(1) Potassium ferrocyanide (2) Phenolphthalein
(3) Dimethyl glyoxime (4) EDTA
- Match the following
Complex
(A) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (P) reacts with H_2S gas
(B) $\text{K}_3[\text{Cu}(\text{CN})_4]$ (Q) estimation of hardness
(C) $[\text{Ca}(\text{EDTA})]^{2-}$ (R) removal of lead poisoning
(D) $\text{K}_2[\text{Cd}(\text{CN})_4]$ (S) anti cancer drug
(T) coordination number of metal is = 4
(1) (A)-S, T, (B) T, (C) Q, R (D) P, T
(2) (A)-S (B) T, (C) Q, R (D) P, T
(3) (A)-P, (B) T, (C) Q, R (D) P, T

5. A person suffering from lead poisoning should be fed with :-
 (1) Hypo (2) Cis-platin (3) $[\text{Ca}(\text{EDTA})]^{2-}$ (4) DMG
6. Zeigler natta catalyst is :
 (1) $[\text{RhCl}(\text{PPh}_3)_3]$ (2) $\text{Al}(\text{C}_2\text{H}_5)_3 + \text{TiCl}_4$ (3) $\text{K}(\text{PtCl}_3(\eta^2 - \text{C}_2\text{H}_4))$ (4) Pt/Rh
7. In which of the following process OMC is formed with the help of other OMC :
 (1) $\text{SnCl}_4 + \text{Butyl lithium} \longrightarrow \text{Bu}_4\text{Sn} + 4\text{LiCl}$
 (2) $\text{C}_2\text{H}_5\text{I} + 4\text{Pb}/\text{Na (alloy)} \longrightarrow (\text{C}_2\text{H}_5)_4\text{Pb} + 4\text{NaI} + 3\text{Pb}$
 (3) $\text{Ni} + 4\text{CO} \longrightarrow [\text{Ni}(\text{CO})_4]$
 (4) $\text{K}_2[\text{PtCl}_4] + \text{CH}_2 = \text{CH}_2 \longrightarrow \text{K}[\text{PtCl}_3(\text{C}_2\text{H}_4)] + \text{KCl}$
8. Which is/are organometallic compound :-
 (I) Grignard reagent (II) Sodium methoxide (III) trimethyl boron (IV) TEL
 (1) Only I (2) I & II (3) I, II, IV (4) I, III, IV
9. Which of the following is an organometallic compound :
 (1) $\text{Ti}(\text{OCOCH}_3)_4$ (2) $\text{Ti}(\text{OC}_6\text{H}_5)_4$ (3) $\text{Ti}(\text{OC}_2\text{H}_5)_3\text{C}_2\text{H}_5$ (4) $\text{Ti}(\text{OC}_2\text{H}_5)_4$

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	4	3	1			2	3	3	2
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	4	3	3	1	4	1	1	3	4
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	4	4	2	2	3	1	3	3	3	1
BEGINNER'S BOX-4	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	1	3	2	3		4	4	3	1
BEGINNER'S BOX-5	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	1	2	2	1,2	1	4	4	2	4	4
BEGINNER'S BOX-6	Que.	1	2	3	4	5	6	7	8	9	
	Ans.	4	2	3	1	3	2	1	4	3	

NCERT QUESTIONS (REASONING)

Q.1 Explain the term Degenerate orbitals :

Ans. Orbitals which have same energy in a subshell are known as degenerate orbitals.

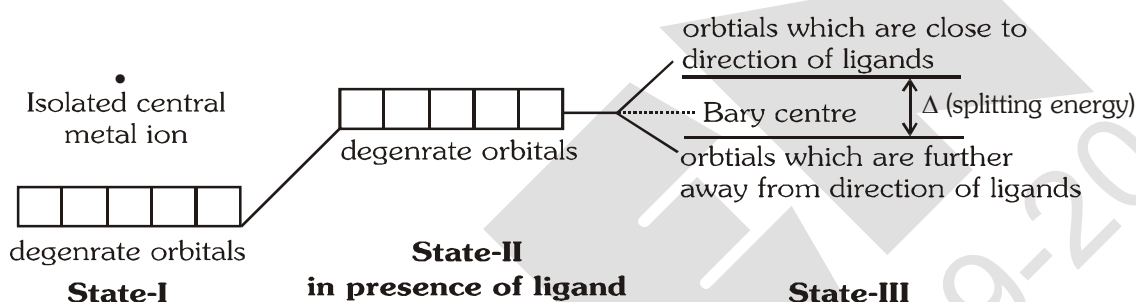
Q.2 What is crystal field splitting ?

Ans. According to CFT the interaction between a transition metal and ligands arises from the attraction between the positively charged metal cation and negative charge of ligand.

As a ligand approaches the metal ion, the electrons of ligand will be closer to some of the d-orbitals and farther away from other causing a loss of degeneracy.

The electrons in the d-orbitals and those in the ligand repel each other due to repulsion between like charges. Thus the d-electrons closer to the ligands will have a higher energy than those further away as a results in the d-orbitals splitting in energy.

This loss of degeneracy of d-orbital is known as crystal field splitting.



The state I represents degeneracy of all the five d-orbitals in the isolated central ion. The state II represents hypothetical degeneracy of all the orbitals at a higher energy level if the negative charge of all the ligands is assumed to be uniformly affecting the electrons in the d-orbitals of the metal ion. The state III represents crystal field splitting.

Q.3 Define (a) crystal field stabilisation energy (b) Pairing energy ?

Ans. (a) Crystal field stabilisation energy : The lowering in the energy of a transition metal ion in a given ligand environment due to crystal field effects.

(b) Pairing energy : The energy required to pair the electrons.

Q.4 Explain the term

(a) Inner orbital complex and outer orbital complexes ?

(b) Low spin and high spin complexes ?

Ans. (a) The empty 'd' orbitals involved in hybridisation may be inner (n-1)d or outer "nd" orbitals and these complexes are called as **Inner orbital complexes** and **outer orbital complexes** respectively.

For example in d^2sp^3 hybridisation (n-1)d, ns and np orbitals are mixed it forms inner orbital complex and in sp^3d^2 ns, np and nd orbitals are mixed so it forms outer orbitals complex.

(b) When the strong field ligand approaches to metal ion value to splitting energy (Δ) is greater than, pairing energy, so it is unfavourable to put electron into high energy orbitals. Therefore, the lower energy orbitals are completely filled before population of the upper sets starts according to the Aufbau principle. Such type of complexes are called low spin complex.

For low spin complexes

splitting energy (Δ) > pairing energy (P)

Weak field ligand causes a small splitting of the d-orbitals where splitting energy is less than pairing energy. It is easier to put electrons into the higher energy set of orbitals than to pair up in the same low energy orbitals because two electrons in the same orbitals repel each other. So one electron is put into each of the five d-orbitals before any pairing occur in accordance with hund rule.

Such complexes are known as high spin complex.

For high spin complexes splitting energy (Δ) < pairing energy (P)

Q.5 In octahedral complex if central metal have configuration d^1, d^2, d^3 always make inner orbital complex, Why ?

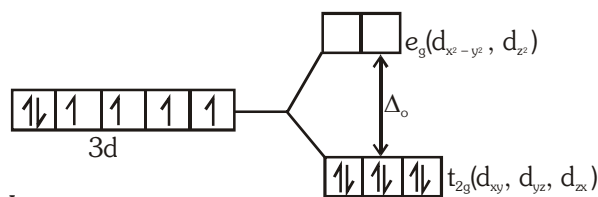
Ans. Central metal ion which have d^1, d^2, d^3 configuration have at least two vacant orbitals in any ligand environment or with any central metal ion with any oxidation state. So always make inner orbital complex.

Q.6 Explain the formation of $\text{Na}_4[\text{Fe}(\text{CN})_6]$ and $\text{Na}_4[\text{FeF}_6]$? Show which is low spin and which is high spin complex and also calculate the Crystal field stabilisation energy (CFSE) ?

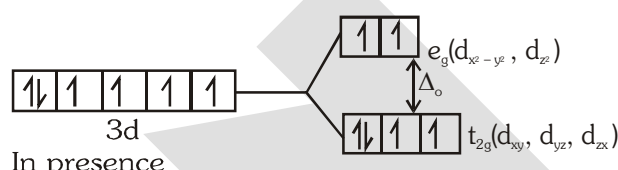
Ans. In given compounds CN^- is strong field ligands and F^- is weak field ligand and in both compounds ions is in +2 oxidation state d^6 configuration.

So in $\text{Na}_4[\text{Fe}(\text{CN})_6]$

$\text{Na}_4[\text{FeF}_6]$



In presence of ligand



In presence of ligand

$$\text{For octahedral CFSE} = \left[-0.4(n_{t_{2g}}) + 0.6(n_{e_g}) \right] \Delta_o + nP$$

where $n_{t_{2g}}$ = number of electron in t_{2g} orbitals

n_{e_g} = number of electron in e_g orbitals

$$\text{Na}_4[\text{Fe}(\text{CN})_6] \text{ CFSE} = -2.4 \Delta_o + 3P. \quad [n_{t_{2g}} = 6, n_{e_g} = 0]$$

$$\text{Na}_4[\text{Fe}(\text{F})_6] \text{ CFSE} = -0.4 \Delta_o + P [n_{t_{2g}} = 4, n_{e_g} = 2]$$

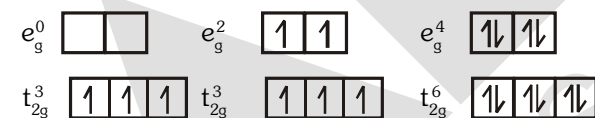
where P = pairing energy to pair up electron.

Q.7 Calculate the paramagnetism into following compound ?



Ans. In all compound H_2O is a weak field ligand so pairing of electron will not occur in

(I) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (II) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (III) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$



$$n = 3$$

$$n = 5$$

$$n = 0$$

$$\mu = \sqrt{15}$$

$$\mu = \sqrt{35}$$

$$\mu = 0$$

Q.8 Why the d^8 configuration always shows paramagnetism 2.83 B.M in octahedral complex?

Ans. In given d^8 configuration for octahedral complex for both strong field ligand and weak field ligand is always have two unpaired electron.



Strong field ligand

Weak field ligand

$$n = 2$$

$$n = 2$$

$$\mu = \sqrt{n(n+2)} \text{ B.M.} = \sqrt{2(2+2)} = 2.83 \text{ B.M.}$$

EXERCISE-I (Conceptual Questions)

TERMINOLOGY OF COORDINATION COMPOUND

- In the complex ion $[\text{Fe}(\text{EDTA})]^-$ the coordination number and oxidation state of central metal ion is :-
 (1) C. N. = 6 O. N. = +3
 (2) C. N. = 1 O. N. = -1
 (3) C. N. = 4 O. N. = +2
 (4) C. N. = 3 O. N. = +3
- The coordination number and oxidation number of the central metal ion in the complex $[\text{Pt}(\text{en})_2]^{+2}$ is :-
 (1) C. N. = 2, O. N. = +2
 (2) C. N. = 6, O. N. = +4
 (3) C. N. = 4, O. N. = +4
 (4) C. N. = 4, O. N. = +2
- Select bidentate or didentate ligand from the following .
 (1) CO (2) SCN^-
 (3) CH_3COO^- (4) $\text{C}_2\text{O}_4^{2-}$
- The oxidation and coordination number of Pt in $[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]^-$ is respectively :-
 (1) + 1, 3 (2) + 2, 4
 (3) + 3, 6 (4) + 2, 5
- The CN and ON of X in the compound $[\text{X}(\text{SO}_4)(\text{NH}_3)_5]$ will be :-
 (1) 10 and 3 (2) 1 and 6
 (3) 6 and 4 (4) 6 and 2
- What is the oxidation number of Fe in $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ ion?
 (1) +2 (2) +3 (3) +1 (4) 0
- The oxidation state of iron in $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$ is
 (1) +1 (2) +2 (3) +3 (4) zero
- Incorrect statement about DMG:-
 (1) It is tetradentate ligand
 (2) Chelating ligand
 (3) dioxime of diacetyl
 (4) in gravimetric determination of Ni is used

IUPAC - NOMENCLATURE

- $\text{K}_3[\text{Fe}(\text{CN})_6]$ is :-
 (a) Potassium hexacyano ferrate (II)
 (b) Potassium hexacyano ferrate (III)
 (c) Potassium ferri-cyanide
 (d) Hexacyano ferrate (III) potassium
 Correct answer is :-
 (1) Only (a) and (b) (2) Only (b) and (c)
 (3) Only (a) and (c) (4) Only (b) and (d)

Build Up Your Understanding

- Give the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$.
 (1) Bromoaquatetraamine Cobalt (III) nitrate
 (2) Bromoaquatetraaminocobalt (III) nitrate
 (3) Bromoaquatetraamine cobalt (III) nitrate
 (4) Tetraammineaquabromido cobalt (III) nitrate
- Which of the following complex is anion :-
 (1) Fluoro pentaammine cobalt (III)
 (2) Trioxalato ferrate (III)
 (3) Penta Carbonyl iron (0)
 (4) Dichloro diammine platinum
- The chloro-bis (ethylenediamine) nitro cobalt (III) ion is :-
 (1) $[\text{Co}(\text{NO}_2)_2(\text{en})_2\text{Cl}]^+$ (2) $[\text{CoCl}(\text{NO}_2)_2(\text{en})_2]^+$
 (3) $[\text{Co}(\text{NO}_2)\text{Cl}(\text{en})_2]^+$ (4) $[\text{Co}(\text{en})\text{Cl}_2(\text{NO}_2)_2]^+$
- Which of the following complexes is not a chelate
 (1) bis (dimethylglyoximate) nickel(II)
 (2) Potassium ethylenediaminetetrathiocyanato chromate(III)
 (3) Tetrammine dichlorocobalt(III) nitrate
 (4) Trans-diglycinatoplatinum(II)
- The correct IUPAC name of the complex $[\text{Fe}(\text{C}_5\text{H}_5)_2]$ is
 (1) Cyclopentadienyl iron (II)
 (2) Bis (cyclopentadienyl) iron (II)
 (3) Dicyclopentadienyl ferrate (II)
 (4) Ferrocene
- The correct name of $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$ is :-
 (1) Tetraammine dichloro platinum (IV) tetrachloro platinate (II)
 (2) Dichloro tetra ammine platinum (IV) tetrachloro platinate (II)
 (3) Tetrachloro platinum (II) tetraammine platinate(IV)
 (4) Tetrachloro platinum (II) dichloro tetraamine platinate (IV)
- The IUPAC name of $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O})_2(\text{NH}_3)]$ is:-
 (1) Potassiumamminedicyano dioxoperoxochromate(VI)
 (2) Potassiumamminecyanoperoxodioxo chromium (VI)
 (3) Potassiumamminecyanoperoxodioxo chromium (VI)
 (4) Potassiumamminecyanoperoxodioxo chromate (IV)
- The IUPAC name for $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ is :-
 (1) Hexaammine cobalt (III) hexacyanochromate (III)
 (2) Hexacyanochromium cobalt hexaammine (VI)
 (3) Hexaammine cobalt (III) hexacyanochromium (VI)
 (4) Hexacyanochromium (III) hexaammine cobalt (III)

18. The IUPAC name for $[\text{Co}(\text{NCS})(\text{NH}_3)_5]\text{Cl}_2$ is :-
 (1) Pentaammine (thiocyanato-N) cobalt (III) chloride
 (2) Pentaammine (thiocyanato-S) cobalt (III) chloride
 (3) Pentaammine (isothiocyanato-N,S) cobalt (III) chloride
 (4) Pentaammine (mercapto-N) cobalt (III) chloride
19. IUPAC name of $\text{K}_2[\text{OsCl}_5\text{N}]$ will be
 (1) Potassium pentachloroazido osmate (VIII)
 (2) Potassium pentachloroazido osmate (VI)
 (3) Potassium pentachloro nitrido osmate (VI)
 (4) Potassium nitro osmate (III)

WERNER'S THEORY & EAN

20. Which of the following has least conductivity in aqueous solution.
 (1) $\text{Co}(\text{NH}_3)_4\text{Cl}_3$ (2) $\text{Co}(\text{NH}_3)_3\text{Cl}_3$
 (3) $\text{Co}(\text{NH}_3)_5\text{Cl}_2$ (4) $\text{Co}(\text{NH}_3)_6\text{Cl}_3$
21. If EAN of a central metal ion X^{+2} in a complex is 34, and atomic number of X is 28. The number of monodentate ligands present in complex are:-
 (1) 3 (2) 4 (3) 6 (4) 2
22. The EAN of cobalt in the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ is :-
 (1) 27 (2) 36 (3) 33 (4) 35
23. The effective atomic number of Cr (atomic no. 24) in $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is
 (1) 35 (2) 27 (3) 33 (4) 36
24. Which gives only 25% mole of AgCl , when reacts with AgNO_3 :-
 (1) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (2) $\text{PtCl}_4 \cdot 5\text{NH}_3$
 (3) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (4) $\text{PtCl}_4 \cdot 3\text{NH}_3$
25. In the metal carbonyls of general formula $\text{M}(\text{CO})_x$ (Which follows EAN rule) if M is Ni, Fe and Cr the value of x will be respectively:-
 (1) 6, 5, 6 (2) 4, 5, 6
 (3) 4, 4, 5 (4) 4, 6, 6
26. A compound has the empirical formula $\text{CoCl}_3 \cdot 5\text{NH}_3$. When an aqueous solution of this compound is mixed with excess silver nitrate, 2 mol of AgCl precipitate per mol of compound. On reaction with excess HCl , no NH_4^+ is detected. Hence it is
 (1) $[\text{Co}(\text{NH}_3)_5\text{Cl}_2]\text{Cl}$
 (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 (3) $[\text{Co}(\text{NH}_3)_5\text{Cl}_3]$
 (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$
27. Which of the following pair the EAN of central metal atom is not same?
 (1) $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{NH}_3)_6]^{3+}$
 (2) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and $[\text{Cr}(\text{CN})_6]^{3-}$
 (3) $[\text{FeF}_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$
 (4) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$

VALENCE BOND THEORY

28. Which of the following compound is paramagnetic
 (1) Tetracyanonickelate (II) ion
 (2) Tetraamminezinc (II) ion
 (3) Hexamine chromium (III) ion
 (4) Diammine silver (I) ion
29. The shape of the complex $[\text{Ag}(\text{NH}_3)_2]^+$ is :
 (1) Octahedral (2) Square planar
 (3) Tetrahedral (4) Linear
30. Hexafluoroferrate (III) ion is an outer orbital complex. The number of unpaired electrons are
 (1) 1 (2) 5
 (3) 4 (4) Unpredictable
31. The shape of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ is :
 (1) Square planar (2) Pyramidal
 (3) Octahedral (4) Tetrahedral
32. Among the following ions, which one has the highest paramagnetism ?
 (1) $[\text{FeF}_6]^{3-}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
33. In the complex ion ML_6^{n+} , M^{n+} has five d-electrons and L is weak field ligand. According to crystal field theory, the magnetic properties of the complex ion correspond to how many unpaired electrons
 (1) 0 (2) 5 (3) 2 (4) 3
34. A magnetic moment of 1.73 BM will be shown by one among of the following compounds .
 (1) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (2) $[\text{Ni}(\text{CN})_4]^{2-}$
 (3) TiCl_4 (4) $[\text{CoCl}_6]^{3-}$
35. The magnetic property and the shape of $[\text{Cr}(\text{NH}_3)_6]^{3+}$ complex ions are :
 (1) Paramagnetic, Octahedral
 (2) Diamagnetic, square planar
 (3) Paramagnetic, tetrahedral
 (4) None of the above
36. Amongst the following ions which one has the highest paramagnetism
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
37. Which of the following complexes is an inner orbital complex ?
 (1) $[\text{CoF}_6]^{3-}$ (2) $[\text{FeF}_6]^{3-}$
 (3) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
38. The wrong statement is :-
 (1) Halide ligands forms high spin complex
 (2) Strong ligands form low spin complex
 (3) $[\text{FeF}_6]^{3-}$ is inner orbital complex
 (4) $[\text{NiCl}_4]^{2-}$ is outer orbital complex

39. What is Incorrect for $K_4[Fe(CN)_6]$
- (1) O.N of Iron is +2
 - (2) It exhibit diamagnetic character
 - (3) It exhibit paramagnetic character
 - (4) It involved d^2sp^3 hybridisation
40. In which of the following molecules, central atom used orbitals of different principle quantum number in the hybridisation :-
- (1) $[Fe(CO)_5]$
 - (2) IF_7
 - (3) $Ni(CO)_4$
 - (4) XeO_4
41. What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion, $[Cu(NH_3)_4(OH_2)_2]^{2+}$?
- (1) Tetrahedral; + 2
 - (2) Square planar; - 2
 - (3) Linear; + 3
 - (4) Octahedral; + 2
42. For FeF_6^{3-} and $Fe(CN)_6^{3-}$ magnetic moment of the fluoride complex is expected to be:-
- (1) The same as the magnetic moment of the cyanide complex
 - (2) Larger than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
 - (3) Smaller than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
 - (4) Larger than the magnetic moment of the cyanide complex because there are fewer unpaired electrons in the fluoride complex
43. Which of the following contains one unpaired electron in the 4p orbitals :-
- (1) $[Cu(NH_3)_2]^+$
 - (2) $[Cu(NH_3)_4]^{2+}$
 - (3) $[Cu(CN)_4]^{3-}$
 - (4) $[Ni(CN)_4]^{2-}$
44. Which of the following complexes are low spin and diamagnetic ?
- (a) $K_4[Os(CN)_6]$
 - (b) $[Mo(CO)_6]$
 - (c) $[Mn(CN)_6]^{4-}$
- Select the correct answer using the codes given below:
- (1) a, b and c
 - (2) a and b only
 - (3) a and c only
 - (4) b and c only
45. How many unpaired electrons are present in the Brown Ring complex $[Fe(H_2O)_5(NO)]SO_4$
- (1) 4
 - (2) 3
 - (3) 0
 - (4) 5

46. Which of the following statements about $Fe(CO)_5$ is correct?
- (1) It is paramagnetic and high spin complex
 - (2) It is diamagnetic and high spin complex
 - (3) It is diamagnetic and low spin complex
 - (4) It is paramagnetic and low spin complex
47. Which is true for $[Ni(en)_2]^{2+}$, $Z(Ni) = 28$?
- (1) paramagnetism, dsp^2 , square planar, C.N. of Ni = 2
 - (2) diamagnetism, dsp^2 , square planar, C.N. of Ni = 4
 - (3) diamagnetism, sp^3 , tetrahedral, C.N. of Ni = 4
 - (4) paramagnetism, sp^3 , tetrahedral, C.N. of Ni = 4
48. Arrange the following in order of decreasing number of unpaired electrons :
- I : $[Fe(H_2O)_6]^{2+}$ II : $[Fe(CN)_6]^{3-}$
 III : $[Fe(CN)_6]^{4-}$ IV : $[Fe(H_2O)_6]^{3+}$
- (1) IV, I, II, III
 - (2) I, II, III, IV
 - (3) III, II, I, IV
 - (4) II, III, I, IV

CFT, COLOUR OF THE COMPLEX AND STABILITY

49. Which one of these ions absorbs energy from visible spectrum :-
- (1) $[Cu(NH_3)_4]^+$
 - (2) $[Cu(NH_3)_4]^{+2}$
 - (3) $[Zn(H_2O)_6]^{+2}$
 - (4) $[Co(H_2O)_6]^{+3}$
50. In the complex $[Ni(H_2O)_2(NH_3)_4]^{+2}$ the magnetic moment (μ) of Ni is :-
- (1) Zero
 - (2) 2.83 BM
 - (3) 1.73 BM
 - (4) 3.87 BM
51. Which of the following system has maximum number of unpaired electrons :-
- (1) d^5 (Octahedral, low spin)
 - (2) d^8 (Tetrahedral)
 - (3) d^6 (Octahedral, low spin)
 - (4) d^3 (Octahedral)
52. In an octahedral crystal field, the t_{2g} orbitals are
- (1) Raised in energy by $0.4 \Delta_0$
 - (2) Lowered in energy by $0.4 \Delta_0$
 - (3) Raised in energy by $0.6 \Delta_0$
 - (4) Lowered in energy by $0.6 \Delta_0$
53. If $\Delta_0 < P$, the correct electronic configuration for d^4 system will be :-
- (1) $t_{2g}^4 e_g^0$
 - (2) $t_{2g}^3 e_g^1$
 - (3) $t_{2g}^0 e_g^4$
 - (4) $t_{2g}^2 e_g^2$

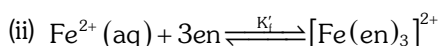
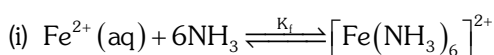
54. Match List-I (Complex ions) with List-II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List-I (Complex ions)	List-II (Number of Unpaired Electrons)
--------------------------	---

A. $[\text{CrF}_6]^{4-}$	i. One
B. $[\text{MnF}_6]^{4-}$	ii. Two
C. $[\text{Cr}(\text{CN})_6]^{4-}$	iii. Three
D. $[\text{Mn}(\text{CN})_6]^{4-}$	iv. Four
	v. Five

Code :	A	B	C	D
(1)	iv	i	ii	v
(2)	ii	v	iii	i
(3)	iv	v	ii	i
(4)	ii	i	iii	v

55. Consider the following complex formation reactions and comment on their formation constant value



- (1) $K_f > K'_f$ (2) $K_f < K'_f$
 (3) $K_f = K'_f$ (4) can not be compared

56. Select most stable complex :-

- (1) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{+3}$
 (3) $[\text{Co}(\text{en})_3]^{+3}$ (4) $[\text{Co}(\text{NH}_3)_4(\text{en})]^{+3}$

57. What will be the theoretical value of 'spin only' magnetic moment when $\text{Fe}(\text{SCN})_3$ reacts with a solution containing F^- ions to yield a complex ?

- (1) 2.83 B.M. (2) 3.87 B.M.
 (3) 5.92 B.M. (4) 1.73 B.M.

58. Which one of the following high-spin complexes has the largest CFSE (Crystal field stabilization energy)?

- (1) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

59. Match List-I (Complex ions) with List-II (CFSE) and select the correct answer using the codes given below the lists :

Column-I	Column-II		
(P) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$	1. $0.6 \Delta_0$		
(Q) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$	2. $0.4 \Delta_0$		
(R) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	3. 0		
(S) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	4. $1.2 \Delta_0$		
P	Q	R	S
(1)	3	2	4
(2)	1	3	4
(3)	4	2	1
(4)	None of these		

ISOMERISM

60. A square planar complex is cis platin $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, shows :

- (1) Geometrical isomerism (2) Optical isomerism
 (3) Linkage isomerism (4) None

61. The two compounds sulphato penta-ammine cobalt (III) bromide and sulphato penta-ammine cobalt (III) chloride represent :

- (1) Linkage isomerism
 (2) Ionisation isomerism
 (3) Co-ordination isomerism
 (4) No isomerism

62. Which of the following complex can not exhibit geometrical isomerism :-

- (1) $[\text{Pt}(\text{NH}_3)_2\text{ClNO}_2]$ (2) $[\text{Pt}(\text{gly})_2]$
 (3) $[\text{Cu}(\text{en})_2]^{+2}$ (4) $[\text{Pt}(\text{H}_2\text{O})(\text{NH}_3)\text{BrCl}]$

63. Which one of the following compounds will exhibit linkage isomerism :-

- (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $[\text{Co}(\text{NH}_3)_3\text{NO}_2]\text{Cl}_2$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (4) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$

64. Out of the following which complex will show geometrical isomerism ?

- (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $\text{Ni}(\text{CO})_4$
 (3) $\text{Na}_3[\text{Ni}(\text{CN})_4]$ (4) $\text{K}[\text{Ag}(\text{CN})_2]$

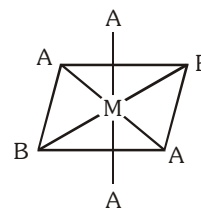
65. Which of the following complexes will show optical isomerism ?

- (1) $[\text{Cr}(\text{NH}_3)_6]^{2+}$ (2) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Pt}(\text{NH}_3)_3\text{Br}]\text{NO}_3$ (4) $[\text{Cr}(\text{en})_3]\text{Cl}_3$

66. The compound $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ represent

- (1) Linkage isomerism
 (2) Hydration isomerism
 (3) Ligand isomerism
 (4) None of these

67. The isomer -



can be marked as -

- (1) Cis isomer (2) Leavo isomer
 (3) Dextro isomer (4) Trans isomer

- 68.** A Planar Complex (Mabcd) gives :-
 (1) Two Optical isomer
 (2) Two geometrical isomer
 (3) Three optical isomer
 (4) Three geometrical isomers
- 69.** Theoretically the No. of geometrical isomers expected for octahedral complex [Mabcdef] is:-
 (1) Zero (2) 30 (3) 15 (4) 9
- 70.** Which of the following complex produce ppt with AgNO_3 and exist in two geometrical isomeric form
 (1) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (2) $\text{PtCl}_2 \cdot 3\text{NH}_3$
 (3) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (4) $\text{PtCl}_4 \cdot 2\text{NH}_3$
- 71.** $\text{Cis}[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Br}_2 \cdot 2\text{H}_2\text{O}$ and $\text{trans}[\text{Pd}(\text{en})_2\text{Br}_2]\text{Cl}_2 \cdot 2\text{H}_2\text{O}$ are :-
 (1) Ionisation isomer
 (2) Geometrical isomer
 (3) Hydrate isomer
 (4) None of these
- 72.** Which of the following octahedral complex have maximum stereoisomers :-
 (1) $[\text{M}(\text{CN})_6]^\pm$ (2) $[\text{M}(\text{en})(\text{CN})_4]^\pm$
 (3) $[\text{M}(\text{en})_2(\text{CN})_2]^\pm$ (4) $[\text{M}(\text{en})_3]^\pm$
- 73.** Which of the following complex does not show geometrical isomerism ?
 (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (2) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
 (3) $[\text{Cr}(\text{en})_3]^{3+}$ (4) $[\text{Pt}(\text{gly})_2]$
- 74.** Select the correct statement :
 (1) Potassium ferrocyanide and potassium ferricyanide can be differentiated by measuring the solid state magnetic moment.
 (2) The complex $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ can be differentiated by adding aqueous solution of barium chloride
 (3) The complex $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{Cl}$ can be differentiated by adding aqueous solution of silver nitrate.
 (4) All of these
- 75.** Which of the following complex compounds does not exhibit geometrical isomerism ?
 (1) $[\text{PtCl}_2(\text{NH}_3)_2]$ (2) $[\text{PdCl}_2\text{Br}]\text{I}$
 (3) $[\text{Pt}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$ (4) $[\text{Pt}(\text{NH}_3)_3(\text{Br})]$
- 76.** The pair of molecules that exhibit geometrical isomerism are
 (1) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ and $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
 (2) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]^{2+}$ and $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$
 (3) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (4) All of the above show geometric isomerism

ORGANOMETALLIC COMPOUNDS

- 77.** Which of the following is π -acid ligand
 (1) NH_3 (2) CO
 (3) gly. (4) ethylene diamine
- 78.** Which of the following statement is/are wrong:-
 (a) Al_4C_3 is an organometallic compound
 (b) Metal carbonyls are organometallic compounds
 (c) TEL is π bonded organometallic compound
 (d) Frankland reagent is σ -bonded organometallic compound
 The answer is:-
 (1) c and d (2) a and c
 (3) a and b (4) All are correct
- 79.** Compounds which contain one or more metal carbon bonds are called :
 (1) Organic compound (2) Complex compound
 (3) Metal carbides (4) OMC compounds.
- 80.** Which one of the following is used as a heterogeneous catalyst ?
 (1) Wilkinson's catalyst (2) Tetraethyl lead
 (3) Zeigler Natta catalyst (4) Grignard's reagent
- 81.** Which of the following statement is true
 (1) FeCO_3 and Fe_3C are organometallic compounds.
 (2) In ferrocene ligand is cyclopentadienyl.
 (3) $\text{Pb}(\text{C}_2\text{H}_5)_4$ is π -bonded OMC
 (4) In zeise salt central metal is Sp^3 hybridised.
- 82.** Synergic bonding involves :-
 (1) The transference of electrons from ligands to metal
 (2) The transference of electrons from filled metal orbitals to anti-bonding orbitals of ligands
 (3) Both the above
 (4) None of these
- 83.** OMC form during purification of a metal is :-
 (1) $\text{Ni}(\text{CO})_4$ (2) $\text{Pb}(\text{C}_2\text{H}_5)_4$
 (3) $\text{Li}-\text{C}_4\text{H}_9$ (4) $\text{Na}_2[\text{Ni}(\text{CN})_4]$
- 84.** Which of the following is not an organometallic compound :-
 (1) $(\text{C}_2\text{H}_5)_2\text{Zn}$ (2) $\text{CH}_3\text{B}(\text{OCH}_3)_2$
 (3) $\text{B}(\text{OCH}_3)_3$ (4) $\text{Ni}(\text{CO})_4$

APPLICATION OF COORDINATION CHEMISTRY

- 85.** The brown ring test for nitrites and nitrates is due to the formation of a complex ion with formula :-
 (1) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Fe}(\text{H}_2\text{O})(\text{NO})_5]^{2-}$ (4) $[\text{Fe}(\text{NO})(\text{CN})_5]^{2+}$
- 86.** In a ferric salt on adding KCN a prussian blue is obtained which is :-
 (1) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (2) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_4$
 (3) $\text{FeSO}_4[\text{Fe}(\text{CN})_6]$ (4) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

87. Hypo is used in photography because it is :-
 (1) A strong reducing agent
 (2) A strong oxidising agent
 (3) A strong Complexing agent
 (4) Photo sensitive Compound
88. The solubility of AgBr in hypo solution is due to the formation of :-
 (1) Ag_2SO_3 (2) $\text{Ag}_2\text{S}_2\text{O}_3$
 (3) $[\text{Ag}(\text{S}_2\text{O}_3)]^-$ (4) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
89. Which of the following is related to Nessler's reagent?
 (1) $\text{PtCl}_4 + \text{KCl} \rightarrow$ (2) $\text{AgCl} + \text{NH}_3 \rightarrow$
 (3) $\text{AgBr} + \text{Na}_2\text{S}_2\text{O}_3 \rightarrow$ (4) $\text{HgI}_2 + \text{KI} \rightarrow$
90. $\text{K}_4[\text{Fe}(\text{CN})_6]$ reacts with FeCl_3 to form :-
 (1) $\text{K}_3\text{Fe}(\text{CN})_6$ (2) $\text{K}_4[\text{Fe}(\text{CN})_3\text{Cl}_3]$
 (3) $\text{K}_3[\text{Fe}(\text{CN})_5\text{Cl}]$ (4) $\text{KFe}[\text{Fe}(\text{CN})_6]$
91. A reagent used for identifying nickel ion is :-
 (1) Potassium ferrocyanid (2) Phenolphthalein
 (3) Dimethyl glyoxime (4) EDTA
92. Which one of the following statement is false for nickel-dimethylglyoximate complex ?
 (1) The stability of complex is only due to the presence of intra-molecular hydrogen bonding
 (2) The complex is stable, only because dimethyl glyoxime ligand is a stronger ligand
 (3) The complex is stable as it has five membered chelate rings as well as intra molecular hydrogen bonding
 (4) (1) and (2) both
93. $\text{CuCl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$ Chocolate brown ppt(X)
 Select the correct statement for (X) :
 (1) Its IUPAC name is copper (I) hexacyanoferrate (II)
 (2) It reacts with excess potassium cyanide forming an another soluble complex which has tetrahedral geometry.
 (3) It has 'spin only' magnetic moment equal to $\sqrt{36}$ B.M.
 (4) 2 and 3 both
94. It is an experiment fact that :
 $\text{DMG} + \text{Ni(II)salt} + \text{NH}_4\text{OH} \rightarrow$ Red ppt.
 Which of the following is wrong about this red ppt:
 (1) It is a non-ionic complex
 (2) It involves intra molecular H-bonding
 (3) Ni(II) is sp^3 hybridised
 (4) It is a diamagnetic complex
95. Wilkinson's catalyst react with H_2 to form an octahedral complex in which Rh(Z = 45) has the following electronic configuration in the ligand field $t_{2g}^{2,2,2}, e_g^{0,0}$. Then which of the following is correct about this new complex
 (1) It is paramagnetic
 (2) Its IUPAC name is chlorodihydriodotris (triphenylphosphine) rhodium (III)
 (3) Hybridisation of Rh(I) is d^2sp^3
 (4) None of these

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	4	4	2	4	3	2	1	2	4	2	3	3	2	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	1	3	2	2	2	3	4	2	2	4	3	4	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	1	2	1	1	2	3	3	3	1	4	2	2	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	2	1	2	2	4	2	2	3	2	3	3	4	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	3	2	1	4	2	4	4	3	3	4	3	3	4	4
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	1	2	2	4	3	2	3	1	3	1	4	3	4	4	4
Que.	91	92	93	94	95										
Ans.	3	4	2	3	2										

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

- Copper sulphate dissolves in excess of KCN to give
 (1) $\text{Cu}(\text{CN})_2$ (2) CuCN
 (3) $[\text{Cu}(\text{CN})_4]^{3-}$ (4) $[\text{Cu}(\text{CN})_4]^{2-}$
- $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]\text{Cl}$ exhibits
 (1) linkage isomerism, geometrical isomerism and optical isomerism
 (2) linkage isomerism, ionization isomerism and optical isomerism
 (3) linkage isomerism, ionization isomerism and geometrical isomerism
 (4) ionization isomerism, geometrical isomerism and optical isomerism

AIIMS 2006

- The pair in which both species have same magnetic moment (spin only value) is –
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, $[\text{CoCl}_4]^{2-}$
 (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
 (4) $[\text{CoCl}_4]^{2-}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- The number of possible isomers of an octahedral complex $[\text{Co}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]^-$ is –
 (1) 1 (2) 2 (3) 3 (4) 4
- The ligands in anti cancer drug cisplatin are
 (1) NH_3 , Cl (2) NH_3 , H_2O
 (3) Cl , H_2O (4) NO , Cl
- Given below, catalyst and corresponding process/reaction are matched. The mismatch is –
 (1) $[\text{RhCl}(\text{PPh}_3)_2]$: Hydrogenation
 (2) $\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$: Polymerization
 (3) V_2O_5 : Haber–Bosch process
 (4) Nickel: Hydrogentaion
- Among the following the species having square planar geometry for central atom are
 (i) XeF_4 (ii) SF_4
 (iii) $[\text{NiCl}_4]^{2-}$ (iv) $[\text{PdCl}_4]^{2-}$
 (1) (i) and (iv) (2) (i) and (ii)
 (3) (ii) and (iii) (4) (iii) and (iv)

AIPMT 2007

- Which of the following will give a pair of enantiomorphs
 (1) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ (2) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (3) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_6]$ (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_2$

- The d electron configurations of Cr^{2+} , Mn^{2+} , Fe^{2+} and Ni^{2+} are $3d^4$, $3d^5$, $3d^6$ and $3d^8$ respectively which one of the following aqua complexes will exhibit the minimum paramagnetic behaviour
 (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$

AIPMT 2008

- Which of the following complexes exhibits the highest paramagnetic behaviour? Where gly = glycine, en = ethylenediamine and bpy = bipyridyl moieties
 (1) $[\text{Co}(\text{OX})_2(\text{H}_2\text{O})_2]^-$ (2) $[\text{Ti}(\text{NH}_3)_6]^{3+}$
 (3) $[\text{V}(\text{gly})_2(\text{OH})_2(\text{NH}_3)_2]^+$ (4) $[\text{Fe}(\text{en})(\text{bpy})(\text{NH}_3)_2]^{2+}$
- In which of the following coordination entities the magnitude of Δ_o (CFSE in octahedral field) will be maximum?
 (1) $[\text{Co}(\text{CN})_6]^{3-}$ (2) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (3) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$

AIPMT 2009

- Which of the following complex ions is expected to absorb visible light?
 (1) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ (2) $[\text{Sc}(\text{H}_2\text{O})_3(\text{NH}_3)_3]^{3+}$
 (3) $[\text{Ti}(\text{en})_2(\text{NH}_3)_2]^{4+}$ (4) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
- Out of TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and NiCl_4^{2-} colourless species are:
 (1) CoF_6^{3-} and NiCl_4^{2-}
 (2) TiF_6^{2-} and CoF_6^{3-}
 (3) Cu_2Cl_2 and NiCl_4^{2-}
 (4) TiF_6^{2-} and Cu_2Cl_2
- Which of the following does not show optical isomerism?
 (1) $[\text{Co}(\text{en})_3]^{3+}$ (2) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 (3) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]^0$ (4) $[\text{Co}(\text{en})\text{Cl}_2(\text{NH}_3)_2]^+$

- Which one of the following complexes is **not** expected to exhibit isomerism :-
 (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $[\text{NiCl}_4]^{2-}$
 (3) $[\text{Ni}(\text{en})_3]^{2+}$ (4) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$

AIPMT 2010

- Which of the following complex ion is not expected to absorb visible light?
 (1) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Ni}(\text{CN})_4]^{2-}$
 (3) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

17. The existence of two different coloured complexes with the composition of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is due to:-
 (1) Ionization isomerism
 (2) Linkage isomerism
 (3) Geometrical isomerism
 (4) Coordination isomerism

18. Crystal field stabilization energy for high spin d^4 octahedral complex is :-
 (1) $-0.6 \Delta_0$ (2) $-1.8 \Delta_0$
 (3) $-1.6 \Delta_0 + P$ (4) $-1.2 \Delta_0$

AIIMS 2010

19. In which of the following compound C-O bond length is maximum :-
 (1) $[\text{Fe}(\text{CO})_5]^-$ (2) $[\text{Ni}(\text{CO})_4]$
 (3) $[\text{Cr}(\text{CO})_6]$ (4) $[\text{Mn}(\text{CO})_5]^+$
20. The CFSE for octahedral $[\text{CoCl}_6]^{4-}$ is 18000 cm^{-1} . The CFSE for tetrahedral $[\text{CoCl}_4]^{2-}$ will be
 (1) $18,000 \text{ cm}^{-1}$ (2) $16,000 \text{ cm}^{-1}$
 (3) $8,000 \text{ cm}^{-1}$ (4) $20,000 \text{ cm}^{-1}$

AIPMT Pre. 2011

21. Of the following complex ions, which is diamagnetic in nature ?
 (1) $[\text{NiCl}_4]^{2-}$
 (2) $[\text{Ni}(\text{CN})_4]^{2-}$
 (3) $[\text{CuCl}_4]^{2-}$
 (4) $[\text{CoF}_6]^{3-}$
22. The complex $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ are the examples of which type of isomerism ?
 (1) Linkage isomerism
 (2) Ionization isomerism
 (3) Coordination isomersim
 (4) Geometrical isomerism
23. The complex $[\text{Pt}(\text{Py})(\text{NH}_3)\text{BrCl}]$ will have how many geometrical isomers ?
 (1) 3 (2) 4
 (3) 0 (4) 2

AIPMT Mains 2011

24. Which of the following carbonyls will have the strongest C-O bond ?
 (1) $\text{Fe}(\text{CO})_5$ (2) $\text{Mn}(\text{CO})_6^+$
 (3) $\text{Cr}(\text{CO})_6$ (4) $\text{V}(\text{CO})_6^-$

25. Which of the following complex compounds will exhibit highest paramagnetic behaviour :-
 (At. No. Ti = 22, Cr = 24, Co = 27, Zn = 30)
 (1) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ (2) $[\text{Ti}(\text{NH}_3)_6]^{3+}$
 (3) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$

AIIMS 2011

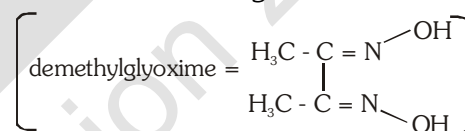
26. The wave length of light absorbed is highest in:
 (1) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ (2) $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{3+}$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Co}(\text{en})_3]^{3+}$
27. Which of the following is diamagnetic:
 (1) $[\text{Cu}(\text{NH}_3)_4]^{+2}$ (2) $[\text{NiCl}_4]^{2-}$
 (3) $[\text{PtCl}_4]^{2-}$ (4) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$

AIPMT Pre. 2012

28. Which one of the following is an outer orbital complex and exhibits paramagnetic behaviour?
 (1) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (3) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (4) $[\text{Zn}(\text{NH}_3)_6]^{2+}$

AIPMT Mains 2012

29. Red precipitate is obtained when ethanol solution of dimethylglyoxime is added to ammoniacal Ni(II). Which of the following statements is not true ?



- (1) Red complex has a tetrahedral geometry.
 (2) Dimethylglyoxime functions as bidentate ligand.
 (3) Red complex has a square planar geometry.
 (4) Complex has symmetrical H-bonding.
30. Low spin complex of d^6 -cation in an octahedral field will have the following energy :-
 (Δ_0 = Crystal field splitting energy in an octahedral field, P = Electron pairing energy)

- (1) $\frac{-2}{5} \Delta_0 + 2P$ (2) $\frac{-2}{5} \Delta_0 + P$
 (3) $\frac{-12}{5} \Delta_0 + P$ (4) $\frac{-12}{5} \Delta_0 + 3P$

NEET-UG 2013

31. A magnetic moment of 1.73 BM will be shown by one among the following :-
 (1) $[\text{CoCl}_6]^{4-}$ (2) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (3) $[\text{Ni}(\text{CN})_4]^{2-}$ (4) TiCl_4

AIPMT 2014

32. Which of the following complexes is used to be as an anticancer agent?
- (1) $\text{mer}[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 - (2) $\text{cis}[\text{PtCl}_2(\text{NH}_3)_2]$
 - (3) $\text{cis-K}_2[\text{PtCl}_2\text{Br}_2]$
 - (4) Na_2CoCl_4

AIIMS 2014

33. Which of the following compound has minimum coordination number :-
- (1) $\text{Co}_2(\text{CO})_8$
 - (2) $\text{Mn}_2(\text{CO})_{10}$
 - (3) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}]$
34. Which of the following compound show both linkage & ionisation isomerism :-
- (1) $[\text{Co}(\text{NH}_3)_5\text{NO}_2](\text{NO}_2)$
 - (2) $[\text{Co}(\text{NO}_2)(\text{Py})_2(\text{NH}_3)_2]\text{NO}_2$
 - (3) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$
 - (4) $[\text{Co}(\text{NO}_2)\text{Br}(\text{NH}_3)_4]\text{Cl}$

AIPMT 2015

35. Cobalt (III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test of chloride ions with silver nitrate at 25°C ?
- (1) $\text{CoCl}_3 \cdot 4\text{NH}_3$
 - (2) $\text{CoCl}_3 \cdot 5\text{NH}_3$
 - (3) $\text{CoCl}_3 \cdot 6\text{NH}_3$
 - (4) $\text{CoCl}_3 \cdot 3\text{NH}_3$
36. Which of these statements about $[\text{Co}(\text{CN})_6]^{3-}$ is true:-
- (1) $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration.
 - (2) $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a high spin configuration.
 - (3) $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration.
 - (4) $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration.

Re-AIPMT 2015

37. The name of complex ion, $[\text{Fe}(\text{CN})_6]^{3-}$ is :-
- (1) Tricyanoferrate (III) ion
 - (2) Hexacyanidoferrate (III) ion
 - (3) Hexacyanoiron (III) ion
 - (4) Hexacyanitoferrate (III) ion
38. The hybridization involved in complex $[\text{Ni}(\text{CN})_4]^{2-}$ is (At.No. Ni = 28)
- (1) d^2sp^2
 - (2) d^2sp^3
 - (3) dsp^2
 - (4) sp^3

39. The sum of coordination number and oxidation number of the metal M in the complex $[\text{M}(\text{en})_2(\text{C}_2\text{O}_4)]\text{Cl}$ (where en is ethylenediamine) is:-
- (1) 7
 - (2) 8
 - (3) 9
 - (4) 6
40. Number of possible isomers for the complex $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ will be : (en = ethylenediamine)
- (1) 3
 - (2) 4
 - (3) 2
 - (4) 1

AIIMS 2015

41. Colour of ruby is due to presence of Cr_2O_3 in Al_2O_3 the reason of colour property
- (1) charge transfer of O^{2-} to Al^{+3}
 - (2) charge transfer of O_2^- to Cr^{3+}
 - (3) charge transfer of O^- to Cr^{3+}
 - (4) d-d transition in Cr^{3+}
42. Electronic configuration of Co in a octahedral complex is $t_{2g}^6 e_g^0$ with a ligand L, than L is
- (1) SCN^-
 - (2) oxalate
 - (3) F^-
 - (4) Cl^-

NEET-I 2016

43. Which of the following has longest C-O bond length? (Free C-O bond length in CO is $1.128\text{\AA}$).
- (1) $\text{Ni}(\text{CO})_4$
 - (2) $[\text{Co}(\text{CO})_4]^\ominus$
 - (3) $[\text{Fe}(\text{CO})_4]^{2-}$
 - (4) $[\text{Mn}(\text{CO})_6]^+$

NEET-II 2016

44. The correct increasing order of trans-effect of the following species is :
- (1) $\text{Br}^- > \text{CN}^- > \text{NH}_3 > \text{C}_6\text{H}_5^-$
 - (2) $\text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^- > \text{NH}_3$
 - (3) $\text{NH}_3 > \text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^-$
 - (4) $\text{CN}^- > \text{C}_6\text{H}_5^- > \text{Br}^- > \text{NH}_3$
45. Jahn-Teller effect **not** observed in high spin complexes of :-
- (1) d^4
 - (2) d^9
 - (3) d^7
 - (4) d^8

AIIMS 2016

46. Which of the following complex is most stable among the following :-
- (1) $\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
 - (2) $[\text{Cu}(\text{NH}_3)_4]^{+2}$
 - (3) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
 - (4) $\text{Na}_3[\text{CoF}_6]$

47. Select the incorrect statement for same type of ligands :-
 (1) Neutral ligands can be strong than anionic ligands
 (2) Chelating ligands are stronger than non chelating
 (3) π -bond containing ligands are classical
 (4) electron pair donating ability depends on electronegativity of donor atom
48. Which of the following ion will form complex with EDTA most rapidly :-
 (1) Bi^{+3} (2) Zn^{+2}
 (3) Cd^{+2} (4) Ag^{+}
49. Regarding brown ring complex formation, which of the following statement is wrong :-
 (1) NO formation takes place
 (2) $\text{Fe}_2(\text{SO}_4)_3$ is used in this process
 (3) Fe^{+2} and NO_3^- both are reduced
 (4) Formula of brown ring complex is $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{+2}$
50. Which of the following homoleptic complex has square planar geometry :-
 (1) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{-3}$ (2) $[\text{Co}(\text{EDTA})]^{-2}$
 (3) $[\text{Pt}(\text{trien})]^{+2}$ (4) $[\text{Ni}(\text{H}_2\text{O})_4]^{+2}$
51. An octahedral complex having formula $\text{Ma}_2\text{b}_2\text{c}_2$, which of the following combination is possible to exhibit optical isomerism.
 (1) $a = \text{Cl}^-$, b & $c = (\text{en})$
 (2) a & b & $c = \text{EDTA}$
 (3) a & $b = \text{C}_2\text{O}_4^{2-}$, $c = \text{en}$
 (4) $a = \text{H}_2\text{O}$, $b = \text{NH}_3$, $c = \text{Cl}^-$

NEET(UG) 2017

52. An example of a sigma bonded organometallic compound is :
 (1) Grignard's reagent
 (2) Ferrocene
 (3) Cobaltocene
 (4) Ruthenocene
53. Pick out the correct statement with respect to $[\text{Mn}(\text{CN})_6]^{3-}$:-
 (1) It is sp^3d^2 hybridised and tetrahedral
 (2) It is d^2sp^3 hybridised and octahedral
 (3) It is dsp^2 hybridised and square planar
 (4) It is sp^3d^2 hybridised and octahedral

54. Correct increasing order for the wavelengths of absorption in the visible region the complexes of Co^{3+} is :-
 (1) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (2) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (4) $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$

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55. If $[\text{CoCl}(\text{NH}_3)_5]^{+2}$ absorb yellow light then what is the colour of the compound.
 (1) Violet (2) Red
 (3) Pale yellow (4) Yellow orange

NEET(UG) 2018

56. The type of isomerism shown by the complex $[\text{CoCl}_2(\text{en})_2]$ is
 (1) Geometrical isomerism
 (2) Coordination isomerism
 (3) Ionization isomerism
 (4) Linkage isomerism
57. The geometry and magnetic behaviour of the complex $[\text{Ni}(\text{CO})_4]$ are
 (1) square planar geometry and diamagnetic
 (2) tetrahedral geometry and diamagnetic
 (3) square planar geometry and paramagnetic
 (4) tetrahedral geometry and paramagnetic
58. Iron carbonyl, $\text{Fe}(\text{CO})_5$ is
 (1) tetranuclear (2) mononuclear
 (3) trinuclear (4) dinuclear

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59. Which of the following forms most stable aqua octahedral complex ion :-
 (1) Mn^{+2} (2) Fe^{+2}
 (3) Co^{+2} (4) Ni^{+2}
60. Which of the following pair are diamagnetic :-
 (1) $[\text{Co}(\text{ox})_3]^{-3}$ & $[\text{Fe}(\text{CN})_6]^{-4}$
 (2) $[\text{Fe}(\text{CN})_6]^{-3}$ & $[\text{Co}(\text{ox})_3]^{-3}$
 (3) $[\text{CoF}_6]^{-3}$ & $[\text{Co}(\text{CN})_6]^{-3}$
 (4) $[\text{CoF}_6]^{-3}$ & $[\text{Fe}(\text{CN})_6]^{-3}$
61. Which of the following species is more reducible :-
 (1) $[\text{V}(\text{CO})_5]$ (2) $[\text{Co}(\text{CO})_6]^{-1}$
 (3) $[\text{Fe}(\text{CO})_5]$ (4) $[\text{Mo}(\text{CO})_6]$
62. Which ligand shows least back bonding :-
 (1) CO (2) C_2H_4
 (3) PPh_3 (4) CN^-

63. $[\text{Cr}(\text{CO})_6]$ react with excess of NO, formed compound will be :-
 (1) $[\text{Cr}(\text{CO})_5(\text{NO})]$
 (2) $[\text{Cr}(\text{CO})_4(\text{NO})]$
 (3) $[\text{Cr}(\text{CO})_4(\text{NO})_2]$
 (4) $[\text{Cr}(\text{NO})_4]$
64. How many of the following behaves as a 1 : 1 type electrolyte ratio in solution - (1) a salt formed by Zeise (2) Wilkinson catalyst (3) Cis-platine (4) Ferrocene (5) Grignard reagent
 (1) 0 (2) 2 (3) 3 (4) 5

65. Which of the following has best π accepting tendency to form synergic bond :-
 (1) CO (2) PPh_3
 (3) NO_2 (4) en

EXERCISE-II (Previous Year Questions)

ANSWER KEY

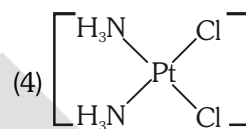
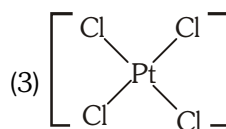
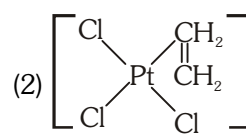
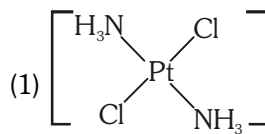
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	2	3	1	3	1	2	2	2	1	4	4	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	1	1	3	2	3	1	2	3	1	3	3	1	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	2	4	4	4	4	2	3	3	1	4	2	3	4	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	3	1	2	3	4	1	2	4	1	1	2	2	4	1
Que.	61	62	63	64	65										
Ans.	1	3	4	2	1										

EXERCISE-III (Analytical Questions)

Check Your Understanding

- A complex compound of cobalt has molecular formula containing five NH_3 molecules, one nitro group and two chlorine atom for one cobalt atom. One mole of this compound produces three moles of ion in aqueous solution. On reaction with excess of AgNO_3 solution two moles of AgCl get precipitated. The ionic formula of the compound.
 - $[\text{Co}(\text{NH}_3)_4 \text{NO}_2 \text{Cl}] \text{NH}_3\text{Cl}$
 - $[\text{Co}(\text{NH}_3)_5 \text{Cl}] \text{ClNO}_2$
 - $[\text{Co}(\text{NH}_3)_5 \text{NO}_2] \text{Cl}_2$
 - $[\text{Co}(\text{NH}_3)_5 (\text{NO}_2) \text{Cl}] \text{Cl}$
- Which of the following is react with AgCl ?
 - KCN
 - NH_4OH
 - $\text{Na}_2\text{S}_2\text{O}_3$
 - All
- Which one of the following shows maximum paramagnetic character ?
 - $[\text{Fe}(\text{CN})_6]^{3-}$
 - $[\text{Fe}(\text{CN})_6]^{4-}$
 - $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
 - $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$
- Which statement is incorrect :-
 - $\text{Ni}(\text{CO})_4$ - Tetrahedral, paramagnetic
 - $[\text{Ni}(\text{CN})_4]^{2-}$ - Square planar, diamagnetic
 - $\text{Ni}(\text{CO})_4$ - Tetrahedral, diamagnetic
 - $[\text{NiCl}_4]^{2-}$ - Tetrahedral, paramagnetic
- CuSO_4 when reacts with KCN forms CuCN , which is insoluble in water. It is soluble in excess of KCN , due to formation of the following complex :-
 - $\text{K}_2[\text{Cu}(\text{CN})_4]$
 - $\text{K}_3[\text{Cu}(\text{CN})_4]$
 - CuCN_2
 - $\text{Cu}[\text{KCu}(\text{CN})_4]$
- Considering H_2O as a weak field ligand, the number of unpaired electrons in $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ will be (At. no. of $\text{Mn} = 25$) :-
 - Four
 - Three
 - Five
 - Two
- Which of the following coordination compounds would exhibit optical isomerism ?
 - tris-(ethylenediamine) cobalt (III) bromide
 - pentaamminenitrocobalt (III) iodide
 - diamminedichloroplatinum (II)
 - trans-dicyanobis (ethylenediamine) chromium (III) chloride
- Among $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$ species, the hybridization states at the Ni atom are, respectively (At. no. of $\text{Ni} = 28$):-
 - dsp^2 , sp^3 , sp^3
 - sp^3 , dsp^2 , dsp^2
 - sp^3 , dsp^2 , sp^3
 - sp^3 , sp^3 , dsp^2

- Which of the following is considered to be an anti-cancer species ?



- Which of the following does not have a metal-carbon bond ?
 - $\text{Ni}(\text{CO})_4$
 - $\text{Al}(\text{OC}_2\text{H}_5)_3$
 - $\text{C}_2\text{H}_5\text{MgBr}$
 - $\text{K}[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]$
- An aqueous solution of CoCl_2 on addition of excess of concentrated HCl turns blue due to formations of :
 - $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]$
 - $[\text{Co}(\text{H}_2\text{O})_2\text{Cl}_4]^{2-}$
 - $[\text{CoCl}_4]^{2-}$
 - $[\text{Co}(\text{H}_2\text{O})_2\text{Cl}_2]$
- The correct order for the wavelength of absorption in the visible region is :
 - $[\text{Ni}(\text{NO}_2)_6]^{4-} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 - $[\text{Ni}(\text{NO}_2)_6]^{4-} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NH}_3)_6]^{2+}$
 - $[\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4-}$
 - $[\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4-}$
- The IUPAC name of the coordination compound $\text{K}_3[\text{Fe}(\text{CN})_6]$ is
 - potassium hexacyanoferrate (II)
 - potassium hexacyanoferrate (III)
 - potassium hexacyanoiron (II)
 - tripotassium hexacyanoiron (III)
- Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour ?
 - $[\text{Cr}(\text{CN})_6]^{3-}$
 - $[\text{Mn}(\text{CN})_6]^{3-}$
 - $[\text{Fe}(\text{CN})_6]^{3-}$
 - $[\text{Co}(\text{CN})_6]^{3-}$
- The value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The correct one is
 - d^4 (in strong ligand filed)
 - d^4 (in weak ligand filed)
 - d^3 (in weak as well as in strong fields)
 - d^5 (in strong ligand filed)

- 16.** A square planar complex is formed by hybridization of which atomic orbitals
 (1) s, p_x, p_y, d_{yz} (2) $s, p_x, p_y, d_{x^2-y^2}$
 (3) s, p_x, p_y, d_{z^2} (4) s, p_x, p_y, d_{xy}
- 17.** The most stable complex among the following is
 (1) $K_3[Al(C_2O_4)_3]$ (2) $[Pt(en)_2]Cl_2$
 (3) $[Ag(NH_3)_2]Cl$ (4) $K_2[Ni(EDTA)]$
- 18.** Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect ?
 (1) Cyanocobalamin is vitamin B_{12} and contains cobalt
 (2) Haemoglobin is the red pigment of blood and contains iron
 (3) Chlorophylls are green pigments in plants and contain calcium
 (4) Carboxypeptidase - A is an enzyme and contains zinc
- 19.** Nickel ($Z=28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[NiX_4]^{2-}$. The number of unpaired electron (s) in the nickel and geometry of this complex ion are, respectively.
 (1) one, square planar (2) two, square planar
 (3) one, tetrahedral (4) two, tetrahedral
- 20.** In $Fe(CO)_5$, the $Fe-C$ bond possesses
 (1) ionic character (2) σ -character only
 (3) π -character only (4) both σ & π character
- 21.** How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion ?
 (1) One (2) Two (3) Six (4) Three
- 22.** The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. No. $Ni=28$)
 (1) 0 (2) 1.73 (3) 2.84 (4) 4.90
- 23.** The coordination number of Ni^{+2} is = 4
 $NiCl_2 + KCN$ (Complex) $\rightarrow$ A (Complex)
 $A + Conc.HCl$ (Maximun) $\rightarrow$ B (Chloro complex)
 The IUPAC name of A and B are
 (1) potassium tetracyanonickelate(II), potassium tetrachloronickelate(II),
 (2) tetracyanopotassiumnickelate(II), tetrachloropotassiumnickelate (II),
 (3) tetracyanonickel (II), tetrachloronickel (II)
 (4) potassiumtetracyanonickel (II), potassiumtetrachloronickel (II).
- 24.** Which one of the following is tridentate ligand
 (1) NO_2^- (2) oxalate ion
 (3) glycinate ion (4) dien
- 25.** Both Co^{3+} and Pt^{4+} have a coordination number of six. Which of the following pairs of complexes will show approximately the same electrical conductance for their 0.001 M aqueous solutions
 (1) $CoCl_3 \cdot 4NH_3$ and $PtCl_4 \cdot 4NH_3$
 (2) $CoCl_3 \cdot 3NH_3$ and $PtCl_4 \cdot 5NH_3$
 (3) $CoCl_3 \cdot 6NH_3$ and $PtCl_4 \cdot 5NH_3$
 (4) $CoCl_3 \cdot 6NH_3$ and $PtCl_4 \cdot 3NH_3$
- 26.** Which of the following has a square planar geometry
 (1) $[PtCl_4]^{2-}$ (2) $[CoCl_4]^{2-}$ (3) $[FeCl_4]^{2-}$ (4) $[NiCl_4]^{2-}$
- 27.** Hybridisation of Ni in $Ni(CO)_4$ is
 (1) dsp^2 (2) dsp^3 (3) d^2sp^3 (4) sp^3
- 28.** Hybridisation and shape of central metal atom of Wilkinson's catalyst is :-
 (1) dsp^2 , Square planer
 (2) sp^3 , Tetrahedral
 (3) sp^3d , Trigonal bipyramidal
 (4) sp^3d^2 , Octahedral
- 29.** Which of the following is not an organo metallic compound :-
 (1) $(C_2H_5)_2Zn$ (2) $CH_3B(OCH_3)_2$
 (3) $B(OCH_3)_3$ (4) $Ni(CO)_4$
- 30.** Which of the following facts about the complex $[Cr(NH_3)_6]Cl_3$ is wrong ?
 (1) The complex is an outer orbital complex
 (2) The complex gives white precipitate with silver nitrate solution
 (3) The complex involves d^2sp^3 hybridisation and is octahedral in shape
 (4) The complex is paramagnetic
- 31.** Among the ligands NH_3 , en , CN^- and CO the correct order of their increasing field strength, is:-
 (1) $CO < NH_3 < en < CN^-$
 (2) $NH_3 < en < CN^- < CO$
 (3) $CN^- < NH_3 < CO < en$
 (4) $en < CN^- < NH_3 < CO$
- 32.** Which one of the following complex ions has geometrical isomers ?
 (1) $[Co(en)_3]^{3+}$ (2) $[Ni(NH_3)_5Br]^+$
 (3) $[Co(NH_3)_2(en)_2]^{3+}$ (4) $[Cr(NH_3)_4(en)]^{3+}$
 [en = ethylenediamine]

33. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide ?
 (1) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$ (2) $[\text{Cr}(\text{en})_3]\text{Br}_3$
 (3) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$ (4) $[\text{Cr}(\text{en})\text{Br}_4]^-$
34. The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelength in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is:
 (1) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$
 (2) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$
 (3) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$
 (4) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$
35. Which of the following statements is/are correct ?
 (1) $[\text{Ag}(\text{NH}_3)_2]^+$ is linear with sp hybridised Ag^+ ion.
 (2) NiCl_4^{2-} , VO_4^{3-} and MnO_4^- have tetrahedral geometry.
 (3) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Pt}(\text{NH}_3)_4]^{2+}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ have dsp^2 hybridisation of the metal ions.
 (4) All the above
36. Which of the following statements is not true about the complex ion $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$?
 (1) It has two geometrical isomers – cis and trans
 (2) Both the cis and trans isomers display optical activity.
 (3) Only the cis isomer displays optical activity.
 (4) Only the cis isomer has non-superimposable mirror image.
37. In Ziegler's salt $\text{C}=\text{C}$ bond length is :-
 Note : $\left\{ \begin{array}{l} \text{C}-\text{C} \text{ bond length in ethane is } 1.54 \text{ \AA} \\ \text{C}=\text{C} \text{ bond length in ethene is } 1.34 \text{ \AA} \\ \text{C}\equiv\text{C} \text{ bond length in ethyne is } 1.20 \text{ \AA} \end{array} \right\}$
 (1) 1.37 Å (2) 1.19 Å (3) 1.87 Å (4) 1.34 Å
38. $[\text{Pt}(\text{NH}_3)(\text{NH}_2\text{OH})(\text{NO}_2)(\text{py})]^+$ will form how many geometrical isomers ?
 (1) 2 (2) 3 (3) 4 (4) 5
39. Which of the following statements is/are correct ?
 (1) The stability constant of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is greater than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$
 (2) Among F^- , Cl^- , Br^- and I^- , F^- forms strongest complexes due to small size.
 (3) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is thermodynamically more stable than $[\text{Zn}(\text{NH}_3)_4]^{2+}$
 (4) All the above
40. All the metal ions contains $t_{2g}^6 e_g^0$ configurations. Which of the following complex will be paramagnetic ?
 (1) $[\text{FeCl}(\text{CN})_4(\text{O}_2)]^{4-}$ (2) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (3) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (4) $[\text{Fe}(\text{CN})_5(\text{O}_2)]^{-5}$
41. $\text{FeCl}_3 + \text{K}_3[\text{Fe}(\text{CN})_6] + \text{H}_2\text{O}_2 \rightarrow \text{Precipitate}$. The colour of the precipitate is :-
 (1) Sky blue (2) Brown
 (3) Prussian blue (4) White

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	3	1	2	3	1	3	4	2	3	1	2	4	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	4	4	1	3	1	4	3	1	4	1	3	1
Que.	31	32	33	34	35	36	37	38	39	40	41				
Ans.	2	3	3	4	4	2	1	2	4	1	3				

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion:** The aqueous solution of $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ is acidic in nature
Reason : It ionises to give a complex ion.
 (1) A (2) B (3) C (4) D
- Assertion :** In the complex $K_2[PtCl_6]$, coordination number of Pt is 6.
Reason : In the complex six coordinate bonds are formed between Pt and chloro ligands.
 (1) A (2) B (3) C (4) D
- Assertion :** Tetrahedral complex do not exhibit geometrical isomerism.
Reason : In tetrahedral complex all the four positions are identical.
 (1) A (2) B (3) C (4) D
- Assertion :** $[Fe(CO)_5]$ is inner orbital complex.
Reason : In the given complex oxidation state of Iron is zero.
 (1) A (2) B (3) C (4) D
- Assertion :** $[Fe(CN)_6]^{3-}$ is paramagnetic in nature.
Reason : $[Fe(CN)_6]^{3-}$ is low spin complex.
 (1) A (2) B (3) C (4) D
- Assertion :** Hexachloroplatinate is a complex anion
Reason: Complex has negatively charged ligands
 (1) A (2) B (3) C (4) D
- Assertion :** $[Ni(CN)_4]^{2-}$ has zero unpaired electron while that of $[NiCl_4]^{2-}$ has two unpaired e^-
Reason: $[Ni(CN)_4]^{2-}$ has strong crystal field while $[NiCl_4]^{2-}$ has weak crystal field
 (1) A (2) B (3) C (4) D
- Assertion :** Cis - $[Fe(en)_2 Cl_2]^+$ can form racemic mixture.
Reason : Cis - $[Fe(en)_2 Cl_2]^+$ is square planar complex .
 (1) A (2) B (3) C (4) D
- Assertion:** Square planar complex Ma_2b_2 has two optical isomers
Reason : Mirror image of Ma_2b_2 is non-super imposable.
 (1) A (2) B (3) C (4) D
- Assertion :** AgI is coloured while AgF is colourless.
Reason : Unpaired e^- is present in AgI
 (1) A (2) B (3) C (4) D
- Assertion :** $[CoF_6]^{3-}$ is high spin complex.
Reason : F^- is strong field ligand.
 (1) A (2) B (3) C (4) D
- Assertion :** Ferrocene is π -bonded organometallic compound.
Reason : Ferrocene is a sandwich compound.
 (1) A (2) B (3) C (4) D
- Assertion :** Solution of Na_2CrO_4 in water is intensely coloured.
Reason: Ox. state of Cr in Na_2CrO_4 is +6.
 (1) A (2) B (3) C (4) D
- Assertion :** Potassium ferrocyanide is diamagnetic whereas potassium ferricyanide is paramagnetic.
Reason : Crystal field splitting in ferrocyanide ion is greater than that of ferricyanide ion.
 (1) A (2) B (3) C (4) D
- Assertion :-** In a mixture of Cd (II) and Cu(II), (Cd^{+2}) gets precipitated in presence of KCN by H_2S .
Reason :- The stability constant of $[Cu(CN)_4]^{3-}$ is greater than $[Cd(CN)_4]^{2-}$
 (1) A (2) B (3) C (4) D
- Assertion :-** aq. solution of $CoCl_2$ is pink in colour. It turns blue in presence of conc. HCl.
Reason :- It is due to formation of $(CoCl_4)^{2-}$.
 (1) A (2) B (3) C (4) D

- 17. Assertion :** Triethylenediamine is a bidentate monoanion.
Reason : Complex containing propylenediamine ligand shows ligand isomerism.
 (1) A (2) B (3) C (4) D
- 18. Assertion :** $[\text{Co}^{\text{III}}(\text{gly})_3]$ is called inner-metallic complex because,
Reason : Both the coordination number and charge of the cation are satisfied simultaneously by ligands.
 (1) A (2) B (3) C (4) D
- 19. Assertion :** All tetrahedral complexes are mainly high spin and the low spin configurations are rarely observed.
Reason : Δ_t is always much smaller even with stronger field ligands and it is never energetically favourable to pair up the electrons.
 (1) A (2) B (3) C (4) D
- 20. Assertion :** NH_2NH_2 although possesses two electron pairs for donation but not acts as a chelating agent.
Reason : The coordination by NH_2NH_2 leads to a three member highly unstable strained ring
 (1) A (2) B (3) C (4) D
- 21. Assertion :** The correct order for the wave length of absorption in the visible region is ; $[\text{Ni}(\text{NO}_2)_6]^{4-} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
Reason : The stability of different complexes depends on the strength of the ligand field of the various ligands.
 (1) A (2) B (3) C (4) D
- 22. Assertion :** The 'spin only' magnetic moment of a green complex, potassium amminetetraacyanidonitrosium chromate(I) is 1.73 BM.
Reason : To have two d-orbitals empty for d^2sp^3 hybridisation, the pairing of electrons take place leaving behind one unpaired electron as CN^- is a stronger ligand.
 (1) A (2) B (3) C (4) D
- 23. Assertion :** Pentaamminethiocyanato-N-chromium(III) tetrachloridozincate(II) is a coloured compound and is an example of ionisation isomerism.
Reason : The compound is paramagnetic and therefore, d-d transition is possible
 (1) A (2) B (3) C (4) D
- 24. Assertion :-** $\text{Cu}[\text{Hg}(\text{SCN})_4]$ and $\text{Hg}[\text{Co}(\text{NCS})_4]$ are isomers.
Reason :- SCN^- is an ambidentate ligand.
 (1) A (2) B (3) C (4) D
- 25. Assertion :-** Coordination number of Pt in Zeise's salt is 5.
Reason :- C_2H_4 act as bidentate ligand
 (1) A (2) B (3) C (4) D
- 26. Assertion :-** Fe^{+3} not used in brown ring test of NO_3^- .
Reason :- NO_3^- is first converted into NO_2 .
 (1) A (2) B (3) C (4) D
- 27. Assertion :** $[\text{Co}(\text{H}_2\text{O})_6]^{+3} \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^{+2}$ changes its colour on reduction.
Reason : Crystal field stabilisation energy increases on reduction
 (1) A (2) B (3) C (4) D
- 28. Assertion :-** Zone refining is based on the fact that impurities are more soluble in molten state than in solid in the presence of oxygen.
Reason :- This method is used to prepare pure metal oxides.
 (1) A (2) B (3) C (4) D
- 29. Assertion :-** If in $[\text{Co}(\text{NH}_3)_6]^{+3}$, NH_3 is replaced by H_2O , same wavelength will be absorbed by the complex :
Reason :- It is a high spin species.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	1	1	2	2	2	1	3	4	3	3	2	2	3	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Ans.	1	4	2	1	1	2	1	4	4	4	3	3	4	4	

METALLURGY

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E

NEET SYLLABUS

General Principles and Processes of Isolation of Elements : Principles and methods of extraction- concentration, oxidation, reduction electrolytic method and refining; occurrence and principles of extraction of aluminium, copper, zinc and iron.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- explain the terms minerals, ores, concentration, benefaction, calcination, roasting, refining, etc.;
- understand the principles of oxidation and reduction as applied to the extraction procedures;
- apply the thermodynamic concepts like that of Gibbs energy and entropy to the principles of extraction of Al, Cu, Zn and Fe;
- explain why reduction of certain oxides like Cu_2O is much easier than that of Fe_2O_3 ;
- explain why CO is a favourable reducing agent at certain temperatures while coke is better in some other cases;
- explain why specific reducing agents are used for the reduction purposes.

An excellent man, like precious metal, is in every way invariable; A villain, like the beams of a balance, is always varying, upwards and downwards.

John Locke

METALLURGY

4.1 INTRODUCTION

Metallurgy : The branch of chemistry which deals with the method of extraction of metals from their ores by profitable means.

Metal : The element which tends to form positive ion is called a metal.

Minerals : The various compounds of metals which occur in the earth's crust and are obtained by mining are called minerals. In earth crust order of abundance of elements is. $O > Si > Al > Fe$

A mineral may be single compound or a mixture.

Ore : The mineral from which a metal can be extracted **profitably** and **easily** is called an ore.

All ores are minerals but all minerals are not ores. (T/F)

Type of Ores :

(I) **Combined Ore** : Metals placed above H in electrochemical series are generally reactive i.e. why they generally found in combined state.

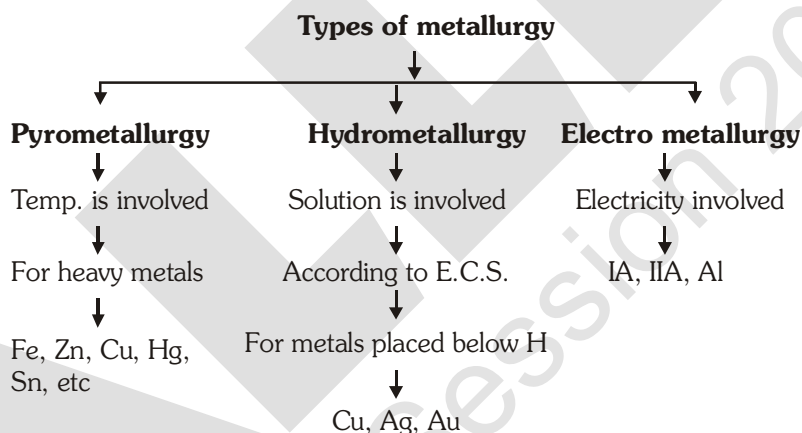
(a) **Halide ore / Sulphate ore / Oxy ore** : Metals are highly reactive ($Li \rightarrow Mg$)

(b) **Oxide ore** : Reactive metal (Al to Sn)

(c) **Sulphide ore** : Metal placed near H or below H. (Pb, Hg, Cu, Ag)

(II) **Native Ore** : Metal placed below H in electrochemical series are generally found in native state.
(Ag, Au, Cu, Pt etc.)

Gangue or matrix : The undesirable impurities present in an ore are called **gangue**.



4.2 STEPS INVOLVED IN THE EXTRACTION OF METALS

The extraction of a metal from its ore is completed in the following four steps.

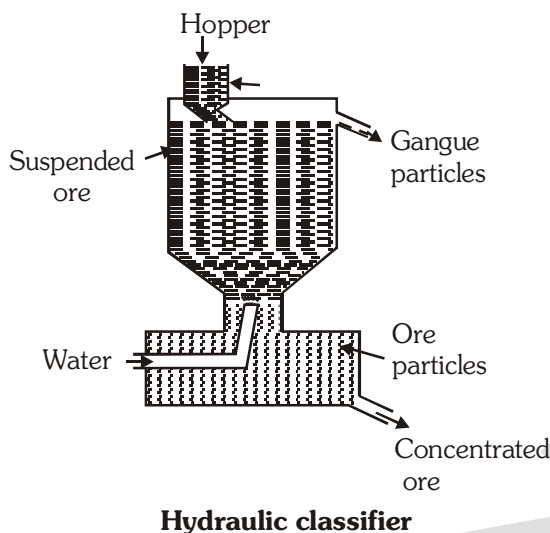
- | | |
|------------------------------|---|
| (A) Concentration of the ore | (B) Conversion of concentrated ore into oxide form. |
| (C) Reduction of the metal | (D) Refining of the metal. |

(A) Concentration of the ore or dressing or benefaction

The removal of impurities from the ore is called its concentration or to increase the concentration of ore in ore sample. Two process – (1)Physical (2) Chemical

(1) PHYSICAL :

(i) **Gravity separation (Levigation)** : This method of concentration of the ore is based on the difference in the specific gravities of the ore and the gangue particles. Powdered ore is agitated with a running stream of water. The lighter gangue particles are taken away by water while heavier ore particles settle down.
Ex. Oxygenated ore



(ii) Froth Floatation method

This method is mainly employed for the concentration of sulphide ores.

The method is based on the different wetting characteristics of the gangue and the sulphide ore with water and oil. The gangue preferentially wetted by water and the ore by oil.

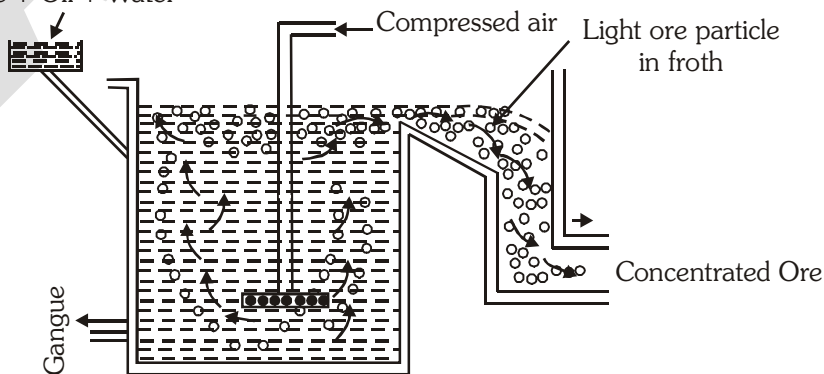
The crushed ore along with water is taken in a floatation cell. Various substances are added depending on the nature of the ore and a current of air is blown in. The substances added are usually of three types.

- (a) **Frothers** :- They generate a stable froth which rises to the top of the tank.
Example of frother is pine oil, Eucalyptus oil, fatty acids etc.
- (b) **Collectors or floating agents** :- These attach themselves by polar group to the granules of the ores which then become water repellent and pass on into the froth.
Example: sodium ethyl xanthate, pine oil and fatty acid.
- (c) **Froth stabilisers** : To stabilise froth.
Ex. Cresol, Aniline etc.
- (d) **Depressants** :- These reagents activate or depress the floatation property and help in the separation of different sulphide ores present in a mixture.
e.g. NaCN.

Impurity of ZnS in PbS ore removed by NaCN



Powdered
Ore + Oil + Water



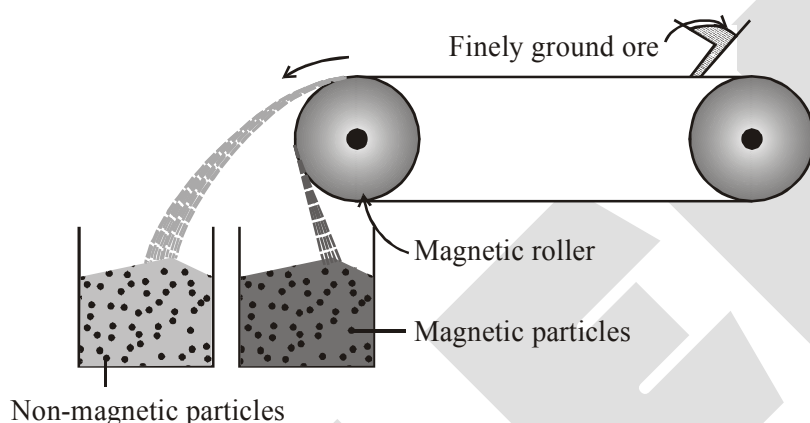
[Froth floatation process]

- Sometimes, it is possible to separate two sulphide ores by adjusting proportion of oil to water or by using 'depressants'.
- For example, in case of an ore containing ZnS and PbS, the depressant used is NaCN. It selectively prevents ZnS from coming to the froth but allows PbS to go along with the froth.

(iii) **Magnetic separation:-**

If either the ore or the gangue (one of these two) is capable of being attracted by a magnetic field, then such separations are carried out (e.g. in case of iron ores)

- e.g.
- SnO_2 having the impurities of $\text{FeWO}_4 + \text{MnWO}_4$ (Wolframite)
 - $\text{FeO} \cdot \text{Cr}_2\text{O}_3$ having the impurities of SiO_2 .

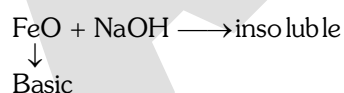
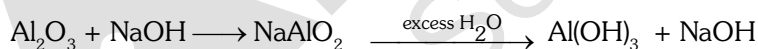


2. **CHEMICAL SEPARATION (LEACHING)** : In this process we use suitable agent which react with ore to form water soluble complex while impurities remain insoluble. Applicable for Al, Ag, Au.

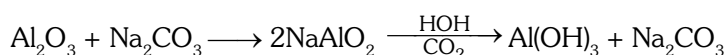
- (a) **Aluminium** :
- Red Bauxite $\rightarrow \text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} + \text{Fe}_2\text{O}_3$ (Major impurity)
 - White Bauxite $\rightarrow \text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} + \text{SiO}_2$ (Major impurity)

(I) **RED BAUXITE** : Two processes

- (i) **Baeyer's process** : NaOH is used.

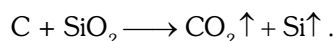
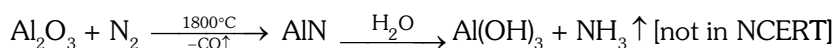


- (ii) **Hall's process** : Na_2CO_3 is used.

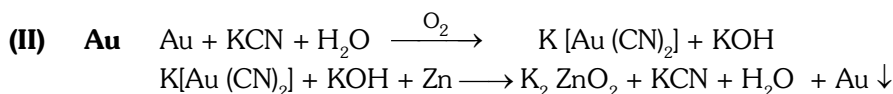
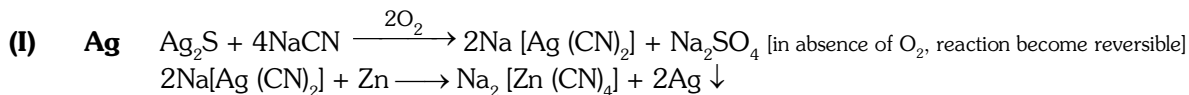


(II) **WHITE BAUXITE** : One process.

Serpeck's process : $(\text{C} + \text{N}_2)$ is used



(b) Ag and Au (CYANIDE PROCESS)

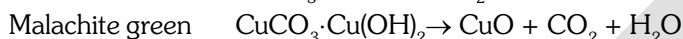
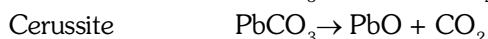
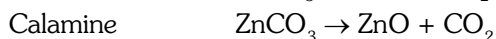
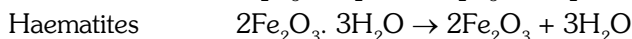
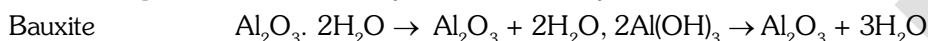


(B) Conversion of concentrated ore into oxide form

Calcination and roasting

(a) **Calcination** : Calcination is a process in which ore is heated, generally in the absence of air, to expel water from a hydrated or hydroxide ore and oxide or carbon dioxide from a carbonate ore at temperature below their melting points.

For Example: All carbonates, hydrated ore and hydroxide ore

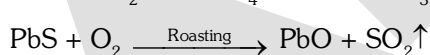
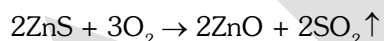


Advantages of Calcination :-

- Moisture is removed.
- Organic matter is destroyed
- The hydroxide and carbonates ores are converted into their oxides.
- The ore become porous and easily workable

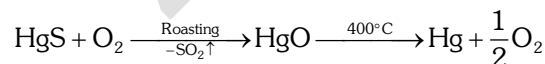
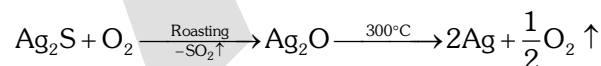
(b) **Roasting** : The removal of the excess sulphur contained in sulphide ores in the form of SO_2 by heating in an excess of air is called roasting.

The concentrated sulphide ore is heated in reverberatory furnace, below its melting point in the presence of an excess of air with or without the addition of an external substance.

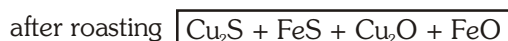
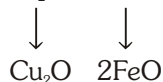
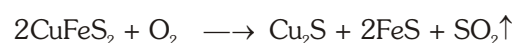


Thermal reduction

Some less stable metal oxide further decompose into metal and oxygen.



Partial roasting



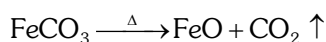
Roasting in Fe metallurgy



Haematite

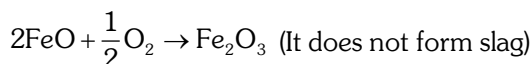


Magnetite

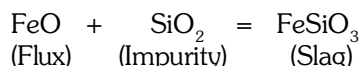


Siderite

Roasting → to prevent wastage of Fe as slag in reduction step

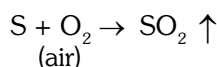


In reduction step



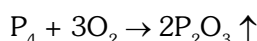
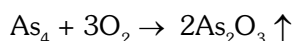
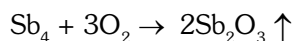
Advantages of Roasting :-

(i) Excess of sulphur is removed as volatile oxide.



(ii) The metal sulphide is converted into metal oxide.

(iii) Impurities of arsenic, antimony & phosphorous are removed as their volatile oxides.



• For PbS, CuS and HgS partial roasting is carried out because these sulphide ore easily convert into metal by auto reduction process.

BEGINNER'S BOX-1

1. Match the following :-

Ore

Main element

(A) Malachite

(P) Fe

(B) Siderite

(Q) Pb

(C) Cerrusite

(R) Zn

(D) Pyrolussite

(S) Cu

(E) Wurtzite

(T) Mn

(1) A-S, B-Q, C-P, D-T, E-R

(2) A-S, B-P, C-Q, D-T, E-R

(3) A-Q, B-S, C-P, D-T, E-R

(4) A-R, B-Q, C-P, D-T, E-S

2. Assertion : Froth floatation is concentration method mainly used for sulphide ores.

Reason : Metal sulphides are highly soluble in water.

(1) A

(2) B

(3) C

(4) D

3. Assertion : In cyanide process, the leaching is accomplished using NaCN.

Reason : CN^- ion reduces Ag, Au & Zn ions in the process.

(1) A

(2) B

(3) C

(4) D

4. Match the ores listed in Column-I with their correct chemical formula listed in Column-II.

Column I

Column II

(A) Cassiterite

(p) FeCO_3

(B) Siderite

(q) SnO_2

(C) Cerussite

(r) PbSO_4

(D) Anglesite

(s) PbCO_3

(1) A-q, B-p, C-s, D-r

(2) A-r, B-p, C-s, D-q

(3) A-p, B-q, C-s, D-r

(4) A-q, B-p, C-r, D-s

5. Which of the following contains both Mg & Ca :
 (1) Magnesite (2) Calamine (3) Carnelite (4) Dolomite
6. Argentite is an ore of:-
 (1) Iron (2) Gold (3) Platinum (4) Silver
7. If Bauxite consists of SiO_2 as impurity, this process is employed :-
 (1) Hall's process (2) Bayer's process (3) Hoopes's process (4) Serpeck's process
8. The impurities present in the ore is called:-
 (1) Slag (2) Flux (3) Alloy (4) Gangue
9. $\text{Ag}_2\text{S} + \text{NaCN} \longrightarrow \text{A}$, $\text{A} + \text{Zn} \longrightarrow \text{B}$
 B is a metal. Hence A and B :-
 (1) $\text{Na}_2[\text{Zn}(\text{CN})_4]$, Zn (2) $\text{Na}[\text{Ag}(\text{CN})_2]$, Ag (3) $\text{Na}[\text{Ag}(\text{CN})_4]$, Ag (4) $\text{Na}_3[\text{Ag}(\text{CN})_4]$, Ag
10. Chemical leaching is useful in the concentration of:-
 (1) bauxite (2) copper pyrites (3) cassiterite (4) None

(C) **Reduction to the metal :**

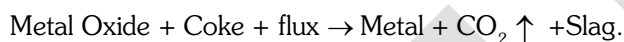
The calcined or roasted ore is then reduced to the metallic state by either of the following method :

(I) **Chemical reduction :**

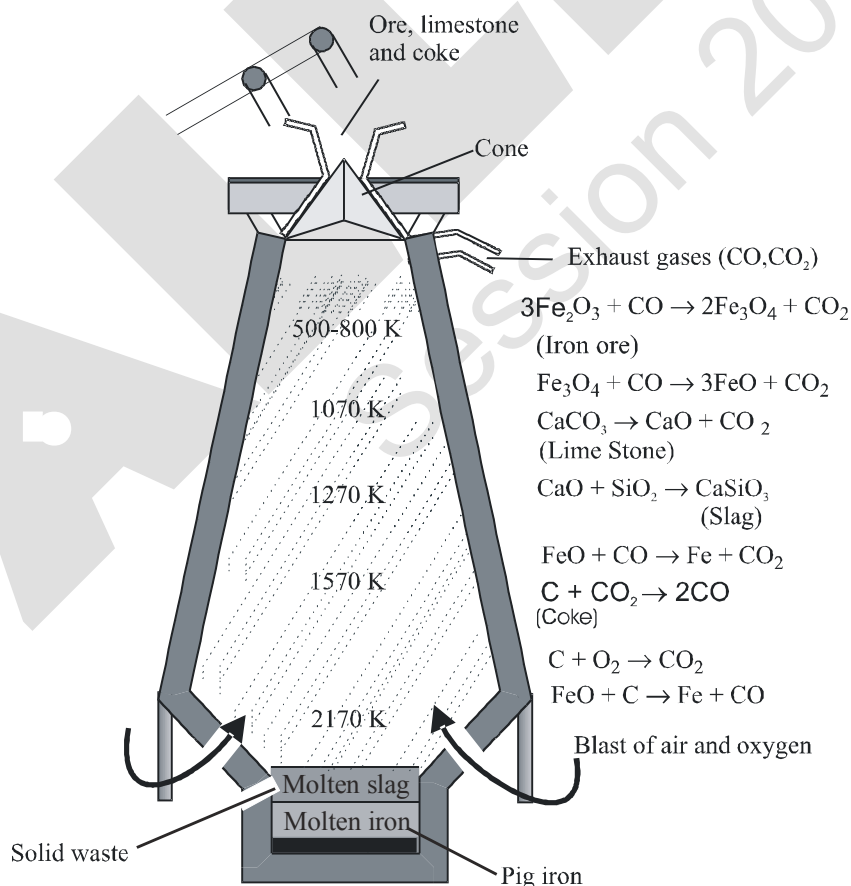
(i) **Reduction by carbon (Smelting)**

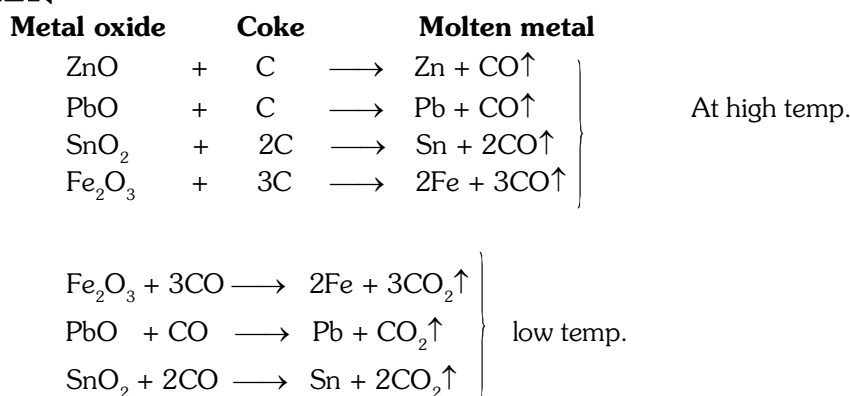
"Reduction of the oxide with carbon at high temperature is known as smelting".

The oxides of less electropositive metals like Pb, Zn, Fe, Sn, Cu etc. are reduced by strongly heating them with coal or coke, in the blast furnace.



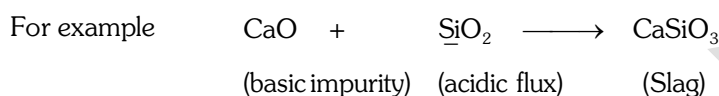
Example - Reaction in blast furnace :-





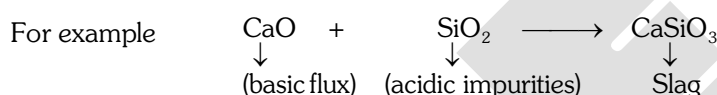
FLUX : Substance used to convert non fusible impurities into fusible one. Three types of flux are used.

(a) **Acidic Flux** : Substance used to remove basic impurities (metal oxide)



Acidic flux are non metal oxide (SiO₂, P₂O₅ etc.)

(b) **Basic flux** : Substance used to remove acidic impurities (non metal oxide)

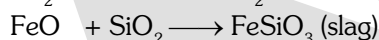
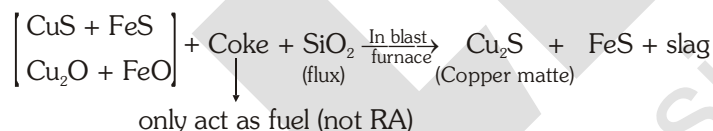


Basic flux are metal oxide. (CaO, MgO, etc.)

(c) **Neutral flux** : Substance used in electrolytic reduction to decrease the fusion temperature and to increase the conductivity of the solution by providing free ions.



Smelting in Cu-metallurgy



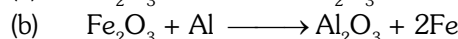
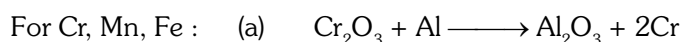
imp flux

(ii) **Alumino thermite process or thermite welding process** : In this process those metal oxide will be reduced which required high temperature and at high temperature carbon react with metal to form metal carbide.

In this process we use aluminium as a reducing agent due to :-

(i) Al has greater affinity towards oxygen as it forms most stable oxide (Al₂O₃)

(ii) This reaction is highly exothermic in nature and once it start it will continue till all the metal oxide is reduced into metal.



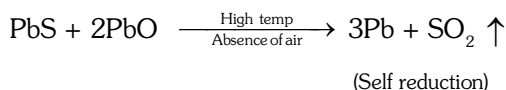
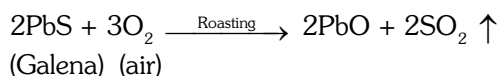
Note : Reaction (b) is used in welding of railway tracks.

(II) Self reduction

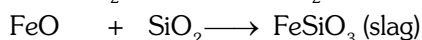
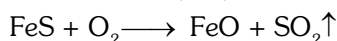
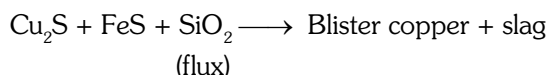
Compounds of certain metals are reduced to metal without using any additional reducing agent. ores of Cu, Pb, Hg etc.

Their sulphide ores are partially roasted to give some oxide. This oxide is now reduced to the metal by the remaining sulphide ore at elevated temperatures in the absence of air. The process is known as self reduction.

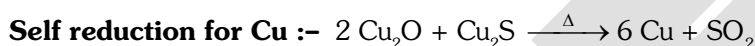
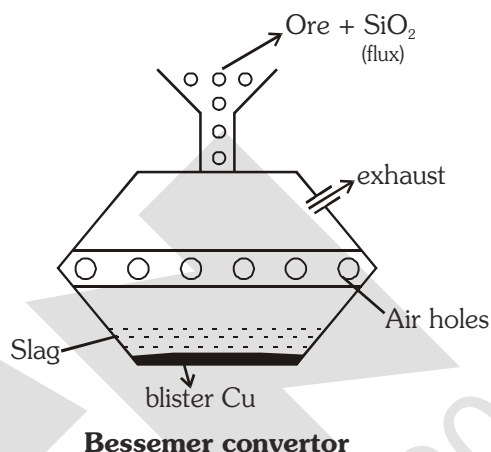
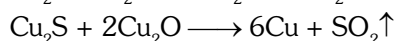
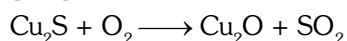
Self reduction for Pb :-



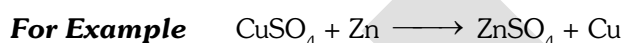
Self reduction in Cu Metallurgy or bessemerisation



gangue flux



(III) **Metal displacement method** : In this method, compound is reacted with a more electropositive & more reactive metal which displaces, the metal from the solution.



Kroll's Process



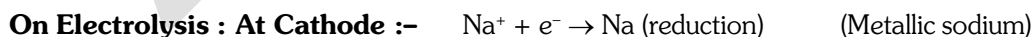
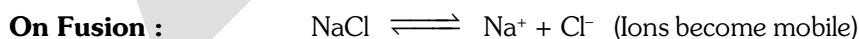
(IV) Electrolytic reduction

This process is mainly used for the extraction of highly electropositive metals. I A, II A & Al

Electrolysis is carried out in a large cell and a small amount of another suitable electrolyte is added which:

- Lowers the melting point of the main electrolyte
 - Enhances its conductivity
 - Reduces corrosion troubles
- e.g. Manufacture of metallic sodium (Down's process)

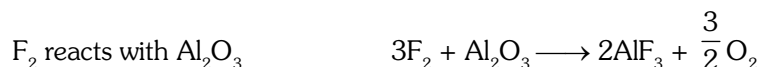
Molten NaCl containing a little CaCl_2 is electrolysed between graphite anode and iron cathode. The various reactions that take place are :



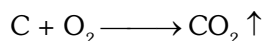
Hall heroult process :

This process is used for extraction of Al from alumina. The extraction of Al from Al_2O_3 is quite difficult because

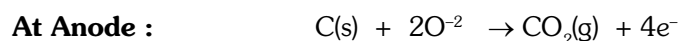
- Fusion temperature of Alumina is quite high (2050°C). Even more than boiling point of Al (1150°C).
- It is a bad conductor of electricity. To overcome these difficulties we mix some amount of neutral flux [$\text{Na}_3\text{AlF}_6 + \text{CaF}_2$]. Neutral flux provides free ions to the solution which decreases the fusion temperature of Alumina from 2050°C to 950°C .



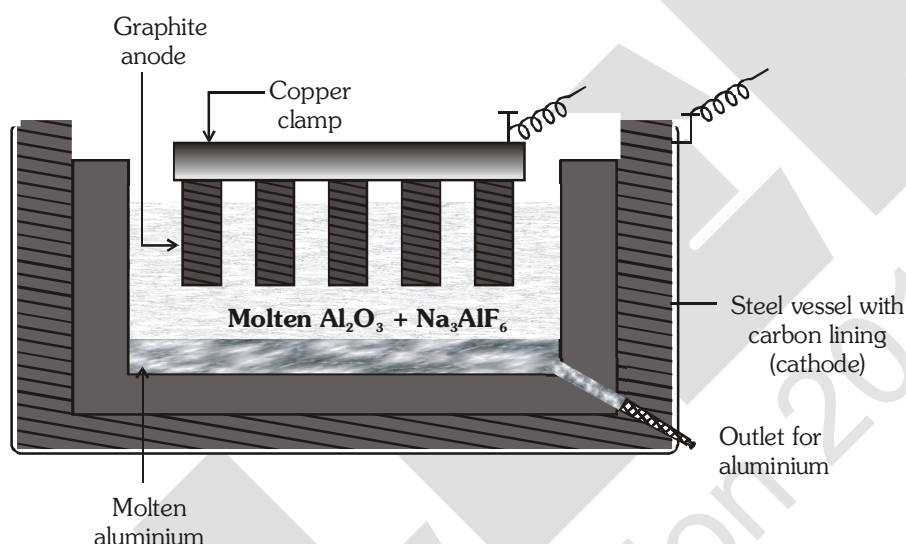
So at anode O_2 gas is liberated which on reaction with graphite anode convert into CO_2 & destroy the anode.



(ii) **From Al_2O_3**



The main drawback of this process is that anode should be changed frequently.



REFINING OF METALS

Metals obtained by the reduction of its compound still contains some objectionable substance and have to be refined. Depending upon the nature of the metal and impurities, the following methods are used for purification of the metals.

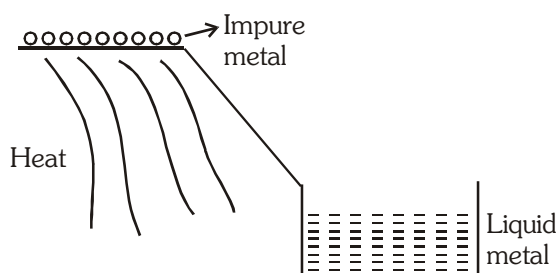
(I) Physical Process :

(i) **Liquation** : This method is used for the refining of metals having low melting point and are associated with high melting impurities.

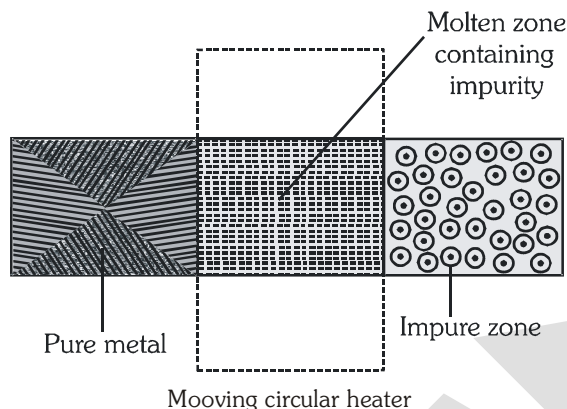
Ex. Pb, Sn, Sb and Bi.

The impure metal is heated on the sloping hearth of a furnace.

The pure metal flows down leaving behind the non-fusible impurities on the hearth.

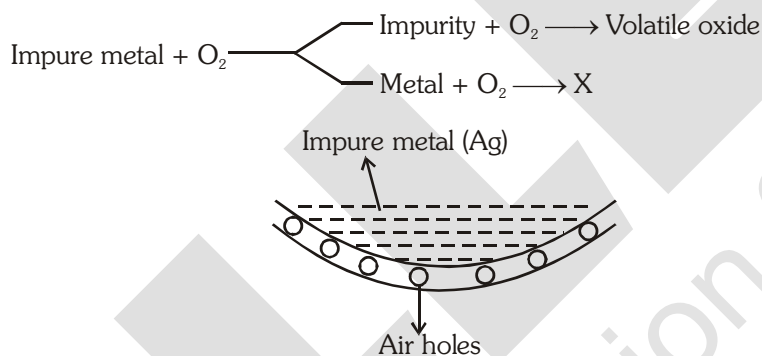


- (ii) **Distillation** : Metals having low boiling point are refined by this method, for example, zinc, cadmium and mercury.
- (iii) **Zone refining** : Metals of very high purity are obtained by zone refining. This refining method is based on the fact that impurities tend to remain dissolved in molten metal.
- Ge, Si and Ga used as semiconductors are refined in this manner.



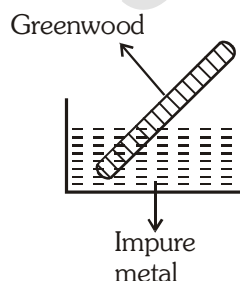
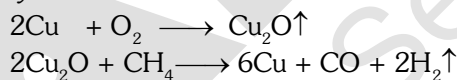
(II) CHEMICAL PROCESS :-

- (i) **Cupellation** : This process is used to purify silver containing the impurities of Pb. This process is used when impurity have greater affinity towards O_2 while metal does not have.

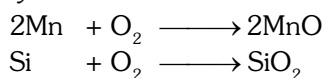


- (ii) **Poling** : Used to purify Cu, Pb.

This process is used for the metal having the impurity of their own oxide. In this process a green wooden pole is heated with molten metal, which provide C and H to metal oxide which reduce impurity of metal oxide to metal.



- (iii) **Bessemerisation** : Actually it is the key principle involve to removal of impurities by oxidation with air being blown oxidise to molten metal.
- Impure metal is heated in a furnace and a blast of compressed air is blown which oxidise the impurity into their oxides and that can be removed in the form of slag.



(III) ELECTRO-REFINING OF METALS

Metals such as Cu, Ag, Zn, Sn, Pb, Al, Ni, Cr are refined by this method.

The impure metal is made the anode of an electrolytic cell, while cathode is thin plate of pure metal.

Electrolyte is the solution of a salt of the metal.

On passing the electric current pure metal from the anode dissolves and gets deposited at the cathode.

The soluble impurities go into the solution while insoluble or less electropositive impurities settle down below the anode as **anode mud** or sludge. For Example

– *Electrorefining of Copper*

Anode	:	Blister copper (98%)
Cathode	:	Pure copper
Electrolyte	:	An aqueous solution of CuSO_4 (15%) + 5% dil H_2SO_4

– *Electrorefining of Silver*

Anode	:	Impure silver
Cathode	:	Pure silver
Electrolyte	:	Aq AgNO_3 + 1% dil HNO_3 on passing electricity impure anode dissolves and pure Ag is deposited at the cathode.

– *Electrorefining of Pb (Bett's process)*

Anode	:	Impure lead.,	Cathode	:	Pure lead.
Electrolyte	:	A mixture of PbSiF_6 and H_2SiF_6			

– *Electrorefining of Al (Hoope's process)*

Anode	:	Impure Al,	Cathode	:	Pure Al
Electrolyte	:	A mixture of Na_3AlF_6 + CaF_2			

(IV) Vapour Phase Refining (Thermal decomposition)

(i) Van - Arkel process :

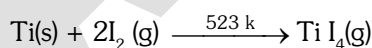
This method is very useful for removing all the oxygen & nitrogen present in the form of impurity.

(i) Employed to get metal in very pure form of small quantities.

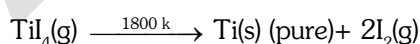
(ii) In this method, the metal is converted into a volatile unstable compound (e.g. iodide), and impurities are not affected during compound formation.

(iii) The compound thus obtained is decomposed to get the pure metal.

(iv) Employed for purification of metals like titanium and zirconium.



Impure



(ii) Mond's process

(i) Nickel is purified by using CO gas. This involves the formation of nickel tetracarbonyl.



BEGINNER'S BOX-2

- Copper matte consists of :-
 - (1) Copper oxide and ferrous sulphide
 - (2) Copper sulphide and ferrous oxide
 - (3) Copper sulphide and ferrous sulphide
 - (4) Copper oxide and ferrous oxide
- Product obtained after Bessemerisation is called as...because.....
 - (1) Concentrated copper ; copper percentage is high
 - (2) Copper matte ; of its appearance
 - (3) Blister copper ; of its appearance
 - (4) Ultra pure copper ; 100 percent copper
- Matte is obtained after this step -
 - (1) Froth floatation
 - (2) Roasting
 - (3) Smelting
 - (4) Refining
- Copper glance istype of ore -
 - (1) Carbonate
 - (2) Sulphide
 - (3) Oxide
 - (4) Sulphate
- High purity copper is obtained by -
 - (1) Zone refining
 - (2) Poling
 - (3) Electrolytic refining
 - (4) Cupellation
- The function of flux during the smelting of ore is:-
 - (1) To make the ore porous
 - (2) To remove gangue
 - (3) To facilitate reduction
 - (4) To facilitate oxidation
- The following equation represents a method of purification of nickel by :-

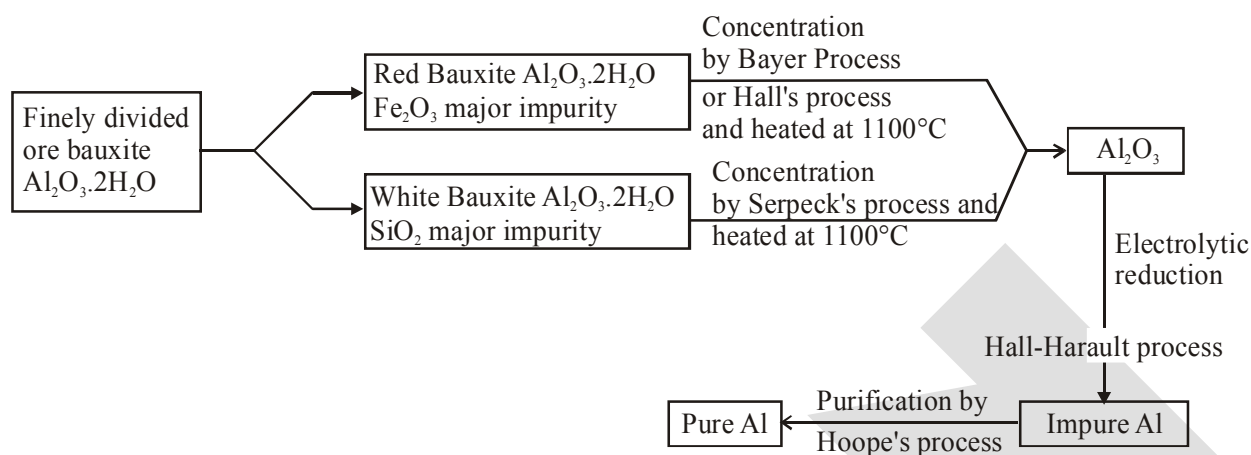
$$\text{Ni (impure)} + 4\text{CO} \xrightarrow{320\text{K}} \text{Ni(CO)}_4$$

$$\xrightarrow{420\text{K}} \text{Ni (pure)} + 4\text{CO}$$
 - (1) Cupellation
 - (2) Mond's process
 - (3) Van Arkel method
 - (4) Zone refining
- The slag obtained during the extraction of copper from copper pyrites is composed of :-
 - (1) Cu_2S
 - (2) CuSiO_3
 - (3) FeSiO_3
 - (4) SiO_2
- Which of the following reaction is a part of Hall's process :-
 - (1) $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$
 - (2) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$
 - (3) $\text{AlN} + 3\text{H}_2\text{O} \rightarrow \text{Al(OH)}_3 + \text{NH}_3$
 - (4) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} + 2\text{Na}_2\text{CO}_3 \rightarrow 2\text{NaAlO}_2 + \text{CO}_2 + 2\text{H}_2\text{O}$
- Which method of purification is represented by the following equations

$$\text{Ti} + 2\text{I}_2 \xrightarrow{523\text{K}} \text{TiI}_4 \xrightarrow{1700\text{K}} \text{Ti} + 2\text{I}_2$$
 - (1) Cupellation
 - (2) Poling
 - (3) Van Arkel
 - (4) Zone refining

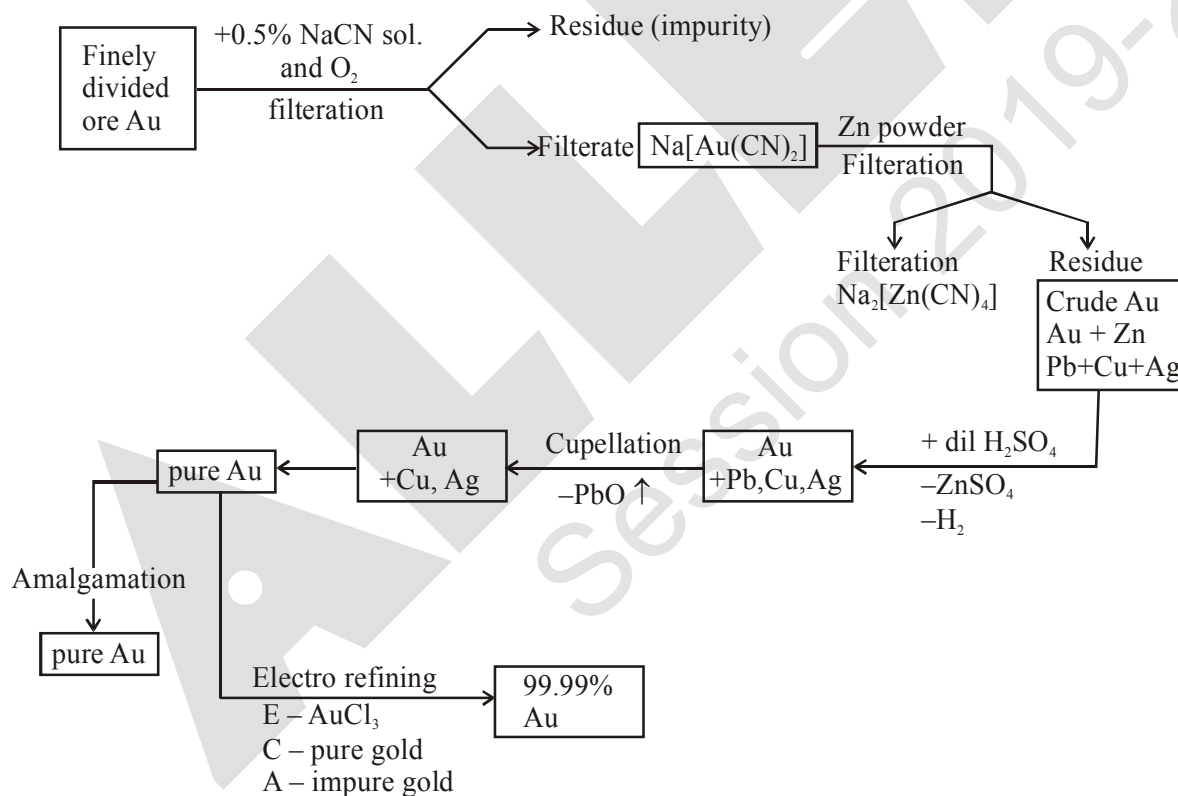
Al

Ore - Bauxite $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$

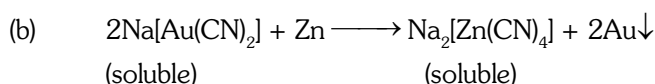
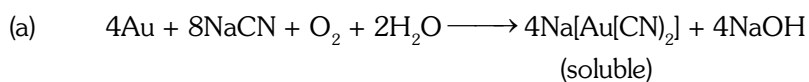


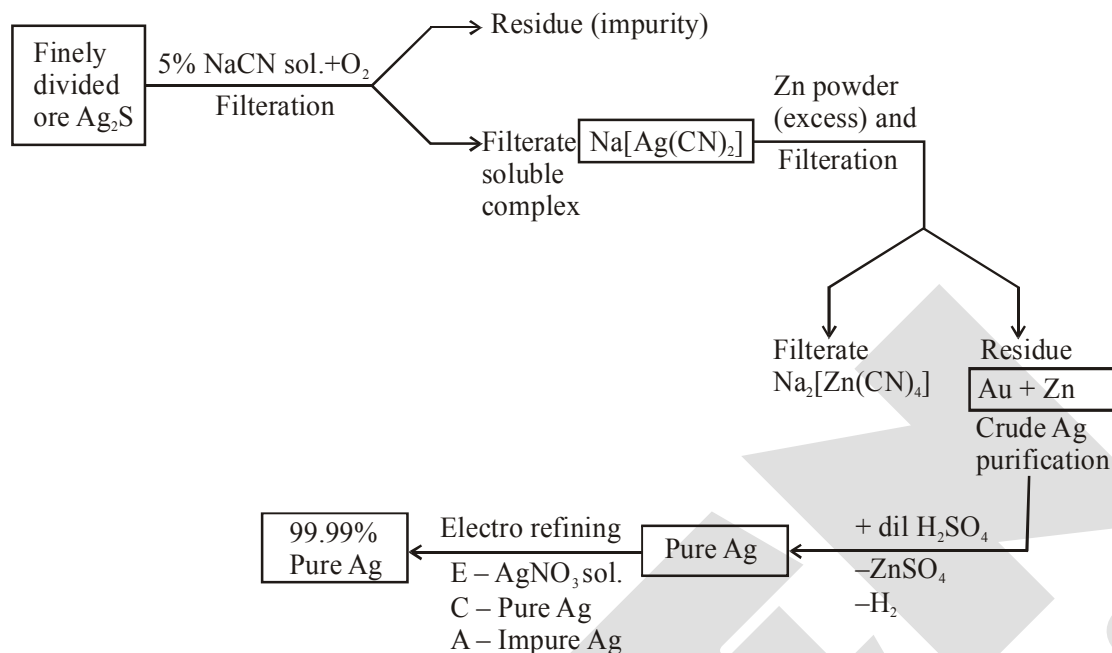
Au

Ore - Auriferous Rock

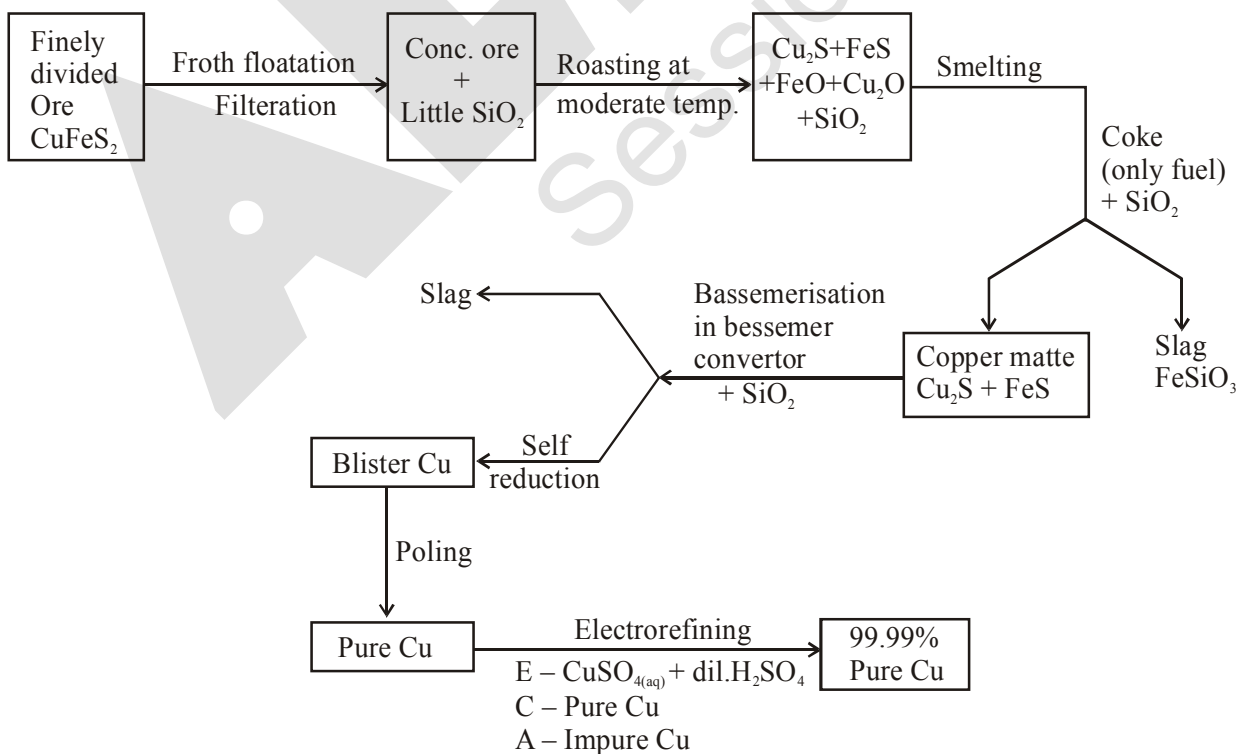


Reaction involved :



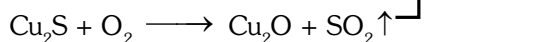
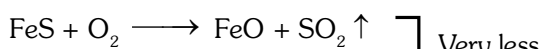
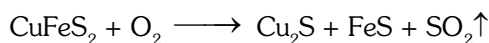
Ag**Ore - Argentite Ag_2S** **Reaction involved :**

- (a) $\text{Ag}_2\text{S} + 4\text{NaCN} \rightleftharpoons 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S} \xrightarrow{+\text{O}_2 + \text{H}_2\text{O}} \text{Na}_2\text{SO}_4 + \text{'S'} + 2\text{NaOH}$
 $\Rightarrow \text{O}_2$ is used to make reaction irreversible which remove Na_2S as $\text{Na}_2\text{SO}_4 + \text{S}$
- (b) $2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \longrightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + \text{Ag} \downarrow$

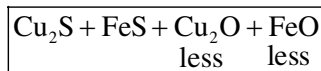
Cu**Ore - CuFeS_2 copper pyrites**

Reaction involved :

(a) Roasting step

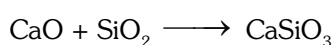
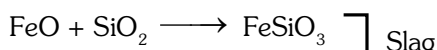
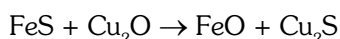


Very less



(b) Smelting step :-

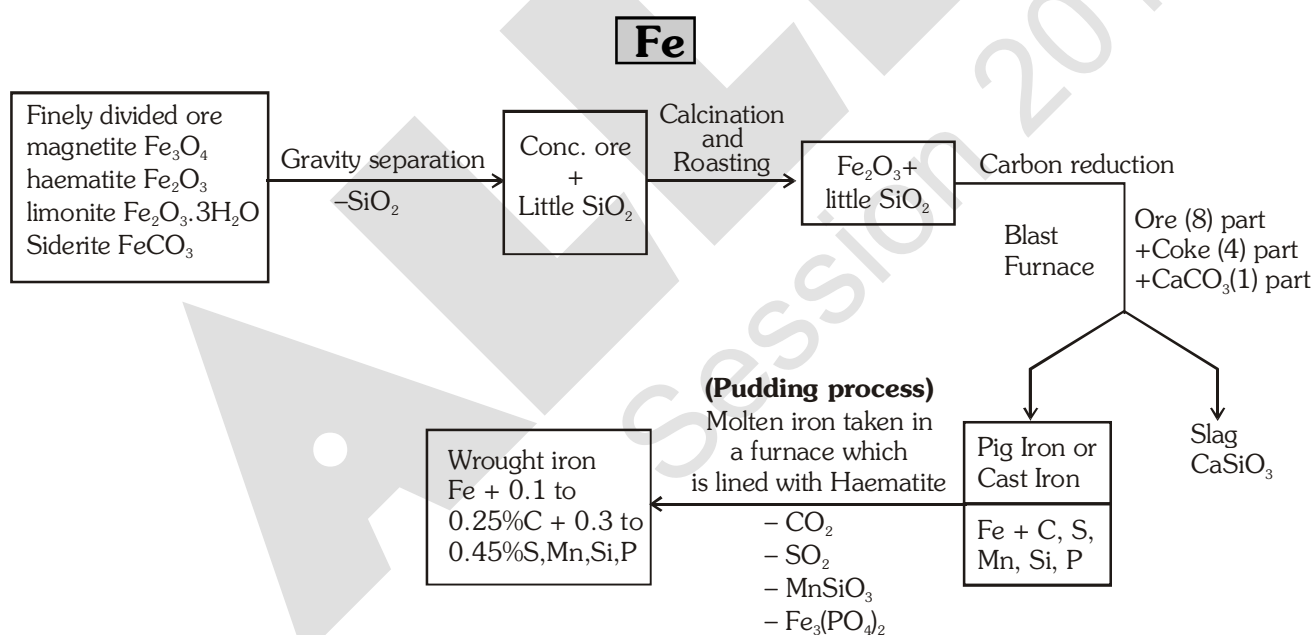
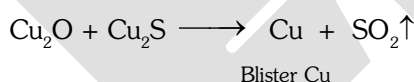
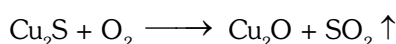
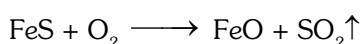
Cu_2S remain unaffected again becoz carbon reduction occurs only for oxide and not for sulphide.



Slag

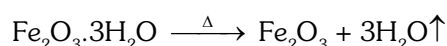
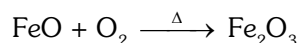
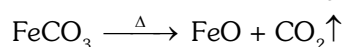
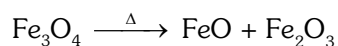


(c) Bassemer converter reaction :-



Reaction involved :

(1) Roasting step :-



FeO reacts with SiO_2 to give FeSiO_3 as slag.

Hence to prevent the formation of FeSiO_3 .

FeO is converted into Fe_2O_3 which does not react with SiO_2 due to high L.E.

GOLDEN KEY POINTS

- Types of Iron :-

Pig iron $\rightarrow$ Cast iron $\rightarrow$ Steel iron $\rightarrow$ Wrought iron

- (i) **Cast iron or pig iron**

It is most impure form of Iron and contains the highest proportion of carbon (2.5 - 4 %) along with traces of S, P, Mn and Si. Cast iron contain 2.5 to 4.3 & pig contain 2.5 to 5%.

- (ii) **Wrought iron (Fibrous iron) or malleable iron**

It is the purest form of iron and contains minimum amount of carbon (0.12 - 0.25%) and less than 5% of other impurities.

- (iii) **Steel**

It is the most important form of iron and finds extensive applications. As far as carbon content (impurity) is concerned it is midway between cast iron and wrought iron, it contains 0.25- 2% carbon. Thus all the three forms of iron differ in their carbon contents, both iron and steel are obtained from cast iron.

Order of M.P. WI > Steel > CI or PI

- Useful gas NH_3 is evolved in the leaching of bauxite by serpeck's process.
- In the electrolytic reduction of Al_2O_3 , cryolite (Na_3AlF_6) is added along with CaF_2 (fluorspar) to—
 - decrease m.p. of Al_2O_3
 - decrease viscosity of electrolyte (CaF_2 is used)
 - increase conductivity
- In the electrolytic reduction, graphite anode gets corroded or finished due to reaction with O_2 liberated at anode, hence it had to be changed periodically.
- In the electrolytic refining (4th step) no electrodes are used. In the Hoope's process molten pure Al is used as cathode and molten impure Al is used an anode.

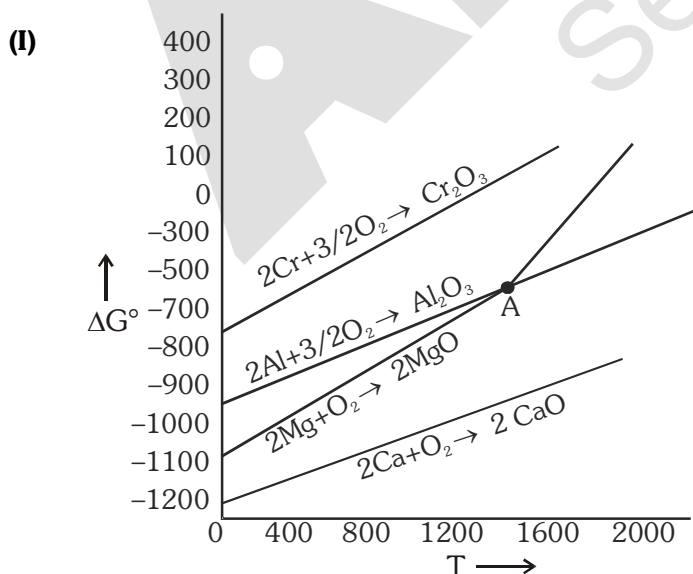
4.3 THERMODYNAMICS OF REDUCTION PROCESSES (ELLINGHAM DIAGRAM)

It is explanation of the feasibility of pyrometallurgical process by using gibbs equation $\Delta G = \Delta H - T\Delta S$

If $\Delta G = -ve$ Process is stable or Spontaneous

$\Delta G = +ve$ or Less $-ve$ then process is Unstable or non-Spontaneous

When pyrometallurgical process contains more than one type of reaction then stability of reaction can be explain by Ellingham diagram. Ellingham diagram contains plot ΔG vs T .



$$\Delta G = -Ve$$



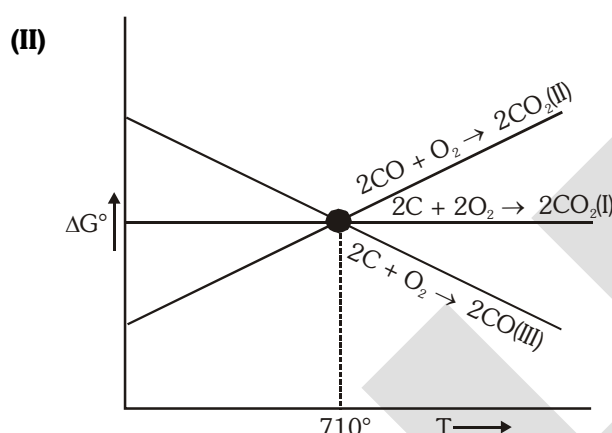
$$\Delta G = +Ve$$



According to Ellingham diagram, the metal below can reduce the oxide of metal above it in the curve, as affinity of metal below for oxygen is more.

Example Al Metal can reduce Cr_2O_3 but can not reduce MgO & CaO .

At very high T after 'A' Point Al metal can reduce MgO because Formation of MgO contains less -ve ΔG .



According to diagram at high T (710° OR above 710°C) Oxidation of C contains more -ve ΔG so at high T 'C' is good Reducing agent.

At Low T (below 710°C) Oxidation of CO contains more -ve ΔG so at Low T, CO is good Reducing agent.

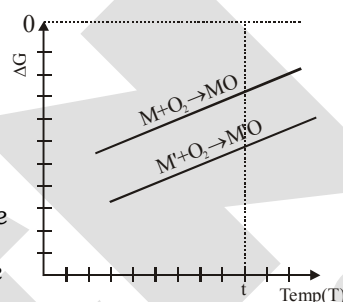
BEGINNER'S BOX-3

- The appropriate flux for removing FeO will be
 (1) SiO_2 (2) CaO (3) MgO (4) Al
- Match the following :-

Metallurgical process	Methods
(A) Copper glance $\rightarrow \text{Cu}$	(P) Leaching by NaOH followed by electrolytic reduction
(B) Bauxite $\rightarrow \text{Al}$	(Q) Froth flotation, partial roasting & self reduction.
(C) Cerussite $\rightarrow \text{Pb}$	(R) Heating in presence of dry HCl followed by electrolysis
(D) $\text{MgCl}_2 \cdot 6\text{H}_2\text{O} \rightarrow \text{Mg}$	(S) Calcination followed by carbon reduction

(1) A-R, B-P, C-S, D-Q (2) A-Q, B-P, C-S, D-R
 (3) A-P, B-Q, C-S, D-R (4) A-S, B-P, C-Q, D-R
- In the metallurgy of iron, during smelting process, CO is main reducing agent at the upper part of blast furnace, because
 (1) It is less concentrated in upper atmosphere
 (2) The CO_2 formed can be easily removed
 (3) In the lower atmosphere, there is no oxide to be reduced
 (4) The entropy change during CO/CO_2 is negative

4. Favourable reducing agent for ZnO/Zn is :-
 (1) C at high temperature (2) CO at high temperature
 (3) Cu at high temperature (4) Cu at low temperature
5. Which of the following is true regarding recovery of copper from its low grade (lean) ores.
 (1) Zn is advantageous economically
 (2) Fe is advantageous economically
 (3) Both Fe and Zn can be used and are equally advantageous
 (4) None of them can be used because Cu is less reactive
6. The slope of a line on Ellingham diagram for a process M/MO is :-
 (1) $-\frac{\Delta G}{R}$ (2) $-\Delta S$ (3) ΔS (4) $-\frac{\Delta S}{T}$
7. The true statement is at temperature 'T'
 (1) $M'(s) + \frac{1}{2}O_2(g) \rightarrow M'O(s)$
 (2) $M'(s) + MO(s) \rightarrow M'O(s) + M(s)$
 (3) $M'O(s) + MO(s) \rightarrow M'O(s) + M(s)$
 (4) $M'(s) + O_2(g) \rightarrow M'O(s)$
8. Auto reduction process is used in extraction of :
 (1) Cu (2) Hg (3) Pb (4) All of these
9. Which is not the correct process-mineral matching in metallurgical extraction.
 (A) Leaching : silver (B) Zone refining : lead.
 (C) Liquation : tin (D) Van Arkel : Zr
10. Consider the following metallurgical processes :-
 (i) Heating impure metal with CO and distilling the resulting volatile carbonyl (boiling point 43°C) and finally decomposing at 150°C to 200°C to get the pure metal.
 (ii) Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged sulphide.
 (iii) Electrolysing the molten electrolyte containing approximately equal amounts of the metal chloride and CaCl_2 to obtain the metal.
 The process used for obtaining sodium, nickel and copper are, respectively,
 (1) (i), (ii) and (iii) (2) (ii), (iii) and (i) (3) (iii), (i) and (ii) (4) (ii), (i) and (iii)



ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	3	3	1	4	4	4	4	2	1
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	3	3	2	3	2	2	3	4	3
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	1	2	4	1	2	2	4	4	2	3

SOME IMPORTANT ALLOY

- | | | | |
|----|---------------------|---|---|
| 1. | Bronze | - | Cu (75-90 %) + Sn (10-25 %) |
| 2. | Brass | - | Cu (60-80 %) + Zn (20-40%) |
| 3. | Gun metal | - | (Cu + Zn + Sn) (87:3:10) |
| 4. | German Silver | - | Cu + Zn + Ni (2:1:1) |
| 5. | Nichrome | - | (Ni + Cr + Fe) |
| 6. | Alloys of steel | | |
| | (a) Chromium steel | | Cr (2-4%) |
| | (b) Nickel Steel | | Ni (3-5%) |
| | (c) Stainless steel | | Cr (12-14%) & Ni (2-4%) Cr forms oxide layer & Protects Iron From Rusting |
| | (d) Invar (इनवार) | | Ni (36%) |

APPENDIX

LIST OF ORES AND THEIR NAMES

TYPES OF ORES	S.N.	FORMULA OF THE ORE	NAME
Oxide Ore	1.	ZnO (Philosopher's Wool)	Zincite
	2.	MnO ₂	Pyrolusite
	3.	SnO ₂	Cassiterite (Tin stone)
	4.	Cu ₂ O	Cuprite (Ruby Copper)
	5.	Fe ₂ O ₃	Haematite
	6.	Al ₂ O ₃ ·2H ₂ O	Bauxite
	7.	FeO·Cr ₂ O ₃	Chromite
	8.	Fe ₃ O ₄	Magnetite
	9.	Fe ₂ O ₃ ·3H ₂ O	Limonite
	10.	TiO ₂	Rutile
Sulphide Ore	1.	ZnS	Zinc Blende (Sphalerite)
	2.	HgS	Cinnabar
	3.	PbS	Galena
	4.	Ag ₂ S	Argentite or Silver glance
	5.	FeS ₂	Iron pyrites (Fool's gold)
	6.	CuFeS ₂ , CuS·FeS	Copper pyrites (Chalcopyrites)
	7.	Cu ₂ S·Ag ₂ S	Copper silver glance
	8.	Ag ₂ S·Sb ₂ S ₃	Pyrargirite (Ruby silver)
	9.	Cu ₂ S (Copper glance)	Chalcocite
Halide Ore	1.	NaCl	Rock Salt
	2.	AgCl	Horn Silver
	3.	CaF ₂	Fluorspar
	4.	AlF ₃ ·3NaF	Cryolite
	5.	KCl·MgCl ₂ ·6H ₂ O	Carnelite
Carbonate Ore	1.	MgCO ₃	Magnesite
	2.	CaCO ₃	Lime stone
	3.	MgCO ₃ ·CaCO ₃	Dolomite
	4.	ZnCO ₃ (Smithsonite)	Calamine
	5.	PbCO ₃	Cerussite
	6.	FeCO ₃	Siderite
	7.	CuCO ₃ ·Cu(OH) ₂	Malachite
	8.	2CuCO ₃ ·Cu(OH) ₂	Azurite
Sulphate Ore	1.	CaSO ₄ ·2H ₂ O	Gypsum
	2.	MgSO ₄ ·7H ₂ O	Epsomite Or Epsom salt
	3.	K ₂ SO ₄ ·Al ₂ (SO ₄) ₃ ·24H ₂ O	Alum
Nitrate Ore	1.	NaNO ₃	Chile- Salt Peter
	2.	KNO ₃	Salt peter or Indian salt peter
Phosphate Ore	1.	Ca ₃ (PO ₄) ₂	Rock Phosphate

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

1. Which of the following is not an ore of Iron:-
(1) Haematite (2) Limonite
(3) Cassiterite (4) Magnetite
2. Aluminium is obtained from Al_2O_3 by this method
(1) Thermal reduction.
(2) Hydro metallurgical method.
(3) Electrolytic reduction.
(4) Reduction by iron.
3. Zinc blende on roasting in air gives :-
(1) Zinc carbonate (2) SO_2 and ZnO
(3) ZnS and $ZnSO_4$ (4) CO_2 and ZnO
4. Litharge is a mineral of:-
(1) Magnesium (2) Lithium
(3) Lead (4) Zinc
5. The oxide cannot be reduced by coke
(1) Cu_2O , ZnO (2) Fe_2O_3 , ZnO
(3) CaO, K_2O (4) PbO, Fe_3O_4
6. Chemical formula of horn silver is:-
(1) Ag_2S (2) AgCl
(3) $AgNO_3$ (4) $Ag_2S \cdot Sb_2S_3$
7. An example of halide ores is:-
(1) Galena (2) Bauxite
(3) Cinnabar (4) Cryolite
8. Which is not a basic flux :-
(1) Silica (2) Lime stone
(3) Calcite (4) Quick lime
9. Iron pyrites ore is concentrated by:-
(1) Froth floatation (2) Electrolysis
(3) Roasting (4) Magnetic separation
10. Which of the following metal is extracted by electrolytic reduction process of its halide are :-
(1) Copper (2) Iron
(3) Sodium (4) Aluminium
11. Which of the following metal can not be extracted by smelting process:-
(1) Lead (2) Zinc
(3) Iron (4) Aluminium
12. The main reducing agent for the extraction of iron metal in a blast furnace is:-
(1) Coke (2) Carbon
(3) Carbon dioxide (4) Carbon monoxide
13. The flux used in extraction of Iron from haematite in the blast furnace is:-
(1) Silica (2) Lime stone
(3) Phosphorus chloride (4) Calcium phosphate
14. The reduction of Cr_2O_3 , by heating it with aluminium is known as:-
(1) Smelting
(2) Roasting
(3) Calcination
(4) Aluminothermite process
15. Which of the following is obtained by hydrometallurgy :-
(1) Copper (2) Gold
(3) Silver (4) All of these
16. Aluminium is purified by :-
(1) Roasting
(2) Sublimation
(3) Electrolytic refining
(4) Reduction with carbon
17. In Goldschmidt thermite process, reducing agent is:-
(1) Fe (2) Na
(3) Ca (4) Al
18. Heating pyrites in air to remove sulphur is known as:-
(1) Roasting (2) Calcination
(3) Smelting (4) Fluxing
19. Liquation process is used for refining:-
(1) Bismuth (2) Lead
(3) Tin (4) All
20. A mineral is called ore if
(1) Metal present in the mineral is costly
(2) A metal can be extracted from it
(3) A metal can be extracted profitably from it
(4) A metal can not be extracted from it
21. Autoredution process is used in the extraction of:-
(1) Cu & Pb (2) Zn & Hg
(3) Cu & Al (4) Fe & Pb
22. In thermite process, thermite mixture is:-
(1) Al powder + sulphide (2) Zn + oxide powder
(3) Na + Oxide (4) Al powder + oxide
23. The process of converting hydrated Alumina into anhydrous Alumina is called:-
(1) Roasting (2) Calcination
(3) Smelting (4) Dressing
24. The metallurgical process in which a metal is obtained in a fused state is called:-
(1) Smelting (2) Roasting
(3) Calcination (4) Froth floatation

- 25.** In the extraction of copper, metal is formed in the Bessemer converter due to reaction:-
 (1) $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$
 (2) $\text{Cu}_2\text{S} \rightarrow 2\text{Cu} + \text{S}$
 (3) $\text{Fe} + \text{Cu}_2\text{O} \rightarrow 2\text{Cu} + \text{FeO}$
 (4) $2\text{Cu}_2\text{O} \rightarrow 4\text{Cu} + \text{O}_2$
- 26.** In the electrolytic refining of copper, Ag and Au are found:-
 (1) On cathode (2) On anode
 (3) In the anodic mud (4) In the cathodic mud
- 27.** Consider :-
 (a) Copper blende = Cu_2O
 (b) Chromite = Magnetic separation.
 (c) Bauxite = $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
 (d) Liquation = Liquid metals e.g. Hg
 Which is/are not correctly matched :-
 (1) (a) only (2) (b) only
 (3) (d) only (4) (a) & (d) both
- 28.** Silver can be separated from lead by :-
 (1) Distillation (2) Amalgamation
 (3) Filtration (4) Cupellation
- 29.** In blast furnace this is acting as reducing agent at lower part :-
 (1) CO (2) H_2
 (3) C (4) None
- 30.** Which of the following metals can not be extracted by carbon reduction process :-
 (1) Pb (2) Al
 (3) Sn (4) Zn
- 31.** The maximum temperature obtained in the....region of the blast furnace used in extraction of iron:-
 (1) Reduction (2) Combustion
 (3) Fusion (4) Slag formation
- 32.** The concentration of chromite ($\text{FeO} \cdot \text{Cr}_2\text{O}_3$) is done by :-
 (1) Leaching process (2) Magnetic separation
 (3) Froth -flotation (4) Calcination
- 33.** Which of the following process involves smelting
 (1) $2\text{PbS} + 3\text{O}_2 \rightarrow 2\text{PbO} + 2\text{SO}_2\uparrow$
 (2) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$
 (3) $\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe} + 2\text{CO}_2$
 (4) $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr} + \text{Heat}$
- 34.** Out of the following, which ores are calcinated during extraction :-
 (a) Copper pyrites (b) Malachite (c) Bauxite
 Correct answer is :-
 (1) a, b, c (2) b, c
 (3) Only a (4) All
- 35.** Which of the following match are incorrect :-
 (a) Goldschmidt aluminothermite process - Cr_2O_3
 (b) Mac Arthur cyanide process - Fe
 (c) Mond process - Ni
 (d) Van Arkel process - Au
 (1) a, c (2) c, d
 (3) b, d (4) a, b
- 36.** Electro metallurgical process (electrolysis of fused salt) is employed to extract :-
 (1) Lead (2) Silver
 (3) Sodium (4) Copper
- 37.** In the extraction of copper from pyrites, iron is removed as:-
 (1) FeSO_4 (2) FeSiO_3
 (3) Fe_3O_4 (4) Fe_2O_3
- 38.** Which one of the following metals can not be extracted by using Al as a reducing agent :-
 (1) Na from Na_2O (2) Cr from Cr_2O_3
 (3) Mn from MnO_2 (4) V from V_2O_5
- 39.** In the electrolytic refining for aluminium extraction the electrolyte used is:-
 (1) Fluorides of Al, Na and Ca
 (2) $\text{Al}(\text{OH})_3$ in NaOH solution
 (3) An aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
 (4) Molten Al_2O_3
- 40.** Which one is mismatched :-
 (1) Poling - refining of copper
 (2) Cupellation - refining of silver
 (3) Smelting - An oxidation process
 (4) Roasting - An oxidation process
- 41.** Which metal can be purified by distillation :-
 (1) Cu (2) Ag
 (3) Fe (4) Hg
- 42.** Carbon cannot be used in the reduction of Al_2O_3 because :-
 (1) it is an expensive
 (2) the enthalpy of formation of CO_2 is more than that of Al_2O_3
 (3) pure carbon is not easily available
 (4) the enthalpy of formation of Al_2O_3 is too high
- 43.** Match list I with list II and select the correct answer using the codes given below the lists
- | List I | | List II | |
|---------------------|---|---------|--|
| A. Van Arkel method | I. Purification of titanium | | |
| B. Solvay process | II. Manufacture of Na_2CO_3 | | |
| C. Cupellation | III. Purification of copper | | |
| D. Poling | IV. Refining of silver | | |
- Codes :**
- | | A | B | C | D |
|-----|-----|----|-----|-----|
| (1) | I | II | IV | III |
| (2) | II | I | III | IV |
| (3) | IV | II | I | III |
| (4) | III | I | II | IV |

44. Anode mud obtained after electrolytic refining of copper contains :-

- (1) Ag (2) Au
(3) Pt (4) All

45. Matte :-

- (1) $\text{Cu}_2\text{S} + \text{FeS}$
(2) $\text{Cu}_2\text{O} + \text{FeS}$
(3) $\text{Cu}_2\text{O} + \text{Cu}_2\text{S}$
(4) $\text{FeS} + \text{SiO}_2$

46. Which of the following reaction is not involved in thermite process :-

- (1) $3\text{Mn}_3\text{O}_4 + 8\text{Al} \longrightarrow 9\text{Mn} + 4\text{Al}_2\text{O}_3$
(2) $\text{Cr}_2\text{O}_3 + 2\text{Al} \longrightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
(3) $2\text{Fe} + \text{Al}_2\text{O}_3 \longrightarrow 2\text{Al} + \text{Fe}_2\text{O}_3$
(4) $\text{B}_2\text{O}_3 + 2\text{Al} \longrightarrow 2\text{B} + \text{Al}_2\text{O}_3$

47. Alumino thermite process is used for the extraction of metals, whose oxides are :-

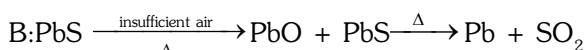
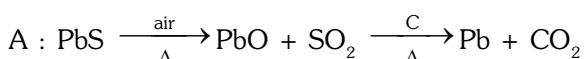
- (1) Strongly acidic
(2) Not easily reduced by carbon
(3) Not easily reduced by hydrogen
(4) Strongly basic

48. Match the following :-

I	II
(A) Calcination	a. $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$
(B) Roasting	b. $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O} \rightarrow \text{Fe}_2\text{O}_3 + n\text{H}_2\text{O}$
(C) Flux	c. $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
(D) Thermite	d. $\text{SiO}_2 + \text{FeO} \rightarrow \text{FeSiO}_3$

A	B	C	D
(1) a	b	c	d
(2) b	a	d	c
(3) d	a	b	c
(4) c	a	b	d

49. Main source of lead is PbS . It is converted to Pb by :-



Self reduction process is :

- (1) A (2) B
(3) both (4) none

50. When haematite ore is burnt in air with coke along with lime at 200°C , the process not only produces steel but also produces an important compound (A), which is useful in making building materials. The compound (A) is

- (1) SiO_2 (2) CaSiO_3
(3) FeO (4) Fe_2O_3

51. Match List-I with List-II and select the correct answer using the codes given below the lists.

List-I (Metals)

List-II

(Process/methods involved in extraction process)

- | | |
|--------|--------------------|
| (a) Au | 1. Self reduction |
| (b) Al | 2. Liquefaction |
| (c) Pb | 3. Electrolysis |
| (d) Sn | 4. Bayer's process |

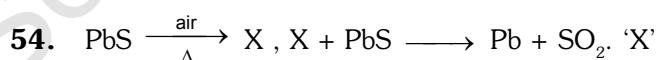
	(a)	(b)	(c)	(d)
(1)	3	1	2	4
(2)	3	4	1	2
(3)	1	2	4	3
(4)	3	2	4	1

52. Main function of the collectors in metallurgy is:

- (1) Stick to the ore and then take it to rise up to the top
(2) Convert the insoluble ore into soluble part
(3) Make the ore hydrophobic
(4) None

53. Reducing agent of haematite in blast-furnace is :-

- (1) Coke in furnace
(2) Coke in upper part and CO in lower part of furnace.
(3) CO in most parts of the furnace
(4) CO in the furnace.



may be :-

- (1) PbO
(2) PbO_2
(3) PbO and PbSO_4
(4) PbO_2 and PbO

55. Which one of the following statements is **incorrect**?

- (1) Tin is extracted by carbon reduction (smelting)
(2) Aluminium is extracted by Hall's process which involves carbon reduction.
(3) Extraction of lead does not involve bessemerisation.
(4) Silver is extracted by cyanide process.

56. Extraction of zinc from zinc blende is achieved by:
- (1) electrolytic reduction
 - (2) roasting followed by reduction with carbon
 - (3) roasting followed by reduction with another metal
 - (4) roasting followed by self-reduction

57. Column - I

Column - II

- | | |
|--|------------------------|
| (1) Metal which occur in the native state in nature is | (P) Hg |
| (2) The oxides of metal that can be commercially reduced by Aluminothermite reduction process is | (Q) Ti |
| (3) van Arkel method is used for preparing ultrapure metal of | (R) Cr |
| (4) Auto reduction process is employed for the sulphide ore of | (S) Ag |
| (1) A-S, B-R, C-Q, D-P | (2) A-R, B-S, C-Q, D-P |
| (3) A-P, B-S, C-Q, D-R | (4) A-Q, B-R, C-S, D-P |

58. Match the following

- | | |
|-----------------|-----------------|
| 1. Zincite | P Sulphide ore |
| 2. Malachite | Q halide ore |
| 3. Horn silver | R Oxide ore |
| 4. Iron pyrites | S Carbonate ore |
- (1) 1 - R; 2 - P; 3 - Q; 4 - S
 (2) 1 - R; 2 - S; 3 - Q; 4 - P
 (3) 1 - S; 2 - R; 3 - P; 4 - Q
 (4) 1 - Q; 2 - S; 3 - P; 4 - R

59. In magnetic separation method which one is true
- (1) either ore is being attracted by magnetic field
 - (2) either gangue is being attracted by magnetic field
 - (3) Both
 - (4) None

60. In froth floatation method depressants are used for-
- (1) to enhance non wettability of mineral particles
 - (2) to make suspension of ore with water
 - (3) to take forth upside
 - (4) to separate two sulphide ore

61. Which reaction shows formation of blistered copper.

- (1) $2\text{FeS} + 3\text{O}_2 \longrightarrow 2\text{FeO} + 2\text{SO}_2\uparrow$
- (2) $2\text{Cu}_2\text{S} + 3\text{O}_2 \longrightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2\uparrow$
- (3) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \longrightarrow 6\text{Cu} + \text{SO}_2\uparrow$
- (4) $\text{Cu}_2\text{O} + \text{C} \longrightarrow 2\text{Cu} + \text{CO}\uparrow$

62. Which reaction is involved in extraction of Ag by cyanide process

- (1) $\text{AgBr} + \text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$
- (2) $\text{AgCl} + \text{NH}_4\text{OH} \longrightarrow [\text{Ag}(\text{NH}_3)_2] \text{Cl}$
- (3) $\text{Ag}_2\text{S} + \text{NaCN} \longrightarrow \text{Na}[\text{Ag}(\text{CN})_2]$
- (4) None

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	2	3	3	2	4	1	1	3	4	4	2	4	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	1	4	3	1	4	2	1	1	3	4	4	3	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	2	3	2	3	3	2	1	1	3	4	4	1	4	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	2	2	2	2	2	1	3	3	2	2	1	2	3	4
Que.	61	62													
Ans.	3	3													

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2007

1. Sulphide ores of metals are usually concentrated by froth floatation process. Which one of the following sulphide ores offers an exception and is concentrated by chemical leaching?
- (1) Sphalerite (2) Argentite
(3) Galena (4) Copper pyrites

AIIMS 2010

2. According to percentage weight arrange the following in descending order in the earth crust
- (1) O_2 , Ca, Mg, S (2) O_2 , S, Mg, Ca
(3) S, Ca, Mg, O_2 (4) Ca, O_2 , Mg, S

AIPMT-Pre 2011

3. Which of the following elements is present as the impurity to the maximum extent in the pig iron ?
- (1) Manganese (2) Carbon
(3) Silicon (4) Phosphorus
4. Which of the following pairs of metals is purified by Van Arkel method ?
- (1) Ga and In (2) Zr and Ti
(3) Ag and Au (4) Ni and Fe

AIPMT-Mains 2011

5. The following reactions take place in the blast furnace in the preparation of impure iron. Identify the reaction pertaining to the formation of the slag:-
- (1) $2C(s) + O_2(g) \rightarrow 2CO(g)$
(2) $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$
(3) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
(4) $CaO(s) + SiO_2(s) \rightarrow CaSiO_3(s)$

AIPMT-Pre 2012

6. Aluminium is extracted from alumina (Al_2O_3) by electrolysis of a molten mixture of:
- (1) $Al_2O_3 + Na_3AlF_6 + CaF_2$
(2) $Al_2O_3 + KF + Na_3AlF_6$
(3) $Al_2O_3 + HF + NaAlF_4$
(4) $Al_2O_3 + CaF_2 + NaAlF_4$

7. In the extraction of copper from its sulphide ore, the metal is finally obtained by the reduction of cuprous oxide with:
- (1) Iron sulphide (FeS)
(2) Carbon monoxide (CO)
(3) Copper (I) sulphide (Cu_2S)
(4) Sulphur dioxide (SO_2)
8. Identify the alloy containing a non-metal as a constituent in it.
- (1) Bell metal (2) Bronze
(3) Invar (4) Steel
9. Which one of the following is a mineral of iron?
- (1) Pyrolusite
(2) Magnetite
(3) Malachite
(4) Cassiterite

AIIMS 2014

10. Which of the following is separated as slag during extraction of Fe in blast furnace :-
- (1) SiO_2 (2) Al_2O_3
(3) CaO (4) MgO

Re-AIPMT 2015

11. In the extraction of copper from its sulphide ore, the metal is finally obtained by the reduction of cuprous oxide with :-
- (1) copper(I) sulphide
(2) sulphur dioxide
(3) iron(II) sulphide
(4) carbon monoxide

AIIMS 2015

12. In extraction of metal by bessemerisation, Cu_2S is converted into :
- (1) Cu_2O (2) CuO
(3) Cu (4) $CuFeO_2$

NEET-I 2016

13. Match items of **Column I** with the items of **Column II** and assign the correct code :

	Column-I		Column-II
(a)	Cyanide process	(i)	Ultrapure Ge
(b)	Froth floatation process	(ii)	Dressing of ZnS
(c)	Electrolytic reduction	(iii)	Extraction of Al
(d)	Zone refining	(iv)	Extraction of Au
		(v)	Purification of Ni

Code :

- | | | | |
|-----------|-------|-------|------|
| (a) | (b) | (c) | (d) |
| (1) (iv) | (ii) | (iii) | (i) |
| (2) (ii) | (iii) | (i) | (v) |
| (3) (i) | (ii) | (iii) | (iv) |
| (4) (iii) | (iv) | (v) | (i) |

NEET-II 2016

14. Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because:
- (1) zinc has lower negative electrode potential than iron
 - (2) zinc has higher negative electrode potential than iron
 - (3) zinc is lighter than iron
 - (4) zinc has lower melting point than iron

NEET(UG) 2017

15. Extraction of gold and silver involves leaching with CN^- ion. Silver is later recovered by :-
- (1) distillation
 - (2) zone refining
 - (3) displacement with Zn
 - (4) liquation

AIIMS 2017

16. Which of the following statement is correct about blast furnace.
- (1) CaCO_3 is used to produce O_2
 - (2) Fe_2O_3 convert into Fe_3O_4
 - (3) Wrought iron can directly be obtain from blast furnace
 - (4) Wrought iron is 100% pure
17. In the fusion of chromite ore with sodium carbonate in excess of air which of the following is not formed.
- (1) Fe_2O_3
 - (2) Na_2CrO_4
 - (3) CO_2
 - (4) $\text{Na}_2\text{Cr}_2\text{O}_7$

NEET(UG) 2018

18. Considering Ellingham diagram, which of the following metals can be used to reduce alumina ?
- (1) Fe
 - (2) Zn
 - (3) Mg
 - (4) Cu

AIIMS 2018

19. In the context of the Hall-Heroult process for the extraction of Al, which of the following statement is correct ?
- (1) the ratio of alumina & cryolite is nearly 1 : 9
 - (2) pure Al is obtained at anode
 - (3) melting point of alumina decreases upto 300 K
 - (4) only carbon monoxide gas is produced at graphite electrode
20. For smelting of Cu correct statements is :-
- (1) Cu is reduced
 - (2) FeS converted in FeO
 - (3) Fe in reduced
 - (4) Cu_2S converted in Cu_2O

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	2	2	4	1	3	4	2	1	1	3	1	2	3
Que.	16	17	18	19	20										
Ans.	2	4	3	1	2										

EXERCISE-III (Analytical Questions)

- In the extraction of silver, Ag_2S is dissolved in :
(1) HCl (2) HNO_3
(3) KCN (4) H_2SO_4
- The method of zone refining of metals is based on the principle of :-
(1) Greater mobility of the pure metal than that of the impurity
(2) Higher melting point of the impurity than that of the pure metal
(3) Greater noble character of the solid metal than that of the impurity
(4) Greater solubility of the impurity in the molten state than in the solid
- Which one of the following ores is best concentrated by froth-flotation method :
(1) Galena (2) Cassiterite
(3) Magnetite (4) Malachite
- Which of the following sulphides when heated strongly in air gives the corresponding metal ?
(1) Cu_2S (2) CuS
(3) FeS (4) HgS
- Stainless steel does not rust because :
(1) Chromium and nickel combine with iron
(2) Chromium forms an oxide layer and protects iron from rusting
(3) Nickel present in it, does not rust
(4) Iron forms a hard chemical compound with chromium present in it
- In the electrolytic method of obtaining aluminium from purified bauxite, cryolite is added to the charge in order to -
(1) Minimise the heat loss due to radiation
(2) Protect aluminium produced from oxygen
(3) dissolve bauxite and render it conductor of electricity
(4) Increases the melting point of bauxite
- Purification of Ge like semiconductor is done by
(1) Cyanide process (2) Van arkel process
(3) Alumino thermite (4) Zone refining
- Which of the following contains both Cu & Fe :-
(1) Chalcocite (2) Chalcopyrites
(3) Malachite (4) Epsum

Check Your Understanding

- The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is:
(1) Cu (2) Cr (3) Ag (4) Ca
- Which series of reactions correctly represents chemical relations related to iron and its compound?
(1) $\text{Fe} \xrightarrow{\text{Cl}_2, \text{heat}} \text{FeCl}_3 \xrightarrow{\text{heat, air}} \text{FeCl}_2 \xrightarrow{\text{Zn}} \text{Fe}$
(2) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{Fe}_3\text{O}_4 \xrightarrow{\text{CO}, 600^\circ\text{C}} \text{FeO} \xrightarrow{\text{CO}, 700^\circ\text{C}} \text{Fe}$
(3) $\text{Fe} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{H}_2\text{SO}_4, \text{O}_2} \text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\text{Heat}} \text{Fe}$
(4) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{FeO} \xrightarrow{\text{dil H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{Heat}} \text{Fe}$
- In which of the following extration no reducing agent is required ?
(1) Iron from haematite
(2) Aluminium from Cinnabar
(3) Magnesium from carnallite
(4) Zinc from zinc blende
- Of the following reduction processes, the correct process(es) is/are :-
(1) $\text{B}_2\text{O}_3 + \text{Al} \xrightarrow{\Delta} \text{B}$
(2) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Cr}$
(3) $\text{TiCl}_4 + \text{Mg} \xrightarrow{\Delta} \text{Ti}$
(4) All the above
- Select the correct statement :-
(1) Dolomite contains both magnesium and calcium
(2) Extraction of lead from galena involves roasting in limited supply of air at moderate temperature.
(3) Extraction of zinc from zinc blende involves roasting followed by reduction with carbon.
(4) All the above
- Which of the following is/are correctly matched ?
(1) Copper - Bessemer converter
(2) Iron - Blast furnance
(3) Chromium - Aluminothermic process
(4) All the above

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Ans.	3	4	1	4	2	3	4	2	4	2	3	4	4	4	

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :** All the ores are mineral
Reason : Ores contains metals in combined state
 (1) A (2) B (3) C (4) D
- Assertion :** Ores are generally converted into oxides, prior to reduction
Reason : Metal oxides can be easily reduced
 (1) A (2) B (3) C (4) D
- Assertion :** In the extraction of Ag the complex $\text{Na}[\text{Ag}(\text{CN})_2]$ is reacted with Zn
Reason : Zn is transition metal.
 (1) A (2) B (3) C (4) D
- Assertion :** In froth floatation process sodium ethyl xanthate is used as floating agent
Reason : Sulphide ores are water soluble
 (1) A (2) B (3) C (4) D
- Assertion :** Cryolite is used in electrolytic extraction of Al from alumina.
Reason : It dissolves alumina, decreases its mpt.
 (1) A (2) B (3) C (4) D
- Assertion :** CuFeS_2 is concentrated by froath floatation method
Reason : CuFeS_2 is main ore of copper
 (1) A (2) B (3) C (4) D
- Assertion :** Wolframite impurities are separated from cassiterite by electromagnetic separation.
Reason : Cassiterite being magnetic is attached by the magnet.
 (1) A (2) B (3) C (4) D
- Assertion :** Lead, tin and bismuth are purified by liquation method.
Reason : Lead, tin and bismuth have low m.p. as compared to impurities.
 (1) A (2) B (3) C (4) D
- Assertion :** In the smelting of copper ore coke is added in the blast furnace.
Reason : Coke reduces, CuO into Cu .
 (1) A (2) B (3) C (4) D
- Assertion :** Extraction of iron metal from iron oxide ore is carried out by heating with coke.
Reason : The reaction $\text{Fe}_2\text{O}_3(\text{s}) \longrightarrow \text{Fe}(\text{s}) + 3/2\text{O}_2(\text{g})$ is a spontaneous process.
 (1) A (2) B (3) C (4) D
- Assertion :-** $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$, this reaction is not possible at room temperature.
Reason :- ΔG value for this reaction is negative.
 (1) A (2) B (3) C (4) D
- Assertion :-** Zone refining is based on the fact that impurities are more soluble in molten state than in solid in the presence of oxygen.
Reason :- This method is used to prepare pure metal oxides.
 (1) A (2) B (3) C (4) D
- Assertion :-** According to ellingam diagram FeO can be easily reduced than Cu_2O by carbon.
Reason :- ΔG° of FeO is -300 kJ where as that of Cu_2O is -450 kJ .
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	
Ans.	2	1	3	3	1	2	3	1	3	3	2	4	4	

IMPORTANT NOTES

IMPORTANT NOTES

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HALOGEN DERIVATIVES

E

S. No.	CONTENTS	Page
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NEET SYLLABUS

Haloalkanes : Nomenclature, nature of C-X bond, physical and chemical properties, **Haloarenes :** Nature of C-X bond, substitution reactions (directive influence of halogen for monosubstituted compounds only), Uses and environmental effect of dichloromethane, trichloromethane, Iodoform, freons, DDT

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- Name haloalkanes and haloarenes according to the IUPAC system of nomenclature from their given structures;
- Describe the reactions involved in the preparation of haloalkanes and haloarenes and understand various reactions that they undergo;
- Correlate the structures of haloalkanes and haloarenes with various types of reactions;
- Use stereochemistry as a tool for understanding the reaction mechanism;
- Appreciate the applications of organo-metallic compounds;
- Highlight the environmental effects of polyhalogen compounds

*"Failure will never overtake me if my
determination to succeed is strong enough"*

A.P.J. Abdul Kalam

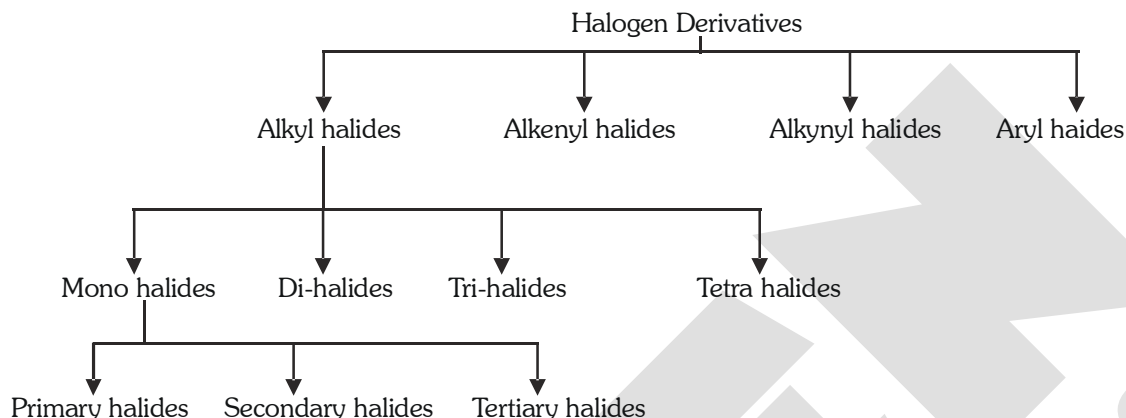
HALOGEN DERIVATIVES

1.0 HALOGEN DERIVATIVES

Compounds derived from hydrocarbons by replacement of one or more H-atoms by corresponding no. of halogen atoms are known as halogen derivatives.

2.0 CLASSIFICATION

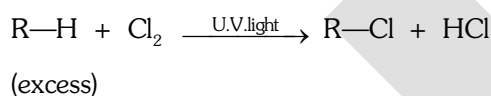
On the basis of nature of hydrocarbon from which they are obtained, halogen derivatives can be classified as :



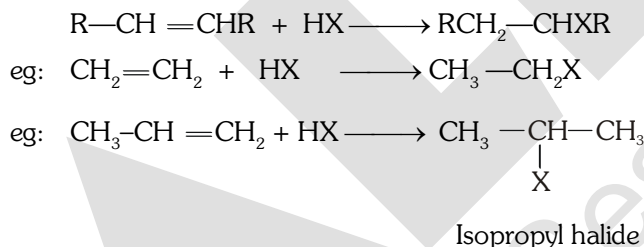
3.0 MONOHALIDES :

3.1 General Methods of Preparation of Monohalides

(1) By direct halogenation of alkanes :

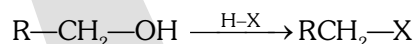


(2) By the addition of H-X on alkenes :

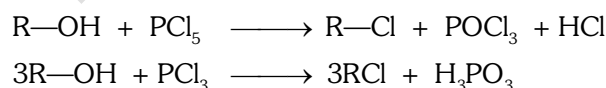


(3) By Alcohols :

(a) By the action of hydrogen halides :

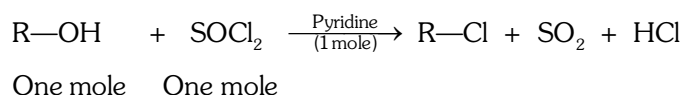


(b) By the action of phosphorous halides :

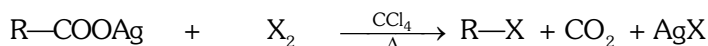


PBr₃ and PI₃ are less stable, thus for bromides (P + Br₂) and for iodides (P + I₂) mixture is used.

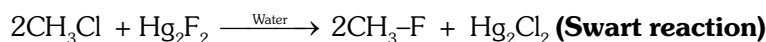
(c) By reaction with thionyl chloride (Darzen's procedure) :



Because of less stability of SOBr₂ and SOI₂, R-Br and RI can not be obtained by this method.

(4) Borodine – Hunsdicker's reaction :Silver salt of (Cl₂ or Br₂)

a fatty acid

(5) By halide exchange :

Note : Finkelstein reaction can only be used to prepare R-I and swart's reaction can only be used to prepare R-F

3.2 Physical Properties

(a) The lower members CH₃F, CH₃Cl, CH₃Br, C₂H₅Cl and C₂H₅F are gases at room temp.

(b) Higher B.P. than parent alkanes.

Decreasing order of B.P. is : **R-I > R-Br > R-Cl > R-F**

among isomeric R-X, decreasing order of B.P. is : **Primary > Secondary > tertiary**

(c) R-F and R-Cl $\longrightarrow$ lighter than water

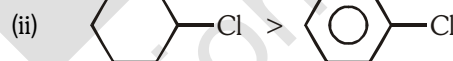
R-Br and R-I $\longrightarrow$ heavier than water

Decreasing order of density is : **R-I > R-Br > R-Cl > R-F**

(d) R-X are polar co-valent compounds but insoluble in water because they can not form H-bonds. They dissolve in organic solvents.

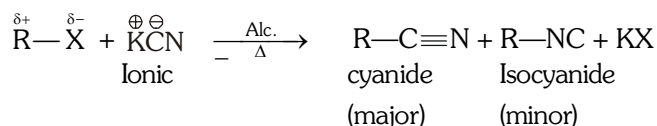
(e) R-X (except R-F) burns with a green flame when interacted with Cu wire. (Beliestein test)

(f) Dipole moment order-

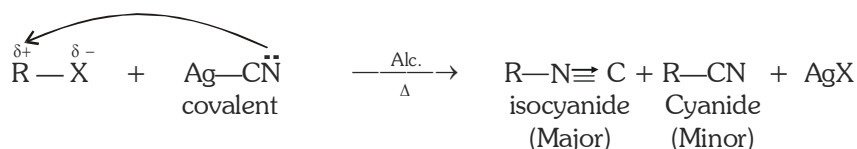
**3.3 Chemical Properties****3.3.1 Nucleophilic substitution reaction (S_N) :**

	Replacement of	Product
(a)	HOH(Boil) X by -OH	R-OH (alcohol) + HX
(b)	KOH aq. X by -OH	R-OH (alcohol) + KX
(c)	R-X $\longrightarrow$ moist Ag ₂ O X by -OH	R-OH (alcohol) + AgX
(d)	KSH alc. X by -SH (Mercapto Gp.)	R-SH + KX Alkane thiol (Mercaptane)
(e)	NaSR' X by SR	RSR' + NaX Thioether
(f)	R' COOAg, Δ X by (R'COO)	R' COOR + AgX (Alkyl alkanoate) ester

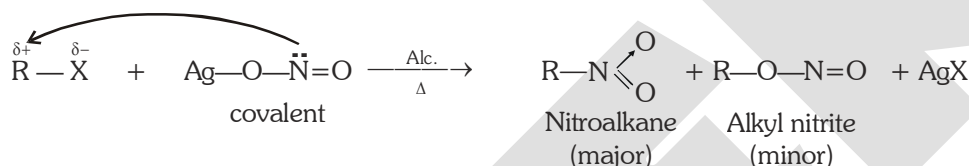
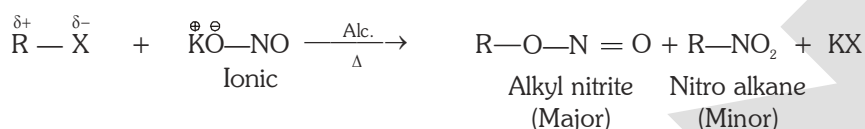
(g) Reaction with KCN and AgCN :



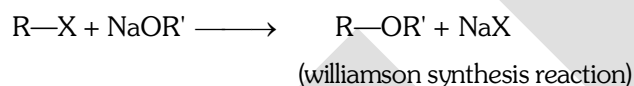
● $\text{:}\ddot{\text{C}}\text{N}^-$ ion is an ambident nucleophile.



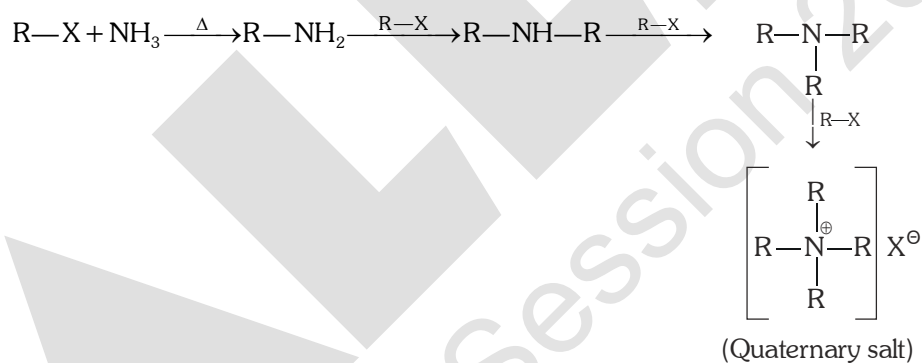
(h) Reaction with KNO_2 and AgNO_2 :



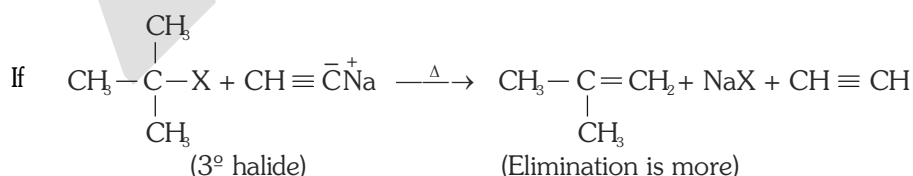
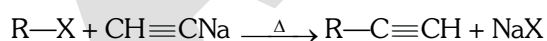
(i) Reaction with NaOR' (Sodium alkoxide) :



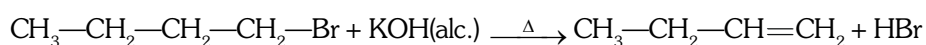
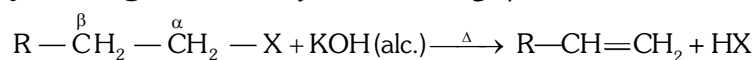
(j) Reaction with NH_3 :

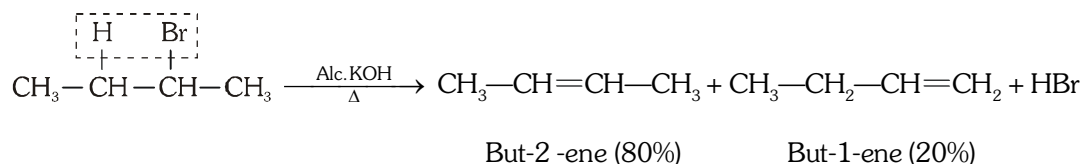


(k) Reaction with $\text{CH}\equiv\text{CNa}$:



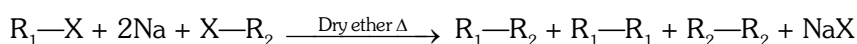
3.3.2 Dehydrohalogenation : Alkyl halides undergo β - elimination on treatment with KOH (alc.) or NaNH_2 .





3.3.3 Wurtz Reaction : $2\text{RX} + 2\text{Na} \xrightarrow{\text{Dry ether}} \text{R}-\text{R} + 2\text{NaX}$

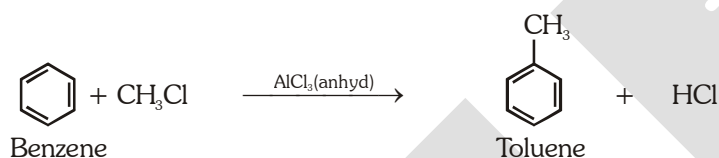
When a mixture of different alkyl halides, ($\text{R}_1 - \text{X}$) and ($\text{R}_2 - \text{X}$) is used a mixture of alkane is formed -



3.3.4 Formation of Organometallic compounds :

- (i) $\text{R}-\text{X} + \text{Mg} \xrightarrow{\text{dry ether}} \text{RMgX}$ (Grignard reagent)
- (ii) $2\text{C}_2\text{H}_5\text{Br} + 2\text{Zn} \xrightarrow{\text{dry ether}} (\text{C}_2\text{H}_5)_2\text{Zn}$ (Frankland reagent) + ZnBr_2
- (iii) $4\text{C}_2\text{H}_5\text{Cl} + 4\text{Na} / \text{Pb} \longrightarrow (\text{C}_2\text{H}_5)_4\text{Pb} + 4\text{NaCl} + 3\text{Pb}$
- Sodium lead Alloy Tetra ethyl lead (used as antiknocking agent)

3.3.5 Friedel - Crafts reaction :



4.0 DIHALIDES

General formula $\text{C}_n\text{H}_{2n}\text{X}_2$. Two H - atom of alkanes, replaced by two halogen atoms to form dihalides.

Dihalides are classified as :

(a) Gem dihalide : The term Gem is derived from geminal means - same position.

Two similar halogen atoms are attached to same C - atom

Ex. CH_3CHX_2 Ethylidene dihalide (1, 1 - Dihalo ethane)

$\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{CHX}_2 \\ \diagdown \\ \text{CH}_3 \end{array}$ Isopropylidene dihalide (2, 2 - Dihalo propane)

(b) Vic dihalides : Vic term from - Vicinal means adjacent C - atoms

Two halogen atoms are attached on adjacent carbon atom.



Vic and Gem dihalides are position isomers.

Ethylene dihalide

(1,2-Dihaloethane)

Propylene dihalide

(1,2-Dihalo propane)

(c) α , ω dihalides : Halogen atoms are attached with terminal C - atom. They are separated by 3 or more C - atoms. They are also known as polymethylene halides.

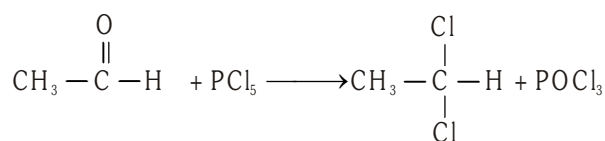
Ex. $\begin{array}{cc} \text{CH}_2 & - & \text{CH}_2 & - & \text{CH}_2 & - & \text{CH}_2 \\ | & & & & & & | \\ \text{X} & & & & & & \text{X} \end{array}$ (1,4-Dihalobutane)

Tetramethylene dihalide

4.1 General Methods of Preparation

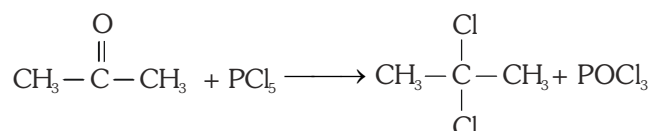
(a) Gem dihalides :

- (i) By the reaction of PCl_5 on carbonyl compound.



Acetaldehyde

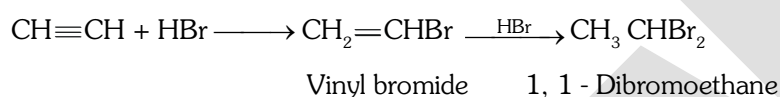
Ethylidene chloride



Acetone

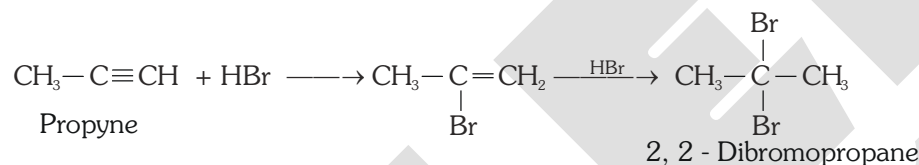
2, 2 - Dichloropropane

- (ii) By addition of halogen acids on alkynes :



Vinyl bromide

1, 1 - Dibromoethane

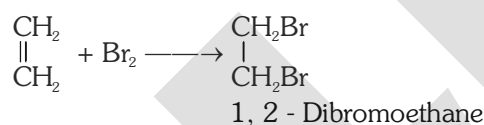


Propyne

2, 2 - Dibromopropane

(b) Vic-dihalides :

- (i) By the addition of halogens to alkenes :

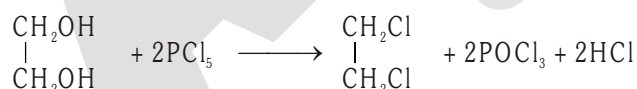


1, 2 - Dibromoethane



1, 2 - Dibromo propane

- (ii) By the action of PCl_5 on glycols :



4.2 Physical Properties

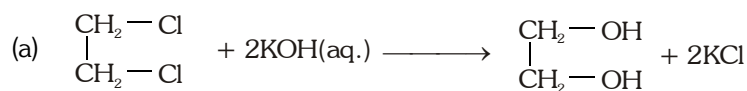
- (i) Lower members are colourless, oily liquids with sweet smell. Higher members are solid.
(ii) These are heavier than water.

4.3 Chemical Properties

(i) Action of KOH(alc.) : (Dehydrohalogenation)



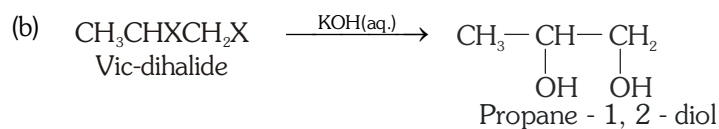
(ii) **Action of KOH(aq.)** : (Hydrolysis) It is a distinction test for gem - and vic - dihalides.



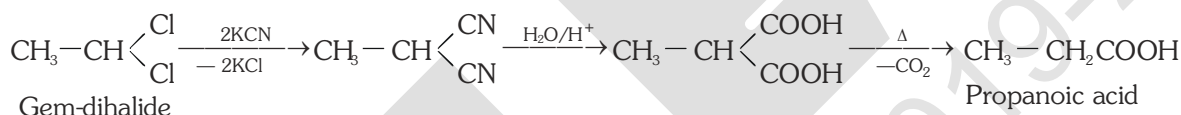
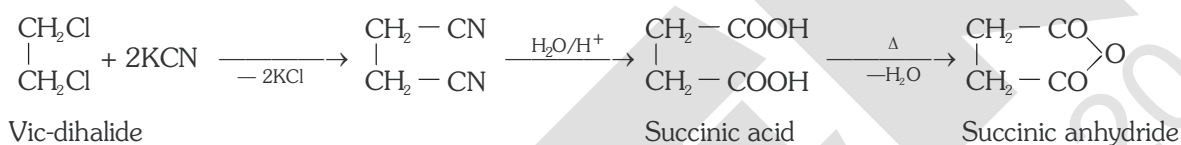
Glycol

Vic-dihalide.

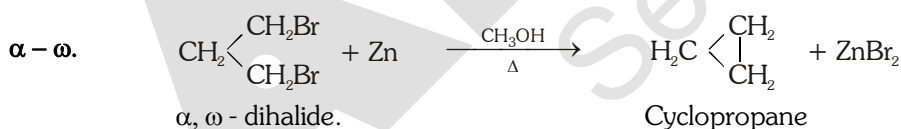
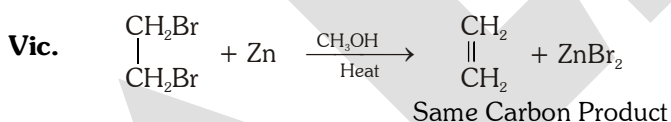
1, 2 - Ethanediol



Gem-dihalides

(iii) **Reaction with KCN** :

(i) - CN group on acid hydrolysis gives - COOH

(ii) Two - COOH group on one C - atom always lose CO₂ to form monocarboxylic acid on heating.(iii) Two - COOH group on vic. C - atom lose H₂O to form cyclic anhydride on heating.(iv) **Dehalogenation** :**BEGINNER'S BOX-1**

1. Which of the following is not organometallic compound

(1) RMgX

(2) R₂Zn

(3) RNa

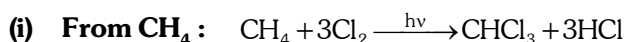
(4) R₂Hg

2. Which is Finkelstein reaction ?

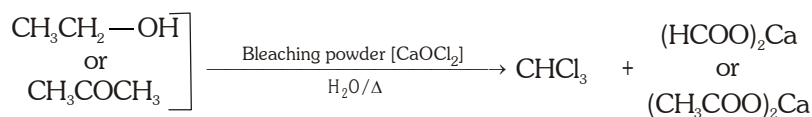
(1) R-X + NaI $\xrightarrow{\text{acetone}}$ (2) R-X + AgF $\longrightarrow$ (3) R-X + NaF $\longrightarrow$ (4) R-F + AgCl $\longrightarrow$

5.0 TRI HALIDES (Haloform CHX_3)

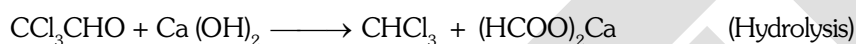
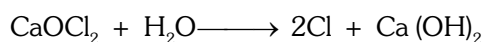
5.1 General Method of Preparation



(ii) By Haloform reaction (lab method) :



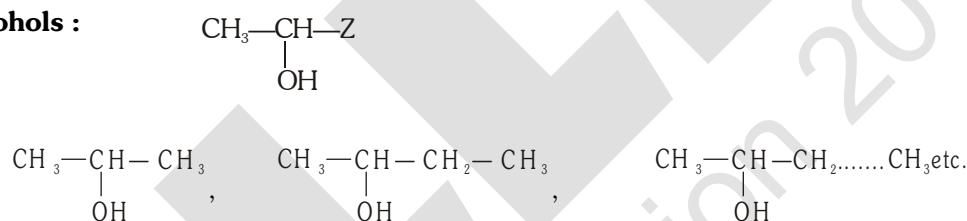
Mechanism :



If CH_3COCH_3 is used then CHCl_3 is formed into 2 steps (Chlorination and Hydrolysis)

(+) ve haloform reaction : Reaction which gives haloform with alkali and X_2 is called as (+) ve haloform reaction.

Ex. Alcohols :



Aldehydes : Only acetaldehyde $\text{CH}_3-\text{C}(=\text{O})-\text{H}$

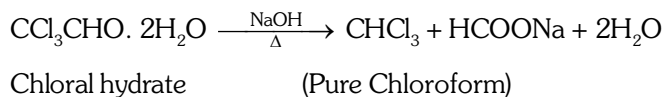
Ketones : $\text{CH}_3-\text{C}(=\text{O})-\text{Z}$

Ex . $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$, $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_3$, $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2\cdots\cdots\text{CH}_3$ etc. (All methyl ketones)

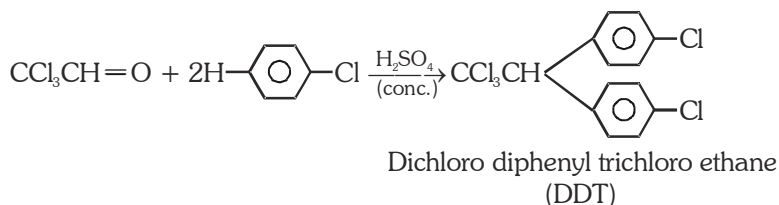
$\text{CH}_3-\text{C}(=\text{O})-\text{COOH}$, $\text{CH}_3-\text{C}(=\text{O})-\text{CHO}$ Also show haloform reaction.

(-) ve haloform reaction : Reaction in which haloforms are not formed with X_2 and alkali.

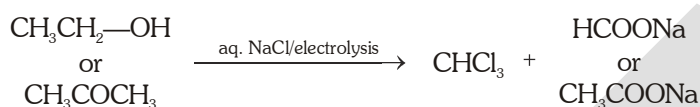
Ex. $\text{CH}_3-\text{C}(=\text{O})-\ddot{\text{Cl}}$, $\text{CH}_3-\text{C}(=\text{O})-\ddot{\text{O}}\text{H}$, $\text{CH}_3-\text{C}(=\text{O})-\ddot{\text{N}}\text{H}_2$, $\text{CH}_3-\text{C}(=\text{O})-\ddot{\text{O}}-\text{CH}_3$. do not show haloform reaction.

(iii) Preparation of pure CHCl_3 :

Chloral can also be used in preparation of D.D.T.



(iv) Industrial preparation :

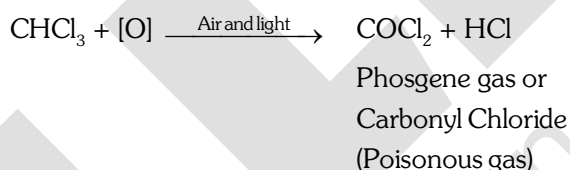


5.2 Physical Properties

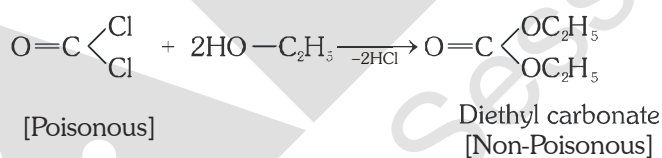
CHCl_3 is colourless and sweet smelling liquid. It's B.P. is 61°C and it is insoluble in H_2O and have density more than H_2O . Chloroform is used as Anaesthetic.

5.3 Chemical Properties

(i) Oxidation :



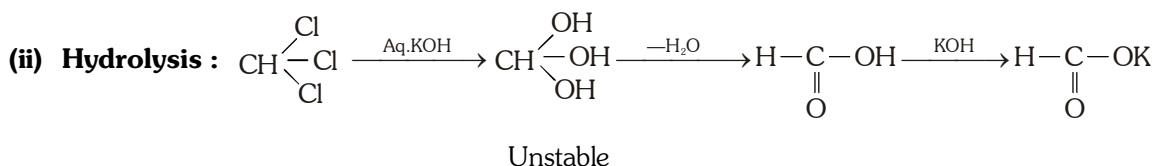
CHCl_3 is stored in dark coloured bottles which are filled upto the brim to prevent oxidation of CHCl_3 into COCl_2 and 1% ethanol is also added to chloroform



GOLDEN KEY POINTS

Test of CHCl_3

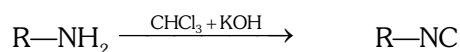
Reagent	Pure CHCl_3	Impure CHCl_3 ($\text{COCl}_2 + \text{HCl}$)
• Blue litmus	No Change	turns into red
• AgNO_3	No reaction	White ppt of AgCl
• Conc. H_2SO_4	No reaction	Yellow solution



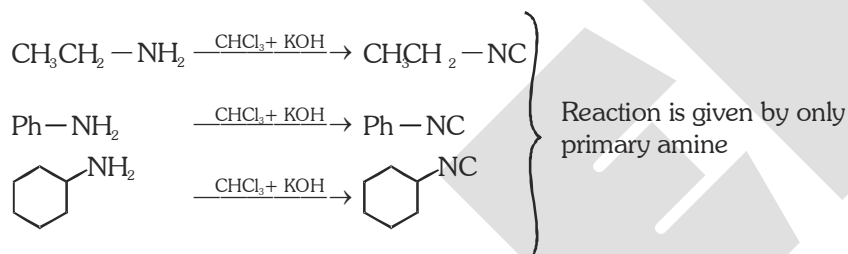
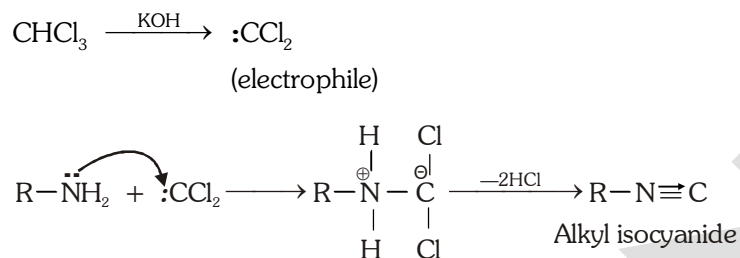
(iii) Carbyl amine reaction or isocyanide test : (Hoffman's carbylamine reaction)

Primary-Amines (Aliphatic or Aromatic) $\xrightarrow{\text{CHCl}_3 + \text{KOH}}$ Isocyanides

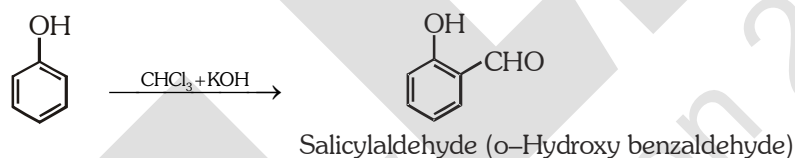
Isocyanides have unpleasant or offensive smell (Isocyanide test).



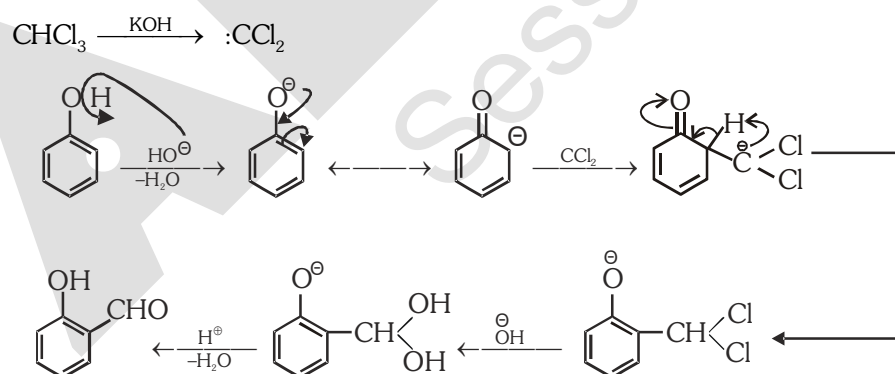
Mechanism :



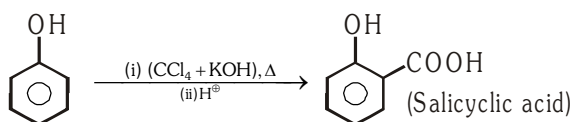
(iv) Reimer Tieman's reaction :

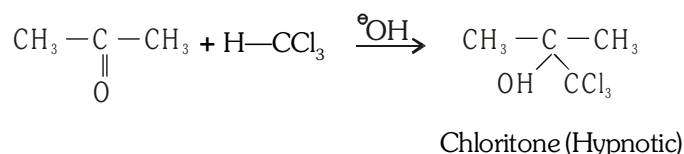
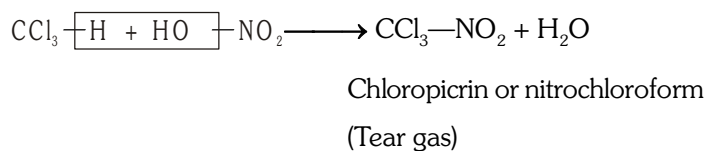
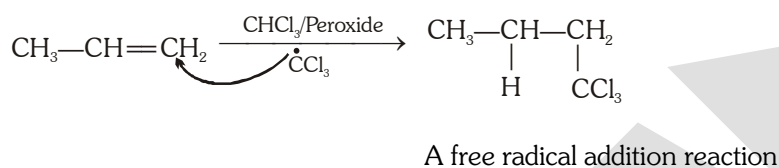


Mechanism : :CCl_2 is neutral attacking electrophile (formed by α, α - elimination reaction)

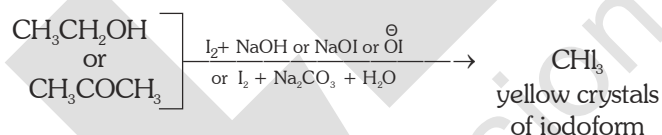


Note : If CCl_4 is used in place of chloroform, salicylic acid is formed as product.



(v) Reaction with CH_3COCH_3 :(vi) Reaction with HNO_3 :(vii) Reaction with $\text{CH}_3\text{CH}=\text{CH}_2$:(viii) Reaction with Ag : $\text{CHCl}_3 \xrightarrow{\text{Ag}} \text{CH}\equiv\text{CH}$ (ix) FCR : $3\text{Ph}-\text{H} + \text{CHCl}_3 \xrightarrow{\text{AlCl}_3} \text{Ph}_3\text{CH}$
Triphenyl methane

GOLDEN KEY POINTS

Iodoform Test

- CHI_3 give yellow ppt. of AgI with AgNO_3 but CHCl_3 does not give AgCl ppt.

Reason : CHI_3 is thermally less stable than CHCl_3 .

Iodoform test can be used to distinguish the following pairs of compounds.

- (i) $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3OH
- (ii) CH_3CHO and $\text{CH}_3\text{CH}_2\text{CHO}$
- (iii) 2-Pentanol and 3-Pentanol
- (iv) Acetophenone and benzophenone
- (v) 2-Propanol and 1-Propanol

Freons

The chlorofluoro derivatives of methane and ethane are called freons.

CF_2Cl_2 – (dichloro difluoro methane)

$\text{C}_2\text{F}_2\text{Cl}_4$ – (Tetrachloro difluoroethane)

Most useful is CF_2Cl_2 (Freon- 12)

Nomenclature of freons :

Freon - cba

$$c = n_{C-1}$$

$$b = n_{H+1}$$

$$a = n_F$$

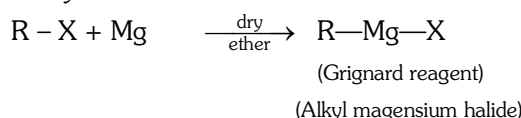


- Excess use of Freons is harmful for Ozone layer (depletion of Ozone layer).

6.0 Grignard Reagent

6.1 General Method of Preparation

Grignard reagents are prepared in the laboratory by the action of alkyl halides on magnesium metal in the presence of dry ether.



The ease of formation of Grignard reagent is in the order $\text{RI} > \text{RBr} > \text{RCl}$

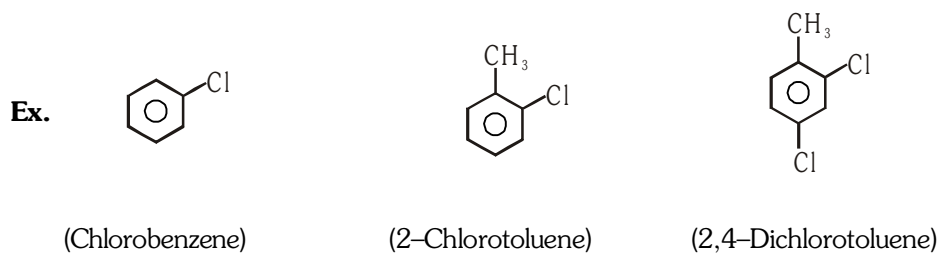
Ether is used to dissolve the Grignard reagent by coordination.

6.2 Chemical Reactions

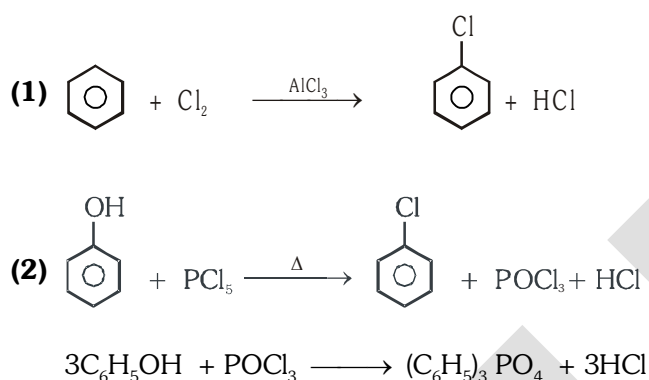
R-Mg-X	(i) O_2 (ii) H_3O^+	R-OH	1° Alcohol
	$\text{H}_2\text{C=O/H}_3\text{O}^+$	$\text{RCH}_2\text{-OH}$	1° Alcohol
	$\begin{array}{c} \text{O} \\ \diagup \quad \diagdown \\ \text{CH}_2\text{-CH}_2 \end{array} / \text{H}_3\text{O}^+$	$\text{R-CH}_2\text{-CH}_2\text{-OH}$	1° Alcohol
	$\text{RCHO/H}_3\text{O}^+$	R_2CHOH	2° Alcohol
	$\text{HCOOEt/H}_3\text{O}^+$	R_2CHOH	2° Alcohol
	$\text{RCOR/H}_2\text{O}$	$\text{R}_3\text{C-OH}$	3° Alcohol
	$\text{RCOOEt/H}_2\text{O (2 : 1)}$	$\text{R}_3\text{C-OH}$	3° Alcohol
	HCOOEt (1 : 1)	RCHO	aldehyde
	$\text{RCN/H}_2\text{O}$	RCOR	Ketone
	RCOOEt (1 : 1)	RCOR	Ketone
	$\text{CO}_2/\text{H}_2\text{O}$	RCOOH	Acids
	$\text{HOH or ROH or NH}_3 \text{ or Ph-OH}$		
	or R-NH_2 or RNH-R		
	or $\text{CH}\equiv\text{CH}$ or Ph-NH_2	R-H	Alkane
	R-X	R-R	Alkane
	R_3N	No reaction	
	$\text{ClCH}_2\text{-CH=CH}_2$	$\text{R-CH}_2\text{-CH=CH}_2$	Alkene
	Cl-NH_2	R-NH_2	1° Amine
	Cl-CN	R-CN	Cyanides
	ClCOOEt	RCOOEt	Ester

7.0 Haloarene

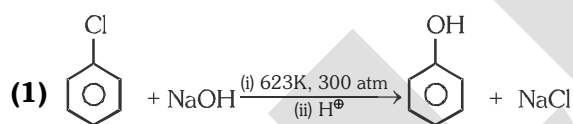
If halogen atom is directly attached to the benzene ring, then compound is called as Haloarene.



7.1 General Methods of Preparation

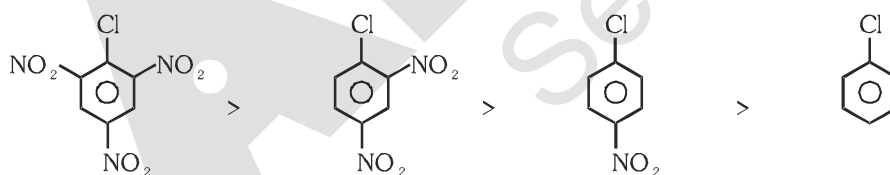


7.2 Chemical Properties

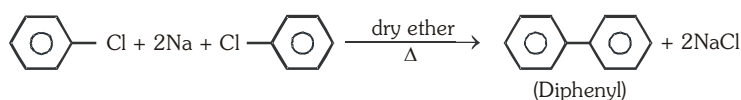


Presence of electron withdrawing group on ring makes the nucleophilic substitution easier.

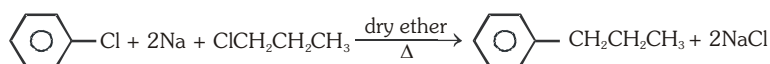
Reactivity Order : (Towards nucleophilic substitution)



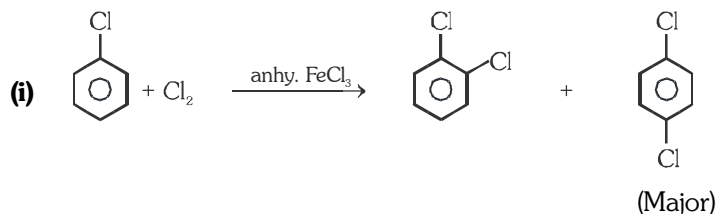
(2) Fittig reaction :



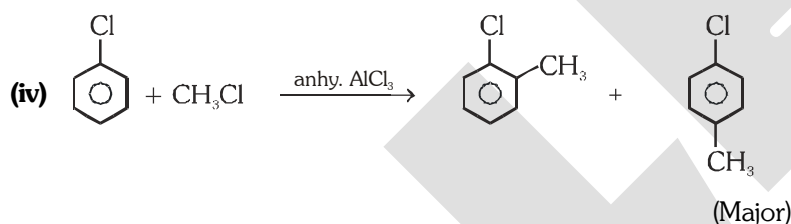
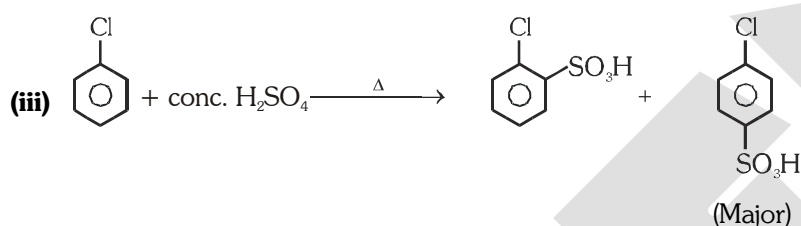
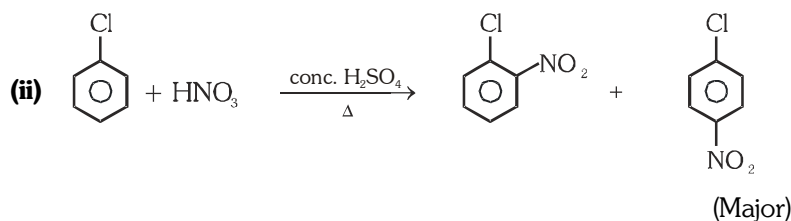
(3) Wurtz fittig reaction :



(4) Electrophilic Substitution Reaction :



Note : Cl is o- and p- directing group.



BEGINNER'S BOX-2

- Which can give haloform with X_2/OH^- ?
 (1) CH_3OH (2) $CH_3-CH_2-CH=O$
 (3) CH_3CHO (4) $CH_3CH_2CH_2OH$
- Which of the following compounds cannot be identified by carbylamine test ?
 (1) $C_6H_5-NH-C_6H_5$ (2) $CH_3CH_2NH_2$
 (3) $CHCl_3$ (4) $C_6H_5-NH_2$
- For preparation of Grignard reagent from haloalkanes which metal is used :-
 (1) Na (2) Mg
 (3) Ca (4) Ag

ANSWER KEY

BEGINNER'S BOX-1		Que.	1	2							
		Ans.	3	1							

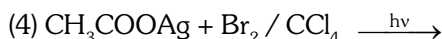
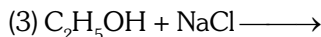
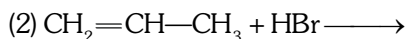
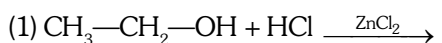
BEGINNER'S BOX-2		Que.	1	2	3						
		Ans.	3	1	2						

EXERCISE-I (Conceptual Questions)

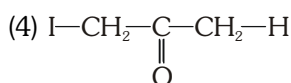
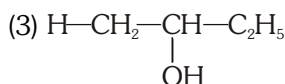
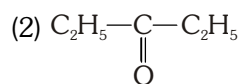
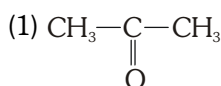
Build Up Your Understanding

GENERAL METHOD OF PREPARATION

1. Alkyl halides can be obtained by all methods excepts



2. Which of the following will not give iodoform test



3. Which of the following product is obtained when bleaching powder is distilled with acetone

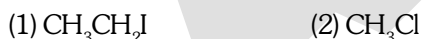


4. Which will give yellow ppt. with iodine and alkali

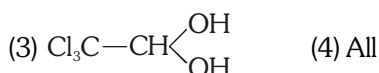
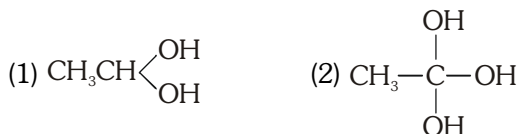


PHYSICAL PROPERTIES

5. Which of the following has the highest boiling point

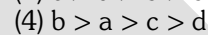
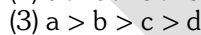
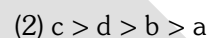
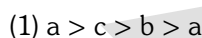
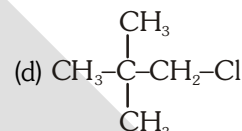
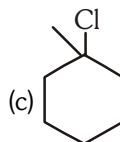
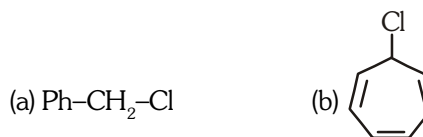


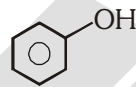
6. A compound containing two —OH groups attached with one carbon atoms is unstable but which one of the following is stable



CHEMICAL PROPERTIES

7. Arrange the following compounds in decreasing order of reactivity in $\text{S}_\text{N}1$ reaction :-



8.  $\xrightarrow{\text{CHCl}_3 + \text{KOH}}$ Product.

about above reaction the incorrect statement is

(1) The name of reaction is Reimer-Tsima's reaction

(2) The intermediate in the reaction is dichloro carbene

(3) The final product is salicylaldehyde

(4) The final product is benzyl chloride

9. The purity of CHCl_3 can be checked by

(1) treating CHCl_3 by NaOH

(2) treating CHCl_3 by HCl

(3) treating CHCl_3 with aq. AgNO_3

(4) treating CHCl_3 by $\text{C}_2\text{H}_5\text{—OH}$

10. Pure CHCl_3 and pure CHI_3 can be distinguished by

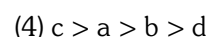
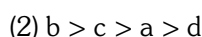
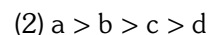
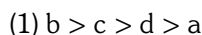
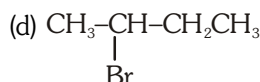
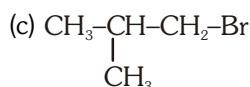
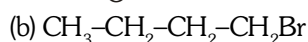
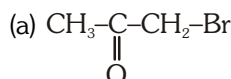
(1) treating with litmus paper

(2) treating with aq. KOH

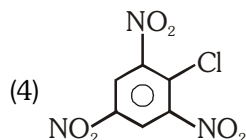
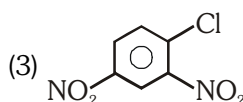
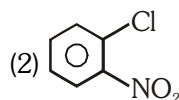
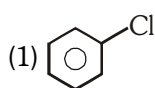
(3) treating with HCl

(4) treating with aq. AgNO_3

11. Arrange the following compound in decreasing order of reactivity in $\text{S}_\text{N}2$ reaction.



12. Which of the following undergoes hydrolysis most easily



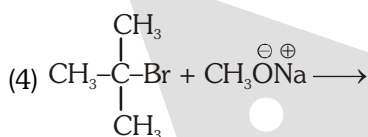
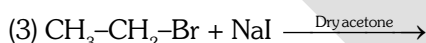
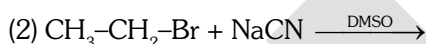
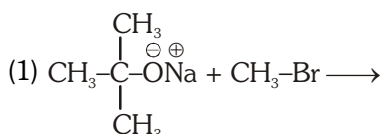
13. Which of the following is used as insecticide

- (1) D.D.T. (2) Chloritone
(3) CHCl_3 (4) All of them

14. Which of the following when heated with KOH and primary amine gives carbylamine test

- (1) CHCl_3 (2) CH_2Cl_2
(3) CH_3OH (4) CCl_4

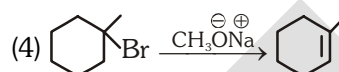
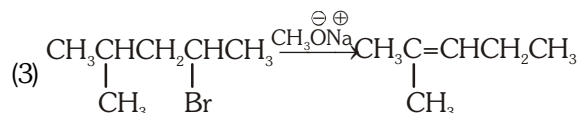
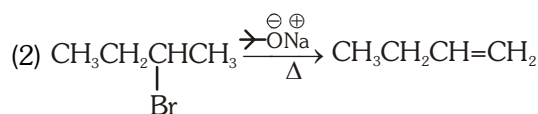
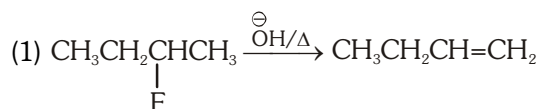
15. Which reaction gives elimination as a major product



16. Iodoform gives a precipitate with AgNO_3 on heating but chloroform does not because

- (1) Iodoform is ionic
(2) Chloroform is covalent
(3) C-I bond in iodoform is weak and C-Cl bond in chloroform is strong
(4) None of the above

17. Which reaction product is wrong (major) product.



18. When alkyl magnesium halide reacts with R-NH_2 , the product is

- (1) R-R (2) R-H
(3) R_2NH (4) R-X

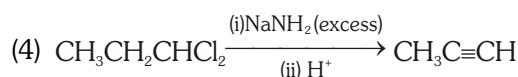
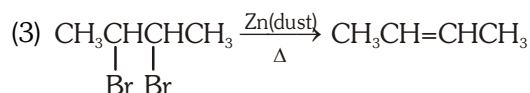
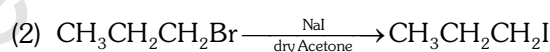
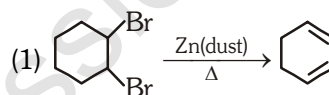
19. Chloroform on reaction with acetone gives:-

- (1) Acetylene (2) Chlorotone
(3) Nitrochloroform (4) Chloroacetone

20. Chloroform reacts with aniline and aqueous KOH gives :-

- (1) $\text{Ph-N} \equiv \text{C}$ (Phenyl isocyanide)
(2) Benzene
(3) Phenyl cyanide
(4) None of these

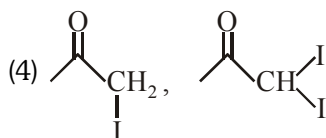
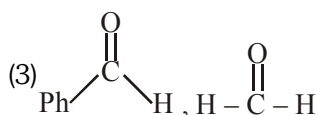
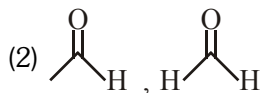
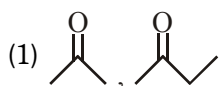
21. Which reaction product is wrong (major) product



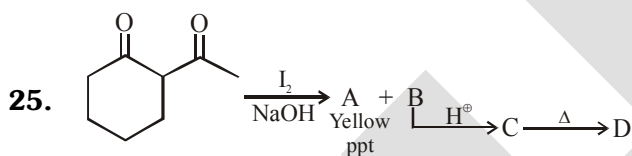
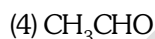
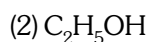
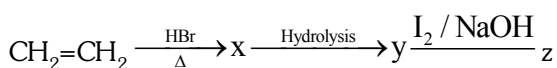
22. Which of the following undergoes nucleophilic substitution by $\text{S}_{\text{N}}1$ mechanism at fastest rate :



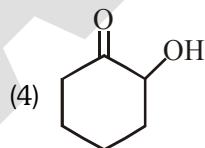
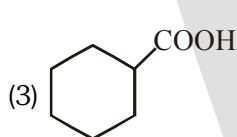
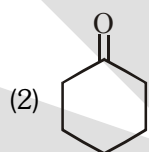
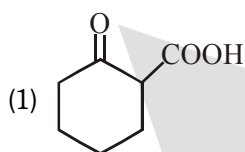
23. Which of the following pair is differentiated by iodoform test?



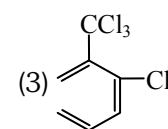
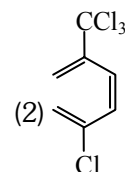
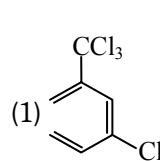
24. Identify z in the following series



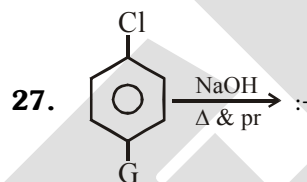
Identify D :-



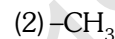
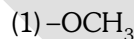
In the above reaction X is



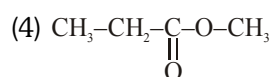
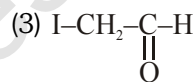
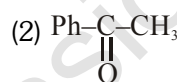
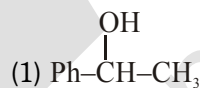
(4) None of these



Rate of reaction is maximum if G is :-



28. Which does not gives iodoform test :-



EXERCISE-I (Conceptual Questions)

ANSWER KEY

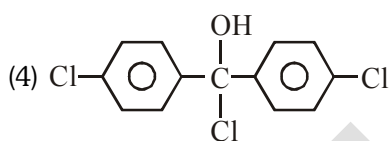
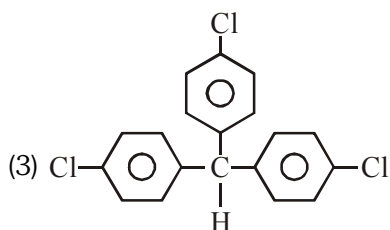
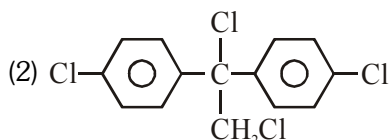
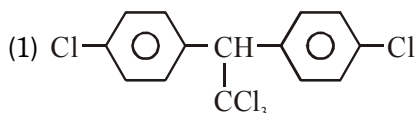
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	1	1	3	4	4	3	4	2	4	1	1	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	3	3	2	2	1	1	3	2	3	2	1	3	4		

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

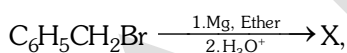
AIPMT 2009

1. Trichloroacetaldehyde, CCl_3CHO reacts with chlorobenzene in presence of sulphuric acid and produces :-



AIPMT 2010

2. In the following reaction



The product 'X' is :-

- (1) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
 (2) $\text{C}_6\text{H}_5\text{CH}_3$
 (3) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{C}_6\text{H}_5$
 (4) $\text{C}_6\text{H}_5\text{CH}_2\text{OCH}_2\text{C}_6\text{H}_5$

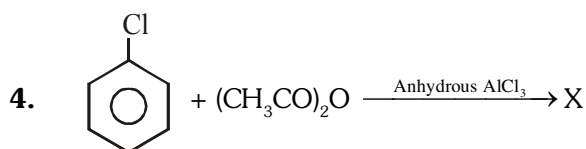
3. Following compounds are given :

- (a) $\text{CH}_3\text{CH}_2\text{OH}$ (b) CH_3COCH_3
 (c) $\text{CH}_3-\underset{\text{CH}_3}{\text{CHOH}}$ (d) CH_3OH

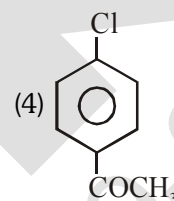
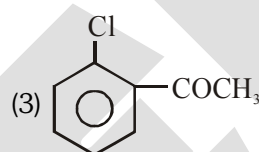
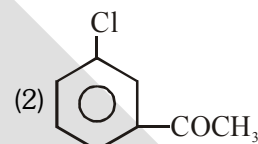
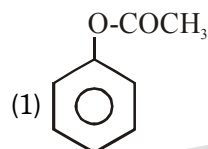
Which of the above compound(s), on being warmed with iodine solution and NaOH , will give iodoform?

- (1) Only (a) (2) (a), (b) and (c)
 (3) (a) and (b) (4) (a), (c) and (d)

AIIMS 2010



What is major product (X) :-



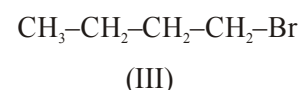
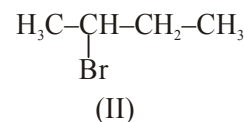
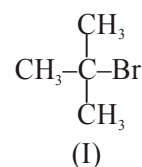
AIPMT Mains 2012

5. Which of the following compounds will give a yellow precipitate with iodine and alkali ?

- (1) Acetamide
 (2) Propan-2-ol
 (3) Acetophenone
 (4) Methyl acetate

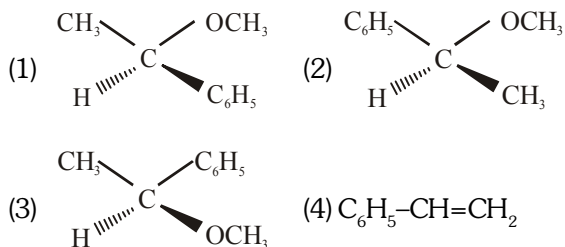
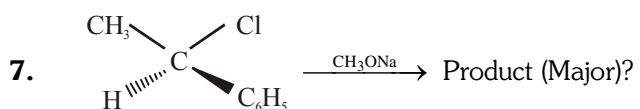
AIIMS 2016

6. Arrange the following compounds in order of their boiling point

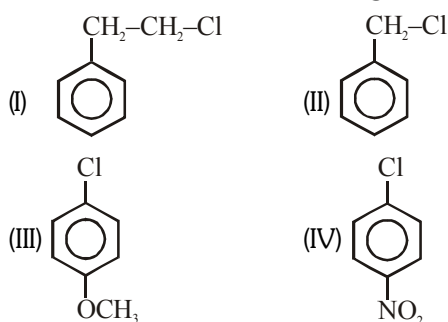


- (1) $\text{I} > \text{II} > \text{III}$ (2) $\text{III} > \text{II} > \text{I}$
 (3) $\text{II} > \text{I} > \text{III}$ (4) $\text{III} > \text{I} > \text{II}$

AIIMS 2017

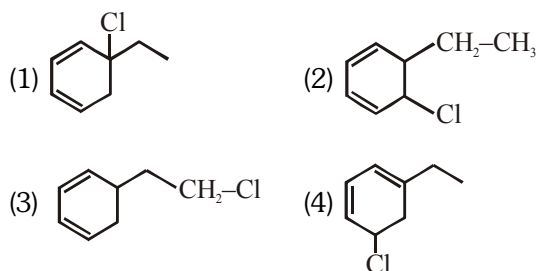
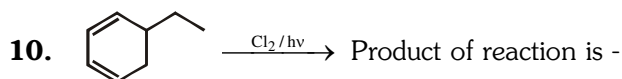
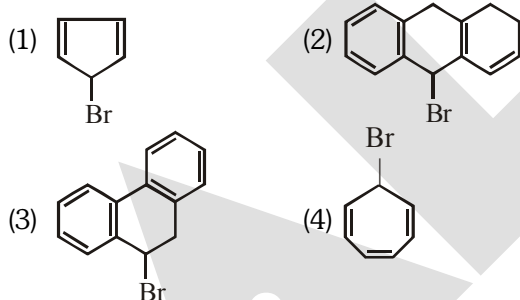


8. Correct order of reactivity towards nucleophilic substitution reaction of following is :-

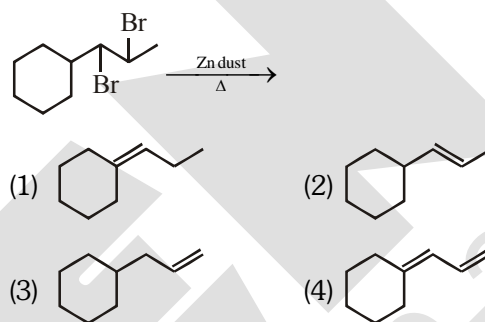


- (1) II > I > IV > III (2) III > IV > I > II
 (3) II > IV > I > II (4) IV > II > III > I

9. Which of the following react fastest with aqueous acetone.



11. Product of following reaction is :-



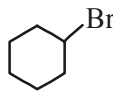
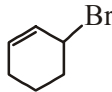
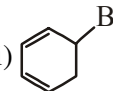
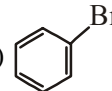
EXERCISE-II (Previous Year Questions)

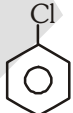
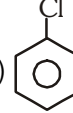
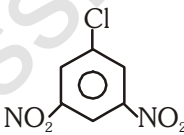
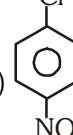
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	1	2	2	4	2,3	2	4	1	4	1	2	

EXERCISE-III (Analytical Questions)

Check Your Understanding

- Which of the following has highest dipole moment:
 (1) CH_3Cl (2) CH_3F
 (3) CH_3Br (4) CH_3I
- The final product in the reaction is
 $\text{Ph}-\text{OH} + \text{CCl}_4 \xrightarrow[(2) \text{H}^+]{(1) \text{KOH}, \Delta} \text{product}$
 (1) Salicyl aldehyde (2) Salicylic acid
 (3) Methyl salicylate (4) Benzyl chloride
- Arrange the following in order of ease of dehydrohalogenation:
 (i)  (ii) 
 (iii)  (iv) 
 (1) (iii) > (iv) > (ii) > (i)
 (2) (iii) > (ii) > (i) > (iv)
 (3) (ii) > (iii) > (i) > (iv)
 (4) (i) > (ii) > (iii) > (iv)
- The product in the following reaction is
 $\text{Ph}-\text{Cl} + \text{Fe} / \text{Br}_2 \longrightarrow \text{Product}$
 (1) o-bromo-chloro benzene
 (2) p-bromo-chloro benzene
 (3) both the above
 (4) 2,4,6-tribromo chloro benzene
- Isobutyl magnesium bromide with dry ether and absolute alcohol gives
 (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{MgBr}$
 (2) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}_3$ and $\text{Mg}(\text{OH})\text{Br}$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$, $\text{CH}_2=\text{CH}_2$ and $\text{Mg}(\text{OH})\text{Br}$
 (4) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{OMgBr}$

- Identify 'Z' in the following reaction series,
 $\text{CH}_3\cdot\text{CH}_2\text{CH}_2\text{Br} \xrightarrow{\text{aq. NaOH}} (\text{X}) \xrightarrow[\text{Heat}]{\text{Al}_2\text{O}_3} (\text{Y})$
 $\xrightarrow{\text{Cl}_2/\text{H}_2\text{O}} (\text{Z})$
 (1) Mixture of $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\underset{\text{Cl}}{\text{CH}_2}$ and $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\underset{\text{OH}}{\text{CH}_2}$
 (2) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\underset{\text{Cl}}{\text{CH}_2}$
 (3) $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\underset{\text{OH}}{\text{CH}_2}$
 (4) $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\underset{\text{Cl}}{\text{CH}_2}$
- Arrange the following compound in increasing order of reactivity towards aromatic nucleophilic substitution reaction.
 (a)  (b) 
 (c)  (d) 
 (1) $c > d > b > a$ (2) $c > b > a > d$
 (3) $d > c > b > a$ (4) $a > d > b > c$
- Chloroform when treated with benzene in presence of anhydrous AlCl_3 , the product formed is
 (1) Chlorobenzene
 (2) Toulene
 (3) Mixture of ortho and para chlorotoluene
 (4) Triphenyl methane

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	1	2	2	3	4	2	3	4

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** Alkyl halides are not soluble in water.
Reason :- Alkyl halide does not form H-bonds with water molecule although alkyl halide is polar in nature.
 (1) A (2) B (3) C (4) D
- Assertion :-** CHCl_3 is more acidic than CHF_3 .
Reason :- Electronegativity of Fluorine is more than chlorine.
 (1) A (2) B (3) C (4) D
- Assertion :-** Isobutanol does **not** give iodoform test.
Reason :- It does not have α -hydrogen.
 (1) A (2) B (3) C (4) D
- Assertion :-** Styrene on reaction with HBr gives 1-bromo-1-phenylethane.
Reason :- Benzyl radical is more stable than alkyl radical.
 (1) A (2) B (3) C (4) D
- Assertion :** Rate of alkaline hydrolysis of methyl chloride to methanol is higher in DMF than in water.
Reason : Hydrolysis of methyl chloride follows second order kinetics.
 (1) A (2) B (3) C (4) D
- Assertion :-** 2-Bromo butane on reaction with alcoholic KOH mainly gives 2-butene.
Reason :- The hydrogen on C_3 carbon is more acidic than hydrogen on C_1 carbon.
 (1) A (2) B (3) C (4) D
- Assertion :-** Chlorobenzene is less reactive than benzene for electrophilic substitution reaction.
Reason :- Resonance effect in chlorobenzene destabilized carbocation intermediate.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7
Ans.	1	2	3	2	1	3	3

OXYGEN CONTAINING COMPOUND

E

S. No.	CONTENTS	Page
1.	Alcohol	21
2.	Phenol	33
3.	Ether	41
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5.	Carboxylic acid	57
6.	Acid derivatives	63
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8.	Exercise-II (Previous Years Questions)	77
9.	Exercise-III (Analytical Questions)	87
10.	Exercise-IV (Assertion & Reason)	91

NEET SYLLABUS

Alcohols : Nomenclature, methods of preparation, physical and chemical properties (of primary alcohols only); identification of primary, secondary and tertiary alcohols, mechanism of dehydration, uses with special reference to methanol and ethanol. **Phenols** : Nomenclature, methods of preparation, physical and chemical properties, acidic nature of phenol, electrophilic substitution reactions, uses of phenols. **Ethers** : Nomenclature, methods of preparation, physical and chemical properties uses. Aldehydes and Ketones : Nomenclature, nature of carbonyl group, methods of preparation, physical and chemical properties and reactivity of alpha hydrogen in aldehydes, Carboxylic acid : Nomenclature, acidic nature, methods of preparation, physical and chemical properties;

OBJECTIVES

After studying this unit, we will be able to :

- Name alcohols, phenols and ethers according to the IUPAC system of nomenclature;
- Discuss the reactions involved in the preparation of alcohols from
(i) alkenes (ii) aldehydes, ketones and carboxylic acids;
- Discuss the reactions involved in the preparation of phenols from
(i) haloarenes (ii) benzene sulphonic acids
(iii) diazonium salts and (iv) cumene;
- Discuss the reactions for preparation of ethers from
(i) alcohols and
(ii) alkyl halides and sodium alkoxides/aryloxides;
- Correlate physical properties of alcohols, phenols and ethers with their structures;
- Discuss chemical reactions of the three classes of compounds on the basis of their functional groups.
- Write the common and IUPAC names of aldehydes, ketones and carboxylic acids;
- Write the structures of the compounds containing functional groups namely carbonyl and carboxyl groups;
- Describe the important methods of preparation and reactions of these classes of compounds;
- Correlate physical properties and chemical reactions of aldehydes, ketones and carboxylic acids, with their structures;
- Explain the mechanism of a few selected reactions of aldehydes and ketones;
- Understand various factors affecting the acidity of carboxylic acids and their reactions;
- Describe the uses of aldehydes, ketones and carboxylic acids.

"If you fail, never give up because FAIL means "first attempt in Learning"

– End is not the end, if fact END means "Efforts never dies"

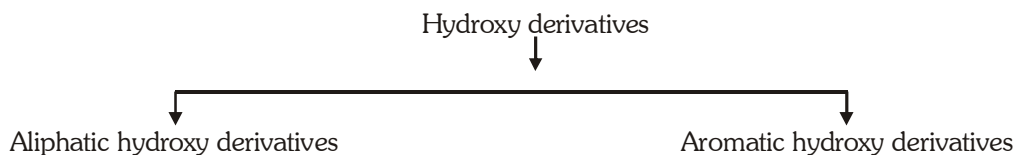
– If you get No as an answer, remember NO means "Next Opportunity"

So Let's be positive. "Dr. A.P.J. Abdul Kalam of feel LIFE"

A.P.J. Abdul Kalam

OXYGEN CONTAINING COMPOUND

1.0 HYDROXY DERIVATIVES



1.1 Aliphatic Hydroxy Derivatives

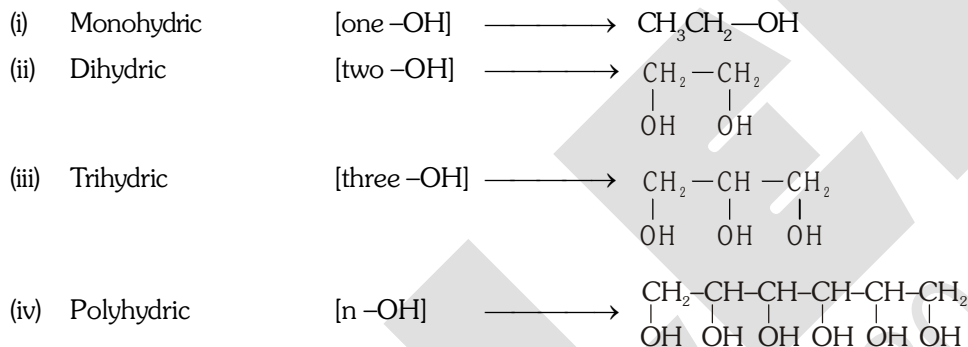
Hydroxy derivatives in which —OH is directly attached to sp^3 'C' (Alcoholic compounds).

1.2 Aromatic Hydroxy Derivatives

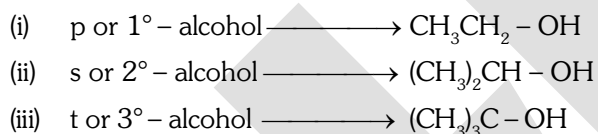
Hydroxy derivatives in which —OH is directly attached to sp^2 'C' or benzene ring (Phenolic compounds).

Aliphatic hydroxy derivatives :

(a) Classification according to number of —OH groups :



(b) Classification according to nature of carbon :



2.0 MONOHYDRIC ALCOHOL

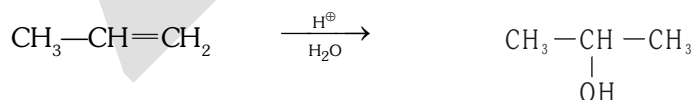
2.1 General Methods of Preparation

(i) From alkanes (By oxidation):

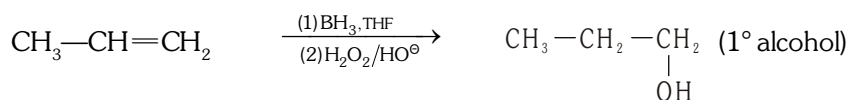


(ii) From alkenes :

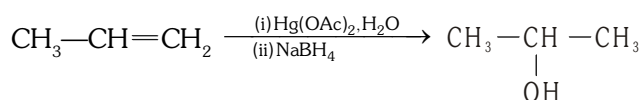
(a) By hydration :

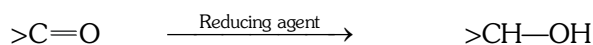


(b) By hydroboration oxidation :

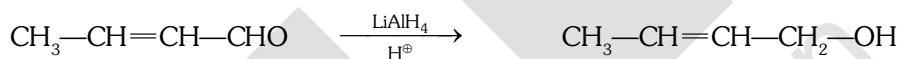
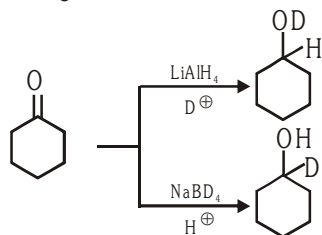
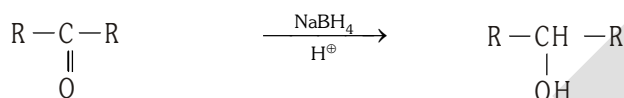
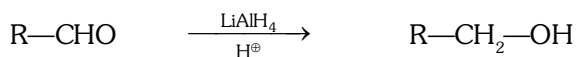
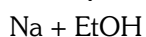
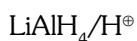


(c) By oxymercuration demercuration :

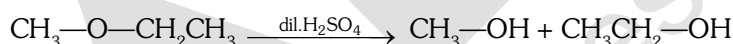
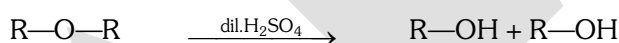
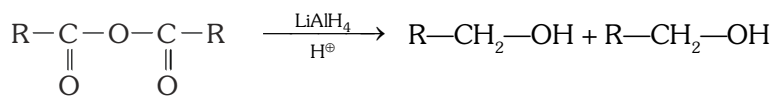
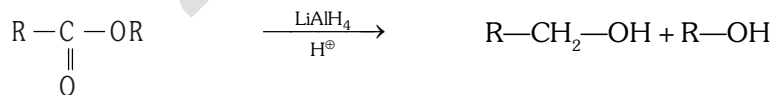
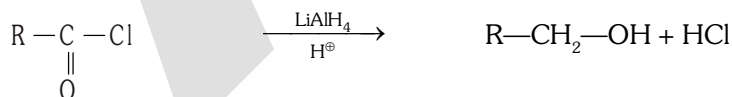
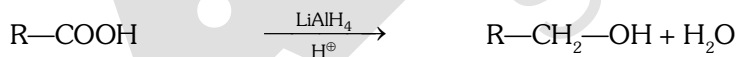
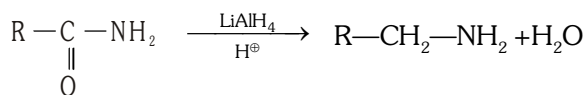


(iii) From alkyl halides (By hydrolysis) :**(iv) From carbonyl compounds (By reduction) :****GOLDEN KEY POINTS**

Reducing agents may be,

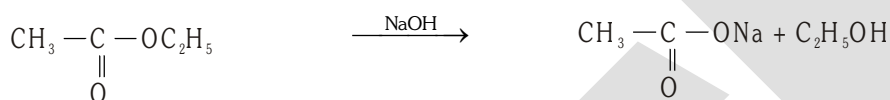
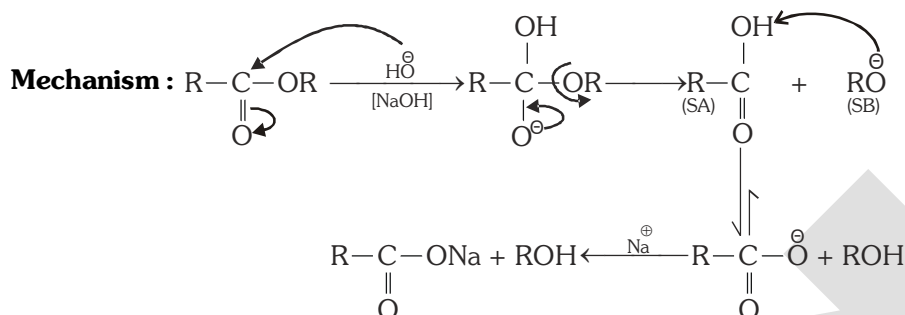
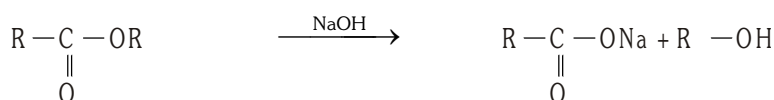


Crotonaldehyde

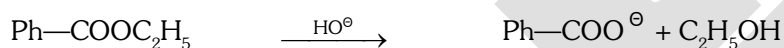
(v) From ethers :**(vi) From acid and derivatives (By reduction) :****Note :** Amide on reduction gives amine not alcohol.

(vii) From esters (By hydrolysis) :

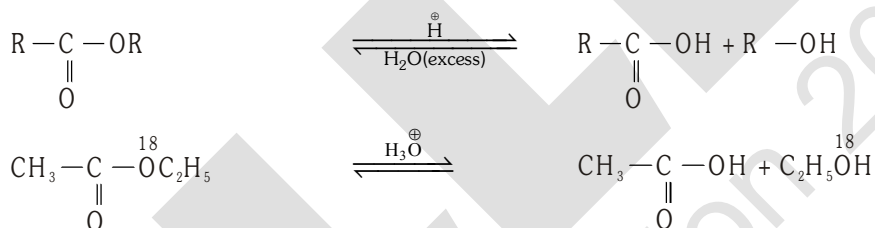
(a) By alkaline hydrolysis :



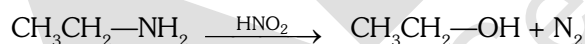
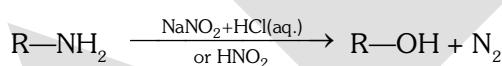
Hydrolysis is Nucleophilic substitution reaction (NSR) and Order of reaction is 2. Alkaline hydrolysis is also called as saponification



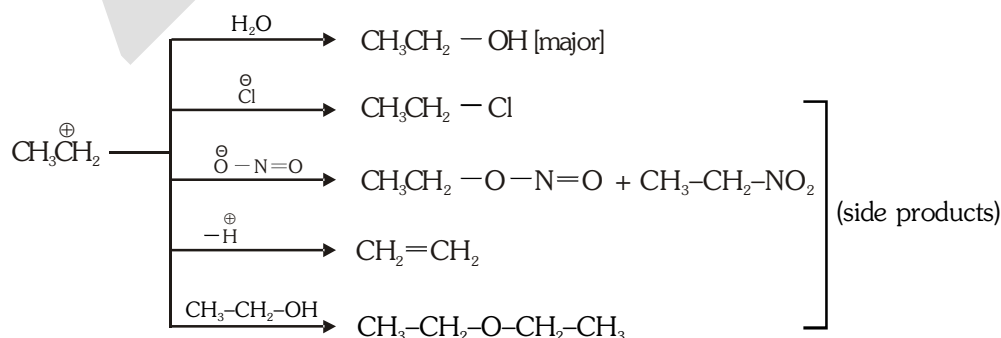
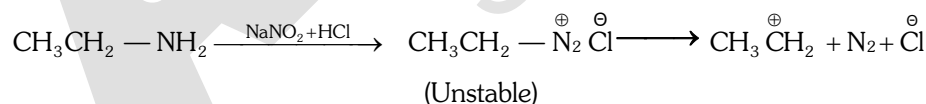
(b) By acidic hydrolysis :



(viii) From p-amines :



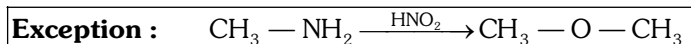
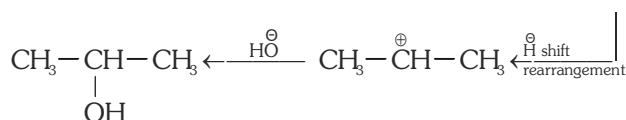
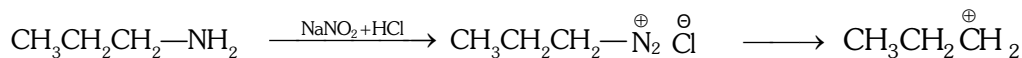
Mechanism :



Intermediate is carbocation so rearrangement may be possible.

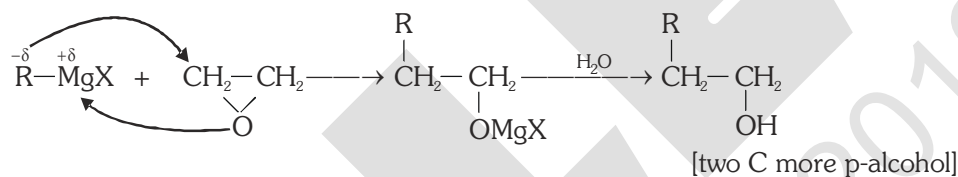
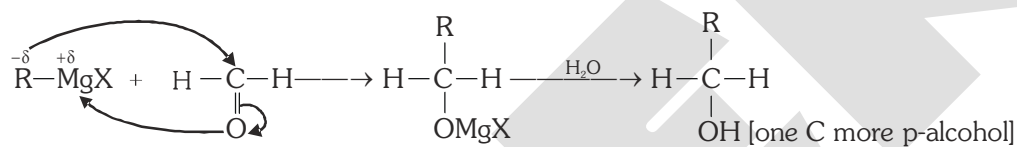
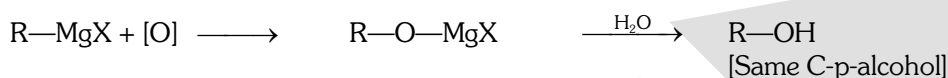
Illustrations

Illustration 1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{—NH}_2 \xrightarrow{\text{NaNO}_2 + \text{HCl}} ?$
Solution. **Mechanism :**

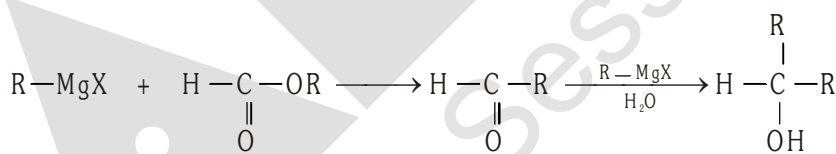
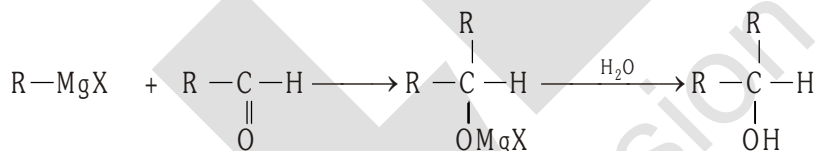


(ix) From Grignard reagent :

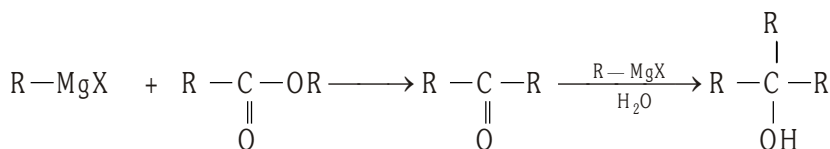
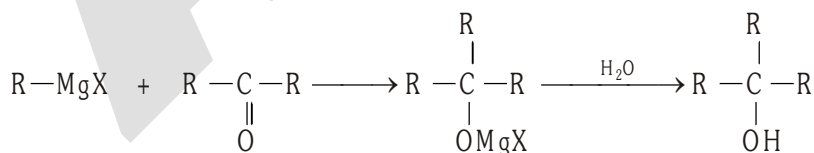
(a) p-alcohol :



(b) s-alcohol :



(c) t-alcohol :



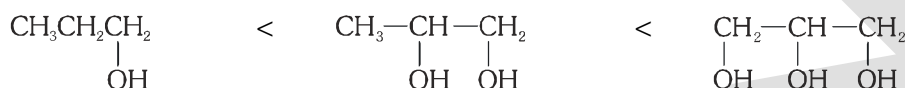
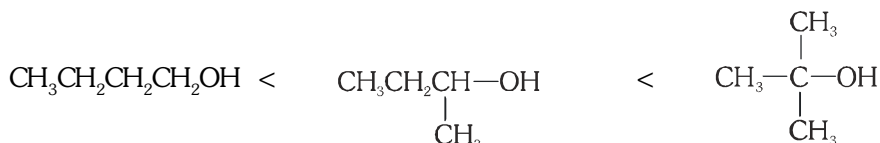
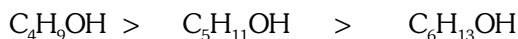
Physical properties :

- (i) C_1 to C_{11} are colourless liquids and higher alcohols are solids.
- (ii) Density of monohydric alcohol is less than H_2O .
- (iii) Density $\propto$ mol. wt. (for monohydric alcohol).

(iv) **Solubility :** C_1 to C_3 and t-butyl alcohol is completely soluble in H_2O due to H-bonding.

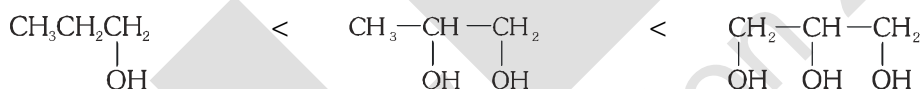
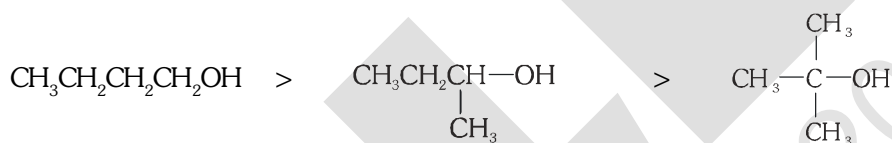
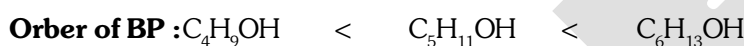
$$\text{solubility} \propto \frac{\text{No. of side chains}}{\text{molecular weight}}$$

Order of solubility :



[Number of OH increases, H-bonding increases]

(v) **Boiling points :** $BP \propto \frac{\text{molecular weight}}{\text{No. of side chains}}$



[Number of OH increases, H-bonding increases]

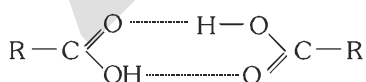
Illustrations

Illustration 2. Boiling point of alcohol is more than corresponding ether. Why ?

Solution. H-bonding in alcohol.

Illustration 3. Boiling point of alcohol is less than corresponding carboxylic acid. Why ?

Solution. Dimer formation in carboxylic acid.

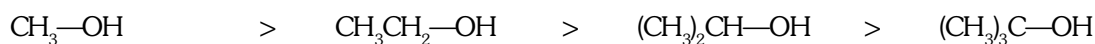


2.3 Chemical Properties

Monohydric alcohol shows following reactions

- (A) Reaction involving cleavage of $O \vdash H$
- (B) Reaction involving cleavage of $C \vdash OH$
- (C) Reaction involving complete molecule of alcohol

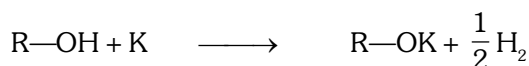
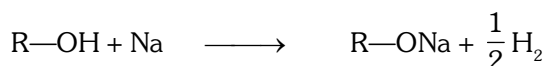
(A) **Reaction involving cleavage of O—H:** Reactivity order (Acidic nature) is



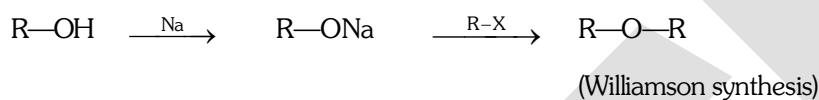
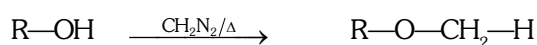
(i) **Acidic nature :**



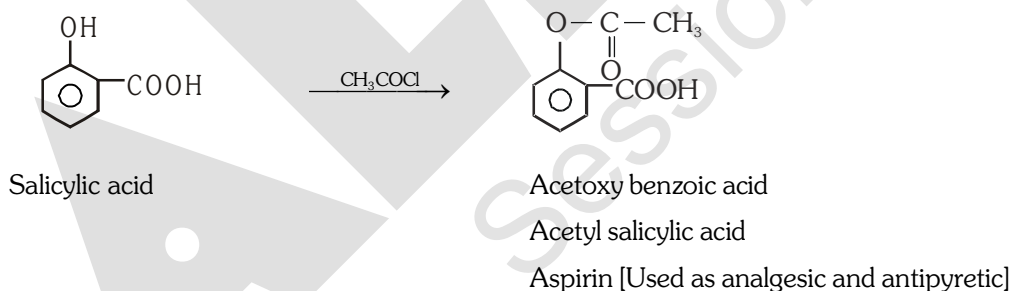
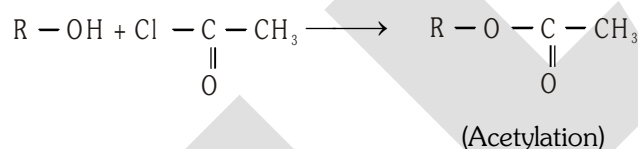
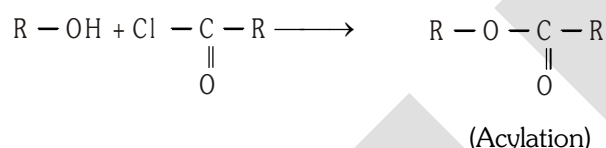
Alcohols are less acidic than H_2O and neutral for litmus paper and give H_2 with active metals (Na, K)



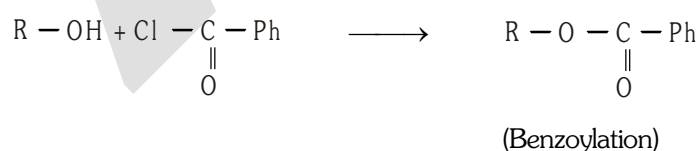
(ii) **Alkylation:**



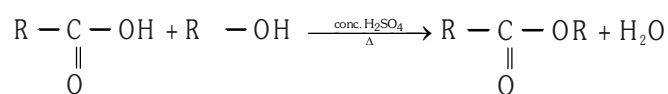
(iii) **Acylation :**



(iv) **Benzoylation : (Schotten Baumann's Reaction) :**

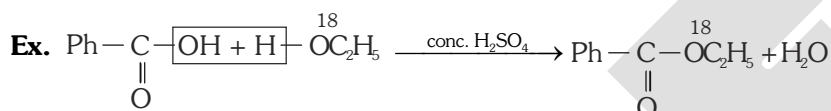
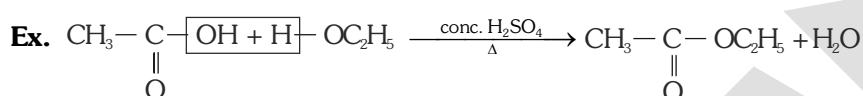
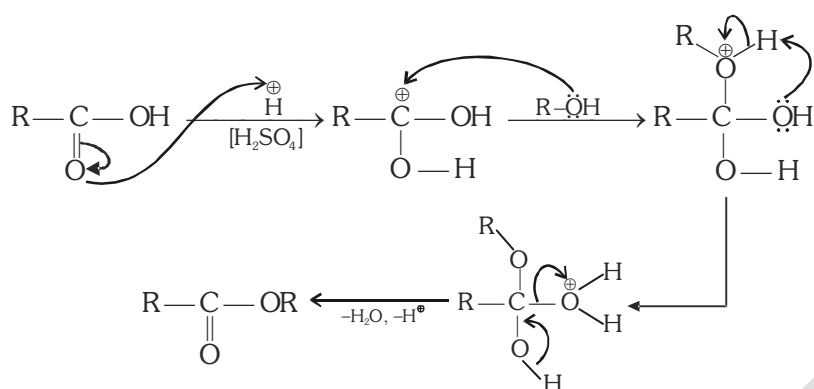


(v) **Esterification :**

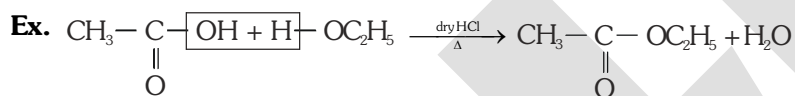


Conc. H_2SO_4 is used as catalyst and dehydrating agent.

Mechanism :

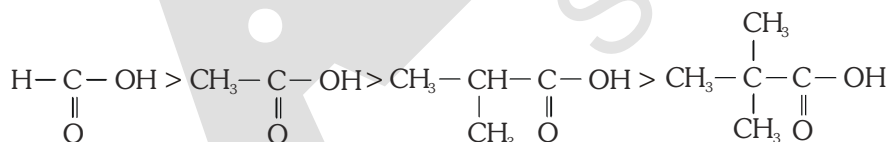


Dry HCl can also be used as dehydrating agent.

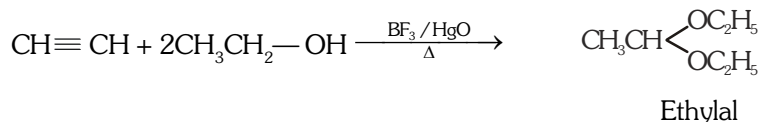
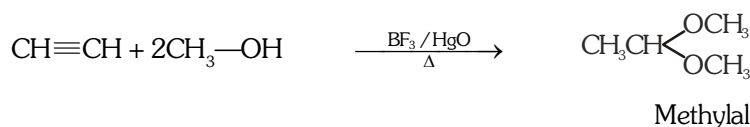


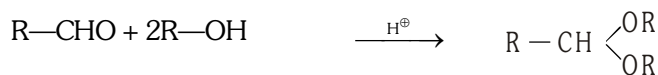
GOLDEN KEY POINTS

- Reactivity for esterification $\propto \frac{1}{\text{Steric hindrance}}$.
- Reactivity of $R-OH$ [If acid is same] : $CH_3-OH > 1^\circ > 2^\circ > 3^\circ$ alcohol
- Reactivity of $RCOOH$ [If alcohol is same] :

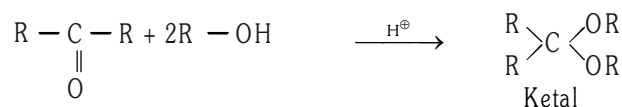


(vi) Reaction with $CH \equiv CH$:

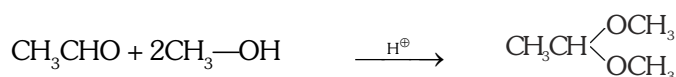


(vii) Reaction with carbonyl compounds :

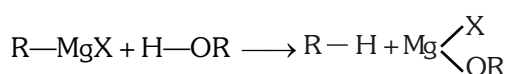
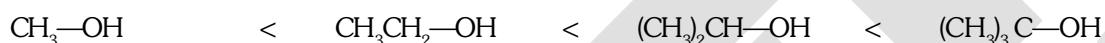
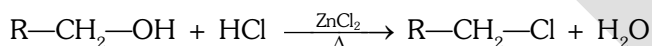
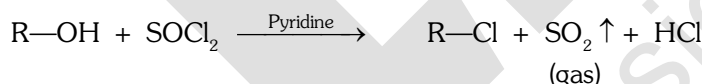
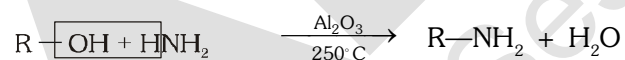
Acetal



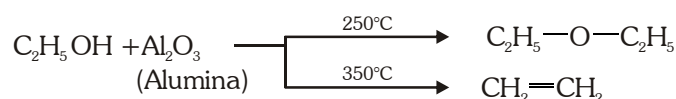
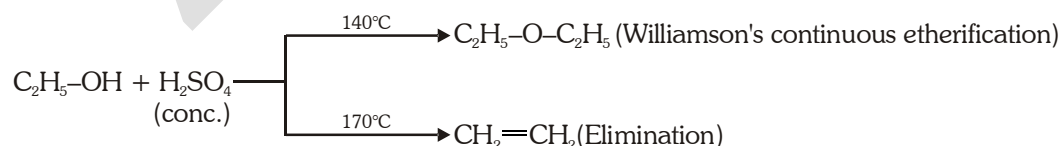
Ketal



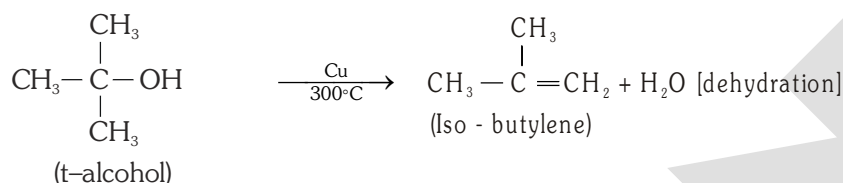
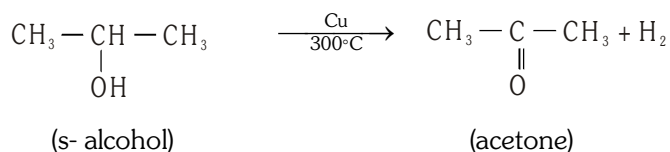
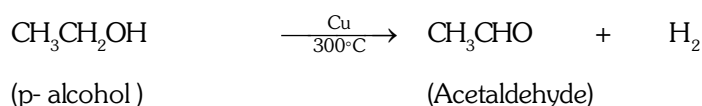
Methylal

(viii) Reaction with Grignard reagent :**(B) Reaction involving cleavage of $C-\dot{O}H$:** Reactivity order or basic nature is**(i) Reaction with halogen acid :**Reactivity order of the acids is $HI > HBr > HCl$ **(ii) Reaction with phosphorous halides :****(iii) Reaction with thionyl chloride ($SOCl_2$) :****(iv) Reaction with NH_3 :** Alumina (Al_2O_3) is used as dehydrating agent.**(C) Reaction involving complete molecule of alcohol :****(i) Dehydration :** Removal of H_2O

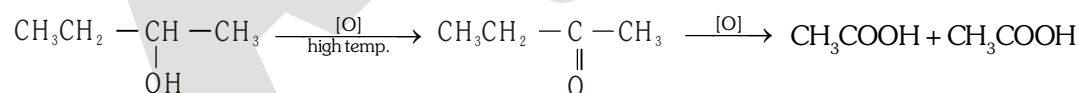
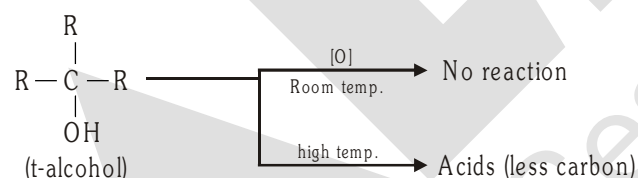
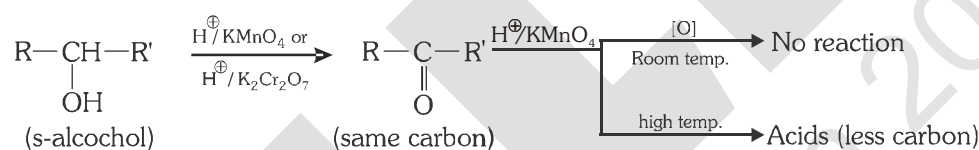
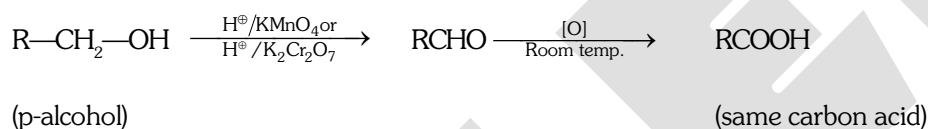
- (a) Intermolecularly removal of H_2O [formation of ether]
 (b) Intramolecularly removal of H_2O [formation of alkene]

Ease of dehydration follows the order : **$3^\circ ROH > 2^\circ ROH > 1^\circ ROH$**

(ii) Catalytic Dehydrogenation :



(iii) Oxidation :



Carbonyl group goes with smaller alkyl group [Popoff's rule suggested for oxidation of unsymmetrical ketones]

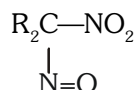
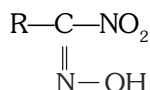
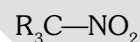
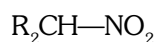
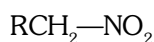
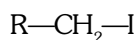
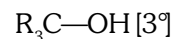
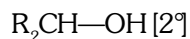
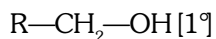
(iv) Distinction between 1°, 2° and 3° alcohols :

(a) **Lucas test :** A mixture of HCl(conc.) and anhydrous ZnCl₂ is called Lucas reagent.

p-alcohol $\xrightarrow{\text{ZnCl}_2 + \text{HCl}}$ No turbidity at room temp. [On heating within 30 minutes.]

s-alcohol $\xrightarrow{\text{ZnCl}_2 + \text{HCl}}$ Turbidity appears within 5 minutes.

t-alcohol $\xrightarrow{\text{ZnCl}_2 + \text{HCl}}$ Turbidity appears within 2-3 sec.

(b) Victor - Meyer test :p-alcohol $\longrightarrow$ Red colours-alcohol $\longrightarrow$ Blue colourt-alcohol $\longrightarrow$ No colour

No reaction

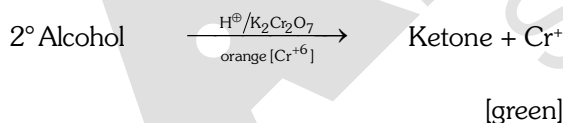
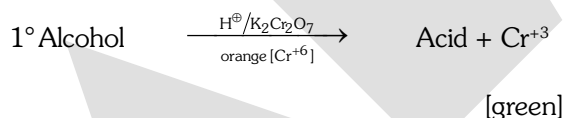
Nitrolic acid (blue)

Pseudonitrol (blue)

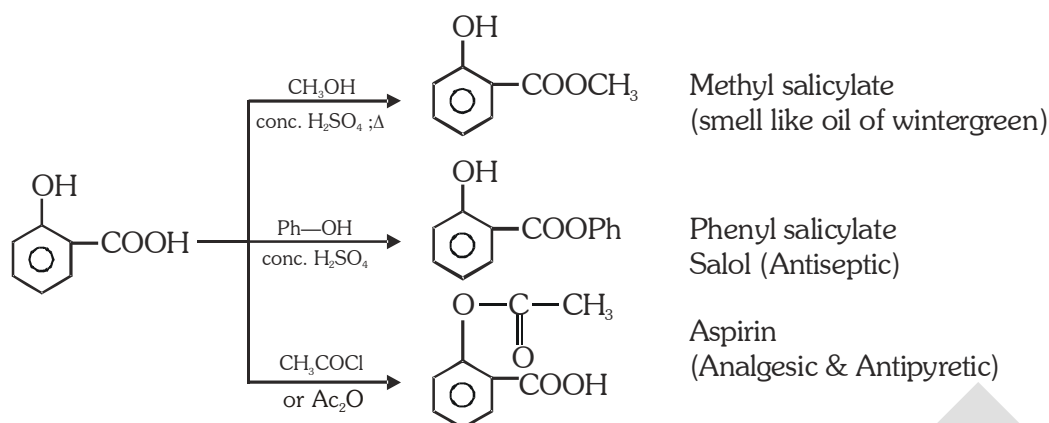
Colourless



Sodium nitrolate (Red) No reaction (Remains blue)

(C) Dichromate test :**(v) Distinction between CH_3-OH and C_2H_5OH**

	CH_3OH	CH_3CH_2OH
B.P.	65°C	78°C
$I_2 + NaOH$	No ppt	Yellow ppt of CHI_3
$Cu/300^\circ C$	Smell of formalin [$HCHO$]	No smell
Salicylic acid	Smell like oil of wintergreen	No smell



BEGINNER'S BOX-1

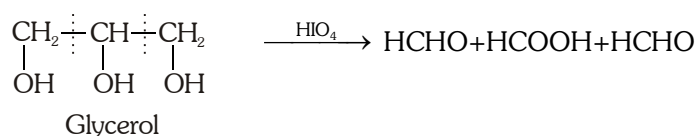
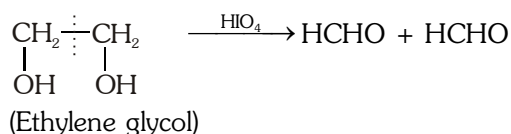
- Which of the following reactions of alkanols does not involve C-O bond breaking
 - $\text{CH}_3\text{CH}_2\text{OH} + \text{SOCl}_2$
 - $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3 + \text{PBr}_3$
 - $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{COOH}$
 - $\text{ROH} + \text{HX}$
- Which of the following alkanols is most soluble in water
 - 1-Butanol
 - 2-Butanol
 - Isobutyl alcohol
 - t-Butyl alcohol
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{PCl}_3} \text{A} \xrightarrow{\text{Alc.KOH}} \text{B} \xrightarrow{\text{H}_3\text{O}^+} \text{C}$
 Find product 'C' is
 - $\text{CH}_3\text{CH}=\text{CH}_2$
 - $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$
 - $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$

Important facts about alcohols

- Toxicity [ethyl alcohol < Iso propyl alcohol < methyl alcohol]
- Absolute alcohol : Ethyl alcohol - 99.5%-100%
- Power alcohol : Rectified spirit + C_6H_6 + Petrol for generation of power
- Methylated spirit : Methanol + Pyridine + mineral naptha + rectified spirit.
- 70% CH_3OH is known as wood spirit.
- 90% $\text{C}_2\text{H}_5\text{OH}$ is known as Raw spirit.
- $\text{C}_2\text{H}_5\text{OH}$ is technically called WASH.
- Rectified spirit contains 95.5% alcohol and 4.5% H_2O .

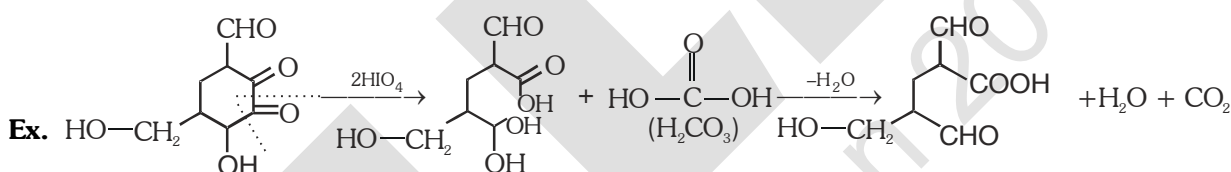
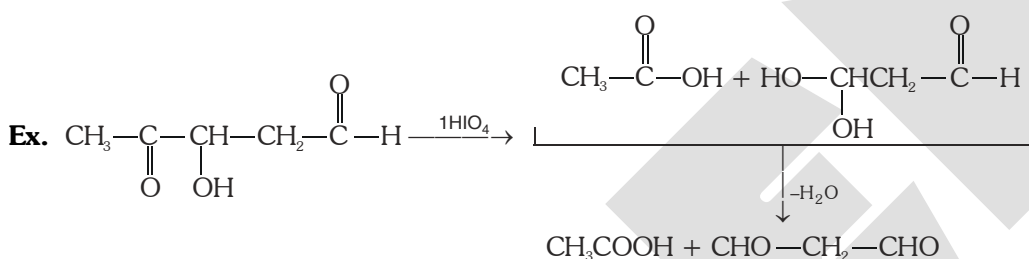
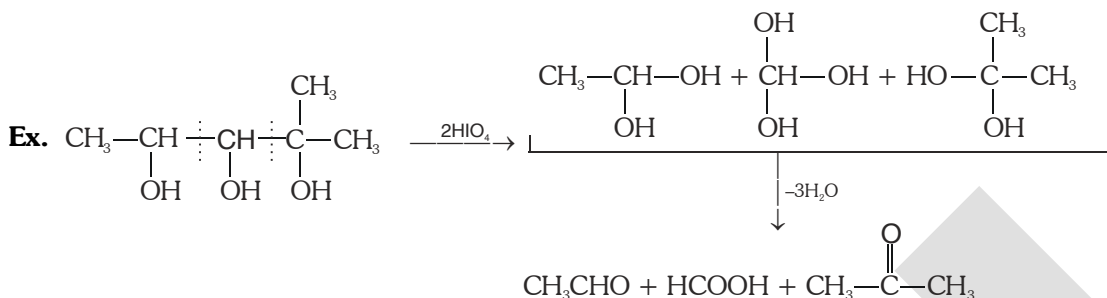
GOLDEN KEY POINTS

Oxidation by HIO_4 [per iodic acid] or $(\text{CH}_3\text{COO})_4\text{Pb}$ [lead tetraacetate] :

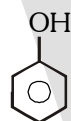


Condition for oxidation by HIO_4 or $(\text{CH}_3\text{COO})_4\text{Pb}$

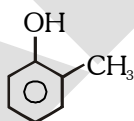
- (i) At least 2 $-\text{OH}$ or 2 $>\text{C}=\text{O}$ or 1 $-\text{OH}$ and 1 $>\text{C}=\text{O}$ should be at vicinal carbons.
- (ii) One HIO_4 breaks one $\text{C}-\text{C}$ bond and adds one $-\text{OH}$ to each carbon.

**4.0 AROMATIC HYDROXY DERIVATIVES**

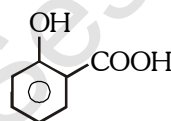
Phenolic compounds : Compounds in which $-\text{OH}$ group is directly attached to sp^2c [Benzene ring]



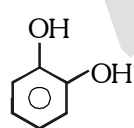
Phenol



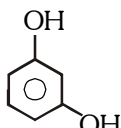
o-Cresol



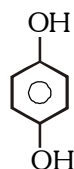
Salicylic acid



Catechol



Resorcinol



Quinol

All phenolic compounds give colour with neutral FeCl_3 .

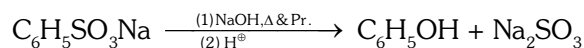


PHENOL (C₆H₅OH)

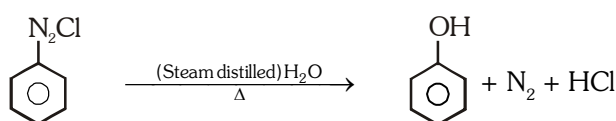
Phenol is also known as carboic acid. In phenol —OH group is attached with sp² hybridised carbon.

4.1 General Methods of Preparation

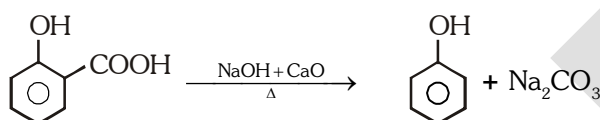
- (1) **From benzene sulphonic acid** : When sodium salt of benzene sulphonic acid is fused with NaOH phenol is obtained.



- (2) **From benzene diazonium chloride** : When benzene diazonium chloride solution is warmed with water, phenol is obtained with evolution of nitrogen.

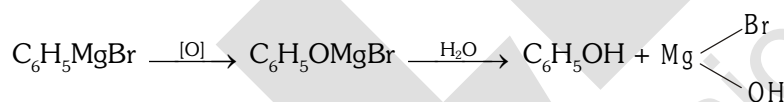


- (3) **By distilling a phenolic acid with sodalime (decarboxylation):**

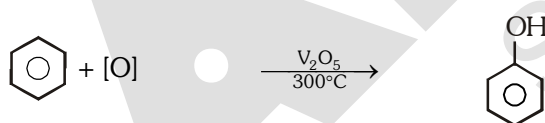


Salicylic acid

- (4) **From Grignard reagent** : The Grignard reagent on reaction with oxygen and subsequent hydrolysis yields phenol.



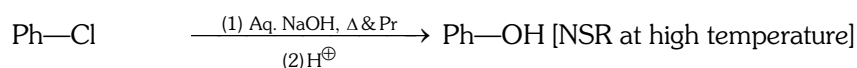
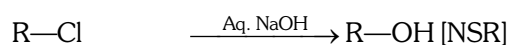
- (5) **From benzene :**



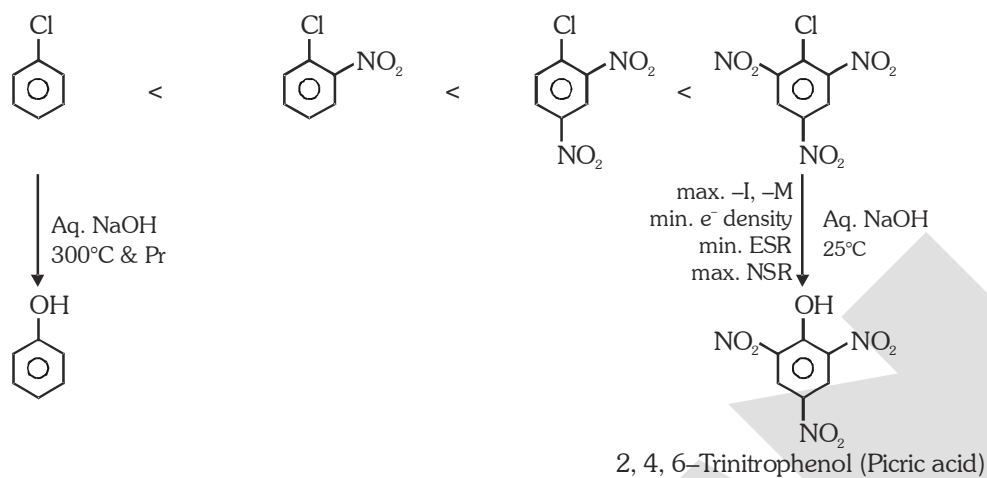
- (6) **From chloro benzene :**



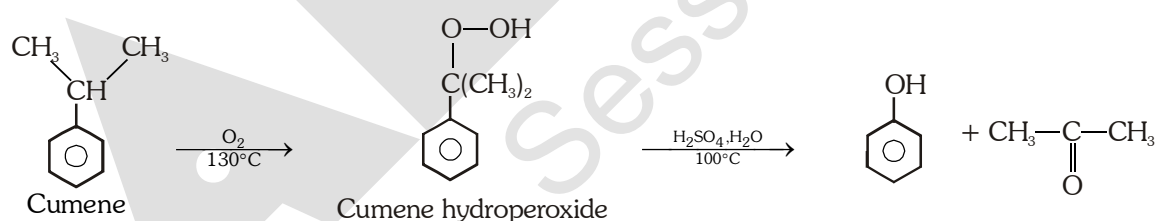
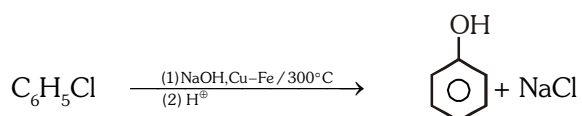
Stable by resonance



Order for NSR :

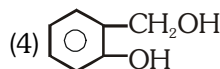
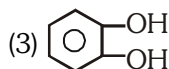
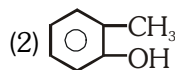
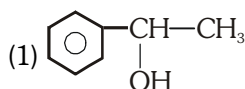
**(7) Industrial preparation of phenol:** Phenol can be prepared commercially by :

- (a) Cumene
- (b) Dow's process

(a) From cumene (Isopropyl benzene) : Cumene is oxidised with oxygen into cumene hydroperoxide in presence of a catalyst. This is decomposed by dil. H_2SO_4 into phenol and acetone.**(b) Dow process :** This process involves alkaline hydrolysis of chloro benzene-(obtained by above process) followed by acidification.

BEGINNER'S BOX-2

1. Which of the following compounds does not show phenolic properties :-



2. The number of dihydric phenols possible with the molecular formula $C_6H_6O_2$ is :-

(1) 2

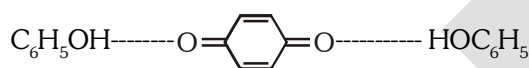
(2) 3

(3) 4

(4) 5

4.2 Physical Properties

- (i) Phenol is a colourless, crystalline solid.
- (ii) It attains pink colour on exposure to air and light. (slow oxidation)



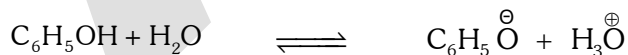
Phenoquinone (pink colour)

- (iii) It is poisonous in nature but acts as antiseptic and disinfectant.
- (iv) Phenol is slightly soluble in water, readily soluble in organic solvents.
- (v) Solubility of phenol in water is much lower than alcohols because of larger hydrocarbon part in the molecule.
- (vi) Due to intermolecular H-Bonding, phenol has relatively high boiling point than the corresponding hydrocarbons, aryl halides.

4.3 Chemical Properties

(A) Reactions due to -OH group :

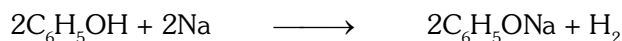
- (i) **Acidic Nature** : Phenol is a weak acid. The acidic nature of phenol is due to the formation of stable phenoxide ion in solution. The phenoxide ion is stable due to resonance. The negative charge is spread through out the benzene ring which is stabilising factor in the phenoxide ion. Electron withdrawing groups ($-NO_2$, $-Cl$) increase the acidity of phenol while electron releasing groups ($-CH_3$ etc.) decrease the acidity of phenol.



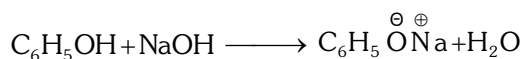
Phenol is stronger acid than alcohols but weaker than the carboxylic acids and even carbonic acid.

GOLDEN KEY POINTS

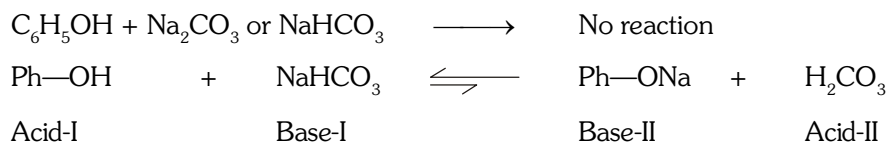
- The acidic nature of phenol is observed as the following:
 - (i) Phenol changes blue litmus to red.
 - (ii) Highly electro positive metals react with phenol.



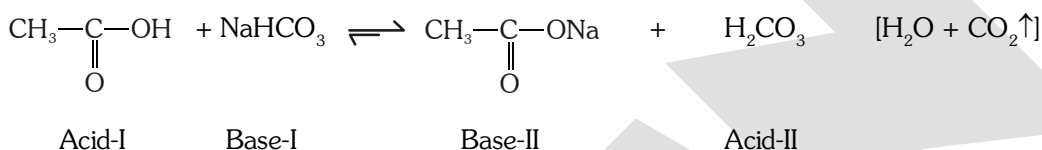
(iii) Phenol reacts with strong alkalies to form phenoxides.



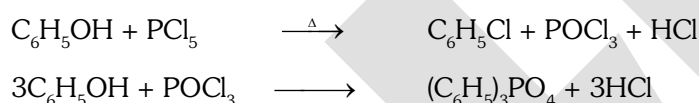
(iv) However phenol does not decompose Na_2CO_3 or NaHCO_3 because phenol is weaker acid than carbonic acid.



Note : Acetic acid reacts with NaHCO_3 and gives effervescence of CO_2 .



(ii) Reaction with PCl_5 : Phenol reacts with PCl_5 to form chloro benzene. POCl_3 formed as biproduct reacts with phenol to form triphenyl phosphate.

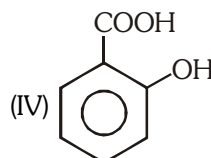
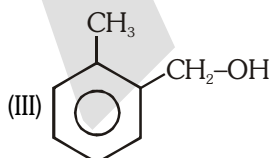
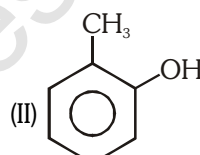
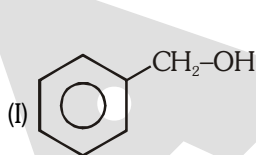


(iii) Reaction with Zn dust: When phenol is distilled with zinc dust benzene is obtained.



Illustrations

Illustration 4. In which of the following compound deoxygenation is possible when heated with Zn



(1) I, II, III

(2) I, III, IV

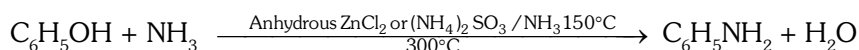
(3) II, IV

(4) II, III

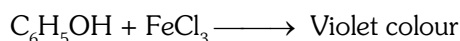
Solution

Ans.(3) Observed in phenolic compound.

(iv) Reaction with NH_3 (Bucherer reaction): Phenol reacts with NH_3 in presence of anhydrous ZnCl_2 to form aniline.

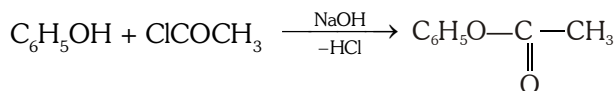


- (v) **Reaction with FeCl_3 :** Phenol gives violet colouration with FeCl_3 solution (neutral) due to formation of a complex.

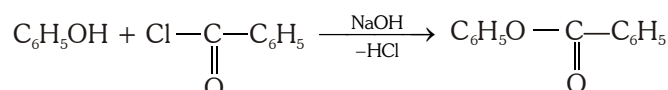


This reaction is used to differentiate phenol from alcohols.

- (vi) **Acetylation :** Phenol reacts with acid chlorides or acid anhydrides in alkali solution to form phenyl esters.

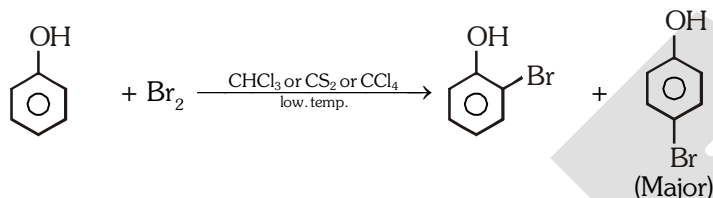


- (vii) **Benzoylation (Schotten-Baumann reaction)**

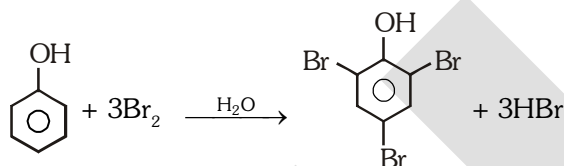


- (B) **Reaction of Benzene Ring :** The $-\text{OH}$ group is ortho and para directing. It activates the benzene nucleus.

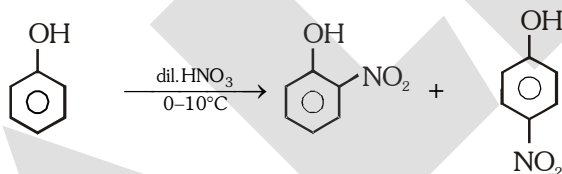
- (i) **Halogenation :** Phenol reacts with bromine in CCl_4 to form mixture of o- and p-bromo phenol.



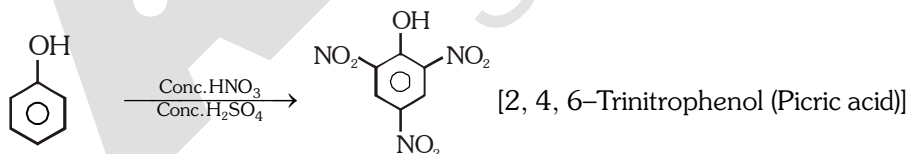
Phenol reacts with bromine water to form a white ppt. of 2,4,6-tribromo phenol. (Test for phenol)



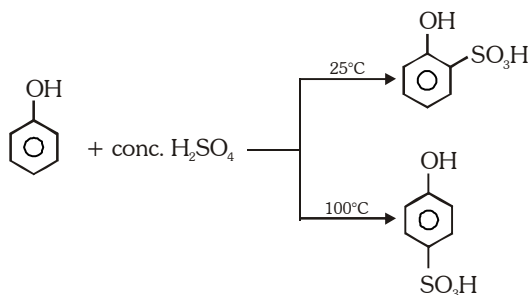
- (ii) **Nitration :** Phenol reacts with dil. HNO_3 at $0^\circ\text{--}10^\circ\text{C}$ to form o- and p- nitro phenols.



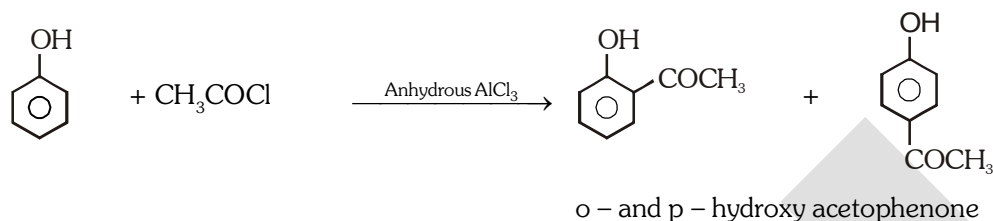
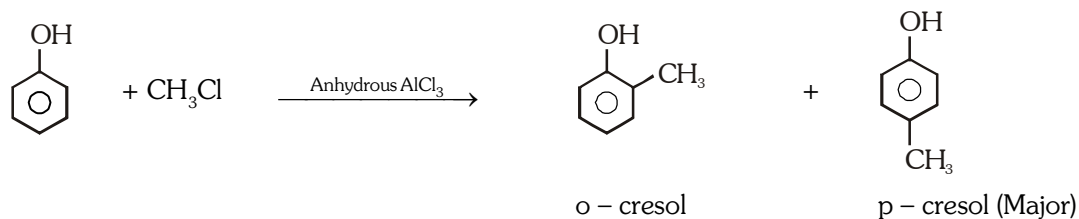
When phenol is treated with nitrating mixture it forms 2,4,6- trinitro phenol (picric acid) but it is not good method to form picric acid because nitric acid oxidise phenol into p-Bezoquinone



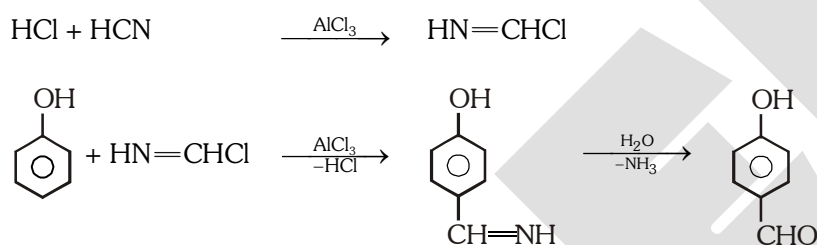
- (iii) **Sulphonation:** Phenol reacts with fuming H_2SO_4 to form o- and p-hydroxy benzene sulphonic acid at different temperatures.



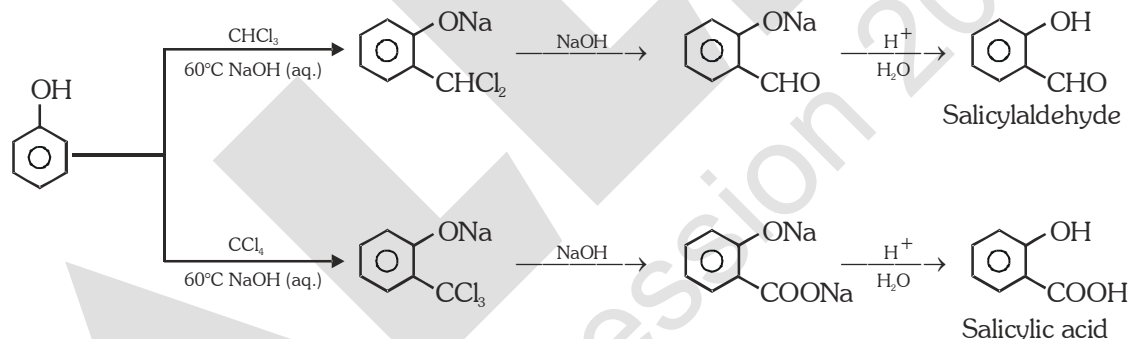
(iv) Friedel - Craft 's reaction:



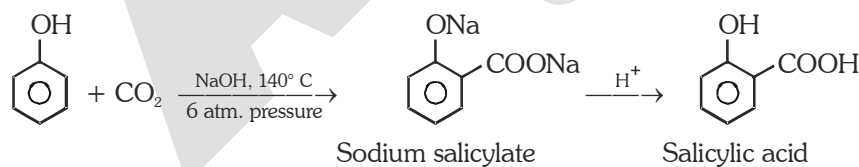
(v) **Gattermann aldehyde synthesis :** When phenol is treated with liquid HCN and HCl gas in presence of anhydrous AlCl_3 it yields mainly p- hydroxy benzaldehyde (formylation)



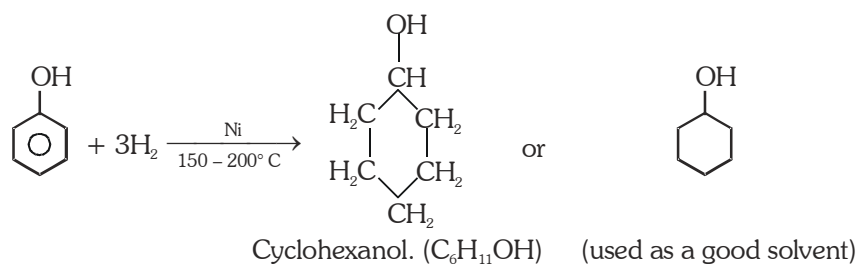
(vi) **Reimer-Tiemann reaction** : Phenol on refluxing with chloroform and NaOH (aqueous) followed by acid hydrolysis yields o-hydroxy benzaldehyde. When CCl_4 is used salicylic acid is formed.



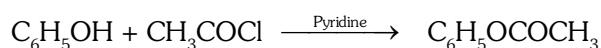
(vii) Kolbe's Schmidt reaction : It involves the reaction of $\text{C}_6\text{H}_5\text{OH}$ with CO_2 and NaOH at 140°C followed by acidification to form salicylic acid.



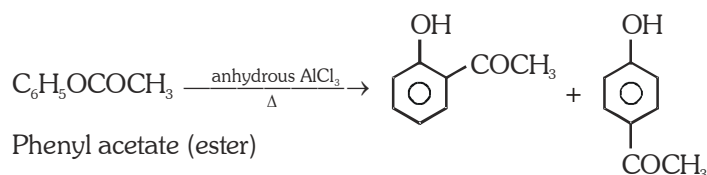
(viii) Hydrogenation: Phenol when hydrogenated in presence of Ni at 150-200°C forms cyclohexanol.



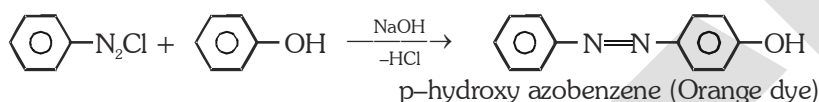
(ix) Fries rearrangement reaction:



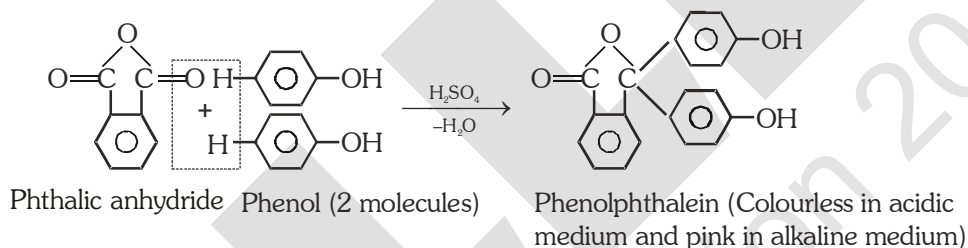
Phenyl Acetate



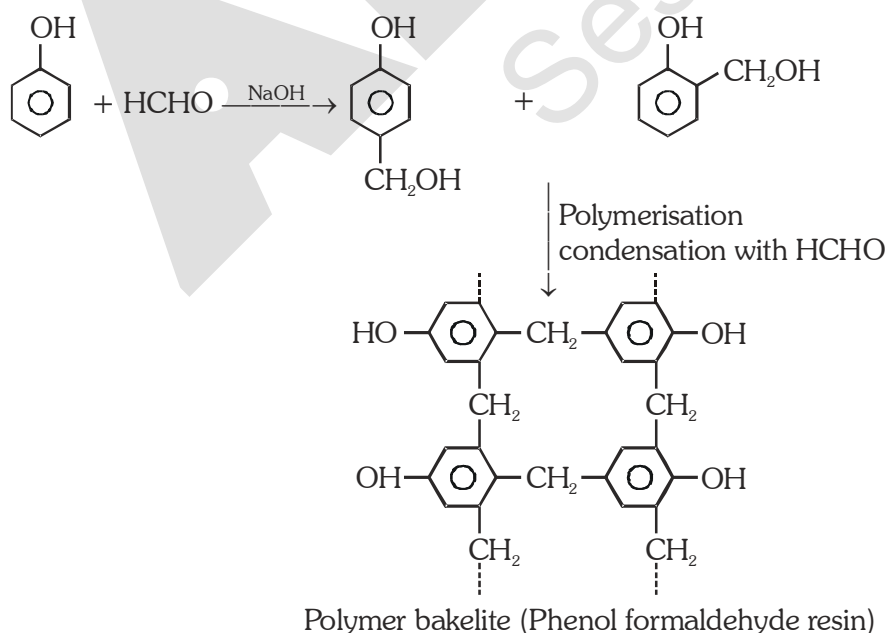
(x) **Coupling reactions:** Phenol couples with benzene diazonium chloride in presence of an alkaline solution to form a dye (p- hydroxy azobenzene) orange dye.

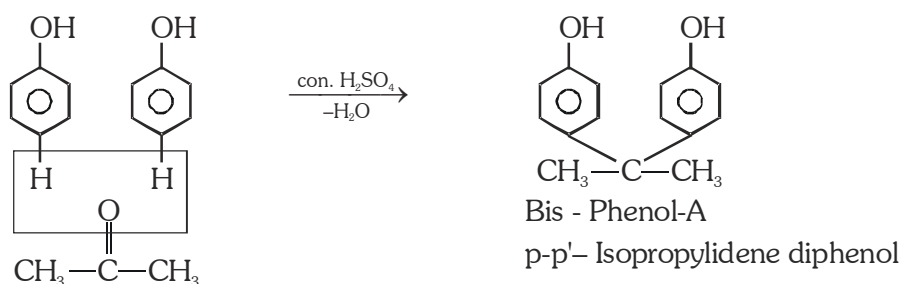
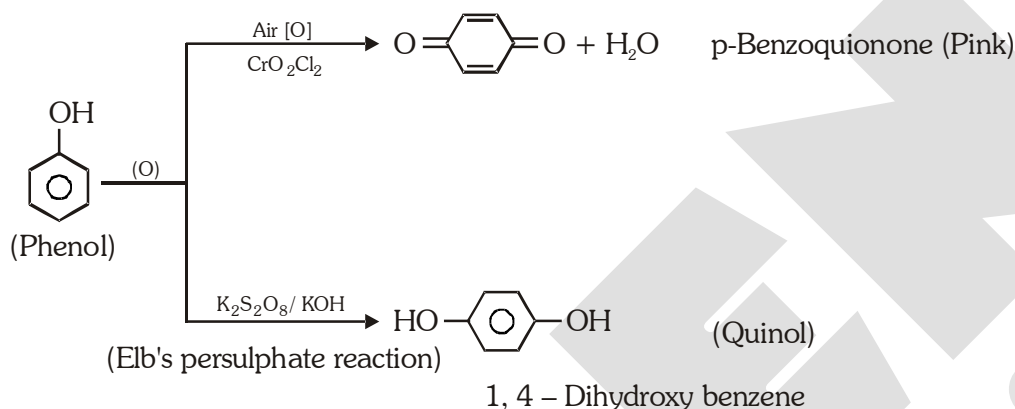


Phenol couples with phthalic anhydride in presence of conc. H_2SO_4 to form a dye (phenolphthalein) used as an indicator.



(xi) **Lederer Manasse (Condensation with formaldehyde) :** Phenol condenses with HCHO (excess) in presence of NaOH or weak acid (H^+) to form a polymer known as bakelite (resin).



(xii) Reaction with acetone: (Condensation with acetone)**(xiii) Oxidation:****BEGINNER'S BOX-3**

- Reimer-Tiemann formylation reaction involves addition of :-
 - (1) Chloroform on phenoxide ion
 - (2) Trichloromethyl carbanion on phenoxide ion
 - (3) Dichlorocarbene on phenoxide ion
 - (4) Hydroxide ion on phenol
- Phenol reacts with benzenediazonium chloride solution to form a compound of the structure :-
 - (1)
 - (2)
 - (3)
 - (4)
- Phenol $\xrightarrow[\text{conc. H}_2\text{SO}_4]{\text{NaNO}_2}$ Green colour $\xrightarrow{\text{H}_2\text{O}}$ red colour $\xrightarrow{\text{NaOH}}$ Blue colour
 This reaction is associated with the name of :-
 - (1) Gattermann
 - (2) Hofmann
 - (3) Liebermann
 - (4) Reimer-Tiemann

Test of Phenol :

- (1) Phenol turns blue litmus to red.
- (2) Aqueous solution of phenol gives a violet colour with a drop of ferric chloride.
- (3) Phenol gives Liebermann's nitroso test.
- (4) Aqueous solution of phenol gives a white ppt. of 2,4,6-tribromophenol with bromine water.
- (5) Phenol combines with phthalic anhydride in presence of conc. H_2SO_4 to form phenolphthalein which gives pink colour with alkali.

Differences between phenol and alcohol (C₂H₅OH) :

- (1) Phenol is more acidic than aliphatic alcohol due to resonance in phenoxide ion.
- (2) Phenol gives violet colour with FeCl₃ while aliphatic alcohol does not give.
- (3) Phenol gives triphenyl phosphate with PCl₅ while aliphatic alcohol does not.
- (4) Phenol on oxidation gives quinone while alcohol gives aldehyde or ketone and acids.

Uses of Phenol :

Phenol is used :

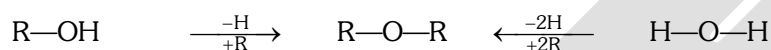
- (1) As an antiseptic in soaps and lotions. "Dettol" (mixture of chloroxylenol and terpineol)
- (2) In manufacture of azodyes, phenolphthalein, picric acid (explosive), cyclohexanol (Solvent for rubber), plastics (bakelite) etc.
- (3) In manufacture of drugs like aspirin, salol, phenacetin etc.
- (4) As preservative for ink.

5.0 ETHER

R—O—R (Dialkyl ether), alkoxy alkane. It's General formula is C_nH_{2n+2}O.

CH₃—O—CH₂CH₃ (Methoxy ethane) or Ethyl methyl ether

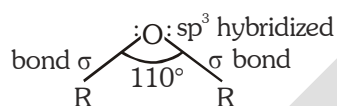
Ether is monoalkyl derivative of R—OH and dialkyl derivative of H₂O



Classification : They may be classified as :

- (a) Simple or symmetrical ether. e.g. R—O—R
- (b) Mixed or unsymmetrical ether e.g. R—O—R'

Structure :

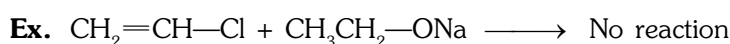
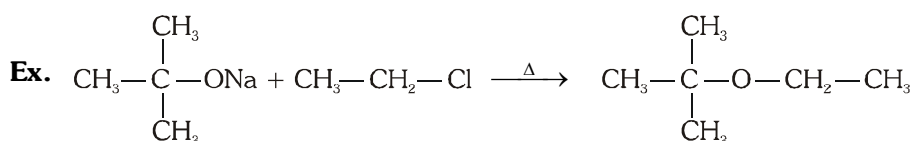
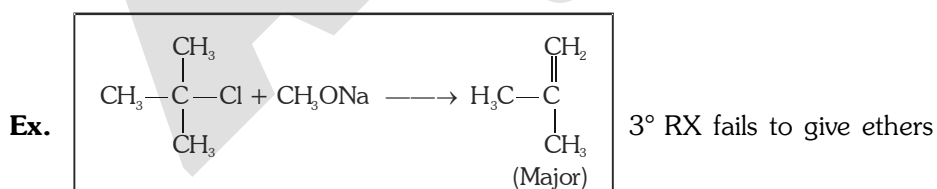
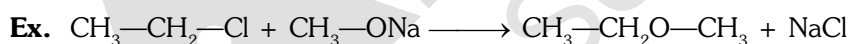
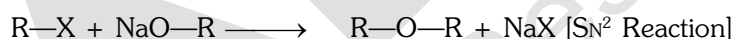


The molecule of ether is bent due to presence of lone pair. The bond angle is 110°. It is greater than that of water (105°) due to the repulsion between bulkier alkyl groups. Due to bent structure, it possesses dipole moment and hence are polar molecules.

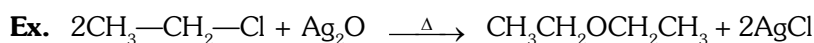
5.1 General Methods of Preparation

(a) From alkyl halides :

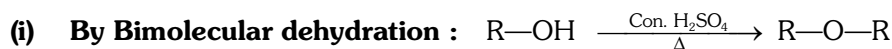
(i) By Williamson's synthesis :



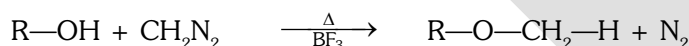
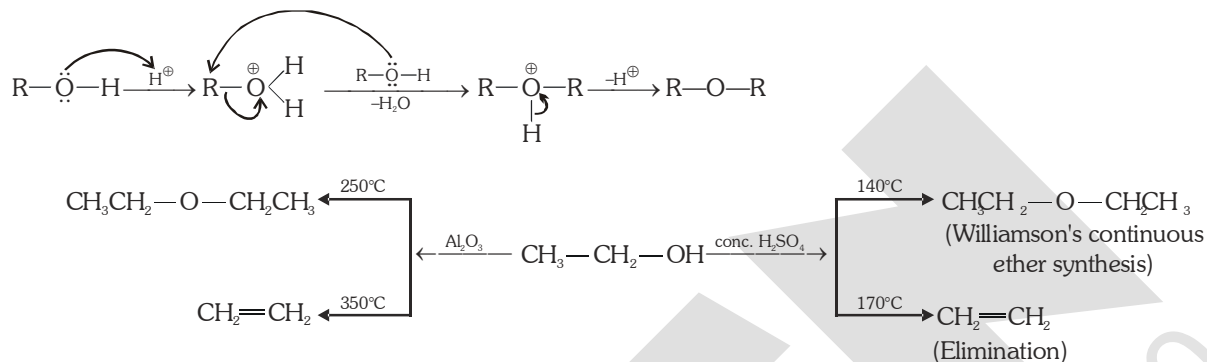
[Stable by Resonance]



(b) From R-OH:



Mechanism :



5.2 Physical Properties

- CH_3OCH_3 , $\text{CH}_3\text{OCH}_2\text{CH}_3$ are gases and higher ethers are volatile liquids.
- Ethers are less polar.
- Ethers are less soluble in H_2O .
- Ethers have less BP than corresponding alcohol.

Illustrations

Illustration 5. Ethers are less soluble in H_2O . Why?

Solution. Due to less polarity, it forms weaker H-Bonding with H_2O .

Illustration 6. Ethers have less BP than corresponding alcohol. Why?

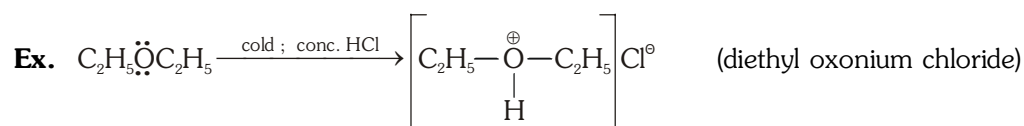
Solution. No H-Bonding in ether molecules.

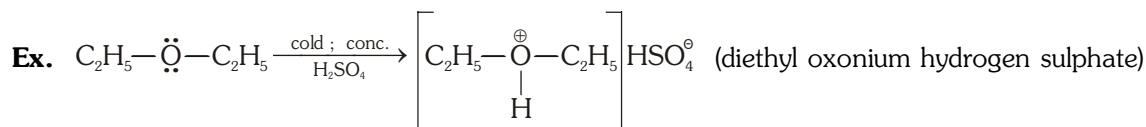
5.3 Chemical properties

Ethers are less polar so less reactive and do not react with active metals [Na, K], cold dil. acid, oxidising and reducing agent. They do not have any active functional group.

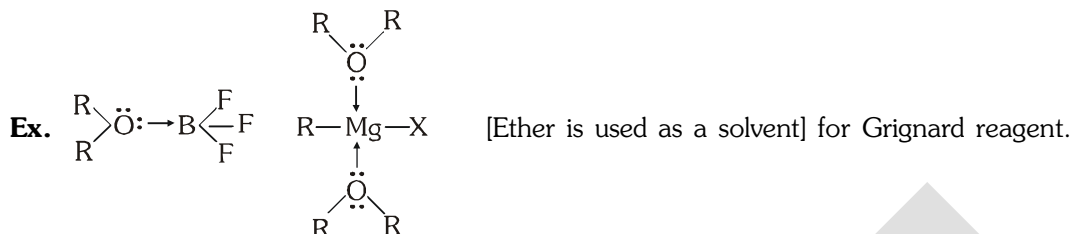
(1) Basic nature : Due to presence of $\ell.p$ on oxygen atom ether behave as lewis base

Ethers react with cold conc. acid and form oxonium salts.

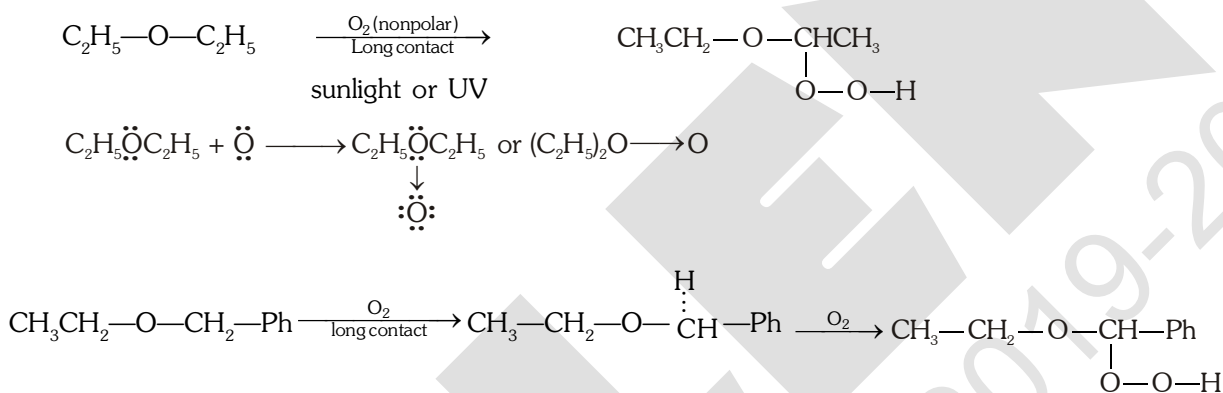




Ethers form dative bond with Lewis acids like BF_3 , AlCl_3 , RMgX etc.



(2) Formation of peroxides : Ether add up atmospheric oxygen or ozonised oxygen. It is explained by Free radical mechanism as intermediate is free radical.



Peroxides are unstable and explosives.

GOLDEN KEY POINTS

Test for peroxides

Ether (peroxide) $\xrightarrow{\text{FeSO}_4/\text{KCN}} \text{Red colour}$

Ether (Peroxides) + $\text{Fe}^{+2} \longrightarrow \text{Fe}^{+3} \xrightarrow{\text{CNS}^-} \text{Fe}(\text{CNS})_3$
(Red)

(3) Reaction with PCl_5 : $\text{ROR} + \text{PCl}_5 \xrightarrow{\text{heat}} 2\text{RCl} + \text{POCl}_3$

(4) Reduction : $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3 \xrightarrow[\text{heat}]{\text{Red P} + \text{HI}} 2\text{CH}_3\text{CH}_3$

(5) Reaction with HX : $\text{R}-\text{O}-\text{R}' + \text{HI} \longrightarrow \text{R}-\text{OH} + \text{R}'-\text{I}$

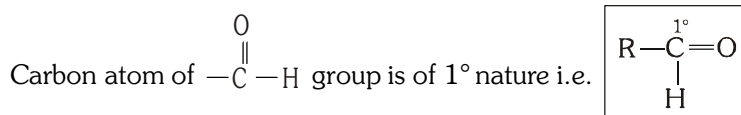
Uses of ether :

- General anaesthetic agent.
- Solvent for oil, fats, resins, Grignard reagent.
- For providing inert & moist free medium to organic reaction e.g. Wurtz reactions.
- In perfumery.
- Di-isopropyl ether $\longrightarrow$ In petrol as an antiknock comp.
- Mixture of alcohol and ether is used as a substitute of petrol. Trade name "**Natalite**"

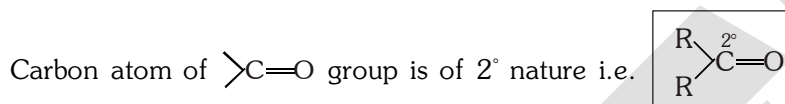
6.0 CARBONYL COMPOUNDS

Organic Compounds having $>\text{C}=\text{O}$ group are called carbonyl compounds and $>\text{C}=\text{O}$ group is known as carbonyl group. Its general formula is $\text{C}_n\text{H}_{2n}\text{O}$ ($n = 1, 2, 3, \dots$) Carbonyl compounds are grouped into two categories.

(a) Aldehydes : Aldehyde group is $\text{—}\overset{\text{O}}{\underset{\text{||}}{\text{C}}}\text{—H}$ (also known as formyl group). It is a monovalent group



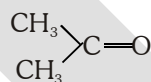
(b) Ketones : The carbonyl group ($>\text{C}=\text{O}$) is a Ketonic group when its both the valencies are satisfied by alkyl group. It is a bivalent group.



Ketones are further classified as :

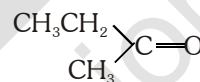
- (i) **Simple or Symmetrical ketones :** Having two similar alkyl groups. $\begin{array}{c} \text{R} \diagup \text{C=O} \\ \text{R} \diagdown \end{array}$
- (ii) **Mixed or unsymmetrical ketones :** Having two different alkyl groups. $\begin{array}{c} \text{R} \diagup \text{C=O} \\ \text{R}' \diagdown \end{array}$

Ex. (Ketones) : Symmetrical

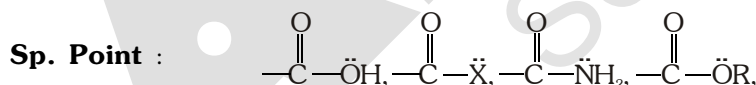


(Acetone or Dimethyl ketone)
Propanone

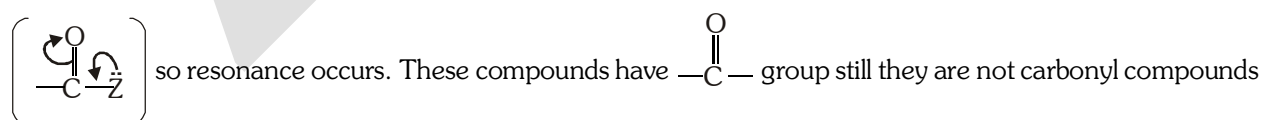
Unsymmetrical



(Ethyl methyl ketone)
Butanone

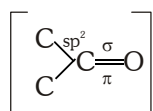


In all the compounds given above, lone pair of electrons and double bond are in conjugate system.



because these compounds have characteristic reactions different from carbonyl compounds.

Structure : In $>\text{C}=\text{O}$ compounds C-atom is sp^2 hybridised which forms two σ bonds and one π bond. The unhybridised atomic orbital of C-atom and the parallel 2p orbital of oxygen forms the π bond in $>\text{C}=\text{O}$ group



The C—C—O / H—C—O bond angle is of 120°

Due to electro-negativity difference in C & O atoms, the $>\text{C}=\text{O}$ group is polar.

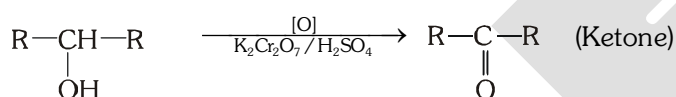
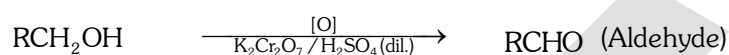
$\text{>C}^{\delta+}=\text{O}^{\delta-}$ Hence aldehydes and Ketones possess considerable dipole moment.

6.1 General Methods of Preparation

(A) For both Aldehydes and Ketones

(1) By Oxidation of Alcohols :

(a) **By $\text{K}_2\text{Cr}_2\text{O}_7$ / H_2SO_4** : Oxidation of primary alcohols gives aldehyde and oxidation of secondary alcohols gives Ketones.

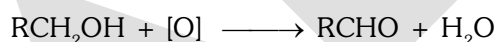


Aldehydes are quite susceptible to further oxidation to acids -



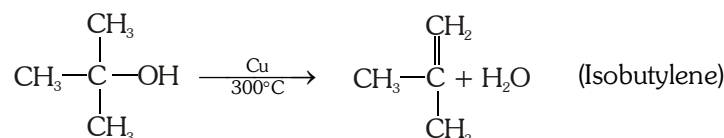
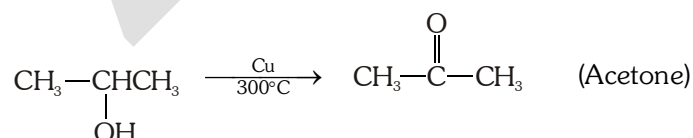
Thus oxidation of primary alcohols is made at the temperature much above the boiling point of aldehyde and thus aldehydes are vapourised out and prevented from being oxidised.

(b) **Mild Oxidising Agent** : 1° alcohols will get oxidised with CrO_3 / Pyridine, (**collin's reagent**) or P.C.C (Pyridinium chloro chromate $\text{CrO}_3 + \text{C}_5\text{H}_5\text{N} + \text{HCl}$) to aldehyde and 2° alcohols to ketone.

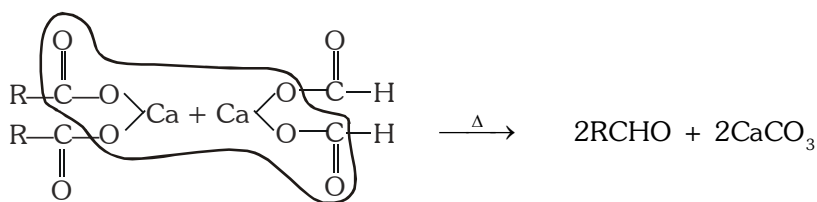


By this reaction, **good yield of aldehyde** is possible.

(2) Dehydrogenation of alcohols :



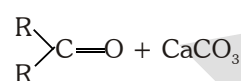
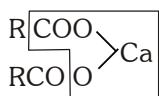
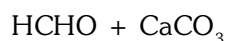
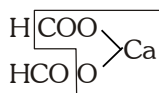
(3) By dry distillation of Ca-salts of carboxylic acid :



Calcium alkanoate

Calcium formate

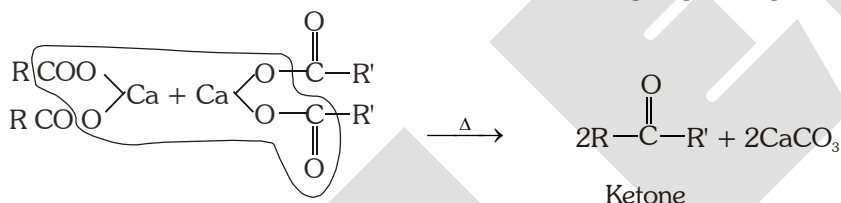
(R — C — R and HCHO are also formed)

$$\begin{array}{c} \text{O} \\ \parallel \\ \text{C} \end{array}$$


Calcium-alkanoate

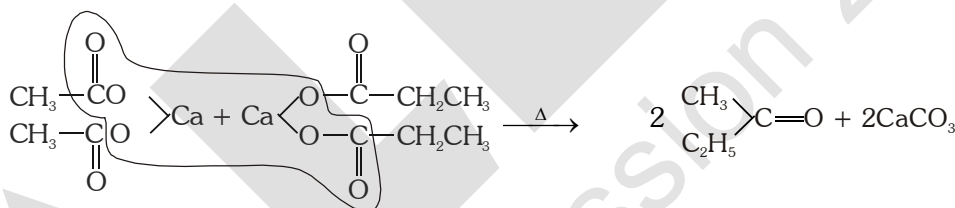
Ketone

Calcium salts of acids other than formic acid on heating together give unsymmetrical ketone



Ketone

To prepare ethyl methyl ketone Calcium acetate and Calcium propionate are used :

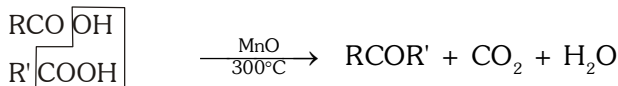
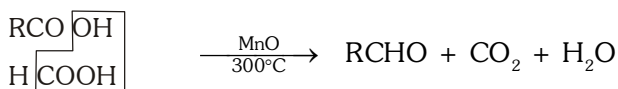
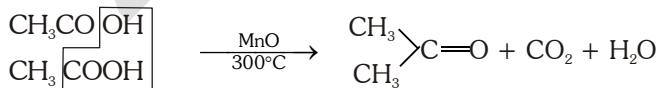
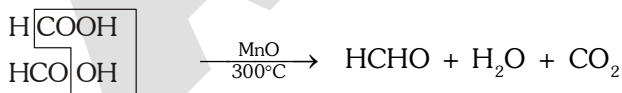


Calcium Acetate

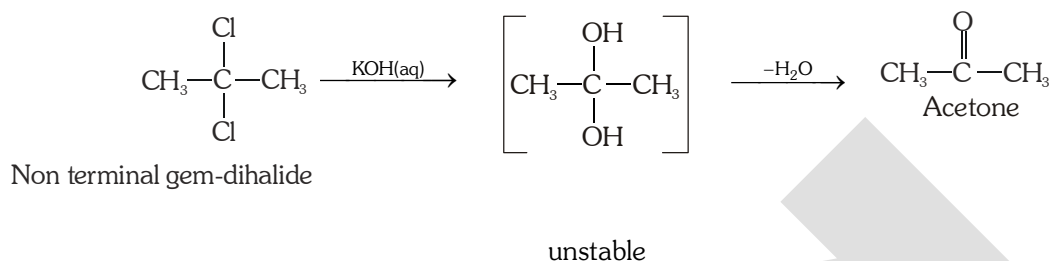
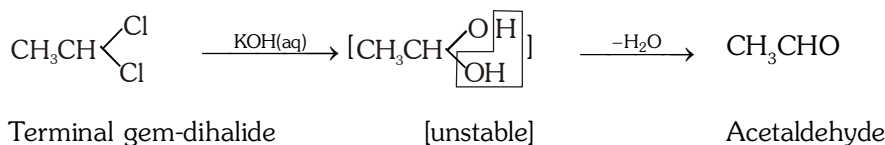
Calcium propionate

Ethyl methyl ketone

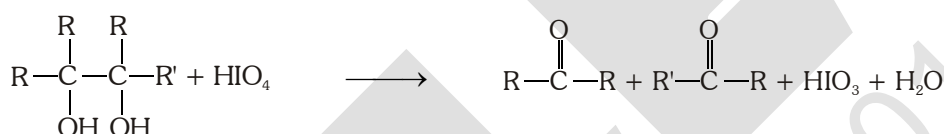
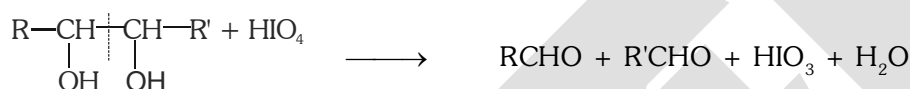
(4) By Thermal decomposition of carboxylic acids : Vapour of carboxylic acids when passed over MnO/300°C give carbonyl compounds



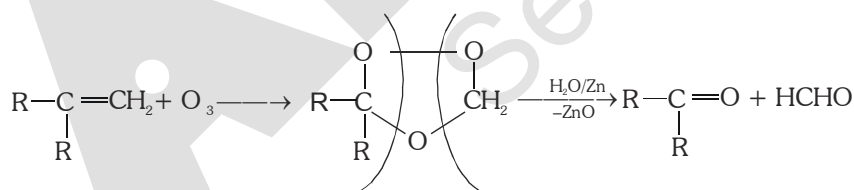
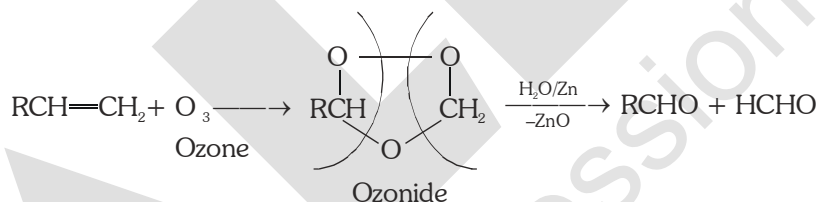
- (5) **By Hydrolysis of gem dihalides :** Terminal gem-dihalides on hydrolysis give aldehydes while the non-terminal gem-dihalides give ketones.



- (6) **By Oxidation of diols :** With periodic acid (HIO_4) or lead tetra acetate $(\text{CH}_3\text{COO})_4\text{Pb}$ vicinal diols get oxidised to form carbonyl compounds

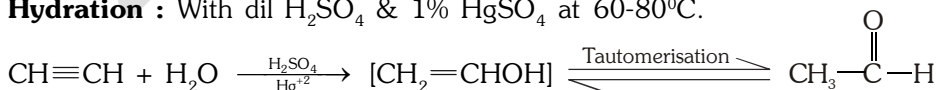


- (7) **By Ozonolysis of alkenes :** This reaction is used to determine the position of double bond in alkene. Zn is used to decompose H_2O_2 formed during hydrolysis.

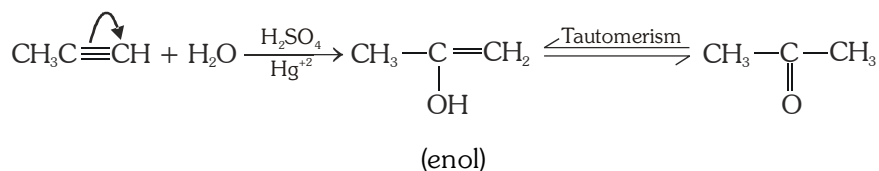


- (8) **From Alkyne :**

- (a) **Hydration :** With dil H_2SO_4 & 1% HgSO_4 at $60-80^\circ\text{C}$.

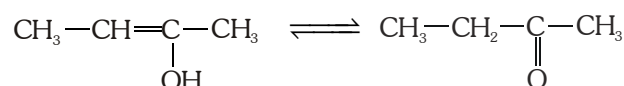
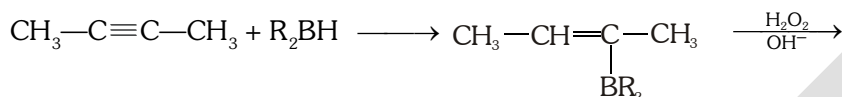
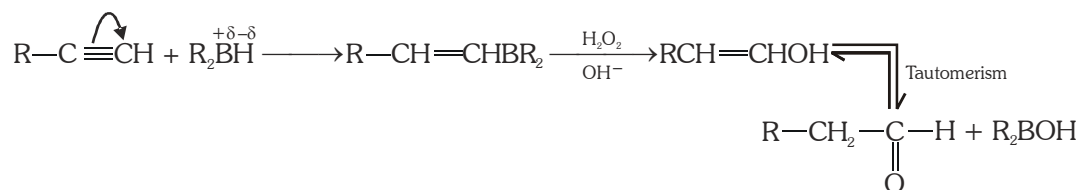


Other alkynes mainly give ketone :



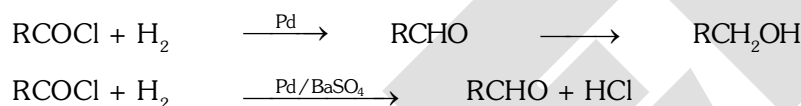
(b) **Hydroboration** : Reaction with B_2H_6 or R_2BH give alkenyl dialkyl borane.

1-alkyne gives $\longrightarrow$ aldehyde
other alkynes $\longrightarrow$ ketone

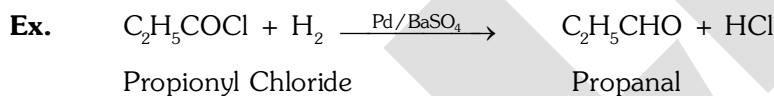


(B) For Aldehydes only

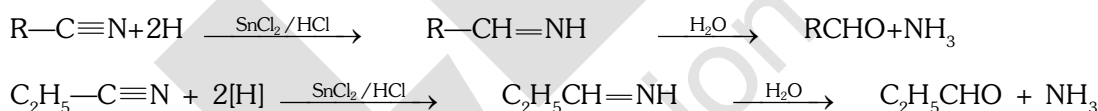
(1) Rosenmund's reduction :



$BaSO_4$ controls the further reduction of aldehyde to alcohols. Formaldehyde can not be prepared by this method.

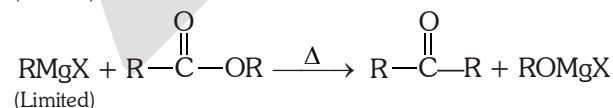
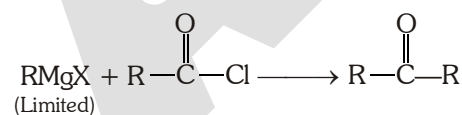
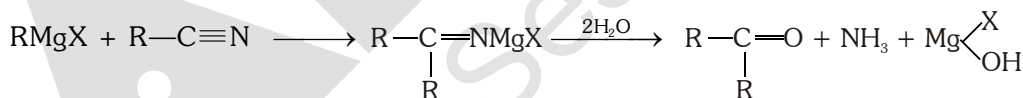


(2) Stephen's reduction :



(C) For Ketones only

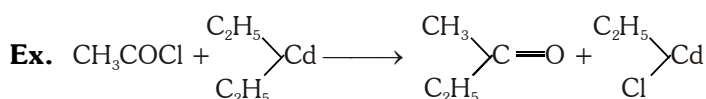
(1) From Grignard's reagent :



(2) From dialkyl Cadmium : $R'CdR'$ (dialkyl Cadmium) is an organometallic compound.



This reaction is superior than Grignard Reaction because the ketones formed, further reacts with Grignard reagent to form 3° alcohols.



6.2 Physical Properties

State : Only formaldehyde is gas, all other carbonyl compounds upto C_{11} are liquids and C_{12} & onwards solid.

Solubility : C_1 to C_3 (formaldehyde, acetaldehyde and propionaldehyde) and acetone are freely soluble in water due to polarity of $\overset{\delta+}{>C}=\overset{\delta-}{O}$ bond and can form H—bond with water molecule. C_5 onwards are insoluble in water.



H-bonding

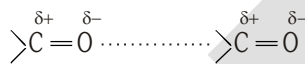
$$\text{Solubility} \propto \frac{1}{\text{Molecular weight}}$$

Boiling point :

$$\text{Boiling point} \propto \text{Molecular weight}$$

Boiling point order is - **Alcohol > Ketone > Aldehydes > Alkane** (of comparable molecular mass)

This is because in alcohols intermolecular H-bonding is present but in carbonyl compounds H-bonding doesn't exist, instead dipole-dipole & vander waal force of attraction is present. Alkanes are non polar.



Density : Density of carbonyl compounds is lower than water.

BEGINNER'S BOX-4

- Acids do not give the characteristic reactions of $C=O$ group because of :-
 (1) Dimerisation (2) Resonance (3) Cyclic structures (4) Attached alkyl radical
- The vapour density of a compound is 45. Its molecular formula will be -
 (1) C_2H_5O (2) $C_3H_6O_2$ (3) $C_4H_{10}O$ (4) $C_5H_{14}O$
- 23 gm of sodium of reaction with methyl alcohol gives :
 (1) Half mole of H_2 (2) One mole of H_2 (3) One mole of O_2 (4) none

6.3 Chemical Properties

Carbonyl compounds undergo following reactions :

6.3.1 Nucleophilic addition reactions (Already discussed in Reaction Mechanism Part-2)

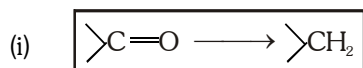
6.3.2 Other reactions

6.3.3 Reactions of only aldehyde

6.3.4 Reactions of only ketones

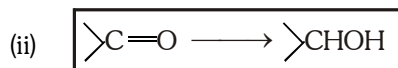
6.3.2 Other reactions

(a) **Reduction** : The nature of product depends upon the reducing agent used.



Reducing agents are

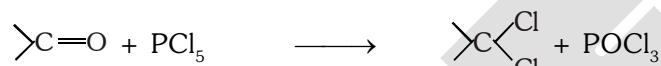
- Red P/HI at 150°C
- Zn-Hg/HCl [Clemensen's reduction]
- (i) N_2H_4 (ii) OH^-/Δ [Wolf Kishner reduction]



Reducing agents are

- Metal + H_2
- LiAlH_4
- NaBH_4
- $\text{Na} + \text{C}_2\text{H}_5\text{OH}$

(b) **Reaction with PCl_5 & SOCl_2 :**

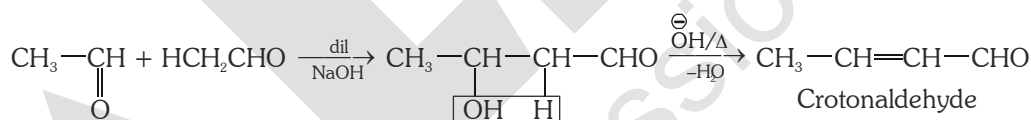


Phosphorus penta chloride



Thionyl chloride

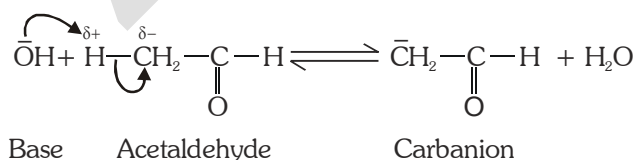
(c) **Aldol Condensation** : Carbonyl compounds which contain α -H atoms undergo condensation with dil. NaOH to give aldol. Aldol contains both alcoholic and carbonyl group, which on heating in alkaline medium gets converted into α, β -unsaturated carbonyl compound.



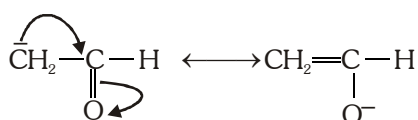
Mechanism of aldol condensation : It takes place in the following two stages :

- Formation of Carbanion
- Combination of carbanion with other carbonyl molecule.

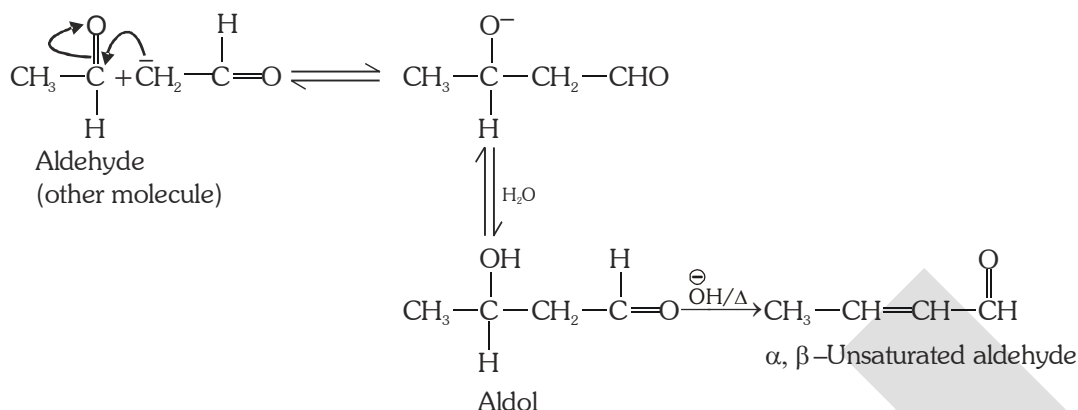
(i) **Formation of Carbanion** : α -H atom of $>\text{C}=\text{O}$ group are quite acidic which can be removed easily as proton, by a base



Carbanion thus formed is stable because of resonance -

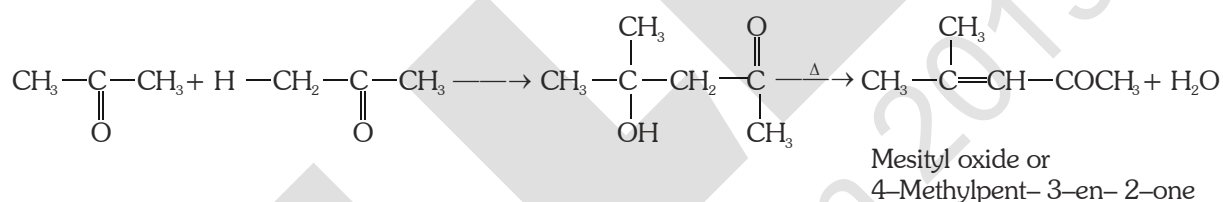


(ii) Combination of carbanion with other carbonyl molecule :

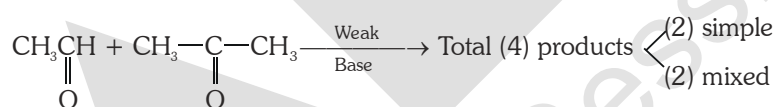


Identical carbonyl compounds $\longrightarrow$ Simple or self aldol condensation.
 Different carbonyl compounds $\longrightarrow$ Mixed or crossed aldol condensation.

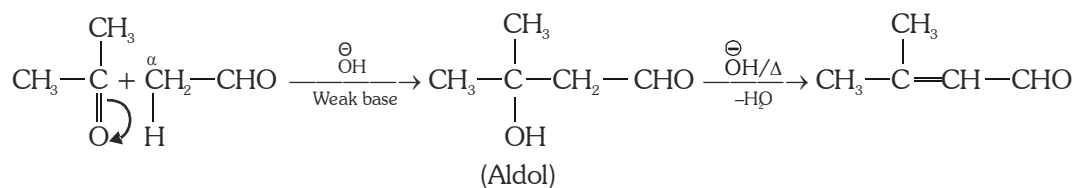
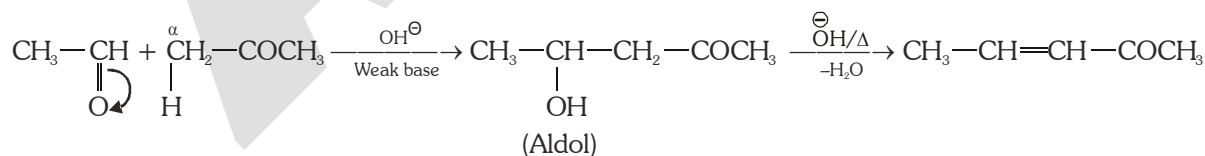
Simple or Self condensation :



Mixed or Crossed aldol Condensation :



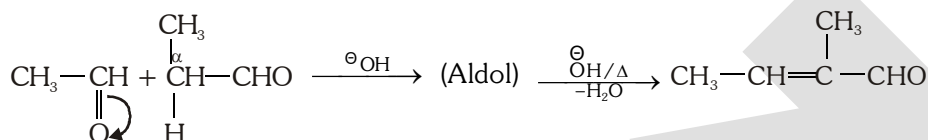
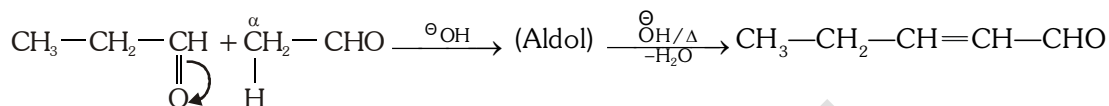
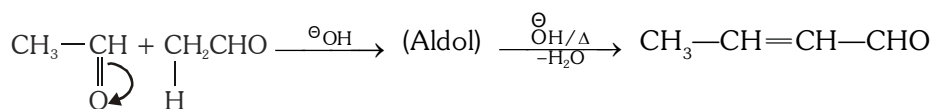
Mixed aldol condensation products of the above reaction are :



Illustrations

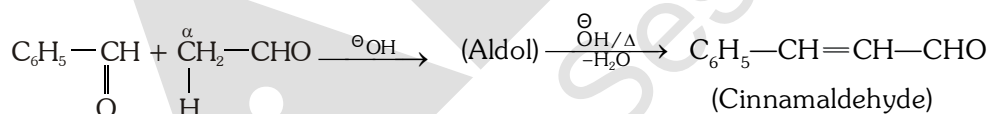
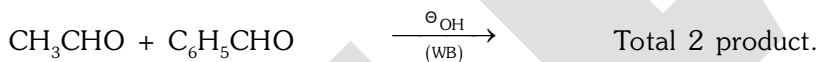
Illustration 7. $\text{CH}_3\text{CHO} + \text{CH}_3\text{CH}_2\text{CHO} \xrightarrow[\text{(WB)}]{^\ominus\text{OH}}$ Total 4 products. Write structure of products ?

Solution.

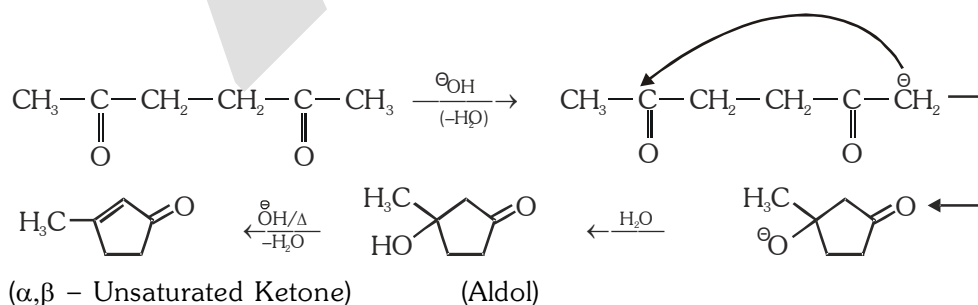


GOLDEN KEY POINTS

- If in crossed aldol condensation reaction, only one carbonyl compound have α -H then total two products are formed.



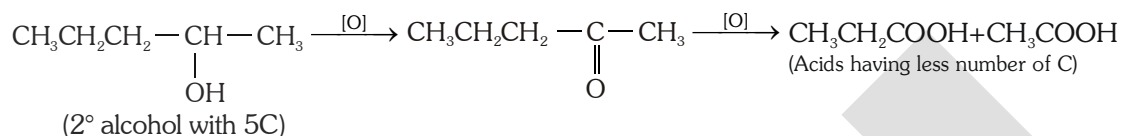
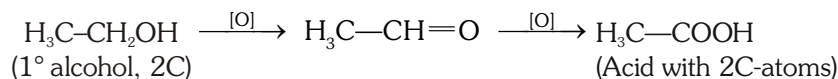
- Intramolecular** aldol condensation :



Here one more product having 3 membered ring is also possible, but 5 membered ring is more stable than 3 membered ring so above product is formed as a major product.

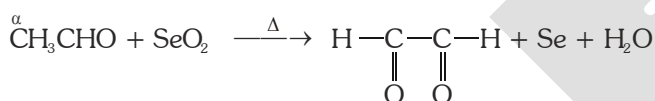
(d) Oxidation reactions :

- (i) **By $K_2Cr_2O_7/H_2SO_4$** : On oxidation with $K_2Cr_2O_7/H_2SO_4$ 1° alc. gives aldehyde, which on further oxidation gives acid with same number of carbons. If 2° alcohol is oxidised at elevated temperature using $KMnO_4/H^+$, it gets oxidised to give acids with less number of C-atom.

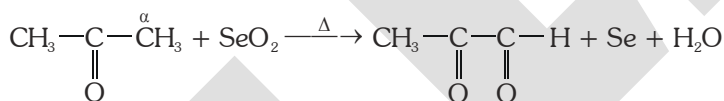


- (ii) **SeO_2 (Selenium dioxide)** : Ketones or aldehydes on oxidation with SeO_2 give dicarbonyl compounds. This reaction is possible only in compounds containing $\alpha-CH_2-$ unit.

HCHO doesn't show this reaction.



Glyoxal

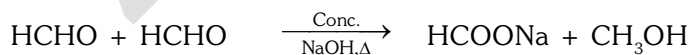


Methyl glyoxal (Pyruvaldehyde)

6.3.3 Reactions of only aldehydes :

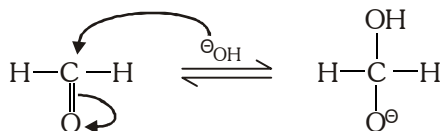
- (a) **Cannizzaro's reaction** : Those aldehydes which do not contain $\alpha-H$ atom give this reaction, with conc. NaOH or KOH; Products are Salt of carboxylic acid + alcohol

In this reaction one molecule of carbonyl compounds is oxidised to acid, while other is reduced to alcohol, such type of reactions are called redox reaction.

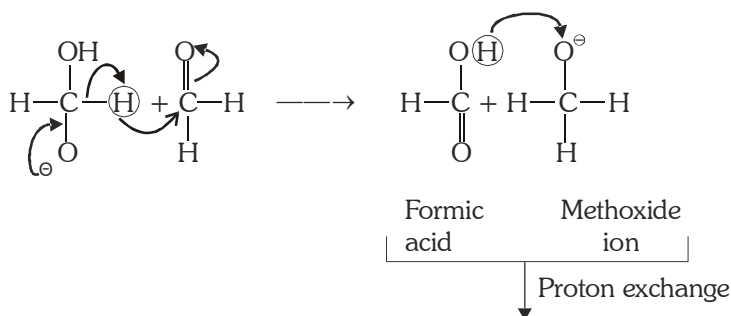


Mechanism involved in cannizzaro's reaction :

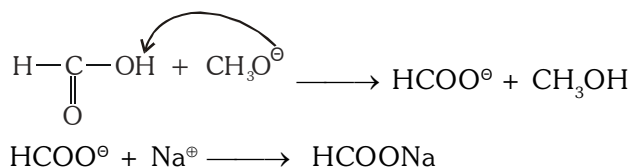
- (a) Rapid reversible addition of $\bar{O}H$ to one molecule of HCHO.



(b) Transfer of hydride ion H^- to second molecule of HCHO

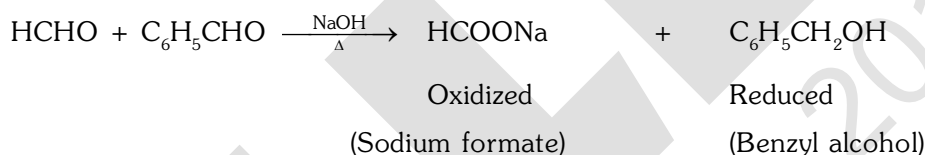


(c) Proton exchange

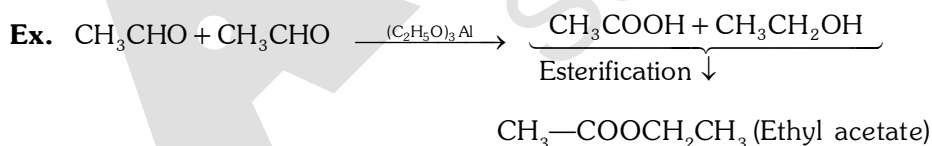
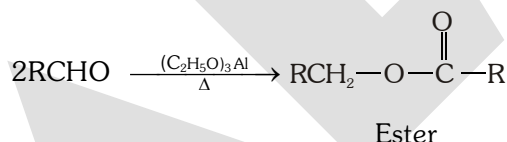


When molecules are same $\longrightarrow$ Simple cannizaro reaction (disproportionation reaction)
 Two different molecules $\longrightarrow$ Mixed cannizaro reaction (Redox reaction)

In mixed or crossed cannizaro reaction more reactive aldehyde is oxidised and less reactive aldehyde is reduced.

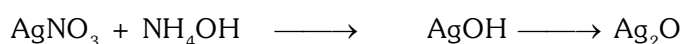
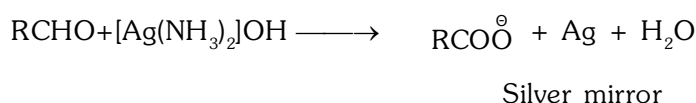


(b) Tischenko reaction : It is a modified cannizaro reaction. All aldehydes undergo this reaction in presence of $(\text{C}_2\text{H}_5\text{O})_3\text{Al}$, to form ester.



(c) Reducing character : Aldehydes are easily oxidised so they are strong reducing agents.

(i) Tollen's reagent : It oxidises aldehydes. Tollen's reagent is ammonical silver nitrate solution ($\text{AgNO}_3 + \text{NH}_4\text{OH}$) $\longrightarrow$ $[\text{Ag}(\text{NH}_3)_2]\text{OH}$

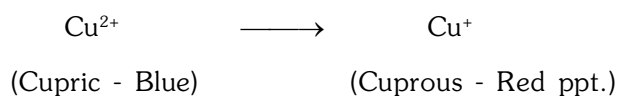
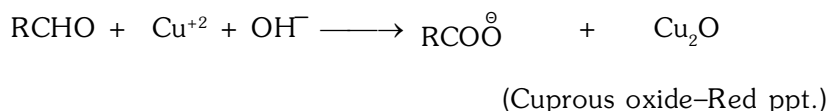


- (ii) **Fehling's solution :** It is a mixture of aqueous CuSO_4 , NaOH and sodium potassium tartarate.

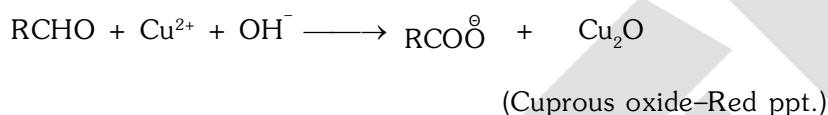
Fehling solution A- (aq.) solution of CuSO_4

Fehling solution B- Roschelle salt (Sodium potassium tartarate + NaOH)

Fehling solution A + Fehlings solution B(Dark blue colour of cupric tartarate)



- (iii) **Benedict's solution :** It is a mixture of CuSO_4 + sodium citrate + NaOH . It provides Cu^{+2} . It is reduced by aldehyde to give red ppt of cuprous oxide.

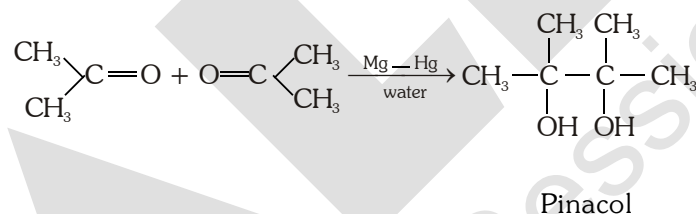


- (iv) **Schiff's reagent :** Dilute solution of p-rosaniline hydrochloride or magenta dye, is a pink coloured dye and is known as schiff' dye.

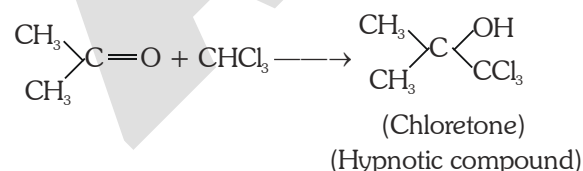
Its pink colour is discharged by passing SO_2 gas and the colourless solution obtained is called schiff's reagent, Aldehyde reacts with this reagent to restore the pink colour.

6.3.4 Reaction of Only Ketones

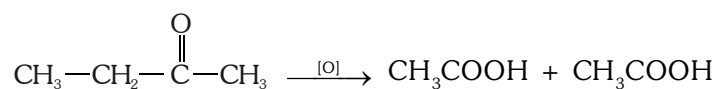
- (1) **Reduction :** Acetone is reduced by magnesium amalgam and water to give pinacol.



- (2) **Reaction with chloroform :**

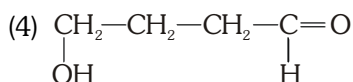
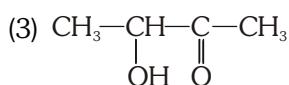
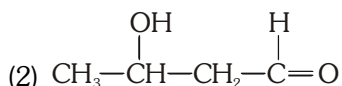
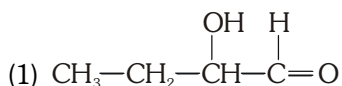


- (3) **Oxidation reaction :** According to popoff's rule $>\text{C}=\text{O}$ group stays with smaller alkyl group.



BEGINNER'S BOX-5

1. The compounds capable in reaction with Tollen's reagent is (are):
 (1) Formaldehyde (2) Formic acid (3) Acetaldehyde (4) All the above
2. Acetaldehyde reacts with NaOH to form :-



3. Fehling's solution is a :-
 (1) Solution of magenta dye bleached by SO_2
 (2) Ammonical solution of AgNO_3
 (3) Mixture of a solution of CuSO_4 and a solution of caustic soda and sodium potassium tartarate
 (4) Alcoholic solution of 2,4-dinitrophenylhydrazine

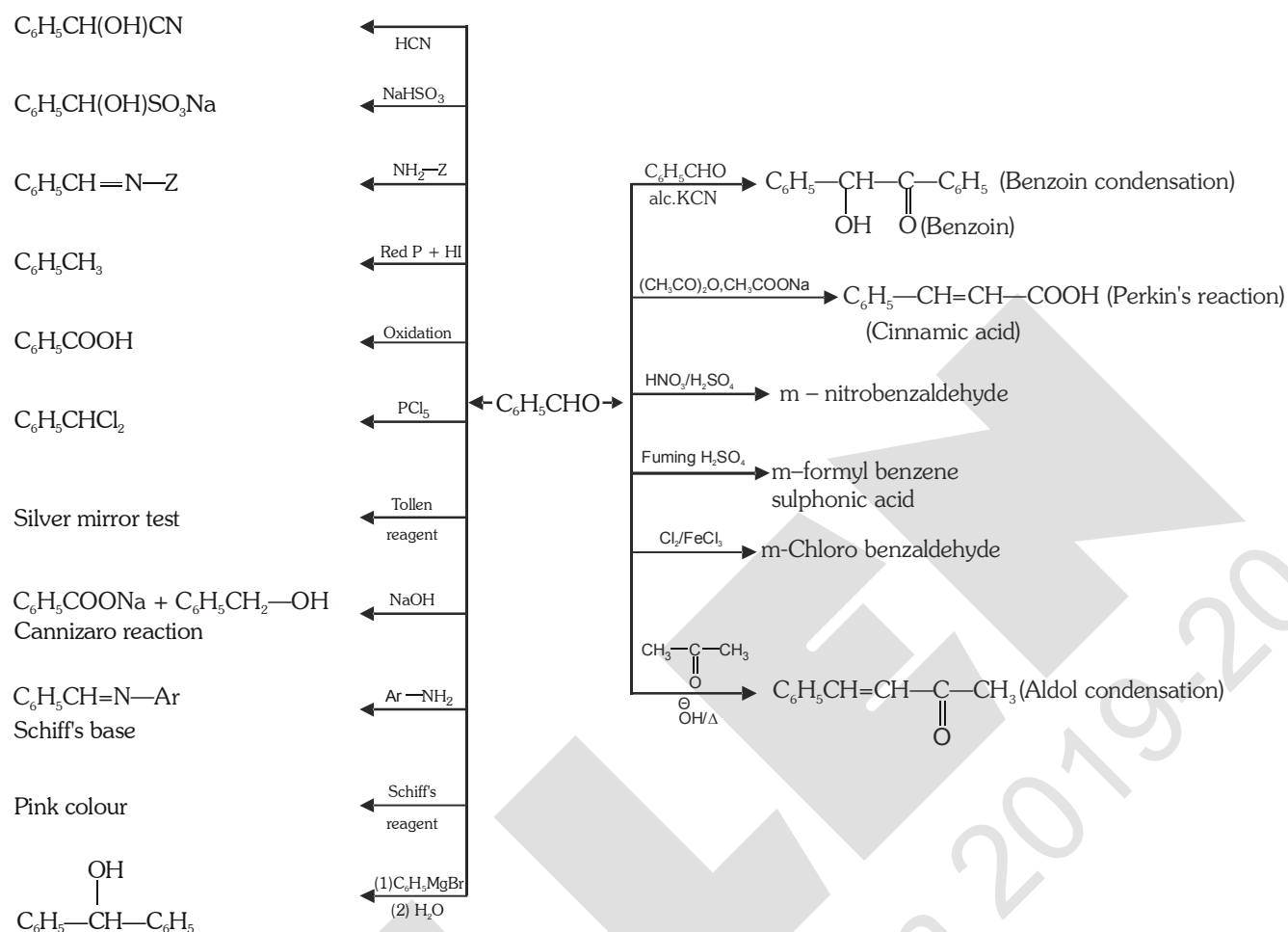
7.0 BENZALDEHYDE ($\text{C}_6\text{H}_5\text{CHO}$) [OIL OF BITTER ALMONDS (COMPONENT OF BITTER ALMOND)]

7.1 General Methods of Preparation

C_6H_6	$\text{CO/HCl} + \text{ZnCl}_2$	(Gattermann-koch aldehyde synthesis)
$\text{C}_6\text{H}_5\text{CH}_3$	(1) CrO_2Cl_2 (2) H_2O	(Etard reaction)
$\text{C}_6\text{H}_5\text{COCl}$	Pd/BaSO_4	(Rosenmund reaction)
$\text{C}_6\text{H}_5\text{CN}$	(i) SnCl_2/HCl (ii) H_2O	(Stephen's reaction)
$(\text{C}_6\text{H}_5\text{COO})_2\text{Ca}$	$(\text{HCOO})_2\text{Ca}/\Delta$	
$\text{C}_6\text{H}_5\text{CHCl}_2$	aq. KOH	
$\text{C}_6\text{H}_5\text{CH}_2\text{OH}$	$[\text{O}]$ (controlled oxidation)	
$\text{C}_6\text{H}_5\text{MgBr}$	HCOOC_2H_5	
C_6H_6	(1) $\text{HCN/HCl} + \text{ZnCl}_2$ (2) H_2O	(Gattermann aldehyde synthesis)

$\rightarrow \text{C}_6\text{H}_5\text{CHO}$

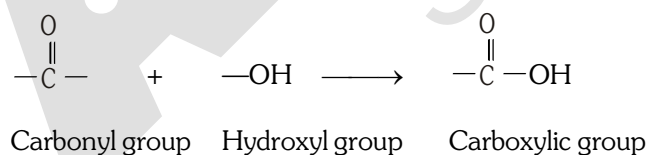
7.2 Chemical Properties



8.0 CARBOXYLIC ACID

Organic compounds having -COOH group are called Carboxylic acids. This functional group is composed of

Carbonyl (>C=O) and hydroxyl (-OH) group.



The properties of the carboxylic group are not simply the combined properties of these two groups, but it has its own distinctive properties. The acidic nature of carboxylic acids is due to the presence of replaceable H-atom in the Carboxylic group. The general formula is $C_nH_{2n}O_2$.

Classification :

Monocarboxylic acid (RCOOH) : Having one carboxylic group, also called monobasic acid.

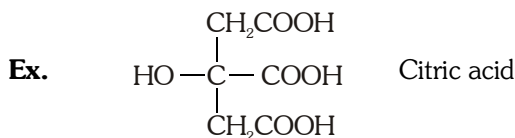
General formula - $C_nH_{2n}O_2$ ($n = 1, 2, 3, \dots$). Higher mono carboxylic acids are called **fatty acids**.

Ex. CH_3COOH acetic acid

Dicarboxylic acid : Having two carboxylic groups, also called dibasic acid.



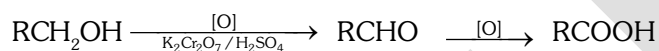
Tricarboxylic acid : Having three carboxylic groups also called tribasic acid.



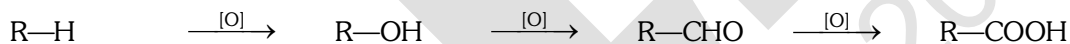
Structure : The carbon atom of $-\text{COOH}$ group is sp^2 hybridised, this C-atom is in centre and thus bond angle around C-atom is 120° .

8.1 General Methods of Preparation

8.1.1 By Oxidation of alcohols & carbonyl compounds : Oxidation is carried out by acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4 .



Acids are third oxidation products of alkane.



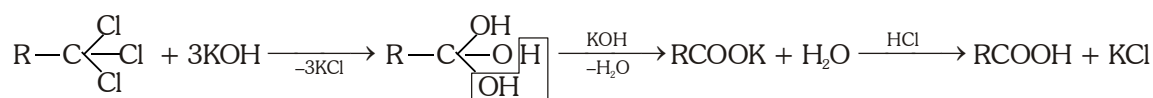
GOLDEN KEY POINTS

- Aldehyde on oxidation give acids having same no. of C-atoms.
- Ketones on oxidation give acids having less no. of C-atoms.
- Oxidation of Ketones occurs on prolonged treatment with strong oxidising agent.

8.1.2 By hydrolysis of alkane nitriles or cyanides : Complete hydrolysis takes place in acidic medium (dil. HCl). In alkaline medium there is partial hydrolysis.



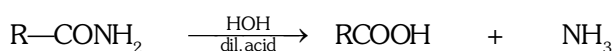
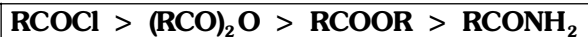
8.1.3 By alkaline hydrolysis of 1, 1, 1-trihaloalkane :



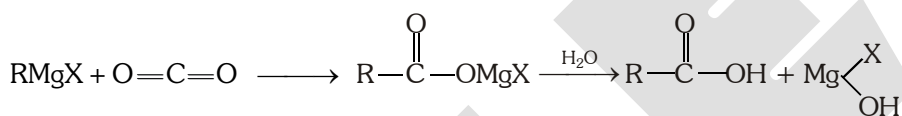
8.1.4 By hydrolysis of acid derivatives :



Reactivity order of acid derivatives :



8.1.5 From Grignard's reagent :



Carbon dioxide

Solid CO_2 (dry ice) is used

8.2 Physical Properties

Carboxylic acids from C_1-C_4 are completely soluble in water.

$$\text{Solubility} \propto \frac{1}{\text{molecular weight}}$$

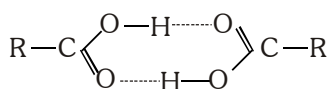
Solubility is due to intermolecular H - bonding with water molecules.

Boiling point : $B. P. \& M. P. \propto \text{Molecular weight,}$

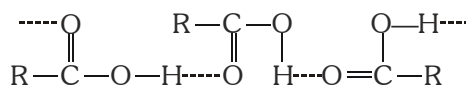
Acids > alcohol

This is because in acids two oxygen atoms take part in H - bonding (while in alcohol only one O - atom takes part).

In vapour or soluble state lower acids occur in the form of dimers.



But in liquid state it exists in polymer form.

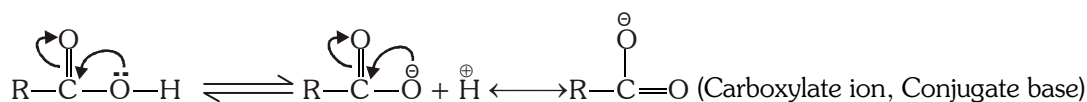


Melting point : Acids having even number of C-atoms have higher melting point as compared to having odd number of C - atoms. The carboxyl group and terminal methyl group in even member acids lie on opposite side to provide more close packing in crystal lattice which results in high melting point.

8.3 Chemical Properties

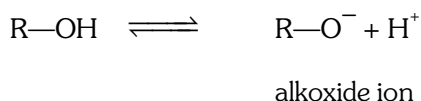
8.3.1 Reaction due to - H atom of - COOH group

(a) Acidic character :



Carboxylate ion is stabilised by resonance so carboxylic acids show considerable acidic character.

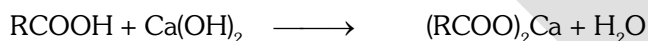
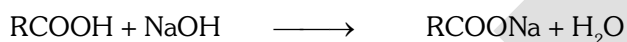
In case of alcohol alkoxide ion is not stabilised so they are neutral.



- (a) Carboxylic acids turn blue litmus to red.
 (b) Addition of carboxylic acid to NaHCO_3 in water gives out effervescence of CO_2 .



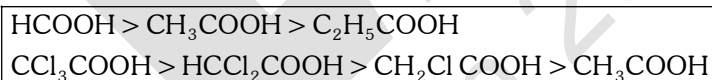
- (c) Form salts with alkalis.



- (d) Action of metals.



The acidic character order is :



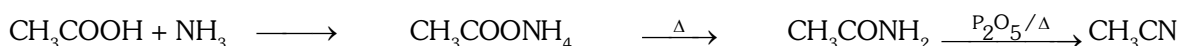
Acidic character may be explained on the basis of I effect and resonance.

- (b) **Reaction with CH_2N_2** : Methyl ester can be prepared by this method.

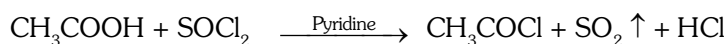
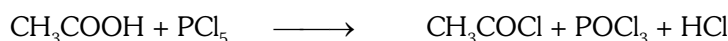


8.3.2 Reaction due to - OH group

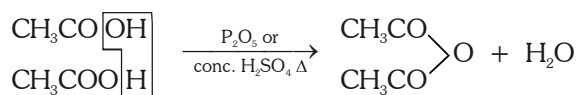
(a) Esterification :

(b) Reaction with NH_3 :

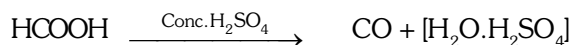
(c) Reaction with thionyl chloride :

(d) Reaction with PCl_5 :

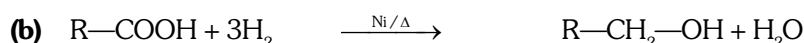
(e) **Reaction with P_2O_5 (dehydrating agent) :**



HCOOH is dehydrated by Conc. H_2SO_4



8.3.3 Reaction due to $\text{—}\overset{\text{O}}{\parallel}{\text{C}}\text{—}$ group of COOH :

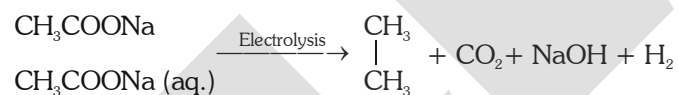


8.3.4 Reaction due to —COOH group :

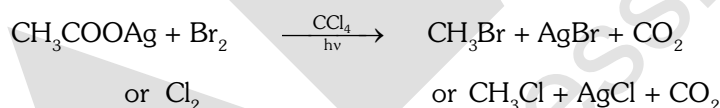
(a) **Decarboxylation :**



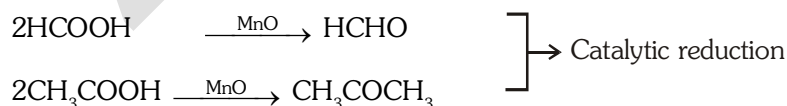
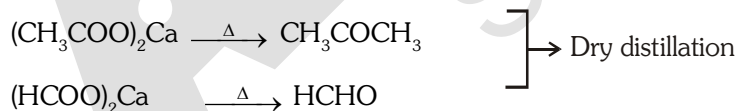
(b) **Kolbe's electrolysis :**



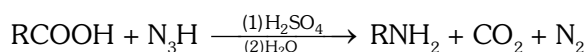
(c) **Hunsdiecker reaction :**



(d) **Formation of carbonyl compounds :**



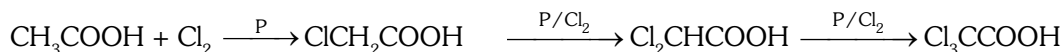
(e) **Schmidt reaction :** The amine formed has one C - atom less than the reactant acid.



Hydrazoic acid

8.3.5 Reaction due to alkyl (R) group :

(a) **Halogenation [HVZ reaction]** : Hell volhard Zelinsky reaction] : In this reaction α - H atoms are replaced by halogen atoms.



GOLDEN KEY POINTS

Test for HCOOH and CH₃COOH

Test	HCOOH	CH ₃ COOH
1. Reducing character		
Reducing agents -		
Tollen reagent	Silver mirror	×
Fehling solution	Cu ₂ O red ppt.	×
K ₂ Cr ₂ O ₇	Cr ⁺³ (green)	×
2. Decarboxylation.	Na ₂ CO ₃ + H ₂	CH ₄

Uses of Formic Acid :

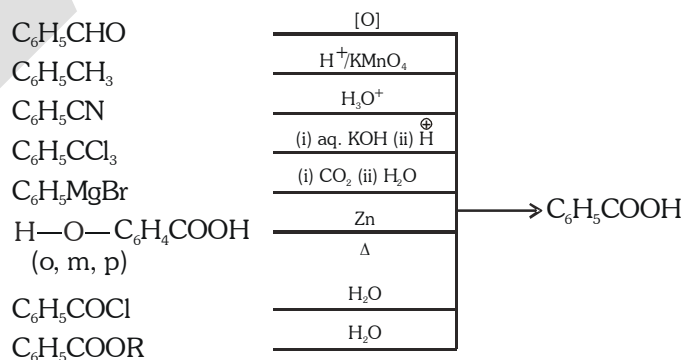
- (i) As an antiseptic
- (ii) For preservation of fruits.
- (iii) For leather tanning.
- (iv) In dyeing wool and cotton fabrics.

Uses of Acetic Acid :

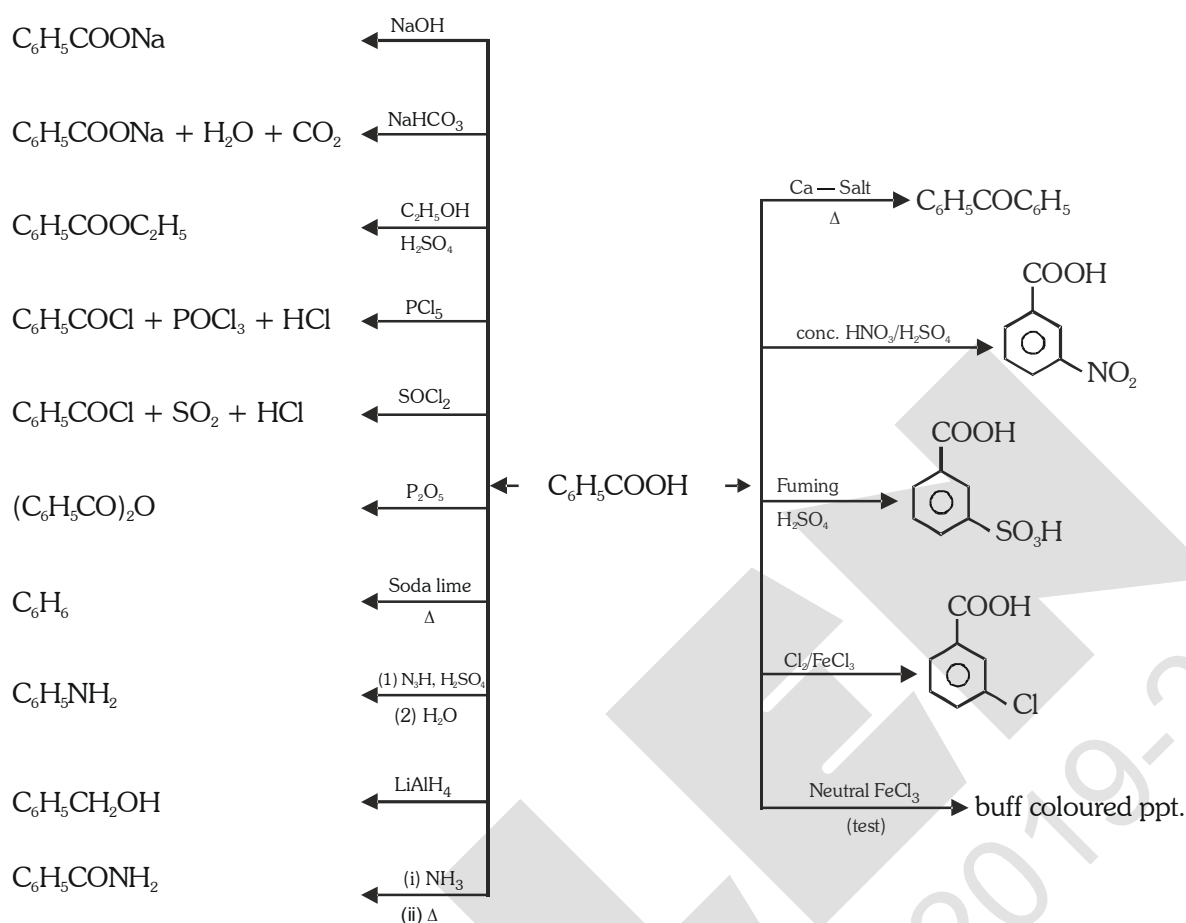
- (i) Vinegar (6 - 10% solution) used as **table acid** and manufacture of pickles.
- (ii) For manufacture of rubber from latex and casein from milk CH₃COOH is used as coagulant.
- (iii) As a solvent and laboratory reagent.

9.0 BENZOIC ACID (C₆H₅COOH)

9.1 General Method of Preparation :

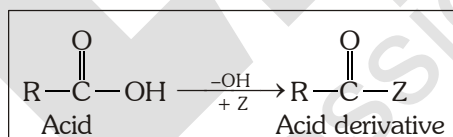


9.2 Chemical properties



10.0 ACID DERIVATIVES

Replacement of $-OH$ group from a carboxylic group ($-COOH$) by a nucleophile like Cl^- , CH_3COO^- , $C_2H_5O^-$, NH_2^- , forms acid derivatives.

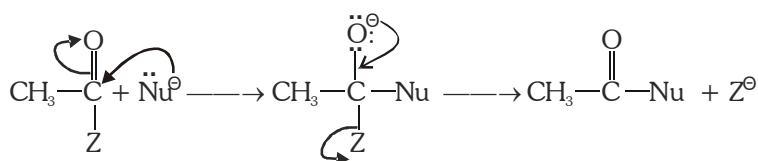


$R-\overset{\overset{O}{\parallel}}{C}-$ is Acyl group and Z is nucleophile $Cl^\ominus$, $CH_3COO^\ominus$, $C_2H_5O^\ominus$, $NH_2^\ominus$

Ex. $CH_3-\overset{\overset{O}{\parallel}}{C}-$ Acetyl group

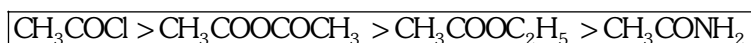
Characteristic reaction for acid derivatives is nucleophilic substitution reaction :

Mechanism :



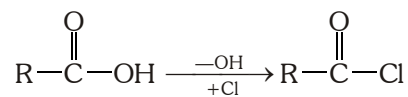
In this reaction Z is leaving group. Weak bases are good leaving groups.

Reactivity order :



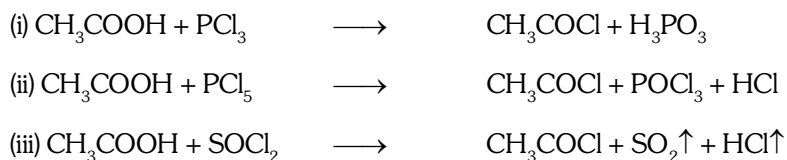
10.1 Acetyl Chloride

Replacement of —OH group from a —COOH group by Cl— atom gives acid chloride.

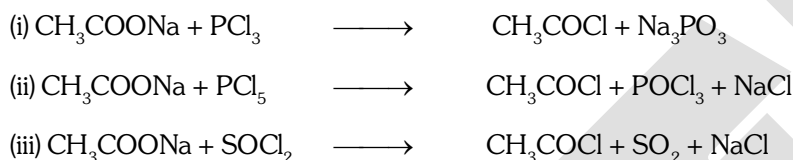


10.1.1 General Method of Preparation :

(a) By heating CH_3COOH with PCl_3 , PCl_5 & SOCl_2 :



(b) By heating the salt of acids with PCl_3 , PCl_5 or SOCl_2 :



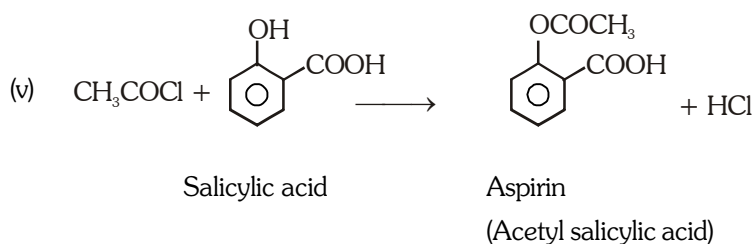
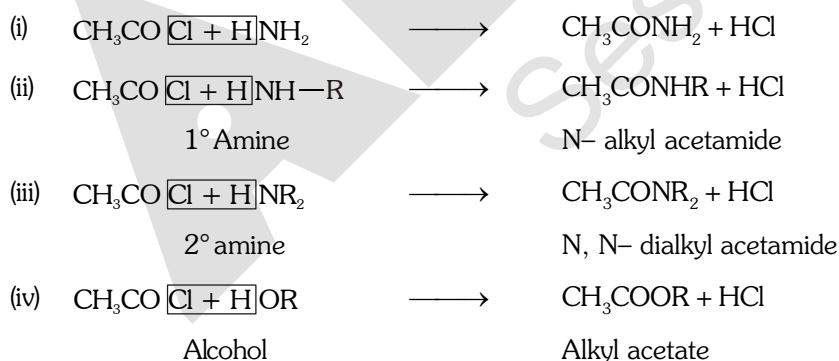
10.1.2 Physical properties :

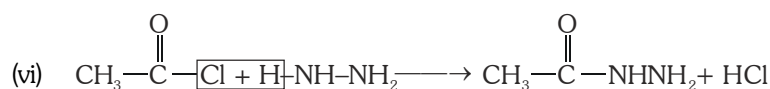
- (i) Pungent smelling liquid.
- (ii) Boiling point 52°C .
- (iii) Soluble in organic solvent, slowly soluble in water.
- (iv) It produces fumes in moist air due to the formation of HCl.

10.1.3 Chemical properties :

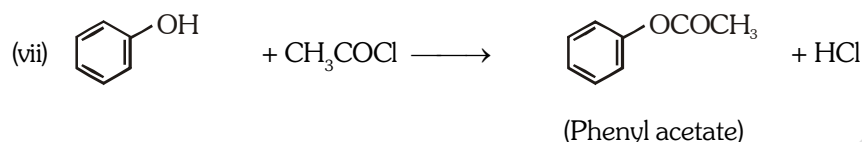
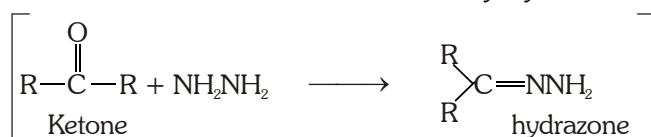
(a) Hydrolysis : $\text{CH}_3\text{CO}[\text{Cl} + \text{H}]\text{OH} \longrightarrow \text{CH}_3\text{COOH} + \text{HCl}$

(b) Reaction with active H - containing compounds (Acetylation) :

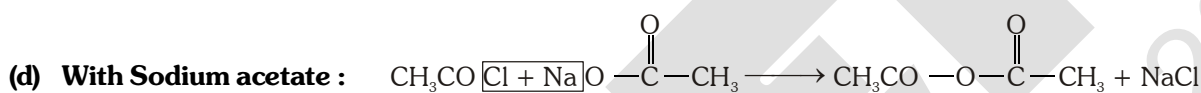
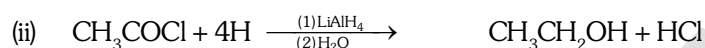




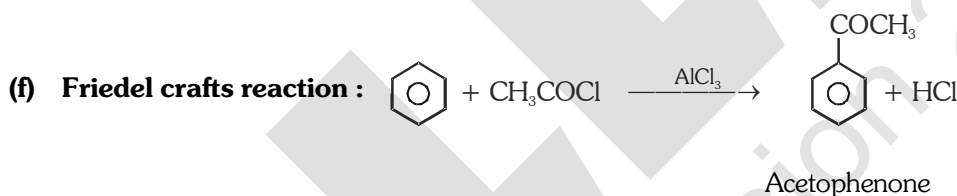
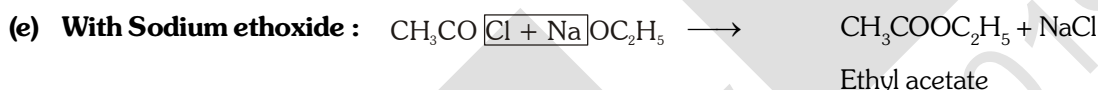
Acetyl hydrazide



(c) Reduction :



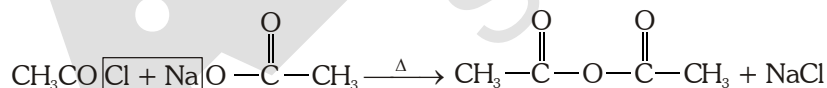
Acetic anhydride



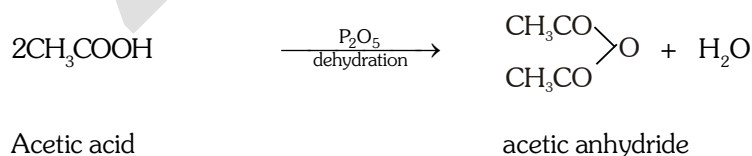
10.2 Acetic Anhydride (Ethanoic Anhydride)

10.2.1 General Method of Preparation :

(a) By heating acetyl Chloride with anhydrous sodium acetate [Lab. Method] :



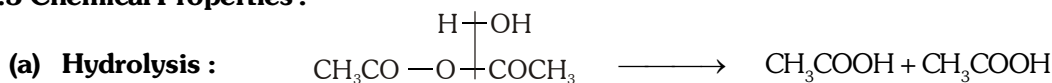
(b) By Dehydration of acetic acid :



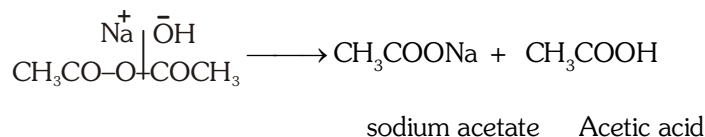
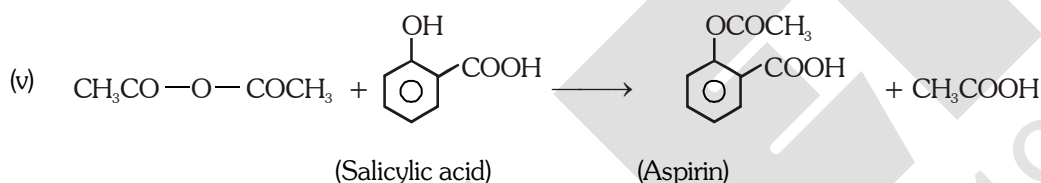
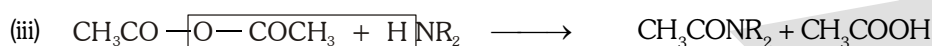
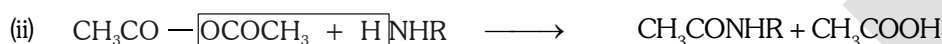
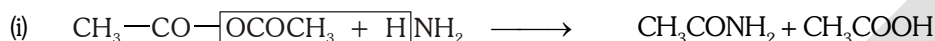
10.2.2 Physical Properties :

- (i) It is pungent smelling liquid.
- (ii) Sparingly soluble in water, soluble in ether & alcohol.
- (iii) Boiling point 139°C.

10.2.3 Chemical Properties :



Hydrolysis order : **Alkaline > Acidic > neutral**

(b) **Reaction with active H - containing compounds (Acetylation) :**(c) **Reduction :** With LiAlH_4 in ether gives ethyl alcohol(d) **Reaction with PCl_5 & SOCl_2 :**

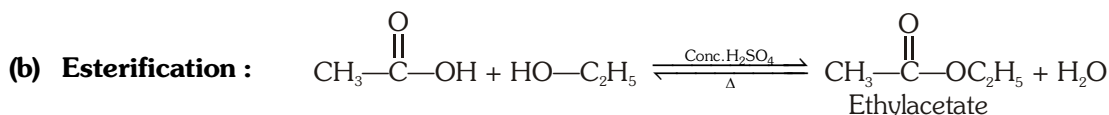
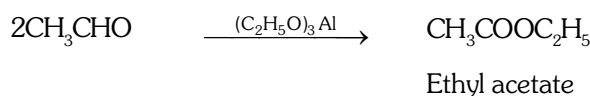
10.2.4 Uses

- (i) As an acetylating agent
- (ii) In the manufacture of cellulose acetate, aspirin, phenacetin, acetamide, & acetophenone etc.
- (iii) For detection and estimation of hydroxyl and amino group.

10.3 Ethyl acetate

10.3.1 General Method of Preparation :

(a) **Tischenko reaction :** By treating acetaldehyde with aluminium ethoxide. (Modified cannizaro reaction)



BEGINNER'S BOX-6

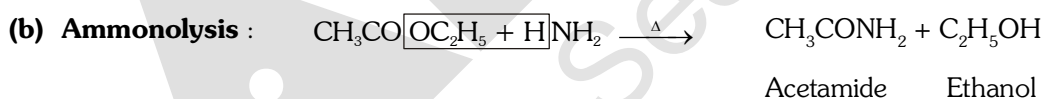
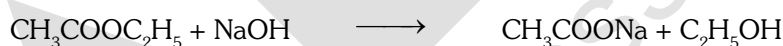
- Alkanoic acids can be prepared by hydrolysis of:-
 - (1) Trihaloalkanes
 - (2) 1,1,1-trihaloalkanes
 - (3) Grignard reagents
 - (4) Ketones
- Acids have much higher boiling points than isomeric esters because :-
 - (1) Acids form dimers by H-Bonding
 - (2) Acids are volatile in steam
 - (3) Esters are non-volatile
 - (4) Acids can ionise to give protons in aqueous solution
- Which of the following compounds can form intermolecular H-bonds :-
 - (1) Ethyl acetate
 - (2) Methyl formate
 - (3) Acetamide
 - (4) Acetic anhydride
- Reaction of ethyl acetate with sodium ethoxide gives acetoacetic ester. This reaction is known with the name of :-
 - (1) Claisen
 - (2) Clemmensen
 - (3) Cannizaro
 - (4) Etard

10.3.2 Physical Properties

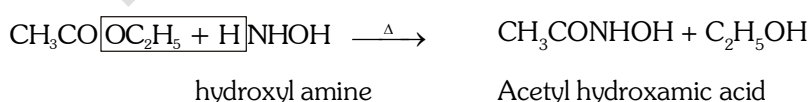
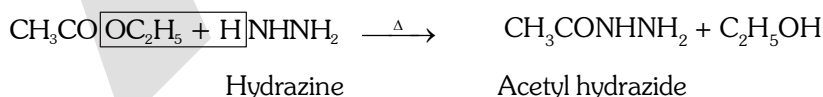
- Fruity smell liquid.
- Boiling point 77°C .
- Slightly soluble in water, soluble in organic solvent.

10.3.3 Chemical Properties :

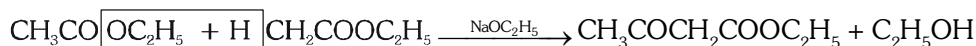
(a) **Hydrolysis** : In acidic medium reaction is reversible and in alkaline medium reaction is irreversible.



(c) **Reaction with NH_2NH_2 & NH_2OH :**



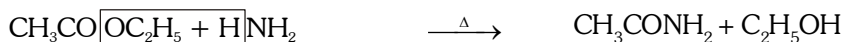
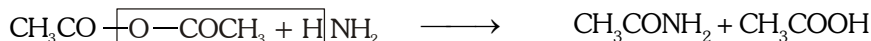
(e) **Claisen condensation** :



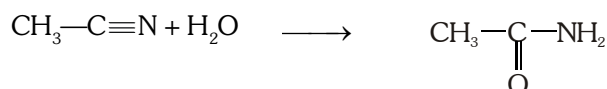
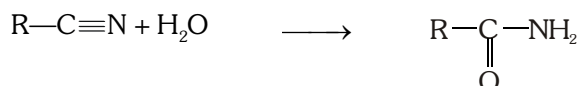
10.4 Acetamide

10.4.1 General Method of Preparation :

(a) Ammonolysis of acid derivatives :

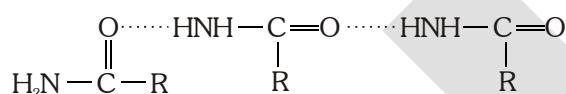


(b) Hydrolysis of alkyl cyanides : By the partial hydrolysis of alkyl cyanides.



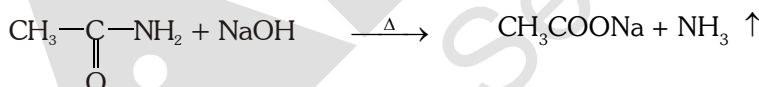
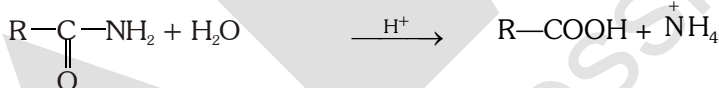
10.4.2 Physical Properties :

- Acetamide is white crystalline solid.
- Pure acetamide is odourless.
- Impure acetamide - Smell like mouse.
- Lower amides are soluble in water, due to H-bonding.
- Their higher M. P. and B. P. are due to intermolecular H-bonding.

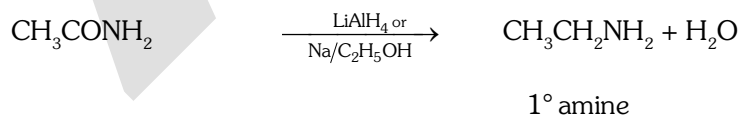


10.4.3 Chemical Properties :

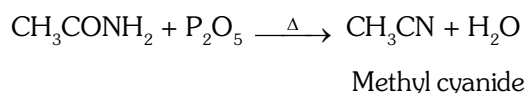
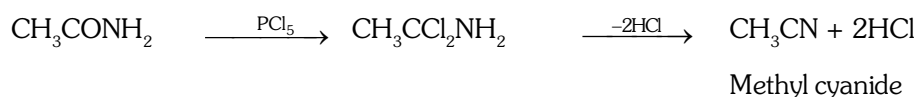
(a) **Hydrolysis** : Amides are hydrolysed rapidly in acidic medium. In alkaline medium hydrolysis is carried out in temperature condition.



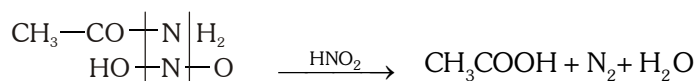
(b) **Reduction to primary amines** : Amine contains same number of carbon atoms.



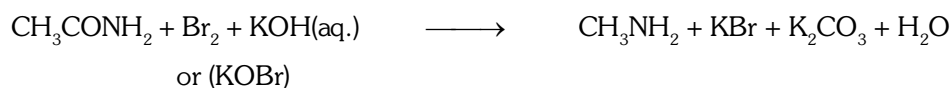
(c) **Dehydration with PCl₅, SOCl₂ or P₂O₅** :



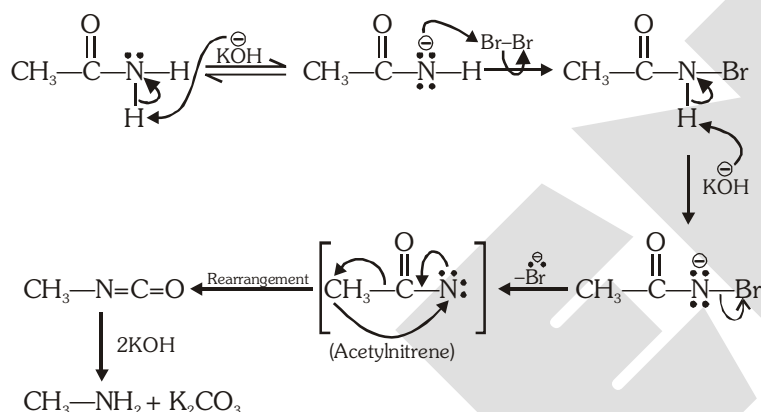
(d) Reaction with nitrous acid :



(e) **Hoffmann's bromamide degradation reaction :** Amides on reaction with bromine, and NaOH or KOH yield primary amines, having one C-atom less than the amides.



Mechanism :



ANSWER KEY

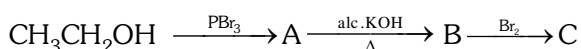
BEGINNER'S BOX-1	Que.	1	2	3						
	Ans.	3	4	2						
BEGINNER'S BOX-2	Que.	1	2							
	Ans.	1	2							
BEGINNER'S BOX-3	Que.	1	2	3						
	Ans.	3	2	3						
BEGINNER'S BOX-4	Que.	1	2	3						
	Ans.	2	4	1						
BEGINNER'S BOX-5	Que.	1	2	3						
	Ans.	4	2	3						
BEGINNER'S BOX-6	Que.	1	2	3	4					
	Ans.	2	1	3	1					

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

ALCOHOL

- 1.** The compound A, B and C in the reaction sequence



are given by the set

- (1) $\text{C}_2\text{H}_5\text{Br}$, $\text{CH}_3\text{CH}_2\text{OH}$, CH_3CHBr_2 .
- (2) $\text{C}_2\text{H}_5\text{Br}$, $\text{CH}\equiv\text{CH}$, $\text{CH}_2=\text{CHBr}$
- (3) $\text{C}_2\text{H}_5\text{Br}$, $\text{CH}_2=\text{CH}_2$, $\text{CH}_2\text{Br}-\text{CH}_2\text{Br}$
- (4) $\text{C}_2\text{H}_5\text{Br}$, $\text{CH}_3\text{CH}_2\text{OH}$, $\text{BrCH}_2-\text{CH}_2\text{Br}$

- 2.** Primary amines can be converted into alkanols by reaction with

- (1) Aqueous HCl (2) Tilden's reagent
(3) NaNO_2 and dil. HCl (4) Hypochlorous acid

- 3.** Which of the following alcohols gives a red colour in Victor Meyer test

- (1) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$ (2) $\begin{array}{c} \text{CH}_3\text{-CH-OH} \\ | \\ \text{CH}_3 \end{array}$
- (3) $(\text{CH}_3)_3\text{C-OH}$ (4) $\begin{array}{c} \text{CH}_3\text{-CH-CH}_2\text{-CH}_3 \\ | \\ \text{OH} \end{array}$

4. Methanol on heating with salicylic acid and a few drops of conc. H_2SO_4 gives the smell of

- (1) Bitter almonds (2) Oil of wintergreen
(3) Rotten eggs (4) Mustard oil

5. Which of the following does not turn orange colour of chromic acid to green

- (1) 1° alcohol (2) 2° alcohol
(3) 3° alcohol (4) Allyl alcohol

- 6.** p, s and t-alcohols can be distinguished by :-

- (1) Reimer-Tiemann reaction
- (2) Tollen's reagent
- (3) Lucas test
- (4) Lassaigne's test

- 7.** Consider the following reaction :

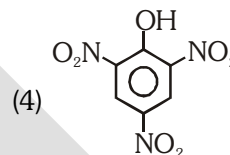
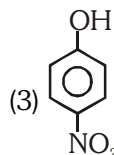
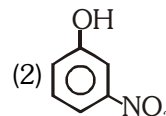
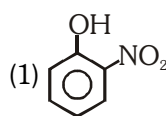


Among the following, which one cannot be formed as a product under any conditions ?

- (1) $\text{C}_2\text{H}_5\text{OSO}_3\text{H}$
- (2) $\text{H}_2\text{C}=\text{CH}_2$
- (3) $\text{HC}\equiv\text{CH}$
- (4) $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$

PHENOL

- 8.** Nitration of phenol with conc. nitric acid gives :-

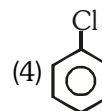
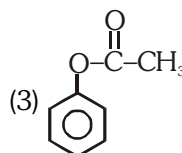
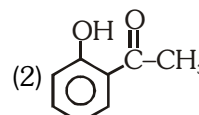
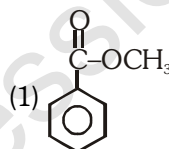
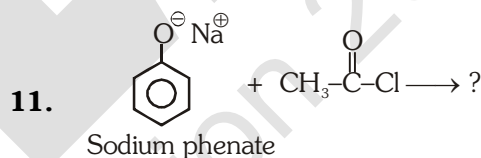


- 9.** Deoxygenation of phenol can be achieved by distillation with :-

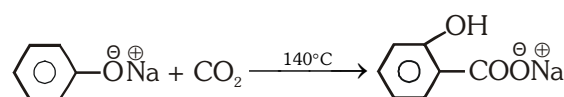
- (1) Raney nickel
- (2) Lithium aluminium hydride
- (3) Sodium borohydride
- (4) Zinc dust

- 10.** Which of the following compounds shows intramolecular hydrogen bonding :-

- (1) p-Nitrophenol (2) Ethanol
(3) o-Nitrophenol (4) Methanamine



- 12.** The reaction

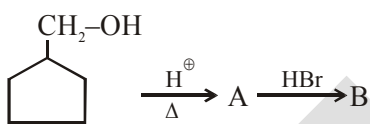
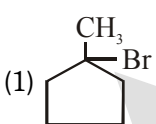
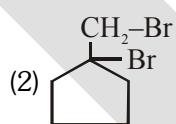
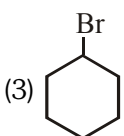
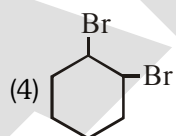
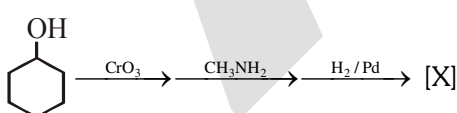
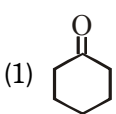
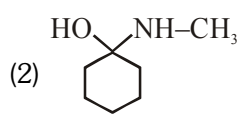
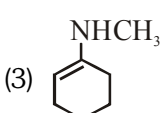
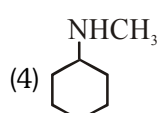


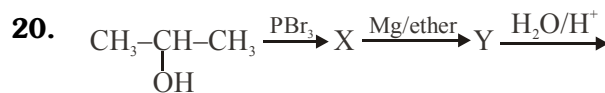
is called :-

- (1) Schotten Bauman reaction
- (2) Kolbe Schmidt reaction
- (3) Reimer–Tiemann reaction
- (4) Lederer–Manasse reaction

13. Phenol can be distinguished from ethanol by reactions with the following except :-
 (1) Iodine and alkali (2) Ferric chloride
 (3) Acetyl chloride (4) Bromine water
14. Phenol on treatment with methyl chloride in the presence of anhydrous AlCl_3 gives chiefly :-
 (1) o-cresol (2) m-cresol
 (3) anisole (4) p-cresol
15. Phenol on heating with NaNO_2 and a few drops of conc. H_2SO_4 mainly gives :-
 (1) p-nitrophenol (2) p-nitrosophenol
 (3) o-nitrophenol (4) m-nitrosophenol
16. Phenol and benzoic acid are distinguished by :-
 (1) Lucas reagent (2) Victor Meyer test
 (3) Caustic soda (4) Sodium bicarbonate
17. Phenol can be distinguished from ethanol by the reaction with
 (1) Na (2) Br_2/Water
 (3) Neutral AlCl_3 (4) 1 and 2 Both

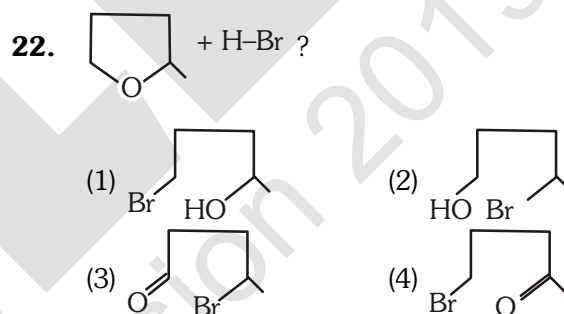
ETHER

18. 
 What is the structure of B :-
 (1)  (2) 
 (3)  (4) 
19. 
 Product [X] of given reaction is :-
 (1)  (2) 
 (3)  (4) 



The final product is :-

- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{OH}$
 (2) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{O}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
 (4) $\text{CH}_3-\text{CH}_2-\text{CH}_3$
21. In the reaction sequence
 $\text{A} \xrightarrow{\text{HBr}} \text{B} \xrightarrow{\text{C}_2\text{H}_5\text{ONa}} \text{Ethoxyethane}$,
 A and B are :-
 (1) C_2H_6 , $\text{C}_2\text{H}_5\text{Br}$
 (2) CH_4 , CH_3Br
 (3) $\text{CH}_2=\text{CH}_2$, $\text{C}_2\text{H}_5\text{Br}$
 (4) $\text{CH}\equiv\text{CH}$, $\text{CH}_2=\text{CHBr}$



23. $\text{CH}_3-\text{CH}_2-\text{OH} + \text{Ph}-\text{CH}_2-\text{OH} \xrightarrow[140^\circ\text{C}]{\text{H}^+}$ of which
 is not obtained?
 (1) $\text{CH}_3-\text{CH}_2-\text{OCH}_2-\text{CH}_3$
 (2) $\text{Ph}-\text{CH}_2-\text{OCH}_2-\text{Ph}$
 (3) $\text{Ph}-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$
 (4) $\text{Ph}-\text{CH}_2-\text{O}-\text{CH}_2-\text{O}-\text{CH}_3$

24. Oxonium ion of ether has the structure :-

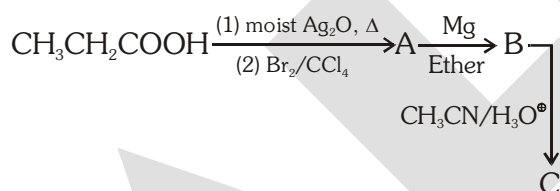
- (1) $\text{C}_2\text{H}_5-\text{O}-\underset{\text{CH}_3}{\text{CH}}-\overset{\oplus}{\text{O}}-\text{H}$
 (2) $\text{CH}_3-\text{CH}_2-\overset{\oplus}{\text{O}}-\underset{\text{H}}{\text{CH}}_2-\text{CH}_3$
 (3) $(\text{C}_2\text{H}_5)_2\text{O} \rightarrow \text{O}$
 (4) $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_2-\overset{\oplus}{\text{O}}-\text{H}$

25. What will happen when diethyl ether treated with hot and conc. HI :-
 (1) Ethyl alcohol (2) Ethyl iodide
 (3) Methyl iodide (4) Methanol

26. Which of the following does not react with aq. NaOH :-
 (1) Phenol (2) Benzoic acid
 (3) CH_3COOH (4) $\text{CH}_3\text{-O-C}_6\text{H}_5$

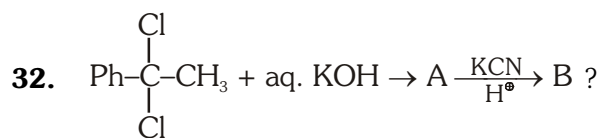
CARBONYL COMPOUNDS

27. Acetaldehyde on warming with Fehling's solution gives a red precipitate of :-
 (1) Elemental copper
 (2) Cuprous oxide
 (3) Cupric oxide
 (4) Mixture of all of the above
28. Acetone does not form :-
 (1) A phenylhydrazone with phenylhydrazine
 (2) A sodium bisulphite adduct with sodium bisulphite
 (3) A silver mirror with Tollen's reagent
 (4) An oxime with hydroxylamine
29. CH_3CHO and CH_3COCH_3 can not be distinguished by :-
 (1) Fehling solution (2) Grignard reagent
 (3) Schiff's reagent (4) Tollen's reagent
30. In the reaction

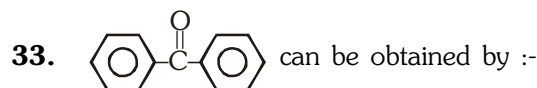


The product C is -

- (1) $\text{CH}_3\text{CH}_2\text{-CH(OH)-CH}_3$
 (2) $\text{CH}_3\text{CH}_2\text{-CN}$
 (3) $\text{CH}_3\text{CH}_2\text{-C(=O)-CH}_3$
 (4) $\text{CH}_3\text{-CH}_2\text{-CH(COOH)-CH}_3$
31. Acetone is obtained by the hydrolysis of the addition product of methyl magnesium iodide and :-
 (1) HCHO (2) CH_3CHO
 (3) CH_3COCH_3 (4) $\text{CH}_3\text{-C}\equiv\text{N}$



- (1) 50% d + 50% l (2) 80% d + 20% l
 (3) Meso compound (4) optically active



- (1) $\text{C}_6\text{H}_5\text{-COCl} + (\text{Ph})_2\text{Cd}$
 (2) $\text{C}_6\text{H}_5\text{-COCl} + \text{C}_6\text{H}_5\text{OH}$
 (3) $\text{C}_6\text{H}_6 + \text{CO} + \text{ZnCl}_2 + \text{HCN}$
 (4) None of the above

34. Which does not react with NaHSO_3 .

- (1) Ph-C(=O)-H
 (2) $\text{CH}_3\text{-O-C}_6\text{H}_4\text{-C(=O)-H}$
 (3) $\text{CH}_3\text{-C}_6\text{H}_4\text{-CH}_2\text{-C(=O)-Ph}$
 (4) $\text{C}_6\text{H}_{11}\text{CHO}$

35. Ketones can be prepared by :-

- (1) Rosenmund reduction
 (2) Etard reaction
 (3) Cannizzaro reaction
 (4) Friedel-Craft reaction

36. Carbonyl compounds are best purified by :-

- (1) Steam distillation
 (2) Hydrolysis of sodium bisulphite adducts
 (3) Fractional crystallisation
 (4) Sublimation

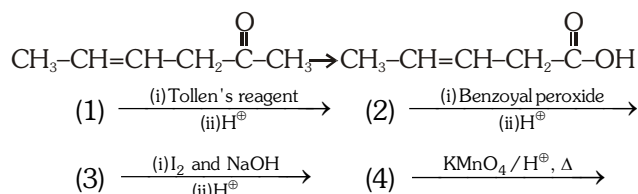
37. Carbonyl compounds readily undergo :-

- (1) Nucleophilic substitutions
 (2) Electrophilic addition reactions
 (3) Nucleophilic addition reactions
 (4) Free radical substitution reactions

38. $\text{CH}_3\text{-}\overset{\text{O}}{\parallel}\text{C-CH}_3$ and $\text{CH}_3\text{-}\overset{\text{O}}{\parallel}\text{C-H}$ are readily distinguished by their reaction with :-
 (1) Iodine and alkali
 (2) 2,4-dinitrophenylhydrazine
 (3) Tollen's reagent
 (4) All the above

39. Formaldehyde and acetaldehyde are readily distinguished by reaction with :-
 (1) A solution of 2,4-dinitrophenylhydrazine
 (2) Fehling's solution
 (3) Tollen's reagent
 (4) Iodine and alkali

40. Which is the most suitable reagent for the following conversion

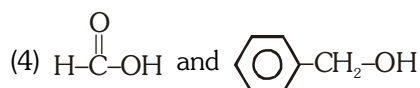
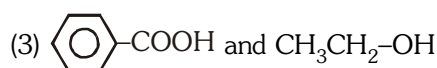
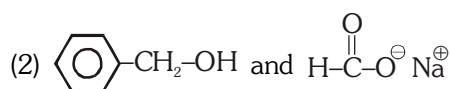
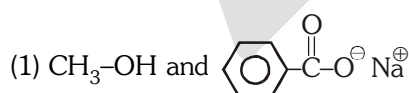


41. Formaldehyde reacts with conc. alkali to form :-
 (1) A resinous mass
 (2) Formic acid
 (3) A mixture of methanol and sodium formate
 (4) Methanol

42. Which of the following compounds does not give aldol condensation :-
 (1) CH_3CHO (2) $\text{CH}_3\text{CH}_2\text{CHO}$
 (3) HCHO (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$

43. Cannizzaro reaction is given by :-
 (1) Aldehydes containing α -hydrogen atoms
 (2) Aldehydes as well as ketones containing α -hydrogen atoms
 (3) Aldehydes not containing α -hydrogen atoms
 (4) Aldehydes containing β -hydrogen atoms

44. Benzaldehyde reacts with formaldehyde in the presence of alkali to form :-



45. Which of the following can be converted to $\text{CH}_3\text{-CH=CH-CHO}$:-

- (1) Acetone (2) Acetaldehyde
 (3) Propanaldehyde (4) Formaldehyde

46. The product of reaction with primary amine and aldehyde is -

- (1) $\text{R}-\overset{\text{O}}{\parallel}\text{C-OH}$ (2) R-ONO
 (3) R'-CH=N-R (4) R-NO_2

47. Brady's reagent is

- (1) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
 (2) $\text{KMnO}_4/\text{NaIO}_4$
 (3) $\text{O}_2\text{N}-\text{C}_6\text{H}_3(\text{NO}_2)_2\text{-NH.NH}_2$
 (4) $\text{O}_2\text{N}-\text{C}_6\text{H}_3(\text{NO}_2)_2\text{-F}$

48. A compound with molecular formula $\text{C}_3\text{H}_6\text{O}$, not gives silver mirror with Tollen's reagent but forms oxime with hydroxyl amine. Compound will be -

- (1) $\text{CH}_2=\text{CH-CH}_2\text{-OH}$
 (2) $\text{CH}_3\text{CH}_2\text{CHO}$
 (3) $\text{CH}_2=\text{CH-O-CH}_3$
 (4) CH_3COCH_3

49. Aldehyde and ketone are distinguished by reagent

- (1) Fehling solution (2) H_2SO_4
 (3) NaHSO_3 (4) NH_3

50. Carbonyl group is converted into methylene group by -

- (1) Acidic reduction
 (2) Raney Ni
 (3) Basic hydrolysis
 (4) Normal Hydrogenation

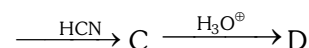
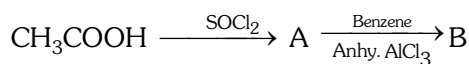
51. When acetaldol is treated with excess of acid then unsaturated product will be :-

- (1) Alcohol (2) Aldehyde
 (3) Acid (4) Alkyl halide

52. The reagent used for the separation of acetaldehyde from acetophenone is -

- (1) NaHSO_3 (2) $\text{C}_6\text{H}_5\text{NHNH}_2$
 (3) NH_2OH (4) $\text{NaOH} + \text{I}_2$

61. In a set of reactions acetic acid yielded a product D



The structure of D would be –

- (1)
- (2)
- (3)
- (4)

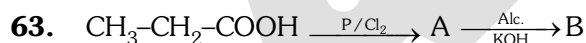
ACID DERIVATIVE

62. The compounds A and B in the reaction sequence



are given by the set :-

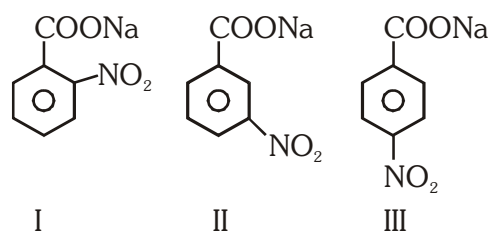
- (1) $\text{CH}_3\text{CO}-\text{O}-\text{COCH}_3$, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
 (2) $\text{CH}_3\text{CO}-\text{O}-\text{COCH}_3$, $\text{C}_6\text{H}_5\text{OCOCH}_3$
 (3) CH_3COCH_3 , $\text{C}_6\text{H}_5\text{OCOCH}_3$
 (4) $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{O}-\text{C}(=\text{O})-\text{CH}_3$,
 $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{C}_6\text{H}_5$



structure of B is :-

- (1) $\text{CH}_2=\text{CH}-\text{COOH}$
 (2) $\text{CH}_3-\text{CH}(\text{Cl})-\text{COOH}$
 (3) $\text{CH}_2(\text{Cl})-\text{CH}_2-\text{COOH}$
 (4) $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{Cl}$

64. Correct reactivity order of decarboxylation is



- (1) I > II > III
 (2) I > III > II
 (3) III > II > I
 (4) III > I > II

65. Which is most reactive towards hydrolysis.

- (1)
- (2)
- (3)
- (4)

66. Which of the following reagents may be used to distinguish between phenol and benzoic acid ?

- (1) Victor-Mayer test
 (2) Neutral FeCl_3
 (3) Aqueous NaOH
 (4) Tollen's reagent

67. Acyl chlorides undergo :-

- (1) Nucleophilic addition reactions
 (2) Nucleophilic substitution reactions
 (3) Electrophilic substitution reactions
 (4) Electrophilic addition reactions

68. The reaction of ethanol on acetic anhydride is an example of :-

- (1) Nucleophilic addition
 (2) Nucleophilic substitution
 (3) Electrophilic addition
 (4) Free radical substitution

69. The reduction of acetamide gives :-

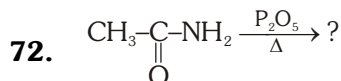
- (1) $\text{CH}_3\text{CH}_2\text{NH}_2$
 (2) $(\text{CH}_3)_2\text{CHNH}_2$
 (3) $(\text{CH}_3)_3\text{CNH}_2$
 (4) $(\text{CH}_3\text{CH}_2)_2\text{NH}$

70. Which is used in preparation of aldehyde by rosenmund reduction

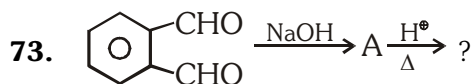
- (1) Ester (2) Acid
 (3) Acid halide (4) Alcohol

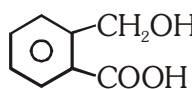
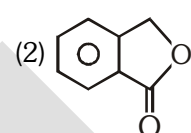
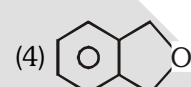
71. Which is maximum reactive :-

- (1) CH_3CONH_2 (2) $\text{CH}_3\text{COOCH}_3$
 (3) CH_3COCl (4) $\text{CH}_3\text{COOC}_2\text{H}_5$



- (1) CH_3COOH (2) CH_3CN
 (3) CH_3CH_3 (4) CH_3CHO



- (1)  (2) 
 (3)  (4) 

EXERCISE-I (Conceptual Questions)

ANSWER KEY

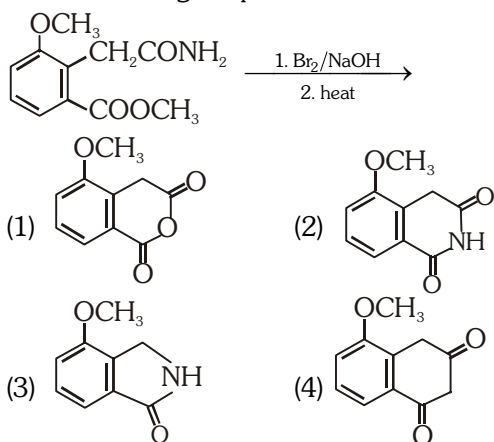
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	1	2	3	3	3	4	4	3	3	2	3	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	3	4	4	3	1	4	2	2	4	2	3	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	1	1	3	4	2	3	3	4	3	3	3	3	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	3	4	1	1	2	1	4	2	2	2	2	1	2	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73		
Ans.	1	2	1	1	2	2	2	2	1	3	3	2	2		

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

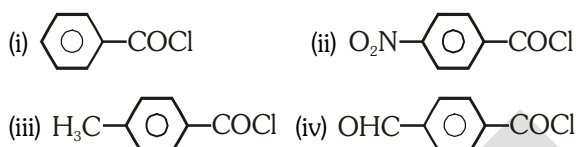
AIIMS 2006

1. The following sequence of reactions gives



AIPMT 2007

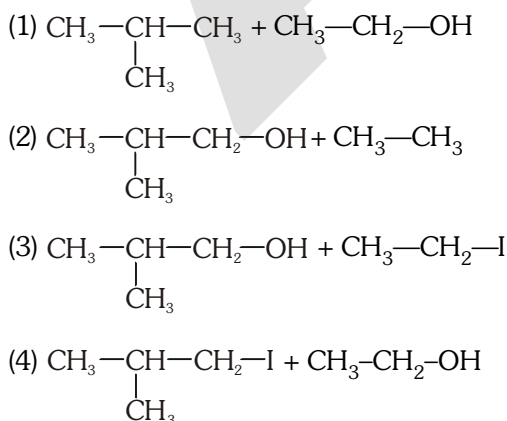
2. Consider the following compounds:



The correct order of reactivity towards hydrolysis is:-

- (1) (i) > (ii) > (iii) > (iv)
 (2) (iv) > (ii) > (i) > (iii)
- (3) (ii) > (iv) > (i) > (iii)
 (4) (ii) > (iv) > (iii) > (i)
3. Which one of the following on treatment with 50% aq. NaOH yields the corresponding alcohol and acid
- (1) Cc1ccccc1C=O
 (2) CCCC=O
- (3) CC(=O)C
 (4) CCC=O

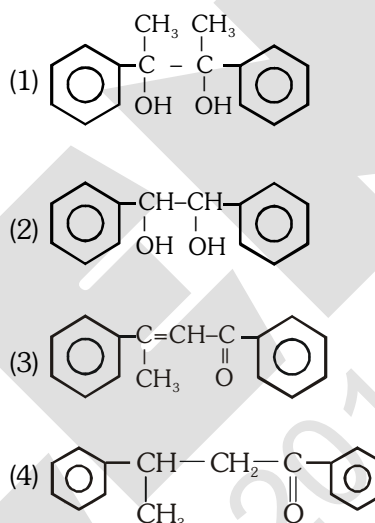
4. In the reaction : CC(C)COCC + HI $\longrightarrow$ which of following compounds will be formed :



5. The product formed in aldol reaction is :-
- (1) a β -hydroxy aldehyde or ketone
 (2) an α -hydroxy aldehyde or ketone
 (3) an α , β -unsaturated ester
 (4) a β -hydroxy acid

AIPMT 2008

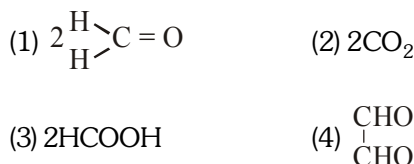
6. Acetophenone when reacted with a base, C2H5ONa, yields a stable compound which has the structure :-



7. A strong base can abstract an α -hydrogen from
- (1) Ketone
 (2) Alkane
- (3) Alkene
 (4) Amine

AIPMT 2009

8. H2COH.CH2OH on heating with periodic acid gives :-



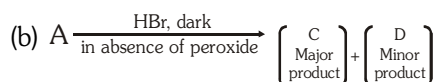
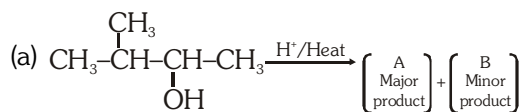
9. Consider the following reaction,
- ethanol $\xrightarrow{\text{PBr}_3}$ X $\xrightarrow{\text{alc. KOH}}$ Y
- (i) H2SO4 room temperature
 (ii) H2O
- $\longrightarrow$ Z ;

the product Z is :-

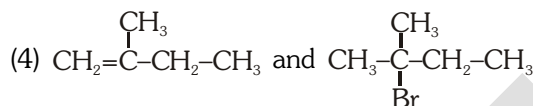
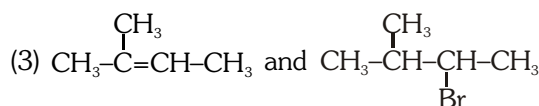
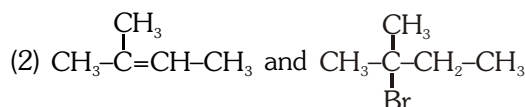
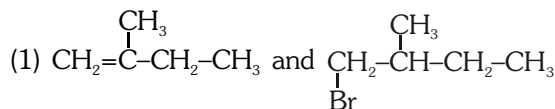
- (1) CH3CH2OH
 (2) CH2=CH2
 (3) CH3CH2-O-CH2-CH3
 (4) CH3-CH2-O-SO3H

AIPMT Pre. 2011

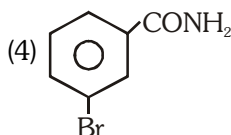
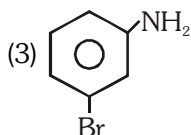
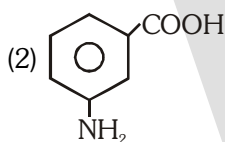
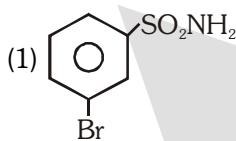
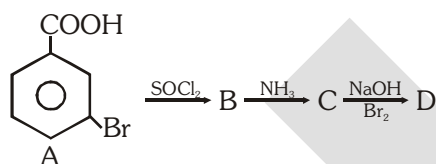
18. In the following reactions,



the major products (A) and (C) are respectively :-



19. In a set of reactions m-bromobenzoic acid gave a product D. Identify the product D



20. Clemmensen reduction of a ketone is carried out in the presence of which of the following ?

- (1) Glycol with KOH
- (2) Zn-Hg with HCl
- (3) LiAlH_4
- (4) H_2 and Pt as catalyst

AIPMT Mains 2011

21. An organic compound 'A' on treatment with NH_3 gives 'B' which on heating gives 'C'. 'C' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is :-

- (1) $\text{CH}_3\text{CH}_2\text{COOH}$
- (2) CH_3COOH
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- (4) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}\text{COOH}$

22. Match the compounds given in List-I with List-II and select the suitable option using the code given below.

List-I	List-II
(a) Benzaldehyde	(i) Phenolphthalein
(b) Phthalic anhydride	(ii) Benzoin condensation
(c) Phenyl benzoate	(iii) Oil of wintergreen
(d) Methyl salicylate	(iv) Fries rearrangement

Code :

(a)	(b)	(c)	(d)
(1) (ii)	(i)	(iv)	(iii)
(2) (iv)	(i)	(iii)	(ii)
(3) (iv)	(ii)	(iii)	(i)
(4) (ii)	(iii)	(iv)	(i)

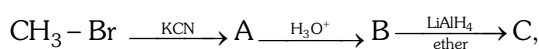
AIIMS 2011

23. Decarboxylation occurs with maximum rate in:

- (1) CH_3COOH
- (2) $\text{C}_6\text{H}_5-\text{COOH}$
- (3) $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$
- (4) $\text{CH}_3\text{COCH}_2\text{COOH}$

AIPMT Pre. 2012

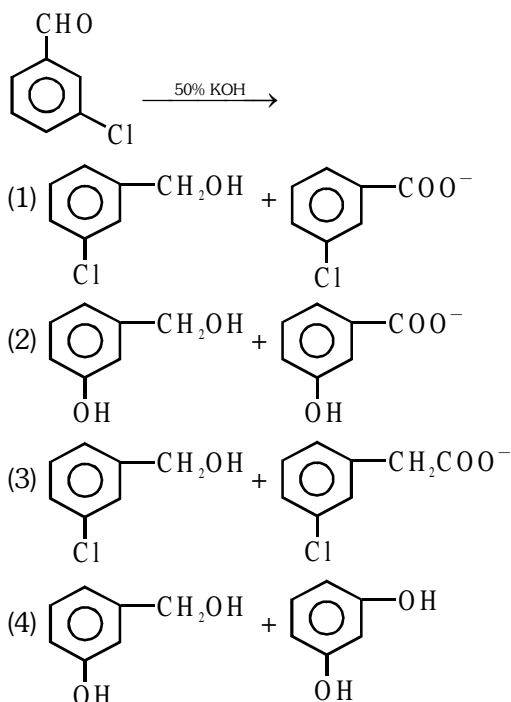
24. In the following sequence of reactions



the end product (C) is:

- (1) Acetaldehyde
- (2) Ethyl alcohol
- (3) Acetone
- (4) Methane

25. Predict the products in the given reaction.

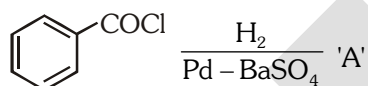


26. CH_3CHO and $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$ can be distinguished chemically by:

- (1) Tollen's reagent test (2) Fehling solution test
(3) Benedict test (4) Iodoform test

AIPMT Mains 2012

27. Consider the following reaction :



The product 'A' is :

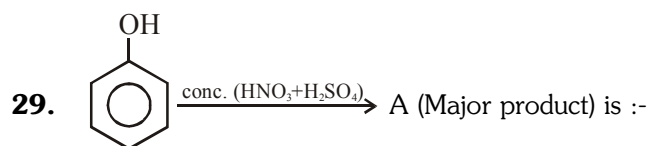
- (1) $\text{C}_6\text{H}_5\text{COCH}_3$ (2) $\text{C}_6\text{H}_5\text{Cl}$
(3) $\text{C}_6\text{H}_5\text{CHO}$ (4) $\text{C}_6\text{H}_5\text{OH}$

NEET UG 2013

28. Reaction by which Benzaldehyde cannot be prepared :-

- (1) + Zn/Hg and conc. HCl
- (2) + CrO_2Cl_2 in CS_2 followed by H_3O^+
- (3) + H_2 in presence of Pd + BaSO_4
- (4) + $\text{CO} + \text{HCl}$ in presence of anhydrous AlCl_3

AIIMS 2013



- (1) (2)
(3) (4)

30. Which of the following reaction will not produce ethylene glycol :-

- (1) $\xrightarrow[\text{Heat}]{\text{H}_3\text{O}^+}$
- (2) $\text{ClCH}_2\text{CH}_2\text{Cl} \xrightarrow[\text{heat}]{\text{OH}^-}$
- (3) $\text{CH}_2=\text{CH}_2 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) CH}_3\text{COOH}}$
- (4) $\text{CH}_2=\text{CH}_2 \xrightarrow[\text{cold}]{\text{alk. KMnO}_4}$

31. Salicylic acid can be easily prepared by reaction between :-

- (1) Phenol and CO_2
(2) Benzoic acid and H_2O_2
(3) Benzene diazonium chloride and CO_2
(4) Phenol and formic acid

AIPMT 2014

32. Among the following sets of reactants which one produces anisole?

- (1) CH_3CHO ; RMgX
(2) $\text{C}_6\text{H}_5\text{OH}$; NaOH ; CH_3I
(3) $\text{C}_6\text{H}_5\text{OH}$; neutral FeCl_3
(4) $\text{C}_6\text{H}_5 - \text{CH}_3$; CH_3COCl ; AlCl_3

33. Which of the following will not be soluble in sodium hydrogen carbonate?

- (1) 2, 4, 6-trinitrophenol
- (2) Benzoic acid
- (3) o-Nitrophenol
- (4) Benzenesulphonic acid

AIIMS 2014

34. Which of the following reagents cannot be used to prepare acetophenone :-

- (1) $C_6H_5COCl + (CH_3)_2Cd$
- (2) $C_6H_5COCl + CH_3MgBr$
- (3) $C_6H_5CN + C_2H_5MgBr$
- (4) $C_6H_6 + (CH_3CO)_2O/AlCl_3$

AIPMT 2015

35. An organic compound 'X' having molecular formula $C_5H_{10}O$ yields phenyl hydrazone and gives negative response to the Iodoform test and Tollen's test. It produces n-pentane on reduction. 'X' could be :-

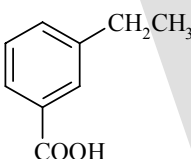
- (1) 2-pentanone
- (2) 3-pentanone
- (3) n-amyl alcohol
- (4) pentanal

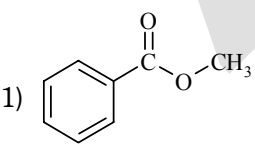
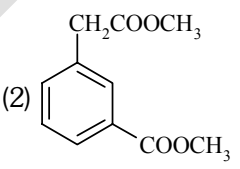
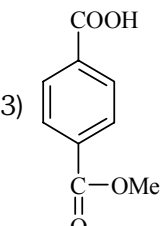
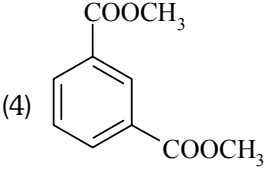
RE-AIPMT 2015

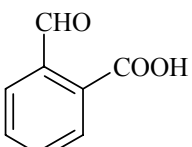
36. Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group ?

- (1) $-CHCl_2$
- (2) $-CHO$
- (3) $-CH_2Cl$
- (4) $-COOH$

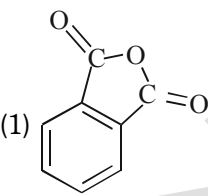
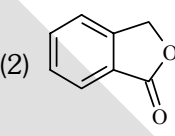
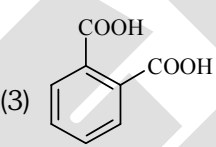
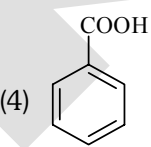
AIIMS 2015

37.  $\xrightarrow[\text{(ii) } CH_3-I, OH^-]{\text{(i) } KMnO_4, KOH/\Delta}$ products is

- (1) 
- (2) 
- (3) 
- (4) 

38.  $\xrightarrow[\text{(ii) } \Delta / H^+]{\text{(i) } [Ag(NH_3)_2]^+ OH^-}$ product.

Product of above reaction is

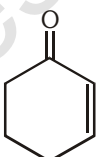
- (1) 
- (2) 
- (3) 
- (4) 

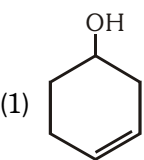
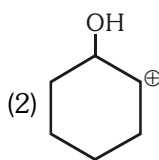
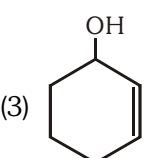
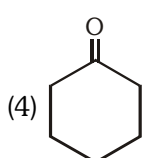
39. Correct order of boiling point is

- (i) $CH_3(CH_2)_3OH$
 - (ii) $C_2H_5-O-C_2H_5$
 - (iii) $CH_3-(CH_2)_3CH_3$
 - (iv) $CH_3-(CH_2)_3-Cl$
- (1) i > iv > ii > iii
 - (2) iii > ii > iv > i
 - (3) i > ii > iii > iv
 - (4) iv > iii > ii > i

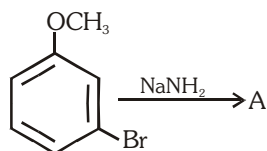
NEET-II 2016

40. The **correct** structure of the product A formed in the reaction

 $\xrightarrow[\text{Pd/carbon, ethanol}]{H_2(\text{gas, 1 atmosphere})}$ A is :-

- (1) 
- (2) 
- (3) 
- (4) 

48. Identify A and predict the type of reaction

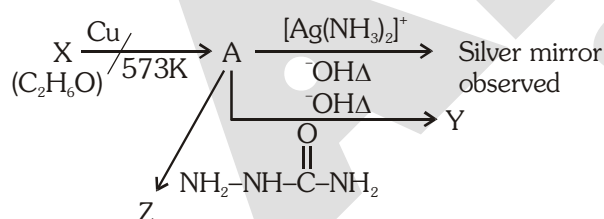


- (1) and elimination addition reaction
- (2) and cine substitution reaction
- (3) and cine substitution reaction
- (4) and substitution reaction

49. Of the following, which is the product formed when cyclohexanone undergoes aldol condensation followed by heating ?

- (1) (2)
- (3) (4)

50. Consider the reactions :-



Identify A, X, Y and Z

- (1) A-Methoxymethane, X-Ethanol, Y-Ethanoic acid, Z-Semicarbazide.
- (2) A-Ethanal, X-Ethanol, Y-But-2-enal, Z-Semicarbazone
- (3) A-Ethanol, X-Acetaldehyde, Y-Butanone, Z-Hydrazone
- (4) A-Methoxymethane, X-Ethanoic acid, Y-Acetate ion, Z-hydrazine

AIIMS 2017

51. Product is :-

- (1) (2)
- (3) (4)

52. Which of the following is the better reactant for the production of m-cresol.

- (1) (2)
- (3) (4)

53. $\text{Ph-CHO} + (\text{CH}_3\text{CO})_2\text{O} \xrightarrow[\text{CH}_3\text{COOH}, \Delta]{\text{CH}_3\text{COO}^\ominus\text{Na}^\oplus} \text{A} \xrightarrow[\text{H}_2\text{O}]{\text{LiAlH}_4} \text{B}$, B is :-

- (1) $\text{Ph-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$
- (2) $\text{Ph-CH=CH-CH}_2\text{-OH}$
- (3) $\text{Ph-CH}_2\text{-CH}_2\text{-COOH}$
- (4) Ph-CH=CH-CHO

54. $\text{CH}_3\text{-CHO} \xrightarrow{\text{NaOBr}} \text{A} + \text{B} \xrightarrow{\text{LiAlH}_4} \text{C}$, C is:-

- (1) $\text{CH}_3\text{-OH}$ (2) CH_3CHO
- (3) HCHO (4) HCOOH

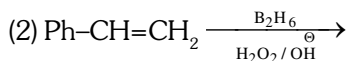
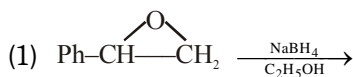
55. Reagent can not be used to convert alcohol into alkyl halide

- (1) Red P/ Br_2 (2) KI / acetone
- (3) SOCl_2 (4) HCl/ZnCl_2

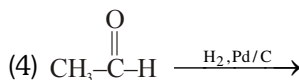
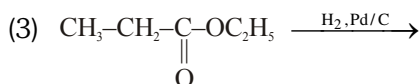
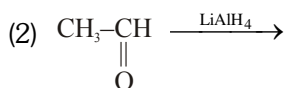
56. Product is

- (1) (2)
- (3) (4)

57. $\text{Ph}-\text{CH}=\text{CH}_2 \xrightarrow{\text{Reagent}} \text{Ph}-\text{CH}_2-\text{CH}_2-\text{OH}$
Which reagent is suitable for given conversion.



58. Which of the following cannot form ethanol?



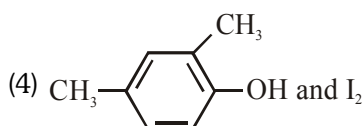
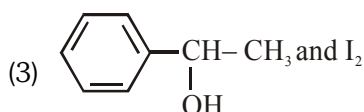
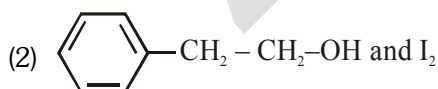
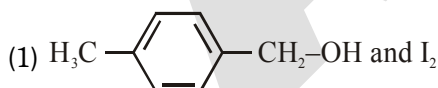
NEET(UG) 2018

59. Carboxylic acid have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their

- (1) formation of intramolecular H-bonding
- (2) formation of carboxylate ion
- (3) more extensive association of carboxylic acid via van der Waals force of attraction
- (4) formation of intermolecular H-bonding.

60. Compound A, $\text{C}_8\text{H}_{10}\text{O}$, is found to react with NaOI (produced by reacting Y with NaOH) and yields a yellow precipitate with characteristic smell.

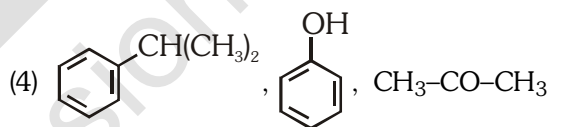
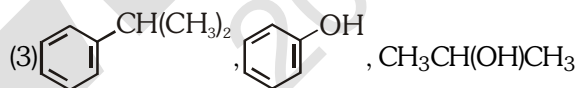
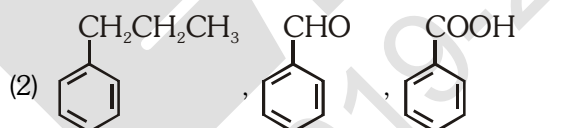
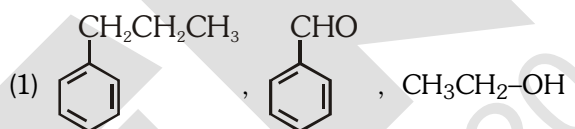
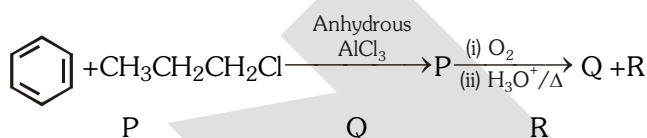
A and Y are respectively



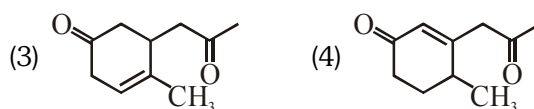
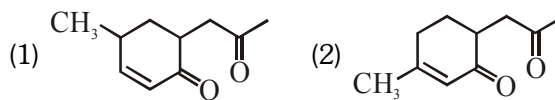
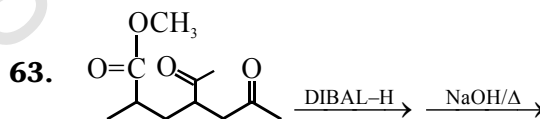
61. The compound A on treatment with Na gives B, and with PCl_5 gives C. B and C react together to give diethyl ether. A, B and C are in the order

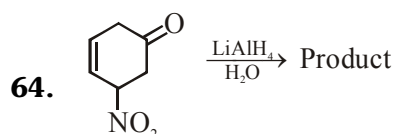
- (1) $\text{C}_2\text{H}_5\text{OH}$, C_2H_6 , $\text{C}_2\text{H}_5\text{Cl}$
- (2) $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_2\text{H}_5\text{Cl}$, $\text{C}_2\text{H}_5\text{ONa}$
- (3) $\text{C}_2\text{H}_5\text{Cl}$, C_2H_6 , $\text{C}_2\text{H}_5\text{OH}$
- (4) $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_2\text{H}_5\text{ONa}$, $\text{C}_2\text{H}_5\text{Cl}$

62. Identify the major products P, Q and R in the following sequence of reaction:

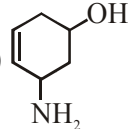
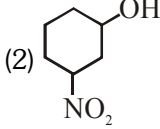
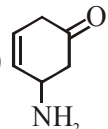
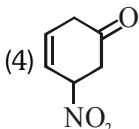


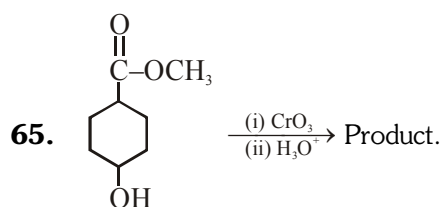
AIIMS 2018



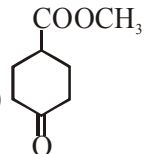
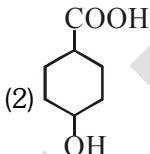
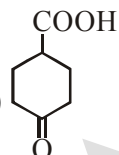
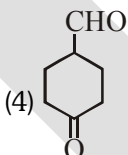


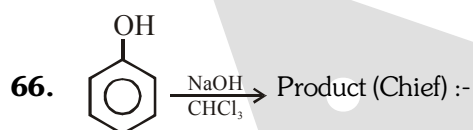
Product of the above reaction is

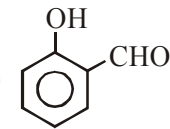
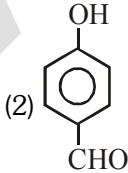
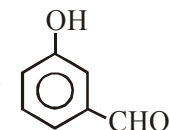
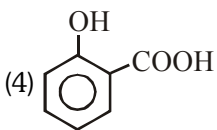
- (1)  (2) 
 (3)  (4) 

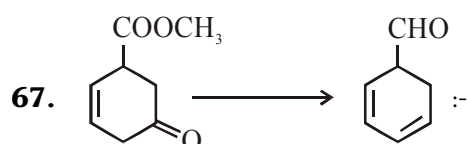


Product of reaction is :-

- (1)  (2) 
 (3)  (4) 

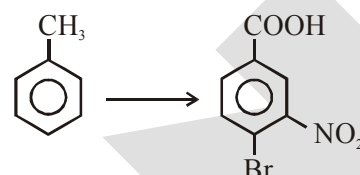


- (1)  (2) 
 (3)  (4) 

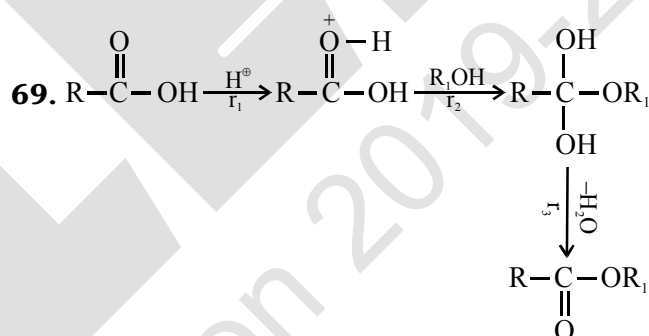


- (1) NaBH_4 , DIBAL-H, $\text{H}_3\text{O}^+/\Delta$
 (2) DIBAL-H, $\text{H}_3\text{O}^+/\Delta$, NaBH_4
 (3) B_2H_6 , NaBH_4 , $\text{H}_3\text{O}^+/\Delta$
 (4) $\text{H}_3\text{O}^+/\Delta$, DIBAL-H, NaBH_4

68. Correct sequence of reagents to get the given: product-

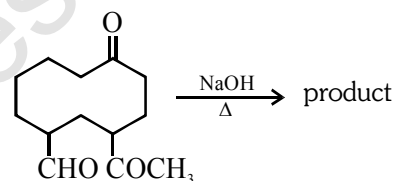


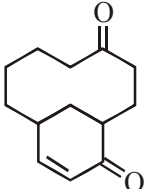
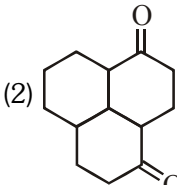
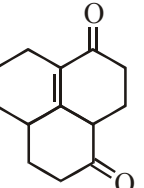
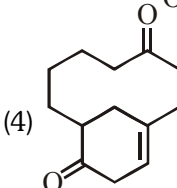
- (1) Br_2/Fe , $\text{HNO}_3/\text{H}_2\text{SO}_4$, KMnO_4/H^+
 (2) KMnO_4/H^+ , $\text{Br}_2/\text{FeBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$
 (3) KMnO_4/H^+ , $\text{HNO}_3/\text{H}_2\text{SO}_4$, $\text{Br}_2/\text{FeBr}_3$
 (4) $\text{Br}_2/\text{FeBr}_3$, KMnO_4/H^+ , $\text{HNO}_3/\text{H}_2\text{SO}_4$

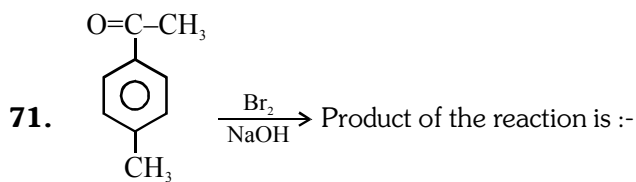


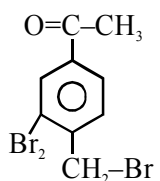
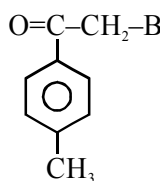
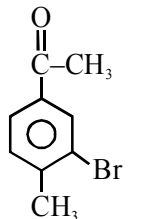
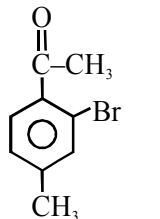
- (1) Fast, Slow, Fast (2) Slow, Fast, Fast
 (3) Fast, Fast, Slow (4) Slow, Fast, Slow

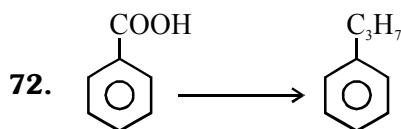
70. For the reaction :-



- (1)  (2) 
 (3)  (4) 

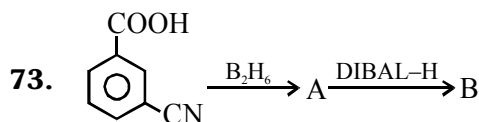


- (1)  (2) 
- (3)  (4) 

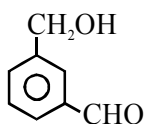
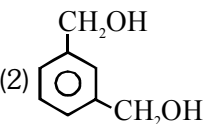
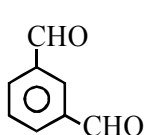
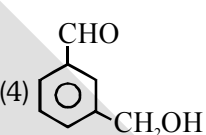


Select the correct sequence of reagents to carry out the above conversion :-

- (1) $\text{SOCl}_2, \text{Cd}(\text{C}_2\text{H}_5)_2, \text{NH}_2\text{-NH}_2/\text{OH}^\ominus$
 (2) $\text{SOCl}_2, \text{C}_2\text{H}_5\text{MgBr}, \text{Zn-Hg/HCl}$
 (3) $\text{SOCl}_2, \text{CH}_3\text{-Li}, \text{NH}_2\text{-NH}_2/\text{OH}^\ominus$
 (4) $\text{Cd}(\text{CH}_3)_2, \text{SOCl}_2, \text{NH}_2\text{-NH}_2/\text{OH}^\ominus$



The product 'B' is :-

- (1)  (2) 
- (3)  (4) 

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	1	3	1	3	1	1	1	1	4	1	3	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	2	3	2	1	1	4	2	1	4	3	1	3	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	2	3	3	2	2	4	1	1	4	1	2	2	2	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	4	1	2	1	1	1	1	2	1	2	1	4	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73		
Ans.	4	4	1	1	3	1	1	1	1	1	2	1	1		

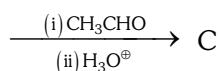
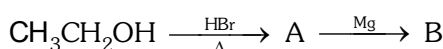
EXERCISE-III (Analytical Questions)

Check Your Understanding

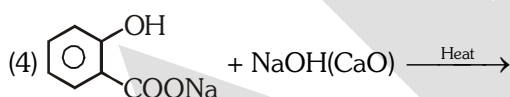
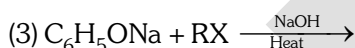
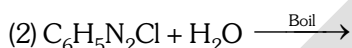
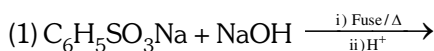
- Methanol can be distinguished from ethanol by the following except
 (1) Reaction with iodine and alkali
 (2) Reaction with salicylic acid and H_2SO_4
 (3) Reaction with Lucas reagent
 (4) Boiling point

- Ethanol on heating with acetic acid in the presence of a few drops of sulphuric acid gives the smell of
 (1) Oil of wintergreen (2) Oil of mustard
 (3) An ester (4) Oil of bitter almonds

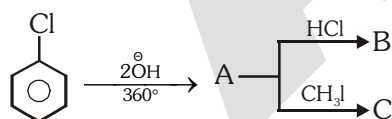
- The compounds A, B and C in the reaction sequence are given by the set :-



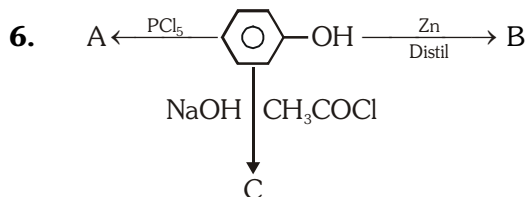
- (1) $\text{CH}_3\text{CH}_2\text{Br}$, $\text{CH}_3\text{CH}_2\text{MgBr}$, $(\text{CH}_3)_3\text{C}-\text{OH}$
 (2) $\text{CH}_3\text{CH}_2\text{Br}$, $(\text{CH}_3\text{CH}_2)_2\text{Mg}$, $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
 (3) $\text{CH}_3\text{CH}_2\text{Br}$, $\text{CH}_3\text{CH}_2\text{MgBr}$, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
 (4) CH_3CHBr_2 , $\text{CH}_3\text{CH}(\text{MgBr})_2$, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- Which of the following reactions will not lead to a phenol :-



- The structures of the compounds / ions A, B and C in the reaction sequence are given by the set :-



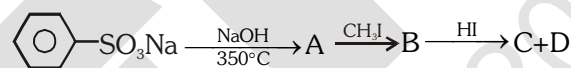
- (1) $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_6\text{H}_5\text{Cl}$, $\text{C}_6\text{H}_5\text{CH}_3$
 (2) $\text{C}_6\text{H}_5\text{O}^-$, $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_6\text{H}_5\text{OCH}_3$
 (3) $\text{C}_6\text{H}_5\text{O}^-$, $\text{C}_6\text{H}_5\text{Cl}$, $\text{C}_6\text{H}_5\text{I}$
 (4) $\text{C}_6\text{H}_5\text{O}^-$, $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_6\text{H}_5\text{I}$



The compounds A, B and C in the above reaction sequence are :-

- (1) Chlorobenzene, benzene, methyl benzoate
 (2) Triphenyl phosphate, benzene, phenyl acetate
 (3) Benzyl chloride, benzene, phenyl acetate
 (4) Benzyl chloride, benzene, phenylacetyl chloride

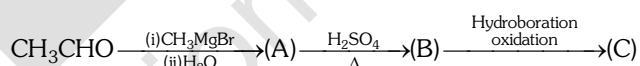
- In the reaction sequence —



A, B, C and D are given by the set :-

- (1) Sodium phenate, anisole, $\text{C}_6\text{H}_5\text{I}$, CH_3OH
 (2) Sodium phenate, phenetole, $\text{C}_2\text{H}_5\text{I}$, $\text{C}_6\text{H}_5\text{OH}$
 (3) Sodium phenate, anisole, $\text{C}_6\text{H}_5\text{OH}$, CH_3I
 (4) Sodium phenate, phenetole, $\text{C}_6\text{H}_5\text{I}$, $\text{C}_2\text{H}_5\text{OH}$

- Compound A and C in the following reaction are



- (1) Identical (2) Functional isomer
 (3) Positional isomer (4) Optical isomer

- Which of the following aldehydes does not form iodoform on heating with iodine and alkali :-

- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{H}$ (2) ICH_2CHO

- (3) $\text{CH}_3-\text{CH}_2-\text{CHO}$ (4) $\text{CH}_3-\text{CH}(\text{OH})-\text{C}(=\text{O})-\text{H}$

- A carbonyl compound gives pink colour with Schiff's reagent and a yellow precipitate when boiled with iodine and caustic alkali. It also gives a red precipitate with Fehling's solution. It is likely to be:-
 (1) Formaldehyde (2) Propionaldehyde
 (3) Acetaldehyde (4) Crotonaldehyde

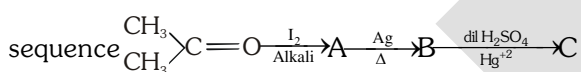
11. A carbonyl compound gives a positive iodoform test but does not reduce Tollen's reagent or Fehling's solution. It forms a cyanohydrin with HCN, which on hydrolysis gives a hydroxy acid with a methyl side chain. The compound is :-

(1) Acetaldehyde
(2) Propionaldehyde
(3) Acetone
(4) Crotonaldehyde

12. Which of the following statement is wrong :-

(1) All methyl ketones give a positive iodoform test
(2) Acetaldehyde is the only aldehyde that gives iodoform test
(3) All secondary alcohols give positive iodoform test
(4) Any alcohol that can be oxidised to an acetyl group gives a positive iodoform test

13. The compounds A, B and C in the reaction

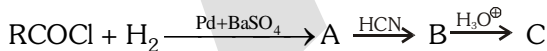


are given by the set :-

(1) CHI_3 , $\text{H}_2\text{C}=\text{CH}_2$, $\text{CH}_3\text{CH}_2\text{-OH}$
(2) CHI_3 , $\text{HC}\equiv\text{CH}$, CH_3CHO
(3) CHI_3 , $\text{CH}_3\text{-C}\equiv\text{CH}$, CH_3COCH_3

(4) CHI_3 , $\text{CH}_3\text{-CH}(\text{OH})\text{-CH}_3$, $\text{CH}_3\text{-C}(\text{O})\text{-CH}_3$

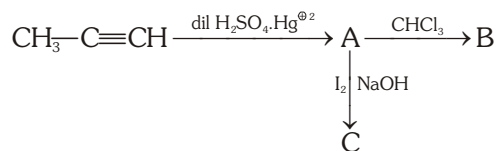
14. In the reaction sequence



A, B and C are given by the set :-

(1) RCHO , $\text{RCH}(\text{OH})\text{CN}$, $\text{RCH}(\text{OH})\text{CH}_2\text{NH}_2$
(2) RCHO , $\text{RCH}(\text{OH})\text{CN}$, $\text{RCH}(\text{OH})\text{COOH}$
(3) RCHO , $\text{R}-\text{C}(=\text{O})-\text{CH}_2-\text{CN}$, $\text{R}-\text{C}(=\text{O})-\text{CH}_2-\text{C}(=\text{O})-\text{OH}$
(4) RCHO , $\text{R}-\text{CH}_2-\text{CN}$, $\text{R}-\text{CH}_2-\text{COOH}$

15. In the reaction sequence



A, B and C are given by the set :-

(1) $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$, CHI_3
(2) CH_3COCH_3 , $\text{CCl}_3\text{-C}(=\text{O})\text{-CH}_3$, CHI_3
(3) CH_3COCH_3 , $\text{CCl}_3\text{-C}(\text{OH})(\text{CH}_3)_2$, CHI_3
(4) $\text{CH}_3\text{CH}_2\text{CHO}$, $\text{CCl}_3\text{-CH}(\text{OH})\text{-CH}_2\text{-CH}_3$, CHI_3

16. Match the reactions given in column-I with the suitable reagent given in column-II

Column-I

Column-II

(i) Benzophenone $\rightarrow$ Diphenylmethane
(ii) Benzaldehyde $\rightarrow$ 1-Phenyl ethanol
(iii) Cyclohexanone $\rightarrow$ Cyclohexanol
(iv) Phenylbenzoate $\rightarrow$ Benzaldehyde

(a) LiAlH_4

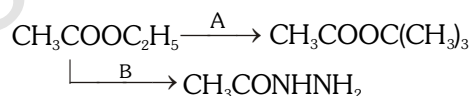
(b) Dibal-H

(c) $\text{Zn}(\text{Hg})/\text{conc. HCl}$

(d) CH_3MgBr

(1) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)
(2) (i)-(c), (ii)-(d), (iii)-(b), (iv)-(a)
(3) (i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)
(4) (i)-(d), (ii)-(a), (iii)-(c), (iv)-(b)

17. The reagents A and B in the reaction sequence



are given by the set :-

(1) $\text{CH}_3\text{-CH}(\text{OH})\text{-CH}_3$, $\text{H}_2\text{N-NH}_2$
(2) $\text{CH}_3\text{-CH}(\text{OH})\text{-CH}_3$, $\text{H}_2\text{N-OH}$
(3) $\text{CH}_3\text{-C}(\text{OH})(\text{CH}_3)_2$, $\text{H}_2\text{N-NH}_2$
(4) $\text{CH}_3\text{-C}(\text{OH})(\text{CH}_3)_2$, $\text{H}_2\text{N-OH}$

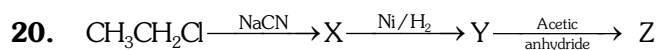
18. Methyl amine reacts with acetyl chloride and forms :-

- (1) CH_3NH_2
- (2) CH_3NHNa
- (3) $\text{CH}_3\text{NHCOCH}_3$
- (4) $(\text{CH}_3)_2\text{NCOCH}_3$

19. Intermediates formed during reaction of



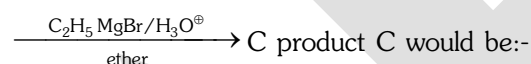
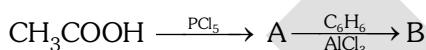
- (1) RCONHBr and RNCO
- (2) RNHCOBr and RNCO
- (3) $\text{RNH}-\text{Br}$ and RCONHBr
- (4) RCONBr_2



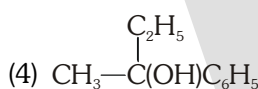
Z in the above reaction sequence is :-

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NHCOCH}_3$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONHCH}_3$
- (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONHCOCH}_3$

21. In a set of the given reactions, acetic acid yielded a product C.



- (1) $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$
- (2) $\text{CH}_3\text{COC}_6\text{H}_5$
- (3) $\text{CH}_3\text{CH}(\text{OH})\text{C}_6\text{H}_5$

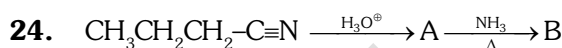


22. CH_3CHO and CH_3COCH_3 can not be distinguished by :-

- (1) Fehling solution
- (2) Grignard reagent
- (3) Schiff's reagent
- (4) Tollen's reagent

23. When propan-1-ol is dehydrated, the product obtained is again hydrated and further oxidised to form a compound. The compound is -

- (1) $\text{HC}\equiv\text{C}-\text{CH}_2-\text{OH}$
- (2) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$
- (3) $\begin{array}{c} \text{CH}_2-\text{CH}_2 \\ | \quad | \\ \text{OH} \quad \text{OH} \end{array}$
- (4) $\begin{array}{c} \text{CH}_2-\text{CH}-\text{CH}_3 \\ | \quad | \\ \text{OH} \quad \text{OH} \end{array}$



Product 'B' is :-

- (1) $\text{CH}_3-\text{CH}_2\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$
- (2) $\text{CH}_3\text{CH}_2\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{NC}$
- (4) $\text{CH}_3\text{CH}_2\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$

25. Sodium ethoxide has reacted with ethanoyl chloride. The compound that is produced in the above reaction is :-

- (1) Ethyl chloride
- (2) Ethyl ethanoate
- (3) Diethyl ether
- (4) 2-Butanone

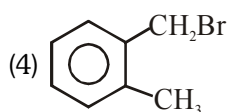
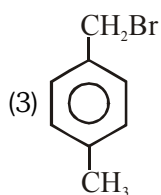
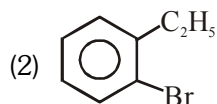
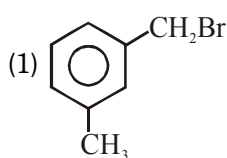
26. Phenol is heated with a solution of mixture of KBr and KBrO_3 . The major product obtained in the above reaction is :-

- (1) 4-Bromophenol
- (2) 2,4,6-Tribromophenol
- (3) 2-Bromophenol
- (4) 3-Bromophenol

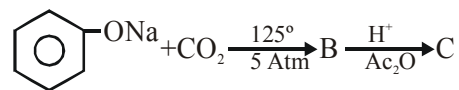
27. Trichloroacetaldehyde was subjected to Cannizzaro's reaction by using NaOH. The mixture of the products contains sodium trichloroacetate and another compound. The other compound is:-

- (1) 2,2,2-Trichloropropanol
- (2) Chloroform
- (3) 2,2,2-Trichloroethanol
- (4) Trichloromethanol

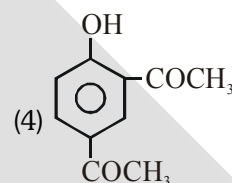
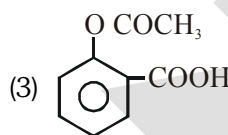
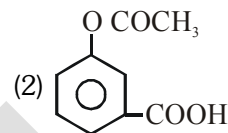
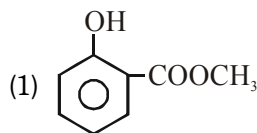
28. Compound (A), C_8H_9Br , gives a yellow precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A):



29. Sodium phenoxide when heated with CO_2 under pressure at $125^\circ C$ yields a product which on acetylation produces C



The major product C would be :



EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	3	3	2	2	3	3	3	3	3	3	2	2	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Ans.	1	3	3	1	1	4	2	2	4	2	2	3	4	3	

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** Chlorobenzene and benzyl chloride can be distinguished by treatment with AgNO_3 .
Reason :- Chlorobenzene does not give white ppt with AgNO_3 due to resonance.
 (1) A (2) B (3) C (4) D
- Assertion :-** Benzyl alcohol turns blue litmus red.
Reason :- Benzyl alcohol is not an isomer of p-cresol.
 (1) A (2) B (3) C (4) D
- Assertion :-** Phenol is stronger acid than alcohols.
Reason :- Phenol is stabilized by resonance whereas alcohols are not.
 (1) A (2) B (3) C (4) D
- Assertion :-** Addition of CH_3OH to phenyl magnesium bromide gives benzene.
Reason :- CH_3OH is a stronger acid than benzene.
 (1) A (2) B (3) C (4) D
- Assertion :-** Ethers behave as bases in the presence of mineral acids.
Reason :- Ethers have lone pairs of electrons on oxygen.
 (1) A (2) B (3) C (4) D
- Assertion :-** Phenols give o- and p-Nitrophenol on nitration with conc. HNO_3 and H_2SO_4 mixture.
Reason :- -OH group in phenol shows -M effect.
 (1) A (2) B (3) C (4) D
- Assertion :-** Formic acid is a stronger acid than benzoic acid.
Reason :- pK_a of formic acid is higher than that of benzoic acid.
 (1) A (2) B (3) C (4) D
- Assertion :-** Ketones on reaction with Grignard reagents, followed by hydrolysis yield tertiary alcohols.
Reason :- All ketones show haloform reaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** $>\text{C}=\text{O}$ group is present both in aldehydes and acid derivatives.
Reason :- Aldehyde gives nucleophilic addition across $>\text{C}=\text{O}$ bond but acid derivatives do not exhibit such reactions.
 (1) A (2) B (3) C (4) D
- Assertion :-** Benzamide and methyl benzoate are derivatives of benzoic acid.
Reason :- Benzamide is less easily hydrolysed as compared to methyl benzoate.
 (1) A (2) B (3) C (4) D
- Assertion :-** Dehydration of alcohols always takes place in basic medium.
Reason :- OH^- is a good leaving group.
 (1) A (2) B (3) C (4) D
- Assertion :-** Acetamide on reaction with KOH and bromine gives acetic acid.
Reason :- Bromine catalyses hydrolysis of acetamide.
 (1) A (2) B (3) C (4) D
- Assertion :-** Mixture of benzaldehyde and acetaldehyde in hot alkaline medium gives cinnamaldehyde.
Reason :- Benzaldehyde is a stronger electrophile than acetaldehyde.
 (1) A (2) B (3) C (4) D

- 14. Assertion :-** Acetophenone shows aldol condensation.
Reason :- Acetophenone contains carbonyl group.
 (1) A (2) B (3) C (4) D
- 15. Assertion :-** Benzyl phenyl ether forms phenol and benzyl iodide with HI
Reason :- Because benzyl carbocations is more stable than phenyl carbocation.
 (1) A (2) B (3) C (4) D
- 16. Assertion :-** m-Nitrophenol is less acidic than p-nitrophenol.
Reason :- p-Nitrophenol has intermolecular-H-bonding.
 (1) A (2) B (3) C (4) D
- 17. Assertion :-** Benzene sulphonic acid on heating with NaOH gives phenoxide.
Reason :- Sulphonic group is a poor leaving group.
 (1) A (2) B (3) C (4) D
- 18. Assertion :-** Acetamide on reaction with Br_2/NaOH give $\text{CH}_3\text{-NH}_2$.
Reason :- Carbon of amide functional group becomes part of Na_2CO_3 .
 (1) A (2) B (3) C (4) D
- 19. Assertion :-** $\text{Ph-O-C}_2\text{H}_5$ can be prepared from PhONa and ethyl iodide.
Reason :- Aryl halides are less reactive substrates for nucleophilic substitution reaction.
 (1) A (2) B (3) C (4) D
- 20. Assertion :-** $\text{C}_6\text{H}_5\text{CH(OH)CH}_3$ gives benzoic acid with NaOBr followed by hydrolysis.
Reason :- First dehydration occur then oxidation
 (1) A (2) B (3) C (4) D
- 21. Assertion :-** Ethyl acetoacetate forms butane-1,4-diol with NaBH_4 .
Reason :- NaBH_4 can reduces ketone and esters group.
 (1) A (2) B (3) C (4) D
- 22. Assertion :-** Methoxy ethane has less boiling point than propanol.
Reason :- Methoxy ethane is more polar than propanol.
 (1) A (2) B (3) C (4) D
- 23. Assertion :-** Isopropyl benzene reacts with O_2 and acid gives phenol.
Reason :- Initially isopropyl benzene converts into 2-phenyl propan-2-ol.
 (1) A (2) B (3) C (4) D
- 24. Assertion :-** Benzaldehyde with alkaline CuSO_4 gives red-brown precipitate.
Reason :- Red precipitate is due to Cu_2O .
 (1) A (2) B (3) C (4) D
- 25. Assertion :-** Methoxy ethane has more boiling point than propanol.
Reason :- Attraction is more in methoxy ethane than propanol.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	4	2	1	1	4	3	3	2	2	4	4	3	2	1
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	2	3	1	1	3	4	3	3	4	4					

NITROGEN CONTAINING COMPOUND

E

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NEET SYLLABUS

Amines : Nomenclature, classification, structure, methods of preparation, physical and chemical properties, uses, identification of primary secondary and tertiary amines. **Cyanides and Isocyanides** (will be mentioned at relevant places) **Diazonium salt** : preparation, chemical reactions and importance in synthetic organic chemistry.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- Describe amines as derivatives of ammonia having a pyramidal structure;
- Classify amines as primary, secondary and tertiary;
- Name amines by common names and IUPAC system;
- Describe some of the important methods of preparation of amines;
- Explain the properties of amines;
- Distinguish between primary, secondary and tertiary amines;
- Describe the method of preparation of diazonium salts and their importance in the synthesis of a series of aromatic compounds including azo dyes.

*"Learning gives creativity
Creativity leads to thinking
Thinking provides knowledge
Knowledge makes you great."*

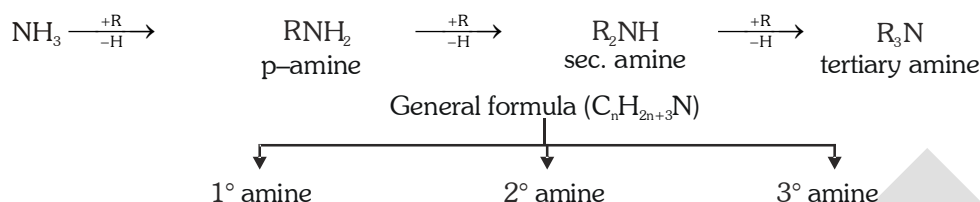
A.P.J. Abdul Kalam

NITROGEN CONTAINING COMPOUND

1.0 Amines

Amines are derivatives of ammonia in which one or more hydrogen atoms are replaced by alkyl group(s).

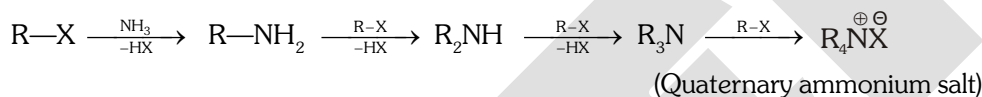
Amines are classified as primary, secondary and tertiary depending on the number of alkyl groups attached to nitrogen atom.



2.0 General methods of preparation

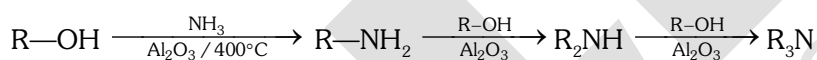
(1) Ammonolysis of alkyl halides and alcohols :

(a) **From Ammonolysis of alkyl halides [Hofmann's ammonolysis]** : When an aqueous solution of ammonia is heated with alkyl halide all the three types of amines and quaternary ammonium salt are formed.



If ammonia is taken in excess, 1° amine is the main product.

(b) Ammonolysis of alcohols : When ROH and NH_3 are passed over Al_2O_3 or ThO_2 at 350°C all the three types of amines are formed.



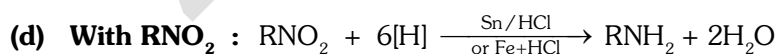
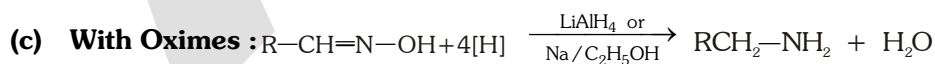
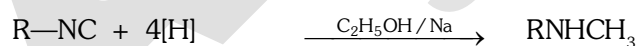
- Quaternary ammonium hydroxide is not formed.
- If excess of ammonia is used, then main product will be primary amine.

(2) By reduction :



This reaction (b) is called mendius reaction.

The reduction of alkyl isocyanides gives secondary amines.



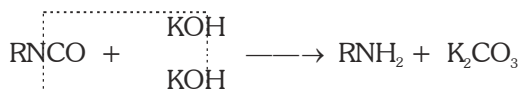
Sn/HCl is used in laboratory preparation

(3) By hydrolysis of :

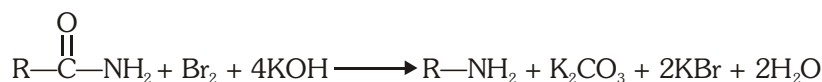
(a) **R—NC** : Alkyl isocyanide undergoes hydrolysis with mineral acid and forms alkyl amine.



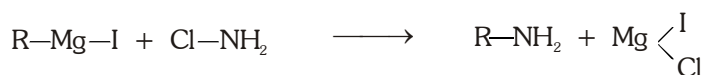
(b) **RNCO** : Alkyl isocyanate undergoes hydrolysis on heating with KOH and forms alkyl amine.



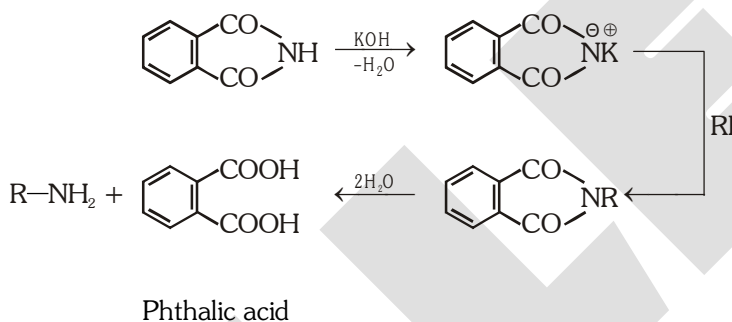
(4) **By Hofmann's bromamide reaction (Hofmann's Hypobromite reaction)** : This is a general method for the conversion of alkanamides into primary amines having one less carbon.



(5) **From Grignard reagent** : Alkyl magnesium iodide reacts with chloramine to yield alkyl amine.

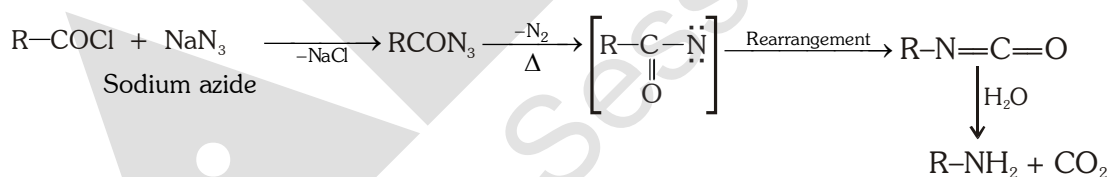


(6) **Gabriel phthalimide synthesis** : Phthalimide is first treated with KOH to obtain potassium phthalimide which is then treated with alkyl iodide. Then alkyl phthalimide on hydrolysis yields alkylamine. This method is used in the formation of pure aliphatic primary amines.

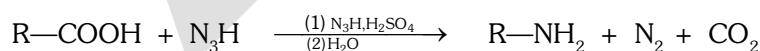


* Aniline is not formed by this reaction.

(7) **Curtius reaction** :

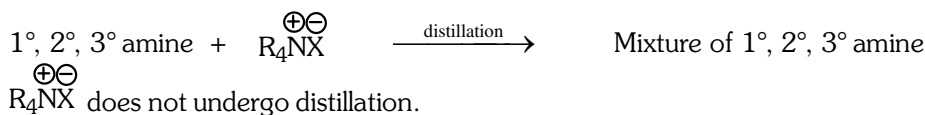


(8) **Schmidt reaction** : In presence of conc. H₂SO₄ alkanic acid reacts with hydrazoic acid (N₃H) followed by hydrolysis to yield alkylamine.



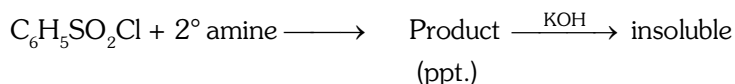
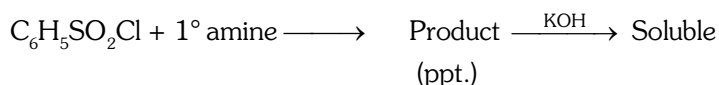
GOLDEN KEY POINTS

• Separation of 1°, 2° and 3° amines



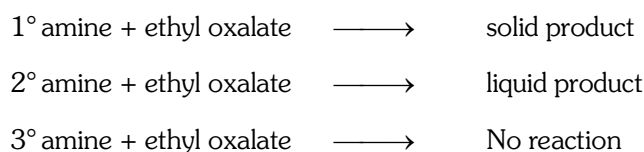
Mixture of 1°, 2°, 3° amine can be separated by following methods.

- (i) **Fractional distillation** : The mixture of amines may be separated by fractional distillation because their boiling points are quite different. It is used in industry.
- (ii) **Hinsberg method** : In this method mixture of amines is separated by using benzene sulphonyl chloride (Hinsberg's reagent).



3° amine does not react with benzene sulphonyl chloride. (No ppt. formed)

- (iii) **Hofmann method** : In this method mixture of amines is separated by using ethyl oxalate.



3.0 PHYSICAL PROPERTIES

- (i) CH_3NH_2 is gas and $\text{C}_2\text{H}_5\text{NH}_2$ is a volatile liquid.
- (ii) Higher amines have fishy smell.
- (iii) H-Bonding (weaker as compared to $\text{H}-\text{O}-\text{H}$).



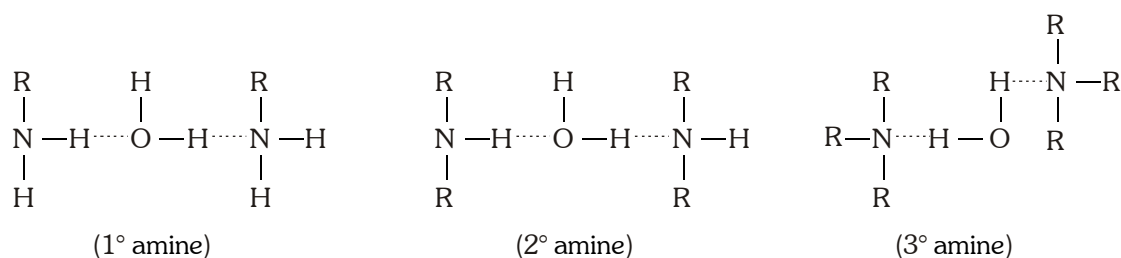
In 3° amine (due to absence of H-atom) H-bonding is not possible.

- (iv) **Boiling point** : Due to weak intermolecular H-bonding the B.P. of 1° and 2° amines are lower than those of alcohols of comparable molecular weight. The boiling point of 3° amines which form no H-bonds are near to those of alkanes of comparable molecular weight.

Boiling point $\propto$ molecular weight

Order of B.P. :	1° amine > 2° amine > 3° amine
so order of volatility :	3° amine > 2° amine > 1° amine

- (v) **Solubility** : Low molecular weight amines are soluble in water. The water solubility of amines decreases with increasing size of alkyl group.



Order of solubility $\longrightarrow$ p- amine > s- amine > t- amine

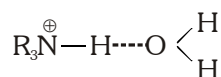
4.0 Chemical properties

- (i) Basic character of amines is due to the presence of lone pair of electrons on the N - atom.
- (ii) Basic strength depends on electron donating tendency.
- (iii) Order of basic character in aqueous solution : $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$
 $(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3$

GOLDEN KEY POINTS

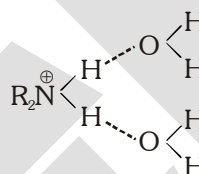
- Tertiary amine is less basic than secondary due to following reasons :
 - (i) **Steric hindrance** : In tertiary amines (R_3N), three alkyl groups attached to N are bulkier and as such exert steric hindrance.
 - (ii) **Decrease in hydration** :

In tertiary amine



Protonated t-amine can form H-bonding with water molecule only at one point
[less stable]

In secondary amine



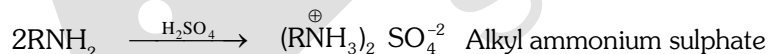
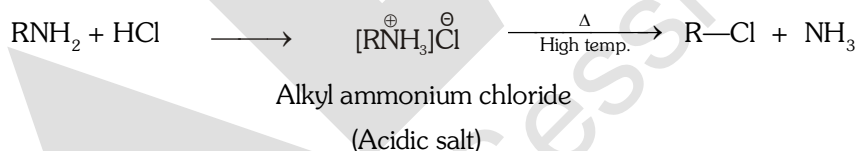
Protonated s-amine can form H-bonding with water molecules at two points
(more stable)

Conjugate acid of 3° amine are less stable as compare to 2° amine due to low hydration so less basic.

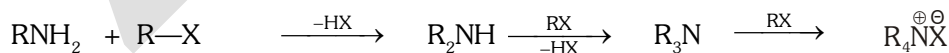
- The basic strength of aniline is less than aliphatic amines as the lone pair of electron present on N-atom interact with the delocalized π - orbital of benzene ring. Hence it is less available for protonation on N-atom.

(1) Reactions showing basic nature :

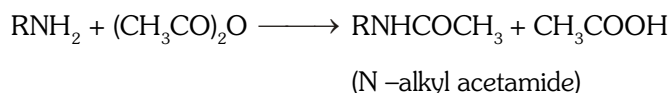
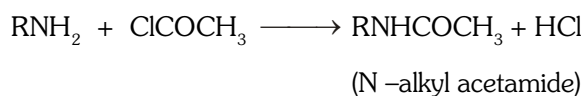
It reacts with acids to form salts.



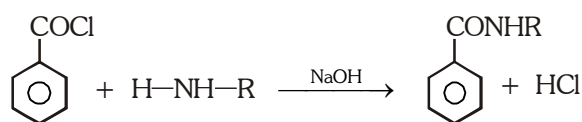
- (2) **Reaction with alkyl halides** : Alkyl amine reacts with alkyl halides and form sec., ter. amines and quaternary ammonium salt.



- (3) **Acetylation** : Acetylation takes place when alkyl amine combines with acetyl chloride or acetic anhydride.



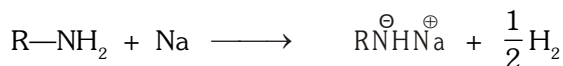
(4) Benzoylation (Schotten Baumann reaction) :



Benzoylchloride

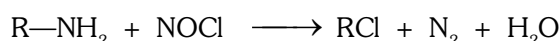
N-alkyl benzamide

(5) Acidic nature : Amines are very weak acids only 1° and 2° amines show acidic nature with active metals.

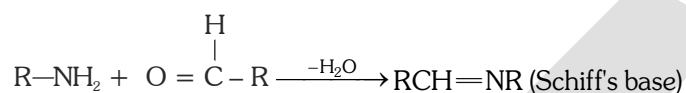


N- alkyl sodamide

(6) Reaction with Tilden reagent :



(7) Reaction with aldehydes :

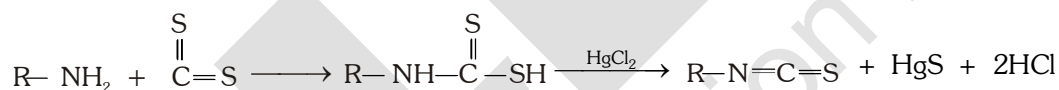


(8) Carbylamine Reaction (Isocyanide test) : When alkyl amine heated with chloroform and alc. KOH alkyl isocyanide is formed which has very bad smell.

This test is also given by aniline . This is a test for p- amines.



(9) Hofmann's mustard oil test : When alkyl amine is heated with carbon disulphide and mercuric chloride alkyl isothiocyanate is formed which has smell like mustard oil.



Alkyl isothiocyanate

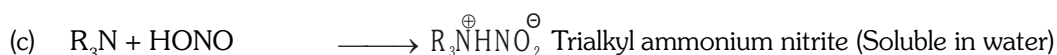


(10) Reaction with HNO₂ :

(a) Primary amines react with nitrous acid to produce nitrogen gas [seen as bubbles]

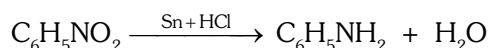
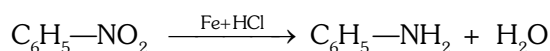
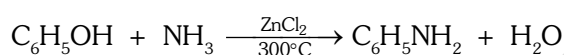
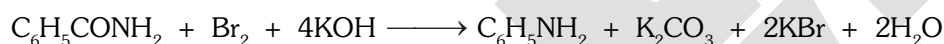
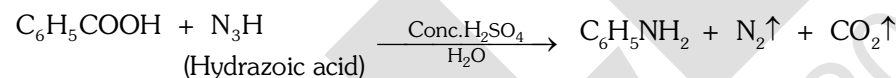


Dialkyl nitroso amine (Yellow oily layer)



Points to Remember :

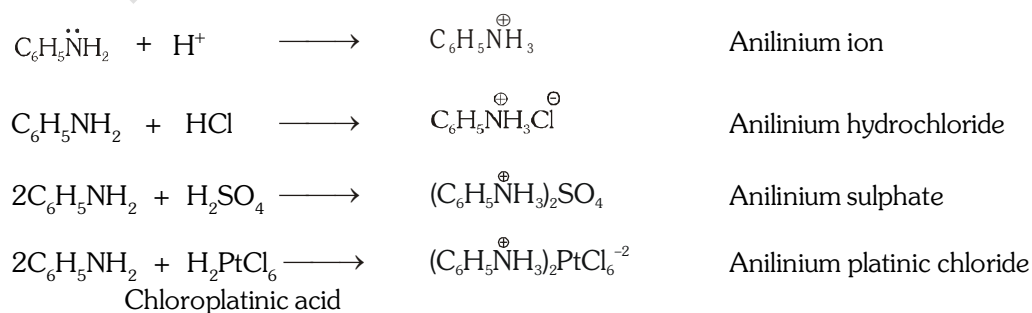
- (i) Nitrosoamines are carcinogens (Cancer causing agents)
- (ii) Amines can have chiral N-atom but cannot be resolved into enantiomeric forms because of rapid inversion of one enantiomeric form into the other.
- (iii) The Schiff's bases formed by reaction of 1° amines and aldehyde/ketones are also called anils.
- (iv) The mixture of 1°, 2°, 3° amines can be distinguished by Hofmann's test or Hinsberg's reagent or nitrous acid test.

5.0 ANILINE (C₆H₅NH₂)**5.1 General Methods of Preparation****(1) Lab method:****(2) Industrial method :****(3) From Phenol :****(4) From benzamide (Hofmann's bromamide reaction):****(5) From benzoic acid (Schmidt reaction) :****(6) From Grignard reagent :****(7) From phenyl isocyanide :****5.2 Physical Properties**

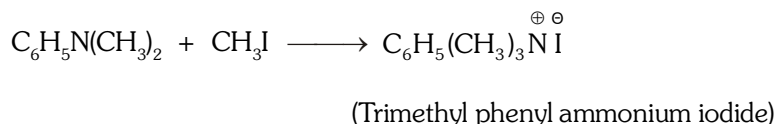
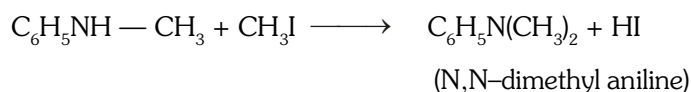
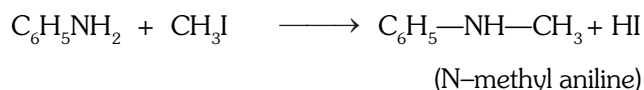
- (i) Fresh, aniline is a colourless oily liquid. On standing the colour becomes dark brown due to action of air and light.
- (ii) It's B.P. is 183°C.
- (iii) It is heavier than water.
- (iv) It has characteristic unpleasant odour. It is toxic in nature.

Reactions due to —NH₂ group

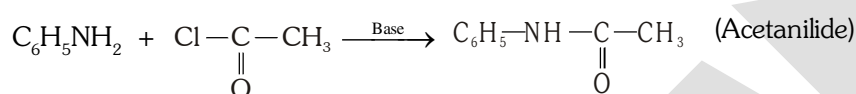
(1) Basic nature : Aniline is weak base but it forms salt with strong acids. It accepts a proton.



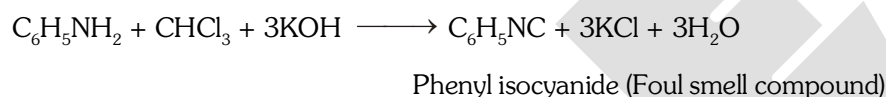
- (2) **Alkylation** : Aniline reacts with alkyl halides forming secondary, tertiary and quaternary ammonium salts depending on the concentration of alkyl halides.



- (3) **Acylation** : Aniline reacts with acid chlorides or anhydrides to form corresponding amides called anilides. [The reaction of $\text{C}_6\text{H}_5\text{NH}_2$ with benzoyl chloride is example of "**Schotten Baumann reaction**"]



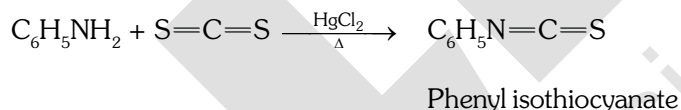
- (4) **Carbylamine reaction** :



Note : (1) Intermediate species is dichloro carbene [$:\text{CCl}_2$].

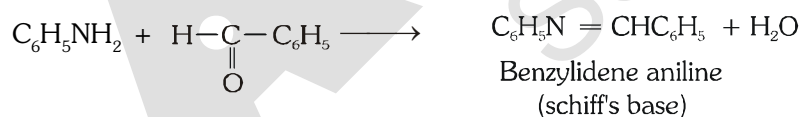
(2) This is test of aniline and other primary amine, known as **Isocynide test**.

- (5) **Hoffmann's mustard oil reaction** : When aniline is heated with alc. CS_2 and excess of HgCl_2 phenyl isothiocyanate having a characteristic smell of mustard oil is formed.

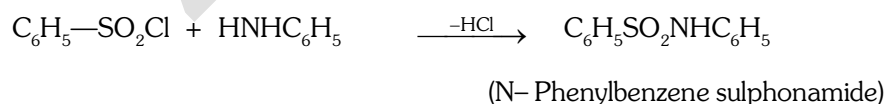


This is a test of aniline and other primary amines.

- (6) **Reaction with aldehydes** : Aniline condenses with aldehydes to form schiff's base.



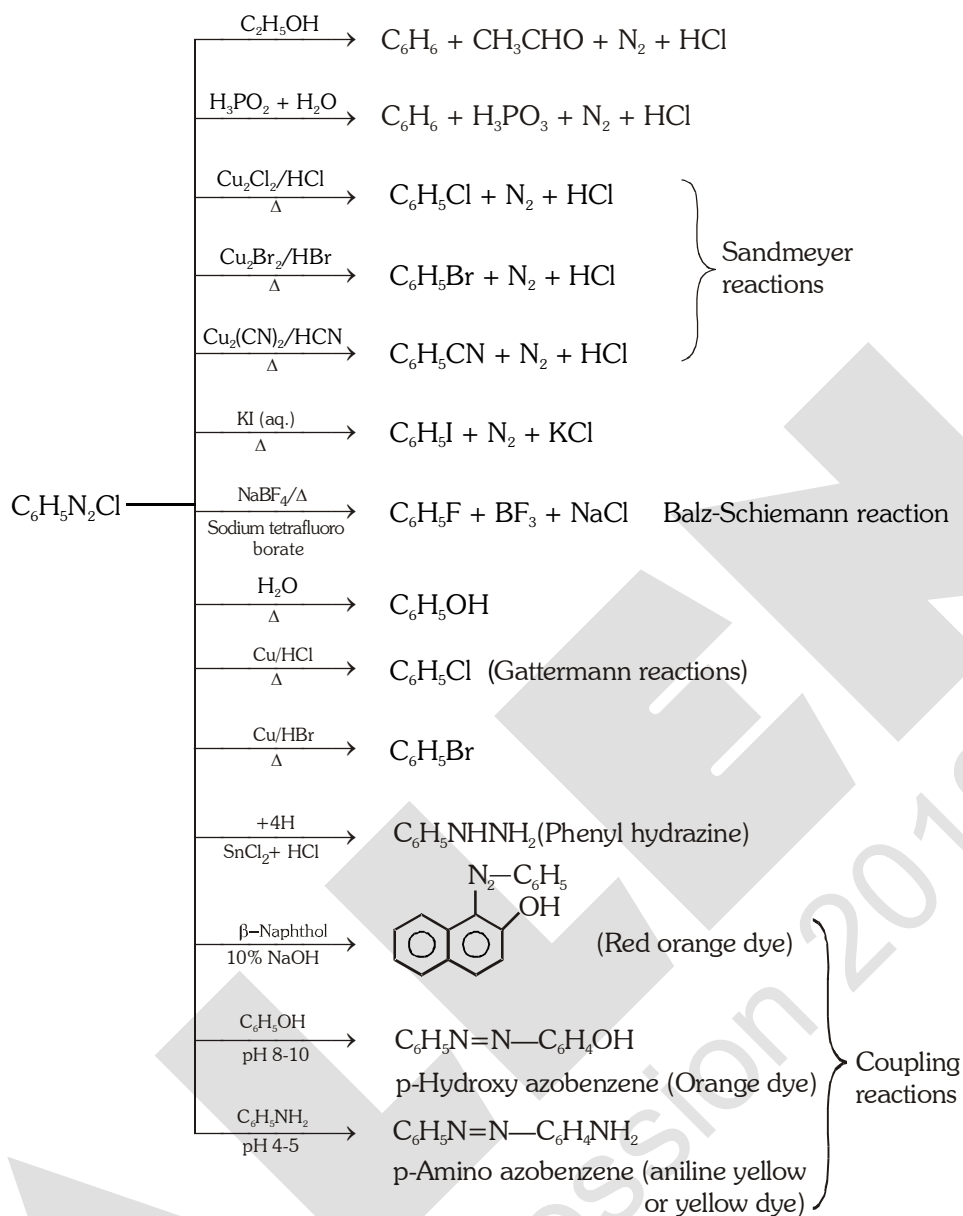
- (7) **Reaction with Hinsberg's reagent** :



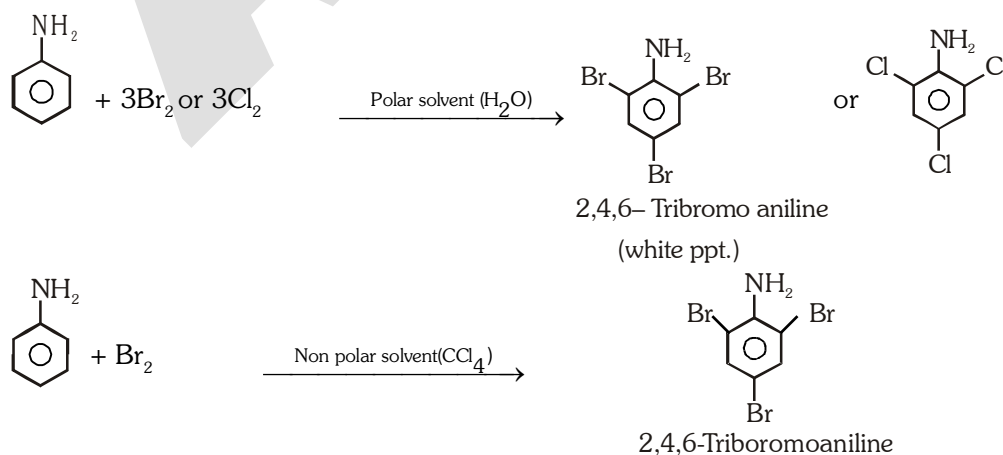
- (8) **Diazotisation** : Diazotisation is a reaction in which ice cooled solution of aniline in an inorganic acid reacts with sodium nitrite solution leading to the formation of diazonium salt.



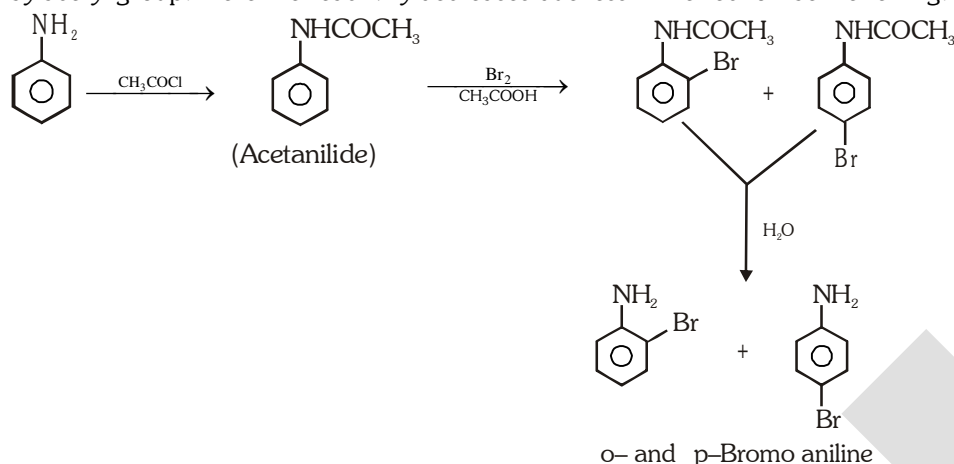
Benzene diazonium chloride is a useful synthetic reagent. It is used in the preparation of many organic compounds



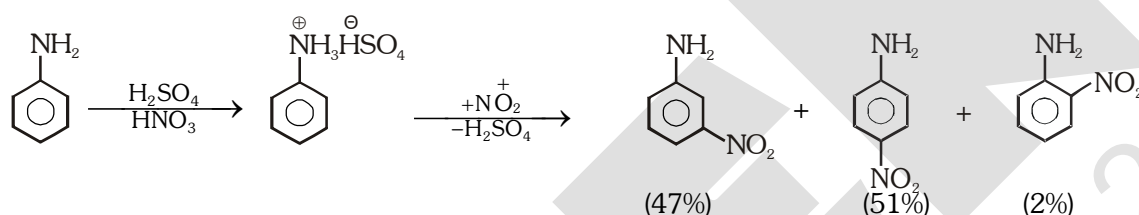
(1) Halogenation : In polar and nonpolar medium Chlorine and bromine react with aniline and form trichloro and tribromo aniline respectively.



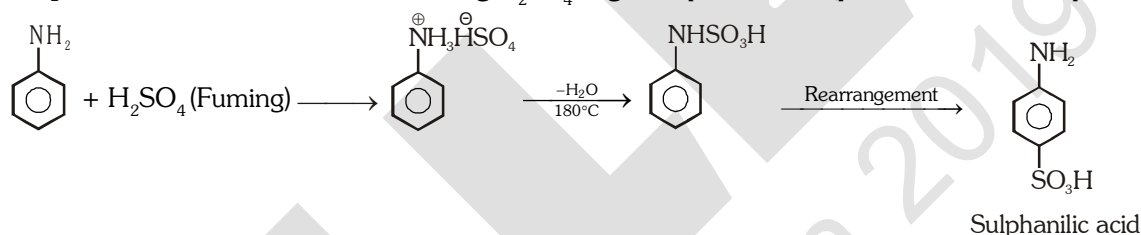
Note : However, monobromo or chloro derivative of aniline can be prepared if $-NH_2$ group is first protected by acetyl group. Here the reactivity decreases due less +M effect on benzene ring.



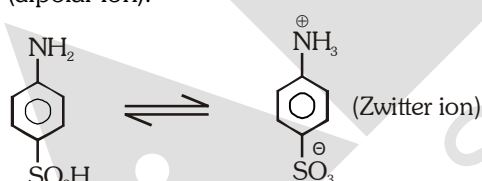
(2) Nitration :



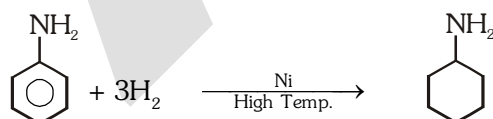
(3) Sulphonation : Aniline reacts with fuming H_2SO_4 to give sulphanilic acid. (p-Amino-benzene sulphonic acid)



- This process is called baking.
- Sulphanilic acid is an important intermediate in the manufacturing of dyes and drugs.
- The compounds in which both proton donating & proton accepting groups present are called amphotile (dipolar ion).

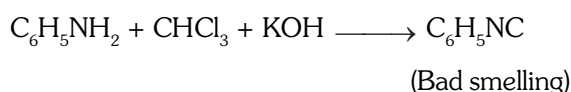


(4) Catalytic hydrogenation : Aniline undergoes hydrogenation in presence of Ni at high temp. to form cyclohexanamine.



5.3 Tests of Aniline

(i) Carbylamine test : Aniline gives carbylamine test or Isocyanide test.



(ii) Dye test : Aniline is first diazotised. On adding alkaline solution. of β -naphthol to the diazotised product a red-orange dye is formed.

(iii) On heating with bromine water, a white ppt. is formed.

BEGINNER'S BOX-1

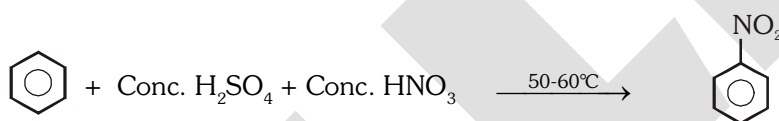
- Which one of following reaction is Schotten–Baumann reaction
 (1) Acetylation of RNH_2 (2) Acylation of RNH_2
 (3) Benzoylation of RNH_2 (4) All of them
- Which of the following pair will yield primary amine on hydrolysis
 (1) CH_3NCO , CH_3NC (2) CH_3CN , CH_3NC
 (3) $(\text{CH}_3)_2\text{NH}$, $\text{CH}_3\text{—CH=NOH}$ (4) None of the above
- Methylamine on treatment with chloroform and ethanolic caustic alkali gives foul smelling compound, the compound is
 (1) CH_3NCO (2) CH_3CNO (3) CH_3CN (4) CH_3NC

6.0 NITRO BENZENE [$\text{C}_6\text{H}_5\text{NO}_2$]

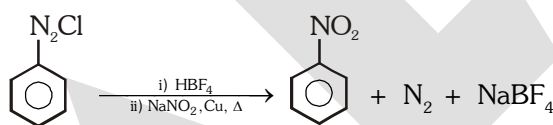
It is also called as artificial oil of bitter almonds or oil of mirbane as its odour is like that of bitter almonds.

6.1 General Methods of Preparation

(1) Lab method :



(2) From diazonium salt :



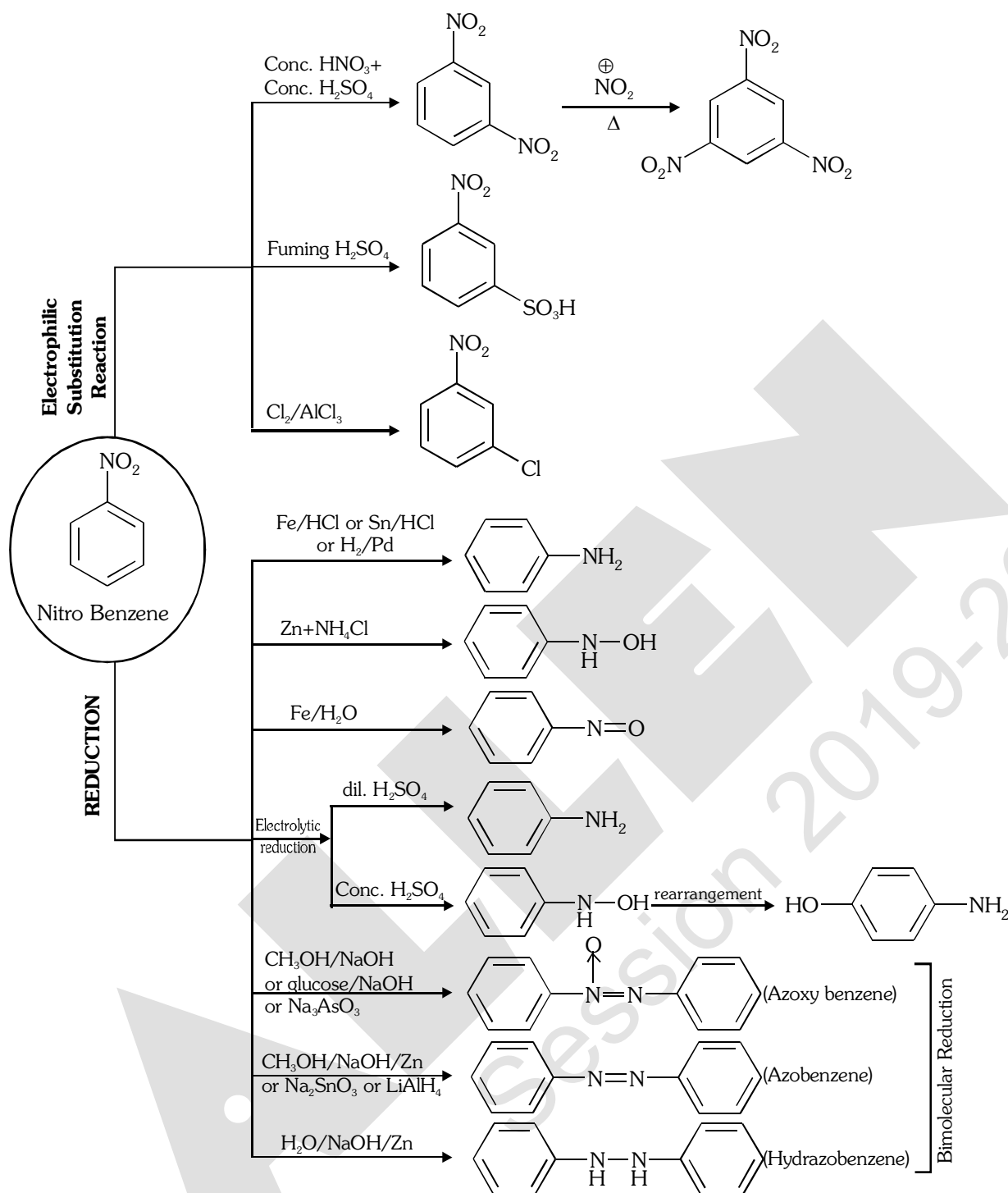
6.2 Physical Properties

- Nitro benzene is light yellow oily liquid
- It has smell of bitter almonds
- It is steam volatile. Its vapours are poisonous in nature.
- It is heavier than water
- It's B. P. is 211°C
- Smell of nitro benzene and benzaldehyde is same

6.3 Chemical Properties

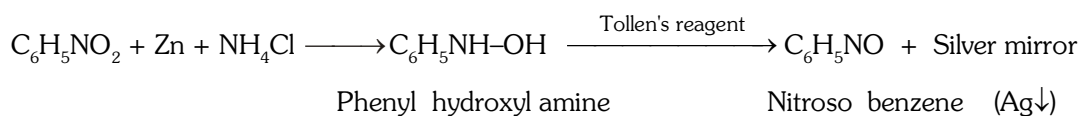
Nitrobenzene shows following chemical reactions –

- Reactions due to NO_2 group.
- Reactions due to benzene ring : (A) Electrophilic substitution (B) Nucleophilic substitution



6.4 Test of Nitrobenzene

Mullikan Barker Test : Ethanolic solution of nitrobenzene is treated with zinc dust and NH_4Cl solution. The mixture is heated and filter in a test tube containing Tollen's reagent a grey or black precipitate (Ag mirror) is formed.

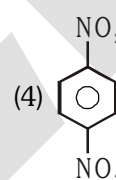
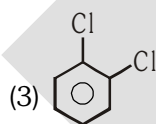
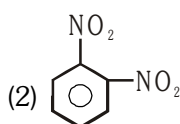
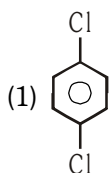


6.5 Uses

- As a solvent
- In manufacture of aniline and azo dye

BEGINNER'S BOX-2

- Nitration of which of the following compound is difficult :-
 (1) Benzene (2) Nitrobenzene (3) Toluene (4) Phenol
- Nitration of nitrobenzene in presence of fuming nitric acid will generate a :-
 (1) Solid product (2) Gaseous product (3) Semi-solid product (4) Liquid product
- $C_6H_5NO_2 \xrightarrow{Sn/HCl} A \xrightarrow{NaNO_2/HCl} B$; To obtain benzene from B, the suitable reagent is :
 (1) $SnCl_2 + HCl$ (2) H_3PO_2 (3) $C_2H_5N_2Cl$ (4) Methanol
- When nitrobenzene is heated with conc. HNO_3 and conc. H_2SO_4 the product would be obtained :-
 (1) T.N.T. (2) D.N.B. (3) D.D.T. (4) T.E.L.
- Which of the following has the maximum value of dipole moment ?



ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3							
	Ans.	4	1	4							

BEGINNER'S BOX-2	Que.	1	2	3	4	5					
	Ans.	2	1	2	2	2					

EXERCISE-I (Conceptual Questions)

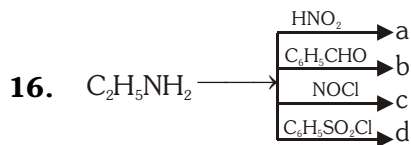
Build Up Your Understanding

AMINE

- Among the following which one is not formed in Hoffmann degradation
 (1) RNCO (2) $\text{R}-\ddot{\text{N}}\text{H}_2$
 (3) RCONHBr (4) RNC
- $\text{CH}_3\text{CH}_2\text{CONH}_2 \xrightarrow[\text{Br}_2]{\text{NaOH}} \text{A}$,
 Aqueous solution of A
 (1) Turns blue litmus to red
 (2) Turns red litmus to blue
 (3) Does not affect the litmus
 (4) Decolourise the litmus
- Ethanamine can be obtained if the following compound is heated with $[\text{KOH} + \text{Br}_2]$
 (1) Ethanamide (2) Methanamide
 (3) Propionamide (4) All the above
- $\text{CH}_3\text{CONH}_2 \xrightarrow[\text{I}]{\text{P}_2\text{O}_5} \text{A} \xrightarrow[\text{II}]{\text{Na/EtOH}} \text{B}$
 Reaction II is called
 (1) Clemensen
 (2) Stephen
 (3) Mendius
 (4) Bauveault-blank reduction
- CH_3CONH_2 , Br_2 & KOH give CH_3NH_2 as the product. The intermediates of the reaction are :-
 (a) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NHBr}$ (b) $\text{CH}_3-\text{N}=\text{C}=\text{O}$
 (c) CH_3NHBr (d) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{N}(\text{Br})_2$
 The correct answer is :-
 (1) a, b (2) a, c (3) b, d (4) c, d
- In which case alkylamine is not formed :-
 (1) $\text{R}-\text{X} + \text{NH}_3 \longrightarrow$
 (2) $\text{R}-\text{CH}=\text{NOH} + [\text{H}] \xrightarrow[\text{alc.}]{\text{Na}}$
 (3) $\text{R}-\text{CN} + \text{H}_2\text{O} \xrightarrow{\text{H}^+}$
 (4) $\text{RCONH}_2 + 4[\text{H}] \xrightarrow{\text{LiAlH}_4}$

- Tertiary amine is obtained in the reaction :-
 (1) Aniline $\xrightarrow{\text{CH}_3\text{I}} \xrightarrow{\text{CH}_3\text{I}}$
 (2) Aniline $\xrightarrow{\text{CH}_3\text{I}}$
 (3) Nitrobenzene $\xrightarrow{\text{Sn/HCl}}$
 (4) None of the above
- $\text{C}_2\text{H}_5\text{NH}_2$ cannot be prepared by the reduction of
 (1) $\text{C}_2\text{H}_5\text{NO}_2$ (2) $\text{CH}_3\text{CH}=\text{NOH}$
 (3) $\text{C}_2\text{H}_5\text{NC}$ (4) CH_3CN
- Gabriel reaction for the synthesis of amines, involves the use of
 (1) 1° amide (2) 2° amide
 (3) Imides (4) Aliphatic amide
- Gabriel phthalimide reaction is used in the synthesis of
 (1) Primary aromatic amines
 (2) Secondary amines
 (3) Primary aliphatic amines
 (4) Tertiary amines
- The reaction : $[\text{C}_2\text{H}_5\text{Br} + \text{NH}_3]$ is in fact an example of
 (1) Ammonolysis only
 (2) Nucleophilic substitution only
 (3) Ammonolysis as well as nucleophilic substitution
 (4) None
- Melting points are normally the highest for
 (1) Tertiary amides (2) Secondary amides
 (3) Primary amides (4) Amines
- Solubility of ethylamine in water is due to
 (1) Low molecular weight
 (2) Ethyl group is present in ethyl alcohol
 (3) Formation of H-bonding with water
 (4) Being a derivative of ammonia
- Which of the following compound liberates CO_2 when treated with NaHCO_3
 (1) $\text{CH}_3\text{COCH}_2\text{NH}_2$ (2) CH_3NH_2
 (3) $(\text{CH}_3)_4\text{N}^+\text{OH}^-$ (4) $\text{CH}_3\text{N}^+\text{H}_3\text{Cl}^-$

15. The product obtained by the alkaline hydrolysis of $C_2H_5-N=C=O$ when treated with t-butyl magnesiumbromide, the compound obtained will be
- (1) t-butylamine (2) n-butylamine
(3) Isobutane (4) n-butane

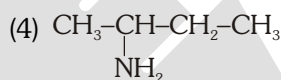
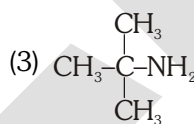
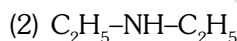
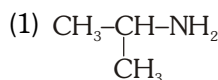


Which product is a Schiff's base :-

- (1) a (2) b (3) c (4) d
17. Acidic nature of amino group is shown by the reaction :-
- (1) $R-NH_2 + NOCl \rightarrow RCl + N_2 + H_2O$
(2) $2RNH_2 + 2Na \rightarrow 2RNH.Na + H_2$
(3) $R.CH_2NH_2 + HNO_2 \rightarrow R.CH_2OH + N_2 + H_2O$
(4) $R.NH_2 + HCl \rightarrow R\overset{\oplus}{N}H_3 \overset{\ominus}{Cl}$
18. The reagent used in the conversion of $C_2H_5NH_2$ to C_2H_5Cl would be
- (1) SO_2Cl_2 (2) $SOCl_2$
(3) $NOCl$ (4) All
19. Hydrogen attached to nitrogen is released in the reaction
- (1) $RCONH_2 + NaNH_2$
(2) $RNH_2 + Na$
(3) Both the above
(4) None of the above
20. If primary amines are treated with ketones the product is
- (1) Urea (2) Guanidine
(3) Amide (4) Schiff's base
21. Reactants of reaction – I are CH_3CONH_2 , KOH , Br_2
Reactants of reaction-II are CH_3NH_2 , $CHCl_3$, KOH
The intermediate species of reaction-I and reaction-II are respectively
- (1) Carbonium ion, carbene
(2) Nitrene, carbene
(3) Carbene, nitrene
(4) Carbocation, carbanion

22. N, N-dimethyl acetamide is obtained in the reaction
- (1) Acetyl chloride and methanamine
(2) Acetyl chloride and ethanamine
(3) Acetyl chloride and dimethylamine
(4) Acetyl chloride and diethylamine

23. This compound does not respond to carbylamine reaction :-



24. Blue litmus can be turned to red by the compound
- (1) ROH
(2) RNH_2
(3) $R\overset{\oplus}{N}H_3 \overset{\ominus}{OH}$
(4) $R\overset{\oplus}{N}H_3 \overset{\ominus}{Cl}$
25. Match List I with List II and select the correct answer using the codes given below :-

List I	List II
[Reagent]	[Used as test reagent for]
A. Ammonical $AgNO_3$	a. Primary amine
B. HIO_4	b. Aldehyde
C. Cold dil. Alkaline $KMnO_4$	c. Vicinal-OH groups
D. Chloroform + $NaOH$	d. Double bond

Codes :

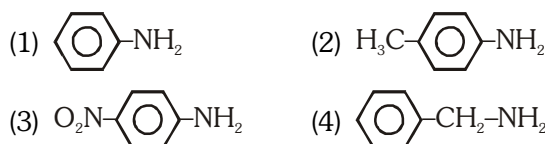
	A	B	C	D
(1)	b	c	a	d
(2)	d	b	a	c
(3)	b	c	d	a
(4)	d	c	b	a

26. Which one of the following aromatic amino compound gives alcohol with HNO_2 :-
 (1) N,N-Dimethylaniline (2) Benzylamine
 (3) N-methylaniline (4) Aniline

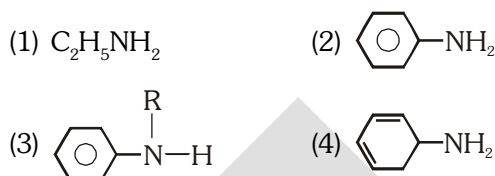
ANILINE

27. Electrolytic reduction of nitrobenzene in weakly acidic medium gives
 (1) Aniline
 (2) p-Hydroxyaniline
 (3) N-Phenylhydroxylamine
 (4) Nitrosobenzene
28. $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{\text{Br}_2/\text{CCl}_4} ?$ The product is :-
 (1) Only o- bromoaniline
 (2) 2, 4, 6-tribromoaniline
 (3) o-and p-bromoaniline
 (4) Only p-bromoaniline
29. Before proceeding for the nitration of aminobenzene, the NH_2 group is first protected by:-
 (1) Alkylation
 (2) Acetylation
 (3) Formylation
 (4) Chloromethylation
30. Aniline is purified by :-
 (1) Azeotropic distillation
 (2) Steam distillation
 (3) distillation in presence of magnesium
 (4) Fractional distillation
31. Reaction $\text{C}_6\text{H}_5\text{NH}_2 + \text{HAuCl}_4 \longrightarrow$
 $[\text{C}_6\text{H}_5\text{NH}_3^+]\text{AuCl}_4^-$ shows ... behaviour of aniline :-
 (1) Acidic (2) Neutral
 (3) Basic (4) Amphoteric
32. Aniline on treatment with bromine water yields white precipitate of :-
 (1) o-Bromoaniline
 (2) p-Bromoniline
 (3) 2, 4, 6-Tribromoaniline
 (4) m-Bromoaniline

33. Which compound does not show diazo reaction:-



34. Which of the following amines give N-nitroso derivative with NaNO_2 and HCl :-



35. Which of the following involves nitrene as an intermediate ?

- (1) Carbylamine reaction
 (2) Hofmann bromamide reaction
 (3) Reimer Tiemann reaction
 (4) Friedal crafts reaction

36. Which of the following does not reduce Tollen's reagent :-

- (1) CH_3CHO (2) HCOOH
 (3) $\text{C}_6\text{H}_5\text{NHOH}$ (4) $\text{C}_6\text{H}_5\text{NH}_2$

37. Aniline can be obtained by :-

- (1) Benzoyl chloride and ammonia
 (2) Reduction of benzamide
 (3) Phenol and ammonia in presence of ZnCl_2
 (4) Benzoic anhydride and ammonia

38. Aniline on direct nitration produces :-

- (1) o-Nitroaniline (2) m-Nitroaniline
 (3) p-Nitroaniline (4) All

39. Nitration of acetanilide followed by hydrolysis gives

- (1) o-Nitroaniline (2) p-Nitroaniline
 (3) o- & p-Nitroaniline (4) o-Nitroanilinium ion

40. $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow[0-5^\circ\text{C}]{\text{NaNO}_2/\text{HCl}} \text{A}$, Which is the incorrect structure of the product 'A' :-

- (1) $\text{C}_6\text{H}_5-\text{N}=\text{N}-\text{Cl}$ (2) $[\text{C}_6\text{H}_5\text{N}_2]^+\text{Cl}^-$
 (3) $[\text{C}_6\text{H}_5-\text{N}^+\equiv\text{N}]\text{Cl}^-$ (4) $[\text{C}_6\text{H}_5-\text{N}^+\equiv\text{N}]\text{Cl}^-$

41. Chloroform and ethanolic KOH is used as a reagent in the following reaction :-

- Hoffmann carbylamine reaction
- Hoffmann degradation reaction
- Reimer-Tiemann reaction
- Hoffmann mustard oil reaction

Code is :-

- Only for a
- Only for a and b
- Only for b and d
- Only for a and c

42. Acetanilide when treated with bromine in acetic acid mainly gives :-

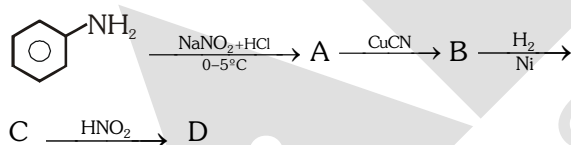
- o-Bromoacetanilide
- N-Bromoacetanilide
- p-Bromoacetanilide
- m-Bromoacetanilide

43. Aromatic nitriles (ArCN) are not prepared by reaction

- $\text{ArX} + \text{KCN}$
- $\text{ArN}_2^+ + \text{CuCN}$
- $\text{ArCONH}_2 + \text{P}_2\text{O}_5$
- $\text{ArCONH}_2 + \text{SOCl}_2$

NITRO GROUP, CYANIDE & ISOCYANIDE

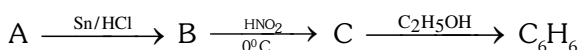
44. Aniline in a set of reactions yielded end product D



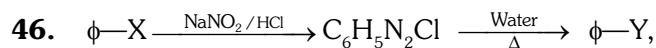
The structure of the product D would be

- $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
- $\text{C}_6\text{H}_5\text{NHOH}$
- $\text{C}_6\text{H}_5\text{NHCH}_2\text{CH}_3$

45. In the reaction sequence identify the functional group present in A, B, C :-



- NO_2 , NH_2 , $\text{N}=\text{N}$
- NO_2 , NH_2 , OH
- $-\text{OH}$, $-\text{NH}_2$, $-\text{NO}$
- $-\text{NH}_2$, $-\text{NO}_2$, $-\text{N}=\text{N}-$



In the above sequence X and Y are :-

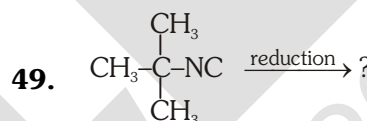
- o-, p- and m-directing
- o-, p- and o-, p-directing
- m and m directing
- m and o, p directing

47. Which of the following compound gives an explosive on decarboxylation :-

- 2,4, 6-Trinitrobenzoic acid
- 2, 4-Dinitrobenzoic acid
- o-Aminobenzoic acid
- o-Hydroxybenzoic acid

48. The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was:-

- Methylisocyanate
- Methylamine
- Ammonia
- Phosgene



- $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{NH}_2 \\ | \\ \text{CH}_3 \end{array}$
- $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{NH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{NH}-\text{CH}_2\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- None

50. Reaction of RCN with sodium and alcohol leads to the formation of :-

- RCONH_2
- $\text{RCOO}^-\text{NH}_4^+$
- RCH_2NH_2
- $\text{R}(\text{CH}_2)_3\text{NH}_2$

51. $C_6H_5NO_2 \xrightarrow{SnCl_2/HCl} A \xrightarrow[0^\circ C]{NaNO_2/HCl} B$; In the

above sequence Benzene from B, is suitably obtained by using :-

- (1) Ethanol
- (2) H_3PO_2
- (3) Both the above
- (4) Methanol

52. Which reagent is used to get iodo benzene from benzene diazonium acid sulphate $[C_6H_5N_2HSO_4]$:

- (1) $CuBr, \Delta$
- (2) Cu powder + HI
- (3) KI, Δ
- (4) None

53. Which of the following is used as a solvent in the Friedel-Crafts reaction :-

- (1) Toluene
- (2) Nitrobenzene
- (3) Benzene
- (4) Aniline

54. Match list I with II and choose the correct answer from the codes given below :-

List I

- | | |
|---------------------|---|
| (A) Aniline | a. Used in making azo dyes |
| (B) Nitrobenzene | b. Sulpha drug |
| (C) Sulphanilamide | c. Solvent in the Friedel Crafts reaction |
| (D) Trinitrotoluene | d. Used as explosive |

List II

Code is :-

	A	B	C	D
(1)	a	c	b	d
(2)	a	b	c	d
(3)	c	d	a	b
(4)	d	c	b	a

55. In the Sandmeyer's reaction, $-N=N-X$ group of diazonium salt is replaced by :-

- (1) Halide group
- (2) Nitro group
- (3) $-OH$ group
- (4) $-NHNH_2$ group

EXERCISE-I (Conceptual Questions)

ANSWER KEY

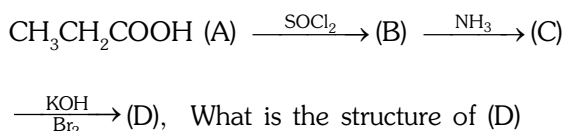
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Ans.	4	2	3	3	1	3	1	3	3	3	3	3	3	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	3	3	4	2	3	2	4	3	2	1	2	2	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	3	4	3	2	4	3	4	3	1	4	3	1	1	1
Que.	46	47	48	49	50	51	52	53	54	55					
Ans.	2	1	1	2	3	3	3	2	1	1					

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

1. In a set of reactions propionic acid yielded a compound (D) ;



- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ (2) $\text{CH}_3\text{CH}_2\text{CONH}_2$
(3) $\text{CH}_3\text{CH}_2\text{NHCH}_3$ (4) $\text{CH}_3\text{CH}_2\text{NH}_2$

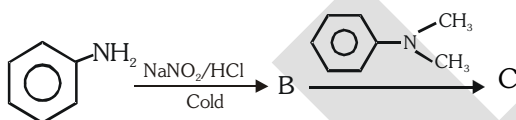
AIPMT 2007

2. Which one of the following on reduction with LiAlH_4 yields a secondary amine

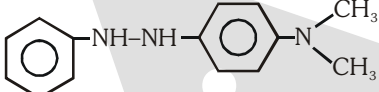
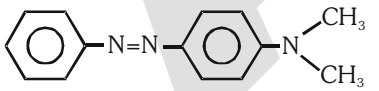
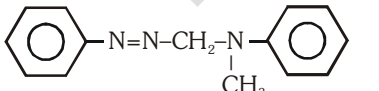
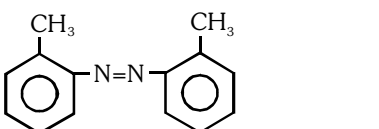
- (1) Methyl isocyanide
(2) Acetamide
(3) Methyl cyanide
(4) Nitro ethane

AIPMT 2008

3. In a reaction of aniline a coloured product C was obtained.

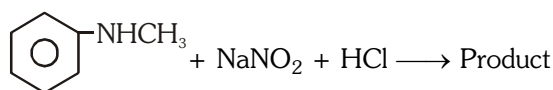


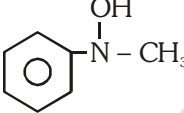
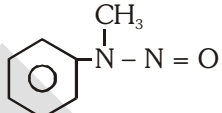
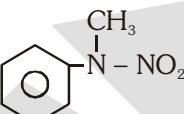
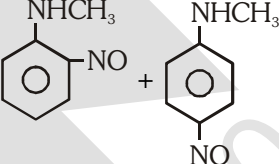
The structure C would be :

- (1) 
(2) 
(3) 
(4) 

AIPMT 2009

4. Predict the product :



- (1)  (2) 
(3)  (4) 

AIPMT 2010

5. Which of the following statements about primary amines is 'False' ?

- (1) Alkyl amines are stronger bases than ammonia
(2) Alkyl amines are stronger bases than aryl amines
(3) Alkyl amines react with nitrous acid to produce alcohols
(4) Aryl amines react with nitrous acid to produce phenols

6. Acetamide is treated with the following reagents separately. Which one of these would yield methyl amine ?

- (1) PCl_5 (2) NaOH/Br_2
(3) Sodalime (4) Hot conc. H_2SO_4

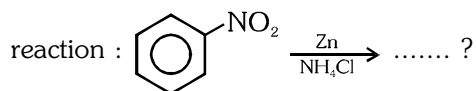
AIIMS 2010

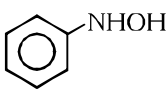
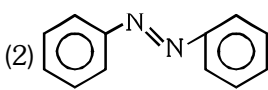
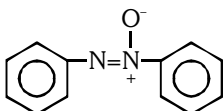
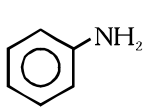
7. Which of the following gives a soluble complex on reaction with benzene sulphonyl chloride (which of the following compounds after reacting with benzene sulphonyl chloride is soluble in alkali solution) :-

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
(2) $\text{CH}_3\text{CH}_2\text{NHCH}_3$
(3) $\text{CH}_3\text{CH}_2\text{OH}$
(4) $(\text{CH}_3)_3\text{N}$

AIPMT Pre. 2011

8. What is the product obtained in the following



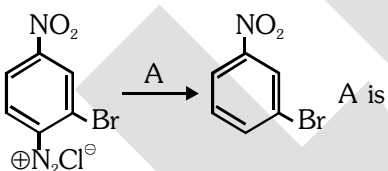
- (1)  (2) 
 (3)  (4) 

AIPMT Mains 2012

9. An organic compound C_3H_9N (A), when treated with nitrous acid, gave an alcohol and N_2 gas was evolved. (A) on warming with $CHCl_3$ and caustic potash gave (C) which on reduction gave isopropylmethylamine. Predict the structure of (A):

- (1) $CH_3 - \overset{\text{CH}_3}{\underset{|}{N}} - CH_3$ (2) $CH_3CH_2CH_2-NH_2$
 (3) $\begin{matrix} CH_3 \\ | \\ CH_3 \end{matrix} CH - NH_2$ (4) $CH_3CH_2-NH-CH_3$

NEET UG 2013

10. In the reaction  A is

- (1) H^+/H_2O (2) $HgSO_4/H_2SO_4$
 (3) Cu_2Cl_2 (4) H_3PO_2 and H_2O

11. Nitrobenzene on reaction with conc. HNO_3/H_2SO_4 at $80-100^\circ C$ forms which one of the following products ?

- (1) 1, 2, 4-Trinitrobenzene
 (2) 1, 2-Dinitrobenzene
 (3) 1, 3-Dinitrobenzene
 (4) 1, 4-Dinitrobenzene

AIIMS 2013

12. Reaction of aniline with HNO_2 followed by treatment of dilute acid gives :-

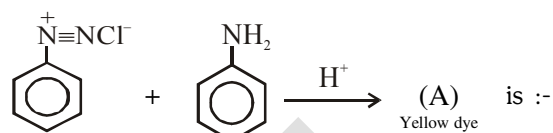
- (1) C_6H_5NHOH (2) C_6H_5OH
 (3) $C_6H_5NHNH_2$ (4) C_6H_6

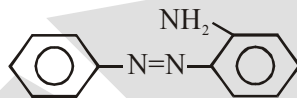
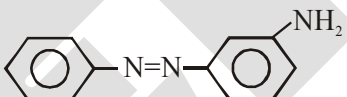
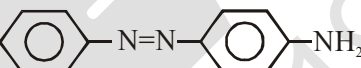
13. Which of the following will give carbylamine test?

- (1) CH_3NH_2 (2) CH_3NHCH_3
 (3) $(CH_3)_3N$ (4) CH_3CONH_2

AIPMT 2014

14. In the following reaction, the product (A)



- (1) 
 (2) 
 (3) 
 (4) 

15. Which of the following will be most stable diazonium salt $RN_2^+X^-$?

- (1) $CH_3N_2^+X^-$ (2) $C_6H_5N_2^+X^-$
 (3) $CH_3CH_2N_2^+X^-$ (4) $C_6H_5CH_2N_2^+X^-$

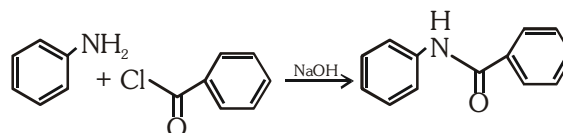
AIPMT 2015

16. The electrolytic reduction of nitrobenzene in strongly acidic medium produces :-

- (1) Azoxybenzene (2) Azobenzene
 (3) Aniline (4) p-Aminophenol

RE-AIPMT 2015

17. The following reaction



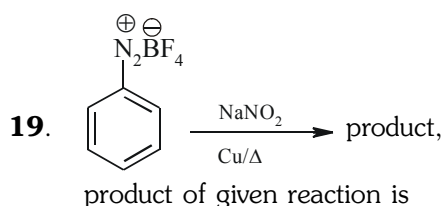
is known by the name :

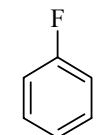
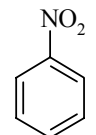
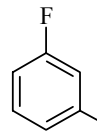
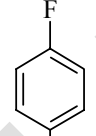
- (1) Acetylation reaction
 (2) Schotten-Baumann reaction
 (3) Friedel-Craft's reaction
 (4) Perkin's reaction

18. Method by which Aniline cannot be prepared is :-

- (1) reduction of nitrobenzene with H_2/Pd in ethanol
- (2) potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
- (3) hydrolysis of phenylisocyanide with acidic solution
- (4) degradation of benzamide with bromine in alkaline solution

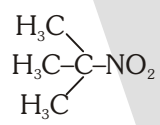
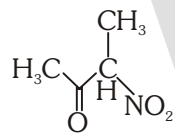
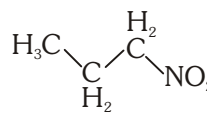
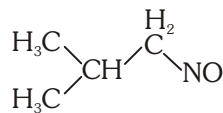
AIIMS 2015



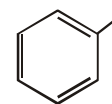
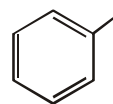
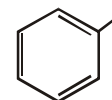
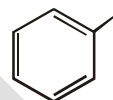
- (1) 
- (2) 
- (3) 
- (4) 

NEET-II 2016

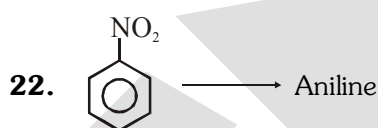
20. Which one of the following nitro-compounds does not react with nitrous acid ?

- (1) 
- (2) 
- (3) 
- (4) 

21. A given nitrogen-containing aromatic compound (A) reacts with Sn/HCl , followed by HNO_2 to give an unstable compound (B). (B), on treatment with phenol, forms a beautiful coloured compound (C) with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound (A) is :-

- (1) 
- (2) 
- (3) 
- (4) 

AIIMS 2016



Aniline will not be prepared in presence of :-

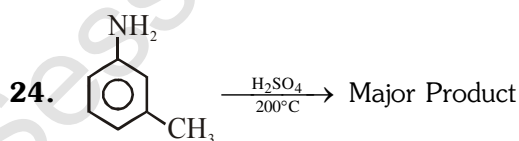
- (1) $Sn + HCl$
- (2) $Fe + HCl$
- (3) $Zn + NH_4Cl$
- (4) H_2/Pd

NEET(UG) 2017

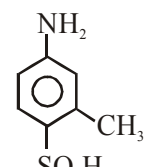
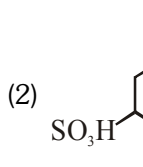
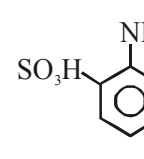
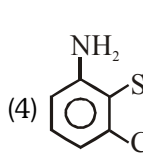
23. Which of the following reactions is appropriate for converting acetamide to methanamine ?

- (1) Hoffmann hypobromamide reaction
- (2) Stephens reaction
- (3) Gabriels phthalimide synthesis
- (4) Carbylamine reaction

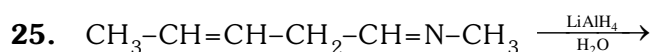
AIIMS 2017



Major product of this reaction is :-

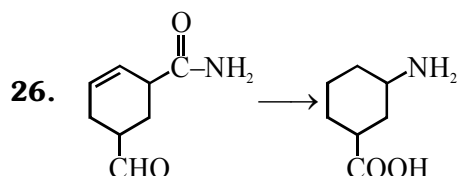
- (1) 
- (2) 
- (3) 
- (4) 

AIIMS 2018



product.

- (1) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{NH}-\text{CH}_3$
- (2) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{NH}-\text{CH}_3$
- (3) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}=\text{N}-\text{CH}_3$
- (4) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}=\text{N}-\text{CH}_3$



Select the correct sequence of reagents for the above conversion.

- (1) H_2 -Pd/c ; Br_2/KOH , Tollen's Reagent
- (2) Tollen's Reagent, Br_2/KOH , H_2 -Pd-c
- (3) Br_2/KOH , Tollen's Reagent, Br_2/KOH
- (4) H_2 -Pd-c, Tollen's Reagent, Br_2/KOH

27. Which of the following is incorrect statement regarding diazo salt :-

- (1) On reaction with $\text{C}_2\text{H}_5\text{OH}$, it form benzene
- (2) on reaction with HCN/CuCN it forms 
- (3) It gives nitrobenzene with NaNO_2
- (4) It gives iodobenzene with KI/Δ

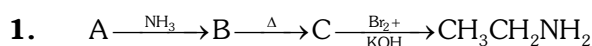
EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	2	2	4	2	1	1	3	4	3	2	1	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27			
Ans.	4	2	2	2	1	4	3	1	1	1	2	3			

EXERCISE-III (Analytical Questions)

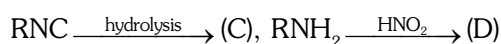
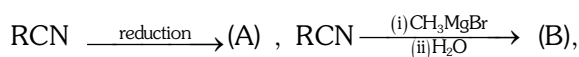
Check Your Understanding



A is :-

- (1) CH_3COOH
- (2) $CH_3CH_2CH_2COOH$
- (3) $CH_3-\underset{\text{CH}_3}{\text{CH}}-COOH$
- (4) CH_3CH_2COOH

2. The correct set of the products obtained in the following reactions

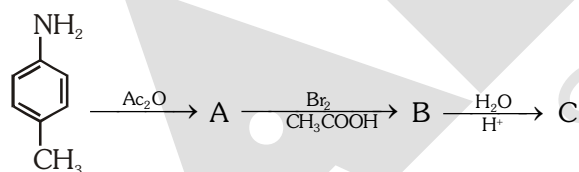


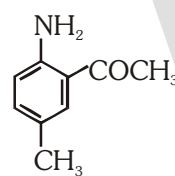
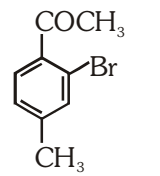
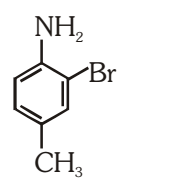
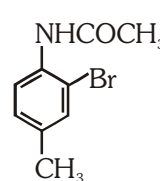
The answer is

A B C D

- (1) 2°Amine Methyl ketone 1°Amine Alcohol
- (2) 1°Amine Methyl ketone 1°Amine Alcohol
- (3) 2°Amine Methyl ketone 2°Amine Acid
- (4) 1°Amine Methyl ketone 2°Amine Aldehyde

3. The final product C, obtained in this reaction would be

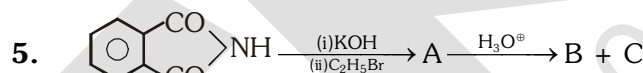


- (1) 
- (2) 
- (3) 
- (4) 



A, B, C respectively in the above sequence are

- (1) Ethane amine, methane nitrile and diethyl amine
- (2) Carbyl amino ethane, ethane amine and primary amine
- (3) Ethyl isocyanide, ethyl amine and methyl isocyanate
- (4) Carbylamino ethane, ethanamine and ethyl methyl amine



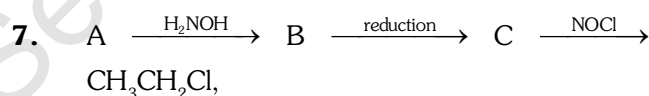
B and C in the above sequence are

- (1) Benzoic acid + aniline
- (2) Phthalic acid + ethylamine
- (3) Phthalic acid + aniline
- (4) Benzoic acid + ethylamine

6. The end-product in the reaction sequence would be :



- (1) Ethyl cyanide
- (2) Ethyl amine
- (3) Methyl amine
- (4) Acetamide



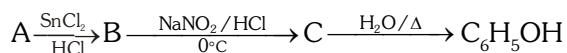
A and C in the above sequence respectively are:-

- (1) Methanal, Methylamine
- (2) Acetone, ethanamine
- (3) Ethanal, dimethylamine
- (4) Acetaldehyde, ethylamine

8. A compound of mol. wt. 180 gm is acetylated to give a compound of mol. wt. 390. The number of amino groups in the compound are :-

- (1) 2
- (2) 4
- (3) 5
- (4) 6

9. In the reaction sequence A, B and C are :-



- (1) Benzene, nitrobenzene, aniline
- (2) Nitrobenzene, aniline and azo-compound
- (3) Nitrobenzene, benzene, aniline
- (4) Benzene, amino compound, aniline

10. Which one of the following tests can be used to identify primary amino group in a given organic compound

- (1) Iodoform test
- (2) Victor Meyer's test
- (3) Carbylamine reaction
- (4) Libermann's reaction

11. At 0°C temperature reaction of aniline with HNO_2 and subsequent treatment with alkaline β -naphthol solution produces a precipitate whose colour would be :-

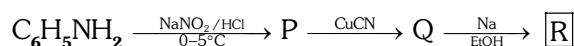
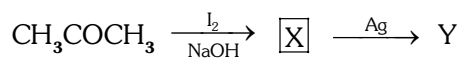
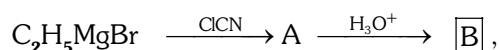
- (1) Black
- (2) Purple
- (3) White
- (4) Orange

12. $\text{CH}_3\text{Cl} \xrightarrow{\text{KCN}} \xrightarrow{+4\text{H}} \text{A} \xrightarrow{\text{NaNO}_2/\text{HCl}} \text{Ethanol} + ?$

Apart from ethanol as the main product, the other products would be

- (1) $\text{H}_2\text{C}=\text{CH}_2$
- (2) $\text{CH}_3\text{CH}_2\text{-ONO}$
- (3) $\text{CH}_3\text{CH}_2\text{-Cl}$
- (4) All the above

13. Identify B, X and R respectively in the following sequence of reactions



- (1) $\text{C}_2\text{H}_5\text{COOH}$, CHI_3 , $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
- (2) $\text{C}_2\text{H}_5\text{COOH}$, CH_3I , $\text{C}_6\text{H}_5\text{COOH}$
- (3) $\text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$, CH_3I , $\text{C}_6\text{H}_5\text{COOH}$
- (4) $\text{C}_2\text{H}_5\text{COOH}$, $\text{C}_2\text{H}_5\text{I}$, $\text{C}_6\text{H}_5\text{CONH}_2$

14. Using Fe/HCl , which one of the following reaction is possible [Here $\phi = \text{C}_6\text{H}_5$]

- (1) $\phi\text{-NO}_2 \rightarrow \phi\text{-NH-NH-}\phi$
- (2) $\phi\text{-NO}_2 \rightarrow \phi\text{-NH}_2$
- (3) $\phi\text{-NO}_2 \rightarrow \phi\text{-NHOH}$
- (4) $\phi\text{-NO}_2 \rightarrow \phi\text{-N=N-}\phi$

15. An aromatic compound A on treatment with $\text{Zn}/\text{NH}_4\text{Cl}$, and subsequent filtration in ammonical silver nitrate solution gives a black precipitate. Hence compound A bears the group :-

- (1) -COOH
- (2) -NO_2
- (3) -CHO
- (4) -NH_2

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	2	3	4	2	2	4	3	2	3	4	4	1	2	2

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion :-** Boiling point of trimethyl amine is higher than that of n-propyl amine.

Reason :- H-bonding is stronger in tertiary amines.

- (1) A (2) B (3) C (4) D

2. **Assertion :-** $\text{H}-\text{C}-\text{N}-\text{CH}_3$ and $\begin{array}{c} \text{O} \\ \parallel \\ \text{CH}_3 \end{array}$

$\text{CH}_3-\text{C}-\text{NH}-\text{CH}_3$ are metamers.

Reason :- There is adjustment of $-\text{CH}_2-$ group in the molecule.

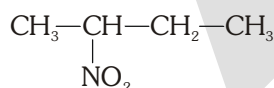
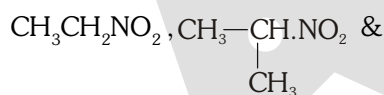
- (1) A (2) B (3) C (4) D

3. **Assertion :-** Pyrole is alicyclic compound.

Reason :- It is aliphatic in nature and has cyclic chain.

- (1) A (2) B (3) C (4) D

4. **Assertion :-** The following compound are given below –



all three compounds are soluble in NaOH.

Reason :- All above compound have lower boiling than CH_3NO_2 .

- (1) A (2) B (3) C (4) D

5. **Assertion :-** Acetonitrile is another name of ethanenitrile.

Reason :- α -H atom of acetonitrile exhibit acidic character.

- (1) A (2) B (3) C (4) D

6. **Assertion :-** Benzene diazonium chloride on reaction with ethanol gives benzene.

Reason :- Ethanol is good oxidising agent.

- (1) A (2) B (3) C (4) D

7. **Assertion :-** Diazonium salt is prepared at low temperature.

Reason :- Diazonium salt is hydrolysed to give phenol.

- (1) A (2) B (3) C (4) D

8. **Assertion :-** Carbylamine reaction involves reaction between 1° amine and chloroform in basic medium to give isocyanide.

Reason :- Methylene ($:\text{CH}_2$) carbene is formed as a reaction intermediate.

- (1) A (2) B (3) C (4) D

9. **Assertion :-** Suitable method for preparation of primary amine is when haloalkane is treated with excess of NH_3 .

Reason :- In this reaction primary, secondary, tertiary amines are formed.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

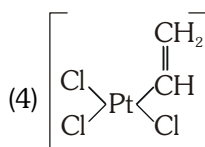
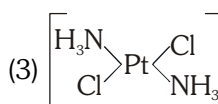
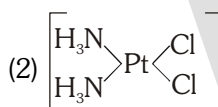
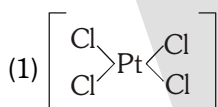
Que.	1	2	3	4	5	6	7	8	9
Ans.	4	4	4	3	2	3	1	3	3

EXERCISE-I (Conceptual Questions)

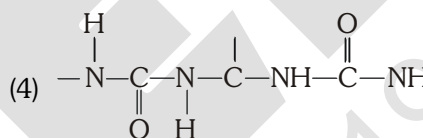
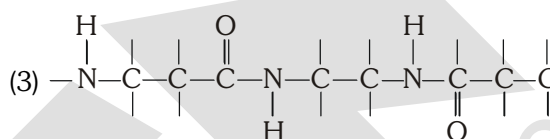
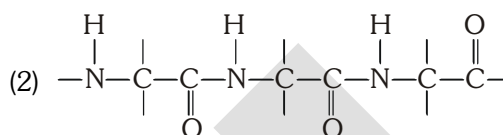
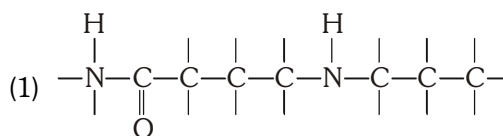
Build Up Your Understanding

BIOMOLECULES

- The nitrogenous base having two possible hydrogen bonding sites is –
 (1) Thymine (2) Cytosine
 (3) Guanine (4) None
- The correct statement in respect of protein haemoglobin is that it –
 (1) Maintains blood sugar level
 (2) Acts as an oxygen carrier in the blood
 (3) Forms antibodies and offers resistance to diseases
 (4) acts as a catalyst for biological reactions
- A sequence of how many nucleotides in messenger RNA makes a codon for an amino acid ?
 (1) 4 (2) 1
 (3) 2 (4) 3
- The hormone that helps in the conversion of glucose to glycogen is –
 (1) Bile acids (2) Adrenaline
 (3) Insulin (4) Cortisone
- The helical structure of protein is stabilized by –
 (1) Hydrogen bonds (2) ether bonds
 (3) peptide bonds (4) dipeptide bonds
- Which of the following is considered to be an anticancer species



- Which of the following structures represents the peptide chain –



- Which functional group participates in disulphide bond formation in proteins –
 (1) Thioether (2) Thiol
 (3) Thioester (4) Thiolactone
- The cell membranes are mainly composed of –
 (1) Phospholipids (2) Proteins
 (3) Fats (4) Carbohydrates
- Which is simplest amino acid –
 (1) Alanine (2) Asparagine
 (3) Glycine (4) Tyrosine
- Which of the following biomolecules is insoluble in water –
 (1) α -Keratin (2) Haemoglobin
 (3) Ribonuclease (4) Adenine
- Which one of the following statements is true for protein synthesis (translation) –
 (1) Amino acids are directly recognized by m-RNA
 (2) The third base of the codon is less specific
 (3) Only one codon codes for an amino acid
 (4) Every t-RNA molecule has more than one amino acid attachment site

- 13.** The presence or absence of hydroxy group on which carbon atom of sugar differentiates RNA and DNA ?
 (1) 3rd (2) 4th (3) 1st (4) 2nd
- 14.** The change in the optical rotation of freshly prepared solution of glucose is known as :-
 (1) tautomerism
 (2) racemisation
 (3) specific rotation
 (4) mutarotation
- 15.** Synthesis of each molecule of glucose in photosynthesis involves :-
 (1) 18 molecules of ATP
 (2) 10 molecules of ATP
 (3) 8 molecules of ATP
 (4) 6 molecules of ATP
- 16.** Which one of the following bases is not present in DNA ?
 (1) Cytosine (2) Thymine
 (3) Quinoline (4) Adenine
- 17.** Lysine; $\text{H}_2\text{N}-(\text{CH}_2)_4-\underset{\text{NH}_2}{\text{CH}}-\text{COOH}$ is :-
 (1) α -Amino acid
 (2) γ -amino acid
 (3) Amino acid synthesised
 (4) β -Amino acid
- 18.** In fibrous protein, polypeptide chains are held together by :-
 (1) Vander waals forces
 (2) Disulphide linkage
 (3) Electrostatic forces attraction
 (4) None of these
- 19.** Which is correct in following -
 (1) Monosaccharides also known as sugars
 (2) Polysaccharides are non sugars
 (3) Maltose and Lactose are reducing sugar
 (4) All of these
- 20.** Structure of glycogen is similar to :-
 (1) Alylose (2) Amylopectin
 (3) Cellulose (4) Glucose
- 21.** Which of the following gives osazone different from the other three :-
 (1) Glucose (2) Mannose
 (3) Galactose (4) Fructose
- 22.** Anomers of glucose (α -form & β -form) are differ in the stereochemistry at which carbon -
 (1) C-1 (2) C-2
 (3) C-3 (4) All of these
- 23.** Sucrose in presence of invertase on hydrolysis gave -
 (1) Glucose (2) Fructose
 (3) Ethyl alcohol (4) 1 & 2 both
- 24.** Which of the following B group vitamins can be stored in our body.
 (1) Vitamin B₁ (2) Vitamin B₂
 (3) Vitamin B₆ (4) Vitamin B₁₂
- 25.** Which of the following are polysaccharides -
 (a) Starch (b) cellulose
 (c) dextrans (d) glycogen
 (1) a, b & c (2) a, b, d
 (3) a & c (4) a, b, c, d
- 26.** Cellulose can not be tested by followings -
 (1) Fehling's solution (2) Tollen's reagent
 (3) Both of these (4) None of these
- 27.** On hydrolysis of proteins, the product is/are -
 (1) Amino acids (2) Peptides
 (3) Enzymes (4) 1 & 2 both
- 28.** Which of the following do not have hemiacetal group:-
 (1) Fructose (2) Maltose
 (3) Sucrose (4) Glucose
- 29.** In amino acids, more number of amino than carboxyl groups makes it -
 (1) acidic (2) Basic
 (3) Neutral (4) None of these
- 30.** Which amino acids are called non essential -
 (1) those which can be synthesized in the body.
 (2) those which have more amino groups as compared to carboxyl groups
 (3) those which have equal number of amino acid and carboxyl groups
 (4) None of these

- 31.** Which of the following is not essential amino acid –
(1) Serine (2) Lysine
(3) Threonine (4) Tryptophan
- 32.** In acidic & alkaline solution amino acids exists as a –
(1) Positive ion & negative ion respectively
(2) Negative ion & positive ions respectively
(3) Neutral in both medium
(4) None of these
- 33.** In which of following shapes are found in tertiary structure of proteins –
(1) Fibrous (2) Globular
(3) Both of these (4) None of these
- 34.** The example of globular protein is
(1) Silk (2) Collagen
(3) Haemoglobin (4) All of these
- 35.** If a native protein is subjected to physical or chemical treatment which may disrupt its various forms without affecting its primary structure, are called –
(1) Inactive protein
(2) Denatured protein
(3) Both of these
(4) None of these
- 36.** The coagulation of egg protein while boiling of egg is called –
(1) Reversible protein denaturation
(2) Irreversible protein denaturation
(3) Renaturation
(4) None of these
- 37.** An enzyme molecule may contain protein and non-protein part. Non-protein part is known as:-
(1) Holoenzyme (2) Cofactor
(3) Inverted enzyme (4) None of these
- 38.** Cofactor which gets attached to the enzyme at the time of reaction are known as
(1) Coenzymes (2) Apoenzyme
(3) Prosthetic group (4) None of these
- 39.** Mainly DNA is localized in –
(1) Cytoplasm (2) Nucleus
(3) Mitochondria (4) Chloroplasts
- 40.** RNA contains following pyrimidine bases –
(a) Thymine (b) Uracil
(c) Cytosine (d) Adenine
(1) b & c (2) a, b, c
(3) a, b, d (4) All of these
- 41.** DNA molecules can duplicate themselves-called –
(1) Replication
(2) Translation
(3) Transcription
(4) None of these
- 42.** Which of the following carbohydrates are branched polymer of glucose.
(1) Glycogen (2) Amylopectin
(3) Cellulose (4) Both (1) & (2)
- 43.** Vitamin K
(1) Is phyloquinone
(2) Soluble in oils and fats
(3) Deficiency lenthens the blood clotting
(4) All of these
- 44.** Which of the following is not an amino acid
(1) Histidine (2) Benzidine
(3) Alanine (4) Proline
- 45.** Vitamin C is called :
(1) Antisterility (2) Antiscurvy
(3) both of these (4) None of these
- 46.** Reducing sugars are one which :
(1) reduce fehling's solutions
(2) not reduce tollen's reagent
(3) Have bonded aldehydic or ketonic groups
(4) All of these
- 47.** Multiple deficiencies caused by lack of more than one vitamin are more common in human beings. This condition of vitamin deficiency is known as
(1) avitaminosis (2) xerophthalmia
(3) convulsions (4) None of these

48. Which vitamin is synthesized in human body from carotene

- (1) Vitamin-A (2) Vitamin-C
(3) Vitamin-K (4) All of these

49. Which of the following is correct about H-bonding in nucleotide –

- (1) A-T G-C (2) A-G T-C
(3) G-T A-C (4) A-A T-T

POLYMER

50. Which of the following is monomer unit of polystyrene :-

- (1) $-\text{CH}_2-\underset{\text{CN}}{\text{CH}}-$ (2) $-\text{CF}_2-\text{CF}_2-$
(3) $-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-$ (4) $-\underset{\text{C}_6\text{H}_5}{\text{CH}}-\text{CH}_2-$

51. Weakest intermolecular forces are present in :-

- (1) Neoprene (2) Terylene
(3) Polystyrene (4) Bakelite

52. Thermosetting polymer, Bakelite is formed by the reaction of phenol with :-

- (1) $\text{CH}_3\text{CH}_2\text{CHO}$ (2) CH_3CHO
(3) HCHO (4) HCOOH

53. Which one is classified as a condensation polymer?

- (1) Teflon (2) Acrylonitrile
(3) Dacron (4) Neoprene

54. Novolac is a :

- (1) linear polymer of urea and formaldehyde
(2) crosslink polymer of urea and formaldehyde
(3) linear polymer of phenol and formaldehyde
(4) crosslink polymer of phenol and formaldehyde

55. Which of the following is not a semisynthetic polymer.

- (1) cis-polyisoprene (2) cellulose nitrate
(3) cellulose acetate (4) vulcanised rubber

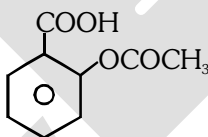
CHEMISTRY IN EVERYDAY LIFE

56. Aspirin is also known as –

- (1) Methyl salicylic acid (2) acetyl salicylate
(3) Methyl salicylate (4) Acetyl salicylic acid

57. Paracetamol is :-

- (1) Analgesic (2) Antipyretic
(3) Both (4) Antiseptic

58.  is used as

- (1) Antiseptic (2) Analgesic
(3) Antibiotic (4) Micro organism

59. Which among the following is a tranquilizer ?

- (1) Equanil (2) promethazine
(3) Omeprazole (4) Cimetidine

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	4	3	1	2	2	2	1	3	1	2	4	4	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	2	4	2	3	1	4	4	4	3	4	3	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	1	3	3	2	2	2	1	2	1	1	4	4	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Ans.	1	1	1	1	4	1	3	3	3	1	4	3	2	1	

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

- Which one of the following is a peptide hormone –
(1) Glucagon (2) Testosterone
(3) Thyroxin (4) Adrenaline
- During the process of digestion, the proteins present in food materials are hydrolysed to amino acids. The two enzymes involved in the process –



are respectively –

- (1) Amylase and Maltase
- (2) Diastase and Lipase
- (3) Pepsin and Trypsin
- (4) Invertase and Zymase

AIIMS 2006

- The pair in which both species have iron is –
(1) Nitrogenase, cytochromes
(2) Carboxypeptidase, haemoglobin
(3) Haemocyanin, nitrogenase
(4) Haemoglobin, cytochromes
- Thymine is –
(1) 5-Methyluracil (2) 4-Methyluracil
(3) 3-Methyluracil (4) 1-Methyluracil
- Lysine is least soluble in water in the pH range
(1) 3 to 4 (2) 5 to 6
(3) 6 to 7 (4) 8 to 9
- Methyl- α -D-glucoside and methyl- β -D-glucoside are –
(1) Epimers
(2) Anomers
(3) Enantiomers
(4) Conformational diastereomers

AIPMT 2007

- Which one of the following vitamins is water-soluble –
(1) Vitamin A (2) Vitamin B
(3) Vitamin E (4) Vitamin K
- RNA and DNA are chiral molecules, their chirality is due to –
(1) D-sugar Component
(2) L-sugar component
(3) Chiral bases
(4) Chiral phosphate ester units

- Which one of the following polymers is prepared by condensation polymerization
(1) Styrene (2) Nylon-66
(3) Teflon (4) Rubber

AIPMT 2008

- In DNA, the complimentary bases are :
(1) Adenine and thymine ; guanine and uracil
(2) Adenine and guanine ; thymine and cytosine
(3) Uracil and adenine ; cytosine and guanine
(4) Adenine and thymine ; guanine and cytosine
- Which one of the following is an amine hormone :
(1) Oxypurin (2) Insulin
(3) Progesterone (4) Thyroxine
- Which of the following statement is not true :
(1) Natural rubber has the trans-configuration at every double bond.
(2) Buna-S is a copolymer of butadiene and styrene.
(3) Natural rubber is a 1, 4-polymer of isoprene.
(4) In vulcanization, the formation of sulphur bridges between different chains make rubber harder and stronger.
- Green chemistry means such reaction which :
(1) reduce the use and production of hazardous chemicals.
(2) are related to the depletion of ozone layer
(3) study the reaction in plants
(4) produce colour during reactions

AIPMT 2009

- Which one of the following is employed as a tranquilizer ?
(1) Chlorpheninamine (2) Equanil
(3) Naproxen (4) Tetracycline
- Structures of some common polymers are given. Which one is not correctly presented ?
(1) Nylon 66 $\text{---}[\text{NH}(\text{CH}_2)_6\text{NHCO}(\text{CH}_2)_4\text{CO}]_n\text{---}$
(2) Teflon $\text{---}(\text{CF}_2 - \text{CF}_2)_n\text{---}$
(3) Neoprene $\left[\text{---CH}_2 - \underset{\text{Cl}}{\text{C}} = \text{CH} - \text{CH}_2 - \text{CH}_2\text{---} \right]_n$
(4) Terylene $\left\{ \text{OC} - \text{C}_6\text{H}_4 - \text{COOCH}_2 - \text{CH}_2 - \text{O} \right\}_n$
- The segment of DNA which acts as the instrumental manual for the synthesis of the protein is :-
(1) Nucleoside (2) Nucleotide
(3) Ribose (4) Gene

17. Which of the following hormones contains iodine ?
 (1) Thyroxine (2) Insulin
 (3) Testosterone (4) Adrenaline

AIPMT 2010

18. Which one of the following is employed as a tranquilizer drug ?
 (1) Mifepristone (2) Promethazine
 (3) Valium (4) Naproxen
19. Which one of the following does not exhibit the phenomenon of mutarotation ?
 (1) (-) Fructose (2) (+) Sucrose
 (3) (+) Lactose (4) (+) Maltose
20. Which of the following structures represents Neoprene polymer ?
 (1) $\left(\text{CH}_2 - \underset{\text{C}_6\text{H}_5}{\text{CH}} \right)_n$ (2) $\left(\text{CH}_2 - \underset{\text{Cl}}{\text{C}} = \text{CH} - \text{CH}_2 \right)_n$
 (3) $\left(\text{CH}_2 - \underset{\text{CN}}{\text{CH}} \right)_n$ (4) $\left(\text{CH}_2 - \underset{\text{Cl}}{\text{CH}} \right)_n$

AIPMT Main 2010

21. Fructose reduces Tollen's reagent due to :-
 (1) primary alcoholic group
 (2) secondary alcoholic group
 (3) enolisation of fructose followed by conversion to aldehyde by base.
 (4) asymmetric carbons

AIIMS 2010

22. The most basic α -amino acid is :-
 (1) Lysine (2) Histidine
 (3) Glutamine (4) Glycine
23. Units of lactose are :-
 (1) α -D Glucose and α -D-Glucose
 (2) α -D Glucose and β -D galactose
 (3) β -D Glucose and α -D galactose
 (4) β -D-Glucose and β -D galactose

AIPMT Pre. 2011

24. Which one of the following statements is not true regarding (+) Lactose ?
 (1) On hydrolysis (+) Lactose gives equal amount of D(+) glucose and D(+) galactose
 (2) (+) Lactose is a β -glycoside formed by the union of a molecule of D(+) glucose and union of a molecule of D(+) glucose and a molecule of D(+) galactose
 (3) (+) Lactose is a reducing sugar and does not exhibit mutarotation
 (4) (+) Lactose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ contains 8-OH groups

25. Which one of the following is employed as Antihistamine ?
 (1) Chloramphenicol (2) Diphenyl hydramine
 (3) Norethindrone (4) Omeprazole

26. Of the following which one is classified as polyester polymer ?
 (1) Terylene (2) Bakelite
 (3) Malamine (4) Nylon-66

AIPMT Mains 2011

27. Which of the following is not a fat soluble vitamin ?
 (1) Vitamin A (2) Vitamin B complex
 (3) Vitamin D (4) Vitamin E
28. Which of the statements about 'Denaturation' given below are correct ?
Statements
 (a) Denaturation of proteins causes loss of secondary and tertiary structures of the protein
 (b) Denaturation leads to the conversion of double strand of DNA into single strand.
 (c) Denaturation affects primary structure which gets distorted
Options :
 (1) (a), (b) and (c) (2) (b) and (c)
 (3) (a) and (c) (4) (a) and (b)

AIIMS 2011

29. Find the hydrolysis product when a phosphodiester bonds of nucleotide breaks:
 (1) 3-OH-deoxyribose-5 PO_4^{3-}
 (2) 5-OH-deoxyribose-3- PO_4^{3-}
 (3) 2-OH-deoxyribose-2- PO_4^{3-}
 (4) 4-OH-deoxyribose-2- PO_4^{3-}
30. Find the hydrolysis product of maltose
 (1) α -D Glucose + α -D-Glucose
 (2) α -D Glucose + α -D-Fructose
 (3) α -D Glucose + α -D-galactose
 (4) α -D Fructose + α -D-Galactose

AIPMT Pre 2012

31. Deficiency of vitamin B_1 causes the disease
 (1) Cheilosis (2) Sterility
 (3) Convulsions (4) Beri-Beri

32. Which one of the following sets of monosaccharides forms sucrose?

- (1) β -D-Glucopyranose and α -D-fructofuranose
- (2) α -D-Glucopyranose and β -D-fructopyranose
- (3) α -D-Galactopyranose and α -D-Glucopyranose
- (4) α -D-Glucopyranose and β -D-fructofuranose

33. Which one of the following is not a condensation polymer?

- (1) Dacron
- (2) Neoprene
- (3) Melamine
- (4) Glyptal

34. Which of the following statements is false?

- (1) The repeat unit in natural rubber is isoprene
- (2) Both starch and cellulose are polymers of glucose
- (3) Artificial silk is derived from cellulose
- (4) Nylon-66 is an example of elastomer

AIPMT Mains 2012

35. Which one of the following sets forms the biodegradable polymer ?

- (1) $\text{HO} - \text{CH}_2 - \text{CH}_2 - \text{OH}$ &



- (2) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH}_2$ and $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$

- (3) $\text{CH}_2 = \text{CH} - \text{CN}$ and $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$

- (4) $\text{H}_2\text{N} - \text{CH}_2 - \text{COOH}$ and $\text{H}_2\text{N} - (\text{CH}_2)_5 - \text{COOH}$

36. Chloroamphenicol is an :-

- (1) Antiseptic and disinfectant
- (2) Antibiotic broad spectrum
- (3) Antifertility drug
- (4) Antihistaminic

NEET UG 2013

37. Nylon is an example of :-

- (1) Polythene
- (2) Polyester
- (3) Polysaccharide
- (4) Polyamide

38. Antiseptics and disinfectants either kill or prevent growth of microorganisms. Identify which of the following statements is **not true** :-

- (1) Disinfectants harm the living tissues
- (2) A 0.2% solution of phenol is an antiseptic while 1% solution acts as a disinfectant
- (3) Chlorine and Iodine are used as strong disinfectants
- (4) Dilute solutions of Boric acid and Hydrogen Peroxide are strong antiseptics

39. Which is the monomer of Neoprene in the following ?

- (1) $\text{CH}_2 = \text{CH} - \text{C} \equiv \text{CH}$
- (2) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$
- (3) $\text{CH}_2 = \underset{\text{CH}_3}{\text{C}} - \text{CH} = \text{CH}_2$
- (4) $\text{CH}_2 = \underset{\text{Cl}}{\text{C}} - \text{CH} = \text{CH}_2$

AIIMS 2013

40. Which of the following is non reducing sugar:-

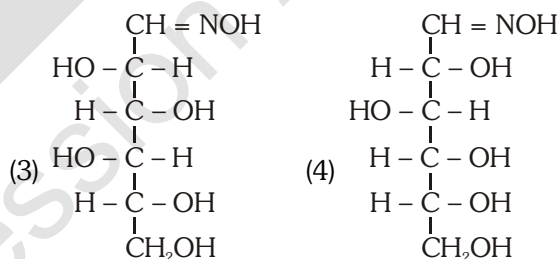
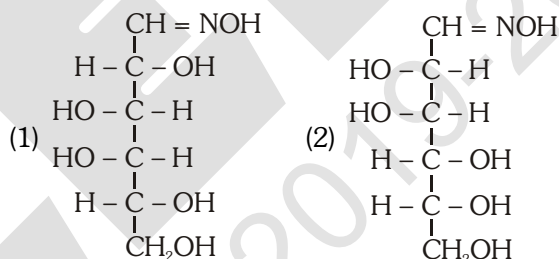
- (1) Sucrose
- (2) Maltose
- (3) Lactose
- (4) Mannose

AIPMT 2014

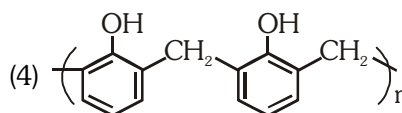
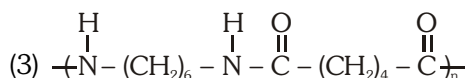
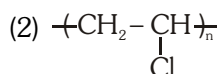
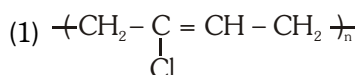
41. Artificial sweetner which is stable under cold conditions only is :-

- (1) Saccharine
- (2) Sucralose
- (3) Aspartame
- (4) Alitame

42. D (+) glucose reacts with hydroxylamine and yields an oxime. The structure of the oxime would be :



43. Which one of the following is an example of a thermosetting polymer?



44. Which of the following organic compounds polymerizes to form the polyester Dacron?
- (1) Propylene and para $\text{HO} - (\text{C}_6\text{H}_4) - \text{OH}$
 - (2) Benzoic acid and ethanol
 - (3) Terephthalic acid and ethylene glycol
 - (4) Benzoic acid and para $\text{HO} - (\text{C}_6\text{H}_4) - \text{OH}$

AIIMS 2014

45. "Maltose" is made up of ?
- (1) α -D-Glucose + α -D-Glucose
 - (2) α -D-Glucose + β -D-Glucose
 - (3) α -D-Glucose + β -D-Fructose
 - (4) α -D-Glucose + α -D-Galactose
46. Nylon 2-Nylon 6 is the copolymer of :-
- (1) Glycine + Adipic acid
 - (2) Glycine + Amino caproic acid
 - (3) Adipic acid + Hexamethylene diamine
 - (4) Amino caproic acid + Alanine
47. RNA and DNA are made up of:-
- (1) Sugar, Phosphate and vitamin A
 - (2) Sugar, Phosphate and bases
 - (3) Nucleic Acid, Vitamins and sugar
 - (4) Sugar, Amino acid and Phosphate

Re-AIPMT 2015

48. Caprolactam is used for the manufacture of :
- (1) Terylene
 - (2) Nylon - 6, 6
 - (3) Nylon - 6
 - (4) Teflon

AIPMT 2015

49. Bithional is generally added to the soaps as an additive to function as a/an :-
- (1) Dryer
 - (2) Buffering agent
 - (3) Antiseptic
 - (4) Softner
50. Biodegradable polymer which can be produced from glycine and aminocaproic acid is :-
- (1) PHBV
 - (2) Buna - N
 - (3) Nylon 6, 6
 - (4) Nylon 2- nylon 6

NEET-I 2016

51. In a protein molecule various amino acids are linked together by :
- (1) α -glycosidic bond
 - (2) β -glycosidic bond
 - (3) peptide bond
 - (4) dative bond

52. The **correct** statement regarding RNA and DNA, respectively is :

- (1) The sugar component in RNA is arabinose and the sugar component in DNA is 2'-deoxyribose.
- (2) The sugar component in RNA is ribose and the sugar component in DNA is 2'-deoxyribose.
- (3) The sugar component in RNA is arabinose
- (4) The sugar component in RNA is 2'-deoxyribose and the sugar component in DNA is arabinose.

53. Which one given below is a non-reducing sugar ?

- (1) Maltose
- (2) Lactose
- (3) Glucose
- (4) Sucrose

54. Natural rubber has

- (1) All cis-configuration
- (2) All trans-configuration
- (3) Alternate cis-and trans-configuration
- (4) Random cis-and trans-configuration

55. Which of the following is an analgesic ?

- (1) Novalgin
- (2) Penicillin
- (3) Streptomycin
- (4) Chloromycetin

NEET-II 2016

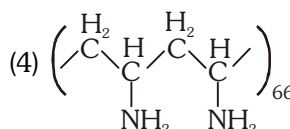
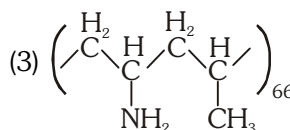
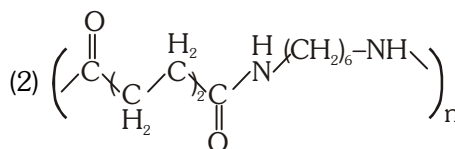
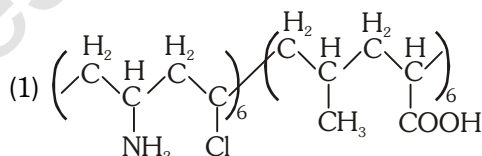
56. The central dogma of molecular genetics states that the genetic information flows from :-

- (1) $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{Proteins}$
- (2) $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{Carbohydrates}$
- (3) $\text{Amino acids} \rightarrow \text{Proteins} \rightarrow \text{DNA}$
- (4) $\text{DNA} \rightarrow \text{Carbohydrates} \rightarrow \text{Proteins}$

57. Which one of the following compounds shows the presence of intramolecular hydrogen bond ?

- (1) Cellulose
- (2) Concentrated acetic acid
- (3) H_2O_2
- (4) HCN

58. Which one of the following structures represents nylon 6,6 polymer ?



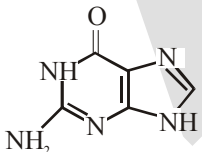
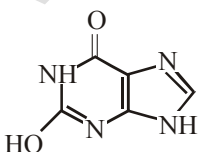
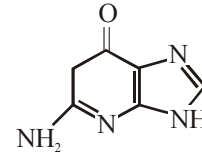
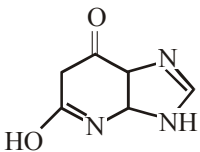
AIIMS 2016

59. Arrange the given compounds in order of their basic strength.
 (i) Histidine (ii) Lysine (iii) Arginine
 (1) $i > ii > iii$ (2) $ii > i > iii$
 (3) $iii > ii > i$ (4) $i > iii > ii$
60. Identify incorrect statement regarding glucose:-
 (1) Pentaacetate of glucose does not react with hydroxyl amine
 (2) Glucose forms cyclic hemiacetal
 (3) Glucose exist in two crystalline forms
 (4) α -form of glucose has higher melting point than its β -form.
61. Amoxycillin is :-
 (1) Broad spectrum antibiotic
 (2) Bacteriostatic
 (3) Synthetic modification of penicillin
 (4) 1 & 3 both

NEET(UG) 2017

62. Mixture of chloroxylenol and terpineol acts as :
 (1) antiseptic (2) antipyretic
 (3) antibiotic (4) analgesic
63. Which of the following statements is not correct :-
 (1) Ovalbumin is a simple food reserve in egg-white
 (2) Blood proteins thrombin and fibrinogen are involved in blood clotting
 (3) Denaturation makes the proteins more active
 (4) Insulin maintains sugar level in the blood of a human body

AIIMS 2017

64. Which of the following monomer can involved in polymerisation.
 (1) $Al(CH_3)_3$ (2) $B(CH_3)_3$
 (3) $LiCH_3$ (4) $As(CH_3)_3$
65. Correct structure of Guanine is
 (1)  (2) 
 (3)  (4) 
66. There are four nitrogen bases in DNA so how many amino acids are present in peptides of Human body?
 (1) 60 (2) 20 (3) 15 (4) 40

67. Which of the following statement is not correct w.r.t sucrose

- (1) Sucrose is made of β -D-Glucose and β -D-fructose
 (2) Hydrolysis of sucrose gives dextrorotatory glucose and laevo rotatory fructose
 (3) Whole molecule is dextrorotatory before hydrolysis
 (4) Sucrose on hydrolysis gives equimolar mixture of D(+) glucose & D(-) fructose

NEET(UG) 2018

68. The difference between amylose and amylopectin is
 (1) Amylopectin have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ α -linkage
 (2) Amylose have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ β -linkage
 (3) Amylopectin have $1 \rightarrow 4$ α -linkage and $1 \rightarrow 6$ β -linkage
 (4) Amylose is made up of glucose and galactose
69. Which of the following compounds can form a zwitterion ?
 (1) Aniline (2) Acetanilide
 (3) Benzoic acid (4) Glycine
70. Regarding cross-linked or network polymers, which of the following statements is **incorrect**?
 (1) They contain covalent bonds between various linear polymer chains.
 (2) They are formed from bi-and tri-functional monomers.
 (3) Examples are bakelite and melamine.
 (4) They contain strong covalent bonds in their polymer chains.

AIIMS 2018

71. Which given statement is incorrect
 (1) Fructose is reducing sugar
 (2) Cellulose is polymer of β -D-(+)-glucose
 (3) DNA is polymer of D-ribose
 (4) Amylopectin is insoluble in H_2O
72. Match the following

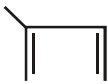
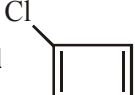
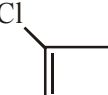
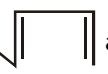
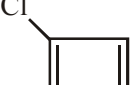
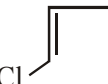
	Column-I		Column-II
a	3-Hydroxybutanoic acid	p	Biodegradable
b	Bakelite	q	Chloroprene
c	Neoprene	r	Phthalic acid
d	Glyptal	s	Phenol

- (1) p s q r
 (2) p q r s
 (3) r s q p
 (4) s q p r

73. Which of the following are essential amino acids:-

- (1) Tryptophan, Lysine, Histidine
- (2) Serine, Lysine, Tryptophan
- (3) Serine, Histidine, Tryptophan
- (4) Serine, Histidine, Lysine

74. Which of the following are monomers of natural rubber and neoprene respectively:-

- (1)  and 
- (2)  and 
- (3)  and 
- (4)  and 

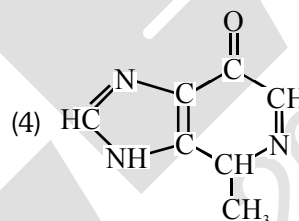
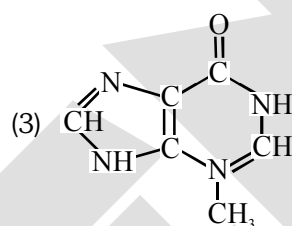
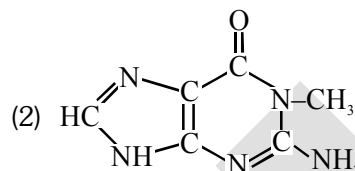
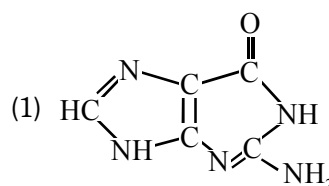
75. Among given polypeptide chain reacts with acetic anhydride and pyridine then which of the following after degradation by enzyme gives N-acetylated :-
Trp – Lys – Gly – Asn – Ala

- (1) Trp + Lys + Asn
- (2) Trp + Lys
- (3) Trp + Ala
- (4) Trp + Lys + Gly

76. Find out the incorrect statement :-

- (1) Iodine is antiseptic
- (2) Chillisalt peter is rich source of iodine
- (3) Histamine is a potent vasodilator
- (4) Chlordiazepoxide is mild tranquilizer suitable for reliving tention

77. Which of the following is the correct structure of Guanine :-



78. Which of the following statement is incorrect ?

- (1) Serotonin is a tranquilizer
- (2) Novestrol is an anti-fertility drug
- (3) Chloramphenicol is narrow spectrum antibiotic drug
- (4) Competitive inhibitor binds with enzyme

EXERCISE-II (Previous Year Questions)

ANSWER KEY

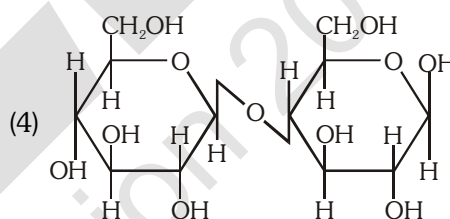
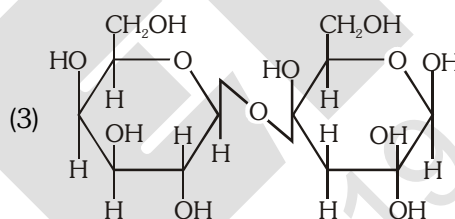
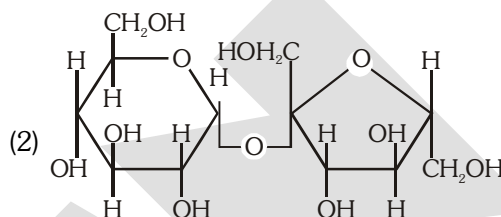
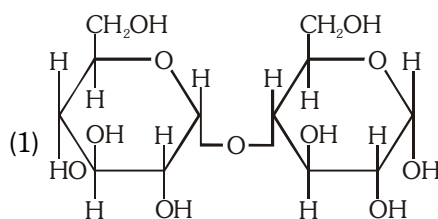
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Ans.	1	3	4	1	4	2	2	1	2	4	4	1	1	2	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	3	2	2	3	1	4	3	2	1	2	4	1	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	4	2	4	4	2	4	4	4	1	3	4	4	3	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	2	3	3	4	3	2	4	1	1	1	1	2	3	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	1	3	1	1	2	1	1	4	4	3	1	1	1	2
Que.	76	77	78												
Ans.	2	1	3												

EXERCISE-III (Analytical Questions)

Check Your Understanding

- Glucose on heating with dilute sodium hydroxide undergoes in a reversible isomerisation (known as Lobry de Bruyn-van Eikenstein rearrangement) and gives –
 - D-glucose
 - D-mannose
 - D-fructose
 - All of these
- In which of the following properties of an open chain structure of glucose could not be explained by Baeyer –
 - Glucose contain aldehyde group, but does not give schiff's test
 - Glucose does not react with sodium hydrogen sulphite and ammonia
 - The pentaacetate of glucose does not react with hydroxy amine indicates absence of $-\text{CHO}$ group
 - All of these
- When glucose was crystallized from a concentrated solution at 30°C & 98°C . It gave α & β form of glucose respectively what are the angle of rotation $[\alpha_D]$ for both forms –
 - (+) 111° & (+) 19.2°
 - (+) 19.2° & (–) 111°
 - (+) 111° & (–) 19.2°
 - (–) 111° & (–) 19.2°
- Dinucleotide is obtained by joining two nucleotides together by phosphodiester linkage. Between which carbon atoms of pentose sugar of nucleotides are these linkages presents.
 - 5' and 3'
 - 1' and 5'
 - 5' and 5'
 - 3' and 3'
- Which of the following statements is not true about glucose.
 - It is an aldohexase
 - On heating with HI it forms h-hexane
 - It is present in furanose form
 - It does not give 2,4-DNP test
- Which statement is incorrect in following –
 - Sucrose is dextrorotatory
 - After hydrolysis of sucrose, dextrorotatory glucose and laevorotatory fructose will obtained
 - Laevorotation of fructose is more than dextrorotation of glucose
 - Cane sugar gives nonequimolar mixture of D-(+)-glucose and D-(–)-fructose

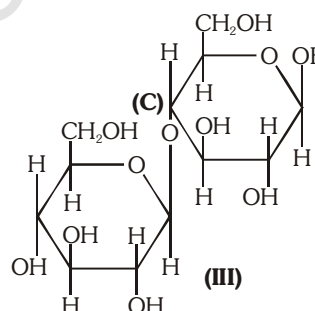
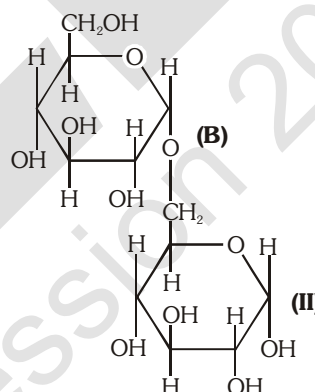
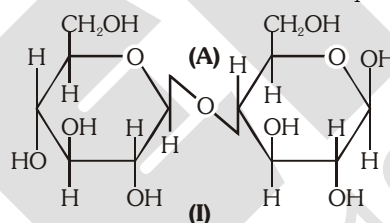
- Which of the following disaccharide is a non-reducing sugar :-



- On hydrolysis of one mole of maltose two moles of D-glucose are obtained. These two glucose units are linked together through a α -glycoside linkage between –
 - C-2 of one unit and C-4 of another unit
 - C-1 of one unit and C-2 of another unit
 - C-1 of one unit and C-4 of another unit
 - C-2 of one unit and C-3 of another unit
- Which statement is/are incorrect –
 - Natural starch has approximately 10-20% of amylose and 80-90% of amylopectin.
 - Amylose is water soluble and gives blue colour with iodine
 - Amylopectin is a branched chain polysaccharides insoluble in water and does not give blue colour with iodine
 - Starch is not hydrolysed by enzyme amylase present in saliva

10. Which of the following statement is true for proteins –
 (1) Proteins are high molecular mass complex biopolymers of amino acids present in all living cells
 (2) Protein may contains phosphorous & iodine
 (3) Proteins may contains iron, copper, zinc and manganese
 (4) All of these
11. Which of following are essential amino acids –
 (a) Aspartic acid (b) Leucine
 (c) Valine (d) Glycine
 (e) Alanine
 (1) a, b & c (2) a, b, d & e
 (3) b & c (4) a, c, d & e
12. At a certain hydrogen ion concentration (pH), the dipolar ion exists as a neutral ion and does not migrate to either electrode. What is the name of pH point for amino acids –
 (1) isoelectric point (2) isoprotonic point
 (3) isotonic point (4) None of these
13. Which of the following reactions of glucose can be explained only by its cyclic structure.
 (1) Glucose forms penta acetate
 (2) Glucose reacts with hydroxylamine to form an oxime
 (3) Penta acetate of glucose does not react with hydroxylamine
 (4) Glucose is oxidised by nitric acid to gluconic acid
14. Protein found in biological system with definite configuration and biological activity is called –
 (1) Amino acids (2) Native protein
 (3) Conformer protein (4) Inactive protein
15. If disruptive agent is removed, the proteins recovers its original physical and chemical properties and biological activity, this process is known as –
 (1) Denaturation (2) Renaturation
 (3) Transcription (4) Translation
16. Which of following statement is not true for enzyme –
 (1) Some enzymes can be non proteins also
 (2) Prosthetic groups which get attached to enzyme at the time of reaction are known as cofactor.
 (3) Enzymes provides lower activation pathways there by increasing the rate of reaction
 (4) None of these

17. Which of following statements is incorrect –
 (1) Proteins have a polyamide chain while nucleic acids contains a poly phosphate ester chain
 (2) Nucleic acids are long chain polymers of nucleotides
 (3) RNA is major source of genetic information which is copied into a DNA molecules
 (4) Proteins are synthesized in a process involving translation of RNA.
18. Complete hydrolysis of DNA or RNA yields following –
 (1) Ribose in RNA & deoxyribose in DNA
 (2) Hetrocyclic nitrogenous purines base
 (3) Hetrocyclic nitrogenous pyrimidines
 (4) All of these
19. Three structures are given below in which two glucose units are linked. Which of these linkages between glucose units are between C_1 and C_4 and which linkages are between C_1 and C_6 .



- (1) (A) is between C_1 and C_4 ; (B) and (C) are between C_1 and C_6
 (2) (A) and (B) are between C_1 and C_4 ; (C) is between C_1 and C_6
 (3) (A) and (C) are between C_1 and C_4 ; (B) is between C_1 and C_6
 (4) (A) and (C) are between C_1 and C_6 ; (B) is between C_1 and C_4

- 20.** Nucleosides are –
 (1) Base + sugar = Nucleoside
 (2) N-glycosides of purine or pyrimidine bases with pentose sugar
 (3) Both of these
 (4) None of these
- 21.** RNA molecules in turn directs the synthesis of specific proteins which are characteristic of each kind of organism. This process is known as –
 (1) Translation (2) Transcription
 (3) Replication (4) None of these
- 22.** A single stand of DNA can act as a template on which a molecule of RNA is synthesized in a specific manner. This process is known as –
 (1) Replication (2) Transcription
 (3) Translation (4) None of these
- 23.** Which statements is/are correct –
 (1) Transcription is used as a synonym for RNA synthesis
 (2) Translation is used as a synonym for protein synthesis
 (3) Translation is unidirectional but transcription can sometimes be reversed
 (4) All of these
- 24.** Insulin is known as hypoglycemic factor because
 (1) It inhibits anabolic processes and promotes catabolic processes
 (2) It decreases glucose concentration in blood
 (3) It does not facilitate entry of glucose into the cells
 (4) None of these
- 25.** Which is correct statement –
 (1) Starch is polymer of α -glucose
 (2) Amylose is a component of cellulose
 (3) Proteins are composed of only one type of amino acid
 (4) In cyclic structure of fructose, there are four carbons and one oxygen atom
- 26.** $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\ddot{\text{N}}\text{H}- \end{array}$ (peptide bond) which statement is incorrect about peptide bond –
 (1) C–N bond length in proteins is longer than usual bond length of C–N bond
 (2) Spectroscopic analysis show planar structure of $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}- \end{array}$ group
 (3) C–N bond length in proteins is smaller than usual bond length of C–N bond
 (4) None of above

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	1	1	3	4	2	3	4	4	3	1	3	2	2
Que.	16	17	18	19	20	21	22	23	24	25	26				
Ans.	2	3	4	3	3	1	2	4	2	1	1				

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** D-(+) Glucose and L-(-) fructose are enantiomers.
Reason :- Glucose molecule is mirror image of fructose molecule.
 (1) A (2) B (3) C (4) D
- Assertion :-** Teflon has high thermal stability and chemical inertness.
Reason :- Teflon is highly cross-linked polymer.
 (1) A (2) B (3) C (4) D
- Assertion :-** Polypropylene is an addition polymer.
Reason :- The monomer unit of polypropylene is $\text{CH}_3\text{-CH=CH}_2$
 (1) A (2) B (3) C (4) D
- Assertion :-** Carboxypeptidase is an exopeptidase.
Reason :- It cleaves the N-terminal bond.
 (1) A (2) B (3) C (4) D
- Assertion :-** Natural rubber is cis-1,4-polyisoprene.
Reason :- Isoprene is 2-methyl-1,3-butadiene.
 (1) A (2) B (3) C (4) D
- Assertion :-** Maltose is composed of galactose.
Reason :- Glucose and galactose differ from C_3 position.
 (1) A (2) B (3) C (4) D
- Assertion :-** β -pleated sheet structure of protein shows maximum extension.
Reason :- Intermolecular hydrogen bonding is present in them
 (1) A (2) B (3) C (4) D
- Assertion :-** All enzymes are made up of proteins which have three dimensional structure.
Reason :- Secondary structure of protein are sequence of aminoacids.
 (1) A (2) B (3) C (4) D
- Assertion :-** Melamine resin has secondary amine functionality.
Reason :- Melamine resin is condensation polymer of melamine and acetaldehyde.
 (1) A (2) B (3) C (4) D
- Assertion :-** Sucrose and saccharin have nearly same sweetness index.
Reason :- Both compounds are the component of carbohydrate.
 (1) A (2) B (3) C (4) D
- Assertion :-** Glucose does not give 2,4-DNP test.
Reason :- Glucose does not have free carbonyl group.
 (1) A (2) B (3) C (4) D
- Assertion :-** Tryptophan is less basic than Histidine
Reason :- Tryptophan has sp^3 hybridised nitrogen but Histidine has sp^2 hybridisation.
 (1) A (2) B (3) C (4) D
- Assertion :-** Bakelite is a thermosetting polymer.
Reason :- It has highly branched structure.
 (1) A (2) B (3) C (4) D
- Assertion :-** Calcium salt of fatty acid can be used as soap.
Reason :- They are water soluble.
 (1) A (2) B (3) C (4) D

- 15. Assertion :-** Aspirin is used to reduce the chance of heart attack.
Reason :- Aspirin increase the rate of prostaglandin synthesis.
 (1) A (2) B (3) C (4) D
- 16. Assertion :-** At neutral pH, Tyrosine act as an acid
Reason :- pK_a of phenol is more than 7.
 (1) A (2) B (3) C (4) D
- 17. Assertion :-** Antihistamine does not effect acid secretion of stomach.
Reason :- Antacid and antihistamine acts on different receptors.
 (1) A (2) B (3) C (4) D
- 18. Assertion :-** Glucose reacts with Br_2 to form saccharic acid.
Reason :- Acid will be formed by oxidation of alcohol with Br_2 .
 (1) A (2) B (3) C (4) D
- 19. Assertion :-** Non competitive inhibitors alters the active site of enzymes.
Reason :- Non competitive inhibitors always bind with active site.
 (1) A (2) B (3) C (4) D
- 20. Assertion :-** Generally RNA are single stranded.
Reason :- They interact with amino acids.
 (1) A (2) B (3) C (4) D
- 21. Assertion :-** Natural Rubber has elastic property.
Reason :- It has linear structure.
 (1) A (2) B (3) C (4) D
- 22. Assertion :-** Cimitidine is an antacid
Reason :- It increases the interaction of histamine with receptors present in stomach wall.
 (1) A (2) B (3) C (4) D
- 23. Assertion :-** α - glucose has lesser melting point than β -glucose.
Reason :- β - glucose is more thermodynamic stable.
 (1) A (2) B (3) C (4) D
- 24. Assertion :-** Sucrose is dextrorotatory but its aqueous solution is laeverotatory.
Reason :- Dextro rotation of glucose is less than laevo rotation of fructose.
 (1) A (2) B (3) C (4) D
- 25. Assertion :-** Novolac on heating with formaldehyde gives bakelite.
Reason :- Bakelite is thermosetting polymer and an infusible solid.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	3	2	1	2	4	2	3	3	4	1	3	2	4	3
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	2	4	4	3	2	2	3	2	1	2					

IMPORTANT NOTES

BIOMOLECULES, POLYMER AND CHEMISTRY IN EVERYDAY LIFE

S. No.	CONTENTS	Page
	Biomolecules	
1.	Carbohydrates	117
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NEET SYLLABUS

Biomolecules : Carbohydrates : Classification (aldoses and ketoses), monosaccharide (glucose and fructose), D.L., configuration oligosaccharides (sucrose, lactose, maltose) polysaccharides (starch, cellulose, glycogen) importance. **Proteins :** Elementary idea of – amino acids, peptide bond, polypeptides, proteins, primary structure, secondary structure, tertiary structure and quaternary structure (qualitative idea only), denaturation of proteins, enzymes. **Hormones :** Elementary idea (excluding structure), **Vitamins :** Classification and function. **Nucleic acid :** DNA and RNA

Polymer : Classification :- natural and synthetic, methods of polymerization (addition and condensation), copolymerization. Some important polymers, natural and synthetic like polyesters, bakelite, rubber, biodegradable and non-biodegradable polymers.

Chemistry in Everyday life : Chemical in medicines - analgesics, tranquilizers, antiseptics, disinfectants, antimicrobials, antifertility drugs, antibiotics, antacids, antihistamines. **Chemical in food :** Preservatives, artificial sweetening agents, elementary idea of antioxidants. **Cleansing agents :** Soaps and detergents, cleansing action.

OBJECTIVES

After studying this unit, we will be able to :

- Define the biomolecules like carbohydrates, proteins and nucleic acids;
- Classify carbohydrates, proteins, nucleic acids and vitamins on the basis of their structures;
- Explain the difference between DNA and RNA;
- Appreciate the role of biomolecules in biosystem.
- Explain the terms - monomer, polymer and polymerisation and appreciate their importance;
- Distinguish between various classes of polymers and different types of polymerisation processes;
- Appreciate the formation of polymers from mono- and bifunctional monomer molecules;
- Describe the preparation of some important synthetic polymers and their properties;
- Appreciate the importance of polymers in daily life.
- Visualise the importance of Chemistry in daily life;
- Explain the term 'chemotherapy';
- Describe the basis of classification of drugs;
- Explain drug-target interaction of enzymes and receptors;
- Explain how various types of drugs function in the body;
- Know about artificial sweetening agents and food preservatives;
- Discuss the chemistry of cleansing agents.

*"To succeed in life and achieve results, you must
understand and master three mighty forces – Desire, Belief and Expectation"*

A.P.J. Abdul Kalam

(iii) Polysaccharides :

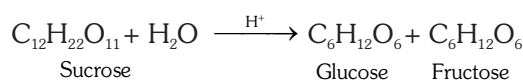
Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharides. Some common examples are starch, cellulose, glycogen, etc. Polysaccharides are not sweet in taste, hence they are also called **non-sugars**.

The carbohydrates may also be classified as either reducing or non-reducing sugars. All those carbohydrates which reduce Fehling's solution and Tollens' reagent are referred to as reducing sugars. All monosaccharides whether aldose or **ketose are reducing sugars**.

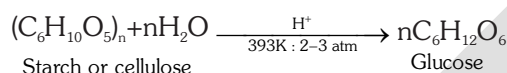
In disaccharides, if the reducing groups of monosaccharides i.e., aldehydic or ketonic groups are bonded, these are **non-reducing sugars e.g. sucrose**. On the other hand, sugars in which these functional groups are free, are called reducing sugars, for example, maltose and lactose.

4.1.2 GLUCOSE (ALDOHEXOSE)**Preparation of Glucose**

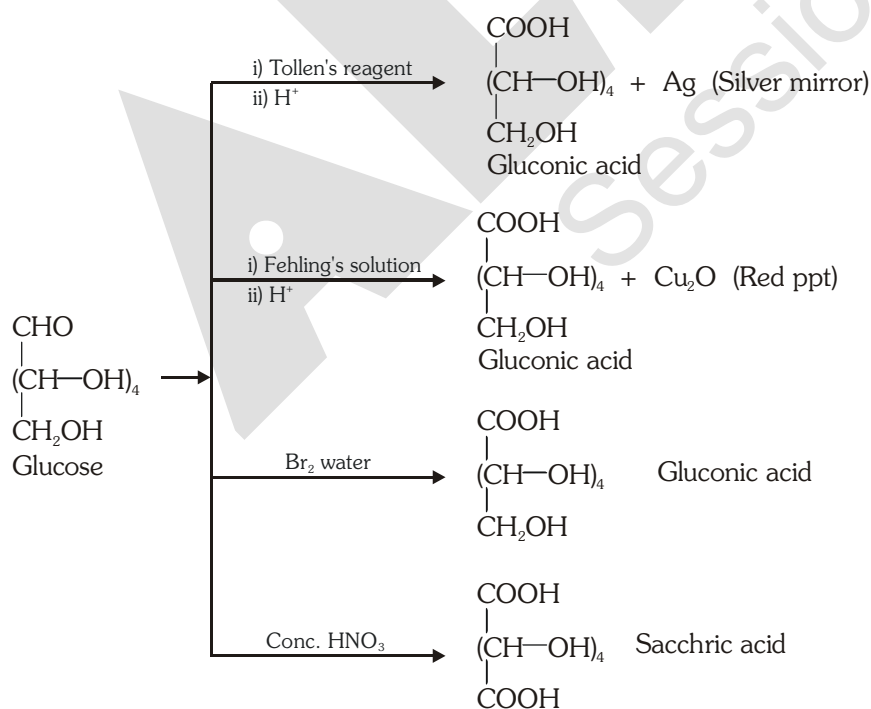
- From sucrose (Cane sugar):** If sucrose is boiled with dilute HCl or H_2SO_4 in alcoholic solution, glucose and fructose are obtained in equal amounts.



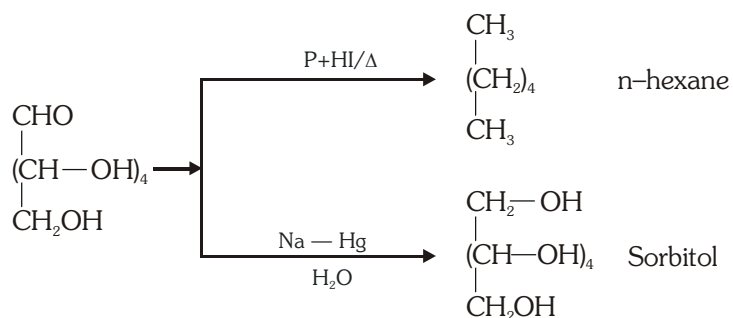
- From starch :** Commercially glucose is obtained by hydrolysis of starch by boiling it with dilute H_2SO_4 at 393 K under pressure.

**Chemical reactions of Glucose :**

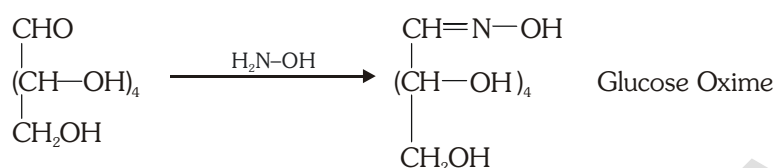
- Oxidation :**



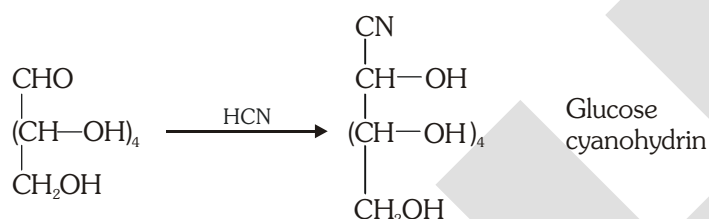
2. Reduction :



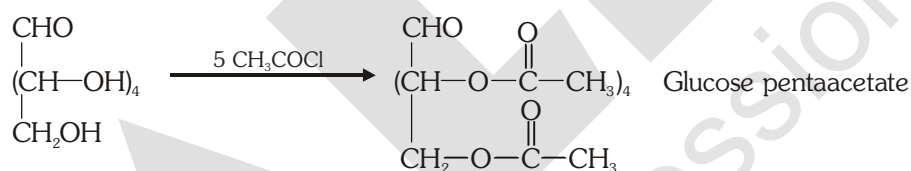
3. Reaction with hydroxyl amine :



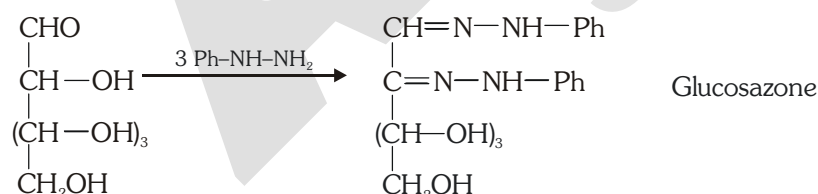
4. Reaction with hydrogen cyanide :



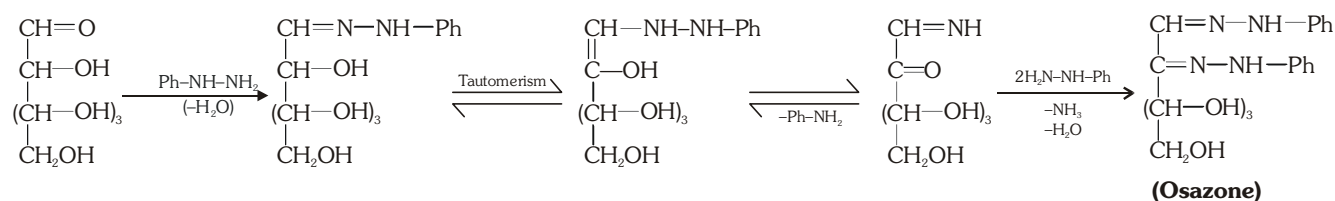
5. Acetylation :



6. Reaction with phenyl hydrazine :

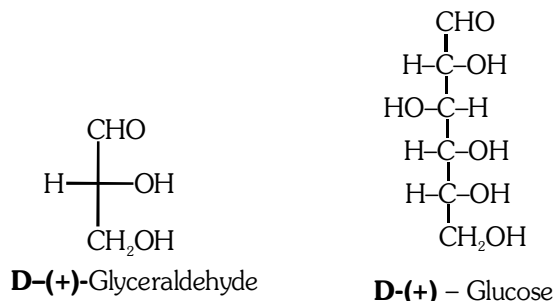


Mechanism :

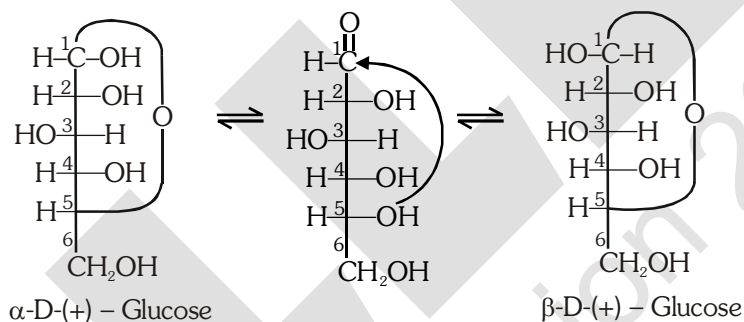


Configuration in monosaccharides :

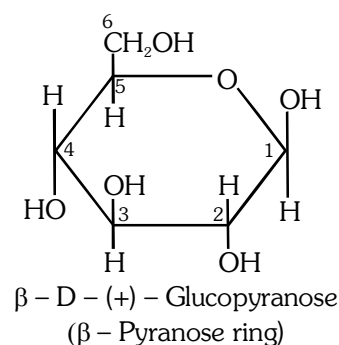
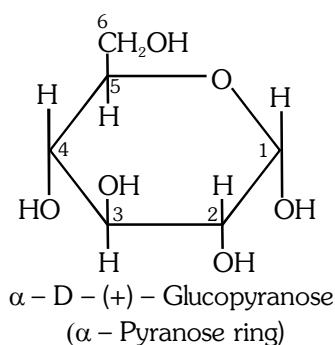
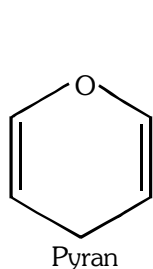
For assigning the configuration of monosaccharides, it is the lowest asymmetric carbon atom (as shown below) which is compared. As in (+) glucose, —OH on the lowest asymmetric carbon is on the right side which is comparable to (+) glyceraldehyde, so it is assigned D-configuration.

**Cyclic structure of Glucose :-**

It was found that glucose forms a six-membered ring in which —OH at C-5 is involved in ring formation. This explains the absence of —CHO group and also existence of glucose in two forms as shown below. These two cyclic forms exist in equilibrium with open chain structure.



The two cyclic hemiacetal forms of glucose differ only in the configuration of the hydroxyl group at C₁, called anomeric carbon (the aldehyde carbon before cyclisation). Such isomers, i.e., α -form and β -form, are called **anomers**. The six membered cyclic structure of glucose is called **pyranose structure** (α - or β -), in analogy with pyran. Pyran is a cyclic organic compound with one oxygen atom and five carbon atoms in the ring. The cyclic structure of glucose is more correctly represented by **Haworth structure** as given below.

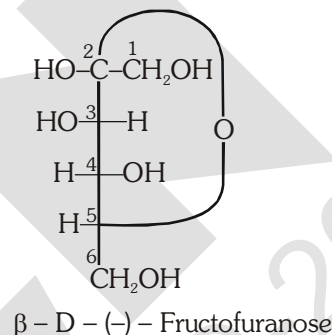
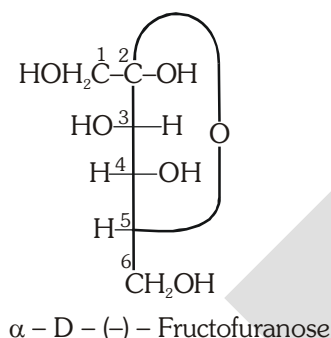
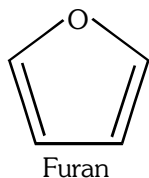
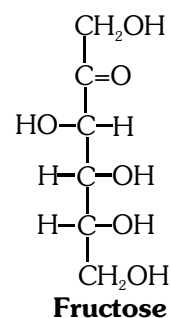


4.1.3 FRUCTOSE (KETOHEXOSE)

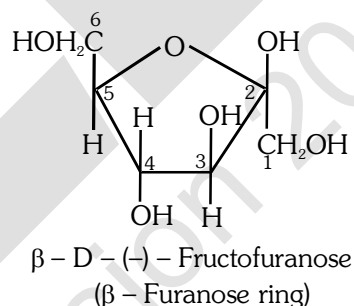
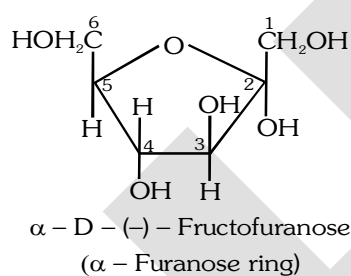
Structure of Fructose :

Fructose also has the **molecular formula $C_6H_{12}O_6$** and on the basis of its reactions it was found to contain **a ketonic functional group at carbon number 2** and six carbons in straight chain as in the case of glucose. It belongs to **D-series and is a laevorotatory compound**. It is appropriately written as **D-(-)-fructose**. Its open chain structure is as shown.

It also exists in two cyclic forms which are obtained by the addition of $-OH$ at C5 to the $(>C=O)$ group. The ring, thus formed is a five membered ring and is named as **furanose** with analogy to the compound furan. Furan is a five membered cyclic compound with one oxygen and four carbon atoms.



The cyclic structures of two anomers of fructose are represented by Haworth structures as given.

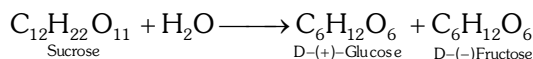


COMPARISON OF GLUCOSE AND FRUCTOSE			
S. No.	Property	Glucose	Fructose
1.	Molecular formula	$C_6H_{12}O_6$	$C_6H_{12}O_6$
2.	Nature	Polyhydroxy aldehyde	Polyhydroxy ketone
3.	Melting point	146°C	102°C
4.	Optical nature	Dextro rotatory	Leavo rotatory
5.	Tollen's reagent	Silver mirror	Silve mirror
6.	Fehling's solution	Red ppt	Red ppt
7.	Molisch test	Violet colour	Violet colour
8.	Phenyl hydrazine	Forms osazone	Forms osazone
9.	Oxidation by conc. HNO_3	Saccharic acid	Mixture of glycollic acid, Tartaric acid and Trihydroxy Gluteric acid

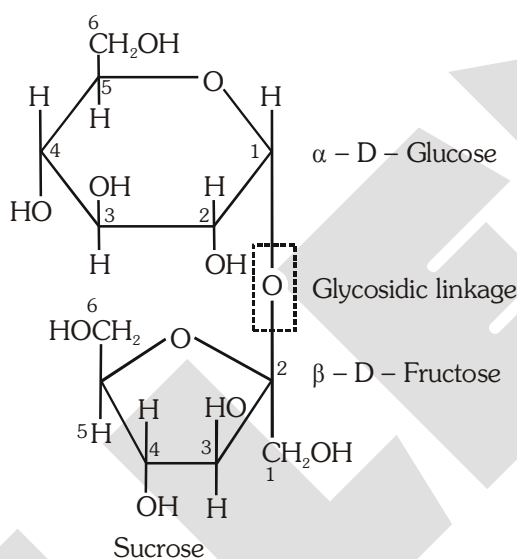
4.1.4 DISACCHARIDES

The two monosaccharides are joined together by an oxide linkage formed by the loss of a water molecule. Such a linkage between two monosaccharide units through oxygen atom is called **glycosidic linkage**.

- (i) **Sucrose** : One of the common disaccharides is **sucrose** which on hydrolysis gives equimolar mixture of D-(+)-glucose and D-(-) fructose.

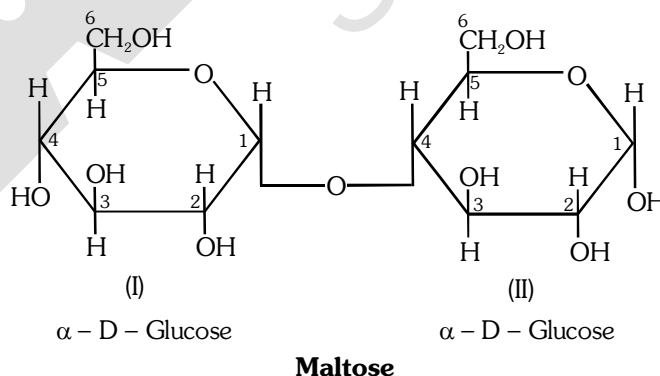


These two monosaccharides are held together by a glycosidic linkage between C₁ of α-glucose and C₂ of β-fructose. Since the reducing groups of glucose and fructose are involved in glycosidic bond formation, sucrose is a **non reducing sugar**.

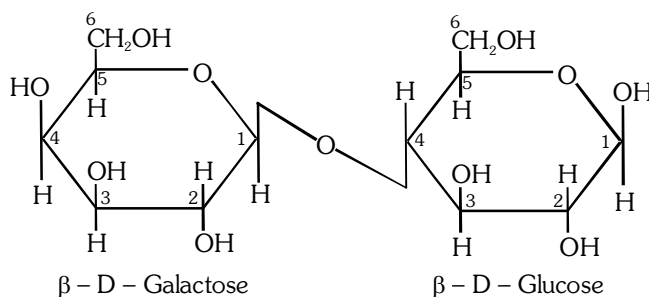


Hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to laevo (-) and the product is named as invert sugar.

- (ii) **Maltose**: Another disaccharide, maltose is composed of two α-D-glucose units in which C₁ of one glucose (I) is linked to C₄ of another glucose unit (II). The free aldehyde group can be produced at C₁ of second glucose in solution and it shows reducing properties so it is a **reducing sugar**.



- (iii) **Lactose** : It is more commonly known as milk sugar since this disaccharide is found in milk. It is composed of β -D-galactose and β -D-glucose. The linkage is between C_1 of galactose and C_4 of glucose. Hence it is also a **reducing sugar**.



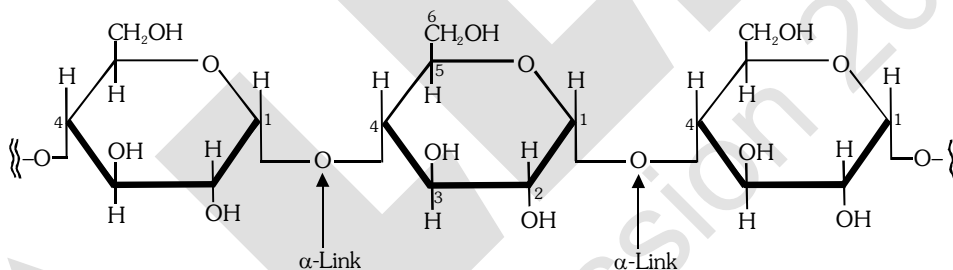
Lactose

4.1.5 POLYSACCHARIDES

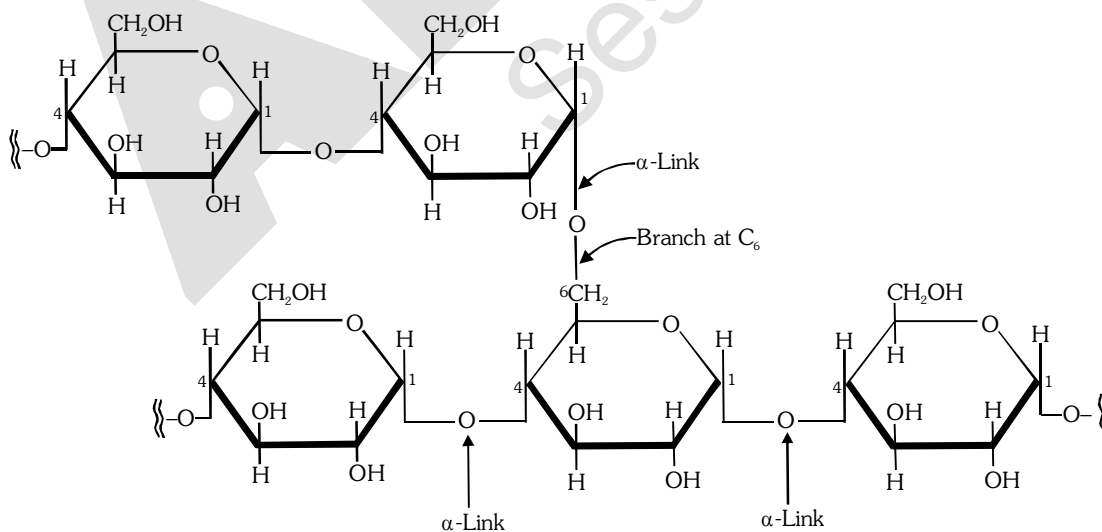
Polysaccharides contain a large number of monosaccharide units joined together by glycosidic linkages. They mainly act as the food storage or structural materials.

- (i) **Starch**: Starch is the main storage polysaccharide of plants. It is the most important dietary source for human beings. High content of starch is found in cereals, roots, tubers and some vegetables. It is a **polymer of α -glucose** and consists of two components—

Amylose and **Amylopectin**. Amylose is hot water soluble component which constitutes about 15-20% of starch. Chemically amylose is a long unbranched chain with 200-1000 α -D-(+)-glucose units held by C_1 - C_4 glycosidic linkage. Amylopectin is insoluble in hot water and constitutes about 80-85% of starch. It is a branched chain polymer of α -D-glucose units in which chain is formed by C_1 - C_4 glycosidic linkage whereas branching occurs by C_1 - C_6 glycosidic linkage.

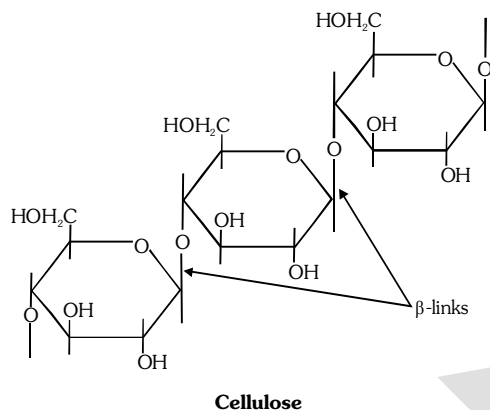


Amylose



Amylopectin

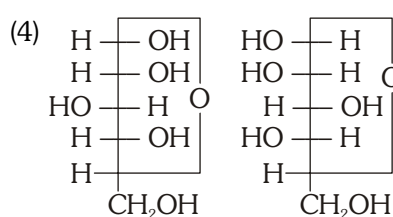
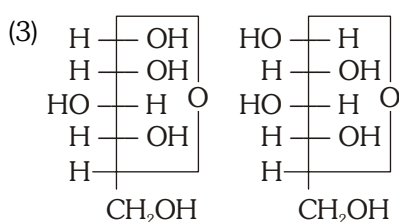
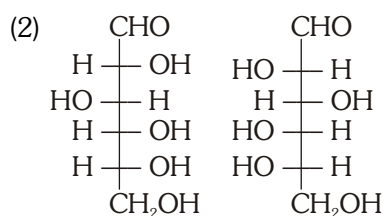
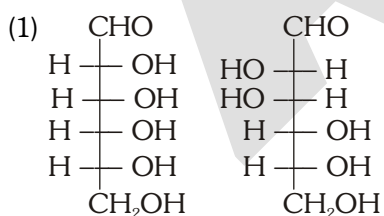
- (ii) **Cellulose** : Cellulose occurs exclusively in plants and it is the most abundant organic substance in plant kingdom. It is a predominant constituent of cell wall of plant cells. Cellulose is a straight chain polysaccharide composed only of **β -D-glucose** units which are joined by glycosidic linkage between C_1 of one glucose unit and C_4 of the next glucose unit.



- (iii) **Glycogen** : The carbohydrates are stored in animal body as glycogen. It is also known as **animal starch** because its structure is similar to amylopectin and is rather more highly branched. It is present in liver, muscles and brain. When the body needs glucose, enzymes break the glycogen down to glucose. Glycogen is also found in yeast and fungi.

BEGINNER'S BOX-1

- Glucose can be tested by following –
 - Tollen's reagent
 - Fehling's solution
 - Both of these
 - None of these
- Glucose on oxidation with nitric acid as well as gluconic acid both gives –
 - Saccharic acid
 - n-Hexane
 - Fructose
 - Glucosazone
- The spontaneous change in specific rotation of an optically active compound is called –
 - Mutarotation
 - Rearrangements
 - Inversion
 - Retention
- Hydrolysis of sucrose brings about a change in sign of rotation from dextro(+) to Laevo(–) and such a sign change is known as –
 - Racemization
 - Inversion
 - Mutarotation
 - None of these
- Which of the following pairs represents anomer:-



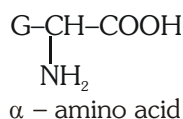
4.2 PROTEINS

The word protein is derived from Greek word, “**proteios**” which means primary or of prime importance.

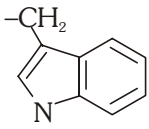
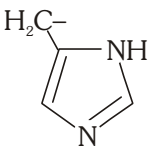
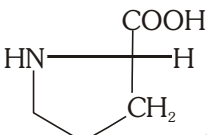
All proteins are polymers of **α -L-amino acids**.

Amino acids : Amino acids contain amino ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) functional groups.

Depending upon the relative position of amino group with respect to carboxyl group, the amino acids can be classified as α , β , γ , δ and so on. Only α -amino acids are obtained on hydrolysis of proteins. They may contain other functional groups also.



Name of the amino acids	Characteristic feature of side chain, G	Three letter symbol	One letter code
1. Glycine	H	Gly	G
2. Alanine	$-\text{CH}_3$	Ala	A
3. Valine*	$(\text{H}_3\text{C})_2\text{CH}-$	Val	V
4. Leucine*	$(\text{H}_3\text{C})_2\text{CH}-\text{CH}_2-$	Leu	L
5. Isoleucine*	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{CH}- \\ \\ \text{CH}_3 \end{array}$	Ile	I
6. Arginine*	$\begin{array}{c} \text{HN}=\text{C}-\text{NH}-(\text{CH}_2)_3- \\ \\ \text{NH}_2 \end{array}$	Arg	R
7. Lysine*	$\text{H}_2\text{N}-(\text{CH}_2)_4-$	Lys	K
8. Glutamic acid	$\text{HOOC}-\text{CH}_2-\text{CH}_2-$	Glu	E
9. Aspartic acid	$\text{HOOC}-\text{CH}_2-$	Asp	D
10. Glutamine	$\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{N}-\text{C}-\text{CH}_2-\text{CH}_2- \end{array}$	Gln	Q
11. Asparagine	$\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{N}-\text{C}-\text{CH}_2- \end{array}$	Asn	N
12. Threonine*	$\text{H}_3\text{C}-\text{CHOH}-$	Thr	T
13. Serine	$\text{HO}-\text{CH}_2-$	Ser	S
14. Cysteine	$\text{HS}-\text{CH}_2-$	Cys	C

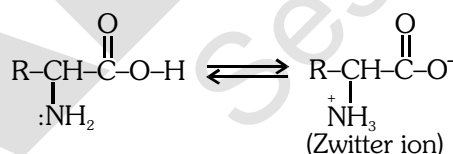
15.	Methionine*	$\text{H}_3\text{C}-\text{S}-\text{CH}_2-\text{CH}_2-$	Met	M
16.	Phenylalanine*	$\text{C}_6\text{H}_5-\text{CH}_2-$	Phe	F
17.	Tyrosine	(p) $\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-$	Tyr	Y
18.	Tryptophan*		Trp	W
19.	Histidine*		His	H
20.	Proline ^(a)		Pro	P

* essential amino acid, a = entire structure

4.2.1 Classification of Amino Acids : Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule.

- The amino acids, which can be synthesised in the body, are known as **non-essential amino acids**. On the other hand, those which cannot be synthesised in the body and must be obtained through diet, are known as **essential amino acids**.

Amino acids are usually colourless, crystalline solids. These are water-soluble, high melting solids and behave like salts rather than simple amines or carboxylic acids. This behaviour is due to the presence of both acidic (carboxyl group) and basic (amino group) groups in the same molecule.



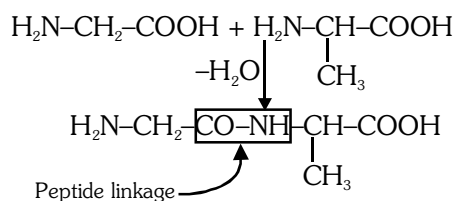
In aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton, giving rise to a dipolar ion known as **zwitter ion**. This is neutral but contains both positive and negative charges.

In zwitter ionic form, amino acids show **amphoteric** behaviour as they react both with acids and bases.

Except glycine, all other naturally occurring α-amino acids are optically active, since the α-carbon atom is asymmetric. These exist both in 'D' and 'L' forms. Most naturally occurring amino acids have L-configuration. **L-Aminoacids** are represented by writing the $-\text{NH}_2$ group on left hand side.

4.2.2 Structure of Proteins :

Proteins are the polymers of α-amino acids and they are connected to each other by **peptide bond** or **peptide linkage**. Chemically, peptide linkage is an amide formed between $-\text{COOH}$ group and $-\text{NH}_2$ group.



Glycylalanine [Gly-Ala]

The reaction between two molecules of similar or different amino acids, proceeds through the combination of the amino group of one molecule with the carboxyl group of the other. This results in the elimination of a water molecule and formation of a **peptide bond** $-\text{CO}-\text{NH}-$.

Proteins can be classified into two types on the basis of their molecular shape.

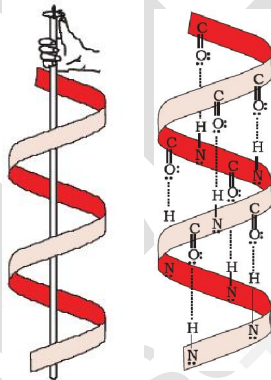
(a) Fibrous proteins

When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed. Such proteins are generally insoluble in water. Some common examples are keratin (present in hair, wool, silk) and myosin (present in muscles), etc.

(b) Globular proteins

This structure results when the chains of polypeptides coil around to give a spherical shape. These are usually soluble in water. Insulin and albumins are the common examples of globular proteins.

Structure and shape of proteins can be studied at four different levels, i.e., primary, secondary, tertiary and quaternary, each level being more complex than the previous one.



- (i) **Primary structure of proteins:** Proteins may have one or more polypeptide chains. Each polypeptide in a protein has amino acids linked with each other in a specific sequence and it is this sequence of amino acids that is said to be the primary structure of that protein. Any change in this primary structure (i.e., the sequence of amino acids) creates a different protein.
- (ii) **Secondary structure of proteins:** The secondary structure of protein refers to the shape in which a long polypeptide chain can exist. They are found to exist in two different types of structures viz. α -helix and β -pleated sheet structure. These structures arise due to the regular folding of the backbone

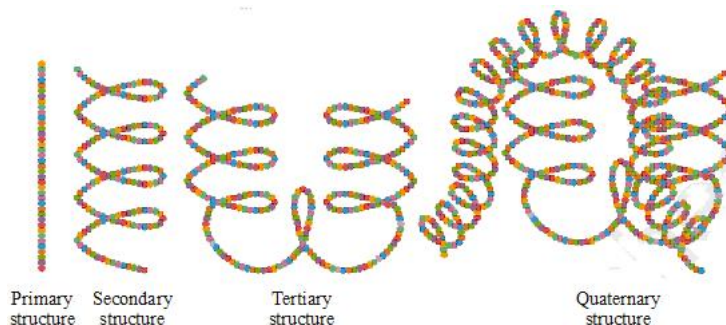
of the polypeptide chain due to hydrogen bonding between $-\text{C}(=\text{O})-$ and $-\text{NH}-$ groups of the peptide bond.

α -Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a right handed screw (helix) with the $-\text{NH}$ group of each amino acid residue hydrogen bonded to the $>\text{C}=\text{O}$ of an adjacent turn of the helix.

In β -structure all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together by intermolecular hydrogen bonds. The structure resembles the pleated folds of drapery and therefore is known as β -pleated sheet.

- (iii) **Tertiary structure of proteins:** The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, van der Waals and electrostatic forces of attraction.
- (iv) **Quaternary structure of proteins :** Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. The spatial arrangement of these subunits with respect to each other is known as quaternary structure.

A diagrammatic representation of all these four structures is given in figure.



4.2.3 Denaturation of Proteins :

Physical change like change in temperature or chemical change like change in pH, the hydrogen bonds are disturbed. Due to this, globules unfold and helix get uncoiled and protein loses its biological activity. This is called **denaturation** of protein. The coagulation of egg white on boiling is a common example of denaturation. Another example is curdling of milk which is caused due to the formation of lactic acid by the bacteria present in milk.

During denaturation secondary and tertiary structures are destroyed but primary structure remains unchanged.

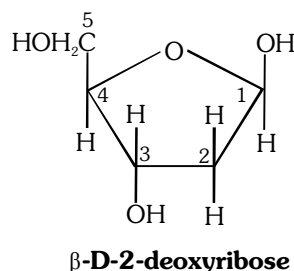
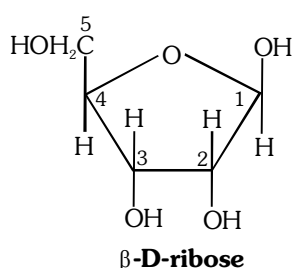
4.3 NUCLEIC ACIDS

Introduction :

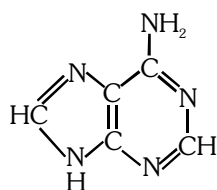
It has been observed that nucleus of a living cell is responsible for the transmission of inherent characters, also called **heredity**. The particles in nucleus of the cell, responsible for heredity, are called chromosomes which are made up of proteins and another type of biomolecules called **nucleic acids**. These are mainly of two types, the **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**. Since nucleic acids are long chain polymers of **nucleotides**, so they are also called polynucleotides.

4.3.1 Chemical composition of Nucleic Acids :

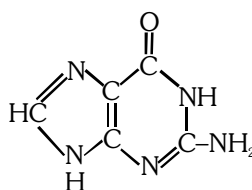
Complete hydrolysis of DNA (or RNA) yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds (called bases). In DNA molecules, the sugar moiety is **β -D-2-deoxyribose** whereas in RNA molecule, **it is β -D-ribose**.



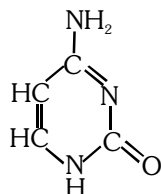
DNA contains four bases viz. **adenine** (A), **guanine** (G), **cytosine** (C) and **thymine** (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is **uracil** (U).



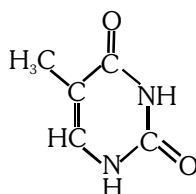
Adenine (A)



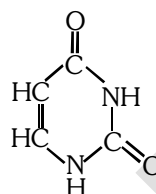
Guanine (G)



Cytosine (C)



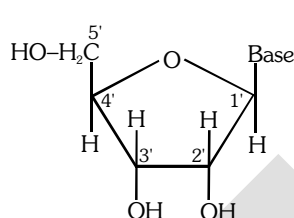
Thymine (T)



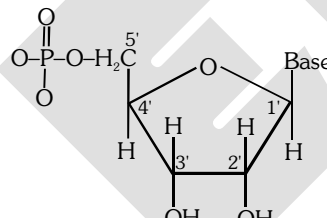
Uracil (U)

4.3.2 Structure of Nucleic Acids :-

A unit formed by the attachment of a base to 1' position of sugar is known as **nucleoside**. In nucleosides, the sugar carbons are numbered as 1', 2', 3', etc. in order to distinguish these from the bases. When nucleoside is linked to phosphoric acid at 5'-position of sugar moiety, we get a nucleotide.



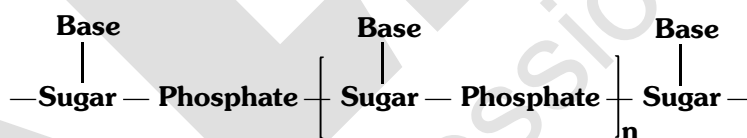
(a)



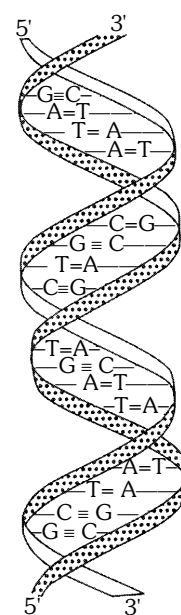
(b)

Structure of (a) a nucleoside and (b) a nucleotide

Nucleotides are joined together by **phosphodiester linkage** between 5' and 3' carbon atoms of the pentose sugar. A simplified version of nucleic acid chain is as shown below. Fig.



Information regarding the sequence of nucleotides in the chain of a nucleic acid is called its primary structure. Nucleic acids have a secondary structure also. James Watson and Francis Crick gave a double strand helix structure for DNA (Fig.). Two nucleic acid chains are wound about each other and held together by hydrogen bonds between pairs of bases. The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine. In secondary structure of RNA, helices are present which are only single stranded. Sometimes they fold back on themselves to form a double helix structure. RNA molecules are of three types and they perform different functions. They are named as **messenger RNA (m-RNA)**, **ribosomal RNA (r-RNA)** and **transfer RNA (t-RNA)**.



Double strand helix structure for DNA

BEGINNER'S BOX-2

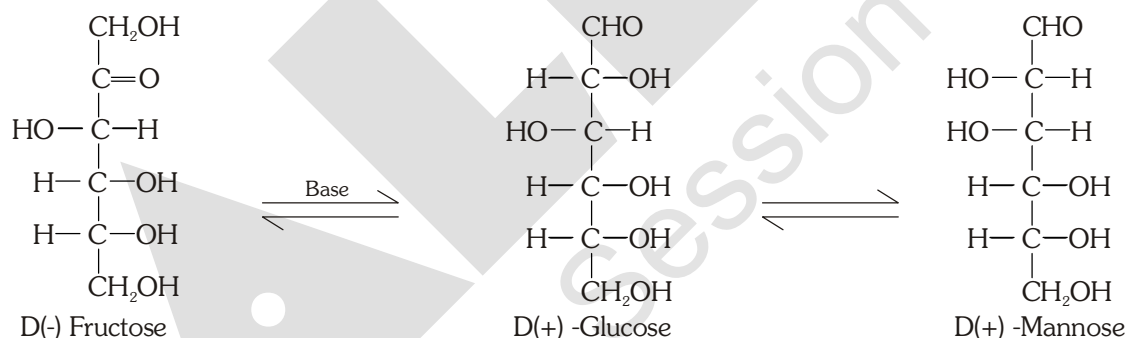
- Zwitter ion is –
 (1) neutral ion (2) positive charge ion (3) negative ion (4) None of these
- In β -sheet of proteins, conformers are held together by (Secondary structure of protein) –
 (1) Intermolecular H-bond (2) Intramolecular H-bond
 (3) Conformers not held together (4) None of these
- DNA contains following purines bases –
 (a) Adenine (b) Guanine (c) Thymine (d) Cytosine
 (1) a, c (2) a, b (3) a, c, d (4) a, b, c, d
- Which of the following statement is/are true –
 (1) DNA has deoxyribose while RNA has ribose sugar (2) DNA contains thymine while RNA has uracil
 (3) Both of these (4) None of these
- RNA on translation gives –
 (1) Protein (2) DNA (3) Both (4) None of these

GOLDEN KEY POINTS

TEST FOR BIOMOLECULES

(I) Lobry de Bruyn van Eikenstein rearrangement :

Glucose contains —CHO group and hence reduces fehling's solution. Fructose, on the other hand, contains >C=O group but still reduces Fehling's solution. This is due to the reason that under basic conditions of the reagent, fructose undergoes a reversible isomerization and is converted into a mixture of glucose and mannose - both of which contain —CHO group and hence reduce fehling's solution.



This conversion of fructose into a mixture of glucose and mannose under basic conditions is called Lobry de Bruyn van Eikenstein rearrangement.

(II) **Test of carbohydrates : Molish test** is used for detection of all types of carbohydrates, i.e., monosaccharides, disaccharides and polysaccharides.

Molisch reagent (1% alcoholic solution of α - naphthol) is added to the aqueous solution of a carbohydrate followed by conc. H_2SO_4 along the sides of the test tube. A violet ring is formed at the junction of the two layers.

(III) **Tests of Proteins:**

Biuret test: An alkaline solution of a protein when treated with a few drops of 1% CuSO_4 solution, produces a violet colouration. The colour is due to the formation of a coordination complex of Cu^{+2} with >C=O and —NH— groups of the peptide linkages.

4.4 VITAMINS

	Vitamin	Sources	Deficiency diseases
(i)	Vitamin-A (Retinol or eye vitamin)	Milk, cod liver oil, butter, carrots, green leaves, tomatoes, eggs, etc.	Night blindness, xerophthalmia (i.e. hardening of cornea of eye) and xerosis.
(ii)	Vitamin-B ₁ Thiamine or Aneurin	Pulses, nuts, green vegetables, Polished rice. yeast and egg yolk.	Beriberi (a disease of nervous system) and loss appetite.
(iii)	Vitamin - B ₂ or Riboflavin or Lactoflavin	Milk, meat, green vegetables and yeast.	Inflammation of tongue or dark red tongue (glossitis), and cheilosis (cracking or fissuring the lips and corners of the mouth)
(iv)	Vitamin-B ₆ or Pyridoxine or Adermine	Rice, bran, yeast, meat, fish, egg, yolk, maize, spinach and lettuce.	Specific dermatitis called acrodynia, pellagra (shrivelled skin) anaemia and convulsions.
(v)	Vitamin-B ₁₂ or Cyanocobalamin	Milk, liver, kidney and eggs.	Inflammation of tongue, mouth etc. and pernicious anaemia.
(vi)	Vitamin-C or L-Ascorbic acid	Citrus fruits, amla, (oranges, lemons), sprouted pulses, germinated grains and leafy vegetables.	Scurvy and brittleness of bones, swelling and bleeding of gums and loosening of teeth.
(vii)	Vitamin D or Ergocalciferol (or antirachitic Vitamin of sunshine vitamin)	Fish liver oil, cod liver oil, milk and eggs.	Rickets (softening and bending of bones) in children, controls Ca and P metabolism.
(viii)	Vitamin-E or Tocopherols (α , β and γ) or Antisterility factor	Eggs, milk, fish, wheat germ, oil, cotton seed oil etc.	Sterility (loss of sexual power and reproduction)
(ix)	Vitamin-H (Biotin)	Yeast, liver, Kidney and milk.	Dermatitis, loss of hair and paralysis.
(x)	Vitamin-K or Phylloquinones or Antihaemorrhagic vitamin	Cabbage, alfalfa, spinach and carrot tops.	Haemorrhage and lengthens time of blood clotting.
(xi)	Coenzyme Q ₁₀	Chloroplasts of green plants and mitochondria of animals	Low order of immunity of body against many diseases

4.5 HORMONES

Hormones are biomolecules which are produced in the ductless (endocrine) glands and are carried to different parts of the body by the blood stream where they control various metabolic processes. These are required in minute quantities and unlike fats and carbohydrates these are not stored in the body but are continuously produced.

Some important hormones along with the sources and functions are summarised below :

NAME	ORGAN OF SECRETION	FUNCTIONS
I. Steroidal hormones		
A. Sex hormones		
(a) Androgens Testosterone)	Testes	Control the development and normal functioning of Androsterone and male sex organs.
(b) Estrogens (Estrone, Estradiol, Estriol)	Ovary	Control the development and normal functioning of female sex organs.
(c) Gestogens (Progesterone)	Corpus luteum	Control the development and maintenance of pregnancy.
B. Adrenal cortex hormones or corticoids (Cortisone, Corticosterone Aldosterone etc.)	Adrenal cortex	Regulate the metabolism of fats, proteins and carbohydrates; control the balance of water and minerals in the body.
II. Peptide hormones		
(i) Oxytocin	Posterior pituitary gland	Controls the contraction of the uterus after child birth and releases milk from the mammary glands.
(ii) Vasopressin	Posterior pituitary gland	Controls the reabsorption of water in the kidney.
(iii) Angiotensin II	Blood plasma of persons with high blood pressure.	Potent vasoconstrictor i.e. contracts the blood vessels.
(iv) Insulin	Pancreas	Controls the metabolism of glucose, maintains glucose level in the blood.
III. Amine hormones		
(i) Adrenaline or Epinephrine	Adrenal medulla	It is an amine compound and was the first hormone to be isolated. Prepares animals and humans for emergency in many ways by raising the pulse rate, blood pressure etc. stimulates the breakdown of liver glycogen into blood glucose and fats into fatty acids during emergency. These properties make adrenaline as one of the most valuable drugs used in medicine.
(ii) Thyroxine	Thyroid gland	Controls metabolism of carbohydrates, lipids and proteins.

GOLDEN KEY POINTS

- Monosaccharides which differ in configuration at first chiral carbon are called anomers. thus, α -D-glucose and β -D-glucose are anomers and so are α -D-fructose and β -D-fructose.
- The C_1 carbon atom in aldoses and C_2 carbon atom in ketoses around which the configuration of anomers differs is called anomeric or the glycosidic carbon.
- Monosaccharides which differ in configuration at any one carbon atom are called epimers. Thus, glucose and mannose which differ in Configuration at C_2 are called C_2 -epimers while glucose and galactose which differ in configuration at C_4 are called C_4 epimers.
- All monosaccharides (aldoses and ketoses) and disaccharides except sucrose reduce fehling's solution, Benedict's solution and Tollen's reagent and hence are called reducing sugars. Others (sucrose, starch, cellulose) which do not reduce these reagents are called non-reducing carbohydrates.
- The spontaneous change of specific rotation with time to an equilibrium value is called mutarotation. All reducing carbohydrates i.e. monosaccharides (glucose, fructose, mannose etc.) and disaccharides (maltose, lactose etc.) undergo mutarotation in aqueous solutions.
- Since glucose (grape sugar) is dextrorotatory, it is also called dextrose. Similarly, fructose being laevorotatory is also called laevulose.
- α -Amino acids are the building blocks of proteins or proteins are the condensation polymers of α -amino acids.
- All the α -amino acids forming proteins have L- configuration while all the naturally occurring carbohydrates have D-configuration.
- DNA undergoes replication and controls transmission of hereditary effects from one generation to the other while RNA controls synthesis of proteins.
- Vitamin D_2 is also called sunshine vitamin since it is obtained by sunlight irradiation of ergosterol present in oils and fats.
- The α -helix structure of proteins is also called 3.6_{13} helix since each turn of the helix has approx. 3.6 amino acids and a 13-member ring is formed by H-bonding.
- The deficiency of protein causes the disease called Kwashiorkor.
- Thiamine is vitamin B_1 while thymine is a pyrimidine base which occurs in DNA.

COMPETITION WINDOW

- All monosaccharides and reducing disaccharides (maltose, lactose etc.) react with three molecules of $C_6H_5NHNH_2$ to form crystalline osazones which are used for their identification and characterization. Further all monosaccharides which differ in configuration at C_1 and C_2 give the same osazone, i.e. glucose, fructose and Mannose.
- Both starch and cellulose are condensation polymers of glucose. Starch is a polymer of α -glucose, whereas cellulose is polymer of β -glucose.
- At isoelectric point, the amino acids primarily exist as the neutral dipolar or zwitterion and hence have the minimum solubility. All amino acid have their own isoelectric point.
- In amino acids. $-COO^\ominus$ group acts as the base while $-NH_3^+$ acts as the acid.

- The biological activity of proteins is due to their secondary and tertiary structures.
- Adenosine (ribose + adenine) is a nucleoside while adenosine monophosphate (AMP), adenosine diphosphate and adenosine triphosphate (ATP) are all nucleotides.
- ATP is called energy currency of the cells. It is a nucleotide containing adenine as the purine base, ribose as the sugar and three inter-linked phosphate groups.
- Energy is stored in living cells in form of ATP.
- Haemoglobin is a globular protein and the red colour of haemoglobin is due to the iron-protoporphyrin complex called the heme.
- Water soluble vitamins are B-complex and C.
- Water insoluble vitamins are A,D,E, and K.
- Vitamins which act as antioxidants are C,D and E.
- Calorific value of various food stuffs follows the Sequence : Fats > Carbohydrates > proteins.

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5					
	Ans.	3	1	1	2	3					

BEGINNER'S BOX-2	Que.	1	2	3	4	5					
	Ans.	1	1	2	3	1					

POLYMER

Polymer is defined as a high molecular weight compound formed by the combination of a large number of one or more types of small molecules (Monomers)

CLASSIFICATION OF POLYMERS :

5.1 BASED ON SOURCE

- (a) Natural polymers (b) Semi-Synthetic polymers (c) Synthetic polymers

(a) **Natural polymers:** These are of natural origin and these are found in plants and animals.

Natural polymers are also called as biopolymers.

Example Proteins (Polymers of amino acids), Polysaccharides (Polymers of mono saccharides), rubber (Polymers of isoprene) silk, wool, starch, cellulose, enzymes, haemoglobin etc.

(b) Semi Synthetic polymers :

Examples Nitro cellulose, cellulose acetate, cellulose xanthate, etc.

(c) **Synthetic Polymers** : These are artificial polymers. For example Polythene, nylon, PVC, bakelite, dacron.

5.2 BASED ON STRUCTURE OF POLYMERS.

(1) Linear polymers

→ These polymers consist of long and straight chains.

→ Ex. Polyvinyl chloride, high density polythene.

(2) Branched chain polymers

→ These polymers contain linear chain having some branches.

Ex. - Low density polythene.

(3) Cross linked polymers

→ Contain strong covalent bond between various linear polymer chains.

Ex. – Bakelite, Melamine

5.3 BASED ON MODE OF POLYMERISATION :

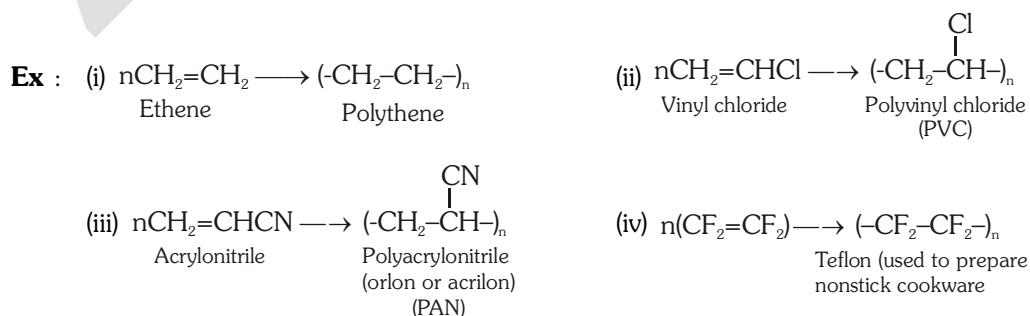
(1) Addition polymer : (Chain growth polymerisation)

→ Formed by the repeated addition of monomer molecules

→ Without elimination of any thing.

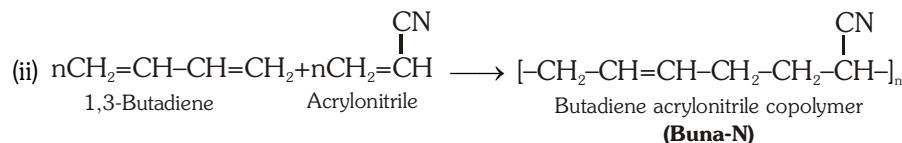
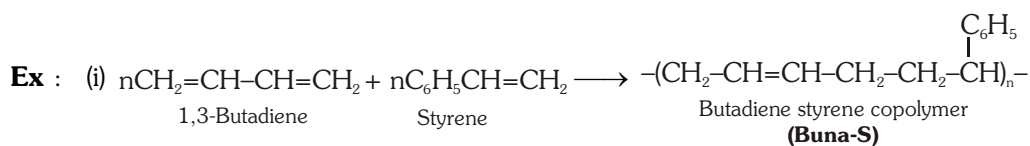
Types :

(a) Homopolymers : Addition polymers formed by the polymerisation of a single monomeric species.

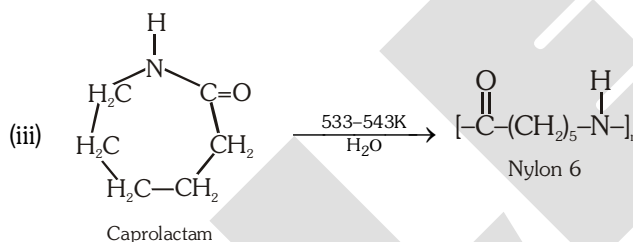
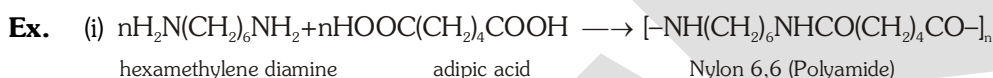
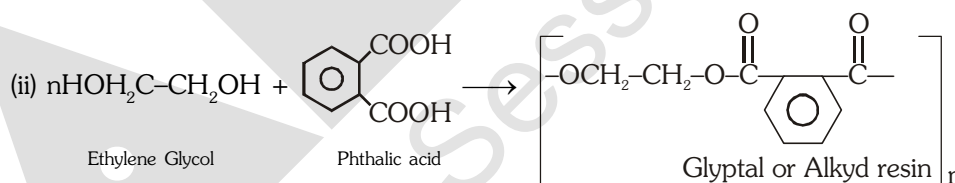
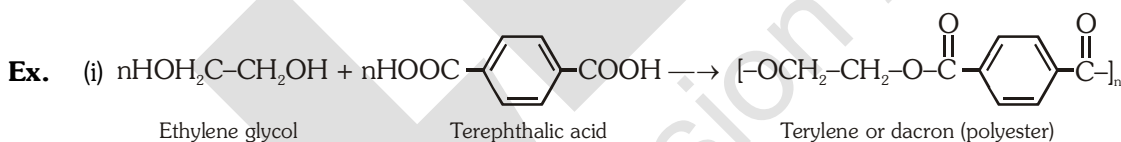
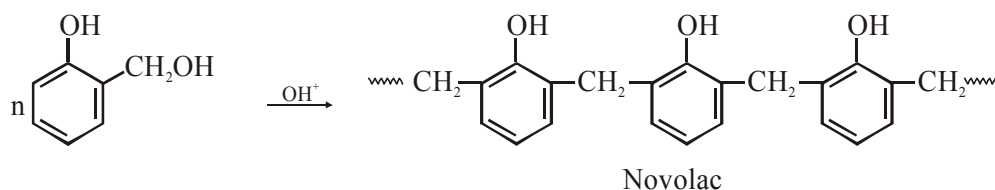
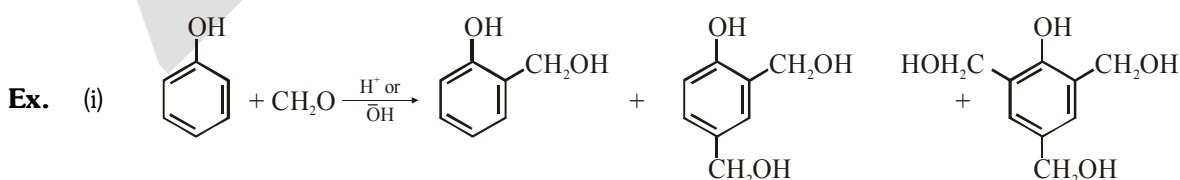


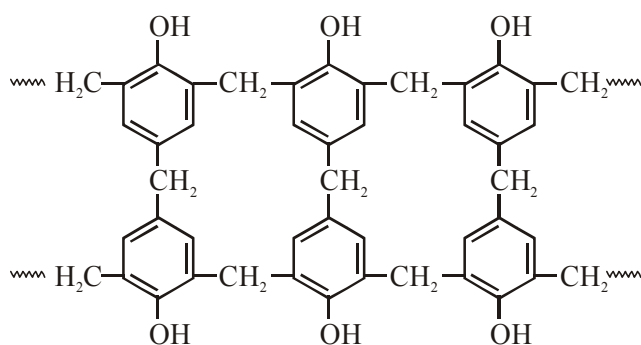
(b) Copolymers :

Addition polymers formed by the polymerisation of two different monomers.

**(2) Condensation polymers : (Step growth polymerisation)**

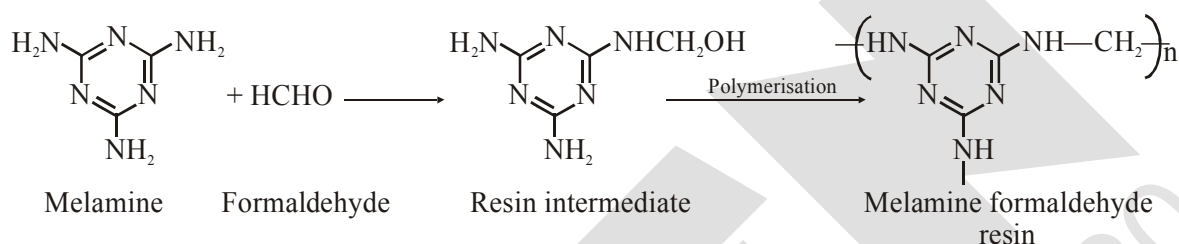
In these polymerisation reaction, the elimination of small molecules such as water, alcohol, hydrogen chloride take place.

(a) Polyamide :**(b) Polyester :****(c) Formaldehyde polymer (Bakelite and related polymers) -****(i) Bakelite :**



Bakelite

(ii) **Melamine - formaldehyde polymer -**



5.4 BASED ON MOLECULAR FORCES :

(1) **Elastomers :**

- The polymer chains are held together by the weakest intermolecular forces.
- These weak binding forces permit the polymer to be stretched.

Ex. : Buna-S, Buna-N, neoprene

(2) **Fibres :**

- Fibres are the thread forming solids which possess high tensile strength and strong intermolecular forces.

Ex. : Nylon 66, polyesters (terylene)

(3) **Thermoplastic :**

- These are the linear or slightly branched long chain molecules capable of repeatedly softening on heating and hardening on cooling.

Ex. : Polythene, Polystyrene ; polyvinyls.

(4) **Thermosetting plastic :**

- These polymers are cross linked or heavily branched molecules.
- These cannot be reused

Ex. : Bakelite, urea - formaldehyde.

Note: Correct order for Intermolecular forces in various classes of polymers.

Thermosetting > Fibres > Thermoplastic > Elastomers.

5.5 PREPARATION OF SOME IMPORTANT POLYMERS.

(1) **Low density polythene :**

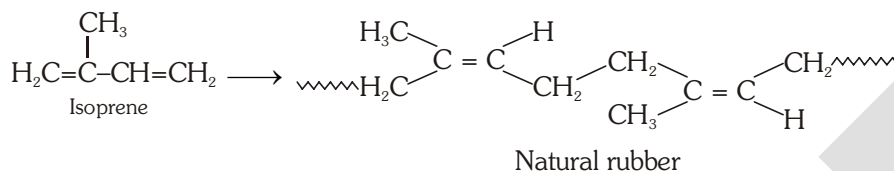
- Formed by free radical addition in presence of peroxide.
- Chemically inert and poor conductor of electricity.
- Used in the insulation of electric wires and squeeze bottles, flexible pipes.

(2) High density polythene :

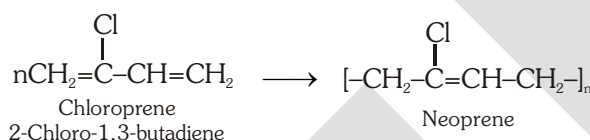
- Formed in presence of titanium tetrachloride (TiCl_4) + triethyl aluminium ($\text{C}_2\text{H}_5)_3\text{Al}$ [Ziegler – Natta catalyst]
- It is chemically inert and more tougher and harder.
- It is used for manufacturing buckets, bottles.

(3) Natural rubber :

- Linear polymer of cis-isoprene (2-methyl-1,3-butadiene) and is also called as **cis-1,4-polyisoprene**



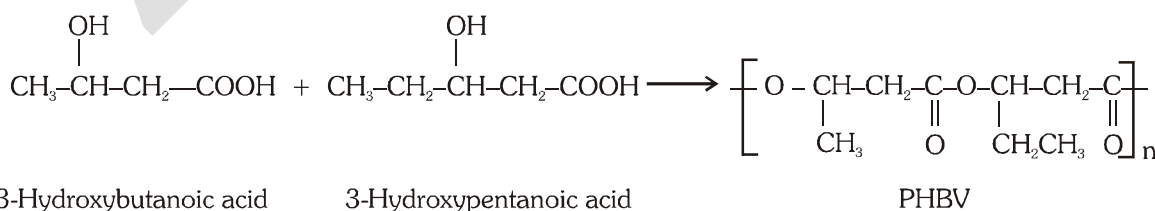
- **Vulcanisation of rubber :** Natural rubber becomes soft at high temperature. To improve upon these physical properties, a process of vulcanisation is carried out. This process consists of heating a mixture of raw rubber with sulphur. On vulcanisation, sulphur forms cross links at the reactive sites of double bond.

(4) Synthetic rubber (Neoprene)**5.6 BIODEGRADABLE POLYMERS****Introduction**

A large number of polymers are quite resistant to the environmental degradation processes and are thus responsible for the accumulation of polymeric solid waste materials. These solid wastes cause acute environmental problems and remain undegraded for quite a long time. In view of the general awareness and concern for the problems created by the polymeric solid wastes, certain new biodegradable synthetic polymers have been designed and developed. These polymers contain functional groups similar to the functional groups present in biopolymers.

Examples :**(1) Poly (β -hydroxybutyrate- β -hydroxy valerate) (PHBV)**

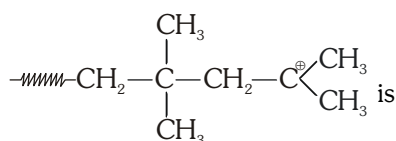
It is obtained by the copolymerisation of 3-hydroxybutanoic acid and 3-hydroxypentanoic acid. PHBV is used in speciality packaging, orthopaedic devices and in controlled release of drugs. PHBV undergoes bacterial degradation in the environment.

**(2) Nylon 2-nylon 6**

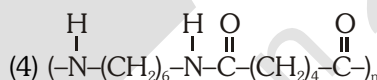
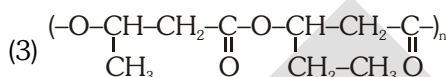
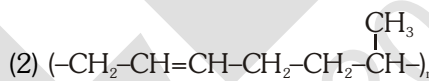
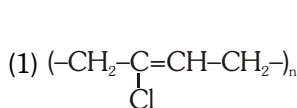
It is an alternating polyamide copolymer of glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$) and amino caproic acid [$\text{H}_2\text{N}(\text{CH}_2)_5\text{COOH}$].

BEGINNER'S BOX-1

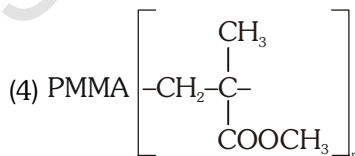
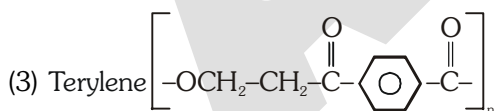
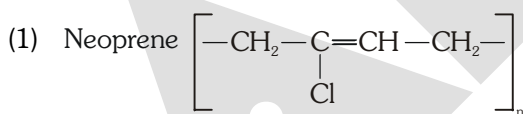
- Which of the following is a biodegradable polymer
 - (1) Cellulose
 - (2) Polythene
 - (3) Polyvinyl chloride
 - (4) Nylon-6
- Which one of the following is a chain growth polymer
 - (1) Nucleic acid
 - (2) Polystyrene
 - (3) Protein
 - (4) Starch
- The monomer of the polymer –



- (1) $\text{CH}_3\text{CH}=\text{CHCH}_3$
- (3) $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)_2$
- 4.** Which of the following polymer is biodegradable



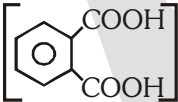
- 5.** Which of the following is not correctly matched –



6. Monomer of $\left[\begin{array}{c} \text{CH}_3 \\ | \\ -\text{C}-\text{CH}_2- \\ | \\ \text{CH}_3 \end{array} \right]_n$ is —

- (1) 2-methyl propene (2) Styrene
(3) Propylene (4) Ethene

MONOMERS AND POLYMERS

S.N.	Monomer	Polymer	Type of Polymers
1.	$\text{CH}_2=\text{CH}_2$ (Ethylene)	Poly ethene	Addition polymer
2.	$\text{CH}_2=\text{CHCH}_3$ (Propylene)	Poly propylene	Addition homo polymer
3.	$\text{CH}_2=\text{CHCl}$ (Vinyl chloride)	Polyvinyl chloride (PVC)	Homopolymer, chain growth
4.	$\text{CH}_2=\text{CH}-\text{C}_6\text{H}_5$ (Styrene)	Polystyrene (styron)	Addition homo polymer, linear chain
5.	$\text{CH}_2=\text{CH}-\text{CN}$ (Acrylonitrile)	Ployacrylonitrile (PAN) or Orlon	Addition homopolymer
6.	$\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ (1,3 Butadiene) + Styrene	BUNA-S rubbers	Addition copolymer
7.	$\text{CH}_2=\text{CHOCOCH}_3$ (Vinyl acetate)	Poly vinyl acetate (PVA)	Addition homopolymer
8.	$\text{CF}_2=\text{CF}_2$ (Tetrafluoro ethylene)	Teflon	Chain growth homopolymer (Nonstick cookwares)
9.	$\text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{CH}=\text{CH}_2$ (Isoprene)	Natural Rubber	Addition homopolymer
10.	$\text{CH}_2=\underset{\text{Cl}}{\text{C}}-\text{CH}=\text{CH}_2$ (Chloroprene)	Neoprene (Artificial Rubber)	Addition homopolymer
11.	Ethylene Glycol + terephthalic acid	Terylene or Dacron (Polyester)	Condensation Copolymer, step growth
12.	Hexamethylene diamine + adipic acid	Nylon-6,6 (Polyamide)	Copolymer, step growth linear
13.	Formaldehyde + urea	Urea formaldehyde resin	Copolymer, step growth
14.	Formaldehyde + Phenol	Bakelite	Copolymer, step growth
15.	Methyl methacrylate $\text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{COOCH}_3$	Poly methyl meth acrylate (PMMA)	Addition homopolymer
16.	Ethylene Glycol + Phthalic acid 	Glyptal or alkyd resin	Copolymer, linear step growth, thermo plastic
17.	Melamine + formaldehyde	Melamine formaldehyde resin	Copolymer, step growth thermosetting polymer
18.	Hexamethylene diamine + sebacic acid	Nylon - 6,10	Copolymer, step growth linear
19.	Caprolactam	Nylon - 6	Homopolymer, step growth linear

ANSWER KEY

BEGINNER'S BOX-1		Que.	1	2	3	4	5	6				
		Ans.	1	2	4	3	3	1				

CHEMISTRY IN EVERY DAY LIFE

6.1 DRUGS

Drugs :

Drugs are the chemicals of low molecular masses which interact with macromolecular target and produce a biological response.

Medicines :

Medicines are the drug which are therapeutic and used for diagnosis, prevention and treatment of diseases.

CLASSES OF DRUGS

(A) Antiseptics :

Which prevent or destroy the growth of the harmful micro organism on living organism externally like wound, cut or skin surface. Common antiseptics are-

Dettol, Savlon, Cetavlon, acriflavin, Iodine, methylene blue, mercurochrome, Furacine, Soframicine

Dettol is a mixture of chloroxenol and terpineol.

Bithional -It is added to soap to impart antiseptic properties

Iodoform is also used as an antiseptic for wounds. Boric acid in dilute aqueous solution is weak antiseptic for eyes.

(B) Disinfectants :

The chemical compounds capable of completely destroying the micro organism on non-living objects are termed as disinfectants. These are toxic to living tissues.

These are utilized for sterilization of floor, toilets instruments and cloths.

Same substance can act as an antiseptic as well as disinfectant by varying the concentration.

eg. 1% solution of phenol in disinfectant while 0.2% solution of phenol is antiseptic.

(C) Analgesics :

The substance which are used to get relief from pain. These are of two types -

(a) Narcotics or habit forming drugs

(b) Non-narcotics or non-addictive

(a) Narcotics : These are alkaloids and mostly opium products, causes sleep and unconsciousness when taken in higher doses.

e.g. Morphine, codeine, heroin

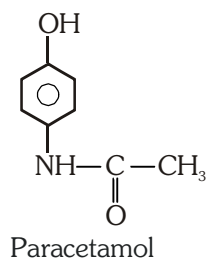
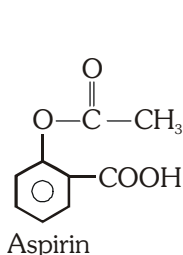
(b) Non-narcotics : Analgesics belonging to this category are effective antipyretics also.

e.g. Aspirin, Novalgin, Ibuprofen and Naproxen

(D) Antipyretics :

To bring down the body temp. in high fever are called antipyretics.

e.g - (a) Aspirin, (b) Analgin (Novalgin), (c) Paracetamol, (d) Phenacetin



(E) Antimalarials :

Drug which is used to prevent or cure malaria.

e.g. Quinine, Chloroquine, Paraquine and Primaquine etc.

(F) Tranquilizers :

The chemical substances which acts on the central nervous system and has a calming effect.

Since these are used for mental diseases so are known as psychotherapeutic drugs.

They are of two types - (a) Sedative or hypnotics (b) Mood elevators

(a) **Sedative** : Reduce nervous tension and promote relaxation. e.g. Reserpine, barbituric acid and its derivatives as luminal, seconal & veronal.

(b) **Mood elevators or Antidepressants** : A drug used for treatment of highly depressed patient, who has lost confidence.

e.g. Benzedrine (amphetamine), Iproniazid and phenelzine

Chlordiazepoxide and meprobamate are relatively mild tranquilizers suitable for relieving tension.

(G) Anesthetics :

These are chemical substances used for producing general or local insensibility to pain and other sensation.

These are of two types (a) General (b) Local

(a) **General** : - Produce unconsciousness and are given at the time of major surgical operations.

e.g. - Gaseous form → Nitrous oxide, ethylene, cyclopropane etc.

Liquid form → Chloroform, divinyl ether and sodium pentothal etc.

(b) **Local anaesthetics**: Produce loss of sensation on a small portion of the body. It is used for minor operations.

e.g. Jelly form → Oxylocain

Spray form → Ethyl chloride

Injection form → Procain

(H) Antibiotics : Initially antibiotic were the chemical substances produced from some micro organism (fungi bacteria or mold) and are used to inhibit the growth or even destroy micro organism.

But now, antibiotics are substances produced wholly or partially by chemical synthesis which in low concentrations inhibits the growth or destroy micro-organism by intervening in their metabolic processes.

These are effective in the treatment of infections diseases.

The range of bacteria or other micro-organisms that are affected by a certain antibiotic is expressed as its spectrum of action.

Three types of spectrum of action :

(i) **Broad spectrum antibiotics** – Antibiotics which kill or inhibit a wide range of gram-positive and gram-negative bacteria. e.g. Ofloxacin, Ampicillin, Amoxycillin, Chlorapheniol

(ii) **Narrow spectrum antibiotics** – Those effective mainly against gram-positive or gram-negative bacteria are narrow spectrum antibiotics. e.g. – penicillin G

(iii) **Limited spectrum antibiotic** – Those effective against a single organism or disease are limited spectrum antibiotics. e.g. – dysidazine – toxic toward certain strain of cancer cell.

Antibiotics have either cidal (killing) effect or a static (inhibitory) effect on microbes. A few example of the two types are –

Bactericidal

Penicillin

Aminoglycosides

Ofloxacin

Bacteriostatic

Erythromycin

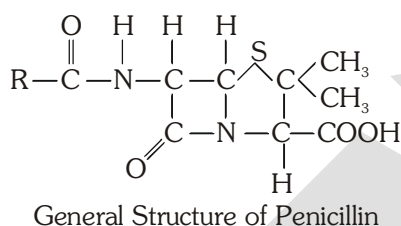
Tetracycline

Chloramphenicol

eg. Salvarsan, Arsphenamine, Prontosil (Sulphanilamide is active part)

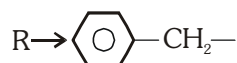
Penicillin - It is highly effective drug for pneumonia, Bronchitis, abscesses, sore throat etc.

Other naturally occurring penicillin -

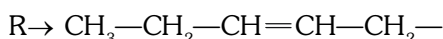


General Structure of Penicillin

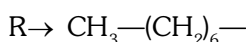
R - May be -



(Penicillin G)



(Penicillin-F)



(Penicillin-K)

Note: Ampicillin & Amoxycillin are synthetic penicillin.

Synthetic antibiotics are Streptomycin, Chloromycetin, Tetracyclins

(I) Antacids

Over production of acid in the stomach causes irritation and pain. In severe cases, ulcers are developed in the stomach. Only treatment for acidity was administration of antacids, such as sodium hydrogencarbonate or a mixture of aluminium and magnesium hydroxide. However, excessive hydrogencarbonate can make the stomach alkaline and trigger the production of even more acid. Metal hydroxides are better alternatives because of being insoluble, these do not increase the pH above neutrality.

A chemical, histamine, stimulates the secretion of pepsin and hydrochloric acid in the stomach. The drug cimetidine (Tegamet), was designed to prevent the interaction of histamine with the receptors present in the stomach wall. This resulted in release of lesser amount of acid. Other e.g. – Ranitidine (Zantac)

(J) Antihistamine

Histamine is responsible for the nasal congestion associated with common cold and allergic response pollen.

Synthetic drugs, brompheniramine (Dimetapp) and terfenadine (Seldane), act as antihistamines. They interfere with the natural action of histamine by competing with histamine for binding sites of receptor where histamine exerts its effect.

Ques. : Why do antihistamines not affect the secretion of acid in stomach ?

Ans. : Reason is that antiallergic and antacid drugs work on different receptors.

6.2 ARTIFICIAL SWEETENING AGENTS

Natural sweeteners, e.g., sucrose add to calorie intake and therefore many people prefer to use artificial sweeteners. Ortho-sulphobenzimide, also called saccharin, is the first popular artificial sweetening agent. It is about 550 times as sweet as cane sugar. It is excreted from the body in urine unchanged. It appears to be entirely inert and harmless when taken. Some other commonly marketed artificial sweeteners and their relative sweetness value in comparison to canesugar is :-

Artificial Sweetner	Sweetness value in Comparison to cane sugar
Aspartame	100
Saccharin	550
Sucralose	600
Alitame	2000

6.3 FOOD PRESERVATIVES

Food preservatives prevent spoilage of food due to microbial growth. The most commonly used preservatives include table salt, sugar, vegetable oils and sodium benzoate. Sodium benzoate is used in limited quantities and is metabolised in the body. Salts of sorbic acid and propanoic acid are also used as preservatives.

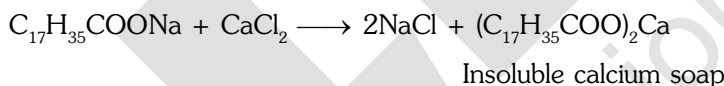
6.4 CLEANSING AGENTS

These are soaps and synthetic detergents. These improve cleansing properties of water. These help in removal of fats which bind other materials to the fabric or skin.

6.4.1 Soaps

Soaps are sodium or potassium salts of long chain fatty acids, e.g., stearic, oleic and palmitic acids. Soaps containing sodium salts are formed by heating fat (i.e., glyceryl ester of fatty acid) with aqueous sodium hydroxide solution. This reaction is known as saponification.

Soap does not work in hard water due to presence of calcium and magnesium ion in hard water. These ions form insoluble calcium & magnesium soap respectively.



These insoluble soap separate as scum in water and are useless as cleansing agent.

Types of soaps

- Toilet soaps • Transparent soaps • Medicated soaps • Shaving soaps • Laundry soaps

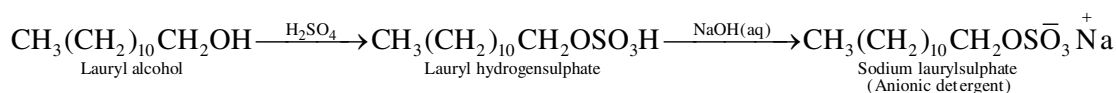
6.4.2 Synthetic Detergents

Synthetic detergents are cleansing agents which have all the properties of soaps, but which actually do not contain any soap. These can be used both in soft and hard water as they give foam even in hard water.

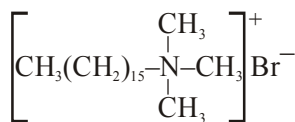
Synthetic detergents are mainly classified into three categories:

- (i) Anionic detergents (ii) Cationic detergents and (iii) Non-ionic detergents

- (i) **Anionic Detergents:** Anionic detergents are sodium salts of sulphonated long chain alcohols or hydrocarbons. Alkyl hydrogensulphates formed by treating long chain alcohols with concentrated sulphuric acid are neutralised with alkali to form anionic detergents. The anionic part of the molecule is involved in the cleansing action. Mostly used for household work and in toothpastes.

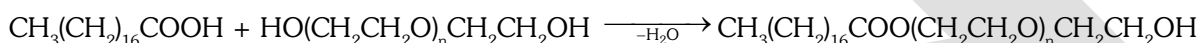


- (ii) **Cationic Detergents:** Cationic detergents are quarternary ammonium salts of amines with acetates, chlorides or bromides as anions. Cationic part possess a long hydrocarbon chain and a positive charge on nitrogen atom. Hence, these are called cationic detergents. Cetyltrimethylammonium bromide is a popular cationic detergent and is used in hair conditioners.



Cetyltrimethyl ammonium bromide

- (iii) **Non-ionic Detergents:** Non-ionic detergents do not contain any ion in their constitution. One such detergent is formed when stearic acid reacts with polyethyleneglycol.



Liquid dish washing detergents are non-ionic type.

GOLDEN KEY POINTS

- Aspirin is used to prevent heart attacks besides being antipyretic and analgesic agents.
- Derivatives of barbituric acid, viz. veronal, amytal, nembutal, luminal and seconal are hypnotic tranquillizer while meprobamate equanil, valium and serotonin are non-hypnotic tranquillizers.
- Soaps, detergents and phospholipids are called surfactants since they lower the surface tension of water.
- All surfactants consist of two characteristic groups, i.e., apolar head group which is water-soluble (hydrophilic group) and a non-polar hydrocarbon tail which is oil-soluble (lyophilic or lipophilic group).
- Sodium soaps are hard while potassium soaps are soft. Therefore, washing soaps are mostly sodium soaps while liquid soaps having creams and toilet soaps are potassium salts.
- Unlike soaps, detergents can be used in hard water. The reason being that magnesium and calcium salts of detergents are soluble in water while those of soaps are insoluble in water.
- Aspirin, phenacetin and paracetamol act both as antipyretics and analgesics.
- The alkaloid morphine and its derivatives such as codeine (morphine methyl ether) and heroin (morphine diacetate) are important narcotic analgesic.
- Aspirin is a non-narcotic analgesic but is toxic to liver. It also undergoes hydrolysis in the stomach producing salicylic acid which causes bleeding from the stomach wall. Therefore, other non-narcotic analgesics such as naproxen, ibuprofen and diclofenac sodium or potassium are preferred to aspirin.
- Enovid E which is a mixture of norethindron (a progestogen) and ethynysestradiol (an estrogen) is the most commonly used oral contraceptive.
- AZT (3'-Azido-3-deoxythymidine) is used against AIDS i.e. HIV-infections.
- Sulpha drugs are effective against bacterial infections.
- Ciprofloxacin and norfloxacin are quinolone based antibacterial drugs.
- Artificial sweetners have no caloric value and hence are useful for diabetic persons.